volume I

The Theory of Money and Financial Institutions

Martin Shubik



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volume 1

Martin Shubik

This is the first volume in a three-volume exposition of Martin Shubik's vision of "mathematical institutional economics"—a term he coined in 1959 to describe the theoretical underpinnings needed for the construction of an economic dynamics. The goal is to develop a process-oriented theory of money and financial institutions that reconciles micro- and macroeconomics, using as a prime tool the theory of games in strategic and extensive form. The approach involves a search for minimal financial institutions that appear as a logical, technological, and institutional necessity, as part of the "rules of the game." Money and financial institutions are assumed to be the basic elements of the network that transmits the sociopolitical imperatives to the economy.

Volume 1 deals with a one-period approach to economic exchange with money, debt, and bankruptcy. Volume 2 explores the new economic features that arise when we consider multiperiod finite and infinite horizon economies. Volume 3 will consider the specific role of financial institutions and government, and formulate the economic financial control problem linking micro- and macroeconomics.

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Volume 1

Martin Shubik

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To my wife Julie and daughter Claire;
to the memory of my parents,
Hillary Belloc, Earl Isaac, Harry Johnson, Oskar Morgenstern, and
Marian Shapley.

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Preface

My thoughts about money and financial institutions began in 1960. I tried without success to make progress until the end of 1970, when I decided to abandon my work on money. I decided to devote my time to my second most important project, embedding the Cournot oligopoly model in a fully closed economic system. Much to my surprise, the second problem held the key to the first. The need to build a process model of price formation in a closed economy calls for the invention of money and other financial instruments and institutions. Money provides the extra degree of freedom needed to provide the leeway for disequilibrium in a closed economic system. The understanding of how its supply is counted, what are its close substitutes, and what are the forces that control its supply is critical to reconciling the studies of micro- and macroeconomics. The essence of understanding money lies in the dynamics.

Money and financial institutions provide the structure to carry process in a disequilibrium economy. They provide a control and evaluation network for the economy. The development of a theory of money and financial institutions provides the first step toward understanding some of the forces behind economic structure. The future development of economics lies in obtaining insights into the control of process in a world in which evaluation, innovation, and disequilibrium are of central concern.

The development of our understanding of how expectations are formed and how innovation is triggered, or how expertise is developed, is critical to devising a new economic dynamics. In common with basic thought in biology and physics, there is a need to understand mass cell, particle, or agent behavior in a stochastic environment subject to some partially controlling agency governed by feedback. The work here is offered as a step toward shifting the stress from an essentially timeless, institution-free, and process-free economic equilibrium to one where process, time, information, and institutions matter.

Money, like language, is a network phenomenon. Its role cannot be considered in isolation. The simplicity or complexity of its function depends on the body of common knowledge, the shared context perceived by individual agents and the communication net. The value of fiat money lies in the dynamics, not in a static timeless equilibrium. It is a property of mass agent stochastic behavior.

The link between optimizing behavior often described in microeconomic theory and evolutionary behavior is provided by "the games within the game." At any point in time the institutions, laws, and customs

provide the rules of the game for individuals who tend to behave as local optimizers. But on the larger horizon, the local optimization of the economic agents combined with chance and social, political, and technological process conspire to change the rules of the game.

Volume 1 deals primarily with a one-period approach to economic exchange with money, debt, and bankruptcy. Volume 2 indicates the nature of the new economic features introduced in considering multiperiod finite and infinite horizon economies. A projected Volume 3 considers the specific role of financial institutions and government and formulates the economic financial control problem linking micro- and macroeconomics. A companion volume with Pradeep Dubey and John Geanakoplos providing a mathematical treatment of strategic market games is projected.

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I am indebted to the support and stimulus provided by my colleagues at the Cowles Foundation at Yale, the Rand Corporation, and the Santa Fe Institute. I count myself fortunate to have never recovered from the stimulus of the intellectual ferment present and friends made at graduate school at Princeton in the 1950s, in particular Lloyd Shapley, Marvin Minsky, John McCarthy, John Nash, Herbert Scarf, and Ralph Gomory in the mathematics department, Colin Pittendrigh in biology, Oskar Morgenstern and Bill Baumol in economics, and Martin Whitman and Djhangir Boushehri in applied finance.

I wish to acknowledge the many colleagues whose joint work and observations have been of great value to me. In particular, Lloyd Shapley and Pradeep Dubey have been joint authors with me of many papers. John Geanakoplos, Ward Whitt, Matthew Sobel, Andreu MasColell, Shuntian Yao, Kofi Nti, John Rogowski, Tom Quint, and Charles Wilson have collaborated with me over the years. More recently, as my concerns have turned toward the possibilities for developing dynamics, the collaboration with Ioannis Karatzas and William Sudderth has been a pleasure. The first steps into microsimulation of these monetary models have been taken with John Miller and Nick Vriend.

Among my former students, research assistants, and later colleagues, Tanya Styblo Beder, Krishna Jawadene, Jingxsi Liu, Siddharta Sahi, Guytag Oh, and Dimitri Tsomocos have worked with me and I have

learned from them. Dimitri in particular made many excellent observations and caught errors in the early versions of the manuscript.

I regard these volumes as preliminary to the eventual development of a dynamics that will change the nature of economic analysis. I believe that the future may lie with development of methods to understand "mass particle behavior" of many heterogeneous agents. Discussions with Brian Arthur, Benoit Mandelbrot, Stuart Kaufmann Atlee Jackson, Robert Shiller, and Per Bak have convinced me that new directions are at hand.

Conversations with Stuart Kaufmann and Leo Buss lead me to believe that the diverse interests of economics and biology in the approach to the elementary understanding of the genesis of organization are closing to the point that interdisciplinary work is fruitful.

The perceptive works of Hyman Minsky and Axel Lejonhufvud and Jim Tobin and my other macroeconomic colleagues at the Cowles Foundation have influenced my thoughts more than they may realize, as have my discussions with Truman Bewley, Tom Rutherford, and Tom Quint.

Tanya Styblo Beder, Martin Whitman, Meyer Berman, and several other friends have helped me to appreciate the world of finance as it is, with constant change and high uncertainty yet with highly specialized boutiques of excellence working successfully in a manner hardly understood by current financial theory.

I owe special thanks to Glena Ames, who with great skill and goodwill has handled the original manscript and trudged through interminable revisions and rerevisions, to Dee Smitkowski for able assistance, and to support for research assistance from the Aequus Foundation.

I CONTEXT

1 Introduction and Background

1.1 Mathematical Institutional Economics

1.1.1 Money and financial institutions

The theory of money and financial institutions focuses primarily on the economizing of trust and the evaluation of information in economic life. We can satisfy the need to economize trust in a world with many mass, anonymous markets by separating the economy into a "real" or transactions sector and a financial sector. Mass, anonymous, trustless trade can be carried out in the real sector with little information processing and evaluation. The financial sector, however, must deal in credit and be able to evaluate the trustworthiness of the individuals and institutions with whom it exchanges commodity or fiat money in return for personal promises.

The economizing of trust is usually set in the context of price-guided economies with an economically powerful government sector. Taken as given is a level of acceptance of "rules of the game" that are made manifest in laws such as those governing property, contracts, bankruptcy, inheritance, taxes, and commerce.

This work will attempt to reconcile micro- and macroeconomic theory through the construction of a process-oriented theory of money and financial institutions. Its prime methodological tool will be the theory of games in strategic and extensive form.

1.1.2 Theory and facts

An existing theory cannot be supplanted by mere facts. It can be supplanted only by other theories that explain whatever was explained correctly by the original theory but go beyond that theory to explain phenomena with which it was unable to cope. One aim of this work is to construct process models of the economy that are as rigorous as general-equilibrium theory but explain more phenomena.

One can, of course, improve on a theory without constructing a truly adequate alternative. Our improvements over general equilibrium will come in the modeling of both mechanism and the carriers of economic process but not particularly in the description of human economic behavior or motivation. The concepts of the utility function and of economic behavior as the conscious optimization of expected utility leave much to be desired, especially in considerations of long-term behavior. It

may be argued that individuals behave "as if" they were maximizing a utility function, but when one contemplates the literature on revealed preference or much of social-welfare theory, one finds that little attention has in fact been paid to the mechanisms and the social environment that intervene between actual behavior and intrinsic preference structures (if such exist). General-equilibrium theory and much of the rest of microeconomic theory are based on models of individuals, or of artificial individuals called firms, attempting to maximize utility or profit. An elegant logical structure can be erected on these flimsy behavioral assumptions. In contrast, much of macroeconomic theory is a patchwork of behavioral assumptions, often designed to provide plausible and operational insight into the crisis of the moment or the policy of the day.

As a science, political economy is still in its infancy. The rational, economic, institution-free individual assumed in its models is an overly simplistic abstraction of a subtle and complex creature who must function with high constraints on perception and ability to calculate, who uses both internal and external memory, and whose goals can hardly be well defined outside the context of the society in which the individual functions. Perhaps in the next few decades, through a combination of experiment and observation, we can create well-defined models of politicoeconomic and socioeconomic behavior. Now, however, we must accept a patchwork of models of motivation and understand that the acceptance or rejection of their value must be based upon a careful delineation of the questions they are meant to answer and the restrictions that must be placed on their domain of applicability.

In much of what follows, I shall make the usual assumptions concerning the existence of well-defined preferences and utility functions, not because I believe in them, but because I want to show that even with these assumptions we can go beyond the results of general-equilibrium theory. Moreover, even without invoking the maximization of utility or using equilibrium as a solution concept, we shall find that the process of setting up playable games (regardless of equilibrium concepts) forces us to invent minimal (financial and governmental) institutions.

When one makes special assumptions concerning thickness of competition and equilibrium, both game-theoretic and general-equilibrium models often predict the same apparently general, noninstitutional outcome. Out of equilibrium, however, institutions appear as a basic necessity for the conveyance of process only in the game-theoretic models.

Even in equilibrium, differences between general-equilibrium theory and the game-theoretic approach emerge when strategic considerations call for specification of items such as cash flow constraints and default penalties, *even though* in equilibrium everyone meets their cash flow requirements and no one defaults.

The economy is the engine of the polity. The financial institutions and instruments of a modern society provide the control mechanism for the economy, and the political, bureaucratic, and other elites serve as the fiduciaries responsible for directing the control mechanism. The accountability of these fiduciaries to the populace varies with the political, social, and legal structure of the society.

The institutions of society in general, and the financial institutions in particular, are the carriers of process. They are the neural network and the sensors of the body economic, guiding the flows of funds, credits, and other financial paper that in turn guide the real goods and services of the economy.

Some time ago, while considering economic dynamics, I coined the term *mathematical institutional economics* (Shubik, 1959b, pp. 1, 8). The economic theories of production and distribution in particular require an understanding of process. Before we reach a point where we can deal adequately with dynamics, however, we can take an intermediate step that involves understanding the purpose of laws, institutions, and customs as part of the structure or "rules of the game" that define and give shape to economic life.

The approach to economic dynamics I shall describe involves a search for minimal financial institutions whose presence emerges as a logical, technological, and institutional necessity when economic activity is described as a playable game.[1]

Those accustomed to manipulating broad abstractions might feel that an insistence on producing models that can be played as games is unnecessarily limiting and somewhat ad hoc. It is the specific thesis developed here that an insistence on models that are defined as playable games creates a logical need to invent all the financial instruments known to society. The essence of money, credit, insurance, commercial banking, central banking, ownership claims, and futures contracts can be approached from a minimalist viewpoint: What is the simplest game that can be constructed in which the properties of these instruments are clearly identifiable?

A theory of money and financial institutions must be able to deal with dynamics. As costs, habits, population, and technology change, the special financial institution of yesterday becomes the public market of today. Private institutions and public markets are both carriers of economic process, and as such they are part of the mechanism of exchange and commerce. Institutional forms evolve with the passage of time, but the essential economic purposes remain and are served in the different institutional manifestations.

1.1.3 On the time scales of the behavioral sciences

The various behavioral sciences all investigate the behavior of individuals and societies, but they differ in the time scales and styles of decision making on which they focus. One might, for example, order them by time scale as follows: biology, anthropology, history, sociology, political science, economics. The smaller the time scale, the more likely it is that models of conscious optimizing behavior will appear. Thus the influence of "rational individual behavior" is strongest in microeconomics and operations research. When the time scale is small—when reaction is required virtually instantaneously—preconditioning and instinct take over from reasoned calculation. When the time scale is in months or years, habit, social convention, and custom may predominate. And of course the list above is not exhaustive; in particular, law, psychology, and demography do not appear. They cannot be easily characterized by time scale, and the nature of decision making in each involves a blend of which *Homo economicus* forms only a small part.

An "economic theory imperialist" might claim that an economic decision-making mode exists for every form of human behavior. While such an approach might provide some operationally useful analogies and insights, by attempting to explain too much, it runs the risk of becoming vacuous. Consider, for example, Samuel Butler's provocative insight "The hen is only an egg's way of making another egg" (Butler, 1878, p. 134). Following this mode of thought, one could argue that the only "true individuals" are genes. A utility-maximizing double helix might see the body in which it is located as a mechanism to be manipulated in order to produce a better double helix. The strategy of this ruthlessly individualistic creature could involve all sorts of subterfuge, including making its host act as though it were altruistic and cooperative.

It is my belief that there is no "best model" to explain the growth of population, the support of the young and the elderly, or the transmission of wealth. There are as many best models as there are well-defined questions to be answered. Economic analysis may help but, on its own, will not answer more than a fraction of the many important questions that can be asked by those concerned with either policy or theory.

In recent years, scholars have presented economic models of the family, marriage, crime, and so on (see Becker, 1975, for example). Some might feel that it is in bad taste to model a child as a consumer durable, but scholarship is not primarily a matter of good or bad taste. The pertinent questions to ask of such a model are: What questions does it answer? How well are those questions answered? How interesting are the questions? And does this approach succeed better than other approaches? That is, can economists answer important questions concerning population control and motivation for family size better than sociologists, demographers, biologists, or others?

Every society has its laws, customs, institutions, and formal and informal organizations. In any given society, tax rates, subsidies, and the price of corn may or may not be easier to change than work hours, habits of frugality, or traditions of craftsmanship. And it may or may not be easier to change work hours than religious beliefs or attitudes toward neighboring societies. Thus, even if economics offers only a very limited set of insights into the understanding of population dynamics, as a matter of policy one might argue that we should concentrate on these insights because they are relevant to the only areas we can control. We can change a tax structure in a year or two, but we can scarcely influence a religious belief in a decade.

This book deals with the role of money and financial institutions as the mechanisms that bind the economy to the polity and society. Taxes and subsidies, savings, wealth, inheritances, and gifts are all controlled in varying degrees and can be manipulated by the command and control system provided by the financial structure of all modern societies.

The institutions of the financial system are the neural network that serves to transmit the imperatives of sociopolitical purpose to the economy.

The reconciliation of macro- and microeconomics calls for the joining together of the more limited and detailed approach of modern microeconomic theory with the political economy of nations.

1.1.4 The invention of institutions

A theory of money and financial institutions requires a blending of anthropology, law, accounting, history, institutional insight, political economy, and mathematical model building. The economic view of human beings must be set appropriately in the context of polity and society. Economic behavior is not all of human behavior, and economists must always be concerned with the boundaries of their models. The context in which the control mechanisms of the economy function must be made explicit. Yet, even though social, political, and technological circumstances vary from society to society, some fruitful generalizations can nevertheless be made.

This book is primarily concerned with political economy, not anthropology or economic history. Thus our fascination with specific institutions or financial instruments such as Lloyds of London or the Baltic Exchange for shipping, auctions *sub hasta*, checks written on the back of a cow, stone money, bars of salt, and so forth must be tempered with a basic interest in what fundamental need these were designed to fill and what purpose they ended up serving.

A key lesson from the study of financial instruments and institutions was noted by Bagehot (1873): instruments and institutions originally invented for one purpose often take on a life of their own. Thus, for example, a warehouse receipt originally used to certify gold on deposit can, by a slight change in the rules to enable endorsement to a third party, become a type of gold-backed money. Government debt, originally incurred to pay for a war, may become sufficiently large that a government can exert control on the economy by manipulating its retirement or refinancing.

Another key lesson is that the name of a financial instrument may remain unchanged over time and circumstances while its properties change. Thus two instruments with different names may have the same economic and financial properties at some points in time.

The title of this book refers to both money and financial institutions because it is not fruitful to study one without the other. "moneyness," whatever that is, is a systemic property that depends upon the rules of the game. Money is what money does, and thus money in France is slightly different from money in the United States because the rules governing its functioning are not the same in the two countries. For many purposes of

application, these slight differences can be ignored, but the decision to ignore them must be recognized as an ad hoc modeling decision and not a matter of a priori reasoning.

Because the operation of financial instruments is best understood in the context of the system in which they operate, we must make an explicit effort to model that system. The laws governing trade and commerce and the economic and financial institutions of a society provide the raw material for the specification of the rules of the game.

The methods of game theory, in which much stress is laid upon the explicit description of rules, provide appropriate means for formalizing economic models that portray the essentials of financial institutions. The rules delimit the types of processes that are feasible. At a high level of abstraction, the basic functions of banks, insurance companies, central banks, treasuries, markets, money, bonds, and other debt and equity instruments should appear as logical or technological necessities. We may, however, need extra detail pertaining to specific societies to make the distinction between commercial banks in England and Japan or between the protection offered by the laws and customs governing the rights of common stockholders in Australia and the United States. A good general theory should be open in a natural way to absorbing specific detail in application. (When a theory that appears to be general and abstract cannot be used to study specific features of the economy, we may suspect the existence of implicit assumptions that disallow certain developments and that cannot be adjusted without radically revising the theory itself. Thus, for example, general-equilibrium theory cannot be easily adjusted to account for oligopolistic sectors.)

Our formulation of exchange and production as a game will provide a context for the emergence of institutions as a natural consequence of trade and taxation. Trade, for example, leads to a need for money. Efficiency then requires credit, which in turn requires a credit or banking system. The specification of a credit system requires laws, customs, and institutions that determine the nature of contracts, seniority, insolvency, bankruptcy, and bank failure.

Another path to modern financial institutions came not from trade but directly from government. Taxation in goods and services by tribal chiefs or kings has been replaced by taxation in money, which instead of going to the sovereign's private treasury goes to a national treasury and is used to finance public works and national wars.

A financial system's essential functions may be fulfilled in different societies by totally different institutions. For example, the care of the elderly may be the responsibility of churches, a social security system, the tribe, the extended family, pension plans, or insurance companies. To be successful, a mathematical institutional economics must allow us to characterize the common economic functions of different institutions while explicitly accounting for arrangements that may be determined by noneconomic considerations.

1.1.5 The theory of money and oligopoly theory

Although this book utilizes technical concepts from several areas of economic theory, it is meant to be self-contained and not particularly mathematical. Thus all technical terms are explained, and references are given for readers who require more detailed explanations.

Microeconomic theory deals with individual optimizing behavior in an open mass-market environment (partial equilibrium), a closed mass market (general equilibrium), an environment with a few large competitors (oligopoly theory and game theory), or an environment with exogenous or outside risk (theories of behavior under uncertainty). The common thread linking almost all microeconomic theories is the view of the more or less rational, goal-oriented individual.[2]

Macroeconomics, far more than much of microeconomics, is the servant of the state and meant to be applied. It consists of a host of ad hoc models and insights formalized in broad aggregate terms to accord with national data-gathering capabilities and to provide insight into, and quick fixes for, the general politicoeconomic problems that beset all modern governments. Whether we consider the "Grand Dynamics" of individuals such as Marx (see Baumol, 1959, for instance) or the development of formal macroeconomic models since Keynes (1936), we find that the emphasis has been far less upon providing a parsimonious, logically complete, static theory of behavior (as in general-equilibrium theory) than upon the dynamics of control. While both macro- and microeconomics have their place, some questions—in particular, most questions concerning money and financial institutions—require a reconciliation of the two approaches.

The fundamental difference between general-equilibrium theory and the theory of the firm or consumer in competition, on the one hand, and oligopoly theory, on the other, is that the first two are nonstrategic while the third is explicitly strategic.

An important open problem in economic theory has been to make the one-sided or open models of oligopolistic competition consistent with the closed economic model that forms the basis of general-equilibrium theory. The well-known descriptions of Cournot (English trans., 1897) and Edgeworth (1925), and the critique of Bertrand (1883), are all formulated in the context of an open or partial-equilibrium economic system. The firms are the strategic entities, and their markets are represented by a demand mechanism. The Cournot analysis was based upon the treatment of quantity or production as the strategic variable in control of the firm. The Bertrand observation and Edgeworth analysis used price as the strategic variable.

A characteristic feature of partial-equilibrium oligopoly analysis is that it is straightforward to build a model of a market with n competitors and with price or production as the strategic variable. When one tries to construct such models within a closed economic system, however, a difficulty appears. In particular, consider a closed trading economy with n types of individuals (where all members of the same type have the same utility function and initial resources) trading in n commodities. Suppose that each type uses price as the strategic variable and that all individuals of the same type possess only one type of good. Thus a trader of type i has an endowment of $(0, 0, \ldots, A, \ldots, 0)$ where A is the ith component of the vector of resources. The difficulty comes from the fact that the price system in a competitive economy trading in n goods has only $n - 1$ degrees of freedom; thus one price must be fixed arbitrarily as a numéraire. In the trading economy just described, however, the strategic roles of the traders are intrinsically symmetric. Thus if we select the price of one commodity ex ante to serve as numéraire, for example, $p_1 = 1$, then all traders other than those of the first type can name prices as part of their strategies in terms of the first good, but traders of the first type have had the strategic freedom to name a price taken away from them. This difficulty can be overcome by adding an extra good that is held by all and that has its price pegged in advance. That is, in order to convert the game described above into a symmetric game, we add an $n + 1$st commodity and set its price $p_{n+1} = 1$; we can then define symmetric strategy sets for all types of traders.

The problem of embedding a model of oligopolistic trade in a closed trading economy is intimately related to the introduction of money in a closed economic system. A system that is strategic for all must be loosely

coupled so that all have room for independent choice. Outcomes must be defined in positions of disequilibrium as well as equilibrium. The extra degree of freedom needed by the system is provided by money, credit, and the bankruptcy laws.

Small variations in the value of the money are an operational necessity for economic dynamics. In equilibrium, the books balance and the apparent need for the extra degree of freedom in the system disappears. Mathematical economic methods that concentrate on the existence of nonstrategic equilibria can avoid the need to consider a role for money because techniques and analysis will be devoted to static and *nonprocess* conditions for the existence of equilibrium prices and trades.

Subsequent chapters develop the process models of the economy as *strategic market games*. Conditions in which these give the same results as the equilibria in general-equilibrium theory are noted. But if the conditions are not precisely appropriate, the coincidence between the noncooperative equilibria and competitive equilibria will not occur. Phenomena such as money shortages and the need for credit will appear.

The essential theme of this work is that a fruitful unification of micro- and macroeconomics can be accomplished if we construct models of the economy with exchange, production, and other factors modeled as playable games of strategy. The very act of modeling economic agents as strategic players in a closed economy will call for the introduction of money and financial institutions to convert the essentially tightly coupled nonprocess model of the general-equilibrium economy to a loosely coupled process model cast as a playable strategic game.

The rules of the game will provide a minimal description of the financial instruments and institutions needed as carriers of process. Before attempting to study the broad sweep of macroeconomic dynamics, we can examine the process models for equilibria. By utilizing the noncooperative equilibrium solution, which dates back to Cournot, we can show that the results of general-equilibrium theory appear as special cases of a far more general class of noncooperative solutions to strategic games.

A minimal structure of the financial instruments and institutions of society is present in any process model of economic activity as an integral part of the definition of the rules of the game. Thus the stress on mathematical institutional economics comes about to a great extent because of logical requirements rather than an interest in specific institutional arrangements.

1.1.6 On the roles of information and evaluation

Underlying much of microeconomics are the assumptions that as a good first-order approximation individuals are well informed and have clear perceptions of value. When Ms. Jones goes to buy a pound of bacon, those assumptions are probably reasonable. When she buys an insurance policy, the assumptions are less true; and when she buys a common stock, they may be even less true.

Economic reality is presented to us for the most part via individual prices, aggregate price and quantity information, and private and public accounting information. The gaps are considerable among individual perceptions of profit and cost, tax accounting definitions, cost accounting definitions, and the definitions of economic theory.

Most of the trade in any modern economy is carried out by fiduciaries or managers handling other people's physical assets or money. In particular, those individuals who control the financial institutions serve at least two basic functions: They provide the perception and evaluation of economic worth, and they control and allocate financial resources.

In a complex modern economy the realities of accounting practices, law in general, and taxation in particular cannot be dismissed by the economic theorist as minor frictions in the system. They are major factors in the determination of economic incentives and economic worth. They are key parts of the gearing mechanism that meshes the economy with technology, on the one hand, and with the sociopolitical structure of the society, on the other.

Those in high positions of public and private finance hold considerable fiduciary responsibility for evaluating economic and societal worth and administrative feasibility. It is important to note that the incentives of these fiduciaries do not match simply those of their aggregate constituency, whether that is symbolized by the "representative stockholder" or the citizen.

1.1.7 History, economic analysis, and expectations

Much of microeconomic theory is ahistorical. As in chess, there is a beginning and end to the game, and context is unimportant. This is not so with macroeconomic theory, or when Marshall describes the growth of an industry. If analytical models are to reflect that there was a yesterday and there may be a tomorrow, care must be taken to specify initial

and terminal boundary conditions or otherwise justify why they do not matter.

Initial conditions reflect history. Terminal conditions reflect our expectations concerning the future. The basic methodological trick used both to finesse the need to know history and to avoid discussing how we use our perception of process to estimate the future is to invent given prior subjective probabilities that summarize our view of the current state of the world and its processes. Aumann (1987), for example, has lumped estimates of strategic behavior together with subjective estimates of the uncertain environment. Bewley (1986b, 1987) has provided formal underpinnings for "Knightian uncertainty," in which the idea that we have well-defined, completely ordered preferences is abandoned.

For the study of many basic economic questions, such as the nature of unemployment, an understanding of the formation of expectations is important. However, there is much to be done even before this step is taken. In particular, even with a grossly oversimplified view of preference and prediction, we can make some observations of value concerning economic process and the role of financial institutions in the economy.

1.2 Methodological Considerations

This book will stress formal model building, with emphasis on the construction of consistent and complete playable game models of monetary and financial mechanisms. Because the level of mathematics used is relatively elementary, few general proofs will be included. Furthermore, it will be argued that the full mathematical investigation suggested by this approach calls for techniques that have not yet been fully developed. One major gap is the lack of a sufficiently general, yet satisfactory, solution concept for games played in extensive form. There are many modifications of the noncooperative equilibrium, but none appears to be universally satisfactory. A second set of difficulties is encountered with infinite-horizon models, where the natural formulation of the problems requires the solution of simultaneous dynamic programs.

The proofs of the more critical assertions made here are contained in a series of joint papers with Dubey, Shapley, and others and are referred to at the appropriate points. A companion volume presenting the mathematical aspects of the theory of strategic market games is projected.

Nevertheless, this book stands alone inasmuch as it presents the basic concepts and theory primarily in verbal form accompanied by relatively simple mathematical models and examples where needed, with references provided for readers who require more detailed explanations.

1.2.1 Necessary conditions for a satisfactory theory of money and financial institutions

There are many features that must be accounted for by any adequate unified theory of society that includes money and financial institutions. For example, fiat money should, in general, not appear directly in the utility functions of individuals unless the modeler can justify this device as a reasonable approximation for the problem at hand. The value of fiat money should be derived as part of the solution emerging from an analysis of its behavior. It has a derived utility, not a direct utility. This is particularly clear in the study of closed, finite-horizon economic systems, in which the value of paper money at the end of the game must be defined.[3]

In partial-equilibrium systems the introduction of money into the preference structures of individuals may be methodologically sound as long as the domain over which this approximation is meant to hold is made clear. For example, in the study of oligopolistic behavior, the firms involved may be large with respect to their markets but small in the context of the economy as a whole; thus it may be a reasonable assumption to treat fiat money as approximately equivalent to "units of utility."

Commodity money should appear directly in the preference structures of individuals in a way that reflects its consumption value. Its dual role as both commodity and money should be reflected in the price structure. For example, if there is a shortage of commodity money and there is no provision for credit or fiat money, its price may rise above its marginal worth as a consumption good, reflecting the shadow price of its capacity constraint on trade.

Assets should not necessarily have direct consumption worth to individuals. The difference between steel plants and strawberries is sufficiently striking that such a distinction should appear in a theory if the extra empirical observation can strengthen the analysis. In applying mathematical methods to the study of political economy, it is important to remember that generality is not an unmixed blessing. It may be spurious where

conditions that obviously apply to the general empirical problem are left out of its mathematization. It might be mind-stretching and more general to consider a world where humans have three, four, or more legs, but for most purposes the assumption that they have two, one, or zero is sufficient.

Labor should naturally play a role as a special commodity. Aside from sociological rationales, there are also important market and financial reasons for this provided slavery is ruled out of consideration. If individuals cannot sell themselves or be destroyed by others without serious consequences, this immediately distinguishes the treatment of human capital in the market for productive assets.

An adequate theory should explain the roles of insolvency and bankruptcy. Eventually it should be able to cast some light on the nature of optimal bankruptcy laws.

Even in an economy with perfect foresight and a money or other loan market, loan limitation conditions should appear naturally. After all, even under the most favorable of circumstances, no banker should be willing to lend an individual more than that individual can pay back. It may be that at equilibrium all individuals will be able to pay back, but in a game of strategy ex post and ex ante rationality do not necessarily match, and for this reason bankers must consider strategies in which they occasionally refuse financing.

The theory should also be rich and flexible enough to accommodate the phenomena of transactions costs, check kiting, and living off the float. It should also be able to accommodate "hyperinflation" or to have within it an explanation of why and how the system breaks down under hyperinflation.

A satisfactory theory should distinguish institutional differences naturally. In many instances they should appear parametrically. Slight variations in the sequence of payments or the cash-flow patterns in a society should be amenable to sensitivity analysis.

The theory should explicitly describe how velocity of circulation and variability in the sequence of trading influence the functioning of the system. Because the system is loosely coupled even under the most stringent of rules, variations in velocity should increase the looseness of the coupling and diminish the possibilities for tight central control.

The theory should be explicit in describing the role of markets and transaction technology. It should specify the sequencing of actions and the nature of information conditions so clearly that the paradoxes nor-

mally associated with perfect foresight or the meaning of rational expectations disappear. Logic tells us that we cannot use current market price to form current market price.

A clear distinction must be made between exogenous and endogenous or strategic uncertainty. The theory must provide a basis for explaining the role of brokers, financial information services, and other information features of a financial system. At present, the adequate modeling of information conditions presents major difficulties.

An adequate theory must be genuinely dynamic and evolutionary, as contrasted with the virtual dynamics or extended static equilibrium that characterizes the treatment of time in general equilibrium.

A basic link between microeconomics and macroeconomics lies in the aggregation of information for various decision-making units. The aggregation of information justifies solution concepts where individual behavior is based on detailed knowledge of one's own actions and adequate knowledge of the joint actions of others in "the market."

An adequate theory of money and financial institutions should help to clarify the purpose and essential product of the institutions. Over time and locale, the institutions may change but the basic functions of finance will still be manifested.

Last, but by no means least, the differences in the functioning of centralized and free enterprise economies should be made clear. The role of competition and the importance or lack of importance of the number of competitors in a market must emerge as part of the theory.

1.2.2 The many roles of money

Money is what money does. Money is part of the rules of the game. The conventional textbook definitions describe some of its functions: medium of exchange, store of value, and standard of worth.

Money helps us overcome the crudities of barter. It should also have certain functional properties associated with efficiency in transportation and information processing. Thus a money should be easy to handle, light, durable, easy to transport, easy to identify, and hard to counterfeit. Jevons (1875, p. 16) noted that money should have utility and value, portability, indestructibility, homogeneity, divisibility, stability of value, and cognizability.

An economy with a monetary system differs from an economy without a monetary system in the fundamental sense that it is governed by a

different system of axioms for trade and commerce. Because there are many differences in monetary, credit, and financial systems, there are many different but closely related economies to be considered. There are certain basic metatheoretical questions that can be asked of all of these. Is there an optimal bankruptcy law? Are there natural or basic forms of banking, treasury, and insurance institutions that must always be present? Does the approach of mathematical institutional economics lead to suggestions for the design of new institutions as well as to the understanding of the old? Once we contemplate such questions, it should become clear that in any attempt to describe or define money it is best to separate out difficulties by isolating money's different properties. Einzig (1948), for example, distinguishes between primitive and nonprimitive monies. It has been a favorite pastime of monetary theorists for several generations to argue about the definition of the money supply. The approach adopted here is to define money in terms of its operations in specific systems. This requires that we specify the rules of the game concerning financial operations and state how the rules concerning the employment of what we call money and other financial instruments influence the strategy sets available to the participants in the economy.

A satisfactory theory of money and financial institutions should distinguish among fiat money, checks, gold certificates, warehouse receipts, IOU notes, and other instruments such as credit cards or lines of credit in a way that shows clearly their common and differentiated functions. The need for such distinctions is not as a minor correction to existing microeconomic theory, or a dressing up of abstraction with institutional detail. It arises as a logical necessity from the study of process.

Money is an institutionalized symbol of trust. When it is considered further from a strategic point of view, other properties appear. In game-theoretic models of the economy three of its critical properties are important: (1) money as a strategic decoupling device; (2) money as a criterion for distinguishing players;[4] and (3) money as a means of generalized side payment.

The introduction of money into an economic system provides an extra degree of freedom in the strategies of economic agents. Exchanges need no longer match in terms of goods alone, as long as they balance in terms of goods plus financial instruments. When the formal modeling is done with care, mysticism and verbal gymnastics concerning what was really meant by Say's law or Walras's law become unnecessary.

Money allows a strategic decoupling of the acts of buying and selling in or out of equilibrium. An economy that uses a money will need to take into account cash-flow effects on economic maximization. A shortage of money could constrain trade. In an economy with complete pairwise markets (anything can be exchanged for anything directly), one can always use one's total wealth for purchases.

What an individual consumer can do with money or other financial paper is not the same as what a commercial bank can do, and the strategies of a commercial bank differ considerably from those of the Federal Reserve Bank, the U.S. Treasury, or the central bank of any nation. The money and financial institutions and instruments of an economy appear as extra rules of the game, but these rules do not necessarily apply symmetrically to all players. Bankers or licensed brokers, for example, are distinguished from ordinary economic agents by sets of special rules created by custom and law.

In partial-equilibrium analysis and in much of game theory, money serves as a measure of welfare or a "utility pill" that appears as a separate linear term in the utility functions of all members of the economy. In practice, this may be a good approximation. The use of money in a sufficiently small neighborhood, in general, may be approximated by regarding money and transferable utility as equivalent. In other models this assumption is not adequate.

1.2.3 On money, coding, and utility

Much of microeconomic theory has been devoted to the mysteries of maximizing high-dimensional utility functions. Money enters as an abstract or unreal commodity to define a budget constraint. Perhaps a more natural view of a modern economy is that money is the reality, and the utility function is the shadow abstraction. Money is a powerful encoding device, which for many economic purposes (both for individuals acting on their own behalf and for the fiduciaries who account for the major part of economic decision making) reduces an extremely high-dimensional problem to two pieces of information: How much will I make now, and what is my evaluation of the worth of my new position in the game? This is in essence the same as the forward description of a dynamic program.

While there are of course many things that money cannot buy, as a first approximation in a limited context, current wealth measured in money

terms and future expected income appears to be a useful guide for much economic activity.

1.2.4 Three approaches to model building

We shall use three highly different approaches to model building: (1) institutional analysis; (2) game theory; and (3) gaming. A central theme is that institutions are the carriers of economic process and provide the links between the economy and other aspects of society. Thus a fruitful way to gain insight to be used in the formulation of an economic dynamics is to examine how various institutions perform their functions. For example, the tâtonnement process for price adjustment described by Walras (English trans., 1877) appears to be modeled after one of the trading methods at the Paris Bourse. The New York Stock Exchange uses a double-auction price-adjustment mechanism. Are these different mechanisms of economic importance?

We use game theory in the model building because many of the most interesting questions concerning the development of the economy involve dynamics and control. Several of the key financial and governmental institutions, by the nature of modern society, are few in number and large in size. Their strategic influence cannot be ignored. Moreover, the methods of game theory provide a natural way to characterize strategic behavior and to illustrate the fine structure of information and the mechanisms of economic exchange and production. This fine structure can be safely ignored when studying static nonstrategic conditions for production and exchange, but is necessary for strategic analysis.

The term *gaming* is used here to stand for the formal construction of playable games whose structure may represent institutions such as auctions, specific types of markets, or even sections of the economy. A playable game is one that can actually be played as an experimental game or a game for teaching or even entertainment without too much difficulty.

The discipline required to construct a playable game calls for an understanding of institutional rules such as how a bid is made or a debt is paid, together with an appreciation of whether the rules can be understood by the players. Most of the institutions of society are operated by individuals of intelligence well below the genius level. An important part of social engineering is to design mechanisms that can be run by those called upon to do the job. The design of playable games connects the abstract models of game theory to the institutional forms of the society.

A further reason for constructing playable games is that they allow for a certain amount of experimental economics. It is instructive to observe how individuals actually play games that represent different markets and bidding mechanisms.

Most of the models presented in the following chapters meet the dual requirements of game theory and gaming. They are well defined mathematically and are playable as games.

1.2.5 On competitive and noncooperative equilibrium

The "competitive-equilibrium solution" for the running of a closed economic system utilizing prices is a misnomer. It could be more accurately called the "price-system solution" to efficient production and allocation in an economy, with no particular reference to competition or central control.

The game-theoretic solution of noncooperative equilibrium, which dates back to Cournot (English trans., 1897) and was generalized by Nash (1951), is far more general than the competitive-equilibrium solution, and it more adequately portrays the concept of competition. An efficient price system may result from a central control board that simply announces prices, or it may be a special limiting case of noncooperative solutions to strategic market games with finitely many players, or it may emerge from a strategic model of the economy with a continuum of small economic agents.

1.2.6 Finance and uncertainty

Many of the important aspects of finance and financial instruments involve risk and uncertainty. Yet when a strategic view of the economy is adopted, it is desirable to distinguish exogenous risk (risk caused by factors outside of the model) from endogenous risk (strategic uncertainty about the actions of other agents).

Our initial analysis of markets, credit, and banking will be carried out primarily without reference to exogenous uncertainty. This reflects a desire to separate difficulties because there is much to be analyzed about production, trade, and transfer of capital in economies even without exogenous uncertainty.

The presence of exogenous risk brings forth new and important phenomena in finance, but the presence of such risk is not a necessary condition for requiring a financial system. Transaction costs alone may provide

a sufficient reason for the invention of such a system. Furthermore, transaction costs may provide a source of the endogenous risk.

The minimization of the cost of trade may require an aggregation of bids and offers, but this is provided by institutions such as markets and clearing houses that are essentially aggregating and disaggregating devices. The traders' names are aggregated in determining the anonymous forces of supply and demand, and the system must remember them and disaggregate them when making or obtaining payments. But aggregation also causes uncertainty: instead of a distribution of bids and offers, the trader sees only mass information from the market.

1.2.7 Institutional sensitivity analysis

Conventional sensitivity analysis in mathematical economics or operations research may involve varying a parameter over a range and studying how the solution changes. For example, sophisticated methods in differential topology now allow economists to trace the behavior of competitive equilibria as the initial resources of the traders are changed (Debreu, 1976).

There is an alternative form of sensitivity analysis that is far more basic to the techniques of modeling in the social sciences. It involves changing variables or modifying the relationships among variables. For example, a fixed average propensity to consume may be replaced by a more complicated function. A still more complicated change might involve the assumptions made about individual preferences.

A model that is extremely sensitive to changes must be as suspect as one that is too insensitive to change. Judging an appropriate degree of sensitivity to change is virtually an art, but one of considerable importance when models are examined that purport to catch the essence of different institutions. In particular, a complex set of different reasons may have been involved in the evolution of various financial instruments and institutions, but once an instrument is in existence, its function and life may be virtually independent of the chance factors that aided its creation.

The question of how a given society started to use paper money is considerably different from that of why a society continues to use paper money. The fish gets on the hook one way, but may get off the hook in a different way. Once a society has had its trade monetized for whatever reasons, the possibilities for demonetization may involve factors totally apart from the original ones that resulted in the adoption of a money.

1.3 Three Types of Mathematical Models

Nonmonetary general-equilibrium theory is essentially a static theory with perfect foresight.[5] There are at least three ways to introduce monetary phenomena into a general-equilibrium system. The first is static and nonprocess; the second and third dynamic and evolutionary. Of the latter, the macroeconomic or behavioristic approach makes explicit use of the role of expectations, while the game-theoretic approach makes use of the concept of a strategy.

1.3.1 Maximization over time given perfect foresight

Suppose there are n individuals in an economy lasting T time periods and there are m_t commodities available during period t. This problem, which involves time in the sense that there are T time periods, can in fact be treated statically. Indeed, even in an economy of infinite duration, if the appropriate items are bounded, the economic maximization problem can be treated as though it were static.

To allow a static treatment, we use the device of perfect foresight. If treated only verbally, the assumption of perfect foresight can lead to an ill-defined and unproductive philosophical debate. If viewed mathematically, the clearly logical but counterfactual structure can be seen. The assumption amounts to the proposition that all individuals are initially supplied with information concerning the prices of all commodities during every period, even into the infinite future. Given this information, individuals are able to calculate their income over the complete existence of the economy. If the free use of credit is assumed, then the individuals maximize their welfare, each subject merely to one overall budget constraint.

The question of the existence of a competitive equilibrium, or of an efficient price system, in this market amounts to asking whether there exists a set of prices $p(t, k_t)$, where $t = 1, 2, \ldots, T$ and $k_t = 1, 2, \ldots, m_t$, such that if all individuals are informed of these prices (the assumption of perfect foresight) and of their initial endowments, each will be able to maximize independently in such a manner that all budget constraints and market balance conditions are satisfied and the outcome is Pareto optimal.

Phrased in the above manner, the idea of perfect foresight is neither particularly mystical nor dynamic. It is merely counterfactual, but a study

based on this assumption could still provide insight. The mathematical question is whether a set of stationary optimal policies exists, not how they evolve.

In the general-equilibrium system with no particular role for a money, the trick used to reduce a problem that appears to be dynamic to one that is basically static involves time-dating commodities so that 1971 wheat and 1972 wheat are regarded as two different commodities, each with a currently existing market where it can be traded.

Uncertainty is handled in the same manner. The number of commodities is formally expanded so that "1984 oranges if there is a flood" and "1984 oranges if the sun shines" become different commodities and are traded in different markets. By using this device, we introduce the paradox of assuming perfect foresight coexisting with exogenous uncertainty. The paradox is immediately resolved, however, when we observe that the assumption of independent markets for all future contingent commodities removes both time and uncertainty from consideration. In game-theoretic terms, the problem, which is presented in extensive form, can be reduced or normalized to a strategic form.[6] The dynamics is essentially removed because all individuals are now left with a single move that is made simultaneously. They all declare a strategy or a plan of action for all time.

The assumption of the simultaneous existence of all futures markets enables individuals to rid themselves of any cash-flow constraints, because assets to be obtained in the far distant future can be liquidated now and the theory implicitly assumes that there are no problems with trust or failure to deliver.

If all futures markets were eliminated and a money and a one-period loan market for money were introduced, then requiring that all transactions be carried out using money would introduce a cash-flow problem. In order to prevent this constraint from lowering welfare, we need a one-period loan market. Longer loans can be achieved by period-by-period refinancing.

The mathematical problem posed by the monetary economy differs from the nonmonetary complete-markets economy inasmuch as a whole host of intertemporal cash-flow constraints must be made explicit and a period-by-period recording of indebtedness is needed. Even with this formulation it is still possible to avoid consideration of dynamics if we make use of perfect foresight and check only for the existence of an appropriate equilibrium. Does there exist a set of prices $p(t, k_t)$, where $t = 1, 2, \ldots, T$

and $k_t = 1, 2, \ldots, m_t$, and the initial price of money is set at 1, and a set of one-period rates of interest r_t for $t = 1, 2, \ldots, T$ such that if individuals are fully informed of their endowments and all of the above, they will be able to maximize independently in such a manner that their actions will all be mutually consistent?

1.3.2 Dynamic programs with expectations

The embedding of monetary trading in the general-equilibrium system while leaving the model static and using perfect foresight does not appear to be in the spirit of what a reasonable theory should look like. But it is highly desirable to divide difficulties before taking on dynamics. A striving for premature realism may confound the analysis with the interaction of the many different aspects of money and financial instruments and may be less instructive than analyzing less realistic but more tractable models.

There is a natural way to convert a static general-equilibrium system into a dynamic evolutionary system, a way that has been used in the study of macroeconomics and in microeconomic behavioral theories of the firm. What we do is replace the assumption of perfect foresight by behavioral mechanisms for the formation of expectations. The system no longer operates as though all future history were given; instead, expectations evolve and are modified as time passes.

The arbitrary feature of this scheme (and indeed a basic arbitrary feature of macroeconomics) is how we select the expectations function. We are called upon to add an item that does not neatly fit under the rubric of the microeconomic maximizer. Methodological considerations require that mechanisms for the formation of expectations be relatively simple, depending upon the body of knowledge in social psychology, business and technological forecasting, and political and economic planning. A behavioral theory of the individual and the firm is also required, as advocated by Simon (1984) and many others. Although a "satisficing" or some other form of the behavioral individual must eventually replace the simple utilitarian model, I do not emphasize this approach because there is much value in setting up the minimal mechanisms for an economic dynamics regardless of the precise description of economic motivation.

1.3.3 Dynamic programs with behavior strategies

A different approach to dynamics is through game theory, where the economy is seen as a game to be played in extensive form and we need to

specify the solution concept to be employed for this game. No broad, generally accepted way to solve general nonconstant-sum games in extensive form exists at present, although certain solutions are suggestive for classes of games with special structures such as markets. In particular, when the information conditions are not too complex and the players are numerous, a dynamic version of a noncooperative equilibrium formed by individuals employing behavioral strategies may be reasonable.

Possibly the most important lesson to be learned in trying to construct formal game-theoretical models of multistage activities involving strategic behavior is that the difficulties in reconciling ex post and ex ante rational behavior are considerable. In theories that concentrate on equilibrium results without concern for disequilibrium, the distinction can be obscured. In multiperson strategic games the linkage of group rationality and individual rationality is at best loose, as can be seen in any study of markets where individual liquidity and group liquidity are at odds.

Game theory has often been regarded as some quintessential extension of the study of individual rational economic human beings that is somehow in contrast with, if not in opposition to, a behavioral view of human activity. There may be elements of truth in this viewpoint when we consider instinctual, habitual, and other subconsciously governed behavior, but the proliferation of potential strategies in a multistage game is such that many legitimate, formally well described feasible strategies appear to have behavioral or even rule-of-thumb counterparts. Thus even threat strategies that appear to involve the emotions are ex ante legitimate strategies and, if they work, may be judged after the fact to have been perfectly rational. If they do not bring the desired result, they may be called irrational. The literature of business cycles is replete with observations that individual rational behavior during bank runs with threats to call loans can snowball into disaster for virtually everyone, but probably not for the few at the head of the queue at the early stages of the panic.[7]

The technical problem to be faced in the use of infinite-horizon game-theoretic models is how to recast the economic models of exchange and production as parallel dynamic programs with many players. This is considered in chapter 23.

1.3.4 Three limits

The treatment of an economy with money and financial institutions as a dynamic game has several important advantages. In particular, it pro-

vides clear models for the study of at least two different types of limiting behavior: the behavior of the system as the time horizon of the market becomes infinite; and the behavior of the system as the number of participants becomes large.

A third important type of limit concerns information. There are two natural bounds that can be described by game-theoretic methods. At one extreme, there is *perfect information*, where all individuals called upon to act are totally informed about all previous actions leading up to the current state. At the other extreme, the individuals know no previous history (except possibly their own actions). The investigation here is directed toward an understanding of modern market economies and, in general, the mass anonymous market where individuals have low information and little communication.

1.4 Many Economic Models with Money and Financial Institutions

Our approach of separating difficulties will lead to the construction of many models of limited scope that will allow us to isolate the various aspects of the functioning of money and financial institutions and study them separately.

The simplest monetary model involves the use of a commodity money in a static noncooperative game embedded in a closed economy. This is considered in chapter 6.

The next level of complication calls for fiat money in an economy with no transaction costs, no exogenous uncertainty, and no money or futures markets, but with oligopolistic uncertainty, where bankruptcy laws must be specified. In examining some simple cases, chapter 11 considers an economy with a money market or other futures contracts as a multistage process. Because those who borrow today may not all be able to pay back tomorrow, bankruptcy law must be specified. This is considered in chapter 11.

The role of assets and production must be investigated from two points of view—as a means of providing the backing for financial paper and as a means of attaining economic growth.

The role of money in economic systems with transactions costs is a separate and critical topic. In a world with high transaction costs, the evolution of money is highly interlinked with the design of efficient trading technologies.

The functioning of a money in a system with exogenous uncertainty requires a separate investigation which is only touched on in this volume. It calls for the consideration of the degree of correlation among the risks present in the economy, the role of time lags, and the interaction between exogenous and endogenous risk. Sophisticated and difficult problems arise concerning moral hazard and adverse selection when both exogenous uncertainty and strategic behavior are present and we can no longer tell whether failure to honor commitments is the result of bad luck or bad faith.

A different set of models and analysis is needed to investigate markets where individuals never have complete information. The work of Radner (1968, 1970, 1979) and of Dubey, Geanakoplos, and Shubik (1987) has given some indication of the difficulties encountered. However, this work hardly scratches the surface of problems involving the role of differentials in the knowledge and understanding of professionals and others as they evaluate and assess risk. (Of course, this describes much of what goes on in deciding how to invest new capital or reorganize the use of old.)

Setting aside uncertainty or complex information patterns, we can produce a set of different yet relevant models by introducing specialized or nonsymmetric players who are allowed by the rules of the game to provide banking services. The special rules imposed on their strategic options may reflect various institutional forms such as reserve banking.

The introduction of a government with taxation, subsidy, lending, and borrowing authority brings us to a new level of complexity. For some purposes, it may be simplest to consider government as an outside agency with a specified set of strategies given in advance (much like the dealer in Blackjack). Otherwise the government must be considered as a large strategic player.

To describe the role of government, we need to specify the relationship between its monetary, fiscal, and other policies and its role as provider of public goods. It is in this specification that the connection between the major concerns of macroeconomic theory and the approach outlined here is strongest.

A theory of international finance involves a completely different set of considerations. To explore fiat money in a single economy, the controller of the game need only introduce one type of poker chip. International finance involves more than one type of chip and one controller. Rules must be developed not only for the small economic agents but for each of

the governments involved. International law is far from perfect, and some of the great loopholes used in illicit and sub rosa economic activity come about when economic agents find they can arbitrage between economic activities that are legal in one jurisdiction but not in another. Whole markets such as the Eurodollar market or gold trading in Hong Kong, Tangiers, or Macao or offshore havens in the Cayman Islands or Jersey owe their existence to the imperfections of the match between national and international law. These special havens may be created to take advantage of a specific set of circumstance but then go on to fulfill other purposes or go out of existence as the opportunity to arbitrage disappears.

This partial list and breakdown of monetary and financial models is given to suggest that it is possible to close the gap between micro- and macroeconomic theorizing in a consistent manner. This is done by recognizing that there is not a single problem to be solved but that different problems must be isolated and analyzed before we can assemble process models of sufficient complexity to forge links of interest and strength between these two highly diverse and differently motivated ways of looking at economic activity.

1.5 Context Rationality and Evolutionary Systems

In any careful attempt to construct process models of a multigenerational economy, it soon becomes clear that the concept of individual optimization must be viewed in a broader context than that of economics or operations research. The "rational optimizing individuals" at most are engaged in context-relevant local optimization. This local optimizing behavior feeds back into the polity and society and modifies them. Thus the dichotomy between economic optimization and evolutionary behavior is not a valid one. The economic system of any society together with locally optimizing agents is evolutionary, with the evolution driven in part by the local optimizations of creatures who may mistake their myopic behavior for the central fact of some grand optimization.

Many years ago Alfred Marshall noted analogies between economics and biology. Boulding (1950) and others have echoed these sentiments. The approach in this book is to search for the points at which the logic of trying to construct explicit models of economic behavior forces us into models of evolution. In particular, money and financial institutions

emerge as the encoders, decoders, evaluators, controllers, and carriers of the economic information of the political economy. The bankruptcy and reorganization rules of the economy, which are highly influenced by the polity and legal system, serve as controls on innovation and economic mutation.

At present, neither economic nor biological theory is sufficiently developed to portray adequately the essential similarities and differences between biological and economic mechanisms in their respective environments. Nevertheless an appreciation of the role of money and financial institutions as representing the information and organizing mechanisms of the economy may help increase our understanding of the relationship between local optimization and evolution.

Notes

1. The one compromise made in this text between the models and the playability of the games is that on occasion games with a continuum of traders are considered. From gaming experience it is conjectured that in the games described, twenty to thirty players are already a good approximation of the "continuum," although the mathematical difficulties proliferate (especially for multiperiod games) if finite numbers of players are considered.

2. In a fundamental way, much of microeconomic theory (general equilibrium, for instance) is part of a *nonsocial* science. The concepts of a mass market and of anonymity allow for the construction of much theory in which only the individual and an impersonal market matter.

3. Note that the societal imposition of bankruptcy laws may implicitly impose a valuation on money.

4. In particular, money provides the government with a means of control over the economy.

5. Section 1.3 can be skipped by those not concerned with the alternatives for mathematical formulation.

6. This is true for finite-horizon problems; the infinite-horizon problems would require some qualification.

7. Eventually, one might hope to develop a type of game-theoretic statistical mechanics in which the correlation of otherwise random action is caused by strategic behavior at a critical time.

2 The Basic Financial Instruments of an Economy

2.1 Financial Instruments and Institutions

Modern trade requires markets that utilize money and financial instruments. We begin by considering trade that uses commodity money. The next step is to introduce various forms of credit, including fiat money. There are, of course, many other financial instruments in a modern economy, and these instruments are created and handled by many different financial institutions. In the United States in the 1970s and 1980s there were around twenty different institutions (the numbers are in constant flux). A rough estimate in the early 1970s indicated over 50,000 financial intermediaries (Kroos and Blyn, 1971). What should be included as a financial intermediary is a matter of definition, but in the 1990s the order of magnitude is about the same, albeit with great changes in composition and the volume of trade.

The institutions include the commercial banks, investment banks, trust companies, mutual savings banks, savings and loans associations, general insurance companies, life insurance companies, investment companies, pension funds, finance companies, mortgage companies, security brokers and dealers, credit unions, federal reserve banks, land banks, various federal lending agencies, and the U.S. Treasury.

These institutions and others like them elsewhere guide the processes of the economic and financial system, regardless of the existence of equilibrium. The physical economy is controlled through *financial dynamics* with the financial system serving as the neural network for the body economic and dictating what happens in disequilibrium.

2.1.1 The weapons in the game

A discussion of the physical flows of goods and services that disregards the dynamics of the divergence between goods and the paper used to control them leads to a technocratic, essentially nonpolitical "engineering" view of the economy. Thus, for example, input-output analysis is useful for planning in any society in situations where demand is more or less given and not subject to large strategic manipulation by independent agents.

Financial paper in the hands of private citizens can give them control of both the consumer goods of a society and the means of production of private and public goods. The financial instruments of a society provide a means for delineating the nature and size of the strategy sets of individual

actors. Do we wish to permit private citizens to save, to hoard, to directly finance factories, or to privately provide for their old age? Their ability to do this depends directly upon the financial instruments at hand. In a society where political decisions determine the direction of investment and the supply of public services, the nature, supply, and distribution of financial instruments and institutions will be different from and more limited than those of a society where socioeconomic and politicoeconomic decision making is decentralized to the citizens at large.

The financial instruments of any society are determined by the desired level of decentralized decision making concerning sociopolitical and economic goals as well as considerations of economic efficiency. Thus, for example, a comparison of the financial instruments and institutions of the United States, Japan, the Soviet Union, or Bulgaria of the 1980s should find considerable divergence. But the divergence may be explained not merely (or even primarily) in terms of "economic efficiency" of development, but in terms of providing the weapons for different levels of decentralization and loose coupling in a dynamic system.

2.1.2 The eight basic instruments[1]

In a dynamic economy with many frictions and transactions costs, imperfect information, errors, limited trust, and aggregated measures, at best there will be some limits to the extension of credit. The magic of the static general equilibrium is not there. If the planning is not done by the state, people need bankers, insurance agents, investment bankers, and other evaluators of individual, corporate, or governmental economic plans.

Individuals do not usually try to guess the details of the price system that will exist over the course of their lives and work out a detailed backward calculation based on a lifetime budget constraint. In practice we make crude estimates and update them as circumstances change. Good theory must eventually acknowledge that the forward updating mechanism is closer to the truth than backward induction and lifetime individual planning.

Instead of being guided by one overall budget constraint, decentralized enterprise economies appear to depend in a large part on spot markets, with trade in money, financial paper, and other contracts that permit limited extensions of credit into the future and constantly revised planning. A centralized economy requires fewer or less complex instruments and less individual but more governmental planning.

Although there may be many institutional variations, I would suggest that eight basic instruments are needed in any modern society to perform the necessary financial functions: (1) services, (2) goods, (3) money, (4) ownership claims, and the four contracts against these, namely (5) service contracts, (6) futures, (7) promissory notes, and (8) contracts on ownership paper.

An adequate model of financial control requires a specification not only of the roles of these basic instruments, but also of the rules, mechanisms, and institutions that control the demand and supply of these items and the motivation and strategic possibilities of the agents who control the institutions involved.

1. *Services:* A service may be regarded as an input that cannot be inventoried, and hence cannot be used as a security against a loan. Because a service is a flow, the quantity of a service is measured by its flow times the duration of the flow.

2. *Goods:* A good is a physical entity whose ownership can be transferred and that can serve as an input into a production or consumption process. For many economic purposes, it is useful to break down the category of goods into consumer, intermediate, and capital goods. For other purposes, the breakdown may be into movables or immovables. Other distinctions include free goods such as air or sunlight, and public goods such as public safety, national museums, parks, and government. From the point of view of finance, however, the key distinction between a service and a good is that the latter (in varying degrees) provides an asset or a store of value against which ownership paper or loans may be issued, whereas the former is a flow and not a store of value.

3. *Money:* Money is *the* liquid asset, and by a combination of law, custom, and history it is a major control asset. It is defined formally in chapter 5. One might wish to distinguish among different forms such as commodity money, specie, and bank money. It will be argued in this book that money, whether a commodity or fiat or even a number kept on the government books, is the only financial asset that does not necessarily have an offsetting liability. The unfortunate custom of talking about bank debt, whether in the form of private bank notes or deposits as money, has added considerably to the confusion and has made it more difficult to appreciate the critical role of fiat or outside money as the surrogate for trust in a modern economy.

4. *Ownership claims:* The main forms of ownership paper are titles to land and buildings, common stock, and ownership certificates for machinery and consumer durables. In a world with no uncertainty, perfect divisibility, liquidity, total honesty, and zero transaction costs, the sale of ownership claims would not provide any particular economic advantage over the sale of an asset and there would tend to be many different ways of carrying out economically identical transactions. In the world as it is, the existence of various states of indivisibility, differences in information, movability, uncertainty, and ownership transfer costs can make the creation of, and trading in, ownership claims important and productive. In this world an asset and a paper claim to the asset can exhibit different economic dynamics.

5. *Service contracts:* A service contract is simply a contract that promises delivery of a service. It may cover a fixed period or have an indefinite horizon (modified by a statement such as "until retirement age"). Salary and wage contracts are probably the most important form of service contracts. Car or apartment rentals, building cleaning, payroll processing, and consultancy retainers are other examples. The only asset backing the service contract of the individual working for wages or a salary is that person's "human capital." Because indentured servitude and slavery are no longer in the rules of the game of most of the civilized world, that capital asset may be of little worth as a security on which to borrow. In contrast, the fleet of cars, the hotels, or the computers of various service firms are far more desirable items with which to obtain financing.

6. *Futures:* A futures contact is a financial instrument that guarantees delivery of a good at some future date. Because the good may not be deliverable at its due date, futures contracts (and for that matter all contracts that link economic activity between two time periods) must be considered as not fully defined unless a complete specification is given of what happens if the economy evolves to a state where the contract cannot be honored. This means specifying rules to cover recontracting, bankruptcy, insolvency, failure to deliver, and so forth.

7. *Promissory notes:* This category encompasses a large array of instruments by which money in the future is promised in return for a money payment now. A government bond, for example, offers the purchaser the government's promise to pay an income stream and possibly a final payment. Care must be taken in defining the payment conditions on these

instruments precisely, especially when they may be automatically rolled over or issued as perpetuities. The British issue of consols provides an example of an issue with no final payment of the capital borrowed. So called bank money is a form of debt, but unlike fiat money it is offset by another financial instrument, such as private IOU notes.

8. *Contracts on ownership paper:* An example of a contract on ownership paper is a *call.* Calls involve an exchange of money for an option to purchase an ownership certificate at a specified price at or before some fixed date in the future. A *put* is an option to sell, A *short sale* is the sale of a stock that is not owned. At the level of abstraction in this book there is no obvious economic necessity for the employment of puts and calls. The economic need for short sales arises when we attempt to complete markets and need to provide enough bets to allow for efficient hedging against uncertainty.

On uncertainty All of these financial instruments can involve some degree of risk. For example, an insurance policy may involve the payment of money now in return for money in the future under a set of contingencies. In a world without uncertainty, a call would be a superfluous instrument. Most promissory notes and service contracts are written in a way that presupposes a low risk of contract nonfulfillment. The institutional variations are considerable, however, and while in main purpose all promissory notes serve the same function, the variety is these institutional manifestations is of great importance to practical affairs. In an uncertain world, even goods and fiat money can be lost or destroyed.

The transition matrix Variations in the conditions of uncertainty, limitations on the length of time over which financial instruments can be issued, changes in physical transaction costs, indivisibility, and data transmission and processing, as well as legal, social, and political changes all provide reasons for the production of a continuous blending of financial instruments and a constant modification and transformation of financial institutions.

Despite this institutional flux, it appears that all instruments can be built up from these basic eight. Furthermore, it is necessary to consider only two time periods "NOW" and "NOW + 1," for all but a few complex instruments involving long-term uncertainty. If refinancing is permitted and there are no transaction costs or uncertainty, then long-term financing can be achieved by rolling over short-term financing.

Table 2.1
Transition matrix

		NOW + 1								
		S	G	M	OC	SC	F	PN	COP	Nothing
NOW	Services (S)									×
	Goods (G)		×							×
	Money (M)			×						×
	Ownership certificates (OC)	?	?	?	×	?	?	?	?	?
	Service contracts (SC)	×	?	?	?	?	?	?	?	?
	Futures (F)	?	×	?	?	?	?	?	?	?
	Promissory notes (PN)	?	?	×	?	?	?	?	?	?
	Contracts on ownership paper (COP)	?	?	?	×	?	?	?	?	?

A transition matrix linking the instruments over two time periods is given in table 2.1. It shows the transformations feasible as the different instruments age and does not specify the actual transformation probabilities that exist in a world of uncertainty. For example, a new 1998 automobile is a good that transforms into a one-year-old car of varying quality depending upon the chances of use, in 1999. It could even be destroyed.

Money may be withdrawn by the issuing agency. A firm can go bankrupt or be reorganized so that all ownership claims are wiped out, or the owner may obtain some amount of money or other financial instruments in the reorganization or liquidation. The writer of a futures or a service contract may fail to meet the contract's obligations in which case the holder, if he has already paid, may be partially or totally wiped out. In the former instance the holder will usually obtain a money repayment or a partial fulfillment of the contract such as "all the copper there is in inventory, plus a monetary settlement for the remaining futures undelivered."

When a promissory note is honored, a monetary payment is made. If it is not honored, there is a possibility that it will become valueless. If it is secured by goods, the goods may then be either appropriated or sold.

The imprecision in the description of the role of time in economic theory is present even here, in table 2.1. Is the purchase of the services of a maid for a morning of cleaning to be regarded as the purchase of a

service contract or the purchase of a service? A reasonable approximation is that the purchase for the immediate period is for the service and not a contract. But the length of time involved brings about qualitative change. As the period lengthens and the stakes become higher, informal, unwritten arrangements are usually replaced by formal written ones, and nonnegotiable paper may be made negotiable. Furthermore, the probability of default in delivery and timing between payment and receipt of services increases.

In table 2.1 the last five rows are filled with question marks. These indicate that in an economy with uncertainty and the possibility of default and reorganization, there is considerable leeway in the economic dynamics. When a financial reorganization takes place, any contract may be transmuted into virtually any other one, with the limits set only by the laws of the land and the bounds on the imagination and financial ingenuity of those involved in the reorganization.

On taxes and subsidies Historically, taxation and subsidies have been owed to or by the king or governmental bodies in the form of claims on personal services, goods, money, or combinations thereof. Tithes, tariffs, octrois, social security benefits, income taxes, sales taxes, death taxes, levies against wealth, and other taxes are all claims against goods, services, or money, and hence can be considered as mixtures of the instruments already noted.

In the United States in the mid-1980s, the agencies involved in issuing and fulfilling these claims included some 70,000 federal, state, and local bodies with taxing power. The mechanics of creating and destroying claims on taxes and subsidies imposes an extra set of accounting conditions and problems in economies where the nature of some of the claims is contingent upon certain events such as death or the sale of assets.

In the modern world almost all taxes are due in money (or other paper such as certain bonds). The choice of working on the roads or fortifications or in the fields of the lord is no longer generally available. The evolution of the monetary economy changed the scope and nature of governmental economic control over the individual.

On the unit of time How does the relevance and usefulness of the classification scheme suggested depend upon the unit of time selected? Due to the cyclical aspects of the world, to the life cycle of individuals, to custom,

law, and human decision-making processes, the year has become they key unit for taxation and accounting. The month and week are the units of choice for paying most permanent employees, while the quarter is the unit for much financial reporting and for the duration of many short-term loans. Seven years has biblical connotations and serves as a statute of limitations in some instances (Deuteronomy 15:1). Seven times seven years is the Old Testament upper limit on bondage (Leviticus 25:10). One might think that in the development of an adequate theory of money and financial institutions, the classification schemes adopted should not be influenced in any conceptually important way by the time units adopted. But whether it be Hicks's "Week" (Hicks, 1946, chapter 9) or the accountant's and tax collector's year, the selection of the units of time should have considerable justification in terms of the facts and rhythms of economic life.

The major change in extending the size of the minimal unit of time studied is to lump the more short-lived goods together with services. For example, if the minimal unit of time were two months as opposed to two minutes, ripe tomatoes would be classed together with services as items that cannot be stored for the next period. If the period is two minutes, the tomatoes can be stored.

The essential issue here is the financial implication of the existence of goods with different life spans. A prime consideration is how suitable they are for posting as security against the issue of various financial instruments. Ripe tomatoes are not a particularly good security.

It is often important to perform a sensitivity analysis or to ask if a change in the unit of time will have a radical or unnatural effect on the model postulated. If the continuous version of an economic model exhibits properties radically different from the finite time increment model, one may suspect that something is unsatisfactory in the formulation of the problem.

One of the mysteries of any theory of money and finance is precisely what is meant by "liquidity." The answer to this question is intimately tied up with the amount of time required to execute a transaction.

On time, pyramids, and complex financial instruments If we permit rollovers of short-term loans, the issuing of long-term loans including perpetuities, and the renegotiation of contracts, we need to take care to avoid paradoxes involving the infinite horizon such as the possibility of

servicing but never repaying debts; or even borrowing more to pay for the servicing. Furthermore, at this point the possibilities for multitiered financing appear. It becomes possible to create chains of paper backed by paper. These reflect new possibilities for distributing risk of different quality,[2] as well as institutional peculiarities based on social and historical phenomena.

A safe procedure for avoiding the paradoxes of books that do not balance is to consider any economic model as being open in time to both past and future history and to close it by providing initial conditions or prehistory and a final "day of judgment" on which all accounts must be settled and all books balanced. Such a procedure may appear to be somewhat unreal, but it has the virtue of enabling the theorist to construct a playable game.

2.1.3 The relative importance and time structure of the basic units

In this section we offer a few statistics that are extremely crude and are meant at best to be qualitatively suggestive for the U.S. economy. Unless otherwise noted the statistics are for the years 1970, 1980, 1990, and 1995 and are gleaned primarily from the *Statistical Abstracts of the United States*. Four dates, mostly ten years apart, have been selected to indicate the magnitude of change in the statistics. A closer update might show considerable changes both in the relative and absolute size of markets and in the invention of new instruments, but because the prime purpose of displaying these data is to "sweeten the intuition" and give some feeling for the orders of magnitude involved, no attempt is made to be immediately current.

Goods and services are what an economic system is about. (Steak still makes better eating than financial instruments.) However, even though the goals of an economy are primarily measured in terms of real goods and services, the control mechanism is best described in terms of financial institutions and instruments, and its function can be characterized in the measurement of some of their stocks and flows.

In 1970 the gross domestic product of the United States was $1035.6 billion, and in 1995 it had reached $7,265.4 billion, with much of its growth attributable to inflation. If we use an index based on 1992 purchasing power, the 1970 GDP becomes $3,397.6 billion and the 1983 GNP becomes $6,742.1 billion.

Table 2.2
A crude breakdown of U.S. GDP (billion $)

	1970	1980	1990	1995
GDP	1,035.6	2,784.2	5,743.8	7,265.4
Wages and compensation	618.1	1,653.9	3,352.8	4,215.4
Transportation and utilities	88.3	242.3	482.1	622.4
Wholesale	72.2	195.3	367.2	484.4
Retail	100.2	245.9	503.5	637.6
Finance, insurance, and real estate	146.6	418.2	1,024.1	1,361.3
Services	120.5	377.3	1,059.4	1,440.3
Manufacturing	249.6	584.4	1,031.4	1,286.3

Source: Yale Social Science Data Archive; Citibase database information provided by DRI/McGraw-Hill.

1. *Services:* All wages are payments for services. Because family services are not monetized, the considerable addition of in-home services does not appear in the accounts. Table 2.2 indicates that around 60 percent of individual income in the United States is from the sale of labor. Household services may add another 20–30 percent that does not go though the markets and thus escapes the scrutiny of the tax collector. The next part of the table indicates the size of the contribution of services that are heavily if not totally devoted to the transaction and administrative technology of the economy. A further entry shows the contribution of manufacturing.

The stock behind the supply of human services is the population, which can be crudely evaluated as long as one does not take the numbers too seriously. An estimate using a Cobb-Douglas function and Kendrick's revised GNP and factor payment values for 1969 yields a value of $4,481.1 billion, which is somewhat above the $2,386.4 billion given for Kendrick's estimate of human capital stock in 1969. Thus, if we view humans only as machines, we can assign them a capital stock value of somewhere between one to two times that of the physical capital stock.

2. *Goods:* Table 2.3 shows the evaluation of physical or real assets. In terms of the analysis in this book, giving the valuation as a monetary number is in essence circular because we are chiefly interested in price formation. We follow this procedure merely to supply some orders of magnitude for the sizes of goods and services in contrast with paper.

Table 2.3
Real assets (billion $)

	1970	1980	1990	1994
Gross	4,428	14,306	25,675	30,925
Private	2,689	9,364	16,871	20,142
Nonresidential equipment	679	2,389	4,506	5,238
Nonresidential structures	790	2,683	4,979	5,772
Residential	1,219	4,292	7,387	9,131
Government	1,060	2,979	4,729	5,593
Equipment	274	488	952	1,200
Structures	785	2,491	3,777	4,394
Federal	424	885	1,477	1,739
Military	276	493	881	1,060
State and local	636	2,094	3,252	3,855
Consumer durable goods	680	1,963	4,075	5,190
Net stock	2,708	8,619	14,947	17,648
Land (1,000,000 acres)	2,271	2,271	2,271	2,271
Private	1,510	1,552	1,622	1,609
Federal	761	719	649	662

Source: *Statistical Abstract of the United States* (1986, no. 743; 1996, no. 364).

Virtually all of these assets have ownership paper written against them and can serve as backing for a credit system. All privately held land requires deeds of ownership. Automobiles and houses require ownership documents, if only for taxation purposes. Corporate ownership must be documented. This leaves essentially consumables, some other personal possessions, and casually accounted for in-process inventories as the only goods lacking specific ownership paper.

Unlike financial paper, which can be created, transferred, and destroyed in a relatively brief time, the real assets of an economy give it a structure that can only be modified over time. A crude standard for the lengths of time involved is the legal write-off times for various assets. The write-off time for land is infinite; for buildings 40–60 years; for farm buildings 25 years; heavy machinery 12 years; office machinery 10 years. The estimated lives for consumer durables and automobiles vary from around 10 to 20 years. Table 2.4 provides estimates from the U.S. Department of Agriculture.

The age structure of producer and consumer durables provides an important and reasonably factual set of data for the economic planner. One must, of course, be extremely careful in attaching significance to any

Table 2.4
Service life expectancy of new appliances

	Years		Years
Freezer	20.4	Color TV	12.0
Refrigerator	15.2	Dishwasher	11.1
Electric clothes dryer	13.7	Washing machine (automatic)	10.8
Gas range	13.5	Black-and-white TV	10.7
Gas clothes dryer	12.8	Electric range	12.1

Source: Ruffin and Tippett (1975).

evaluation of stock involving an overall monetary worth because the fallacy of composition limits its applicability. (If all or even a substantial fraction of owners of land, for example, tried to sell simultaneously, the total worth of land would become far different from the numbers displayed.) If we assume random mass-market trading at a sufficiently high level, however, the evaluation of the worth of stock does provide an operationally useful figure to compare with the outstanding market value of the financial instruments issued against it.

In order to compare the real and paper sectors of an economy, it is desirable to know not merely the worth of physical stocks and flows but also the level of trade in ownership paper and debt (i.e., ownership changes in the secondary markets). This activity provides a mixed signal concerning intertemporal needs for liquidity and a measure of disequilibrium and differences of opinion among the agents.

Approximating on the basis of an estimated mortgage life of 6–7 years, we find a turnover in private real estate of around 15 percent of the stock.[3] Trade in automobiles is about 15–20 percent of the stock of which somewhat more than half is for new products. Replacements of machinery and plant involve around 5 or 6 percent of stock. (See table 2.5).

When trade in consumer goods is considered, for every final sale there may be between three or four derivative sales. In a nonstrategic world in or near equilibrium this would be of little interest; but in a society where individuals have considerable freedom of economic action in all capital markets and much trade takes place using financial instruments, each stage in the trade of goods involves different actors and instruments in the financial control of the physical process.

Even in an individual enterprise economy the distortions between government accounting conventions and economic reality have to be large.

Table 2.5
Replacement of machinery and plant (billion $)

Year	1982	1983	1984
Net capital stock (a)	6,242.2	6,577.6	6,958.6
	Capital consumption		
Allowance (b)	356.4	377.1	403.3
b/a (%)[a]	5.71%	5.73%	5.80%
	Useful life from these data		
(90%/(b/a))	15.76	15.70	15.53
(100%/(b/a))	17.51	17.44	17.25

Source: *Survey of Current Business*, 1986, January.
a. These figures are relatively low partly because all data include structures.

Table 2.6
Occupied housing units by age of structure, 1980 (in thousands)

	%	#
Total	100	80,390
< 5 years	12	9,649
5–10	13	10,451
10–20	20	16,078
20–30	18	14,470
30–40	11	8,843
> 40	26	20,901

Source: *Statistical Abstract of the United States* (1985, table 1317).

Thus the private housing stock contains many houses far beyond their full legal depreciation. Some may even be several hundred years old, with innumerable repairs and renovation.[4] (See table 2.6).

3. *Money:* To the individual in a mass-market society with prices, money represents the ultimate in strategic economic power. It provides instantaneous liquidity or a reasonable approximation thereof. Liquidity is a basically strategic concept reflecting the speed and breadth of an individual's potential control over real and other financial assets. For any agent in a loosely coupled mass dynamic economy with thick markets, money is approximately a transferable utility in the sense that because any sale by A to B, if the parties are small enough, has virtually no influence on price, the prices to all provide a fixed rate of exchange. When, however, behavior is highly correlated and the economy is far from equilibrium, such as in panics or mass reactions to major events that bring about an overall

Table 2.7
U.S. money stock, seasonally adjusted (billion $)

	12/70	12/80	12/90	12/95
Currency + travel checks	49.5	119.3	254.6	381.5
Demand deposit and checkable	164.9	289.6	571.3	747.6
Savings deposits	260.9	400.2	923.8	1137.1
Time deposits:				
Small	151.2	728.5	1172.8	933.7
Large	45.1	260.2	480.9	419.8
Money market mutual funds	n/a	n/a	507.3	n/a
M1	214.4	409	825.8	1129
M2	626.6	1601.1	3279.5	3655
M3	677.2	1992.3	4125.9	4592.5
L	814.8	2330	4976.6	5697.6

Source: Yale Social Science Data Archive; Citibase database information provided by DRI/McGraw-Hill.

change in expectations, then the fallacy of composition is manifested in the dynamics of disequilibrium and as swings in expectations.

Custom and law determine the instruments of individual liquidity. Thus, in a society where gold is a legal means of payment, it will be part of the money supply. Behind custom and law are the enforcement mechanisms, hence what may be liquid for one set of agents may or may not be liquid for another. Garvey (1977, chapter 5) in his description of Soviet monetary policy indicates how the liquidity of the consumers was controlled separately from the productive sector.

Table 2.7 provides an indication of the considerable change in money stock and composition in the United States in a span of twenty-five years.

4. *Ownership claims:* In a world where economic activities are only partially monetized, where conceptual problems concerning the meaning of complex forms of ownership, depreciation, marketability, transactions, and information costs exist, where barter transactions take place and many services (such as housewife services) are rendered without going through a market, it is no surprise that accounting is not an exact science.[5]

In particular, not all assets are represented by ownership paper. The more likely it is that an asset can be used to secure a loan, or that the asset cannot be easily transferred physically but its title can be transferred, the more likely it is that ownership paper will be created. Hence there are deeds, titles, and stock certificates.

When one buys a washing machine or an automobile on consumer credit, the paperwork concerning the validation of ownership will, in general, be done more precisely and carefully than when one buys the item for cash. For smaller items such as toasters and even television sets, many owners do not even bother to keep ownership papers or record serial numbers.

Beyond the stock market, in a decentralized economy such as the United States or Japan, another set of major markets are in real estate. These markets are far more imperfect than the stock market, and the motives for operating in real estate are more diverse than in the stock market. Leaving aside tax laws, which influence motivation considerably, many individuals and firms who buy real estate actually want to use the asset. They live in the houses or operate the manufacturing plants. The separation of ownership from active management is far less in this market than it is in the stock market. The opportunities to be a passive investor trading in real estate debt or ownership paper are more limited than are those to trade in corporate shares or debt. Institutions such as real estate investment trusts (REITS) and mortgage-backed securities have come into being to enlarge the possibilities of passive investments more dissociated from the management of real estate.

5. *Service contracts:* Few service contracts are negotiable instruments. For salary and wage contracts, this is probably related to considerations of bondage because the capital asset providing backing for one side of the contract is a person. In general, service contracts are explicitly bilateral and specialized, and hence are often not fully identical. In contrast, in the stock market trade is essentially anonymous and shares of the same issue are fungible.

Wage contracts provide a considerable linkage between present and future economic activity. The provisions may vary from hourly payments for casual labor to life tenure. Professionals not working for themselves tend to be paid monthly; sales personnel weekly or biweekly; taxi drivers, laborers, and handlers daily or weekly.

6. *Futures:* The economic purpose of a commodities futures market is to allow investors to hedge against changes in price of basic commodities or inputs that may be needed in an industrial process. Thus, for example, in the United States there are well-developed futures markets for wheat, eggs, copper, lead, oil, and many other commodities. The centrally planned

economy of the former Soviet Union had no futures markets. It had to depend on planning agencies to provide the hedging and allocation needed to avoid production stoppages due to supply failure.

Futures cannot usually be purchased with terms longer than one or two years. Longer futures take the form of contracts between specific institutions, such as a thirty-year contract for a coal supply by a power company. Such contracts usually have renegotiation or other forms of flexibility built into them.

A futures contract is essentially an insurance policy covering delivery at an agreed upon price to be paid at delivery. As such, the actual sale of a contract involves the exchange of one piece of paper (the futures contract) for another (money). Only a fraction of the money required to make the final purchase of the actual commodity is put up before delivery. This is the required earnest money.

7. *Promissory notes:* As Ben Franklin noted, "The Creditors are a superstitious Sect, Great Observers of set days and times."[11] By far the greatest linkage between the present and the futures is provided by the market for loans, which includes much of insurance as well as bonds and other forms of debt (although mutual insurance companies provide equity rather than debt positions).

8. *Futures on financial instruments:* Puts, calls, shorts, and warrants are in essence futures contracts on ownership paper. Currency futures provide hedging opportunities on foreign exchange. It was only in April 1973 that an organized exchange, the Chicago Board Option Exchange, was opened to trade financial futures. The CBOE offered a secondary market for paper that previously did not exist or was hardly negotiable before. In its first eight months of operation, 1.1 million contracts were written.

Table 2.8 provides an aggregate view of credit market instruments. The time structure of all of this paper is key to understanding the constraints of the past and present on the future. For example, over 70 percent of V.A. mortgage issues were for 30 years; but if the actual life of a mortgage were 5–10 years, this might indicate a turnover rate for real estate of around 5–10 years. Table 2.9 shows U.S. Treasury debt maturity structure.

A glance at the *Wall Street Journal* indicates that the length of futures (most are under a year), the number of contracts, and the value of trade for agricultural contracts are such that volume is small relative to most financial markets.

Table 2.8
Flow of funds basis

Instrument	1970		1980		1990		1995	
	Billion $	% GNP	Billion $	% GNP	Billion $	% GNP	Billion $	% GNP
U.S. Treasury	291.2	27.5	1,044.4	35.5	3,911.7	67.3	6,018.7	81.5
State and local	144.0	13.6	330.1	11.2	1,184.4	20.4	1,301.1	17.6
Corp. and for. bonds	201.3	19.0	490.6	16.7	1,704.0	29.3	2,759.6	37.3
Mortgages	473.1	44.8	1,451.8	49.3	3,803.7	65.4	4,724.1	63.9
Consumer credit	143.1	13.5	386.7	13.1	812.0	14.0	1,115.5	15.1
Bank loans n.e.c.	151.2	14.3	456.3	15.5	815.0	14.0	946.0	12.8
Open market paper	40.1	3.8	163.8	5.6	609.9	10.5	700.4	9.5
Other loans	102.7	9.7	316.1	10.7	933.5	16.1	985.4	13.3

Source: Annual Statistical Digest of the Federal Reserve Board (1981, Mar. 1991, Nov. 1996).

Table 2.9
Maturity of treasury debt (million $)

	1970		1980		1990		1995	
	Absolute	Rel. %	Absolute	Rel. %	Absolute	Rel. %	Absolute	Rel. %
Less than 1 year	123,423	49.82	297,385	47.72	626,297	34.00	1,002,875	34.93
1–5 Years	82,318	33.23	197,409	31.68	630,144	34.21	1,157,492	40.32
More than 5 years	41,973	9.10	128,391	8.99	558,462	31.79	710,414	24.75

Source: Annual Statistical Digest of the Federal Reserve Board and *Statistical Abstract of the United States* (1987, no. 524).

2.1.4 On trust and finance

Although debentures are not secured by particular assets, their seniority over common stock in general provides an important margin of safety to corporate lenders. Loans to individuals, unless explicitly secured, derive their protection from the courts, custom, and to some extent even force (manifested, for example, though collection agencies for moneylenders). The major markets for consumer loans are for secured loans such as mortgages and automobile loans. Important exceptions are credit cards and short-term retail credit, where credit assessment of the reliability of the individual customer is of considerable importance. Financial structures involving the interchange of paper claims backed by physical goods and the law seem to be designed to optimize the availability of credit for a given level of trust and security. The more security there is, the less the need for trust. A bank loan covered several times by the market value of AT&T stock does not need a large dossier on the personal history of the borrower.

This chapter sketches the features of economies utilizing financial paper, hence it is several steps removed from "theorem proving." Nevertheless, even at this level of generality we can make the following assertion: *For a given level of trust, all other things being equal, an economy with more real assets that can be posted as security will have a financial structure as close or closer to the desired politicoeconomic optimum than an economy with fewer real assets.*

Put even more simply, the borrowers in the economy with more real assets have more hostages available, and (up to a point) hostages can be traded against trust.

A unifying theme in this work is that the financial system enables individuals in a heavily anonymous economy to economize on trust. This is done using money, debt, and other financial interments in situations where the degree of trust may vary considerable. One of the major challenges in the development of a theory of money and financial institutions is to characterize and make operational the concept of differing degrees of trust.

2.1.5 The institutions or actors who create and destroy the basic instruments

The production of goods and services in the United States is primarily controlled by private corporations and individuals. Table 2.10 provides a

Table 2.10
Number of active businesses in the United States (thousands)

	1970	1980	1990
Nonfarm proprietorships	5,770	8,932	14,783
Partnerships	936	1,380	1,554
Corporations	1,665	2,711	3,717

Source: *Statistical Abstract of the United States* (1986, nos. 874, 878, 879; 1997, no. 833).

breakdown of the legal forms. Many of the proprietorships are for professional services. The aggregation conceals the size of and concentration in major industries, with the top 500 nonfinancial corporations accounting for 69.5 percent of corporate sales in 1984 (*Statistical Abstract of the United States*, 1986, p. 532).

The strategic players are firms, some of which may be large and oligopolistic, and individuals who even as employees exert power through both the political system and unions. In 1984 the workforce had 113.5 million members, of whom about 13.7 million were in unions (*Statistical Abstract of the United States*, 1986, pp. 393, 423).

Money: If we count bank debt as money, then among the institutions in the United States that participate in the creation of money are the Federal Reserve System, the Treasury, the commercial banks, and the public in general. A simple picture of causality is not possible. Different groupings may have different potentials for money creation.

An accurate description of the money expansion possibilities in an economy cannot be given by merely describing the individual financial units. "It takes two to tango," and in some instances it may take two or more to create new money. The story of how money can be created or destroyed calls for the specification of the rules of the money game. The minimum list of institutional actors whose role must be described includes: The Federal Reserve System, the Treasury, the tax authorities, and the banking system. In the former Soviet Union the system's volume was regulated directly by state authority through direct controls and the Soviet State Bank and Treasury (Garvey, 1977, p. 39).

Ownership claims: It is sometimes forgotten that the stock market is primarily a secondhand market for negotiable paper. The creation and destruction of ownership paper may frequently involve the use of the

Table 2.11
New security issues of corporations (billion $)

	1970	1980	1990	1995
Stocks	8.1	21.9	40.2	100.9
Bonds	24.366	44.65	298.9	573.2

Source: *Statistical Abstract of the United States* (1986, no. 858; 1997, no. 811).

stock market, but the market itself is a brokerage arrangement and not a creator or destroyer of ownership paper per se.

New stock is created through flotations by new or existing firms. New issues were approximately as shown in table 2.11 (*Statistical Abstract of the United States*, 1986, p. 507). Retirements in 1970 were estimated at $2,411 million. The changes represented around 39,000 new issues and 9,000 retirements. These should be compared with a market value of sales on the exchanges of $131 billion, with a turnover of the order of 20 percent.

Stock ownership paper is wiped out by failure, merger refinancing, or confiscation. It is created by groups who have the power to be believed, as is illustrated by a quote from the South Sea Bubble financing: "A company for carrying out an undertaking of great advantage, but nobody to know what it is."[6] Leaving aside corporations, paper of varying levels of negotiability is constantly being created and destroyed by the changing patterns of real estate holdings and development.

As economies are increasingly monetized and accounting practices and reporting improve, we may expect the amount of ownership paper to increase. For example, a state or city pawnbroker such as the Dorotheum in Vienna creates tickets representing the owners' claims against property that previously may have been unrecorded. The ticket is a compound instrument combining an ownership claim with a futures contract to redeem the pawned goods at a specified price and by a specified time.

Among the goods of sufficient importance to merit separate discussion are houses, automobiles, and major consumer durables. In economies where decentralized markets are used as the means for the distribution of these items, the creation of this paper involves four parties: producers, consumers, specialized financial institutions, and government agencies.

Service contracts: The major parties to the creation of service contacts are individual suppliers of labor, unions, small and large private firms, and institutions and governmental bodies. The security aspects of a civil

service post or a university tenure do to some extent provide extra borrowing power or credit worthiness, but the labor service contract in general, as has already been noted, is not negotiable paper.

Futures: The major parties to the creation and destruction of futures are primary producers and industrial users. The smooth functioning of these new and growing markets has been highly influenced by a combination of computer and communication technology and law.

Promissory notes: The creation of promissory notes, bonds, loans, mortgages, insurance claims, and other debt is a major subject by itself. I shall offer only one small, but in my opinion extremely important, point concerning loans.

Whenever a loan is created, there are at least two directly active parties (and frequently more). There is almost always an indirectly involved, but important, third party—a governmental agency that controls (and frequently changes) the rules. And there are at least two basic types of paper, several subspecies, or mixed instruments involved.

An individual who borrows from a bank usually takes the loan in some mixture of money (or liquidity or "now money"), such as cash and a bank deposit, in exchange for a piece of paper the borrower creates (albeit with the consent of the bank and in conformity to various regulatory and legal requirements). Depending upon the nature of the securing of the loan, other paper such as stock or other ownership claims may be pledged. But the fundamental feature is that when a loan is made, regardless of the type of money the lender supplies, it is the borrower who is the prime creator of a new type of paper, which is a promissory note (an IOU or a future delivery contract for money). If the lender does not control or limit the promises of the borrower, disaster can result. A borrower who receives $100 could, by several strokes of the pen, write out an IOU note promising to pay $1,000,000,000 in a year. If there were no usury laws or other limitations this contract might be legally binding, but the chances of its fulfillment would be slight. When a lender permits a borrower to promise more than the borrower can probably pay, the lender simply increases the chances for insolvency.

Money and promissory notes are different financial instruments, and their direct creators are different institutions and individuals. Because a banking system plays an important risk-evaluation role in an enterprise

economy, it is reasonable to expect that it will apply rules not only to its lending but to what it will permit a borrower to offer. Only with relatively shady banking groups in deals of dubious legality will the offer by a borrower to pay ten times the prime rate improve his chances of getting a loan.

Derivatives: The individual owner of a claim can write a put or call or other derivative. The leverage is large, and the number of players in this game used to be few. Puts and calls are instruments for magnifying the gains or losses to be made from fine differences in perception, and hence are magnifiers of a dynamic adjustment (or disequilibrium?) process. Since the late 1980s the world of exotic instruments generally called derivatives has exploded, along with even finer divisions in risk assessment.

The specific actors in any modern economy can be usefully divided into five broad groups. In a process world where information and evaluation are critical and bureaucracies abound, the differentiating characteristics of the following actors need to be developed:

1. Consumers: passive buyers, voters, savers, who sell little beyond their services.

2. Businesspeople and entrepreneurs: the active investors of private resources.

3. Administrators: the bureaucrats and technicians who are the major carriers of routines and processes.

4. Financial middlemen and financiers: the group whose function and tools include information processing, evaluation, brokerage, and the control of money and financial paper.

5. Politicians: the brokers of power and fiduciaries of the public sector, coordinating (among other factors) public administration finance and business.

A fruitful theory of economic process should help us identify where these different functions emerge as a virtual necessity of the process.

2.1.6 Social and individual contracts

The relationships between the individual and the government, between the individual and banks, and between the individual and other individ-

uals of like power and position are considerably different. This difference is reflected in the nature of the implied or explicit contract between the parties. In particular, the asymmetry in contractual relationships is considerable with respect to money and credit.

Of the eight basic economic instruments described in section 2.1.1, two are nonfinancial (goods and services) and the other six are financial. Two of the latter (money and ownership paper) were presented as not being contracts, and four (futures, service contracts, debt, and options) were presented as contracts. In one sense all six financial instruments are contracts, but especially with money and certificates of individual ownership, the two-party aspect of the instrument is hardly relevant for most operational purposes. This point is enlarged upon further below.

Historically there have been many ways in which new fiat money, fresh from the printing press, has entered the economic system. Fiat money might be regarded as a reflection of society's debt to itself, hence one might believed that it nets to zero. But a consideration of overlapping generations (see Volume 2, chapter 9) will argue against this interpretation. Even if fiat netted to zero for society as a whole, to the individual who holds a unit of fiat and regards it as a debt note of the government, there is little of operational significance to this relationship. The meaning of the other side of the balance sheet is not as clear as when a second individual owes the first money. When an individual presents a dollar bill to the treasury for payment, that individual stands to receive another dollar bill; hence there is no strategic significance to the act. When, however, the individual presents an IOU note for a dollar to the debtor who wrote it, he expects to receive a dollar or to take action against the debtor if the dollar is not forthcoming.

For somewhat different reasons the contractual aspects of ownership paper may also be suppressed much of the time. If an individual happens to be a sole owner of an item or establishment, his ownership paper is merely a certificate establishing his claims. If there is mass joint ownership in the form of common stock, then the paper involves the contractual relations between the corporation and the individual. But there will usually be considerable asymmetry between what the individual noncontrol stockholder can do to the corporation and vice versa. In essence, the individual's strategic power involves hardly more than buying or selling paper in a secondary market.

2.2 Past and Future Instruments

2.2.1 On present and past instruments and institutions

In section 2.1.2 it has been suggested that, in essence, any past, present, or future financial instrument can be characterized as a blend of no more than eight basic instruments. Thus, for example, an insurance policy has elements of ownership paper and debt. The role of chance was not discussed explicitly in the description of table 2.1. The insurance policy involves chance in its purest textbook form. The policy is a lottery ticket in which money payments now may be transmuted into money payments later, depending on a random event. A *tontine* involves a pooling of money that will be paid to the individual who has the good fortune to be the final survivor.

Taxes, technology, population size, law, custom, education, transaction costs, and risk of every sort are basic factors in the generation and demise of financial instruments and institutions. The invention of the treasury bill is attributed to Walter Bagehot in 1877.[7] Post office savings accounts were used in Great Britain chiefly to encourage small savings among a segment of the population who feared banks but were comfortable entering a post office. The institutional role of the post office has diminished as its costs have risen and attitudes have changed. With the changes, post office savings have dwindled. The invention of the certificate of deposit is credited to Walter Wriston. The money market fund did not exist until 1972. The first fund in the United States was the Reserve Fund started in 1972 by Harry Brown. Its competitive advantage was provided by transactions costs and legal restrictions. A small depositor was not in a position to buy a $100,000 certificate paying a higher rate of interest than a savings account, and the law placed a limit on interest payments by banks. Improvements in record keeping and communication offered an opportunity to sell shares in an institution devoted to buying large short-term debt instruments. The rise of the Eurodollar market came about primarily as a way to avoid U.S. banking regulations. A good indication of the variety of new instruments as of the late 1970s was given by Stigum (1978), and since then new instruments have proliferated.

By most measures, economies of more than four hundred years ago were considerably simpler than those of today. An appropriate analogy might be between a bicycle and a jet aircraft. The need for a delicate and

quick-reacting control system is far larger now than then. In his last book on premodern financial institutions, Goldsmith (1987) reinforces the observations of Kindleberger (1984) that modern finance dates to the seventeenth century. There were few financial instruments prior to that time. Instead of national debt, there was the debt of the monarch. Taxes and debt financing were used primarily to support the king and aristocracy and the financing of wars. Goldsmith (1987) suggests that around 50 percent of the revenues of Augustan Rome and of the early Abbasid caliphate went for the armed forces (pp. 55, 79). One-third of expenditures of Periclean Athens and one-half of the expenditures for the Florentine Republic in the early fifteenth century were for the military (pp. 247–248). In Elizabethan England, estimated expenditures varied from one-fifth in the early part of the reign to four-fifths after 1585 (p. 248). Goldsmith concludes with the observation: "in all cases military expenditures were the most important single category and generally accounted for between one-third and two-thirds of the total" (p. 249). He also observes, however, that these has been a great disparity in the percentage of national product to government expenditures, "ranging from a few percent in Augustan Rome and Elizabethan England to between one-fourth and one-third in the Abbasid caliphate, the Ottoman Empire, Mughal India, and Tokugawa Japan" (p. 249).

2.2.2 Future instruments and institutions

The current vogue in banking in the last half on the decade of the 1990s is for new products. Interest rate swaps, hybrid instruments, special credit cards, and mortgage pools are all products of the ability to communicate at little cost and in next to no time with every major financial center in the world. Financial reporting requirements combined with on-line data banks and financial analysis programs have transformed the possibilities for efficient arbitrage. Along with this growth of data and data-processing capability has been a growth in credit evaluation services and other forms of technical financial consulting. Thus, for example, a natural application of integer programming is to the evaluation of the payouts from a mortgage pool with mortgages of many different durations and rates.

The advent of relatively cheap business calculators in the form of the personal computer, and communication developments such as the French P.T.T. Minitel (which converts a home telephone into a powerful communication device), are revolutionizing the instruments and institutions

of finance. As the ability to calculate improves and communication approaches being almost instantaneous, some of the problems and paradoxes both with the theory and practice of finance are highlighted. Precisely what is liquidity, and how is it influenced by the change in technology? Transactions take time to clear when credit instruments are used. Float times recently were as much as four days for clearing on the New York exchange; a settlement day every two weeks on the London Exchange; and thirty days for many credit card and other purchases. Checks may take several days to clear through a banking system, creating a float in which one party parted with liquidity but the other has not yet been credited. What is happening to the float, and who are the winners and losers? In a world where money might be lent for a period of a few hours or less at interest, how will time zones enter the calculations?

The explosion in technology has not been matched by equal changes in law or custom. Change in jurisdiction today and possibly twenty years from now may not cause fewer or qualitatively easier problems than they did twenty years ago. The desires for the total anonymity of cash may be increased rather than decreased by the efficiency of credit cards, data banks, and computers in keeping track of one's movements.

The changes in the internal financial instruments and institutions of a society require adjustment to a single major jurisdiction.[8] But international trade poses strategically different and more complex problems. Great swindles can be based on arbitrage created in dealing under the laws of two or more jurisdictions (see, for example, the history of I.O.S. as recounted by Raw, Page, and Hodgson, 1971). Even more critical, however, is that the possibilities for trade wars, international prosperity, and depression depend on the power and fiat of a handful of powerful nation-states whose international financial connections and reaction times are fast becoming greater and quicker without a commensurate growth in international law.

In reviewing the flood of new financial instruments from the mid-1960s to the 1980s, Miller (1986) suggests that taxes and regulation provided much of the impetus to innovation, but (shades of Bagehot) even if the original impetus is removed, an instrument may survive for other reasons. He also suggests, however, that with fewer "inefficient tax and regulatory structures" there may be a slowing down of innovation. Although I substantially agree with Miller's observations on the impetus given by taxes and regulation to arbitrage via new instruments, finance is part of the

control system of the political economy, and the purely economic criteria of efficiency[9] are not easy to define without a specification of political bias.

The popular financial press emphasizes the new ability to lay off risk with new instruments and information processing. Given the nature of the nation-state, the greater internationalization of trade and finance, and the tendency of many to ignore the fallacy of composition, it is by no means clear whether overall risk to society is being increased or decreased. What is clear is that the growth of new and sophisticated instruments will continue.

2.2.3 Competitive and command economies

In his masterful estimation of the national product of the early Roman Empire, Goldsmith (1984, p. 283) suggests that the expenditures of the imperial government were probably not above 3 percent of national product and on the order of 5 percent for all government units. In India the share of public consumption in 1960 was 7 percent, and Goldsmith notes that this was roughly the ratio in England in 1688 and in the United States and France in 1820. This contrasts with tax revenues as a percentage of gross domestic product in 1983 varying from as low as 27.2 percent for Spain to as high as 50.5 percent for Sweden, with most of the developed countries in between (*Statistical Abstract of the United States*, 1986, p. 846).

A necessary but not sufficient condition for a reasonably laissez-faire or competitive economy is that the government's share in both the production and appropriation of the national product be small. Even with high government revenues, the means of production may still be in the private sector. A command economy can be run with a blend of private ownership, partial price controls, rationing, high taxes, and a high component of government demand. In 1943 and 1944, the United States spent 41.4 percent of GNP on defense and was thus, temporarily, a command economy. Although not much of the structure of the ownership of goods and paper had changed, the strategic possibilities of individual owners in this period were highly constrained.

The key factor distinguishing a centrally planned economy from an enterprise economy is the scope of freedom allowed to the individual's set of economic strategies. Politics enter into the economics inasmuch as

political forces may set the restrictions on economic actions and limit the choices of some or all individuals. Thus a dictatorial state or an empire can be consistent with either a laissez-faire or a command economy.[10]

Accounting documents, nonnegotiable warehouse receipts, and calculations in terms of a numéraire or in terms of quotas on physical units of goods are all part of a command economy. Paper will exist, but it is sterilized if it cannot be traded. The development of secondary markets for paper provides an index of the decentralization of economic freedom.

The more decentralized economic decision making becomes, the greater are the possibilities of swings. The implicit assumptions about the smooth functioning of markets, exchanges, banking, and other financial institutions include a belief in the noncorrelated market behavior of large numbers of small economic agents. But in panics and bull markets, individuals may behave in a manner that indicates a high level of implicit or explicit coordination.

When a government in a more or less laissez-faire economy wants to pursue overall macroeconomic goals, it must pay a price for the decentralization of economic power it has created. The financial instruments in the hands of the public in an enterprise economy make the system far more loosely coupled than in a command economy. This loose coupling makes government control and choreography of the economy far more difficult to achieve than in the command economy.

A brief consideration of the pre-Gorbachev financial structure of the Soviet Union, without even going into detail, raises some fundamental questions concerning the meaning of economic efficiency and the need for and purpose of many financial institutions and instruments.

The fundamental economic concept of efficiency is Pareto optimality, the points at which it should not be possible to rearrange the economic outcome to make all better off. Yet before one can even describe the set of feasible outcomes, it is necessary to specify "the game." What are the process mechanisms; what institutions have been assumed, and how do they restrict the feasibility of outcomes? Institutions are the carriers of process, and there is for this reason no such thing as an institution-free description of a process model of an economy.

The decision to have or not to have the major means of production privately owned cannot be construed merely or even primarily as an economic question. If a society chooses either by vote or force to nationalize most of its productive capacity, then the question to the economist is what

constitutes economic efficiency given the existence of the nationalized sector.

The financial design problem for the Soviet Union was the choice of financial instruments and institutions that would allow it to achieve economic efficiency constrained by its sociopolitical institutions. It is perfectly possible that the Soviet Union (like its czarist predecessor) was grossly inefficient, as has been suggested by Rosefielde (1986). The many political and economic changes in the nation-states that are heirs to the former Soviet Union do indicate that there are necessary trade-offs between political philosophy and economic goals.

The former Soviet Union has been described as having a "monobank-monobudget system" evolved largely through politically determined experimentation, but with some historical connections to the czarist Russia's State Bank (Garvey, 1977, pp. 2–3). The system was a control system where money played a role well beyond Lenin's pronouncement that in his society money was a mere numéraire (Lenin, 1960); but given the structure of Soviet economic planning and limitations on the financial strategic possibilities for the individual out of government, it might be better to describe the last Soviet economic scheme as involving monetary management more than monetary policy. A key consideration was the division of enterprise money from household money and internal finance from foreign trade finance. Liquidity, which is a critical strategic measure in an enterprise economy, appears to have been of little importance to Soviet industry and hardly permitted to the Soviet consumer.[11]

A quotation from a description of Russian banking before the Soviet era suggests some historical antecedents: "In examining the credit system of Russia, the fact that first of all strikes us is that it has always been conducted, if not in accordance with invariable principles, yet at least after a uniform method and one which has conformed to the political policy of the Empire. The State is the great dispenser of favors, the general manager of all affairs of its subjects" Antoine E. Horn, editor-in-chief of the *Journal de St. Petersburg*.

The first banknotes[12] in Russia were issued in the reign of Catherine II on December 29, 1768. Banks of issue were established in St. Petersburg and Moscow. In 1786 they were consolidated with the loan bank of the Russian Empire.

Until 1861 the basis for landed credit in Russia was the number of serfs on an estate.[13] With emancipation, the bank did not have this security.

Under Alexander II, reforms of the whole financial system were instigated, and in 1860 the State Bank was founded. Private joint-stock or nongovernment banks came into being in 1865. In 1883 the "Peasants Bank," a strictly state institution, was founded, and in 1886 the Land Credit Company was transmuted into the "State Aristocracy" or "Titled Nobility Bank," also controlled by the state; it had restrictions against selling foreclosed estates of the nobles to commoners (Horn, 1896, p. 405).

In 1856 the St. Petersburg Bourse was trading three currency exchange rates, two loans, and the securities of sixteen corporations (including the Russo-American Company for ventures in Alaska); it opened only on Tuesdays and Fridays from four to five (Horn, 1896, p. 411). It developed rapidly, and by 1872 it was open six days a week. Its existence was terminated by the October Revolution, and its imposing building by the water was utilized as offices by the Soviet navy.

Lenin wrote shortly before the October Revolution: "Without big banks, socialism would be impossible. The big banks fund the "state apparatus" which we need to bring about socialism, and which we take ready-made from capitalism.... A single State Bank, the biggest of the big, with branches in every rural district, in every factory, will constitute as much as nine-tenths of the socialist apparatus. There will be country-wide bookkeeping, country-wide accounting of the production and distribution of goods; this will be, so to speak, something in the nature of the skeleton of socialist society" Lenin (1960, vol. 26, p. 106 [italics omitted]).

The Soviets were unable to take over the old system as a fully functioning operation. In 1921, the new State Bank of the Russian Soviet Federated Socialist Republic was created under the Commissariat of Finance. In 1930 a fundamental step in removing strategic freedom from enterprises was the abolition of interenterprise credit. This gave the State Bank full control over short-term credit. The State Bank, the Investment Bank, the Bank for Foreign Trade, and the Savings Bank formed the backbone of the Soviet banking system after 1932: reforms from 1965 on indicated increased concern for improving the production incentive system and for introducing somewhat greater flexibility in the uses of pricing and interest rates on investment.

The State Bank collected and disbursed more than half of the national income of the Soviet Union, which acounted for around 50,000 government entities. The savings bank system incorporated into the State Bank after 1963 provided the general populace with limited banking facilities,

with most of the 74,000 offices combined with the post and telegraph offices (Garvey, 1977, p. 65). Private savings appear to have been primarily for large consumer goods or vacations. Lottery bonds added a little fun and gambling to the consumer saver. These bonds, savings deposits, and money were the financial instruments available to the Soviet citizen, although in the major cities there was trade in cooperative apartments. After 1959 the Stroybank handled virtually all long-term capital investment.

The purpose of this quick overview of the Soviet financial system is to provide a contrasting example that highlights the proposition that financial instruments and institutions, their role and their control are essentially manmade and are fundamentally delineated by strategic considerations.

The dissolution of the Soviet Union and the changes undertaken by its inheritors might lead a pure free enterpriser to conclude that in some overriding way the capitalist system has proved to be more "efficient" than the Soviet system. This conclusion, however true, involves a dangerous confusion between political and economic goals of a society. Indeed while most economists agree about the dangers and inefficiency of bureaucracies, it is not easy to separate economic from political change.

The overriding rule concerning the creation and destruction of most financial instruments and institutions appears to be that the needs of an efficient atomistic trading and production technology provide the initial impetus for the creation of instruments. But once they exist, these instruments present governments with opportunities for control and taxation. The interaction between the government controllers on the one side and distributors, producers, and consumers on the other is present in every economy. And control mechanisms lead automatically to the search for evasive maneuvers by consumers, capitalists, and commissars. Financial institutions may thus be agents of change as well as mechanisms for control. They provide the economic context for both political and economic growth.

Notes

1. This section draws from Shubik (1975).

2. The somewhat vague word "quality" is used to emphasize the evaluative aspect of the treatment of financial risk. Probabilities are being assessed, but the dimensions of risk are also being sought out. We understand the deal, but what factors have we left out? Hence consultants, evaluators, appraisers, insurance, and reinsurance.

3. Personal communication, R. Slighton, Chase Manhattan Bank. The *Handbook of Mortgage-Backed Securities*, edited by Frank J. Fabozzi (1985, p. 280) states: "It was discovered that the mortgage had a weighted average life of approximately 12 years."

4. The problem of classifying items as old or new is nicely illustrated by a story told by Richard Ruggles: "This old axe has been in the family for 200 years, but it has had five new blades and seven new handles."

5. A useful exercise for every economist is to read the basic assumptions of generally accepted accounting principles (GAAP) to appreciate the approximations and implicit biases that are employed by the accounting profession in attempting to reflect economic reality.

6. MacKay (1980) reports that £500,000 was to be raised by selling 5,000 shares at £100 each. Some 1,000 subscriptions were sold at £2 each, with £98 to be paid one month later. At this point, the promoter disappeared.

7. See the introduction by Frank C. Genovese to Bagehot ([1873] 1962).

8. In some instances state, county, and other local forces may enter.

9. "Efficiency" usually implies both Pareto optimality and no arbitrage in the appropriate feasible set of outcomes.

10. The term *command economy* is used for an economy with a highly centralized planning structure and considerable restrictions on the economic strategies of individuals. Words such as "democratic," "communist," and "socialist" are set aside as referring to political structures that may influence or control the domain of public and private economic choice. For a description of the Nazi economy, see Schmidt (1984).

11. For a description of the change and growth of a market economy in China, see Chow (1986).

12. These banknotes were not legal tender, but until 1812 they were accepted at par by the Russian Treasury.

13. For an excellent fictional development of this theme, see Gogol's *Dead Souls*.

3 Micro and Macro

3.1 What Are the Questions?

Political economy has many branches and many closely allied topics. A partial listing includes special theories of the firm, consumer behavior, production, oligopoly, labor, agriculture, international trade, welfare, public and corporate finance, and banking, macroeconomic theories of the functioning of the economy as a whole, general-equilibrium theory, economic history, demography, and studies of the interactions of politics and economics.

Each of these topics has its own emphases and its own key questions, and each uses its own set of models of reality, with stress on different elements. A natural question to ask is: Are all of these different approaches to the various economic aspects of society compatible with each other and logically consistent?

Not only are many of the models not consistent, but there are good reasons to argue that consistency is neither to be expected nor necessarily desired. The optimal model needed to answer the type of question posed in industrial organization differs radically from that needed to answer a question concerning the efficient distribution of resources, or a question on social security or control of the money supply.

There are, however, some questions that call for an exploration of the linkage between models that do not match. In particular, the basic question explored here is how the role of money and financial institutions and instruments can be reconciled with microeconomic theories of economic optimization. This requires an exploration of the interface between the microeconomic and macroeconomic views of the economy.

The viewpoint of those developing general-equilibrium theory, welfare economics, and many aspects of mathematical economics has tended to be normative. This contrasts with the prevalent attitudes of macroeconomists, whose major concerns are with positive or empirical economics and with advice giving and the study of policy decisions. Weintraub (1978) offered a stimulating discussion of the differences in these two viewpoints and the need (or lack of need) to reconcile them. No attempt is made here to interpret "what Keynes really meant" (see Leijonhufvud, 1968) and how or why it is compatible with the works of Walras. I will give a brief sketch of some of the basic approaches and methods of microeconomic and macroeconomic theory and suggest that for the purposes at hand some reconciliation of these two bodies of theory is desirable. The way to

effect this reconciliation is through the methods of game theory, which lay stress on the strategic aspects of finance.

3.2 On Microeconomics

Microeconomic theorizing and applications have been split into partial-equilibrium analysis, which may have an institutional context, and general-equilibrium analysis, which does not. These remarks can be best illustrated by considering some specific aspects of microeconomics. We look, then, at the theory of the competitive firm and industry, oligopoly theory, and general-equilibrium theory.

3.2.1 Partial-equilibrium analysis

A cursory glance at virtually any textbook dealing with the theory of the firm shows that an institutional environment is assumed as the basis for the firm's activities. Money and markets are taken for granted. Both verbal and diagrammatic treatments of the behavior of the firm include a role for management and monetary transactions, and the less mathematical texts may discuss the context of the industry and society. The justification given for drawing the short-run average cost function of the one-product firm (see figure 3.1a) as U-shaped is that at low levels of production there are increasing returns to scale as management, organization, and plant are built up and used more effectively. After an optimum utili-

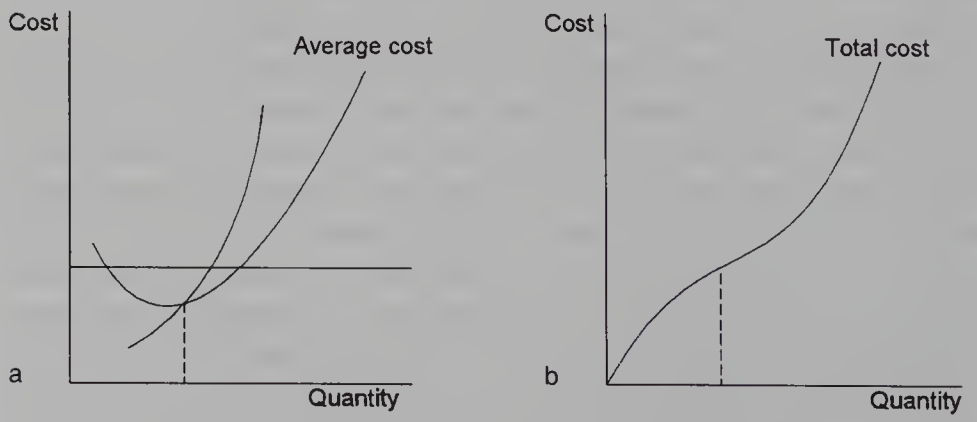


Figure 3.1
Cost curves.

zation has been achieved, decreasing returns set in as "too many cooks begin to spoil the broth," or as unimputed resources limit output.

The U-shaped average cost curve has an institutional story embedded in its shape. It implies an S-shaped total cost curve (as shown in figure 3.1b), which is mathematically far more difficult to work with at a high level of generality than are convex total cost curves.[1] An important technical condition that permitted the development of general-equilibrium theory at a high level of generality is that production is represented by a cone or convex set (see figure 3.2). In the first instance, all traces of the institutional description implicit or explicit in partial-equilibrium theory are removed (even "ownership" of the production set becomes unimportant).

The techniques of partial-equilibrium analysis, whether utilizing diagrams or a full mathematical formulation, are compatible with a verbal imprecision that reflects changes in economic and social conditions and institutions. Thus without precisely spelling out what constitutes an industry or how a workforce is expanded by immigration or improved by education, a set of verbal and diagrammatic snapshots may be presented to suggest, for instance, that the presence of cheap water power and good transportation to populated areas causes growth in the textile industry that attracts skilled workers, which in turn improves the level of education and degree of industriousness of the local population, thereby lowering the long-run average costs of the firms and stimulating a further expansion of the industry.

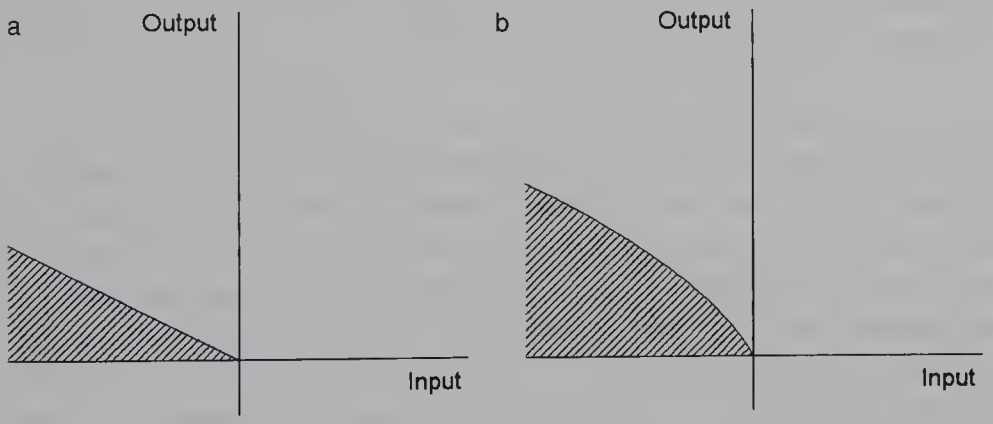


Figure 3.2
Production sets.

This type of argument is heavily qualitative, highly based on institutional insight, and in general adequate to deal only with relatively simple causal sequences. The language of Marshallian partial-equilibrium analysis (Marshall, 1920) is the language of political economy for the high-level businessman, politician, or bureaucrat. Its way of dealing with the imponderables of economic organization, social change, growth of population, spread of information, and knowledge is purposefully vague.

For many purposes this approach is of considerable use. When one requires advice or understanding in order to make a decision that cannot wait until a detailed study is made, a descriptive analysis may be of great value. The price paid for using this approach comes when a question is asked that requires more detail or precision than this type of model can deal with. For example, the operational meaning of an industry poses difficulties when one wishes to design legislation for the control of competition. In a modern society, where firms are run by large bureaucracies, the language of partial-equilibrium economics may not be sufficient to describe the internal decision structure of the bureaucracy. Further more with the growth of communication, network increasing returns are changing the structure of industry.

3.2.2 Oligopoly theory

A branch of microeconomic theory that is often highly mathematical, yet in its manifestation as industrial organization is institutionally oriented, is oligopoly theory. The questions it is designed to answer questions that are more detailed and precise than many of those raised in the theory of the competitive firm. How many firms are required for there to be sufficient competition in a market? What is a market? How can the degree of competition or collusion be measured?

In attempting to mathematize models of oligopolistic competition, economists from Cournot (English trans., 1897) onward have tried to describe process and the dynamics of adjustment among strategically interlinked firms. This has required increasing precision in the formulation of the rules of the game. But these rules provide a rudimentary description of markets and other economic institutions that are the *carriers of economic process.*

In a striking and almost paradoxical way, the development of a satisfactory microeconomic theory of money and financial institutions and the development of oligopoly theory are highly related problems. This point

is illustrated in detail in the models discussed from chapter 6 onward. Here it is noted that the description of oligopolistic competition in general calls for a specification (no matter how abstract) of market mechanisms and the strategic alternatives of the firms. These virtually always involve a description of trade in terms of money and markets. This is not a logical necessity, but an economic fact.

The use of money and credit in organized markets provides an economy with the possibility of designing a transaction technology that is parsimonious in the number of markets needed for trade. Furthermore, it allows for a basic *strategic decoupling of individual behavior*. If bids and offers involve the use of money and credit, then many individuals can act independently and to a great extent anonymously, even taking actions that may ex post turn out to be foolish. The somewhat variable value of money and credit, together with rules governing the failure of individuals to pay back what they owe, provide sufficient flexibility in the system to coordinate any set of actions in or out of equilibrium, regardless of equilibrium considerations.

A key problem in microeconomic theory is how to reconcile the open or partial economy models of noncooperative oligopolistic behavior, which have existed since the time of Cournot, with the closed general-equilibrium model of the economy as a whole. A natural way to achieve this is to introduce markets explicitly into the general-equilibrium model and to have mechanisms involving trade in money and credit designed to carry the dynamics of the system in equilibrium or disequilibrium. But this is tantamount to reformulating the model as a game of strategy. This is in the spirit of oligopoly theory, and in subsequent chapters this program is carried out.

3.2.3 General equilibrium

A detailed discussion of general equilibrium is given is chapter 4. Here a few remarks are offered to illustrate both the power and the danger in economic modeling and its interpretation.

The central questions investigated in the development of general equilibrium involve the existence of efficient production and distribution guided by individual trading at a given set of prices. These can be investigated without explaining how prices come into existence or whether competition is needed to guide individual behavior.

In the mathematical development of general-equilibrium theory, the assumption that the production technology is represented by conic or convex sets (see figure 3.2) to a great extent removes institutional considerations from the model. When production sets are strictly convex, institutions are implicitly included as nontraded goods. When the production technology is linear, firms make zero profits, and hence ownership is irrelevant; but with a strictly convex production technology profits will be made and must be distributed. The firms may be regarded as each owning a private technology and earning a profit or rent that is an imputation of value to the nontraded institution, the firm. The worth is translated into the markets via individually owned shares (the theory provides no analysis for the imputation of value to voting and control).

The modeling price paid for mathematical generality was the suppression of institutions, markets, time, uncertainty, and transaction costs. The gain was the insight into price structure, production, and distribution that arises from extremely simple preference and ownership conditions.

Both the general-equilibrium system and macroeconomic models offer a full accounting scheme for the economy as a whole. Yet money and financial instruments have no role to play in one and are central to the other. In essence, general-equilibrium theory is a multidimensional, noncontrol, noninstitutional economic theory of efficient production and allocation of resources, whereas macroeconomic theory is a low-dimensional, highly institutional theory aimed primarily at solving problems of control in the economy through the network of available financial and governmental institutions.

3.3 On Macroeconomics

Macroeconomic theory is a *control theory*. The reasoning employed involves a blend of assumptions about institutions, sociopolitical features of the economy, demography, and economic dynamics. It is not merely the level of detail that distinguishes macroeconomic theorizing from microeconomic analysis, but the nature of the questions as well. Until recently there was little or no explicitly normative component to macroeconomic analysis.[2] The stress is upon how the overall system works and various aggregates behave, and less so upon why, even though this information might help in the normative design of appropriate governmental policy and control.

Macroeconomic theory tries to explain the level of unemployment, the rate of growth of GNP, the magnitudes and changes in magnitudes of aggregate savings, investment, taxes, income, and consumption and how they are influenced by monetary, fiscal, and other policy.

Institutionally in the United States in the late twentieth century, monetary policy refers more or less to the variables controlled by the Federal Reserve System, and fiscal policy to the instruments of the rest of government, such as taxes, subsidies, and work programs. One aspect of fiscal policy is the control of the path of government debt. Because macroeconomics has a high applied component, the distinctions among instruments tends to be fuzzy, and the time horizon over which they are meant to be effective is rarely specified.

Macroeconomic models are, for the most part, open in time and explicitly formulated in terms of dynamics but closed for the determination of incomes and expenditure in much the same way that the general-equilibrium model is closed, although they frequently include an open foreign sector. Such models often discuss the mechanisms that determine the magnitudes of the variables in the next period as functions of current and previous states.

3.4 Toward a Reconciliation of Micro- and Macroeconomics

There are three stages in the development of a microeconomic dynamics with money and financial institutions. The first stage is to construct fully defined process models in which there is an explicit formulation of the updating mechanisms that determine precisely how strategic actions affect prices and other economic variables. The second stage is to analyze these models for appropriate static or stationary solutions, to see if they are consistent with general-equilibrium theory and with the type of stationary-state analysis that is useful for the related macroeconomic models. If they are not consistent, we must examine the reasons why. The third stage is to develop a satisfactory dynamic theory.

We are in a position to tackle the first two stages, which are preliminary to developing a satisfactory dynamics. There is a temptation to avoid the detail needed to describe process models, but a premature attempt to avoid such detail may result in a failure to understand how and why financial institutions and information processing play a critical role in guiding economic dynamics.

Table 3.1
Models, solutions, and questions

Model type	Solutions	Questions
Extensive form	Behavioral, Bayesian perfect equilibria	Information conditions; individually optimal strategies under various conditions of communication
Strategic form	Maxmin, noncooperative equilibria	Existence of equilibria; individually optimal strategies under conditions of no direct communication
Coalitional form	Cooperative solutions: core, value, bargaining set, and others	Power, fair division, bargaining, and equity

3.4.1 Questions, models, and solutions

The type of question being asked influences the model that is built and the solution concept that is needed. The different relationships among questions, models, and solutions is illustrated in the theory of games, where there are three fundamentally different types of model associated with many different solution concepts. These are illustrated in table 3.1.

The *extensive form* of a game is a natural model for considering details of process. It is central to the investigation of the minimal amount of complexity required in the development of a mechanism to carry a process.

The *strategic form* (most frequently exemplified by matrix games) suppresses much detail but can nevertheless be regarded as a highly coded process model. It is playable as a game. In general, economic interest in games in strategic form has been directed toward the analysis of noncooperative equilibria.

We may regard the noncooperative-equilibrium solution to a game in strategic form as a static solution applied to a process model that is defined *without regard for equilibrium*; that is, the game is an economic mechanism that exists independently from any solution concept.

The *coalitional form* is used primarily to consider the benefits of cooperation and coalition formation. Most of the solution concepts suggested for the coalitional form are normative in nature and are not of primary concern to this volume.

3.4.2 The general-equilibrium model is not a process model

Suppose we assume that in an economy with trade, a set of prices appears exogenously (for example, announced by a central planning board). We also assume that each individual is constrained to maximize his welfare on

the assumption that he has no influence on prices whatsoever and that his budget constraint is given by the worth of his initial assets at the prices announced. If we wish to answer the question of whether there exists a set of prices that, if announced, would balance all supply and demand arising from the constrained optimization of the traders, we do not need a fully defined process model.

Given a set of prices for which there exists excess supply or demand, it is not necessary for the mathematics to work out how supplies are rationed or prices change. For that, a more complex model is required.[3] Instead, one may prove indirectly that a set of market-clearing prices must exist.

3.4.3 The noncooperative solution requires a process model

In order to model a closed economic system as a well-defined game of strategy, we must make explicit all rules concerning price formation, credit, market clearance, and so forth.

There are many ways to construct detailed market process models. In particular, we could model a price-controlled economy by using a distinguished central player who controls all prices in the manner of a central government agency in a price-run socialist economy. In contrast, we could assume that market prices emerge as a result of the actions of all individuals.

It is in the details of the process modeling that the institutions of an economy appear. The specification of price formation results in mathematical constructs that can be identified with market institutions. The specification of money and credit conditions leads to a consideration of banking, insurance, bankruptcy, bank failure, and insurance reserves. In short, markets, money, and financial institutions emerge as necessary conditions in defining a process model. The approach here is to consider *minimal institutions*, the simplest mechanisms required to provide the economic functions needed.

Given the strategic model, the noncooperative-equilibrium solution can be examined for conditions under which the noncooperative equilibria coincide with the competitive equilibria. When this coincidence takes place, the relative prices and distribution of goods will be the same, and the apparent need for a superstructure of markets, financial institutions, and a trading technology disappears. The vital role of money, financial institutions, and markets appears in the guidance and control of an economy in disequilibrium.

Table 3.2
Types of solutions

	General equilibrium	Noncooperative equilibrium	Macroeconomic and other equilibrium behavioral theories
Model	Static	Process or dynamic	Process or dynamic
Solution	Nonstrategic	Static, strategic	Many ad hoc rules of behavior
Comments	Noninstitutional	Markets and financial institutions present	Markets and financial institutions present

3.4.4 Three models and many solutions

The great success of general-equilibrium theory in answering questions concerning the existence of efficient prices and the production and distribution of goods was aided by the parsimonious abstraction of its modeling. A static, nonstrategic solution concept was explored, and the minimal requirements called for a nonprocess model.

It is possible to build a process model without becoming fully involved in dynamics. This is exemplified in the use of the noncooperative-equilibrium solution. Even though the solution concept is static, the modeling requirements call for a full process description. This was already implicit in the work of Cournot, who first explored the relationship between oligopolistic price and competitive price.

Process models can and have been built, but our knowledge of what constitutes a satisfactory description of behavior is still limited. Macroeconomic writings and the behavioral theory of the firm (see, for example, Nelson and Winter, 1982) have many suggestions for the behavior of economic units, but they are for the most part ad hoc, and there is no generally accepted theory of economic dynamics.

Table 3.2 summarizes and compares basic aspects of general-equilibrium and noncooperative-equilibrium microeconomic theories and macroeconomic behavioral theories.

3.5 On Macroeconomic Models

This work is devoted primarily to the development of a theory of money and financial institutions that forges links between micro- and macroeconomics. It utilizes the methods of game theory, primarily the extensive and

strategic forms of a game together with the noncooperative-equilibrium solution. It is a treatise neither on general-equilibrium theory nor on modern macroeconomics. But because they play such important roles, in the remainder of this chapter I shall sketch several basic macroeconomic models. This sketch, combined with the discussion of general equilibrium in chapter 4, will make the discussion minimally self-contained.

3.5.1 Structural and reduced forms

Much of the material presented here can be examined in far greater detail by referring to such excellent textbooks as Dornbush and Fischer 1981, Branson 1989, or the more advanced work of Sargent (1979) and the Nobel lecture of Tobin (1982). Some examples of full-scale macroeconometric models include the Brookings model (Duesenberry et al., 1965), DRI (Eckstein, 1983), and Fair (1974, 1976, 1984).

In essence, a macroeconomic model of an economy is a process description given by a set of equations involving n time-dependent endogenous variables. Denoting the endogenous variables by $y_i(t)$, where $i = 1, \ldots, n$, and the exogenous variables by $x_j(t)$, where $j = 1, \ldots, m$, we may express the *n structural equations* as

$$g_i(y_1(t), \ldots, y_n(t), x_1(t), \ldots, x_m(t)) = 0 \quad (i = 1, \ldots, n), \tag{3.1}$$

where the equations consist of behavior equations, control equations, and accounting identities. A *control equation* contains elements that are strategic choices of the government. In such a form, the system is amenable to simulation; the equations together with initial conditions should provide an appropriate and sufficient description to construct a computer program that will generate future histories of the overall economy contingent on the specification of all exogenous variables. The model is designed to react to exogenous shocks, which can include changes in political climate, taxation policies, and many other social and political factors.

Much of practical economic analysis uses statics and comparative statics, although there are difficulties in going from a static equilibrium analysis even to the most elementary of comparative statics. Without describing the many mathematical and empirical problems involved, I shall simply refer to the "Cobweb" example frequently displayed in elementary textbooks to illustrate the phenomenon of instability when a parametric shift in demand or initial resources sets off an outward spiral-

ing of supplies and prices. Two simple models of economic adjustment are frequently considered, one put forth by Marshall and the other by Walras. The Marshallian process has quantity adjusting to excess demand at a given price, whereas the Walrasian model has price adjusting to remove excess demand. They lead to different market instabilities. Plott and George (1988) have some evidence that, at least in a simple experimental market, the Marshallian process is a better predictor. There is now a considerable literature on temporary general equilibrium and the fix-price method (see Grandmont, 1982, for a survey). Yet the question of the relative speed of adjustment of prices and quantities is ad hoc and can hardly be conjectured without detailed knowledge of the mechanisms and processes under consideration. I shall not discuss stability further except to note that the mathematical problems are considerable, as exemplified by Scarf, 1960, which shows the instability of a general-equilibrium exchange model with a unique equilibrium against slight changes in endowments.

Understanding why financial instruments and institutions exist requires insight into the microeconomics of exchange and production and the macroeconomics of intergenerational distribution and macroeconomic control. This understanding can be obtained without delving into the details of dynamics. Macroeconomic policy advice, however, cannot avoid a more direct contact with dynamics.

Instead of concentrating on macroeconomic dynamics by means of simulation, one might wish to examine the equilibrium conditions among the endogenous variables by solving the equations set (3.1) to express the endogenous variables in terms of the exogenous variables:

$$y_i(t) = h_i(x_1(t), x_2(t), \ldots, x_m(t)) \quad (i = 1, \ldots, n). \tag{3.2}$$

The equations expressed in this manner are referred to as *reduced-form equations* and can be regarded as the static-equilibrium solution of the endogenous variables. Thus, in the sense that all endogenous variables are simultaneously determined in equilibrium, as Tobin (1982) notes, the system may be regarded as a general-equilibrium system. This is not necessarily true because many of the equations do not reflect explicit optimization and in many instances it is not clear that they can be interpreted in terms of conscious optimization. In general, because the equation systems are nonlinear, running the simulation of the structural equations provides the most practical means for solving them.

The reduced-form equations are of immediate use to the policy analyst because the derivatives of the equations with respect to the various control variables give a measure of their effectiveness, referred to as *multipliers*, at least on the margin, in that they indicate the size of the influence a change in a control variable has on the economy.

The empirical and theoretical problems involved in model validation and equation estimation are of central concern to the macroeconomist and econometrician. Econometric problems are not considered further here, except in the caveat at the end of the chapter.

3.5.2 A simple macroeconomic model

A simple model will illustrate some of the problems in connecting macro and micro models. There are many difficulties to be overcome in converting this macro model into a well-defined strategic market game. Here we confine ourselves to a macroeconomic sketch. We use the following notation:

N = total labor supply at time t (the argument t is not expressed);

K = capital stock;

C = consumption;

I = investment;

Y = gross national product;

M = money supply;

P = price level;

r = rate of interest;

W = wages of the employed;

G = government spending;

T = government taxes.

These eleven variables are clearly of concern to the nation as a whole. They are gross aggregates, simultaneously comprehensible and understood to be approximations by the businessperson, union leader, and government administrator.

Many macroeconomic models consist of combinations of these variables, some regarded as endogenous, others as exogenous. Furthermore, there are critical, but not yet specified, families of rabbits hidden in the macroeconomic magician's hat, namely, the assumptions to be made

concerning "expectations." These may include expectations on inflation, changes in the inflation rate, marginal productivity of capital, population growth, government control, or other economic, socioeconomic, or politicoeconomic factors that help provide a picture of a forward-evolving dynamics.

A sound mathematical technique used in the analysis of finite, multistage models is the backward solution, in which the optimal solution to the last stage of the process is calculated first, and this calculation is then used to solve the penultimate stage. This process is continued until a policy for what should be done at the first stage is developed.

Life is lived forward, and planning for the most part is based on our knowledge of the past and our conjectures concerning some limited contingencies about what might happen in the future. While there are techniques in dynamic programming that make backward solutions meaningful even though the horizon may be infinite (see Denardo, 1981), many of us will feel more comfortable with a forward behavioral solution that provides the same policy.[4]

One of the reasons why the stationary-state equilibrium is so attractive a solution to study is that it links statics and dynamics with minimal assumptions about the updating of expectations. An equilibrium can be maintained with consistent expectations. But it is important to note that when there are only a few major control players, such as the Federal Reserve and several government agencies, the gaming aspects of control may be such that no equilibrium may exist. Theories based on "rational expectations" depend heavily for their logical validity upon the presence of only small agents.[5]

Keeping in mind that the central difficulty in linking a resolutely rationalistic, normative, static microtheory is in the formulation of the expectations mechanism, we consider an extremely simple macroeconomic model and interpret it in microeconomic terms. For illustration, we use a six-equation model that is a slightly simplified version of a Keynesian model studied by Sargent (1979):

$Y = f(K, N)$	Production	Technological	(3.3)
$W/P = \delta F/\delta N$	Wages	Behavioral, economic	(3.4)
$C = c(Y - T, r)$	Household consumption	Socioeconomic behavior	(3.5)

$I = I(K, N, r)$	Investment	Behavioral	(3.6)
$C + G + I = Y$	GNP	Accounting	(3.7)
$M/P = m(r, Y)$	Money, interest	Behavioral	(3.8)

We may consider T, G, K, M, and W as exogenous and Y, C, I, N, P, and r as determined by (3.3)–(3.8).

I have these equations labeled from two points of view: their economic content and the type of relation they postulate. The first equation is the production function, whose basis must lie primarily in technological, economic, and institutional understanding of the means of production in the economy. In application, the justification of the nature of aggregation is critical. Thus one may question the value of the lessons learned from an aggregate single-output Cobb-Douglas production function or the difficulties posed by using a fixed-coefficients Leontief model.

The second equation equates real wages to marginal productivity. Thus it embodies the assumptions that, for the economy as a whole, production is approximated by a differentiable function and the firms are competitive rational optimizers with no money illusion who hire up to the margin. Because W is exogenous, we do not have a classical story for wage formation within the model.

The third equation defines a household consumption function. Its arguments and shape depend heavily upon assumptions concerning policies for indefinite-horizon problems involving family formation, life cycles, and bequests in the context of overlapping generations. Taxes appear in the arguments to reflect governmental controls.

The fourth equation encapsulates the investment policy of private industry. Here it is assumed to depend on capital, labor, and the rate of interest. At this point, a host of opportunities for behavioral embellishments are available. Thus, in an investment accelerator model, one might make investment depend on changes in consumption. This raises questions concerning the formation of expectations.

The fifth equation is an accounting identity that includes government spending as a control variable.

The sixth equation defines a relationship among the money supply, price level, and transactions, reflecting precautionary and speculative demands for money.

The verbal description of Keynes (1936, chapter 13) offers a radical solution to some of the central unsolved problems of microeconomic theory as applied to the British economy of the 1930s and seen through the eyes of a "practical man of affairs." In one equation, the complexities of behavior under uncertainty are mean to be encapsulated. Both empirical evidence and theory are challenged.

Although the psychological distance from the broad sweep of macroeconomic descriptions of behavior to the point of view of game theory appears to be great, the possibilities available in a game in extensive form are so large that virtually any behavioral pattern can be reinterpreted as a legitimate strategy. Thus it is possible that many behavioral theories can be reconciled with the concept of ex ante rational behavior, although in any situation that includes large agents the concept of what constitutes ex ante rational behavior is not straightforward.

The purpose of this illustration is to set the stage and make connections to the noncooperative-game-theory analysis that follows. This simple model, together with the Hicks IS-LM apparatus, is presented only to provide intuitive insight into the linking of ideas rather than as a discourse on the relative merits of one popular macroeconomic model or another.

The very phrase "IS-LM" is a statement about equilibrium that can be linked to a microeconomic basis. IS calls for an equilibrium between investment and savings in the real goods markets. LM calls for equilibrium between the demand for and supply of money. This second equilibrium condition is irrelevant to general equilibrium or to a centralized economy in which financial equilibrium can be brought about by fiat, but it is critical to a macroeconomic view of an enterprise economy in which the neural network of a monetary control system is in place. If the monetary system merely manifests itself in prices that are immediately homogeneous of order zero, then the monetary system may be in place but has no control value. In a classical model one can graft money into the system, but as manifested in the mathematical analysis, it is a veil leading to equations that can be separated into two systems—one for paper and the other for real goods, with no economic influence on each other.

The critical observation comes back to Bagehot. A monetary or financial instrument may be introduced for one purpose but end up serving another. The introduction of money to facilitate trade, of places of deposit to guard against theft, of taxes to pay for wars and other expenditures of the king, and of large pools of liquid assets to back insurance funds and

pensions brought into being a control network on the economies of modern societies that was never contemplated by those who introduced the instruments.

Returning to IS-LM, if we specify equation (3.8) as

$$M/P = l(r) + k(Y), \tag{3.9}$$

then we can use figures 3.3a and 3.3b and then overlay them to show the matching of the double equilibrium in the real and monetary market using the IS-LM diagrams of Hicks (1937). (Three sets of equilibrium conditions can actually be shown in a single diagram by plotting the LM and the IS curves and determining their intersection. We omit this extra figure.)

The shape of the LM curve can be obtained from the four-quadrant diagram of figure 3.3a by adding the liquidity and transactions components. The liquidity demand is shown as a function of r in quadrant 2. The transactions demand is shown in quadrant 4 as a function only of GNP. A transaction model suggested by Baumol (1952), using the inventory model developed by Whitin (1953), has transaction needs varying with storage costs and costs of replenishment. Hence the interest rate should play a role to reflect forgone opportunity cost. But here we take the simpler condition of $k(Y)$ instead of $k(Y, r)$. Quadrant 3 shows the nominal money supply divided by the price level. The equilibrium condition requires that liquidity needs plus transaction needs account for the money supply. We can derive some interesting methodological observations from even this simple model.

A point on the LM curve can be constructed by completing a rectangle as follows. We start with A as the required transaction need for money; given the money supply, the liquidity demand plus government expenditure must then be at C. For these to be in equilibrium, the GNP and rate of interest must be consistent with the point D.

When we turn to Keynes's liquidity preference arising from precautionary and speculative motives (Keynes, 1936, p. 170), we find that he is writing about a world with both endogenous and exogenous uncertainty and with incomplete markets in a dynamic setting. In a Walras-Arrow-Debreu-McKenzie world these conditions do not exist, and many adjustments must be made to the theory to accommodate them.[6]

The IS curve provides a locus of equilibria for investment and savings. A similar four-quadrant diagram can be used to trace out the IS curve.

This is shown in figure 3.3b. In quadrant 2, private plus government investment is shown. Private investment is postulated to be a function of the rate of interest r. Quadrant 4 shows saving as a function of GNP. Here it is drawn in a manner different from most elementary textbooks in order to illustrate a few questions concerning the assumptions. Could it be possible that with a sufficiently low GNP, savings become zero? This is illustrated by having the savings curve intersect the axis before zero.

The LM curve provides a locus of equilibria between the supply of and the demand for money. The IS curve provides a locus of equilibria in which investment equals saving. A simultaneous equilibrium of both IS and LM will be at their intersection.

The methodological observations on this simple model are as follows:

1. The equations are a low-dimensional representation of a high-dimensional stochastic phenomenon. The four-quadrant graph only works for some simple functional forms.

2. A control system is present. At the bare minimum there are three sets of players: consumers, firms, and government. Although all consumers and firms may be small in comparison to the overall market, the government is not.

3. The equations are identities or behavioral equations, or they can be control equations.

4. The exogenous variables can be classified broadly as either control variables belonging to the large player, the government, or other major players or other factors not necessarily strictly economic or in control of the explicit players.

5. The control system is a loosely coupled group of monetary and financial institutions and government agencies that have evolved through an intermix of transaction needs, law, historical accident, and explicit planning.

6. In a decentralized society, much of the system has to be loosely coupled. A basic function of money and the credit system is to let error, time lags, and mismatches occur without destroying the system. But the value of loose coupling for decentralization comes through the properties of the system that make it difficult to control precisely.

7. Leaving aside oligopolistic effects, if the government were the only large player, this would be enough to call for Federal Reserve watching,

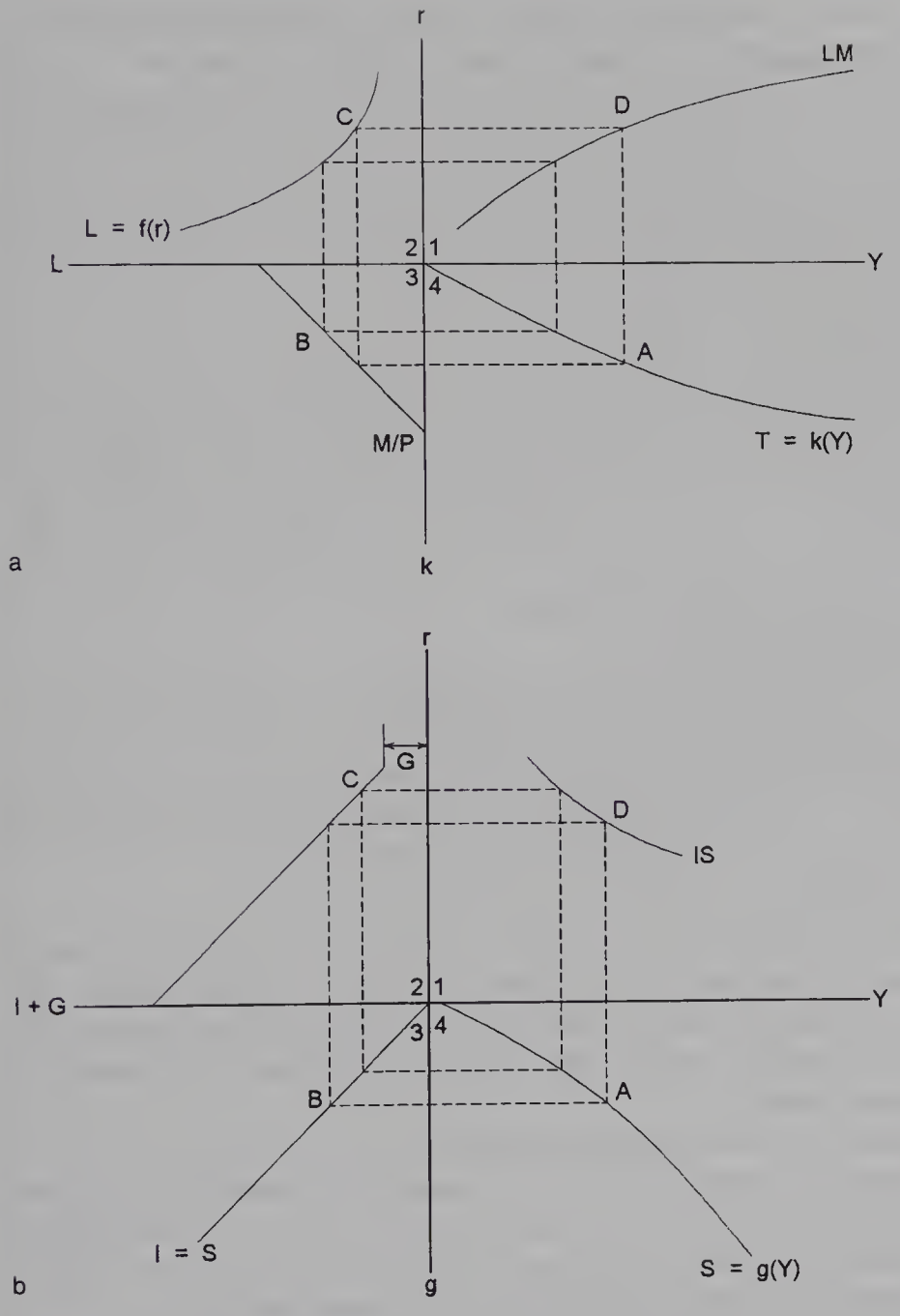


Figure 3.3
a. Construction of the LM curve. b. Construction of the IS curve.

open-mouth operations, and political and social forecasting as an explicit part of the game between the govenment and individual agents.

8. The key contribution of Keynes was not in intuitively guessing at the shape of the consumption function or the liquidity trap. It was in "escaping from old ideas" (Keynes, 1936, preface). The breakout was from a classical model that could be interpreted as a closed economic system. Keynes opened it up and replaced it by an open system with financial and fiscal controls.

9. The equations of macroeconomics involve a host of assumptions about institutions and about individual and social behavior, where validation is an intermix of econometrics and ad hoc theory.[7] Thus, for example, the consumption function could be based on a life-cycle, permanent-income hypothesis or on some other justification. The national product description could be multisector, take into account innovation, be fixed coefficients, or admit flexible substitution. Expectations can be introduced into many of the relations.

10. Keynes's models provided the intellectual structure needed for the growth of the national income accounting system. This alone helped to move macroeconomics a quantum leap closer to the world of affairs and policy. It is easy to underestimate the importance of the role of both public and private accounting in providing material for much of the administrative and planning activities of economic life.

By opening Pandora's box, Keynes showed the way to introduce many important considerations into models of economic process. But the very ease in which new factors can be added—exogenous and endogenous variables shuffled—makes it difficult to avoid a patchwork of often contradictory models.

The transition from the classics to Keynes was in one direction, and the transition from Walras to Arrow, Debreu, and McKenzie was in the opposite direction. While macroeconomics drew closer to power, politics, institutions, and practicality, general-equilibrium economics became more mathematical, precise, abstract, closed, and noninstitutional.

A reconciliation between general-equilibrium theory and macroeconomics requires that the general-equilibrium models be opened up to encompass the economic control system of financial and governmental institutions. But this must be done in a manner that is simultaneously

abstract, general, and institutional. This can be achieved by using strategic market games to remodel the closed economy as a playable game of strategy in which the minimal institutions emerge as a necessity of the modeling.

Keynes was a great publicist. His key book was entitled *The General Theory of Employment, Interest and Money* (1936), which has a magisterial and timeless ring to it. The title seems almost to imply that it is institution-free. Yet, as has already been pointed out in chapter 2, an institution-free dynamic economic theory is a logical impossibility. *The rules of the game carry with them at least an implicit description of institutions.* The lessons of Keynes, if they are to be applied to times and societies other than the one in which he wrote, must be reinterpreted in terms of the local institutions and political contexts.

3.5.3 Models for policy

Large macroeconomic models are now an accepted part of world economic administration and bureaucracy. The heritage of Keynes was not mere theory, but policy institutions and process. The International Monetary Fund, the World Bank, national income accounting groups,[8] and macroeconomic modelers build models based on Keynes's work for direct governmental and business use as well as for academic use.

The earliest formal models for planning appear to be those of Tindbergen (1959). Klein started large-scale modeling in the United States (Klein, 1950; Klein and Goldberger, 1955). The Klein models expressed their outputs in real terms. Klein's original model had twelve equations and was expanded to thirty-two in the Michigan version (Suits, 1962). The large-scale Brookings (Duesenberry et al., 1965), Wharton (Evans and Klein, 1967), and Federal Reserve–MIT (Deleeuw and Gramlich, 1969) models involved around 150, 200, and 175 equations in various versions. The Wharton model was used to provide quarterly forecasts of the economy.

The efforts called for organized teamwork to fit together diverse submodels covering different views of consumption behavior, investment behavior, wage-price relationships, and Leontief input-output methods to provide a sufficient disaggregation of production. The financial control system is far from monolithic. Not only must the Federal Reserve and Treasury be considered, but banks, several varieties of insurance companies, pension funds, and other financial institutions must also be modeled.

More recently, especially for forecasting and commercial use, the Data Resources or DRI model has been expanded to around 800 equations (Eckstein, Green, and Sinai, 1974; Eckstein, 1983). The relationship between these macroeconomic models and both microeconomic and macroeconomic theory poses many problems. The work of Lucas (1976) and others stressing rational-expectations models leads to a need to estimate the parameters of objective functions specified by considerations of optimizing behavior (see Hansen and Sargent, 1980).[9] This approach raises fundamental problems concerning basic concepts of information and how it is processed by rational economic individuals. If we adopt the viewpoint that individuals are *context-rational* in the sense that no one can process anything more than a highly simplified view of the overall system,[10] then there is no reason to believe that expectations must ever reach a state of universal consistency. This point is made by Fair (1984) in his investigations of U.S. and multicountry models. These observations are congenial with the observations of Herbert Simon on "satisficing man" and strike at the roots of the economic optimizing paradigm.

Although this paradigm must eventually be abandoned, at this time it is maintained because there is still much to be deduced or induced from the proposition that, as a first approximation, in many situations people more or less know what they want and how to get it. In particular, the argument developed here is that general-equilibrium theory has presented an elegant counterfactual picture of optimization that is totally nonprocess and noninstitutional, while macroeconomic theory has presented an ad hoc mixture of theory and advice with institutions taken as given. There is thus room for a theory that offers an explanation of the genesis of the institutions, even though it is likely that no purely economic theory is ever going to provide a general dynamics without utilizing specific information about the cultural, social, political, and legal context of the nation whose economy is being studied.

A reading of Fisher, 1931; Keynes, 1930a, 1930b, 1936; Friedman 1969; and many other works[11] could give the unwary the impression that a few magic equations play the same central role in the dynamics of monetary theory that the equations of motion play in Newtonian mechanics. An abstract economic theory of money and financial institutions alone may not be sufficient to provide specific equations of motion for any actual society. Before a satisfactory politicoeconomic theory of dynamics

can be built, we need to appreciate the minimal requirements of the carriers of process.

3.5.4 Policy science and policy confronting science

The developments in general-equilibrium theory and in much of mathematical economics have been away from political economy in the direction of pure economics or even abstract theory of choice.[12] In contrast, macroeconomics remains part of political economy. The institutions count, and political and societal realities provide a context that must be taken into account.

Much of macroeconomic theory and application is of this world, here and now. Against the imperative forces of society, a constant stream of advice is needed to avert ever-looming economic disaster. In order to be effective, the necessary information and advice must be supplied by economists of repute who understand the sociopolitical context in which their advice will be used. The administrative and political realities of transforming advice into policy are such that lags exist in decision time, perception time, and implementation. These lags are as much a part of the macroeconomic control problem as other, more purely economic lags are a part of production, capital plant construction, and inventory management.

The politicoeconomic control system is a multivariate, stochastic, nonlinear, infinite-horizon, parallel, dynamic game with at least one and in general several players of significant size and power. Superimposed on this game are a flock of variables and factors whose analysis must be treated as exogenous to the professional capabilities of economists. In the social sciences, in general, one discipline's variables and causal factors are another's parameters.

At best, a macroeconomic model to be used for policy advice is (even at 800 or 8,000 equations) a highly aggregated assembly of professional best guesses, bolstered by a patchwork of partial empirical evidence and snatches of theory. Even logical consistency becomes difficult to check when the models use more than a handful of equations.[13]

The real and pressing questions of public policy cannot wait for theory to catch up with the needs for advice. Theory depends on insight, and insight on interpretation of the actions and raw data generated by the world we live in. The data we have are generated for the most part in response to real or perceived administrative needs. These needs influence and are highly influenced by theory. Superimposed on this are the costs of

Table 3.3
Political economy

	Socialist-radical	Neo-Keynesian	Keynesian	Neoclassical-Keynesian	Monetarist-neoclassical
Politics	Extreme left	Left of center	Center	Right of center	Extreme right
Money	Real forces emphasized; money a tool for existing power structure	Real forces emphasized; money assumed to accommodate	Money and real forces intimately related	Money matters along with everything else	Only money
Wage rate and income distribution	Wage rate basis of value; income distribution the most important economic question	Money wage is the linchpin of the price level; income distribution very important	Money wage rate fundamental; income distribution question of less importance	Wage rate one of many prices; the resultant of all the demand and supply equations in a general-equilibrium system, income distribution a matter of equity, not of "scientific" inquiry	
Capital theory	Surplus generated by reserve army	Surplus needed over wages (quasi-rents)	Scarcity theory	Marginal productivity theory and well-behaved production functions	
Employment theory	Any level of employment possible: assumes growth in employment over time	Growth with any level of employment possible, although growth at full employment emphasized	Any level of employment possible; full employment desirable	Full employment assumed; unemployment a disequilibrium situation	Full employment assumed in long run; no explicit short-run theory of employment
Inflation	Primarily due to money wage changes, but can also be due to profit changes	Due to money wage or profit margin changes	Due to changes in money wages, productivity, and profit margins	In long run, primarily a monetary phenomenon related to money supply via portfolio decisions; in short run, may be related to Phillips curve	Primarily a monetary phenomenon in the sense of being related to the supply of money via portfolio margin decisions

data gathering and the processing of raw data into interpreted information. Although the explosion in the last fifty years of communication and computational technology has radically changed the possibilities for generating large economic models, the conceptual problems of theory generation and validation remain.

As anyone who has ever served as an expert witness in a complex economic dispute knows, only a thin line separates advocacy from scientific objectivity. The number of implicit assumptions that must be made is almost always unbearably large. This in no way detracts from the value of experts, nor from their honest attempts at objectivity. One works with the best that is available, even if it leaves much to be desired.

An individual's political bent often serves as a selection device providing implicit assumptions for choosing among competing theories when the evidence is not strong enough to provide for a consensus. Table 3.3, which appeared in a semipopular article on post-Keynesian economics, provides an intuitive sketch of possible relationships between sociopolitical viewpoint and choice of macroeconomic model (Davidson, 1980). Although I do not fully subscribe to the categorization offered (in particular, it is possible to understand and accept the technological lessons of the price system and the assumptions of marginal productivity in an appropriate domain of discourse without regard to political coloration), nevertheless it is important to keep in mind that political economy often offers political solutions to economic problems where straight economic analysis tells us that there is no unique, purely economic solution.

Notes

1. For exposition, and possibly also for practical approximation, a full exploration of cubic cost curves might be merited here. Unfortunately, space considerations preclude such discussions.

2. Proponents of specific policies frequently provide norms when using arguments in adversarial debates over political policy, but basic macroeconomic theory offers little normative description of the individual or society, unless one possibly interprets "rational expectations" as the way rational individuals should form their expectations.

3. One might consider, for example, models of temporary general equilibrium (see Grandmont, 1982).

4. See Heyman and Sobel, 1982, and Sanghvi and Sobel, 1976 for a discussion of forward and behavioral dynamic programming and game solutions.

5. See Dubey, Geanakoplos, and Shubik, 1987, for a discussion of the problems with rational expectations and few agents.

6. The critical missing link appears to be the role of expectations in an overlapping-generations model as contrasted with its role in general equilibrium. Geanakoplos and Polemarchakis (1986) develop this in reference to IS-LM.

7. "Ad hoc" is used here as a positive and not a pejorative phrase.

8. There have been at least three major economic accounting revolutions: double-entry bookkeeping, national income accounting, and the accounting for the input-output tables of Leontief.

9. As introduced by Muth (1961), "rational expectations" means nothing more than model-consistent expectations—a property shared by both the general-equilibrium and noncooperative-equilibrium theories.

10. This could be due to individual capacity limitations and costs associated with interpretation, perception, and the nature of the data processing required to convert data and raw sensory inputs into a meaningful cognitive map.

11. See, for example, Harry Johnson's excellent survey (1967) of the state of monetary theory.

12. For example, in spite of the verbal discourse, Arrow's impossibility theorem is not confined in any way to voting. It deals with the aggregation of revealed preferences for any set of outcomes (Arrow, 1951, 1963).

13. See, for example, Brainard and Tobin, 1968, or even the debates on Say's law and Walras's law.

4 General Equilibrium and Beyond

4.1 On General Equilibrium

General-equilibrium theory, as formulated by Cassel (1923), Wald (1951), and von Neumann (1945) and brought to completion by Arrow and Debreu (1954), forms an important landmark in the mathematical development of economic ideas. The "invisible hand" suggested by Adam Smith and explored in greater detail by Walras ([1874] 1954) was a precursor of the more formal inquiry into price, production, and distribution of Hicks (1946) and Samuelson (1947); and this in turn served as the basis for the mathematization of many of the basic concepts behind a price system, as so elegantly presented by Debreu (1959).

The intuitive idea behind the existence of an efficient price system is that under certain conditions if all consumers and producers in an economy are confronted with a set of given fixed prices, each can optimize individually, and the resultant of the many independent acts will be a set of trades in which supply and demand balance in all markets.

The conditions required for the existence of a price system that provides the appropriate signals for efficient, decentralized, individual decision making are not easy to formalize, and the general proof of the existence of an efficient price system requires a relatively high level of mathematical sophistication.

Consider an economy with m commodities or economic goods. Let a_{ij} be the amount of commodity j owned by individual i originally. Suppose that for each commodity j there is a price $p_j \geq 0$. The total value of all of i's possessions at market is given by

$$\sum_{j=1}^{m} p_j a_{ij}.$$

If i can costlessly buy and sell all commodities, then i can choose for final consumption any bundle of goods satisfying

$$\sum_{j=1}^{m} p_j x_{ij} \leq \sum_{j=1}^{m} p_j a_{ij} \tag{4.1}$$

and will attempt to select the most preferred final bundle $(x_{i1}, x_{i2}, \ldots, x_{im})$ among the alternatives feasible. Thus each x_{ij} will be a function of prices

$$x_{ij} = f_{ij}(p_1, p_2, \ldots, p_m). \tag{4.2}$$

We must also consider firms. A firm k's operations can be described by the transformation set

$$T_k(y_{k1}, y_{k2}, \ldots, y_{km}) = 0, \tag{4.3}$$

where if $y_{kj} > 0$, it is an output, and if $y_{kj} < 0$, it is an input. The set described by T is the set of technologically optimal production transformations; that is, it will not be possible to find an array with more outputs, without there being more inputs.

We assume that firms try to maximize profits. Because the model has a finite horizon, the economic concept of profit is clearly defined:

$$\pi_k = \sum_{j=1}^{m} p_j y_{kj}.$$

Each y_{kj} will be a function of all prices:

$$y_{kj} = g_{kj}(p_1, p_2, \ldots, p_m).$$

For any set of prices $(p_1, p_2, \ldots, p_m)$, in any market j we may define the excess demand or supply conditions. Suppose that there exist n consumers and g firms. Then for j excess demand is given by

$$e_j(p_1, p_2, \ldots, p_m) = \sum_{i=1}^{n} x_{ij} - \sum_{i=1}^{n} a_{ij} - \sum_{k=1}^{g} y_{kj}, \tag{4.4}$$

where the right-hand terms represent, respectively, aggregate demand of the consumers, initial aggregate supply, and net aggregate demand (or supply) of the firms. The e_j are homogeneous of order zero in prices; that is,

$$e_j(p_1, p_2, \ldots, p_m) = e_j(kp_1, kp_2, \ldots, kp_m) \quad \text{for } k > 0.$$

In other words, in this system there is no importance attached to any real quantity of money. The key item that counts is the ratio between every pair of prices. There are m commodities in the economy but only $m - 1$ independent prices. If we so chose, we could select the price of one as a numéraire or require the additional condition that all prices sum to one:

$$p_1 = 1 \quad \text{or} \quad \sum_{j=1}^{m} p_j = 1.$$

We need to modify (4.1) to include the income derived by the consumers from the ownership of the firms. If we adopt the not unreasonable convention that the only "natural persons" in the economy are the consumers, then either directly or indirectly they may be considered as owning everything. The easiest way we can model this is to imagine that a consumer i holds s_{ik} percent of the stock of firm k (where $s_{ik} \geq 0$ and $\sum_{i=1}^{n} s_{ik} = 1$). The firms are then regarded as merely profit-maximizing automata who flow through all profits to their owners.

Thus each individual's wealth constraint becomes

$$\sum_{j=1}^{m} p_j x_{ij} \leq \sum_{j=1}^{m} p_j a_{ij} + \sum_{k=1}^{g} s_{ik} \pi_k, \tag{4.5}$$

where the last term is the individual's share of profits. We may call (4.5) the individual's budget, wealth, or income constraint because, at this level of modeling in a static system, no operational differences among wealth, income, or budget have appeared. There is no credit in this system, as is indicated by the requirement of the budget constraint.

Because the system is closed and the value of any set of expenditures planned at any price must equal the value of the individual's initial endowments plus share of profits, in aggregate we have

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p_j x_{ij} = \sum_{i=1}^{n} \sum_{j=1}^{m} p_j a_{ij} + \sum_{k=1}^{g} \sum_{j=1}^{m} p_j y_{kj}.$$

This enables us to express the accounting definition known as *Walras's law*:

$$\sum_{j=1}^{m} p_j e_j \equiv 0. \tag{4.6}$$

In equilibrium, we require that all markets clear or that some goods be free; that is, in every market

$$e_j = 0 \quad \text{and} \quad p_j > 0$$

or (4.7)

$$e_j < 0 \quad \text{and} \quad p_j = 0.$$

The second condition allows for the possibility that some goods of intrinsic value may nevertheless be so abundant in supply relative to demand that their price is zero.

The mathematical methods and the precision of formulation of economic questions have become far more refined over the course of the last twenty years. Among the questions examined have been the possible existence of free goods; constant, increasing, or decreasing returns to scale in production; assumptions concerning preference orderings; whether preferences can be represented by utility functions; whether utility functions can be defined up to an arbitrary linear transformation; and questions concerning the implications of saturation of wants. Problems concerning the existence of boundary equilibria and the conditions required for the uniqueness of equilibrium points have also been considered. An excellent discussion of most of these problems is found in Arrow and Hahn 1971.

Our concern in this book is primarily with the role of money and financial institutions rather than with the precise and delicate details of distribution and production given price. This being the case, after displaying a simple example in section 4.2, we concentrate on a different set of problems in the interpretation of the general-equilibrium model of the economy, namely: the relationship between competition and price; the role of information in price formation; and the relationship between dynamics and statics.

4.2 A Simple Example of a General-Equilibrium System

A simple example with two consumer goods, two types of consumers, and two production functions utilizing two inputs, "land and labor," serves to illustrate the key strengths of the general-equilibrium system, provides a basis for a critique of what is left out, and supplies a starting point from which other models can be built. These process-oriented models will provide the means to integrate money, markets, and financial institutions into our description of an economic system with production and exchange.

We begin with some basic information. There are two types of consumers. A consumer of type i has preferences that can be characterized by a utility function

$$z_i = \Phi_i(x_i, y_i) \quad \text{for } i = 1, 2,$$

where x_i and y_i are the amounts of the final outputs of food and shelter provided by the economy.

For purposes of the example, the utility functions are assumed to be differentiable and increasing in both x_i and y_i. Furthermore, the indifference curves or contours of these functions are convex curves.

The initial endowments of an individual of type i are given by $(a_i, b_i, 0, 0)$, where

$$a_i, b_i > 0 \quad \text{for } i = 1, 2, \quad a_1 + a_2 = A, \quad \text{and} \quad b_1 + b_2 = B,$$

where A and B are the total supplies of land and labor. In this example, we do not consider that either of the initial endowments has immediate worth to the consumer. It follows that they will either be used to produce final goods or not be used at all.

Suppose that the production processes are each characterized by a production function:

$$x = f(u_1, v_1) \quad \text{and} \quad y = g(u_2, v_2).$$

These are assumed to exhibit constant returns to scale; that is, they are homogeneous of order one:

$$f(ku_1, kv_1) = kf(u_1, v_1).$$

We further assume that the production function contours are differentiable and convex curves.

Given the fixed supplies of endowments, we must have

$$u_1 + u_2 \leq A \quad \text{and} \quad v_1 + v_2 \leq B,$$

where $u_1, u_2, v_1, v_2 \geq 0$.

We make use of the classical Edgeworth box to derive the efficient transformation set discussed in section 4.1. We denote it by $T(x, y) = 0$.

The origin in figure 4.1 indicates that $f(0, 0) = 0$. The continuous convex curves represent the isoquantity contours of $f(u, v)$, that is, contours showing a fixed output of food as the appropriate combinations of land and labor are varied. The dotted lines show the isoproduct contours for shelter, where we regard 0 as the origin $g(0, 0) = 0$. Any point in the box is a pair of outputs described by $(x(u_1, v_1), y(A - u_1, B - v_1))$.

The curve given by the tangency between the two sets of isoquantity curves is the efficient transformation set. Elsewhere, it is possible to increase the production of one output without sacrificing any production of the other.

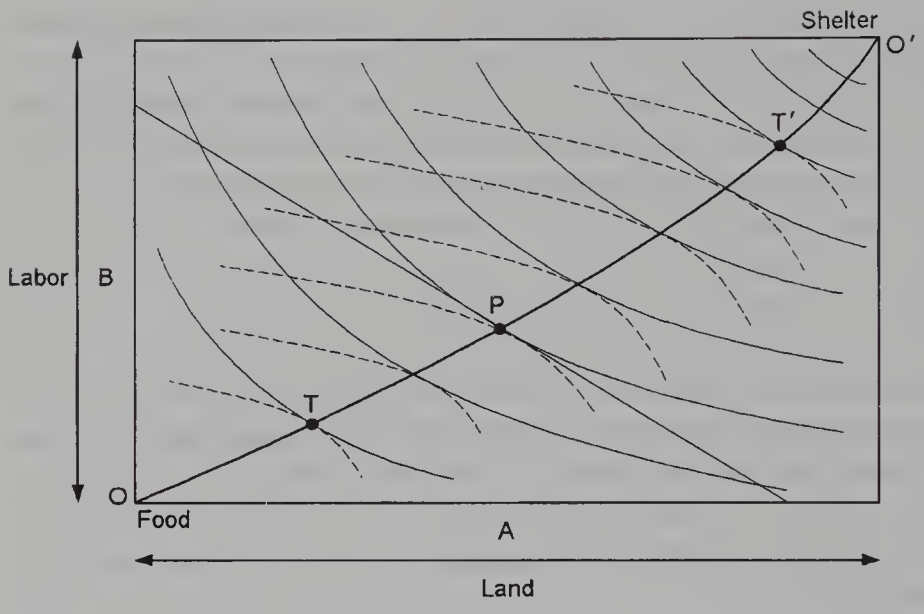


Figure 4.1
Efficient production transformation sets.

Mathematically, points on $OTT'O'$ satisfy the equation[2]

$$\begin{vmatrix} \dfrac{\partial f}{\partial u_1} & \dfrac{\partial f}{\partial v_1} \\ \dfrac{\partial g}{\partial u_2} & \dfrac{\partial g}{\partial v_2} \end{vmatrix} = 0.$$

The curve $OTT'O'$ suggests the efficient set of outputs that could be achieved by the individuals of the society, without offering any insight into how they should select a particular pair of outputs or how they should go about dividing those outputs once they have been selected.

We can introduce the idea of a price system in figure 4.1 and then use this as one way to go about considering the distribution of resources. Consider the line through P tangent to both isoquants at P. This may be regarded as defining the price of labor in terms of land.

If we wish to consider this economy to be guided by a price system, we can introduce four prices:

p_1 = the price of land,

p_2 = the price of labor,

Table 4.1
Equilibrium trade ratios

	Land	Labor	Food	Shelter
Land	1	p_2	p_3	p_4
Labor	$1/p_2$	1	p_3/p_2	p_4/p_2
Food	$1/p_3$	p_2/p_3	1	p_4/p_3
Shelter	$1/p_4$	p_2/p_4	p_3/p_4	1

p_3 = the price of food,

p_4 = the price of shelter.

Because there has been no mention of money in this description, we can define price as a ratio between two quantities of goods. Because there are four goods, there are

$$_4C_2 = \frac{4!}{2!2!} = 6 \text{ ratios.}$$

Without loss of generality, we can set up land as the standard value or numéraire. There are four prices, but only three are independent. By setting $p_1 = 1$ we assign an arbitrary number to the price of an acre of standard land.[3]

Table 4.1 shows all prices and ratios. There are 4^2 or 16 relations exhaustively, but only 6 ratios (all different pairs) of interest and 3 independent prices. The main diagonal gives the price of an item in terms of itself. The first row shows all prices in terms of land, and all other entries can be calculated from this.

The slope of the line through P gives the exchange ratio $p_2/1$ or p_2 between labor and land. If this price were given to a producer of food, so that the producer could regard it as independent of his behavior, in production the producer would select outputs such that

$$\frac{\text{the marginal physical product of labor}}{\text{the marginal physical product of land}} = \frac{\text{price of labor}}{\text{price of land}}$$

or

$$\frac{\partial f/\partial v_1}{\partial f/\partial u_1} = \frac{p_2}{1}.$$

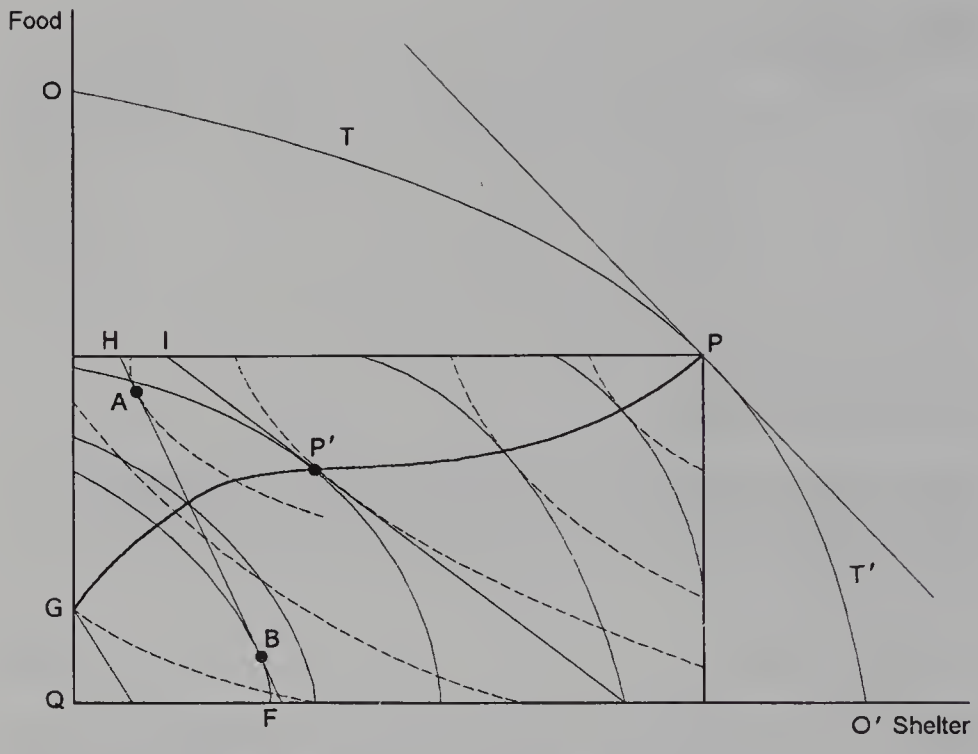


Figure 4.2
Edgeworth box modified for production and exchange.

Similar considerations hold for the producer of shelter, and we can see from figure 4.1 that if the prices of land and labor are given by the line through P, the resource demands of both types of producers would exhaust the supplies of land and labor, giving an output indicated by the point P in figures 4.1 and 4.2. We have at P the condition

$$\frac{\partial f/\partial v_1}{p_2} = \frac{\partial f/\partial u_1}{1}, \qquad \frac{\partial g/\partial v_2}{p_2} = \frac{\partial g/\partial u_2}{1}.$$

If we draw the curve $OTT'O'$ in the output space of food and shelter, we can study distribution in relation to production as follows. In a cooperative society without prices or any particular formula for arriving at distribution, production efficiency conditions call for staying on the boundary $OTT'O'$, but beyond that limitation no criterion of acceptability among the points on this set has been suggested.

We could try to draw a welfare economics rabbit out of a hat by inventing a social-welfare function $W(\Phi_1(x_1, y_1), \Phi_2(x_2, y_2))$, which

might enable us to select some set of most preferred productions and distributions. Unfortunately, a large literature on welfare and social choice shows the difficulties in trying to explain where a social-welfare function comes from, how it is formed, and what it looks like (see, for example, Sen, 1986, and Arrow, 1951, as well as the critique by Little [1950]).

At this point, rather than deal with concepts of social welfare, we restrict ourselves more narrowly to some aspects of distribution and production via a price system. The two types of consumers must allocate the quantities (x, y) of food and shelter among themselves, where (x, y) are the coordinates of P. Given production as fixed, we can construct an Edgeworth box diagram in which the continuous contours represent the indifference curves of a consumer of one type and the dotted curves those of the other type.

The curve $QGP'P$ shows the set of efficient distributions associated with the supply P. Figure 4.2 shows an instance of a point P' with the property that the line IP' is tangent to both sets of indifference curves at P' and has the same slope as the tangent at P.

Suppose that a central government has announced prices $p_1 = 1$ for land and p_2 for labor. Assume that consumers have sold all of their initial endowments and are paid in a unit of account by a bookkeeping transaction. A consumer of type i receives an income

$$I_i = a_i + p_2 b_i.$$

Total income for all will be $I = A + p_2 B$. Suppose that the government is constrained to announce prices for food and shelter of p_3 and p_4 such that

$$p_3 x + p_4 y = I, \qquad \text{where } x = x_1 + x_2, \quad y = y_1 + y_2$$

(that is, the consumers have enough income to buy all products). If we liquidate the initial resources of each consumer at the input prices and let consumers buy outputs at output prices, it is as though we have selected a feasible set of purchases. For example, if HF is the budget constraint of consumer 1 when viewed from the left and of consumer 2 when viewed from the right, the markets will not clear because one will try to trade to point A and the other to B. If the budget constraint is given by line IP', then both will trade to P'. At this point we have efficiency and

$$\frac{\partial T/\partial x}{\partial T/\partial y} = \frac{p_3}{p_4} = \frac{\partial \Phi_1/\partial x_1}{\partial \Phi_1/\partial y_1} = \frac{\partial \Phi_2/\partial x_2}{\partial \Phi_2/\partial y_2}.$$

The problem of production and distribution has been broken into three parts if we use a price system: (1) the efficient production problem illustrated in figure 4.1; (2) the efficient distribution illustrated in figure 4.2; and (3) the welfare problem, which is subsumed by specifying initial ownership claims and requiring that a price system be used.

The triumph both of the analysis (Arrow and Debreu, 1954; Debreu, 1959) and of computation methods (Scarf, 1973) for finding price systems for so-called competitive equilibria is in showing their existence and not in showing how they are arrived at by competition. They show their *efficiency* properties but say nothing about *equity*.

The existence of prices in a general-equilibrium system that yield efficient production and consumption allocations establishes the possibility of using prices to run either a capitalist enterprise economy or a centrally controlled economy with a central planning bureau that announces prices.

At this level of technological abstraction, there is absolutely no paradox in static efficiency conditions being the same for both capitalist and socialist systems (Lerner, 1944; Lange, 1936). It is at the level of institutions, information flows, price-formation behavior, and control that distinctions appear. They are manifested in process.

4.3 A Critique of General Equilibrium

As Koopmans (1977a) has noted, general-equilibrium theory is essentially preinstitutional. The ability to formulate many deep problems about the relationship between prices and efficient production has been of great significance in laying the groundwork for an economics that accommodates process. But this strength should not mask the gap between verbal descriptions of competition and the highly noninstitutional, process-free descriptions of preferences, ownership, and technological relations that make up the basic elements of general-equilibrium theory.

Many important developments of the past twenty years have been directed toward finding mathematical generalizations that weaken assumptions on preferences, take care of boundary equilibria, or cast light on production conditions. A certain amount of work has been done on the stability of equilibrium points under perturbations in excess supply and demand conditions (for a good general summary, see Arrow and Hahn, 1971), but few general results concerning specific full feedback mechanisms for the formation of price are known.

The next step in the development of microeconomics is to build full process models of the market or centralized price system in which economic institutions appear as necessary consequences of fully defining the rules of the game.

The limitations of the general-equilibrium model are as follows:

1. It is basically noninstitutional.

2. It distinguishes too few differentiated economic agents.

3. Although equilibrium conditions are specified, no explanation of price formation is given. The model is static.

4. There is no essential role for money.

5. The agents are modeled nonstrategically. Prices are given, not formed.

6. It is essentially timeless. We could make a simple, pro forma change of notation that treats goods in different time periods as different goods, and hence replaces an optimization problem in M dimensions by one in MT dimensions, where M is the number of goods and T the number of time periods, but this mathematical extension picks up only a few of the substantially new phenomena involved in the consideration of time and process in an economy.

7. The horizon is finite, and the agents appear without context at the start and disappear at the end.

8. The many problems posed by uncertainty in an actual economy are avoided by using the type of device that was used to extend the model to several periods. Suppose that an economy trades in M commodities and that it can be in any one of K states of nature. We can then consider an economy with trade in MK commodities, which may be regarded as options.

9. The proofs for the existence of an efficient price system were independent of the number of competitors, indicating that the mathematical model, though appropriate for the study of efficient prices, is not necessarily appropriate for the study of competitive price formation.

10. It was not designed to study economies with an intermix of monopolistic, oligopolistic, and competitive elements.

11. It is inadequate for the study of nonsymmetric information conditions.

12. It has no satisfactory explanation of unemployment.

13. Bankruptcy plays no role. There is neither birth nor deaths of firms.
14. Expectations are finessed. There is no learning.

The general-equilibrium model does not need markets, banks, or other credit or financial institutions because trust is implicitly perfect. All trade is balanced at the end of the market, and at the start of trade each individual has available implicitly a credit line equal to the individual's net worth "at market," that is, the worth of the individual's initial assets at the final market price.

Firms are not institutions. Production is described by correspondences available to all who have the resources. It is as though the only items necessary to bake a cake were a recipe (free to and immediately understood by all), and the ingredients. The firm as an entity with an internal organization and a management with goals of its own is not included in this abstraction.

In the Debreu model there is really only one major actor and one shadow actor—the consumer and the producer. The consumer has his own preference ordering and tries to maximize his own welfare. The producer is a shadowy manager of a firm that may be owned by others. Even though his interests may conflict with those of others, the producer is modeled as a profit-maximizing perfect fiduciary who flows through the profits to a nonvoting group of stockholders.[4]

In macroeconomic models and in economic life, we frequently distinguish investors, savers, speculators, brokers, bankers, consumers, manufacturers, retailers, and others. It is important to ask at what level of abstraction these distinctions appear or should appear.

The general-equilibrium model is essentially static in the sense that time is handled by merely enlarging the number of variables and renormalizing (Wold, 1969). Work on sequences of economies (Green, 1973; Grandmont, 1982) and the infinite horizon have modified this. Even so, the work of Arrow and Debreu (1954) and Debreu (1959) includes no explicit mechanism for the formation of price; that is, no formal markets exist. It is not surprising that, at this level of theorizing, one cannot distinguish between a competitive economy, where prices arise from competitive behavior, and a socialist economy, where prices are controlled. The difference between them is in the mechanism of price formation, but in this theory the mechanism is left out.

In any complex economy money plays a far more important and subtle role than merely acting as a unit of account, yet there is no role for it in the general-equilibrium model. In recent years Hahn (1965), Starr (1974), Grandmont and Younes (1972), and many others have attempted to introduce money into a modified general-equilibrium model. Although considerable progress has been made, this work has been hampered by the desire to stay close to the original general-equilibrium formulation.

The general-equilibrium theory is nostrategic. Prices appear by magic, and the individual consumer is constrained to maximize welfare given these prices. It is in fact an assumption of the theory that the individual maximizes welfare as though prices were given. This is a *deduction* from the theory when the system is modeled as a game of strategy and solved for its noncooperative equilibria. As noted in subsequent chapters, the mathematical technique of considering a continuum of traders reveals a coincidence between some of the results of noncooperative-equilibrium game theory and general equilibrium.

Limitations 6, 7, and 8—that the general-equilibrium model is essentially timeless, avoids the problems with exogenous uncertainty, and has economic agents who are all present at the start of the economy and disappear at the end—are all artifacts of modeling the economic system in the normal form rather than in the extensive form of a game.

The economic meaning to enlarging the number of commodities traded in an economy with M commodities, T trading periods, and K states of Nature to MKT is that perfect future markets and insurance contracts exist for all times and all circumstances. This is an assumption that merits modification.

Although the idea of studying production and exchange with a fixed number of economic agents certainly appears to suggest a set of useful models for the study of short-run features of the economy, the assumption of a fixed number of active individuals implicitly cuts out a host of factors that connect economic life with the polity and society. Birth and death, nurture, education, old-age support, and inheritance are all ruled out. A further fruitful development of the theory must take these into account. And, as indicated in subsequent chapters, much work is being done along these lines.

Because the general-equilibrium model is examined for the existence of an efficient price system that is independent of the number of traders, it does not offer a means for studying the effect of thin markets and few

competitors. To do this, we would need to be able to show how the presence of different numbers of competitors of different sizes and with different levels of control influences the formation of prices.

A further difficulty in the study of competition with few participants is that there may be an intermix of competition and collaboration among groups. This possibility raises two sets of problems for model building. The first concerns the description of communication conditions, and the second involves what is meant by a solution to a game of strategy. Both difficulties are related to the description of institutions and laws as rules of the game.

Because the general-equilibrium model is not fully formalized as a game of strategy in extensive form, it is somewhat vague as to the information and communication conditions present in the economy. One interpretation that is consistent with the theory is that all individuals trade simultaneously in ignorance of each other's actions. They know only (a) their own endowments for periods $t = 1, 2, \ldots, T$; (b) all prices from periods $t = 1, 2, \ldots, T$; and (c) the size of all dividends to be received in all periods.

Given a completely formulated n-person game, even with each individual constrained to making one move, the size of the partition of information sets becomes enormous. Each configuration represents a somewhat different economy. The general-equilibrium model appears, at most, to cover only symmetric information conditions, that is, situations in which all individuals are equally ignorant of each other's moves and of the actions of nature.

It is important to note the difference between symmetric information concerning the moves of other economic agents and information concerning the moves of nature. In summary, the general-equilibrium system provides considerable insight into the technological and preference requirements for a society to be in a position to use a price system to achieve efficient production and distribution. Because it is not concerned with process, no distinction is made between a centralized price system and one that comes about through competition among many small agents. Furthermore, no insight is provided about the problems of control or the mechanisms required in either an enterprise or a centralized economy. These limitations in no way detract from the theory's contribution to our understanding of the role of price as an efficient decentralizing device. The general-equilibrium system provides a natural basis or departure point

from which to consider control mechanisms. It is in the control mechanisms for economies that can have a varying intermix of competitive or controlled prices that financial instruments play their role.

Notes

1. Here O' is the point where $\mu_2 = A - \mu_1$, $v_2 = B - v_1$.
2. Here $\mu_2 = A - \mu_1$, $v_2 = B - v_1$.
3. Implicit in this description is that all goods are of standard quality and are divisible and fungible; that is, one unit can be substituted for another without restriction.
4. It is hard to give much meaning to stockholder voting in the general-equilibrium model without specifying a great amount of corporate law and running the risk of losing the price system if the laws are not appropriate.

II EXCHANGE AND MONEY

5 The Distribution of Resources, Trade, and Markets

5.1 Methods for Allocating Resources

Much of economics deals with the allocation of individually owned scarce resources. Price theory, which forms a considerable part of microeconomic theory, introduces a price mechanism to solve the problems of production and allocation of resources in a given population.

An important feature of the political process is how it controls the products of jointly owned scarce resources within a society. Thus, for example, a society might use an economic mechanism to solve the problems of production and distribution of automobiles but turn to a political mechanism to solve the distribution problems of heavy military weaponry. In a modern industrial society, it is not easy to make clean differentiations among strictly economic, political, and social problems.

What are the alternatives to the marketplace? Where are they used, and how important are they to our economy? Are they used in all societies? Do they appear in the same mixes in underdeveloped countries as they do in developed countries? How deeply does the democratic form of government depend upon a market structure primarily characterized by small enterprises and a price system? Is there any relationship between the price system and democratic institutions? These questions are critical in helping us understand the development of modern politicoeconomic systems.

The following eight mechanisms appear to be the major societal means for allocating resources:

1. The free market with a price system;
2. Voting;
3. Bidding;
4. Bargaining;
5. Higher authority (fiat or dictatorship);
6. Force, fraud, and deceit;
7. Custom, including gifts and inheritance; and
8. Chance.

These mechanisms are by no means totally independent, and each in fact stands for a gross aggregation of more refined mechanisms. Nevertheless, the list in this crude form is suggestive of the approaches used in every society for resource allocation. Although we do not have a clear characterization of the relative importance of these mechanisms within

our own society, I would assert that they are critical to understanding politicoeconomic and socioeconomic allocation.

They are mostly subject to persuasion and other methods that can change individuals' preferences and hence behavior when allocating resources.

The market mechanism of the impersonal price system, with each individual using money or credit to buy and sell at prices set by a faceless market, appears well suited for solving the technical and administrative problems involved in the decentralized, efficient allocation of many resources. This is especially true for satisfying the desires of consumers for goods and services of a personal nature that can be produced by relatively small individual firms or distributed by government-owned industries.

The smooth functioning of a competitive price system requires many participants on both sides of the market. The supply and demand conditions for small restaurants or tailors meet this requirement. The demand for automobiles is generated by many individuals, but on the other side, there are only a few firms selling automobiles. When there are only a few suppliers of a good, the market is oligopolistic, and there are many good economic reasons to suspect that the price mechanism will not function simply.

Even if there are potentially many suppliers and customers, special considerations may cause a society to prevent development of an open economic market with a competitive price system. Trade in heroin, for example, could potentially function as a price-guided market were it not restricted from doing so.

Voting is a method used to control jointly owned resources, where the actual operation of the resources is overseen by a group of trustees acting as fiduciaries for the owners. Thus, for example, in corporations that sell shares, the stockholders are the owners and are called upon to vote on occasion, but the top executives and directors are responsible for running the corporation. In a similar manner, one may argue that, in voting, the citizens of the United States are appointing trustees or fiduciaries to run their political system. Votes on bond issues, taxation schemes, subsidies, and grants are all examples of the political mechanism being used as a method for defining effective demand for joint goods or services.

The voting mechanism is far more closely related to the formation of societal values than is the market system. Speaking loosely, we might characterize much of the market system as dealing with the effective

demand for individual and family wants, whereas voting systems deal with the aggregation and guidance of societal values.

No country uses its price system alone to determine attitudes toward justice, public health, foreign aid, or education. The price of beans, baby bonnets, and basketballs may be determined by the market without reference to deep political problems, but this is not the case for moon shots, urban development, or decisions on national communications systems. Undoubtedly, individual values are related to societal values, but it is important to recognize that they are not the same.

There is a growing literature on the theory of voting and on political parties operating in societies with voting systems. (See, for example, Arrow, 1951; Farquharson, 1969; Downs, 1957; and Shubik, 1968b, 1970) Voting methods differ widely from society to society, and the implications of these differences have been only partially explored.

Another important device for allocating resources is bidding. Where a few large customers, such as a branch of government or the military, buy whole systems from the few firms capable of producing them, the sealed bid is often used to decide the contract award. The art and antique, fur, used car, and tobacco markets use an open auction, while the Dutch auction is employed for selling tulip bulbs. Bidding and auctioneering appear in the West as far back as Rome and Babylon; the earliest oriental reference I have found is from seventh-century China (Yang, 1950). An interesting survey of auctions has been given by Cassady (1967). In recent years, much mathematical and operations research work has been devoted to the study of bidding, motivated to a great extent by the growing economic importance of the construction of large-scale systems (see Vickrey, 1961; Griesmer, Levitan, and Shubik, 1967; Smith, 1965; Engelbrecht-Wiggans, Shubik, and Stark, 1983; Milgrom and Weber, 1982; and Engelbrecht-Wiggans, 1980). In some instances, bidding procedures may become so complex and the items to be bid on may be so large that a two-stage process is called for, with the first bid for a contract to prepare a study or evaluation of the program on which the second bid will be submitted.

One major difference between the price mechanism and the bidding process should be noted. Implicit in the price mechanism is the assumption that all parties are capable of evaluating the worth of the transaction. Bidding, on the other hand, lays stress upon the importance of evaluation in the economic process.

Bargaining is used as a means of allocation both in the political and in the economic arena. Where it is used most frequently in labor-management relations, and on the international level, where it is used in tariff and trade negotiations, and in disarmament conferences.

Large-scale bargaining is often carried on through intermediaries, and much of the process is devoted to reformulating and clarifying issues as well as specifying the positions of the participants. Such bargaining can be both lengthy and costly, which in part accounts for the growth of professional arbitration. A theoretical analysis of the time and cost aspects of the bargaining process was developed by Cross (1969) and extended by Rubinstein (1982) and others.

There is a considerable difference between two-person and multiperson bargaining. It is fairly well established that two-person bargaining, in spite of its costliness, is a useful and viable means of settling a broad range of problems, but it is not yet clear how the effectiveness of bargaining changes as the number of parties is increased.

In many areas, formal institutions and procedures exist to oversee the bargaining process. The work of Iklé (1964) on how nations negotiate, which stresses the diplomatic aspects of bargaining, is a valuable companion to the work of Cross, which stresses the economic aspects of bargaining whereas the former lays emphasis on its international diplomatic aspects.

What we have discussed thus far could be described as formal bargaining inasmuch as, in general, there are relatively well defined "rules of the game," professionals and intermediaries are often employed, and there is a more or less formal structure to the process. The alternative, informal bargaining, more properly belongs with haggling. Informal bargaining and haggling are of historical and anthropological interest but have little influence on a large modern economy. In many South American, African, and Asian countries, there are still relatively simple markets where one can haggle over how many chickens a pig is worth, and the exchange may be effected by barter. And undoubtedly there are many individuals even in more modern economies who derive considerable joy from bargaining, "buying wholesale," and getting discounts. The process is only uneconomic if we fail to realize that bargain hunting is to some as much of a sport as baseball is to others.

The haggling in an oriental bazaar or a horse-trading market (Böhm-Bawerk, 1923; Shapley and Shubik, 1971; Green, 1971) might be re-

garded as a preliminary step toward the formation of a price system. In the mythology of economic theory, horsetraders rushing from post to post trying to get a better deal narrow the range of prices until finally, with enough buyers and enough sellers and to a good first approximation, only a single price exists for a specific commodity. This view of the market brings out the relationship between haggling, bargaining, and the price system. Haggling is bargaining in a more cultural than formal institutional setting; and the price system may be regarded as a limiting case of multilateral bargaining when the participants are very many, the commodities standardized and plentiful, and the communications good.

That one form of trading merges into another under certain circumstances does not mean that the forms are similar. Quantitative differences often imply qualitative differences, and this appears to be more or less the rule for trade, exchange, and distribution.

Governed by both custom and force, the decision of a dictatorship, a recognized high authority, or an elite group is used in many instances to determine allocation. For example, prizes and scholarships may be awarded and individuals promoted within a hierarchy by fiat decisions. Here, once more, it must be stressed that the decision may be closely related to an evaluation process. Awarding a prize or promoting an individual generally calls for a far more complex and multifaceted evaluation than does buying an automobile.

Wartime or other periods of emergency or scarcity usually give rise to rationing systems, complex mixtures of allocation by fiat and by market price (or in some cases black market price). At the bureaucratic level, rationing and quota systems often generate covert bargaining systems as well. Thus, a Soviet factory manager was often forced into intricate bargains with suppliers as well as the central bureaucrats.

Although it is a far cry from dictatorship, arbitration is also a means of enlisting a third party recognized by the original group as having higher authority. Given that the parties have agreed to submit their dispute to arbitration, unless they set bounds, the outcome depends upon the whim or wisdom of the "higher authority."

Force, fraud, and deceit are often overlooked as methods for allocating resources, even though in a country the size of the United States billions of dollars are probably allocated in this manner (Walter, 1985). There are many definitional problems in the classification of fraud and deceit, especially in the area of "white-collar crime." A distinction must be made

between the force of a dictator or hierarchy and that of a robber or invader: the dictator's force is legitimized within the society, while the robber's or invader's is not.

Although the distribution of goods by "force" is usually not legitimized by society and references to it are for the most part pejorative, there is a minor exception: prizes from physical, intellectual, or other competitions. Boxing matches and chess tournaments both come under this category.

Custom accounts for a large share of allocation (Polanyi, Arensberg, and Pearson, 1957), especially in more traditional societies. A church may be entitled to a tithe, or a king to certain resources. Tipping is basically a custom-determined method for allocating resources. Inheritance customs are extremely influential, and gifts play an important role in distribution.

At this point, it is important to note that not only goods and services are distributed in society. Some of these mechanisms also work well for honor and prestige. Others are not as efficient. Thus fiat, custom, and even voting or politics are better means for obtaining an appropriately legitimized title of nobility than is the price system. The title devaluations in the Austro-Hungarian Empire and in Imperial Brazil provide examples.

Gifts, honor, and prestige all pose problems far beyond those of "normal" economic goods. It has long been stressed by anthropologists (Mauss, 1954) and others that although the exchange relationship and the gift and honors relationship have much in common, they are considerably different.[1] Even though the economist or psychiatrist would immediately say that *something* must be given in return for the gift, both will recognize that what is given in return is paid in a totally different currency, one that may be enormously difficult to characterize.

Last but not least, plain luck should not be overlooked. Some people actually do win lotteries. Others encounter other types of windfalls. Still others specialize in having everything go wrong in their vicinity. Chance plays its role in human affairs. Perceptive and prepared individuals will undoubtedly exploit good fortune better than others, but they still need the good fortune. A mediocre Poker player with good cards stands a better chance of winning the pot than a good Poker player with rotten cards.

In some instances where an impasse is reached with respect to the division of an indivisible article, the allocation may be decided by "tossing for it." One of the most important instances of random allocation is in the distribution of external economies or diseconomies. The grower of roses or the burner of rubber tires may not care which way the wind blows.

Nature tosses the dice and determines who receives the externality (Shapley and Shubik, 1969a). In cases involving negligence torts, the attribution of responsibility for random occurrences is often critical (Bohlen, 1941).

The price system is not God-given. A simple-minded cost-effective analysis that fails to account for the nature and costs of the actual distribution process can easily generate large-scale errors. It has not been demonstrated that the trend in modern economies is necessarily toward a wider role for the price system. Indeed, the problems of large-system implementation and of external economies and diseconomies point to the opposite direction.

5.2 Barter

Alternative methods for distributing resources notwithstanding, the stress in this volume is on competitive or controlled price systems and the financial instruments and institutions used to operate them.

Section 5.4 deals with the formal definition of a market. The distinction made here focuses on when the mechanisms of trade can be best approximated by considering trade as a sequence of face-to-face exchanges in any number of commodities or a set of exchanges involving pairwise trades by individuals. It is important to note, though, that quantitative differences in trade clearly imply qualitative differences. For an economy with just two traders, each with one good for exchange, the classical Edgeworth box offers no distinction concerning the meaning of barter, the market, or monetary exchange. It is only when more goods and more traders are considered that distinctions appear.

Suppose that there are n traders, each of whom has an initial supply of m commodities. Thus i owns $(a_1^i, a_2^i, \ldots, a_m^i)$, and the total supply of commodity j is $a_j = \sum_{i=1}^{n} a_j^i > 0$. At its simplest, in a group of individuals without trust or a formal trading organization, barter may be regarded as a process in which individuals seek each other out and exchange a set of goods desired by one for a set desired by the other. If any group can trade with any group, then we may consider all the *partitions* of the number n, which we denote by $g(n)$, and select any two groups to trade with each other.

The number of partitions of a set of size n is the number of ways it can be split up into smaller sets. Thus, for example, for a set of four in which individual members are indistinguishable, we have

$$4, \quad 3+1, \quad 2+2, \quad 2+1+1, \quad 1+1+1+1,$$

or $g(4) = 5$. If the actual makeup of the coalitions makes a difference, then the number of distinguishable groupings is considerably larger; call it $p(n)$. For the relatively small number 4, we can enumerate all possibilities. If we use the notation (i, j) to stand for a coalition of size 2 composed of traders i and j, we have

$(1,2,3,4)$	1
$(1,2,3)$ and (4); $(1,2,4)$ and (3); $(1,3,4)$ and (2); $(2,3,4)$ and (1)	4
$(1,2)$ and $(3,4)$; $(1,3)$ and $(2,4)$; $(1,4)$ and $(2,3)$	3
$(1,2)$, (3), and (4); $(1,3)$, (2), and (4); $(1,4)$, (2), and (3);	
$(3,4)$, (1), and (2); $(2,4)$, (1), and (3); $(2,3)$, (1), and (4);	6
(1), (2), (3), and (4)	1
	15

or $p(4) = 15$.

If trade is regarded as an exchange between two parties, then the number of trading pairs that can form with four individuals is

Group sizes	Trading pairs
4	0
3, 1	4
2, 2	3
2, 1, 1	12
1, 1, 1, 1	6
	25

A certain amount of work has been done on games in partition-function form (Lucas, 1963), but because partitions proliferate rapidly, search, bargaining, and haggling costs make this form of trade highly improbable in the absence of highly structured mechanisms.[2] Even when anthropologists discuss trade and barter in primitive societies, they generally imply a considerable structure and technology for the mechanisms of trade (Herskovits, 1940, chapter 9). Constructing a game of strategy for a few dozen individuals that leads to efficient trade in reasonable time without many moves and without credit should illustrate the difficulties.

5.2.1 Double coincidence of wants

Limiting ourselves to a trading mechanism for three traders with a failure of the double coincidence of wants, we find that at least the rudimentary aspects of inventorying and brokering appear.

The simplest concept of trade between two individuals has direct value given for direct values. At least since Jevons [1871], however, it has been known that a sequence of trades is sometimes needed to achieve an optimal distribution. Consider the following situation:

Trader A has $(1,0,0)$ and a utility function $x^1 + 2y^1$;
Trader B has $(0,1,0)$ and a utility function $y^2 + 2z^2$;
Trader C has $(0,0,1)$ and a utility function $2x^3 + z^3$,

where x^i, y^i, and z^i are the amounts of goods 1, 2, and 3 held by trader i. The optimal final distribution is $(0,1,0)$ for A, $(0,0,1)$ for B, and $(1,0,0)$ for C. This can be achieved if B is willing to act as "middleman." The first trade is between A and B; they exchange bundles, with A's utility going from 1 to 2 and B's from 1 to 0. Then B trades with C, with B's utility going from 0 to 2 and C's from 1 to 2.

There are six paths involving two trades by which the final optimum can be reached. Figure 5.1 shows a formal game in extensive form for this trade. Selected randomly, the trader to move first can offer to exchange bundles with one of the others or refuse to trade. If he refuses to trade, the next trader is selected randomly and can offer or refuse to trade with the first trader; but if he offers to trade, then the second trader has four alternatives: (1) accept the trade and offer the next trade; (2) accept but refuse further trade; (3) refuse to trade but offer the third trader a trade; or (4) refuse to trade. The third trader can also accept or refuse to trade.

Only one-third of figure 5.1's game tree is drawn in full. It is left to the reader to check that optimal trade can be obtained as a noncooperative equilibrium.

Even when we consider this as a parlor game, one trader must buy to sell, and in the process he lowers his safety level; if his exposure is only for a few seconds, however, we would hardly call him a broker or consider his inventorying costs seriously. With more goods and people, this would be the case.

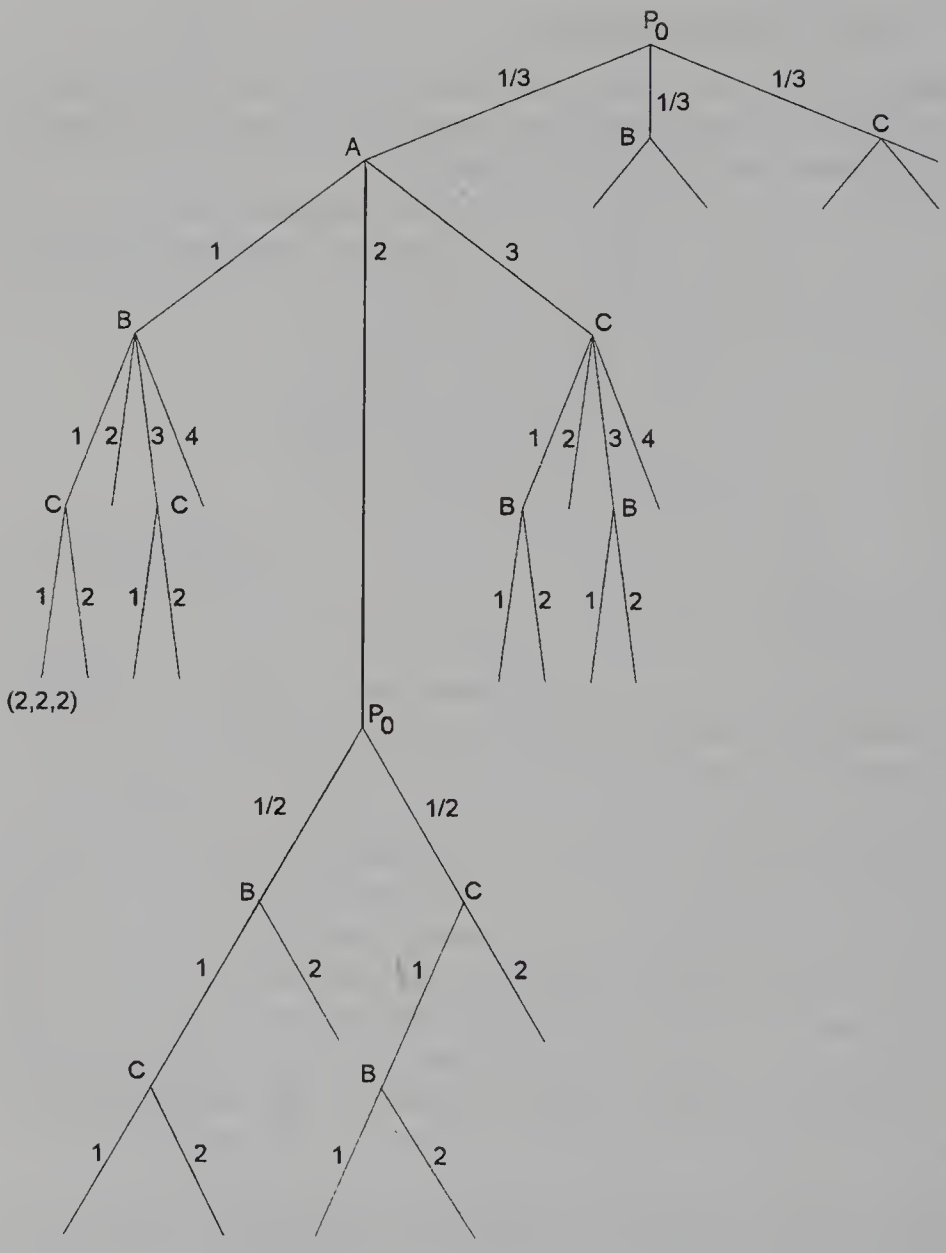


Figure 5.1
Trade in extensive form.

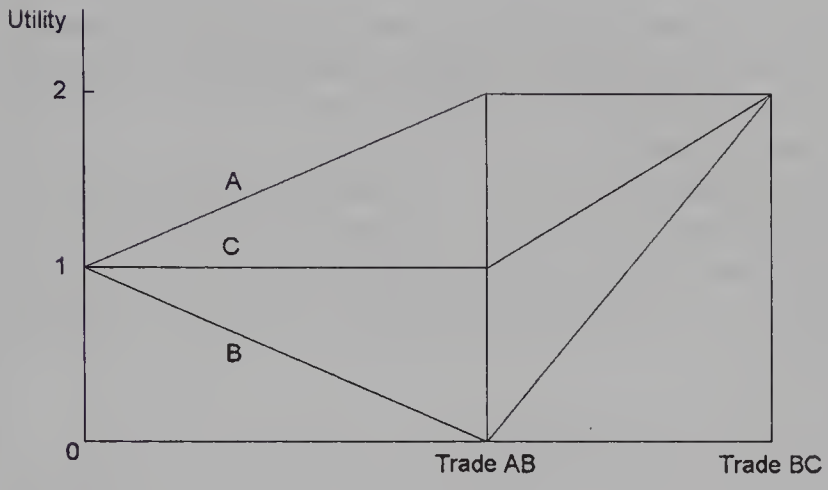


Figure 5.2
Broker safety levels.

Figure 5.2 shows the safety levels or exposures of the three individuals when trader *B* acts as "broker-dealer," exchanging first with trader *A* and then with trader *C*. After the first trade, his utility has fallen.

When there are both many goods and many individuals, there are two important possibilities for aggregation, represented as trading posts staffed by brokers or principal agents acting for their own accounts. Here a broker is someone who facilitates a transaction between two parties, whereas a principal agent is someone who buys from one party to sell to the other. In the example given above, if we called the trader in the middle anything, it would be a principal agent rather than a broker because he exposes himself to the full strategic risk in a sequence of transactions.

At a trading post, be it a post in the New York Stock Exchange, where a specialist "makes a market" in five or six stocks, or weekly open-air market in the Guatemalan highlands, individuals make essentially anonymous bilateral trades. By "anonymous", we mean that the name or affiliation of an individual trading at a particular post does not affect the trade. (In some instances in history, of course, caste considerations rule some individuals out of trade.) Trading post activities include aggregation of information, communication, and quantity. These activities are performed more or less well depending upon the post, the traders, and the goods being traded. If many people trade at the same post, the opportunities for quick communication and general knowledge of supply and demand conditions are good. It is conceivable that an exchange may involve several goods

being simultaneously traded for several goods, but in general trades can be broken down into bilateral trades of one good for another or for money. Without a money, indivisibilities may result in more than one item being offered in exchange for another.

One type of trade that can involve an exchange of large quantities of different goods between two parties over a considerable length of time is an international trade barter agreement, but this is by no means in keeping with the spirit of the interpretation of barter as a form of primitive subsistence trading.

5.2.2 Trading posts for all exchanges

We can assume the existence of trading posts for every pair of commodities. In our simple example of three traders and three goods, three trading posts are required. If the goods are 1, 2, 3, the posts are for $(1,2)$, $(2,3)$, and $(1,3)$; call these a, b, and c.

We can consider a highly simplified game of trade as follows. Assume that the commodity of each individual is completely indivisible. A has three alternatives: to offer nothing or one unit in market a or c. Similar alternatives exist for B and C. Assume that each trading post is run by a broker or a mechanism that merely gathers goods offered for exchange on both sides and clears bids and offers at the post. This is a highly simplistic and perhaps unrealistic mechanism, but it functions for virtually any number of traders with little complication. Unfortunately, as can be seen by looking at the game in extensive form shown in figure 5.1, a single functioning of the market is not sufficient for optimal trade.

The notation in figure 5.3 is as follows. The labeling of 0 with ABC indicates that all players move simultaneously. Each has three alternatives; hence the game can be played in twenty-seven ways. Only nine of the twenty-seven branches are shown in the figure. The alternatives are labeled as follows:

For player A: (1) offer nothing, (2) offer at post a, (3) offer at c;

For player B: (1) offer nothing, (2) offer at post a, (3) offer at b;

For player C: (1) offer nothing, (2) offer at post b, (3) offer at c.

If all select their third moves, A and C exchange and then, when B offers his good at a post where nothing is offered in exchange, we need a rule stating that his good is returned to him. This is not a rule without

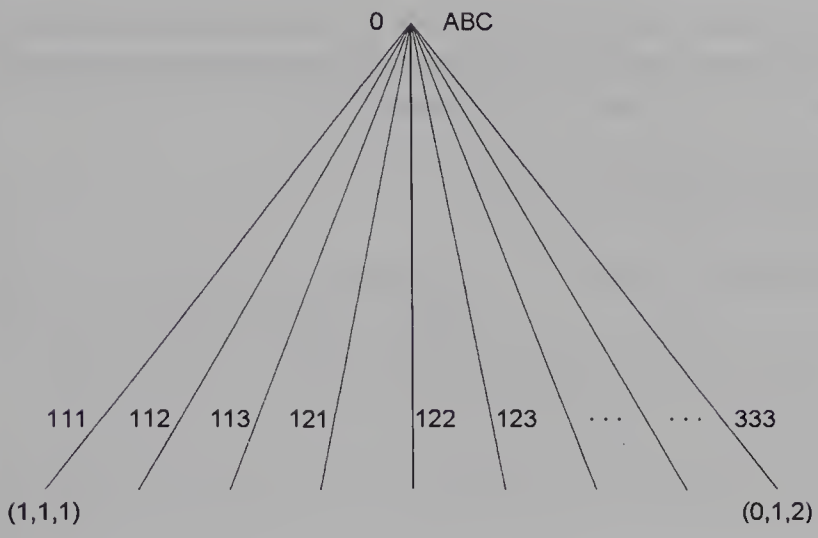


Figure 5.3
Simultaneous trade.

consequences. If B's good is a ripe tomato, it may well rot if it does not find an immediate trade.

It is easy to check that there are several equilibria, in one of which there is no trade and all traders maintain the status quo value of $(1, 1, 1)$. If we enlarge the game to include a second round of trading, the no-trade equilibrium remains but we can also obtain efficient trade, although not with full symmetry. In order to achieve efficient distribution, one trader must expose himself at the end of the first round by buying to sell again in the second round.

The efficient trades are associated with noncooperative-equilibrium points in this game. However, if the commodities were divisible, the efficient equilibria would be destroyed.[3]

Bradley (1973), Starr (1976), Ostroy and Starr (1974), and others have studied the technology of trading procedures. They capture the essence of barter without trading posts by considering encounters between all pairs of traders. Thus, for an economy with n individuals in a round-robin tournament, there are $n(n-1)/2$ encounters in an initial sequence of rounds. There are many interesting combinatorial problems, including calculating the minimum sequence of trades needed to achieve an optimal distribution, the effect of transaction costs, and the role of prices.

In contrast with this work, Clower (1967) considers a barter economy to be one in which there exist $m(m-1)/2$ trading posts. He does, however,

observe that this appears paradoxically to be an economy where all goods are money. The existence of a trading post calls for a higher level of formal organization than is usually implied by barter.

5.3 Marketing and Transaction Costs

Before considering markets per se and the contrivance of money, we note a selection of the details that virtually any society must take care of in distributing the goods and services supplied through the economic process.

Because our study is not primarily directed toward economic anthropology, we make no attempt to trace any of the specific chains of causality that may have led to the formation of the existing market mechanisms. Details of demography, habit and custom, law and tradition conspire with the physical features of goods and with information and control factors to provide the basis for the variety of institutional detail required to understand and control or influence specific markets.

The subjects of markets and marketing have for many years been regarded as peripheral to economic theory. Perhaps this is true in the preinstitutional study of production and distribution, but it is not true when we are considering process.

5.3.1 Sorting, grading, and quality control

How many items are available for sale in an economy such as that of the United States? As a first approximation, we note that the Standard Industrial Classification Manual (OMB, 1972) provides a four-digit industrial classification; not all these numbers are used, but the upper bound is 10,000. The Bureau of the Census breaks some of this information into product classes using a five-digit code; and the Standard Industrial Classification itself utilizes a seven-digit product code. This does not include variations involving size, grade, and other features of product differentiation. For example, a survey of thirty-eight chain stores found 35,000 differently coded items.

The degree of arbitrariness in categorizing the number of commodities in a society is great, and the classification depends upon our legal, technological, and marketing purposes. Any result in mathematical economics that depends in a delicate way upon the number of commodities assumed to exist must therefore be suspect. Thus, if a result holds for

n individuals and m goods and we reclassify the m goods into m' goods (introducing, for example, four grades of apples where previously there were only three), we should expect little change in the solution.

An early precondition to the formalization of market trade and the use of money is the creation and control of weights and measures. A society must have measures for continuously divisible commodities such as grain and be able to specify conditions on size, weight, and composition in order to declare two items fungible.

Kisch (1965, p. 3) suggests that weighing is probably the youngest of the inventions of counting, measuring, and weighing. With the invention of coinage Governments entered as monopolists, frequently with capital punishment for forgery and coin clipping, but it was only much later that the production and control of standard weights and measures became a government monopoly.

Skinner (1967, pp. 2–3) suggests that linear measurement was first developed around 10,000 to 8000 B.C.; capacity measurements came later, possibly for trade in food grains. The earliest weights recorded have been dated at 3800 B.C., and two tomb paintings from around 2650 B.C. are reported to show capacity measures, weights, and two wooden balance beams.

Early Egyptian weighing scenes show balances used for precious metals or temple treasure, but they were not used for commerce until around 1350 B.C. There are paintings of an Egyptian market scene using barter dated around 2500 B.C. (Skinner, 1967, pp. 8–9). It is also possible, though the evidence is not clear, that weights and measures figured in the commercial life of the pre-Aryan civilization of the Indus valley in the towns of Harappa and Mohenjodaro around 2000 B.C. (Skinner, 1967, pp. 10–11).

To this day, in markets such as furs, gems, produce, and wine, the sorting, grading, and matching functions are critical. In the trading of consumer durables such as shoes, clothes, electrical appliances, and automobiles, the clash among quality control, information, and advertising considerations comes to the fore. The existence of specialized magazines such as *Consumer Reports* illustrates this. What are the dimensions of the services provided by a compact disk player or a freezer? Most customers are not in a position to make technically informed judgments about the dimensions or measures of quality, and those who supply the markets deal with both the formulation of what constitutes quality and quality control.

5.3.2 Distribution: Inventorying and transportation

The U.S. Office of Management and Budget (OMB) breaks down the country into some 300 standard metropolitan statistical areas (SMSAs). Spread over a large terrain in 1976 were 313,000 manufacturing establishments, 370,000 distributors, and 1,913,000 retailers serving 66,700,000 households (*Statistical Abstract of the United States*, 1978). By 1984, the numbers had changed to 358,000, 416,000, 1,923,000, and 83,527,000, respectively (*Statistical Abstract of the United States*, 1986).

Clearly, a refrigerator on a factory floor is not a final product to a consumer, whereas a serviced refrigerator installed in the home is. An indication of the importance of distribution is given by the percentage of national income accounted for by distributors, retailers, and services. In 1972 and 1984, these amounts were

	1972	1984
Wholesale	\$54.4 billion (3.2%)	\$164.9 billion (4.6%)
Retail	\$90.2 billion (5.2%)	\$336.9 billion (9.3%)
Services	\$122.3 billion (7.1%)	\$529.4 billion (14.6%)

The GNP for these years was \$1,717.1 billion and \$3,622.8 billion (*Statistical Abstract of the United States*, 1978, 1986).

As Chamberlin (1933) and Hotelling (1929) have noted, once one considers spatial competition or service, oligopolistic or monopolistic competition becomes the rule rather than the exception.

The importance of inventories varies considerably, from a few days to years. Included in inventorying is packaging, preservation, special storage, goods handling, obsolescence, breakage, theft, and the costs of working capital locked up to finance inventory. The last of these is of special concern to us in this book.

In a modern society, the firms or factories carry raw, intermediate, and finished goods inventories on their own accounts and at their own risk. Much of finance and insurance is designed to make this feasible and efficient.

5.3.3 Brokerage, search, information, and advertising

A society's communication network is crucial to the functioning of trade and markets. A key feature in the development of trade in empires are

roads, other transportation, and message-sending facilities. Examples of the importance of roads are provided by the Roman and Incan Empires. The direct ancestor of the modern postal service appears to be the monopoly granted to the house of Taxis around 1560. The first recorded postmaster general in England was Sir Brian Tuke in 1533, although the University of Paris had a postal service in the thirteenth century. In the United States alone, the post office employed 682,600 workers in 1984 (*Statistical Abstract of the United States*, 1986). The telegraph was patented by S. F. B. Morse in 1840 and first used in a line from Washington to Baltimore in 1844, and the telephone was patented by A. G. Bell in 1876 (Brooks, 1975). Edgeworth, in his discussion of the meaning of a competitive market, considers as his idealization that all traders are costlessly connected by telephone (Edgeworth, [1881] 1932). The first clearing system for checks in the United States was formed in 1853 with fifty-two members. The Federal Reserve clearing system dates from 1913. The copying machines, jet aircraft, high-speed digital computers, hand calculators, and communications satellites of the post–World War II era have all represented quantum leaps in the communications systems binding together markets over vast areas of the world. As transportation and communication technology change, so does the role and the nature of risk taking by brokers and other middlemen in trade.

Advertising has always posed a knotty problem to the economic theorist. There is clearly a component in advertising directed toward influencing and forming taste, as opposed to providing information to individuals with given tastes. There is also a considerable industry devoted to the production and utilization of economic and financial data. A rough index of this industry is given by estimates of the number of brokers, dealers, and financial work force, reported by the New York Stock Exchange as 133,000 in 1976 and 409,800 in 1986; the membership of the American Economic Association, which was about 27,000 in 1979 and 20,000 in 1987; and the number of accountants and auditors, estimated at 1,013,000 in 1980 (NYSE *Fact Book*, 1977, 1987; *American Economic Review*, May 1979, p. 385, and May 1987, p. 368; *Statistical Abstract of the United States*, 1986).

Information is produced, processed, and, together with expert assessment, bought and sold. An immediate question to the economist is whether the differential in information, say to those with "inside" sources, is reflected in market prices. This question can be posed mathematically

and answered in the affirmative up to a point (Dubey, Geanakoplos, and Shubik, 1987). Yet, in terms of economic actuality, this question is not as important or as hard to investigate as "What is meant by expert opinion?" and "Why do many individuals buy the expert opinion of one or more economic or financial experts?"

5.3.4 Innovation

A dynamic economy is an equilibrium-seeking system that is rarely, if ever, in an equilibrium state. A key element in the dynamics is innovative activity, a crude estimate of which is given by the amount spent on research and development. In 1984 this was estimated at around 95.9 billion dollars, of which around 33.6 billion was defense- or space-related (*Statistical Abstract of the United States*, 1986). New products require the creation of new markets and the availability of high-risk funds from venture capitalists investment bankers and corporate, central planning, or other governmental funding agencies.

5.3.5 Credit and trust

One of the central features of a monetary and financial system is the evaluation of trust, risk, and credit. A good banker must judge not only honesty, but competence and perception. For example, a good investment banker understands that inventors frequently worry more about seeing their inventions brought to life than about mundane details such as economic viability. The mere fact that an inventor is willing to stake his life's savings along with the money of outside investors is not qualitatively the same as the knowledge of the passengers in a jet liner that the pilot's interest in survival is highly correlated with their interests.

The multiplication of bank loan departments, credit information services, and agencies such as Dun and Bradstreet, on the one hand, and the Securities and Exchange Commission, on the other, is evidence of the growth of private, corporate, and governmental credit evaluation, which calls for considerable disclosure from a variety of economic entities.

5.4 Markets and Money

But this barter introduced the use of money, as might be expected ... for it is not everything which is naturally most useful that is easiest of carriage; for which reason they invented something to exchange with each other which they should mutually

give and take, that being really valuable itself, should have the additional advantage of being of easy conveyance, for the purposes of life, as iron and silver, or anything else of the same nature: and this at first passed in value simply according to its weight or size; but in process of time it had a certain stamp, to save the trouble of weighing, which stamp expressed its value.
—Aristotle, *Politics*

Having sketched a host of special considerations involving the distribution of goods and services, we proceed by making several drastic simplifications. The aspects of distribution that generate activities and costs are

1. Transaction technology,
2. Perceptual problems concerning evaluation, grading, and judging new markets,
3. Credit, trust, and contract enforcement, and
4. Search and data collection and processing.

The problems posed by transaction technology more properly belong to production theory than to monetary theory. Without going into detail to explain why under certain configurations of costs certain markets will not be present, we may study the implications of limited markets by directly specifying the market structure. Thus we may separate questions involving technological limits on the existence of markets from questions concerning the economic implications of the lack of certain markets.

The theoretical problems and difficulties involved in taking into account different levels of perception and skill in the evaluation of economic worth and opportunities are large. No attempt is made here to deal with this grey zone that lies between economics and psychology.

The roles of credit, trust, and contract enforcement are central to the understanding of money and financial institutions. They are related both to information conditions and to rules concerning precommitments to using strategies and carrying out threats. The models in subsequent chapters place stress upon these factors.

Searching for and processing information and data are at the heart of trade. This includes the aggregation of information and the design of markets, financial instruments, and reporting methods to optimize in the tradeoff between the costs of loss of detail and the costs involved in processing and digesting a cascade of information.

Our investigation of money and financial instruments begins with the nature of markets and the market structure of an economy as a whole. We start with Jevons ([1871] 1970), who offers his definition of a market and includes in a footnote a reference to Cournot's definition:

> By a market I shall mean much what commercial men use it to express. Originally a market was a public place in a town where provisions and other objects were exposed for sale, but the word has been generalized, so as to mean any body of persons who are in intimate business relations and carry on extensive transactions in any commodity.... The traders may be spread over a whole town, or region of country, and yet make a market, if they are, by means of fairs, meetings, published price lists, the post office or otherwise, in close communication with each other. Thus, the common expression Money Market denotes no locality: it is applied to the aggregate of those bankers, capitalists and other traders who lend or borrow money, and who constantly exchange information concerning the course of business.[6]
>
> In economics we may usefully adopt this term with a clear and well-defined meaning. By a market I shall mean two or more persons dealing in two or more commodities, whose stocks of those commodities and intentions of exchanging are known to all. It is also essential that the ratio of exchange between any two persons should be known to all the others. It is only so far as this community of knowledge extends that the market extends. Any persons who are not acquainted at the moment with the prevailing ratio of exchange, or whose stocks are not available for want of communication, must not be considered part of the market. Secret or unknown stocks of a commodity must also be considered beyond reach of a market so long as they remain secret and unknown. Every individual must be considered as exchanging from a pure regard to his own requirements or private interests, and there must be perfectly free competition, so that anyone will exchange with anyone else for the slightest apparent advantage. There must be no conspiracies for absorbing and holding supplies to produce unnatural ratios of exchange....
>
> The theoretical conception of a perfect market is more or less completely carried out in practice. It is the work of brokers in any extensive market to organize exchange, so that every purchase shall be made with the most thorough acquaintance with the conditions of the trade. Each broker strives to gain the best knowledge of the conditions of supply and demand, and the earliest intimation of any change. He is in communication with as many other traders as possible, in order to have the widest range of information.

5.4.1 On the properties of markets

Trade and exchange with any forms of money and other financial instruments take place with some form of market structure. Before we examine several specific mechanisms of trade in chapters 6 and 7, we approach the

properties of markets axiomatically; that is, we consider reasonable or desirable features that we might wish to impose upon markets and market structures. Given various limitations on market structure, we define a class of *strategic market games*, which we can study for the behavior of the noncooperative-equilibrium solution.

The work presented here is based upon investigations by Dubey, Mas-Colell, Sahi, and Shubik (Dubey, Mas-Colell, and Shubik, 1980; Dubey, Sahi, and Shubik, 1993; Dubey and Sahi, 1992). The approach to a description of market mechanisms is both axiomatic and taxonomic. Many different institutional structures are used to facilitate trade and exchange in specific items. Elaborate traditions often spring up around market towns, and certain markets are more prestigious than others. Methods of exchange are highly diverse, and information and knowledge conditions vary from market to market. For example, country estate auctions have open sequential bidding in which bidders may be rank amateurs out for fun or professional dealers with clear perceptions of resale prices. Large contracts may be bid on by few firms spending many months to put together secretly submitted simultaneous bids.

In retail markets, the sellers in most mass economies post prices ahead of time; the buyers see the prices and independently decide if they wish to buy. In some markets or stores haggling and tie-in sales are possible. There are still small and large markets employing barter. Many markets do not even have a clearly defined location; an example is the Eurodollar market. The stock market provides an example of an anonymous, mass, double-auction market.

Most markets are far from the idealized, noninstitutional, timeless, price-generating transformations implicitly portrayed in general-equilibrium theory. Our attempt here to characterize some of the desiderata for market mechanisms provides only a small way station toward realism. Our concern is more for considering what certain mechanisms would look like if they were required to satisfy certain reasonable properties than to characterize specific existing markets.

5.4.2 The characterization of a market

1. We take as our basic building blocks n individuals trading in m commodities. We concentrate first on exchange, leaving aside the additional problems posed by production.

2. We characterize each individual i as a member of the set $I_n = (1, 2, \ldots, n)$.[5] Each individual i has an initial endowment a^i of the m commodities, where $a^i = (a_1^i, a_2^i, \ldots, a_m^i)$; here $a^i \in \mathbb{R}_+^m$, the positive orthant in the Euclidian space of m dimensions.

3. A *market* is a mechanism that processes *messages* (defined in point 5 below) and produces *trades*. We call the set of all markets in an economy e the *market structure* of that economy. We denote this set by M_e, where $m_e = |M_e|$ is the number of markets in the economy e.

4. A *trade* is a mapping of a *market input* by i into a *market output* to i. An *individual trade* is a mapping involving one individual i and one market s.

5. A *message* is a statement containing the name of the sender i, the name of the market being addressed, and contents consisting of sentences constructed from a syntax in which nouns are names of goods, markets, and traders, adjectives are quantities, and verbs are bids and offers. Simple bids and offers do not require connectives; but for contingent or complex messages we must also allow "and", "if", "or", "then", and operations such as "less than or equal to."

On the number of markets Any market can involve any subset of traders on either side, trading any subset of goods against another. Hence the number of market structures fast becomes astronomical for even a modest number of traders and goods, even leaving out the multiplicity of exchange mechanisms that might be used to clear a market when groups of goods are being traded for groups of goods.

When a single good is being exchanged for a different single good, there is a natural way to aggregate bids and offers and to form a final market price. The quantities of good i offered by all sellers can be added, and the quantities of good j supplied by all buyers of i can be added. The exchange rate or price of i in terms of j can be defined as the ratio of two quantities. In chapter 6, we consider the number of mechanisms involving the exchange of one commodity for another; with few restrictions on the pattern of trade, this number is small.

When we consider barter or bilateral exchange between two countries, it is easy to conceive of trades involving quantities of five or six items being exchanged for quantities of three or four items. Once we contemplate any form of mass market, it is difficult to construct plausible satisfactory market clearance mechanisms without introducing conditions that lead to the emergence of prices.

Conditions on markets An implicit assumption of modern microeconomic analysis is the presence of a mass, anonymous market mechanism. Here we suggest some explicit assumptions concerning markets.

CONDITION 1 *Openness:* A market is open if everyone is permitted to trade in it. This means that instead of having to specify one of many sets of names at a market, we need only specify all names for any market. (A market that is not open is *exclusive*, for example, an auction by invitation only.)

CONDITION 2 *Anonymity:* A market is anonymous if a permutation of all names has no influence on the outcome.

CONDITION 3 *Aggregation:* We have not yet defined how messages are processed and combined at a market; but whatever the process of aggregating messages is, we can ask that it be *associative* and *commutative*.[6] A further property of an aggregate market is that an individual's trade depends only upon his or her message and the aggregation of the messages of all others.

Conditions on messages

CONDITION 4 *Feasibility:* A message processed at a market will result in a projected trade that requires individual i to deliver a specific bundle of goods to the market. The *feasibility* condition requires that no matter how individual i's message is processed, i will always be in a position to deliver any bundle of goods called for by the market.[7]

CONDITION 5 *Transmittal restriction:* One might wish to restrict individuals to sending at most one message to a market, but this actually poses no extra restriction because a single message may contain contingent clauses to be processed by the market. Hence any two messages that are consistent can be considered as a single message joined with an "and" or an "or".

CONDITION 6 *Orthogonality:* A reasonable restriction to impose upon a message sent to a specific market is that it can refer only to trade that takes place at that particular market. In other words, the variables contained in the message must be ones used in the mechanism of trade at that market. (An example may help to illustrate this restriction. In a market where only oranges are traded for apples, references to contingencies based upon the exchange rate between gold and silver are ruled out.)

CONDITION 7 *Message simplicity:* Given that all messages are composed from a finite list of symbols, we may wish to examine a special class of messages of minimal length—for example, those involving no contingencies.

Further comments on markets

Traders and specialists We concentrate primarily upon markets in which all individuals play symmetric roles. Thus we might require as a specific assumption that there exists only one type of individual. This contrasts with a distribution system in which there are auctioneers, brokers, specialists, and retailers and wholesalers, each of whom plays a specialized role in the technology of exchange. (The existence of these specialties indicates that individuals are limited-capacity data processors; a more refined theory than that offered by the model of economic man would take this into account.)

Information conditions The technology of production and distribution frequently calls for great differences in timing. In some markets (such as mass retailing), it is frequently the case that the suppliers name a price for their goods and the potential buyers simply accept or reject those prices.

By limiting ourselves to the simultaneous sending of messages to all markets, we imply symmetric information concerning the moves of all traders. This limits the possibility for complex strategies based on the availability of information on the moves of others.

Because of the extreme arbitrariness of reaction functions that can be constructed when some traders are provided with information about the moves of others, one might worry that market equilibrium conditions may be destroyed. When we construct strategic market games in chapter 6, we shall show that this is not the case.

Conditions on market structures

CONDITION 8 *Nonredundancy:* A reasonable special condition that cuts down heavily on the number of markets in a market structure is that there exists no more than one market in which a commodity i can be exchanged directly for j. This is plausible if there are no transaction costs. In actuality, there are; hence considerations of space and transportation may call for many markets for the same good.

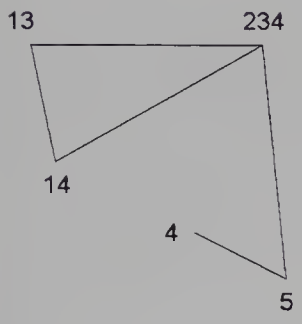


Figure 5.4
Connected market structure.

CONDITION 9 *Connectedness:* A market structure is connected if any good i can eventually be exchanged for any good j through a series of exchanges in a set of markets. Figure 5.4 shows a connected market structure with five markets in which, in order to trade good 1 for good 5, one would have to go through at least two sets of exchanges sequentially in two markets. The labels at each point indicate the commodities available for exchange at that point; a line connecting two points indicates a market for the exchange of combinations of the goods at each end. Thus there are only complex markets shown here, except for the simple market between 4 and 5.

Because we will be primarily concerned with simultaneous trading, we do not further consider exchanges going through chains of commodities.

5.4.3 The definitions of money and credit

In order to define a money, we must first specify a market structure. A *real* or *fiat money* is a commodity or instrument such that there exists a set of markets within the market structure at which the money can be exchanged directly for every other commodity or instrument in the economy.

If we limit consideration to market structures with simple markets, then commodities or instruments can be represented by points, and any line connecting two points represents a simple market. Figures 5.5a to 5.5d illustrate four market structures with four commodities and three, three, five, and six markets, respectively. In the first structure no good is a money; in the second, good 1 is a money; in the third, goods 1 and 3 are monies; and in the fourth, all four goods are monies. It follows immediately that for m

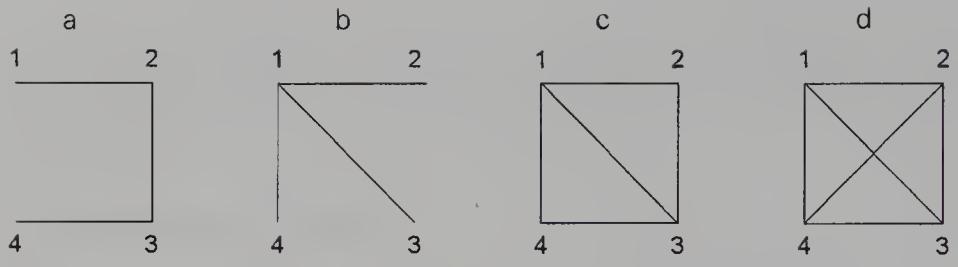


Figure 5.5
No, one, two, or four moneys.

goods there will exist structures such that any number of goods except $m - 1$ can be monies simultaneously.

A money can be a real good such as a bar of gold or a brick of tea or it can be a creation of the state in the form of paper whose acceptance in trade is backed by the law. The essential difference between a *commodity money* and a fiat money is reflected in the preferences of individuals. The difference between a money and a nonmoney is reflected in the strategic opportunities in trade offered by one but not the other. Phrased more concisely, "moneyness" is a property manifested in the strategy sets of traders. The distinction between a commodity money and a fiat money is reflected in the preference orderings of individuals. A fiat money would have essentially no residual worth were it demonetized; a commodity money would retain worth as a commodity even if demonetized.

Fiat money differs basically from a full-valued commodity money in the sense that use of the latter requires no act of faith or extension of credit, whereas use of the former involves belief and trust in, or at least obedience to, the state. Other financial instruments involving credit are created in pairs. A debtor-creditor relationship is created when a bank makes a loan or an individual extends credit to another individual. In a like manner, it might appear that the issue of fiat paper by a central government must be balanced by an opposite entry in the ledgers.

Because of the disparity in strategic power between individuals and central governments, for most purposes the debtor-creditor relationship with respect to the holding of fiat money can be ignored. The presentation of a dollar bill to the U.S. Treasury will, at best, bring its owner a new dollar bill in exchange. Fiat money[8] need not be redeemed. Operationally there is no offset to it.

The asymmetry between banks and private individuals is large and reflected in the rules of the game concerning their respective abilities to

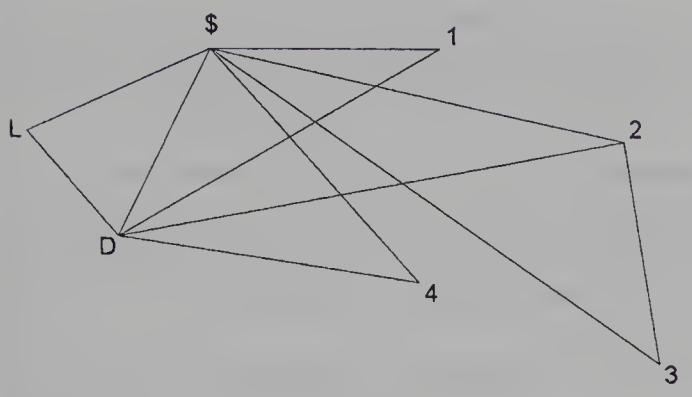


Figure 5.6
Market structure with goods and paper.

create acceptable credit. Even so, unlike the government, it is likely that banks and individuals will eventually be required to settle accounts.

Bank credit (denoted by D in figure 5.6) can be meaningfully defined as a near-money in the sense that in an economy such as that of the United States it can be used almost all of the time to pay for almost everything. In figure 5.6, the connection between D and 3 has been left out to indicate that there might be a market that will take cash only rather than accept checks. The point L stands for individual indebtedness to the banks. Individuals may "monetize" their credit by exchanging indebtedness with the banks, that is, L for D. Sometimes bank loans outstanding may be sold to a third party; this is indicated in the connection between L and \$. The connection between 2 and 3 in figure 5.6 indicates the possibility of a system that has elements of barter in an otherwise monetary economy.

A key element distinguishing a "full-bodied" commodity money (defined formally in chapter 6) from fiat money or from other forms of debt is that is carries its value without need for contract. The distinction among fiat, bank debt, and individual debt comes in the laws of issue and enforcement of contract together with the penalties for failure to fulfill contract. These points are specified in greater detail in chapter 9.

5.4.4 Simple markets and market structures

A *simple market* is a market in which quantities of a single commodity i may be exchanged for another single commodity j.

CONDITION 10 *Moneyness:* A market structure contains a money if there exists one commodity that can be exchanged directly for every other commodity.

CONDITION 11 *Completeness:* A market structure is said to be complete if every good is a money, that is, if any good i can be directly exchanged for any good j. Figure 5.5d shows a complete market structure.

The minimal (in the sense of fewest edges) graph representation of a connected, nonredundant market structure with a money is a graph connecting one of m points to the remaining $m - 1$. Figure 5.5b shows a four-commodity economy with a single money (good 1), and figure 5.5a shows a market structure with no money.

It is clear that, for efficiency or optimality to be achieved, there must be markets for all economic goods. Trading efficiency calls for a selection of appropriately simple efficient trading mechanisms. It is suggested here that an important class of such mechanisms are *simple markets with a minimal connected market structure, that is, a single money*. Furthermore, to a good first approximation they are simple markets processing *simple messages*.

A message is said to be simple if it contains no strings of contingency clauses. A message such as "Sell 1,000 XYZ for $22 a share" is simple. A message to the supermarket manager that the buyer intends to buy 14 cans of beans if the price is 55 cents, 23 cans if the price is between 45 and 54 cents, 28 cans if the price is between 40 and 44 cents, and so on, is a complex message that is in general not processed in most markets.

It is possible that individuals think in terms of complete demand schedules, but this is an open question in psychology.[9] Most market moves, however, are not composed of large contingent messages, even if they have been selected from complex strategies. Indeed, messages with two or three contingencies are rare even in highly sophisticated markets with sophisticated traders and brokers. Connected market structures composed of simple markets processing simple messages, where the markets are open and anonymous and trade is aggregated, capture the essence of trade in a modern monetary, competitive economy.

The foregoing discussion provides a heuristic overview of the considerations that enter any attempt to describe the essence of a mass market. Dubey, Mas-Colell, and Shubik (1980) were able to establish the conditions under which a strategic market game with enough money and a continuum of traders (i.e., enough traders for competition) will have the property that every noncooperative equilibrium (NE) is Walrasian (i.e., is a competitive equilibrium, or CE).[10] The five conditions used are

1. Convexity: Traders have available a convex set of strategies.

2. Anonymity: From the point of view of the market, only the message sent by the trader matters.

3. Continuity (of outcomes with respect to strategies).

4. Aggregation: The trading possibilities of any player are influenced by the messages of the other players only through the mean of those messages (and not, for example, through the variance).

5. Nondegeneracy: It must be possible for individual players to influence their trading possibilities in the market; that is, the actions of each individual must influence his payoff.

The markets discussed in chapter 6 provide examples of price-formation mechanisms with the properties noted. There are, however, some well-known markets, such as the double-auction market and other markets based on the Bertrand-Edgeworth mechanisms, that stress price formation and do not satisfy condition 3. These are considered further in chapter 7. The formal proof is given in the paper noted.

A different set of axioms used by Dubey, Sahi, and Shubik (1993) allows for many rounds of trade and covers both Cournot and Bertrand-Edgeworth price mechanisms. In the first set of axioms prices were not assumed. In this alternative set the assumptions are

1. Price mediation.

2. Nonatomicity.

3. No-trade option.

4. Connectivity.

5.4.5 Primitive and ceremonial monies and fiat and coin

In his study of primitive money, Einzig (1948), never quite offers a concise definition of what he means by a primitive money in contrast with a modern money, even though chapter 1 of book III bears the lead heading "Definition of Primitive Money." The closest he comes is this: "A unit or object conforming to a reasonable degree to some standard of uniformity, which is employed for reckoning or for making a large proportion of the payments customary in the community concerned, and which is accepted in payment largely with the intention of employing it for making payments" (p. 319). He then offers a check list for the classification of primitive money (p. 335):

1. Has it a concrete existence, or is it an abstract unit?
2. Does it assume the form of currency or credit?
3. What material does it consist of, and what form does it assume?
4. Has it full intrinsic value, or is it a fiduciary currency?
5. Is it used exclusively for monetary purposes?
6. Is it a luxury or a necessity?
7. Is its use confined largely to commercial or noncommercial payments?
8. Is its use universal, or is it limited to certain categories of members of the community using it?
9. Is its commercial use universal, or is it confined to certain categories of goods?
10. Is it used for internal or external payments?
11. Is it produced locally, or is it imported?
12. Is its use conventional, or is it imposed by state authority?
13. Is it convertible into other objects?
14. Does it change hands by tale or by measured quantity?
15. Which of the monetary functions does it perform?

I would suggest that this definition is close to money or near-money in a modern sense.

Although there appears to be little to be gained in maintaining a tight distinction between modern and primitive money, anthropological evidence suggests the existence of "ceremonial monies." For example, tokens of value in the Palau islands were used in marriage, divorce, birth, and death rites and on several other occasions, and the renowned stone money of the Yap islanders also seems not to have been a general medium of exchange (Herskovits, 1940, pp. 240, 258).

Although money cannot buy happiness, (and vice versa) in all societies at more or less formalized levels there are relationships among money, prestige, and power. In some senses, it is possible that the formal introduction of a ceremonial currency whose domain of operation is confined to special occasions is a less rather than more primitive approach to extending the domain of economic measurement.

Ceremonial monies, like fiat money, depend for their scope of use upon the laws and customs of society and on society's ability to enforce its

rules. The etymology of the words for money and the names of monetary units indicate the close relationship between the use of money, the growth of law, and the use of weights and measures. In particular, the word "mark" has been explained as a coin or weight unit that has been officially marked.[11]

On commodity money and coin By our definition of money, any commodity that can be exchanged directly for all others is a money. This can be seen most clearly as part of the rules of a game formalized by society. A commodity such as a bar of gold or sheet of copper is not quite a commodity money if law and custom do not require that it be accepted as money. It may be considered a near-money, as suggested in section 5.4.3.

The transition that converts gold,[12] silver, or other commodities into a full money is suggested by Skinner (1967, p. 53) in his definition of coinage: "A coin may be defined as a small ingot of metal which bears a mark of such well recognized authority as to guarantee both its weight and its standard of purity, so that it passes as readily accepted value (i.e., as 'currency'), thus rendering unnecessary the use of the balance and weights." This definition stresses both law and authority, on the one hand, and weights and standards, on the other.

The first coinage is attributed to the kings Gyges and Andys of Lydia in Asia Minor (circa 700–637 B.C.). It was made of electrum, a natural alloy of approximately 73 percent gold and 27 percent silver. The largest coin was the Stater (Skinner, 1967, p. 54).

In the course of his reign from 568 to 554 B.C., King Croesus of Lydia issued coins of pure gold and silver. Coinage quickly spread through the Aegean, the earliest European mint being that on the island Aegina (Skinner, 1967, p. 56). It is of interest that coinage was apparently never invented in the Americas.[13]

5.4.6 Markets and bureaucracies

This chapter has stressed trade and exchange rather than production. When we consider production, we must take sequences of trades and bureaucratic decisions into account as an initial set of goods held by an initial group of owners is transformed into a final set of goods held by the final set of owners.

Modern societies tend to have large, complex industries with many different patterns of vertical and horizontal integration; machines, plants,

and products of great complexity and indivisibility; and labor and other supply contracts of varying lengths of time. When we consider all of these factors together, it becomes extremely difficult to determine when an external market or internal organization provides an optimal form of economic organization. In particular, in most enterprise economies these considerations become especially hard to comprehend with clarity when oligopolistic power, corporate control, and taxes are taken into account (see Williamson, 1975).

5.4.7 A note on institutions

In order to set the context for the discussion of trade, production, and finance in an economy with money and financial institutions, we offer a few aggregate statistics for 1962 and 1982. Figures 5.7 and 5.8 show only the 1962 statistics, but the structure is the same in 1982.

In 1962 (1982), the population of the United States was 186,700,000 (232,345,000). The number of families and unrelated individuals was 54,652,000 (61,019,000). The number of corporations and partnerships in operation at all levels of business was 2,092,000 (4,262,000), or about 1 per 89 (55) people. When we include individuals who claimed some form of self-employment, then in 1962 (1982) the total number of business organizations was 11,382,000 (14,546,000), of which 1,268,000 (2,926,000) were corporations, 832,000 (1,380,000) were partnerships, and 9,183,000 (10,160,000) were proprietorships. These institutions accounted for much of the economic life of the society, but there are others big and important enough that they need to be noted in our overview.

In 1962 (1982) besides 1 federal and 50 state governments, there were around 91,000 (82,290) other public bodies with tax power. These include counties, townships, school districts, and other territorial bodies.

Among the not-for-profit organizations were around 80 (90) religious bodies, each with memberships over 50,000; 30–50 major unions and a total union membership in 1964 (1984) of approximately 17,200,000 (17,300,000); some 2,000 four-year and 1,300 two-year colleges and around 7,000 hospitals (in 1984).

In 1962 (1982), the top 500 industrial firms employed 9,652,000 (14,400,000) people (*Fortune*, May 2, 1983, p. 277). The federal government employed 2,505,000 (2,871,000), while California and New York State and local governments employed 892,000 (1,147,000)[14] and 935,000 (998,000), respectively.

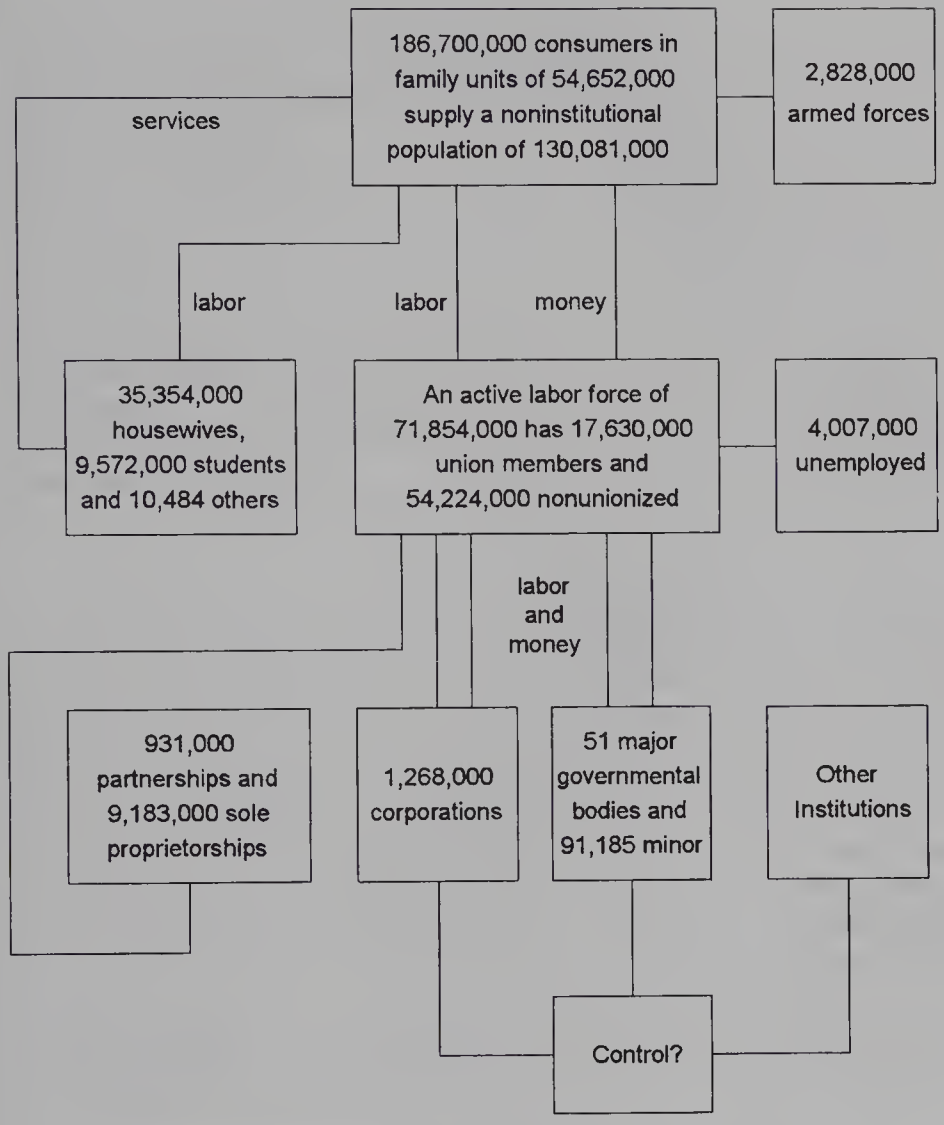


Figure 5.7
Participants in U.S. economy, 1962.

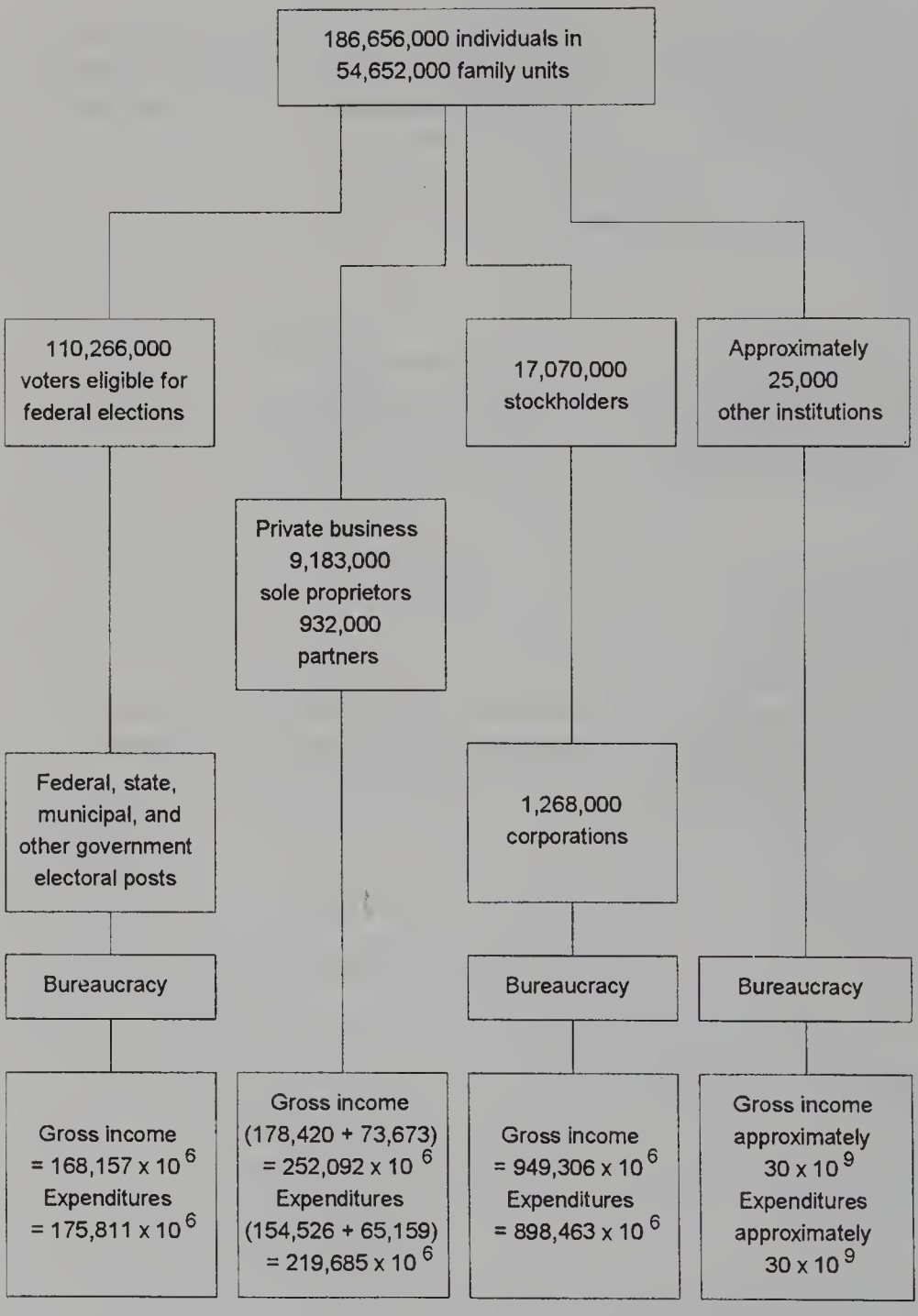


Figure 5.8
Ownership, bureaucracy, and control.

Table 5.1
Workforce structure, 1963

Craftsmen and skilled production workers	8,700,000
Clerical and office workers	10,100,000
Operatives and kindred workers	12,040,000
Farm laborers and foremen	2,270,000
Other laborers	3,560,000
Sales workers	4,345,000
Managers, officials, and proprietors	7,410,000
Professional, technical, and kindred workers	8,040,000
Lawyers	212,000
Accountants	417,000
Bookkeepers	913,000

When we add up the figures on the active labor force in 1962, we face a discrepancy because there must be at the very least 11,120,000 people involved in partnerships and sole proprietorships, yet only 7,332,000 members of the labor force appear to be available. This is due to double counting of people holding combinations of more than one job or business. Figure 5.7 shows the individual as owner, member of the wage-earning labor force, housewife, student, or other.

Ownership and control are related but different in any complex economy. Especially in a society such as the United States, voting mechanisms are used to determine who are the direct controllers or trustees of many assets. Various governmental bodies control around 20 percent, and corporations roughly 35 percent, of all assets. Figure 5.8 shows the ownership and gives a rudimentary indication of how voting and control might enter.

For government bodies, the open buying of votes is illegal. Control of a corporation can be legally acquired by buying voting stock. The control of not-for-profit organizations is in the hands of those who appoint the directors.

There are many other ways in which other aspects of U.S. society could be presented. As the purpose here is merely to "sweeten our intuitions" rather than to try to compete with the *Statistical Abstract* only two more are presented. Table 5.1 indicates the structure of the workforce and the size of various professions in 1963.

By 1980, the number of lawyers passed 500,000 and computer-associated employment was over 1,200,000 (*Statistical Abstract of the Untied States*, 1986, p. 400). By 1991, the number of lawyers and judges employed had reached 772,000; accountants and auditors, 1,446,000; bookkeepers, 1,912,000; and computer professionals and system analysts, 1,800,000 (*Statistical Abstract of the United States*, 1992, p. 392).

Finally, we observe that in 1963 (1984) the financial institutions in the United States consisted of the Treasury; the Federal Reserve and banking systems, with around 14,000 (14,500) banks; approximately 1,400 (2,130) life insurance companies; 6,000 (1,390) savings and loan associations; 20,000 (15,000) credit unions; other government financial agencies such as the Federal Deposit Insurance Agency; finance companies; pawnbrokers; and other private credit agencies. Currently the number of banks is declining sharply while pension and mutual funds have grown considerably.

5.5 Money and Financial Institutions

Money is a network phenomenon. It is extremely difficult and of dubious value to define money in vacuo without specifying the rules of the monetary system and the market and financial institutions. This is not merely an appeal for more economic history and institutionalism; it is at the very basis of the mathematical modeling of economies with a financial structure. The invention of new financial instruments and the creation of new market structures is to the technology of exchange what the invention of new machines and the development of scientific instruments is to production technology.

Some of the appropriate questions to ask of an economy with markets and financial institutions are:

1. How many individuals and what conditions are needed for casual barter to be replaced by barter concentrated at fixed marketplaces?
2. What are the conditions for a commodity money to appear? What is the smallest group of individuals who have used a commodity money?
3. When do specialized traders and brokers appear?
4. When will trading in warehouse receipts start to take place? When will banks evolve?

5. What are the necessary conditions for the appearance of a fiat money? What has been the minimum size of a society with a successful independent fiat money?

6. What are the minimal conditions for the evolution of a credit system?

7. When will formal insurance contracts first appear?

8. When will bankruptcy laws be formulated?

9. What are the conditions for the emergence of a money market and rate of interest?

10. When will forward contracts for trading in commodities come into being?

11. What are the conditions under which stocks, bonds, and similar forms of ownership and debt will appear?

12. What is the minimum size of a society that will support special information services such as credit agencies and expert evaluators?

At the most abstract and simultaneously institutional level, "money" is defined by all of the rules of the game pertaining to operations of the physical entity or bookkeeping ciphers called money in various societies.

Perhaps it would lessen confusion and improve research and communication if we used a differentiated set of terms to stand for "monies" manifesting only a few of the many properties money has in a modern economy beyond that of being exchangeable for all other goods, services, or instruments.

5.5.1 Institutions, trade, and the long and short run

Given that an individual enters an economy with functioning markets and financial institutions, does the individual have major economic choices over how to trade? This is an empirical question to which the answer appears to be no, in general. The markets and financial infrastructure are a vast public good or externality that, except in a period of social or political breakdown (such as revolution or defeat in war), imposes a pattern of doing business on the individual. Although during a hyperinflation we may consider an individual bartering rather than buying a packet of cigarettes, it is difficult to believe that in normal times he will be able to buy cigarettes or pay for a subway ride by barter. Once the system of trade is in existence, to a great extent it takes on the aspects of a multiperson noncooperative game for all concerned. How it came into being

may be virtually irrelevant to determining how economic behavior is manifested in the system.

How the system will change and evolve is a matter of comparative political economy and social change. It poses problems in understanding more the context of economic behavior than economic behavior itself. Money is an emergent property of the dynamics of an economic network.

Notes

1. In virtually all of his writings, Polanyi stresses *reciprocity* and *redistribution* in contrast with market exchange. The first involves friendship, kinship, status, and hierarchy; the second political or religious affiliation; while the third in theory involves none of these (see especially Polanyi, 1944).

2. Mathematical tabulations of partitions are available in the U.S. Department of Commerce, *Handbook of Mathematical Functions* (1964); the brief table below indicates how the numbers increase:

n	$g(n)$
4	5
5	7
10	42
20	627
50	204,226
100	190,569,292

3. The equilibrium point would remain if there were two or more traders of each type. But with only one trader and the goods divisible, then there is no competition on the same side of any trading post, and traders will be tempted to cut back their offers if they attach any value to the item offered. This technical point is taken up in detail in chapter 6.

4. "I find that Cournot has long since defined the economic use of the word 'market,' with admirable brevity and precision, but exactly to the same effect as the text above. He incidentally says in a footnote (*Recherches sur les principes mathématiques de la théorie des richesses*, Paris, 1838, p. 55), 'On sait que les économistes entendent par marché, n'est pas un lieu déterminé où se consomment les achats et les ventes, mais tout un territoire dont les parties sont unies par des rapports de libre commerce, en sorte que les prix s'y nivellent avec facilité et promptitude' (translation by Bacon, Macmillan, 1897, p. 51, note)."

5. Note that no comment has been made about individual *utilities* or *preferences*. In describing market mechanisms, we concentrate on the mechanics of trade, not the motivation for trade. When we describe strategic market games, we must introduce preference structures.

6. Elements of a set are associative if $(a+b)+c=a+(b+c)$ and commutative if $a+b=b+a$. In particular, if a trader submits several bids using agents, we might want this to be the same as if he had bid as a single individual. This is not always the case.

7. The feasibility condition is often violated. Individuals may bid in two or three markets knowing that if they are successful everywhere they cannot honor their bids; lack of success

in some markets gives them a high probability that they can meet their commitments. Contract law may enforce a penalty for failure. There is a small verse to this effect: "He who sells what isn't his'n must pay up or go to prison."

8. See Lerner (1947) for a discussion of fiat and government.

9. It is my opinion that individuals do not think in such terms and that the assumption that they do not is a particularly valuable first approximation for most investigations.

10. As it stands, the statement is not quite accurate. It needs the qualification that the presence of "enough money" is assumed. The full meaning of enough money is discussed in chapter 6 and later chapters. Mathematically, it amounts to having the NEs be interior.

11. In contrast, it has been suggested that the word "dollar" derives from "Thaler," which in turn derives from Joachimsthal in Bohemia, where in the fifteenth century the counts of Schlick coined silver coins known as Joachimsthalers.

12. In a lighter vein, *Larousse Gastronomique* defines gold as follows: "Precious non-rusting metal sometimes used in the Middle Ages in thin leaves for wrapping some pastes and certain roast birds. Gold is still used for this purpose in the Far East" (English edition, 1961, p. 467).

13. The closest to a commodity money in the Americas was cacao beans. Vaillant (1950, p. 132) notes:

> Barter was the only means of exchange, and value was established by desirability and rarity. Money, an exchange medium of fixed value, did not exist. However, something had to be found which would balance an inequality of exchange by being not too valuable to use in adjusting small transactions and at the same time universally wanted. The cacao bean answered this requirement and was easily portable as well. The Aztecs were extremely fond of chocolate (the word itself is of Aztec derivation), so that beans were gladly converted into the national luxury drink. Quills of gold dust were sometimes used as an exchange medium, as were crescent-shaped knives of thin-beaten copper. These last had not the common acceptance or the utility of cacao beans, although they represented easily portable value.

14. Full-time equivalents for 1970 and 1984 (*Statistical Abstract of the United States*, 1971, p. 421; 1986, p. 297).

6 Trade Using a Commodity Money

6.1 Strategic Market Games

In chapter 5, we discussed market mechanisms without reference to individual preferences or to any particular concept of solution for trade within a market structure. In this chapter, we construct a class of games in strategic form and consider the noncooperative equilibrium solution as applied to the class of games defined.

A strategic market game for traders $i \in N$, with endowments $a^i \in \Omega^m$ and utility functions $u^i : \Omega^m \to \mathbb{R}$, is defined by specifying the market structure M and the strategy sets S^i for each $i \in N$[1]. It can be shown that noncooperative equilibria exist for all of the class of games described above (Dubey and Shubik, 1978), where the participants in the markets use only simple messages, under the following three conditions:

1. Each u^i is concave and nondecreasing in each variable.
2. $\sum_{i \in N} a^i > 0$.
3. For any $(j,k) \in M$, there exist at least two traders who have positive endowments of j and desire k, as well as two traders who have positive endowments of k and desire j. (A trader i is said to "desire" commodity j if u^i is a strictly increasing function of the jth variable.)

6.1.1 Games with aggregated strategies

Among the desirable properties of a market, anonymity and aggregation alone are sufficient to provide considerable structure to games in strategic form. Aggregation provides an important link between micro- and macroeconomic models of economic behavior. P_i, the payoff function for an individual i, is generally a function of the n strategies of player i, or $P_i(s_1, s_2, \ldots, s_n)$. If a strategy amounts to naming a quantity of one or more goods, then each individual's payoff depends on only two strategic variables, i's own actions and the sum of the market actions of the others. The simple classical Cournot oligopoly game illustrates this.

Suppose there are n firms, each with total costs of production given by $C_i(q_i)$, where $C_i' > 0$, $C_i'' > 0$ and $C_i(0) = 0$. Suppose that demand in the market for the identical good sold by all firms is given by $p = \varnothing(q)$, where $q = \sum_{i=1}^{n} q_i$, $\varnothing' < 0$, $\varnothing'' > 0$, then the payoff to any firm i is given by

$$P_i = q_i \varnothing(q) - C_i(q_i) \quad \text{for } i = 1, \ldots, n.$$



Figure 6.1
Simultaneous moves in the market.

Every payoff is a function of two variables, the strategy of the firm i and the market as a whole. The equilibrium conditions when viewed from any firm i depend on the profit function and the aggregate behavior of the others. At equilibrium, the only information required is as noted above, whereas away from equilibrium, a firm might easily be guided by its knowledge of the costs of others.

Aggregating information cuts down enormously on the number of strategies for each player, in particular ruling out selective threats in multistage games.[2] Aggregation brings with it anonymity, and "the market" may take on a life of its own.

6.1.2 Brokers and retailers

In most of the models considered here, trade is carried out through single simultaneous moves by all parties (figure 6.1). Although the symmetry of information suggested by such models may reasonably approximate trade in a well-populated double-auction market, it does not appear to mirror mass retail markets. For many items, such as goods in department stores and supermarkets, a price is displayed; the customer faces a search problem of the type noted in section 6.2.5.

The purchase of a house, automobile, or other major consumer durable tends to involve nonsymmetric information and search, on the one hand, and face-to-face communication and bargaining, on the other. Offers and counteroffers are made; the negotiations frequently break off while the buyer looks elsewhere and the seller deals with other buyers. (A market model with nonsymmetric information about the moves of players—where the buyer can find out the prices named by the sellers before having to act—is shown in figure 6.8).

In the purchase and sale of houses and stock, the brokers usually act as middlemen; the risk both in buying and selling lies with the owners. Although brokers often work for a fixed fee or a percentage of the price of the item sold, special incentives such as extra commissions may be negotiated when active marketing or selling is called for.

Trade with brokers, face-to-face bargaining, and most retailing have information conditions far more complex than those in the three basic models discussed in section 6.4 and chapter 7. Fortunately, the simplification of simultaneous moves is an adequate first approximation for all trade. As proved rigorously elsewhere (Dubey and Shubik, 1981a), for any game in extensive form not involving exogenous uncertainty, the equilibrium points of the original game remain equilibrium points in all associated games that have the same moves but more information.

When we introduce more information, such as having customers informed of the retail prices to be charged before they buy, the set of equilibrium points may change (Roberts and Sonneschein, 1977; Dubey and Shubik, 1986). Indeed, as the Edgeworth-Bertrand analysis of price competition duopoly demonstrates, strategic sophistication by all firms may be destabilizing rather than stabilizing (Edgeworth, 1925; Shubik, 1959b, chapter 5). Because it may be regarded as a limiting noncooperative equilibrium in a game with traders so small they are unable to make strategic use of extra information about the moves of others, the Walrasian equilibrium, considered as arising from a game of strategy, is robust when information conditions change.

6.2 On the Number of Types of Markets with Trade in Money

Here we make the restrictions on market mechanisms suggested in chapter 5 even more stringent. The simple messages sent by traders, buyers, or sellers are limited to quantities, prices, or both. The number of different market structures to be expected under certain reasonable restrictions on the nature of trade is limited. In particular, one-move markets are without bargaining, haggling, or recontracting. A move by a trader will be construed as a bid or offer that is accepted or rejected.

6.2.1 Bids and offers

Consider an economy with m commodities, a money (which may or may not be a commodity), and a market having a set of buyers and sellers and a mechanism that matches their bids and offers. We assume that all buyers and sellers bid simultaneously and independently and that no complex contingent bids or offers are considered. We further assume that markets are for single commodities and that the method and mechanism of exchange in each market is the same.

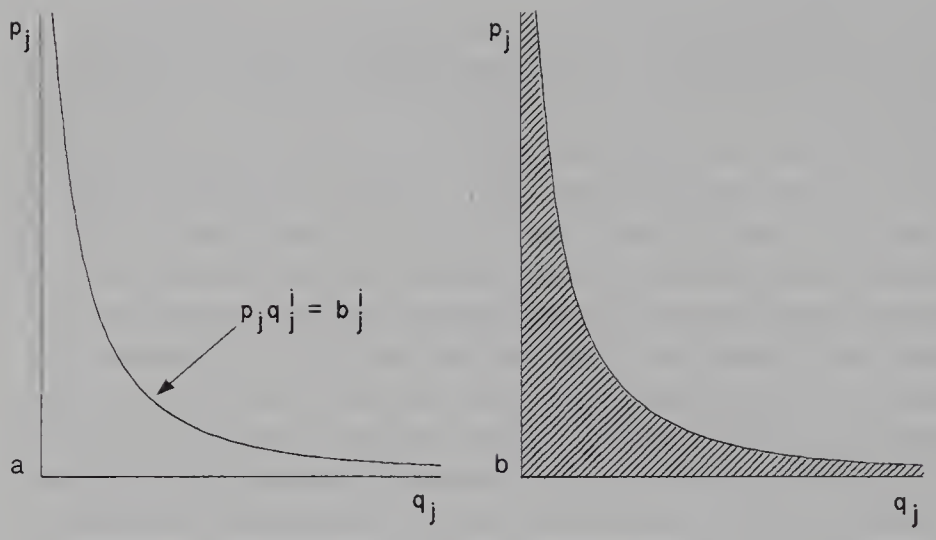


Figure 6.2
Quantity of money bid.

Buyers must use a money to bid. They may specify a price or accept the market price, and they may specify a quantity or buy what they can. Limiting ourselves to money payments and to specifications concerning price and quantity to be purchased, we have four types of bids.

Bids

Money bid A trader i bids an amount of money b_j^i for the jth commodity. The trader has no reserve price and takes what the market offers. This provides an extremely simple quantity bid that enables us to construct a mechanism similar to that of Cournot. The market clearing mechanism will give the trader an amount q_j^i, which lies on the hyperbola $p_j q_j^i = b_j^i$, as shown in figure 6.2a. If the trader bids b_j^i but agrees to trade for less than all of the money bid,[3] then the feasible set of trades includes the shaded area in figure 6.2b. (Imposing upper or lower bounds on prices or quantities acceptable to the bidder may truncate the range of the hyperbola.)

Price-quantity bid Suppose that trader i offers a price p_j^i for an amount q_j^i. It is reasonable to expect that i is willing to buy q_j^i for a price less than p_j^i, hence the outcomes acceptable to i are indicated by the line MB in figure 6.3a. (There is an implicit limit in this bid inasmuch as $q_j^i p_j^i$ must be less than or equal to the credit line and cash of the individual.)

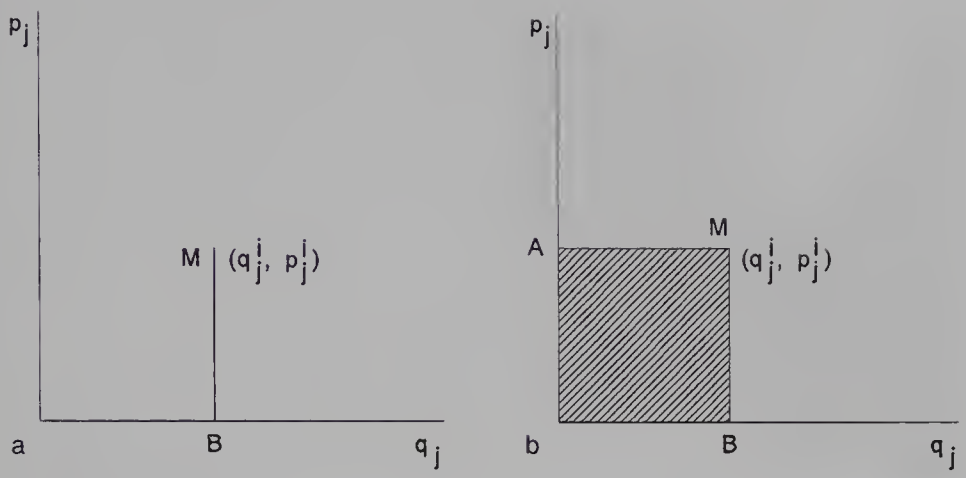


Figure 6.3
Price-quantity bid.

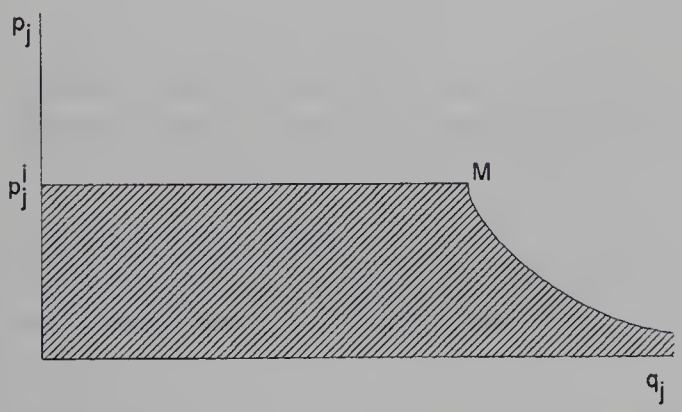


Figure 6.4
Price bid.

Trader i may accept an amount less than q_j^i, if unable to fill i's total order. If i is willing to take a partial fulfillment, then any outcome in the shaded rectangle indicated in figure 6.3b is feasible. If i's bid must be exposed to the seller (as, for example, in an advertisement for purchase), the bid may be at a single price p_j^i for a specific quantity q_j^i (point M in figure 6.3b). A single price bid may be for any amount up to some limit (line AM in figure 6.3b).

Price bid Trader i states a price p_j^i and stands willing to buy any amount i can obtain at p_j^i or less. (This has an implicit bound in it because i's expenditures cannot exceed i's cash and credit.) Any point in figure 6.4 is

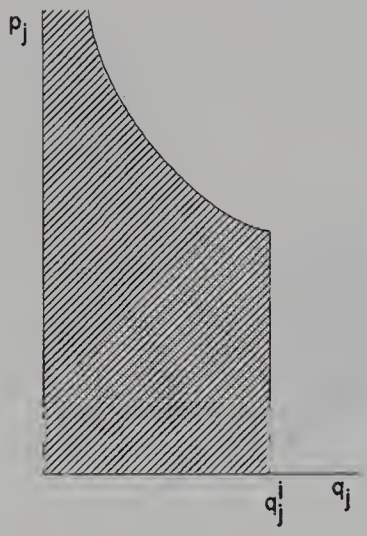


Figure 6.5
Quantity bid.

feasible. The point M is where the credit constraint becomes relevant. (We can modify this bid by introducing an upper or lower bound, or both, on the quantity to be accepted.)

Quantity bid If trader i bids for a quantity q_j^i at any price and credit constraints are introduced, this is the same as the money bid if $p_j q_j^i >$ constraint. If i bids for q_j^i or less, the feasible set of outcomes is that shown in figure 6.5.

Offers The seller or offerer does not have quite the same possibilities as a buyer. It is theoretically possible that a seller i could announce a willingness to sell any amount q_j^i at any price p_j^i so long as $q_j^i p_j^i \geq M^i$, where M^i is i's monetary requirement. However, it appears more reasonable to require the seller to specify an upper bound on the quantity offered of the commodity being sold[4] and to name a price. The buyer more naturally has bids involving not only the price and quantity of the commodity but also the quantity of money the buyer is willing to spend.

Price-quantity offer A seller i may offer q_j^i at p_j^i or more (figure 6.6a). Alternatively, i may offer q_j^i or less at p_j^i or more (figure 6.6b).

Price offer Seller i offers any amount at p_j^i or above, subject to the amount of the commodity i has available. If the seller's offer is disclosed ahead of time to the buyer, then we may expect that any trade will take

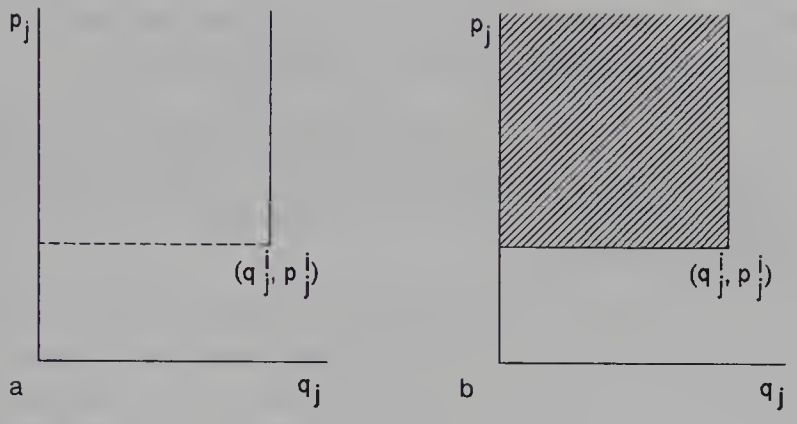


Figure 6.6
Price-quantity offer.

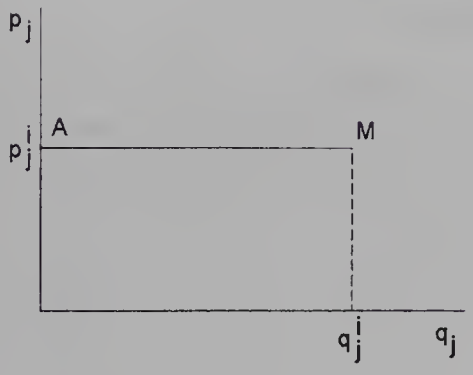


Figure 6.7
Price offer.

place at price p_j^i (figure 6.7, with outcomes restricted to line AM). Most retail sales involve offers of this variety.[5]

Quantity offer Seller i may offer q_j^i at any price (see figure 6.6a for $p_j^i = 0$; see figure 6.6b for the sale of q_j^i or less).

In a single market with simultaneous moves the types of bids and offers noted above can reasonably be considered as inputs to a market clearing mechanism. These are considered further in section 6.2.3.

6.2.2 Credit constraints and transaction costs

In addition to the types of bids and offers discussed, there are a host of minor variations that depend on market technology and on credit

conditions, which appear as ad hoc requirements in any attempt to completely specify "the rules of the game" for any market mechanism.

Credit constraints Trader i may wish to bid beyond i's immediate cash resources. The trader may have to arrange bank credit and bid by certified check before i's bid is considered valid; in some instances, the seller might be willing to extend direct credit.

Transaction costs The transportation, storage, display, packing, and unpacking of many commodities may be extremely expensive. Ripe tomatoes sent to market but not sold are essentially confiscated, in the sense that the costs of retrieval may easily exceed worth. Even a money bid that is not accepted involves costs. The money has been tied up during the market and there may be broker or specialist fees to pay.

6.2.3 Market mechanisms and clearing devices

As defined in chapter 5, a market is a correspondence, a mechanism, or a transformation that takes in bids and offers and transforms them into purchases and sales at a market price. While different institutional conditions may be reflected in the sequencing of bids, information, and communication, here we restrict our enquiry to the three simplest single-move games: (1) buyers and sellers bid and offer simultaneously, and a market mechanism resolves trade; (2) sellers move simultaneously, but their offers are disclosed to the buyers before they move; and (3) buyers bid simultaneously, and their bids are disclosed to the sellers before they move.

Let B^i be the set of bids available to buyer i, and let S^j be the set of offers available to seller j. A market mechanism is a mapping T that transforms the vector of bids and offers $(b^1, b^2, \ldots, b^{n_1}, s^1, s^2, \ldots, s^{n_2})$, where n_1 is the number of bidders and n_2 the number of sellers, into a vector of trades and prices.

More precisely, let us consider n individuals where i has an endowment a_j^i of good j (where $j = 1, \ldots, m$) and an endowment of money of M^i. Let there be m individual markets, one for each commodity j where that commodity is exchanged for money.

However, a reasonable limitation, is to assume that individuals are not simultaneously buyers and sellers of the same commodity at the same time. This rules out the oligopolistic stock market phenomenon of "wash sales," where an individual buys and sells in order to transmit a false sig-

nal of market activity. Although the assumption of no wash sales may be convenient for some analyses, in general, it would be preferable to deduce the conditions under which an individual is not simultaneously a buyer and seller of the same commodity.

If there were no overall credit constraints, in each market j an individual i would have a set of moves, $B_j^i \times S_j^i$, where a move could be described by a pair (b_j^i, s_j^i). If we wished to assume that an individual is at most a buyer or seller, we could add the condition $b_j^i s_j^i = 0$.

In each market j there is a mechanism such that

$$\begin{aligned}(b_j^1, s_j^1, \ldots, b_j^n, s_j^n; a_j^1, \ldots, a_j^n; M^1, \ldots, M^n) \\ \xrightarrow{T_j} (p_j^1, p_j^2, \ldots, a_j^1 + x_j^1 - z_j^1, \ldots, a_j^n + x_j^n - z_j^n; \\ M^1 - p_j^1 x_j^1 + p_j^1 z_j^1, \ldots, M^n - p_j^n x_j^n + p_j^n z_j^n),\end{aligned} \tag{6.1}$$

which transforms all bids, offers, and initial holdings in market j into prices, amounts bought (x_j^i), and amounts sold (z_j^i) of all commodities. This also enables us to calculate net changes in money holdings. A reasonable restriction on the mechanism is that all trade is required to take place at the same price. This requires that $p_j^i = p_j$ for $i = 1, \ldots, n$.

In general, we cannot assume that bids in one market are independent of bids in another. There is at least a credit interlinkage, which may or may not be independent of the nature of the bids (for example, different "margin" requirements may make i's credit line a function of i's bids). Without credit constraints the set of all moves in all markets for i is

$$\prod_{j=1}^{m} (B_j^i \times S_j^i). \tag{6.2}$$

With a credit constraint, the size of the set of feasible moves must be appropriately restricted.

6.2.4 Moves, strategies, and contingencies

A move is not generally the same as a *strategy*—a plan an individual i uses to select i's moves as a function of the information available when i is called on to move. If, however, we limit ourselves to markets with simultaneous moves by the buyers and sellers and with symmetric knowledge by all about the states of nature, then even though the market clearing

mechanism may be complicated and the bids complex, they cannot reflect knowledge of the moves of others. There is no recontracting. Moves and strategies are identical.

If one set of individuals moves first and these moves are announced before the others move, then the strategies of the latter group will call for moves to be selected *contingent* on the behavior of the former. Strategies are more complicated than moves.

6.2.5 Information and search

There have been numerous papers on search models of trade where individuals check on different employment opportunities sequentially, or go from gas station to gas station, car dealer to car dealer, or realtor to realtor (see Spence, 1974, for example). When these models are viewed from the perspective of markets with strategic traders, they clearly lack symmetric information conditions. The search process involves complicated plans depending on what the potential buyer learns after examining offers by some sellers.

For example, consider buyer i, wanting to buy a unit of a single commodity, facing two sellers. We may imagine that all sellers announce a price simultaneously. Buyer i does not know these prices. At a cost of c per inspection, i can check on a single price; i then has the choice to buy or to sample again. Buyer i places a value of K on the item. Suppose the payoff to a seller j is p_j if j sells a unit at price p_j; otherwise, it is 0. The payoff to buyer i is as follows:

$$P_B = \begin{cases} 0, -c, -2c & \text{if no purchase after 0, 1, or 2 searches} \\ K - p_j - c \text{ or } K - p_j - 2c & \text{if purchase from } j \text{ after 1 or 2 searches.} \end{cases}$$

If both sellers have been inspected, we may wish to assume either that there is or that there is not an extra cost to return from the second to buy at the first. In this example, assume no extra cost. The game tree in figure 6.8 is drawn as if the price bids were discrete.[6]

The sellers simultaneously each set a price. Buyer i can opt out with a score of 0 or pay c and decide to find out the price of seller 1 or of seller 2 (branches 1, 2, 3 in first move). At i's second move, i can opt out with a score of $-c$, buy with a score of $K - p_1 - c$ or $K - p_2 - c$, or obtain more information (branches 1, 3, and 2 in second move). At i's third move, i has complete information and must abstain from buying, which gives i a

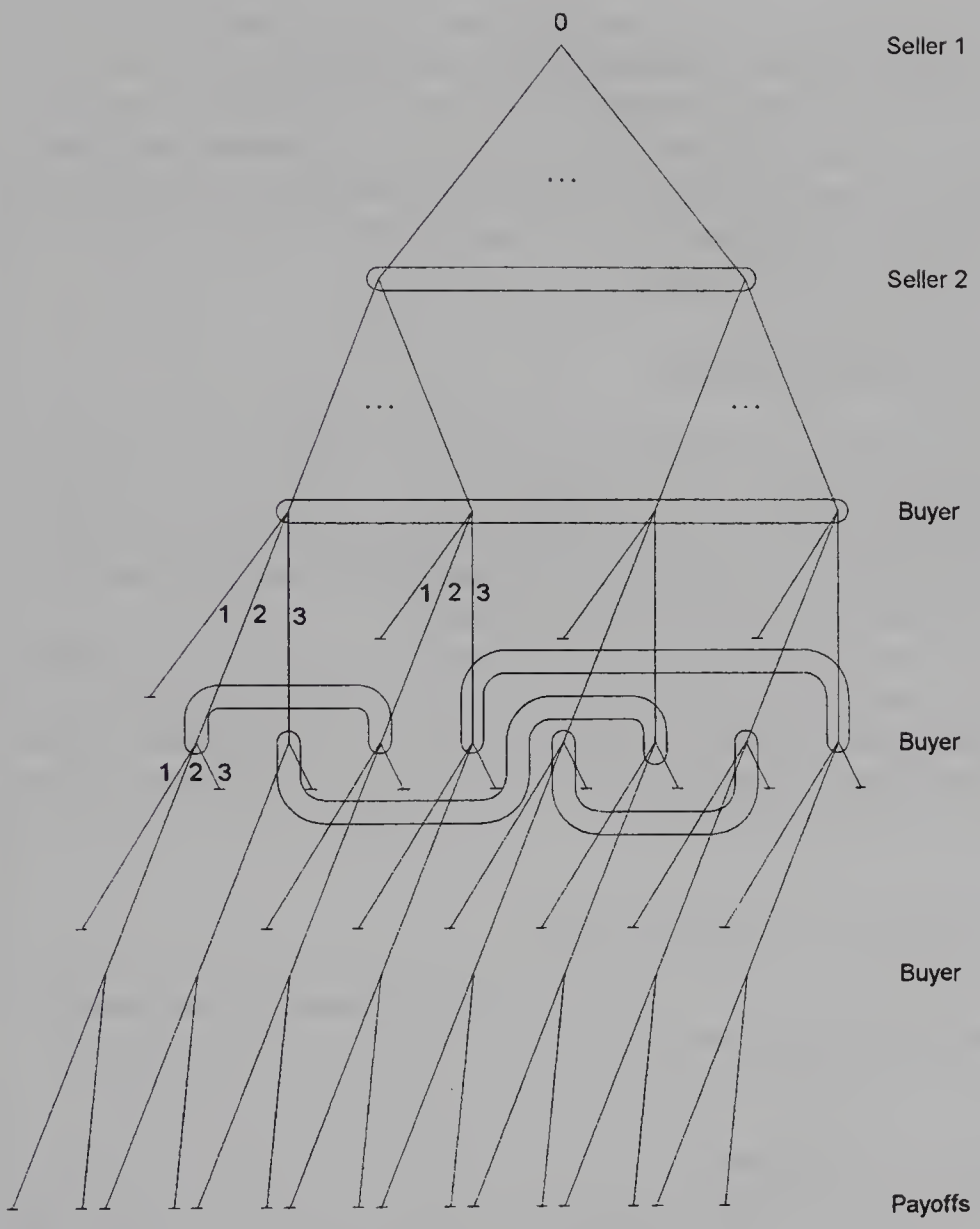


Figure 6.8
Purchase with search.

score of $-2c$, or i can buy and obtain a score of $K - \min(p_1, p_2) - 2c$ (if $p_1 = p_2$, the buyer randomizes between the sellers).

This type of game, in which the buyer has a potentially complex sampling strategy, illustrates three basic features of markets with search: nonsymmetry of information conditions, the possibility of the purchase of additional information, and the recognition by all parties of the nonsymmetry in information.

6.3 On Different Markets

As we have seen, when there is symmetry in strategic information (i.e., bidders and sellers make their bids or offers in ignorance of each others' moves), there are four types of bids and three types of offers to consider. Leaving aside minor variants, this would give us twelve possibilities in any market combining bid and offer types.

The market mechanism must then transform these bids and offers into prices and final goods and money distributions. Of the twelve combinations, many are similar. Here and in chapter 7, we consider three of them in detail because they appear to cover the salient features of the various bid and offer possibilities.

6.4 The Sell-All Market Structure

In a free-enterprise system we obviously need highly definite and stable rules of the game, especially as to money.
—Henry C. Simons (1936)

6.4.1 The basic model

To describe a well-defined game of exchange in which a specific commodity is used as a means of payment, we must spell out how the prices are formed. The classic general equilibrium model is sufficient to study the existence of prices (often not unique) *at equilibrium*. For a proper game model, however, we need rules that determine the prices for positions of disequilibrium as well. With every player free to make an independent decision, the model must yield a well-defined output for every set of inputs.

There are several types of price-forming mechanisms to be considered, each placing different restrictions on the strategic possibilities. In particu-

lar, the traders might control only the quantities offered or demanded, or they might name reservation prices, price ranges, or even complete demand curves for their individual transactions. Multistage bargaining might be introduced, or a centralized procedure that converts a set of unilateral price declarations into a unique, marketwide price for each good.[7] Here, because we are interested in general, anonymous exchange, with marketwide prices but with a minimum of ad hoc institutional detail, we adopt what is essentially a generalization of Cournot's original approach. Although the strategic variables are *quantities*, not prices, they include quantities of the special good that serves as "cash" or "trading money." Indeed, in this simple version, *only* that good is subject to strategic choice.

Assume that there are n traders trading in $m+1$ goods, where the $(m+1)$st good has a special operational role, in addition to its possible utility in consumption. We attribute to each trader an initial bundle of goods[8]

$$a^i = (a_1^i, \ldots, a_m^i, a_{m+1}^i),$$

where $a_j^i \geq 0$ for all $j = 1, \ldots, m+1$ and $a_j^i > 0$ for at least two traders and a concave utility function

$$u^i = (x_1^i, \ldots, x_m^i, x_{m+1}^i).$$

We emphasize that u^i need not actually depend on x_{m+1}^i; the possibility of a fiat money is not excluded.

The general procedure will be for the traders to put up quantities of the first m goods to be sold, and simultaneously to put up quantities of the $(m+1)$st good to buy them, all at prices determined by the marketwide supply and demand for each good. For expository purposes, our prototype version requires the traders to offer for sale *all* their holdings of the first m goods, though they need not spend all of their $(m+1)$st good. A trader i may (and usually will) buy some of i's own goods back, but they must go through the market. In other words, in this version of the model, trader i does not own i's initial bundle outright; i merely owns a *claim on the proceeds* when the bundle is sold.[9]

Let us imagine m separate "trading posts," one for each of the first m commodities, where the total supplies $(\bar{a}_1, \ldots, \bar{a}_m)$ have been deposited for sale "on consignment" (see figure 6.9). Trader i makes bids by

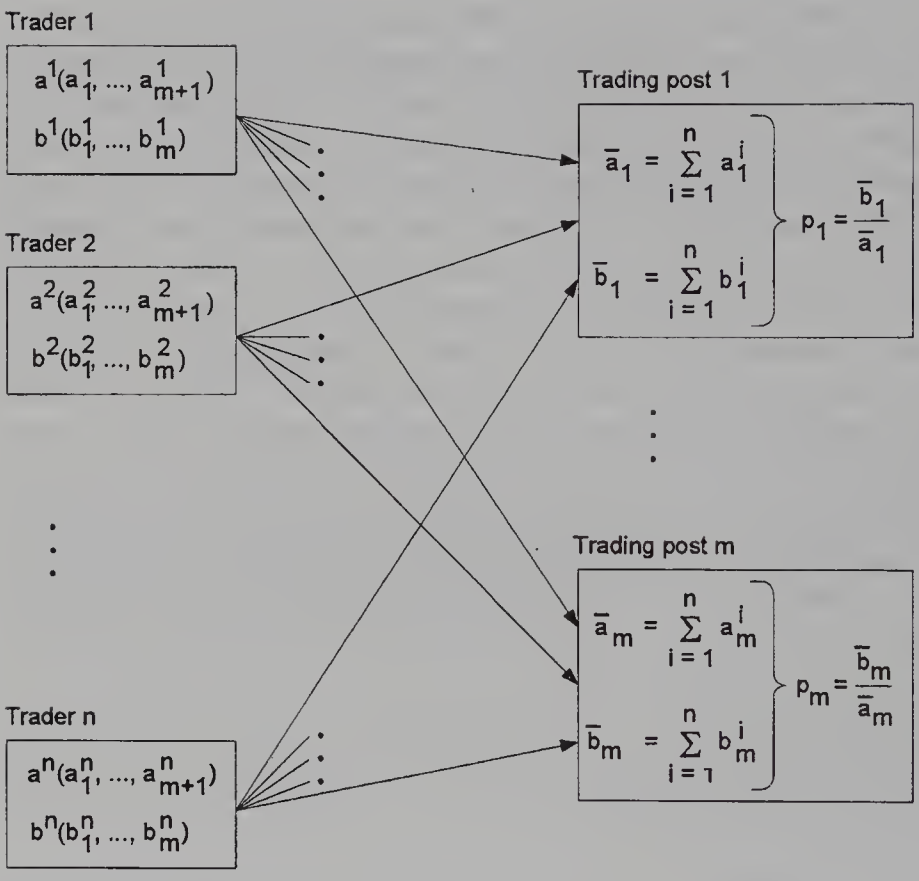


Figure 6.9
Trade at the trading posts.

allocating amounts b^i_j of i's $(m+1)$st commodity among the m trading posts, $j = 1, \ldots, m$. We denote i's *strategy*, in the game-theoretic sense, by the vector $b^i = (b^i_1, \ldots, b^i_m)$. There are a number of possible rules governing the permitted range of bids. In the simplest case, with no credit of any kind, the limits on b^i-are given by

$$\sum_{j=1}^{m} b^i_j \leq a^i_{m+1}, \quad \text{and} \quad b^i_j \geq 0, \quad j = 1, \ldots, m.$$

The interpretation of this spending limit is that the traders are required to pay *cash in advance*. More generally, we might allow them to defer payment, either in anticipation of receipts or under some other credit arrangement that would have to be made explicit in the model.[10]

Price formation The prices now emerge in a natural way, as a result of the simultaneous bids of all buyers; we define

$$p_j = \bar{b}_j/\bar{a}_j, \quad j = 1, \ldots, m.$$

Thus bids precede prices. Traders allocate their budgets *fiscally*, committing quantities of their means of payment to the purchase of each good without definite knowledge of what the price per unit will be. At an equilibrium, this will not matter because prices will be what the traders expect them to be. In a multiperiod context, moreover, the traders will know the previous prices and may expect that fluctuations in individual behavior in a mass market will not change prices by much. Nevertheless, any deviation from expectations will result in changing the quantity of goods received, rather than the quantities of cash spent. In practice, if one allocates a portion of one's budget for purchase of a certain good in a mass market, this will be different—but not too different—from a decision to buy a specific amount at an unspecified price. It is a matter of letting one's stomach rather than one's purse absorb the surprises.

The prices in our model are so determined that they will exactly balance the books at each trading post. The amount of the jth good that the ith trader receives in return for i's bid b_j^i is[11]

$$x_j^i = \begin{cases} b_j^i/p_j & \text{if } p_j > 0,\ j = 1, \ldots, m, \\ 0 & \text{if } p_j = 0,\ j = 1, \ldots, m, \end{cases}$$

and i's final amount of the $(m+1)$st good, taking into account of i's sales as well as purchases, is

$$x_{m+1}^i = a_{m+1}^i - \sum_{j=1}^{m} b_j^i + \sum_{j=1}^{m} a_j^i p_j.$$

In the game-theoretic sense, i's *payoff* must be expressed as a function of all the traders' strategies; accordingly, we write

$$\Pi^i(b^1, \ldots, b^i, \ldots, b^n) = u^i(x_1^i, \ldots, x_{m+1}^i),$$

where the x's depend on the b's according to the three preceding display equations. That Π^i is concave as a function of b^i, for each i, is important for the existence proof.

Because of the mechanism for price formation and the anonymous allocation of sales, all traders pay the same price for the same good, although

the operation of the market system gives rise to what have been called "pecuniary externalities", in the sense that the prices paid by any one trader are dependent on the monetary actions of the others (see, for example, Viner, 1931, or Shubik, 1971). In the classic barter market, this is not so; trading pairs or groups can form and exchange goods unaffected by the actions of others.

The noncooperative solution A *noncooperative* or "best-response" *equilibrium* has been defined in game theory as a set of strategy choices by the players with the property that no player, given the choices of the others as fixed, can gain by changing his own choice (Nash, 1951).[12] Specialized to the present model, the solution concept will be recognized by economists as a close relative of the classic Cournot oligopoly solution (see Manas, 1972). It will consist of an n-tuple of strategies,

$$\hat{b} = (\hat{b}^1, \hat{b}^2, \ldots, \hat{b}^n)$$

such that for each i, the function $\Pi^i(\hat{b}^1, \ldots, b^i, \ldots, \hat{b}^n)$, considered as a function of b^i alone, is maximized at $b^i = \hat{b}^i$.

Theorem 6.1 for the noncooperative equilibrium (NE) may be proved by an application of the Kakutani fixed-point theorem to the "best-response" adjustment process or transformation: $b \to b'$, where each b'_i is a best response by player i to the choices $(b^j : j \neq i)$ of the other players. The details of the proof, which are rather lengthy, are given elsewhere (Dubey and Shubik, 1978).[13]

THEOREM 6.1 For each trader, $i = 1, \ldots, n$, let u^i be continuous, concave and nondecreasing. For each good, $j = 1, \ldots, m$, let there be at least two traders with positive initial endowments of good $m + 1$ whose utility for good j is strictly increasing. Then a noncooperative equilibrium exists.

6.4.2 The Edgeworth box

The case $m = 1$, $n = 2$ lends itself to simple two-dimensional descriptive analysis, based on the familiar Edgeworth box. To avoid the confusion of too many lines and curves in one place, a sequence of diagrams is constructed, using the same labels as far as possible. As we shall see, much of this geometry will apply also to the general case, with many goods and traders, because of the way in which the operation of the market "decouples" most of the interactions among traders and among goods.

In figure 6.10, the first trader's holdings are measured up and to the right of the point 0^1, while the second trader's are measured down and to the left of O^2. The dimensions of the box are $\bar{a}_2$ high by $\bar{a}_1$ wide. The point R represents a typical *initial allocation*; thus we have $a_1^1 = |M^1R|$, $a_2^1 = |G^1R|$, and so on.[14] The points S^1 and S^2 represent typical *strategy choices*. Because we are requiring all of good 1 to be sent to market, we have $b_2^1 = |M^1S^1|$ and $b_2^2 = |M^2S^2|$, respectively. (If we did not, the strategies would be arbitrary points in the rectangles $RM^1O^1G^1$ and $RM^2O^2G^2$, respectively.)

Now consider the line joining S^1 and S^2. Its slope is $(b_2^1 + b_2^2)/(a_1^1 + a_1^2)$, which is just the price p_1, and it divides the line M^1M^2 into two segments, equal in length to x_1^1 and x_1^2, the amounts of good 1 purchased by traders 1 and 2, respectively. Similarly, it divides the line G^1G^2 into segments equal to their final holdings x_2^1, x_2^2 of the monetary commodity, or "cash." The *final allocation* is therefore represented by the point F. The vector RF represents the actual transaction that takes place; its slope is naturally equal (in absolute value) to the price p_1.

Figure 6.11 shows the effect of holding S^2 fixed while varying S^1. As S^1 moves over the interval from M^1 to O^1, the point F traces out the curve A^1RB^1. This curve is a portion of a hyperbola whose asymptotes are the horizontal and vertical axes through S^2. Similarly, the curves A^2FB^2 is traced if we hold S^1 fixed and vary S^2. The endpoints A^1, A^2 correspond to zero bids, while the endpoints B^1, B^2 reflect the upper limit on the amount a trader can bid. If we select a different bid for trader 2, say $\hat{S}^2$, we will get a different curve (B^2A^2) similar in shape to A^1RB^1. (It happens in this case that because we did not allow trader 2 enough cash to be able to buy back his original holding when trader 1 plays S^1, the curve A^2B^2 stops short of the point R.)

These traces are comparable to the price rays or "budget sets" that confront a trader in the classical Walrasian model with its fixed prices. The difference is that in the present case, because the price is not constant but reacts to variations in a trader's own decisions, we get a curve instead of a line. The curve is concave, as one would expect, so that if a trader i increases i's purchase, i drives the price up, while if i bids less the price falls. The connection between the two approaches may be illustrated in figure 6.11 by supposing the second trader's holdings and bid to be very large, pushing the points O^2 and S^2 far off the page. The first trader's hyperbolic "budget set" would then approximate a straight line through

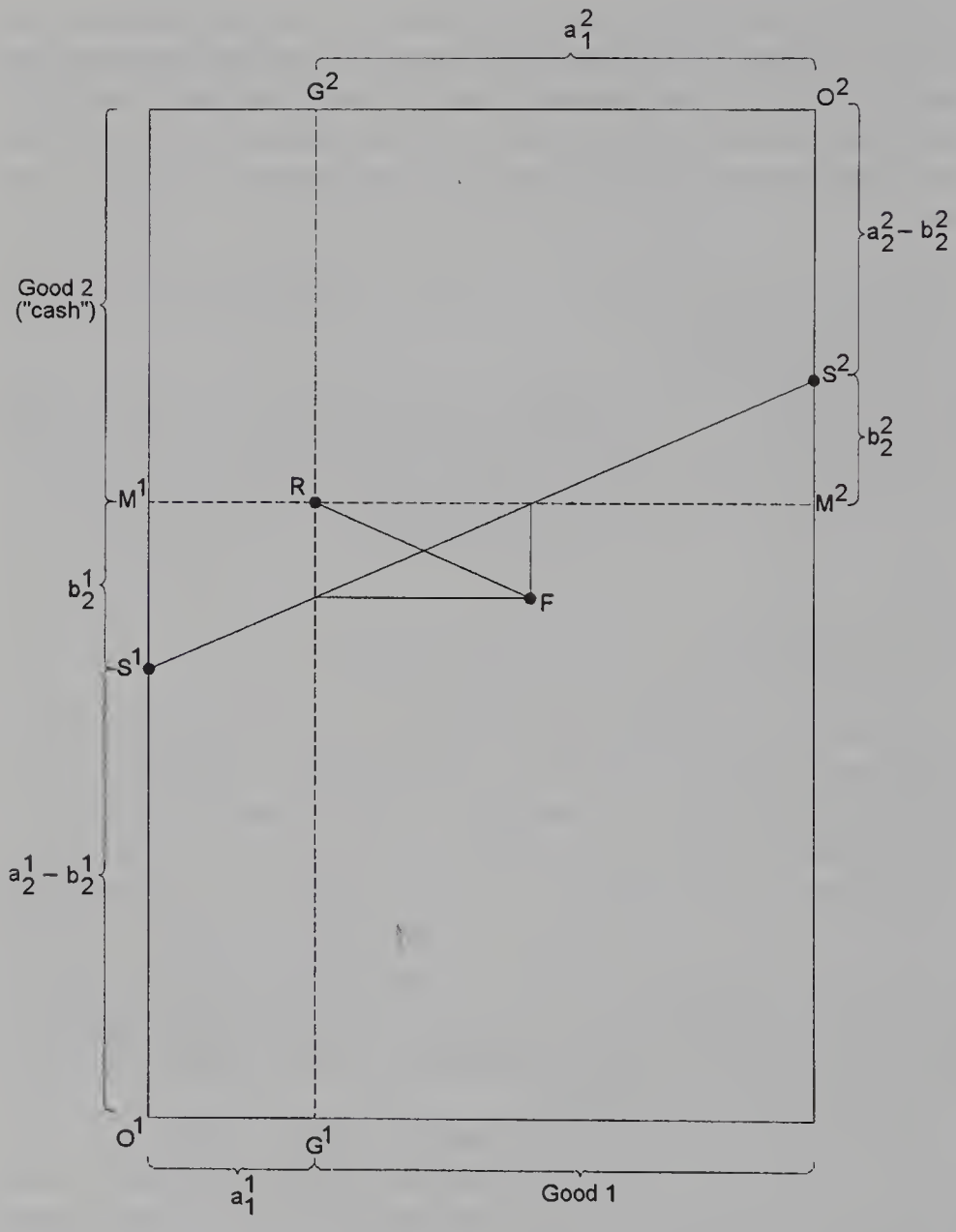


Figure 6.10
Price formation.

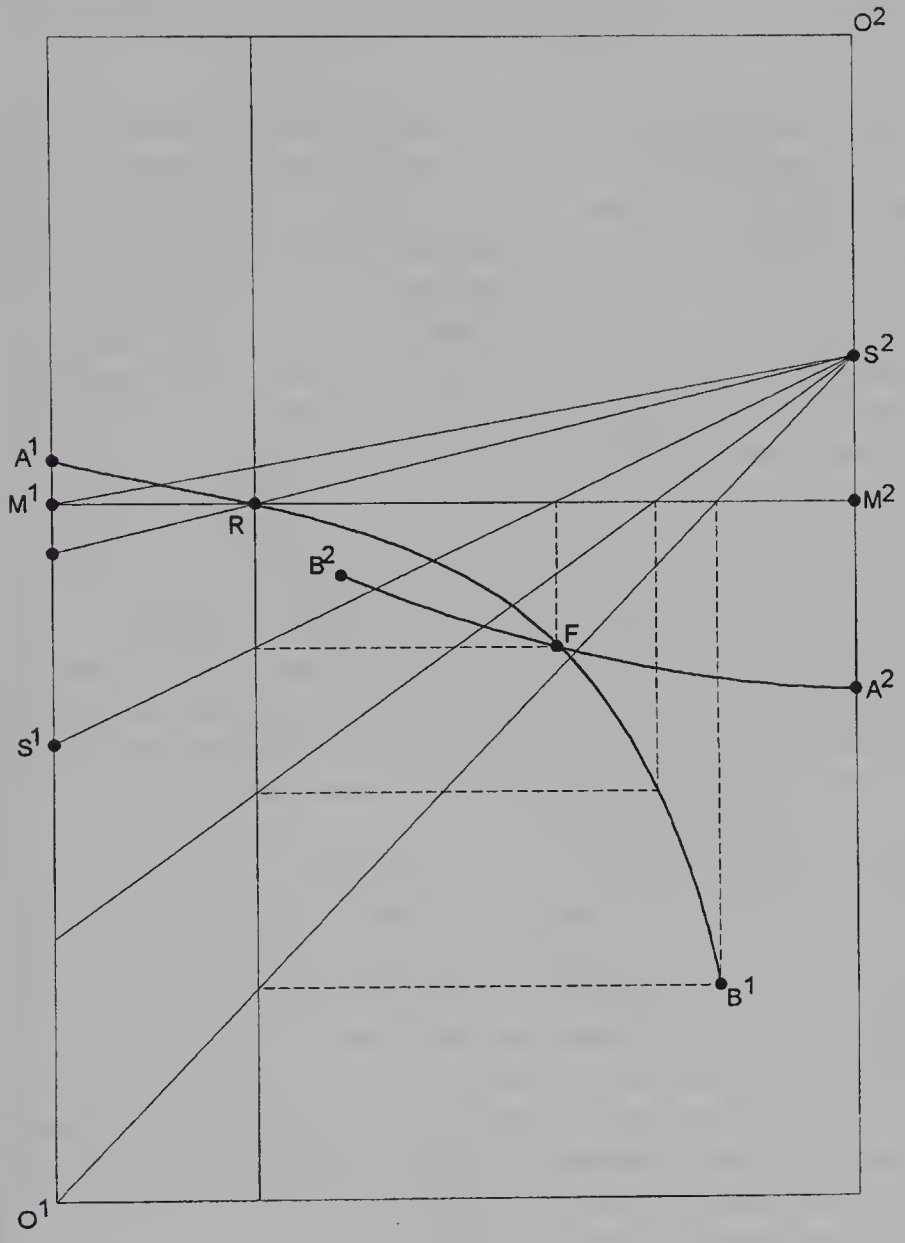


Figure 6.11
Budget sets.

R and F, reflecting the fact that the first trader now has little influence on the price.

Two kinds of equilibrium We are now ready to add the traders' preferences to the mechanics of the rules of exchange by superimposing the contours of the utility functions u^1 and u^2. As shown in figure 6.12, S^1 happens to be the *best response* to S^2 because A^1B^1 is tangent at F to the contour of u^1. (Note that the curvature of A^1B^1 is such that there is always a unique point of tangency.) Similarly, S^2 is the best response to S^1 because A^2B^2 is tangent at F to one of the contours of u^2. Thus figure 6.12 illustrates a noncooperative or "Nash" equilibrium (NE) for the market; neither trader, knowing the strategy of the other, would wish to change.

A striking feature of this kind of equilibrium is its nonoptimality. Because the curve A^1B^1 and A^2B^2 are not generally tangent to each other, the point F cannot be expected to be Pareto optimal or "efficient."[15] In effect, the traders are working with unequal *marginal* prices, represented by the unequal slopes of A^1B^1 and A^2B^2 at F. Any outcome in the shaded region would be preferred by both traders to the NE allocation at F. In particular, they would both benefit from increased trade at the *average* price p_1, represented by the slope of RF.

The reader familiar with the Edgeworth diagram will recognize the *contract curve* $C^1EP^2C^2$ in figure 6.12, which is a subset of the more extensive *Pareto set* $O^1P^1C^1EP^2C^2O^2$. The competitive equilibrium (CE) is represented by the competitive price ray RE, which is tangent to both indifference curves at the competitive allocation E. The situation illustrated seems to be typical: there is less volume of trade at the NE than at the CE. But the reverse is also possible. In fact, by a somewhat contorted but perfectly legitimate arrangement of the indifference contours, we can make any point outside the dotted lines in figure 6.12 the location of a unique CE, while keeping a unique NE at F.

It is also not difficult to construct utility functions for which several distinct strategy pairs (S_1, S_2) are in noncooperative equilibrium, and which may or may not be accompanied by a corresponding multiplicity of CEs; the two kinds of solution are not directly interlinked. But their general mathematical properties are quite similar. For example, we would expect that even for general m and n, if the utility functions are smooth and if the initial allocation is chosen according to a nonatomic probability

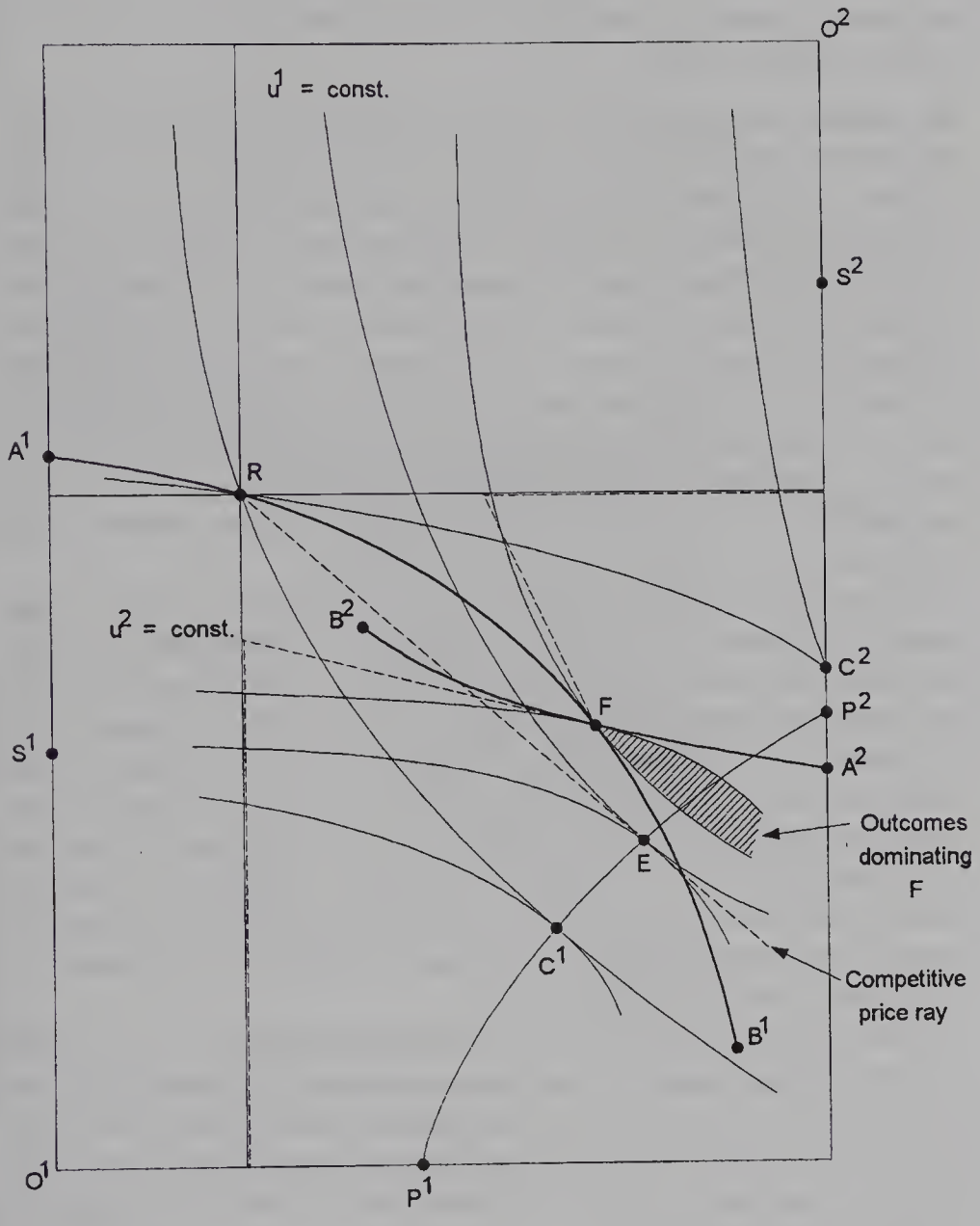


Figure 6.12
Noncooperative (F) and competitive (E) equilibrium.

distribution, then with probability 1 there will be a finite, odd number of NEs (cf. Debreu, 1970).

The feasible set and credit The set of feasible outcomes can be determined by holding one trader's bid at its upper extreme O^i or lower extreme M^i and sweeping through the strategies of the other trader. The feasible set is shown in figure 6.13 (solid shading); for example, the outcome L results if both traders bid their upper limit.[16] This feasible set has nothing to do with rationality or motivation: it merely describes those outcomes that the mechanism *can* be made to produce. It necessarily contains all NE allocations, but as for the CE allocations, we know a priori only that they lie in the quadrants of the Edgeworth box "northwest" or "southeast" of the point R. It may well be that, under our present trading rules, some or all of the CE allocations fall outside of the feasible set.

By introducing *credit*, we can enlarge the feasible set, permitting the traders to bid more of the "means of payment" commodity than they actually possess as cash on hand. In diagrammatic terms, the first trader, say, could be allowed the segment C^1M^1 instead of O^1M^1 (figure 6.13). The length O^1C^1 has a simple interpretation in terms of banking: it is the largest amount of good 2 that could be loaned to trader 1 with the certainty that trader 1 will be able to repay. To see this, consider that the worst case is $S^1 = C^1$, $S^2 = M^2$, with trader 1 bidding the limit and trader 2 bidding nothing. This leads to the outcome Q^2, at which trader 1 is just barely solvent. A similar "conservative" credit limit C^2 for trader 2 can be determined in the same way.[17]

If we alter the rules of the game to enable both traders to use this conservative banking credit, the feasible set is extended, as indicated by the stripes in figure 6.13. For example, the point L' results if both traders bid up to their new limits. Because the new feasible set still does not fill up the two basic quadrants, however, we cannot be sure that any competitive outcomes will be attainable.

With more liberal credit, the feasible set would cover a larger part of the two basic rectangles, and eventually all of them, but it would also include areas above and below the Edgeworth box, representing situations in which one or the other trader is "caught short." If we regard credit as the issue of a financial instrument, the interpretation of a point outside of the box is simple and familiar—it amounts to stating that after trade a

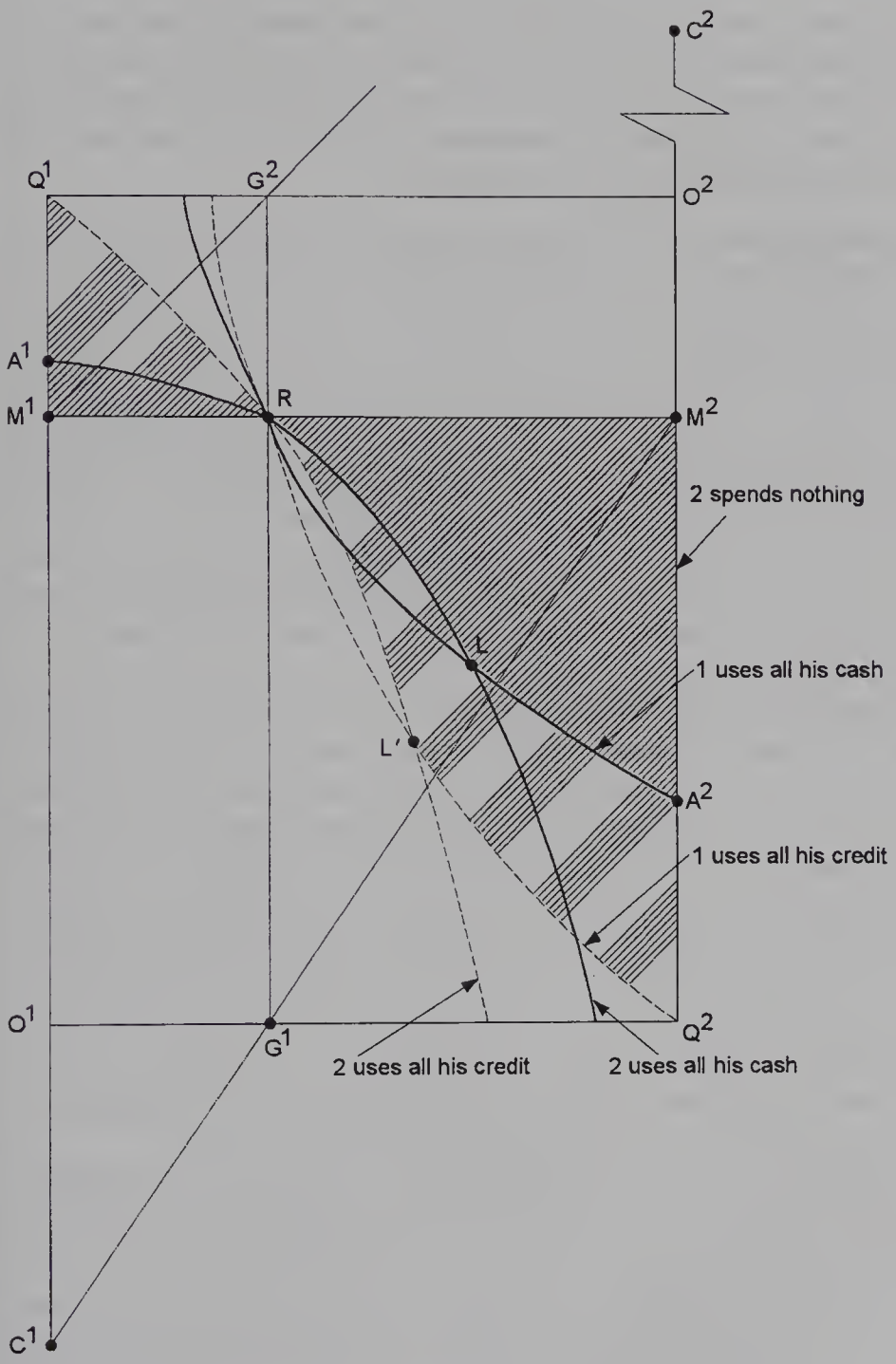


Figure 6.13
Feasible set with and without credit.

trader ends up with none of the monetary commodity in hand and with outstanding claims the trader cannot meet. To complete the model, we must either describe the utility to any player at such an outcome or add further moves to the game corresponding to bankruptcy proceedings. (We turn to the discussion of credit in chapter 11.)

Individual rationality In the present version of our model we are requiring that all of the initial endowments, except for the "payment" commodity, be put up for sale (but see chapter 7, where the requirement is removed). Consequently, the concept of ownership is somewhat different from that usually assumed in microeconomics, and is closer to that in law. All goods are monetized, and trade is virtually anonymous. Although the economy is an accountant's dream, by forcing goods to pass through the market, ownership rights are weakened. In particular, it may not be possible for traders to recover the bundle they originally start with, or obtain an equivalent bundle. This contrasts with the classical "Edgeworth" and modern "core" models, where traders can always defend their initial holdings if they so desire.

The defensive possibilities are easily illustrated. In figure 6.14, the first trader is fortunate, having relatively little of good 1, which must go to market, and relatively much of good 2, which he can use at his discretion. He cannot, of course, protect his actual initial holdings because this would mean being able to enforce the point R regardless of the other trader's choice of S^2. But he can force the outcome to have *at least as much utility* for him as R. In fact, by playing S^1 as shown, he restricts the possible outcomes to the set A^2B^2, which lies entirely on the "high" side of the indifference curve $u^1 = c^1$, as drawn. This is the best he can do defensively: S^1 is what would be called his "max-min" strategy, c^1 is his "max-min" payoff. We see that moving S^1 either up or down would move B^2 into a region of lower payoff, either toward R or toward L.

The second trader, on the other hand, has a relatively poor defensive position. The highest u^2 indifference curve that completely contains one of the A^1B^1 sets is the curve $u^2 = c^2$, which is distinctly inferior to R for her. There are two critical points: the endpoint A^1 and the tangency point near B^1. As S^2 moves, the A^1B^1 curve rotates on R. Because raising S^2 would make A^1 worse, while lowering S^2 would make the critical point near B^1 worse, S^2 is her "max-min" strategy.

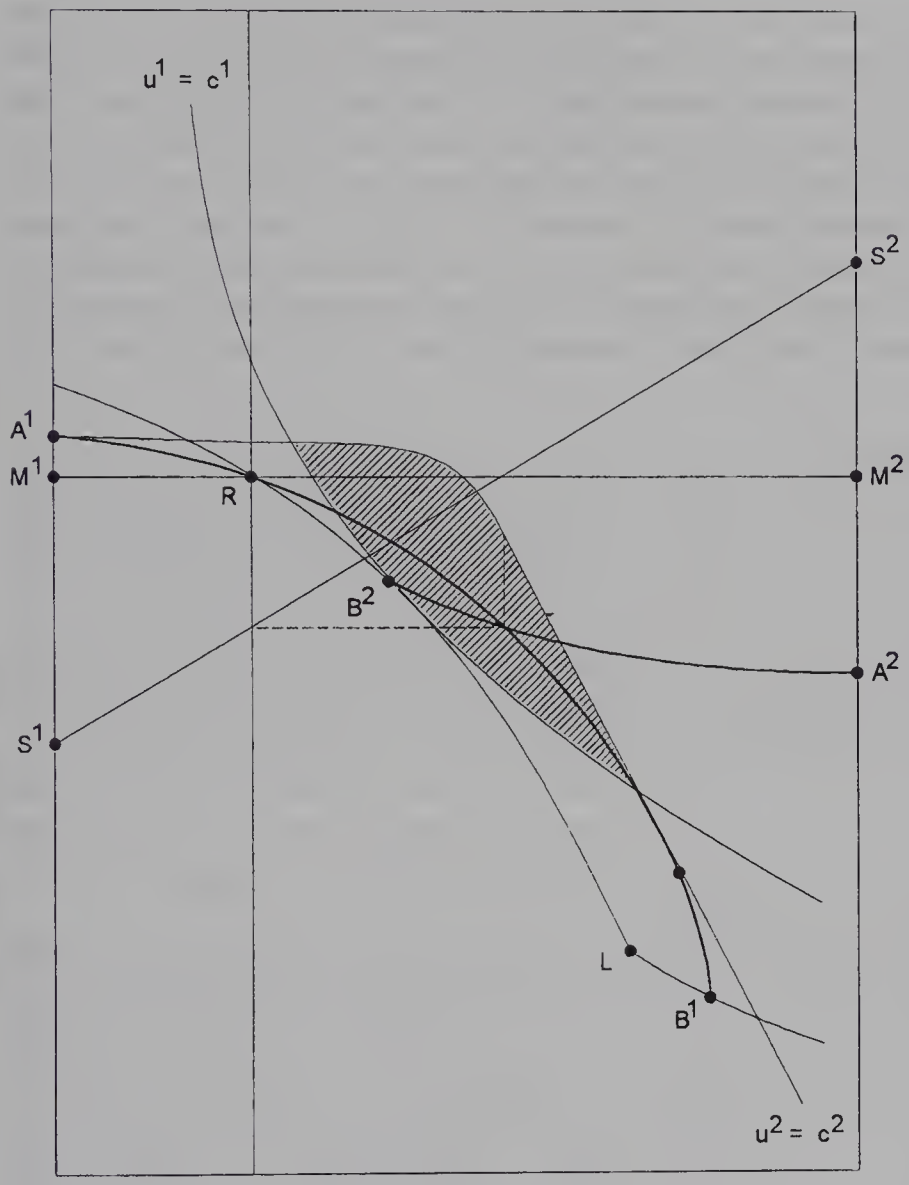


Figure 6.14
Max-min strategies and the individual rationality zone.

The shaded area in figure 6.14 represents the so-called individually rational allocations, bounded by the curves $u^1 = c^1$ and $u^2 = c^2$. The *feasible* individually rational allocations are just those lying beneath the line RM^2, and it is a theorem that every NE solution will lie in that region. In contrast, the Walras-Edgeworth individually rational zone, which contains every CE solution, is defined by the two indifference curves through R (not shown). It is easy to see that this region could be entirely disjoint from the other. This is not too surprising because the two regions arise from games with the same economic data but different rules of trade.

Many traders The same diagrams can be used, one-sidedly, when there are more than two traders. This is because our model has the "aggregation property," with both strategies and outcomes being *additive* over traders. Viewed by the other traders, a set or coalition of traders acting together is hardly distinguishable from a single, larger trader. In the diagrams, we may regard the point S^2 not as a single bid but the vector sum of bids by traders $2, 3, \ldots, n$, using R as the "origin" from which the vectors are defined. The first trader may notice a quantitative, but not qualitative difference from the two-person case: because the other traders combined may have far more resources than he does alone, the distance of S^2 from R and S^1 may make the slope of S^1S^2 (i.e., the price) almost insensitive to his choice.

It might appear at first glance that the *feasible set*, when there are many traders, would continue to be a sizable fraction of the set of outcomes feasible under unrestricted, Edgeworthian barter. But this is not the case. Let us count dimensions. The unrestricted allocation space has dimension $(m + 1)(n - 1)$.[18] Even if $m = 1$, there are $(2n - 2)$ dimensions of possible outcomes if trade is not required to pass through our "trading post" mechanism, compared to at most n dimensions that can arise when n traders bid by each selecting a point on a line as his strategy. Thus, starting at $n = 3$, the feasible set is only a lower-dimensional surface in the set of outcomes that would be possible under unrestricted trading.

Many commodities Because the trading posts in effect operate independently of each other, one can also use these diagrams when there are more than two commodities (i.e., more than one trading post). But there are two ways in which the markets remain intercoupled: (1) through the spending limits, which apply to all bids combined, so that a trader's upper

bound at each trading post will depend on what the trader spends at the others, and (2) through the utility functions, which in general reflect complementarities, substitution effects, and so on, among the different goods, so that the utility maps we superimpose on each Edgeworth box will depend on what is happening in the others. The second effect (as in the classical model) can only be visualized with hyperdimensional eyesight, but the first is still within reach of our diagrams.

Figure 6.15 illustrates a three-commodity situation. The two "boxes" are erected at right angles to each other, in the back of the three-dimensional figure (solid lines). The initial point, R, projects to R_1 in the first box and R_2 in the second.[19] We may conveniently regard R as the vector sum of M^2R_1 and M^2R_2. The second trader's pair of bids b_1^2 and b_2^2 are shown at S_1^2 and S_2^2, and their sum at S^2. Thus $|M^2S^2| = |M^2S_1^2| + |M^2S_2^2|$. The unspent balance is represented by the vertical bar O^2S^2.

The first trader's bids are shown at S_1^1 and S_2^1. As S_1^1 varies over the permitted range $O_1^1M_1^1$, i, the first marketplace sweeps out the curve through R_1 and F_1 in the usual way. Similarly, the curve through R_2 and F_2 describes the first trader's possibilities in the second marketplace. But his joint choice of S_1^1, S_2^1 is restricted by the spending limit

$$|M_1^1S_1^1| + |M_2^1S_2^1| \leq |M_1^1O_1^1| = |M_2^1O_2^1|.$$

The case actually illustrated has the first trader spending most of his cash on good 1 and the rest on good 2. (Had he held back some, we would have shown it by a vertical bar at O^1.) The resulting final allocation is found by taking the vector sum of M^2F_1 and M^2F_2, which yields the point F.

Now suppose the first trader changes his allocation of cash between goods 1 and 2, causing the points S_1^1 and S_2^1 to move in opposite directions. The point F will then trace out a curve in space, which we have tried to suggest by the open dots perched above the curve's projection on the base. This projection on the base shows the trade-off between goods 1 and 2 and a fixed amount of cash bid by the first trader; note that it is concave to his origin O^1. If the first trader bids less than the maximum allowed (strict inequality above), a curved triangular surface is generated, extending out from F and the open dots toward the viewer and rising to an apex directly above O^1. This locus, not shown, indicates ways in which the first trader can obtain more of good 3 (which is the commodity money) at the expense of goods 1, 2, or both; it is concave to the origin O^1 and its

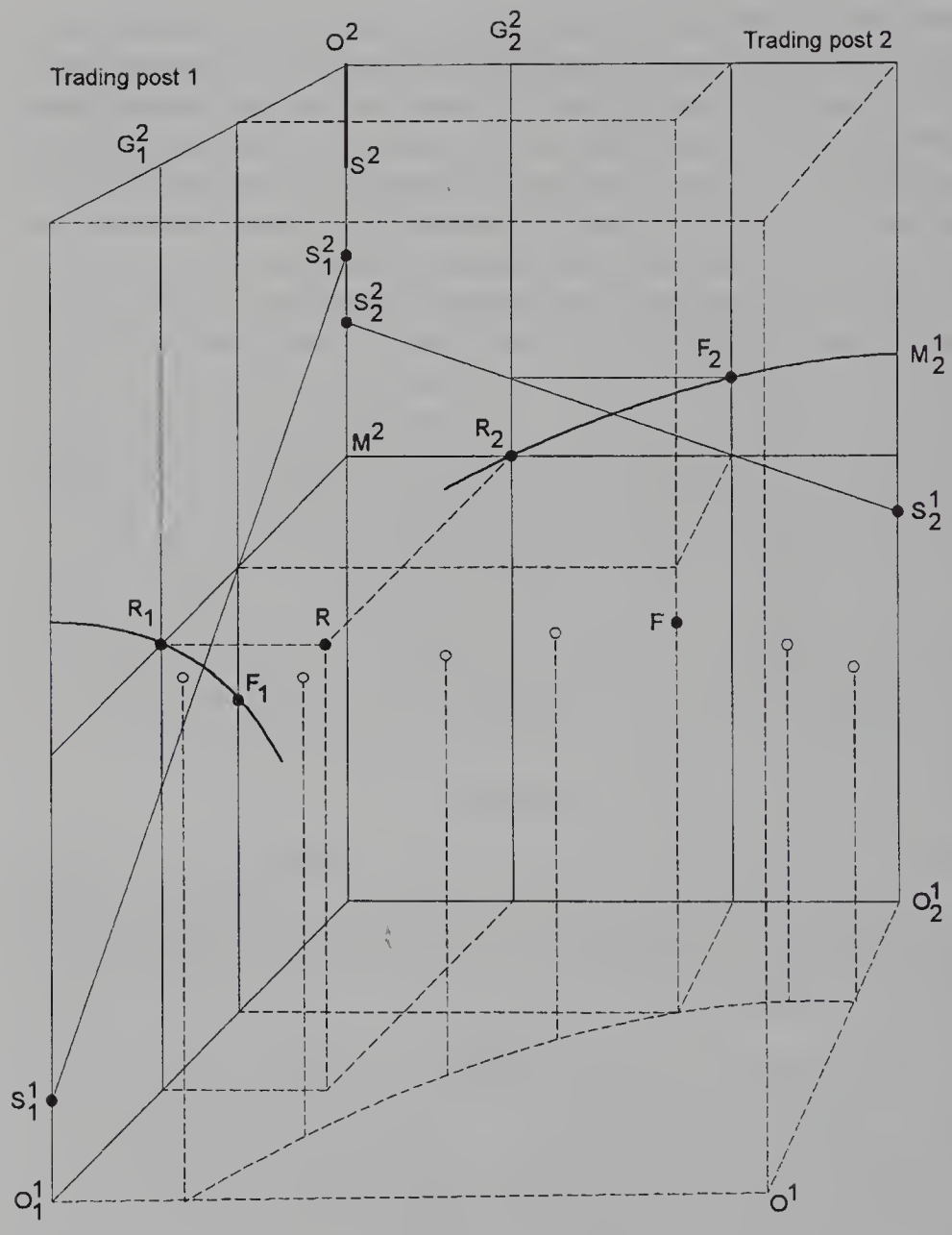


Figure 6.15
Two trading posts.

projection is the large, roughly triangular region in the base of the diagram. The first trader's best response to the second trader's given strategy (S_1^2, S_2^2) is determined by the relationship of this surface to the indifference surfaces of u^1, in the three-dimensional commodity space with origin at O^1.

6.4.3 Replication

Inquiries into the behavior of economic models with large numbers of participants often make sweeping assumptions of symmetry, in the hope of keeping the models mathematically tractable and easy to visualize while capturing at least some of the characteristic effects of large numbers. A favorite technique, involving a high but not total degree of symmetry, is called "replication." Imagine a basic economic system juxtaposed to a large number of identical replicas of itself; then take away all barriers and form a "common market." Equivalently, assume that all traders in the full model are drawn from a small number of *types*, with an equal number of individuals of each type. Traders of the same type have identical endowments and identical tastes, but they are not constrained to act alike. That is, they are not members of a bloc or cartel but remain independent decision makers.[20]

The number of members of a type—the "replication number"—provides the modelers with a simple size parameter that can be varied without calling for additional data. Of course, we do not intend to treat replication as though it were some kind of actual expansion process, like growth by homogeneous population increase, or by accretions of similar countries to a common market. Although replication should be considered only as a technical device of comparative statics, as such it has repeatedly proved its worth in explorations of the size effect (see Edgeworth, 1881; Shubik, 1959a, 1968b; Debreu and Scarf, 1963; Shapley and Shubik, 1966, 1969b; Debreu, 1975; Owen, 1975; and Shapley, 1975b).

A simple case The effect of replication on our model is illustrated in figure 6.16. We have $m = 1$ and $n = 2k$, where k is the replication number. The point S^1 represents the *average* of the bids of the k traders of type 1, as does S^2 for type 2.[21] The point F therefore indicates the average final bundles for the two types.

Let us focus on a typical member of type 1. His own bid might be, say, at $\tilde{S}^1$. To discover his personal final bundle, $\tilde{F}$, we must draw the line

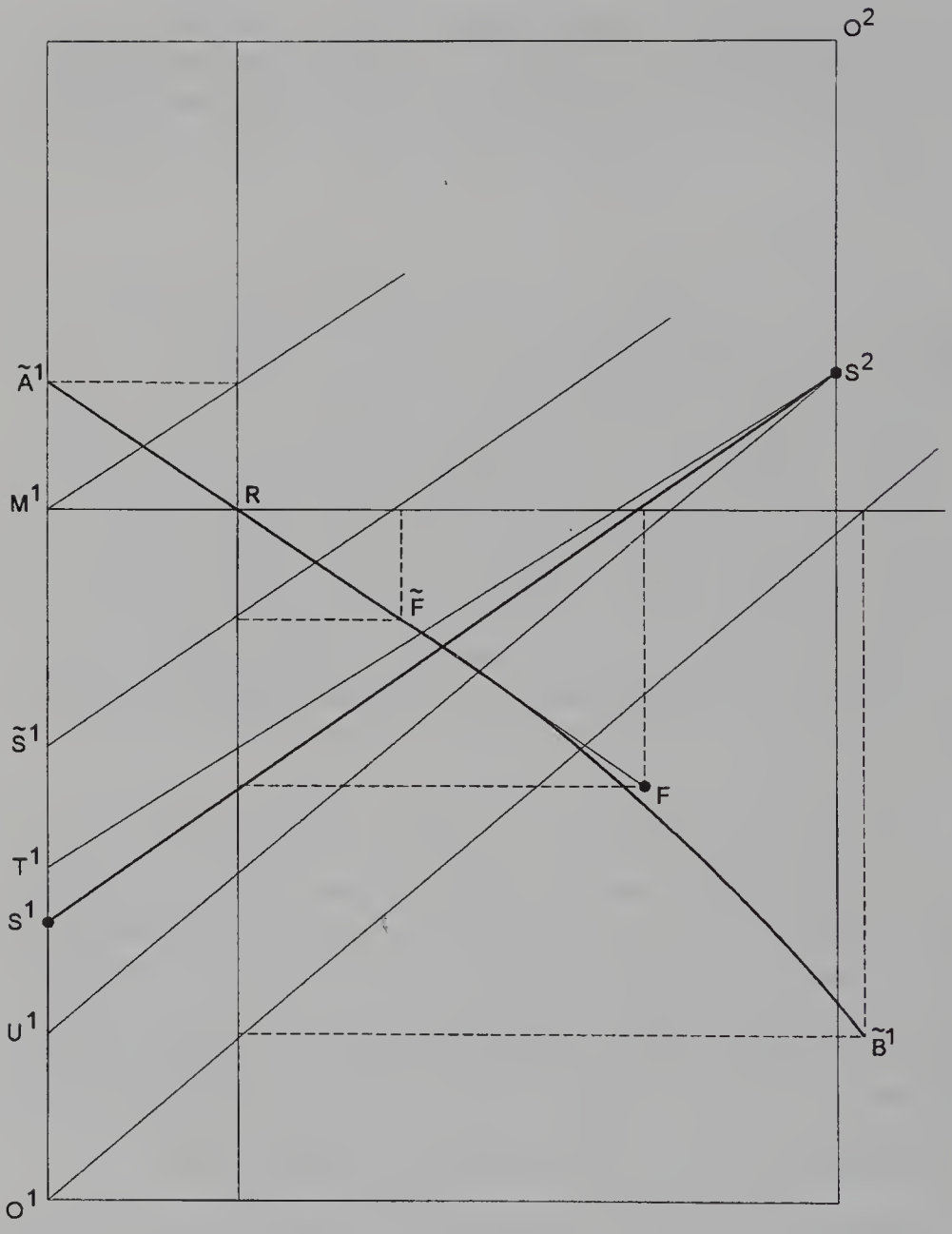


Figure 6.16
Price formation in a replicated market.

through $\tilde{S}^1$ that is *parallel* to S^1S^2; we then locate the point $\tilde{F}$ in the usual way. Of course, $\tilde{F}$ necessarily falls on the straight line through R and F because all transactions take place at the same point.

Now suppose our trader changes his bid from $\tilde{S}^1$ to M^1, that is, decides to buy nothing. This depresses the average bid for type 1, moving it from S^1 to T^1, where $|S^1T^1|/|\tilde{S}^1M^1| = 1/k$. Because the new price is the slope of T^1S^2, our trader's new final bundle is $\tilde{A}^1$, determined by the line through M^1 parallel to T^1S^2. Similarly, at the other extreme, if he decides to bid the limit, the result is $\tilde{B}^1$, determined by the line through O^1 parallel to U^1S^2, where $|U^1S^1|/|\tilde{S}^1O^1| = 1/k$. We should not be surprised to see $\tilde{B}^1$ fall outside the box. This means merely that our trader happens to have enough "cash" to buy more than the combined initial bundles of one trader of each type.

The locus of possible final bundles for our variable trader is the curve $\tilde{A}^1R\tilde{F}\tilde{B}^1$, shown in figure 6.16 for the case $k = 4$. The flattening of the curve, as compared to the curve A^1RFB^1 in figure 6.12 (which is the case $k = 1$), clearly shows the diminished influence of a single individual on price after replication. Indeed, the price variation in the present case is encompassed by the thin pencil of sloping lines within the angle $U^1S^2T^1$, which represents the bounds on the ability of the single trader to move price. In the limit, as $k \to \infty$, the curve becomes the fixed-price "budget set" of the classical CE model—but with one important difference: it is truncated at the $\tilde{B}^1$ endpoint instead of extending to the horizontal axis through O^1, in recognition of the limited quantity of the payment commodity available.[22]

The general case For the replicated form of the general model presented in section 6.4.1, we consider a market with T types of traders and k of each type; thus $n = kT$. The initial holding of good j by trader s of type t will be denoted d_j^{ts}, which we can shorten to a_j^t since it does not depend on s. The total amount of good j (summed over both t and s, will be denoted a_j, as before. A typical strategy will be denoted $b^{ts} = (b_1^{ts}, \ldots, b_m^{ts})$. The defining equations of section 6.4.1 then become

$$p_j = \bar{b}_j/\bar{a}_j,\ j = 1, \ldots, m, \quad \text{where } \bar{b}_j = \sum_{t=1}^{T}\sum_{j=1}^{k} b_j^{ts},\ \bar{a}_j = \sum_{t=1}^{T}\sum_{j=1}^{k} a_j^{ts};$$

$$x_j^{ts} = b_j^{ts}/p_j,\ j = 1, \ldots, m \quad \text{and} \quad p_j \neq 0;$$

$$x_j^{ts} = 0,\ j = 1, \ldots, m \quad \text{and} \quad p_j = 0;$$

$$x_{m+1}^{ts} = a_{m+1}^{ts} - \sum_{j=1}^{m} b_j^{ts} + \sum_{j=1}^{m} b_j^{ts} p_j;$$

$$\Pi^{ts}(b^{11}, \ldots, b^{1k}, b^{21}, \ldots, b^{Tk}) = u^t(x_1^{ts}, \ldots, x_{m+1}^{ts}),$$

where $u^t(x) = u^{ts}(x)$ is the common utility function for members of type t.

Suppose that an NE has been found in which all traders of the same type make the same bids: $b^{ts} = b^t$, $s = 1, \ldots, k$.[23] It is instructive to look at the equilibrium conditions for the "interior" case, namely, where all b_j^t are positive and

$$\sum_{j=1}^{m} b_j^i < a_{m+1}^t \quad (t = 1, \ldots, T).$$

Assuming that the concave functions $u^t(x)$ are differentiable, let $u_j^t(x)$ denote their partial derivatives with respect to x_j. Then setting $du^t/db_j^{ts} = 0$ gives us

$$u_j^t(x^{ts})\left[\frac{1}{p_j} - \frac{b_j^{ts}}{(p)^2 \bar{a}_j}\right] = u_{m+1}^t(x^{ts})\left[1 - \frac{a_j^{ts}}{\bar{a}_j}\right],$$

which must hold at $b = \hat{b}$. Because prices are in terms of good $m+1$ as numéraire, we can introduce $p_{m+1} = 1$ and rewrite the condition for a symmetric, interior NE as follows:

$$\frac{u_j^t(\hat{x}^t)}{\hat{p}_j}\left[1 - \frac{\hat{b}_j^t}{\bar{b}_j}\right] = \frac{u_{m+1}^t(\hat{x}^t)}{\hat{p}_{m+1}}\left[1 - \frac{a_j^t}{\bar{a}_j}\right], \quad j = 1, \ldots, m,$$

where $\hat{p}$ and $\hat{x}$ are the prices and allocation corresponding to $\hat{b}$.

The expressions in curly brackets reveal the effect of oligopoly, but note that they all lie between 1 and $1 - 1/k$. In the limit, therefore, these conditions for NE reduce to the conditions for CE, with the prices of the goods proportional to the marginal utilities of every trader. Thus we may state a convergence theorem as follows:[24]

THEOREM 6.2 Assume that for infinitely many values of k the market has a symmetric, interior NE, and let $\hat{p}^{(k)}$ be the corresponding m-vector of prices. Let $\tilde{p}$ be any limit point of the $\hat{p}^{(k)}$, and define $\tilde{p}_{m+1} = 1$. Then the

$m+1$ prices $\tilde{p}_1, \ldots, \tilde{p}_m, \tilde{p}_{m+1}$ will be competitive for the market (for any value of k); that is, an allocation x will exist for which $\tilde{\tilde{x}} = a$ and, for each s and t, $\tilde{x}^{ts}$ maximizes $u^t(x^{ts})$ subject to $x^{ts} \geq 0$ and

$$\sum_{j=1}^{m+1} \tilde{p}_j(x_j^{ts} - a_j^{ts}) = 0.$$

It should be noted that the NE approaches the CE "from below," that is, through outcomes that are not in general Pareto optimal. This contrasts with the convergence of cooperative solutions like the core and the value, which are by definition Pareto optimal all the way (see, for example, Debreu and Scarf, 1963; Shapley and Shubik, 1969b; Debreu, 1975; Owen, 1975; and Shapley, 1975b).

The type of convergence revealed in theorem 6.2 depends crucially on having interior solutions, with no trader up against his spending limit. All must have enough cash (or credit, in an extended model), where the meaning of "enough" depends on the payment commodity's marginal utility relative to that of the other goods. Of course, there can never be "enough" fiat money in this sense (in a one-period model), for the NE will always have all traders spending the limit and wishing they could spend more. If the CE is considered socially desirable (as one road to the Pareto optimum, if for no other reason), then a society whose trading system resembles our replicated model should make sure that the means of payment is something that is both widely available and generally desirable, either as a consumption item or as a means of payment in future periods.

6.4.4 Transactions needs and enough money

When there are only two commodities being traded, both are money. In our discussion above, we forced a distinction between the two commodities, essentially for didactic purposes. The Edgeworth box provides the simplest illustration of trade where one commodity is up for sale and the other is used as the means of exchange.

Our simplification entailed us imagining that there was a strategic limitation imposed on all traders, who were required to offer for sale all of one commodity. In general, this is not the case. Farmers may hold back from the market milk their families intend to drink. As unrealistic as the sell-all model may appear to be, it has important economic and taxation properties. All goods must pass through the market, thus every period all

wealth in the economy is evaluated in the market. Nothing is hidden from the tax collector or the national accountants. In reality, wealth concealment is an everyday feature of economic life in all societies.

In general, if there is unrestricted trade in two commodities, that is, traders can hold back goods from the market, there is never any need for credit because all points in the Edgeworth box can be achieved. This is not true where one of three or more commodities serves as money.

When a society is using a commodity money, we may say that it has enough money if the competitive equilibrium of the associated Walrasian system can be achieved without the need for credit. It must be stressed that these are feasibility conditions and *not* equilibrium conditions. Just because obtaining a competitive equilibrium distribution is feasible does not imply that it is strategically desirable.

The conditions for enough money are dependent on the commodity selected to be the money and on its distribution among the members of an economy. The conditions for an exchange economy can be stated formally. Consider an economy with n individuals, each individual i with a bundle of resources $(a_1^i, a_2^i, \ldots, a_{m+1}^i)$ where $m+1$ is the number of goods and

$$a_j = \sum_{j=1}^{n} a_j^i > 0; \quad a_j^i > 0 \ \forall i = 1, \ldots, n.$$

The standard conditions for a Walrasian equilibrium are that the set of prices $(p_1, \ldots, p_m)$ form an equilibrium (where we have set $p_{m+1} = 1$, i.e., the $m+1$st commodity is used as a numeraire).

Each individual maximizes

$$\varnothing_i(x_1^i, x_2^i, \ldots, x_{m+1}^i)$$

subject to

$$\sum_{j=1}^{m+1} p_j(a_j^i - x_j^i) = 0,$$

where

$$\sum_{j=1}^{m}(a_j^i - x_j^i) = 0, \quad x_j^i \geq 0.$$

To the Walrasian conditions, we add an extra *cash expenditure* condition for each individual:

$$\sum_{j=1}^{m} p_j x_j^i \leq a_{m+1}^i.$$

This condition must be satisfied to attain the Walrasian equilibria for trade in a market where all goods are put up for sale, without resorting to credit in the associated strategic market game.

A sufficient condition for the existence of enough money is that the commodity selected as the money always maintains a finite positive worth as a commodity for all individuals; that is, every individual i has a marginal utility for the monetary commodity (considered only as a commodity) greater than or equal to some preassigned number.

Suppose we consider a sequence of economies where each individual i has an initial supply of resources $(a_1^i, \ldots, a_m^i, ka_{m+1}^i)$. As we increase k, the economies change. There will be a large enough k that that economy will have enough money.

Not long after the invention of money, coinage was debased.[25] Because, in modern economies, the purchase power of coins generally exceeds the market price of their constituents, most coinage is partly commodity money and partly fiat.

A commodity money may be considered to be full-bodied if its value when demonetized would be the same as when monetized. If, the condition on the other hand,

$$\sum_{j=1}^{m} p_j x_j^i \leq a_{m+1}^i$$

fails to hold (where the x_j^i are purchases at the Walrasian equilibrium) and there is no credit, an extra shadow price for money, higher than its barter commodity value appears. The Walrasian equilibria cannot be achieved, and there is a shortage of money, whose price then exceeds its worth as a commodity, and which commands a premium in the cash-constrained noncooperative equilibrium due to its strategic properties as a money.

Like the methods of trade, the needs for money vary. Thus if individuals are permitted to hold back goods from the market, the cash expenditure constraint changes to

$$\sum_{j=1}^{m} \max(0, (x_j^i - a_j^i) \leq a_{m+1}^i.$$

6.4.5 A continuum of traders

In our discussion of an exchange economy with a finite number of traders, we observed that as the number of traders became large (in an appropriate way), the noncooperative equilibria approached the competitive equilibria. As long as the numbers are finite and there are no transaction costs or saturation, an individual maintains some small influence over price. The appropriate mathematical method to help forge the links between the ideas of the competitive economy and strategic market games is to approximate competition among the many by competition among an uncountable number of insignificantly small individuals. This can be done by building a measure-theoretic model of competition. Dubey and Shapley (1994) have done this, establishing an equivalence relationship between the NEs of the game and the CEs of the Walrasian model, although they note conceptual difficulties in treating the aggregation of strategies.

The continuum models have the advantage that there can be many individuals and they can all be different. While discussing the emergence of competitive price in the shrinking of the contract curve (in modern terms, the "core"), Edgeworth remarked that he did not like the idea of exact replication, but felt that one should be able to treat all individuals as somewhat different—"not equally but nearly equally, natured" (Edgeworth, 1881, p. 40).

The decision to use a finite strategic market game model or a continuum model must be based both on mathematical tractability and on the adequacy of the proposed approximation in answering the question at hand. As a first approximation for questions concerning government policy and producers and consumers as a whole, a finite-measure model with one player facing a mass of small players appears to have some merit.

Notes

1. N is the set of integers up to $n = |N|$; Ω^m is the positive orthant of the m-dimensional Cartesian space $\mathbb{R}^m$. Given an n-tuple of strategies, the payoff to i is the utility of i's final bundle in the allocation produced by the strategies.

2. That is, i cannot aim directly at j, but can only act through the mass market.

3. This could happen if an arbitrary market mechanism chose to accept a fraction of each bid prior to calculating the clearing price.

4. A form of retail swindling ("bait and switch") involves advertising an item at a low price without disclosing that the item is in highly limited supply to attract customers to sell them something else.

5. The point to be stressed here is that the information conditions make the more flexible bid (of p_j^i or above) essentially of no advantage over the bid $\bar{p}_j^i$. Even minute communication or transmission costs rule out the more complicated bid in favor of the simpler. Thus stores do not advertise a radio for sale at "$100 or more."

6. A vertex of the tree not in an information set is a point of perfect information. All information sets should be labeled with the name of the player called on to choose. For simplicity in this example, the names are on the side of the tree. The starting vertex of the game is also labeled 0.

7. Levitan and Shubik (1971, 1972) have given examples of different strategy spaces and Pareto surfaces that are possible in oligopolistic models.

8. Superscripts will consistently be used to denote traders. Summation over traders will be denoted by a horizontal bar, $\bar{a}_j$ means $\sum_{j=1}^{n} a_j^i$. The model in this chapter is based heavily on Shapley and Shubik, 1977; see also Shubik, 1973, and Shapley, 1976.

9. This simplifying condition, which is not essential to our general approach, is by no means as unreasonable as might appear at first glance; in a multiperiod context, it would amount to requiring that all goods in the economy be "monetized" every trading period.

10. The simplest way to model credit is to remove the spending limit entirely, while extending the domain of the utility function to include negative amounts of the $(m+1)$st commodity so as to provide a suitable *disutility* for being caught short of cash when the sales and purchases are all added up. To close the model, an outside source of cash would have to be postulated to cover the payments due the other traders when one trader defaults. In a dynamic, multiperiod context, we can conceive of durable goods, other than the $(m+1)$st "payment" commodity, that are carried forward and used as security for the granting of credit for commercial loans.

11. Note that $x_j^i = 0$ implies $b_j^i = 0$. Thus a trader receives nothing if and only if the trader bids nothing.

12. The term *best-response equilibrium* is due to Robert Wilson. Although Nash (1951) included the possibility of "mixed strategies," that is, strategies chosen by randomizing devices the players construct, these have no plausible interpretation in our present model.

13. Based in part upon unpublished notes of Shapley.

14. The notation $|PQ|$ indicates the length of the segment PQ.

15. Indeed, Pareto optimality in a NE can only occur at corners of the indifference curves, or in special cases such as $F = R$ or $F = B^2$.

16. Most boundary points are attainable, but those on the line M^1M^2 are not, except for R itself. There is, however, an exceptional "null" outcome in the Edgeworth box, in which the first trader bids and gets the point M^1 and the second trader bids and gets the point M^2. As long as both traders bid except at R, the outcome will be off the line M^1M^2. Our rule is that if a trader bids nothing, the trader gets nothing, even if the price is zero. So if both traders bid nothing, the entire stock of good 1 is lost. (We may imagine that it goes to an otherwise unnoticed "scavenger," who makes infinitesimal bids in the hope of making a killing.) This exceptional outcome is manifestly undesirable and unstable, and it has no real effect on the solutions of the game under the hypotheses in theorem 6.1, though it causes technical difficulties in the proof.

17. It will become clear in sections 6.4.2 and 6.4.3 that when there are many traders, the amount of "conservative credit is likely to be negligible because it depends on the amount of a trader's own money that will surely come back to the trader from his own bids. In this respect our two-trader diagrams may be misleading. We may also remark that "conservative" credit does not require the aid of a central clearinghouse for the trader's promissory notes.

18. The "+1" here is for the monetary good, while the "−1" is for the condition that the sum of the holdings is constant.

19. The labels are generally consistent with the previous diagrams, but with subscripts added to distinguish the two trading posts.

20. This distinction can be very important in the game-theoretic approach, but is less important in a behavioristic theory. For example, as we shall see, replication changes the NE, but not the CE.

21. More generally, if there were many types, then the analogue of S^2 would not be the average, but $1/k$ times the sum of all bids from traders not of type 1.

22. It may be observed that the "conservative" line of credit for an individual shrinks to zero as k increases. A simple calculation shows that, in the present case, the amount of such credit is equal to $a_2^1 + a_2^2/(k(a_2^1 + a_2^2) - a_2^1)$.

23. The conditions given in theorem 8.1 (see section 8.4.1) are sufficient to guarantee the existence of such a symmetric NE. It is quite possible also to have nonsymmetric NEs, whose presence we have no clear way of ruling out.

24. A somewhat more general form of this theorem will be found in Shapley, 1976.

25. One of the earliest recorded inflations caused by debasement of the currency by the government appears to have been that of the Roman Empire under Augustus.

7 Market Mechanisms with Simple Markets

7.1 The Bid-Offer Market

Let us briefly summarize the sell-all market structure described in section 6.4. There are n individuals, each with an endowment of $(m+1)$ commodities. Trader i has resources $(a_1^i, a_2^i, \ldots, a_{m+1}^i)$ and preferences represented by a concave utility function $u^i(x_1^i, x_2^i, \ldots, x_{m+1}^i)$. Each trader i is required by the rules of the game to offer all of i's endowments of the first m commodities for sale in the m markets. The $(m+1)$st commodity is used to bid on the other m. Because the total supplies of all goods except money must pass through the markets, the meaning of ownership is modified to imply that ownership of a commodity is title to the money receipt from the sale of the commodity rather than the commodity itself. This somewhat unrealistic assumption makes the initial analysis easier. In the previous model, a strategy by an individual trader could be represented by a vector of dimension m representing bids in the m markets. The model now constructed permits all traders to decide how much to offer to the market.

Because at each trading post j the amount for sale is $a_j = \sum_{i=1}^n a_j^i$ and the amount of money bid for good j is $b_j = \sum_{i=1}^n b_j^i$, price is

$$p_j = b_j/a_j. \tag{7.1}$$

The amount of commodity j obtained by i is

$$x_j^i = \begin{cases} b_j^i a_j/b_j, & \text{if } b_j > 0 \\ 0, & \text{if } b_j = 0 \text{ for } j = 1, \ldots, m \end{cases} \tag{7.2}$$

and

$$x_{m+1}^i = a_{m+1}^i - \sum_{j=1}^m b_j^i + \sum_{j=1}^m a_j^i b_j/a_j. \tag{7.3}$$

For this model, Shapley and Shubik (1977) have shown that a noncooperative equilibrium in pure strategies always exists and that, if there is "enough money," there will exist noncooperative equilibria for a sufficiently high replication of the market that lies within an arbitrarily close region of competitive equilibrium points of the economy.[1]

7.1.1 The "offer for sale" model

It may be argued that for many markets and certainly for those involving production and the carrying of inventory that the restriction forcing

traders to offer all of their goods for sale is unreasonable. This restriction is relaxed here. Instead, we assume that a trader can be a seller, a buyer, both, or neither in each market.

A strategy for a trader i is vector (s^i, b^i) of dimension $2m$ such that

$$b_j^i \geq 0, \quad 0 \leq s_j^i \leq a_j^i \quad \text{for } j = 1, \ldots, m, \tag{7.4}$$

where b_j^i is the bid for and s_j^i is the offer of good j by i

$$\sum_{j=1}^{m} b_j^i \leq a_{m+1}^i, \tag{7.5}$$

where the amount of commodity j in the final possession of trader i is

$$x_j^i = \begin{cases} a_j^i - s_j^i + b_j^i s_j / b_j, & \text{if } b_j > 0 \text{ where } s_j = \sum_{j=1}^{n} s_j^i \\ a_j^i - s_j^i, & \text{if } b_j = 0 \text{ for } j = 1, \ldots, m \end{cases} \tag{7.6}$$

and

$$x_{m+1}^i = a_{m+1}^i - \sum_{j=1}^{m} b_j^i + \sum_{j=1}^{m} s_j^i p_j, \tag{7.7}$$

where $p_j = b_j / s_j$ and $p_j = 0$ if $s_j = 0$.

7.1.2 The strategy set

To give some insight into the nature of the strategy sets of a trader and the relationship of these sets to final outcomes, we consider a market trading in a commodity money with one commodity being bought or sold. In figure 7.1, the point with coordinates (a, M) indicates the initial endowment of a trader. The trader's strategy set is a rectangle given by $0 \leq s \leq a$ and $0 \leq b \leq M$. Any point in this rectangle is a strategy.[2] We wish to study how the strategy (s, b) transforms the initial point (a, M). Suppose that all other traders have offered Q units of the good and a total of B units of money. Then a bid of (s, b) takes the initial holding of (a, M) to a final holding of

$$(a - s + b/p, M - b + sp),$$

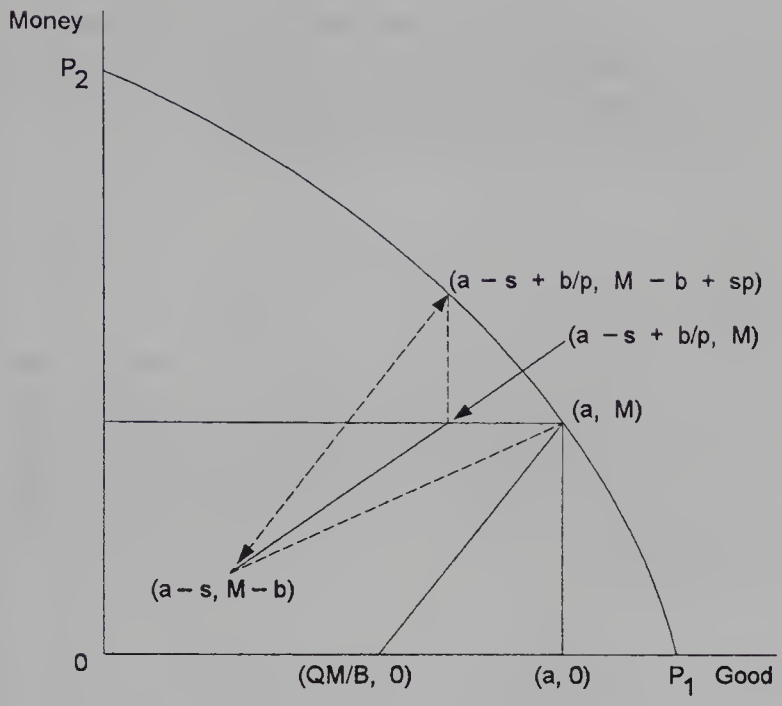


Figure 7.1
Initial and final holdings.

where

$$p = \frac{B+b}{Q+s}(Q > 0). \tag{7.8}$$

We may observe that a bid of $(s - b/p, 0)$ would also map[3] (a, M) onto $(a - s + b/p, M - b + sp)$.

The bids that map (a, M) onto itself are given by

$$a = a - s + b/p \quad \text{and} \quad M = M - b + sp$$

or

$$b = s\left(\frac{B+b}{Q+s}\right) \quad \text{or} \quad Qb = Bs. \tag{7.9}$$

All bids on the line connecting $(QM/B, 0)$ to (a, M) map (a, M) onto itself. Any strategy to the right of this line maps (a, M) onto a point on the curve P_1P_2 to the right.

All points on the curve P_1P_2 can be obtained by bids on the "L-shaped" boundary of the strategy set, that is, bids of the form $(s, 0)$ or $(0, b)$ where $0 \leq s \leq a$ and $0 \leq b \leq M$. If we let (q, m) be a point on the curve P_1P_2, then the equation for this surface is given by

$$m = M + \frac{(a-q)B}{Q+a-q}, \tag{7.10}$$

which is a concave curve.

Likewise, we can generalize these observations for m commodities and a money. Suppose the initial holdings of trader i are given by $(a_1^i, \ldots, a_m^i, M^i)$. All others together have offered $(Q_1, \ldots, Q_m)$ for sale and bid $(B_1, \ldots, B_m)$. A strategy by trader i is a point in a $2m$-dimensional rectangular set of the form $(s_1^i, \ldots, s_m^i, b_1^i, \ldots, b_m^i)$. The initial holdings are transformed to final holdings of

$$\left(a_1^i - s_1^i + b_1^i/p_1, \ldots, a_m^i - s_m^i + b_m^i/p_m; M^i - \sum_{j=1}^{m} b_j^i + \sum_{j=1}^{m} s_j^i p_j\right). \tag{7.11}$$

We may derive the equation for the set of final holdings that can be reached by all bids. Let $(q_1^i, q_2^i, \ldots, q_m^i, m^i)$ be a point in the set of final holdings. Then the equation of this surface is given by

$$m^i = M^i + \sum_{j=1}^{m} \left\{ \frac{B_j(a_j^i - q_j^i)}{Q_j + a_j^i - q_j^i} \right\}, \tag{7.12}$$

which is also concave. Whereas in the example with one good any final outcome could be achieved by a line of strategies (as shown in figure 7.1), here any outcome can be achieved by the set of strategies formed by the product of m lines.[4]

To confirm that (7.12) is the surface of final allocations produced by using any strategy, we can consider an arbitrary strategy (b_j^i, s_j^i) and show that it produces a final allocation as in (7.11). Substituting these values for q_j^i and m^i into (7.12), we obtain

$$\sum_{j=1}^{m} (s_j^i p_j - b_j^i) = \sum_{j=1}^{m} \left\{ \frac{\frac{B_j}{p_j}\{s_j^i p_j - b_j^i\}}{Q_j = s_j^i - \frac{b_j^i}{p_j}} \right\}$$

$$= \sum_{j=1}^{m} \left\{ \frac{\dfrac{B_j}{p_j}\{s_j^i p_j - b_j^i\}}{\dfrac{(Q_j + s_j^i)(b_j^i + B_j) - b_j^i(Q_j + s_j^i)}{b_j^i + B_j}} \right\}$$

$$= \sum_{j=1}^{m} \left\{ \frac{\dfrac{B_j}{p_j}\{s_j^i p_j - b_j^i\}}{\dfrac{B_j}{p_j}} \right\} = \sum_{j=1}^{m} (s_j^i p_j - b_j^i). \tag{7.13}$$

We can also show how the availability of banking enlarges the strategy set of a trader. Suppose that the trader has resources of (a, M) and that the bank (say through its economic activity forecast) is able to accurately estimate the aggregate bids and offers of all other traders. The meaning of the granting of credit is that the trader i can bid a sum $\sum_{j=1}^{m} b_j^i > M^i$. Given that the bank has a forecast of what aggregate bids and offers are going to be, it can calculate the effect of trader i on the market even if i is a large oligopolist. Figure 7.2 shows this in curve P_1P_2.

The point E which represents the strategy $(0, M)$ maps the endowment (a, M) into the point P_1, which is a final endowment of $(a + M(Q/B + M), 0)$. If the trader could borrow, then all strategies of the form $(s, M + (B + M/Q)s)$ for $0 \leq s \leq a$ also map (a, M) to P_1. Thus trader i could borrow as much as $(B + M)/Q$ and still be able to pay back i's loan after trade. This amount is represented by the length HI; the strategy set of the trader without borrowing is given by the rectangle $EFGH$. With borrowing that i will be able to pay back, the strategy set becomes the trapezoid $EFGHI$.

7.1.3 Preferences

From figure 7.2, we can make two observations about preferences and this type of bidding model. First, because the curve P_1P_2 is strictly concave, and given the aggregate strategies of the others, there is a unique maximum, denoted by point N, that trader i can achieve.[5]

Second, if there were only one commodity for trade and that money were fiat, the indifference contours for trader i would be parallel to the money axis; hence the optimum would occur at the point P_1. But this is assuming that other traders would offer positive amounts of goods while trader i would offer no goods and all i's fiat money. It is easy to check

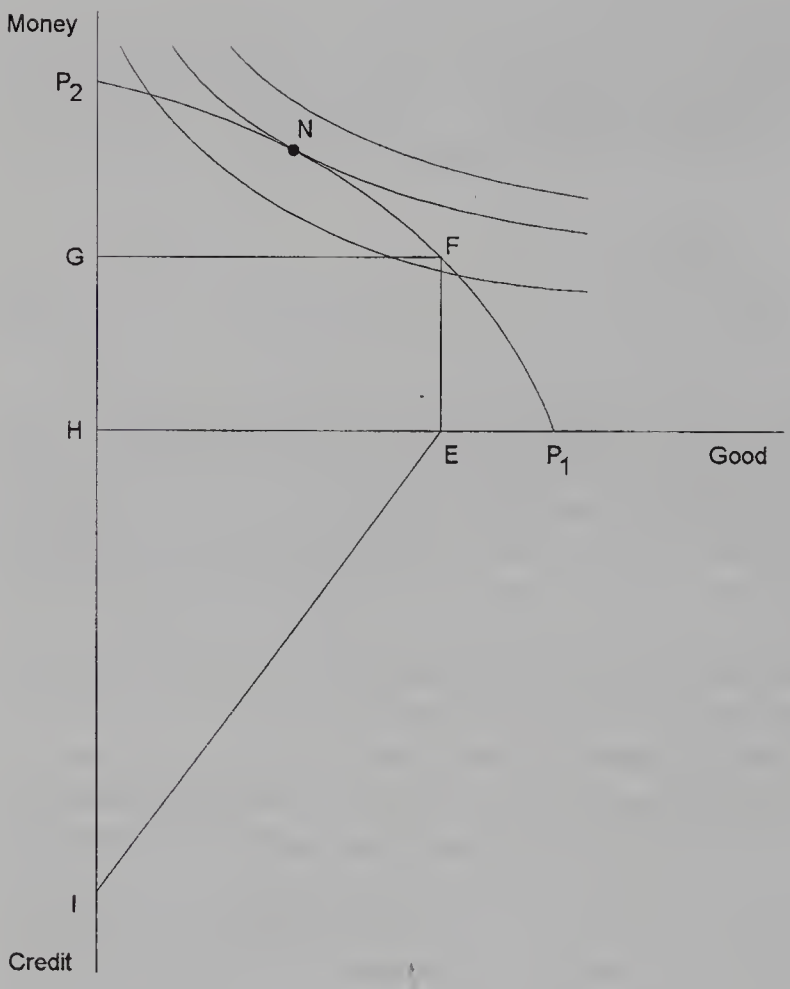


Figure 7.2
Oligopolistic budget set.

that when there is only one good for sale and trade is in fiat, no goods will be offered in the one period model and no trade will take place. This is not true for more than one commodity provided the appropriate bankruptcy laws accompany the use of fiat for trade. (This point is discussed in detail in chapter 11.)

7.1.4 A simple example

To examine the nature of this mechanism, let as consider a simple two-trader example of the "offer for sale" model before discussing the general results. It is important to distinguish clearly the three features that fully define the game, namely, (1) the bids and offers, which describe the moves;

(2) the markets, which transform moves to outcomes; and (3) the information conditions, which influence the domain of the strategies.

Here we assume that the trader can both bid and offer in the same market. The reader might wonder if this is a reasonable condition. Why not assume directly that if a trader wishes to buy a commodity, the trader will not simultaneously sell it? Rather than make this assumption, we prefer either to prove it or to show that it is false. In a world with taxes, transaction costs, thin markets, and other imperfections (especially in information conditions), it is not uncommon to have a trader buy and sell the same item in the same time period. For example, within the year, one might sell a stock and buy it back a month later in order to establish a tax loss while maintaining one's inventory.

Consider two sets of traders n, each trading two goods, one of which may be regarded as money, and differing only in their endowments. They all have utility functions of the form

$$u^i = \log x_i + \log m_i. \tag{7.14}$$

Trader i of type 1 has resources of (10, 30), and trader j of type 2 has resources of (30, 10). We might expect that a trader of type 1 will try to maximize his utility by bidding for the good, while one of type 2 will offer it for sale. Final allocations will be given by

$$u^1 = \log\left(10 + \frac{b_i}{b}q\right) + \log(30 - b_i) \tag{7.15}$$

and

$$u^2 = \log(30 - q_j) + \log\left(10 + \frac{q_j}{q}b\right). \tag{7.16}$$

The first-order conditions for optimization call for

$$\frac{1}{10 + \dfrac{b_i q}{b}}\left[q\left(\frac{b - b_i}{b^2}\right)\right] = \frac{1}{30 - b_i} \tag{7.17}$$

and

$$\frac{1}{30 - q_j} = \frac{1}{10 + \dfrac{q_j b}{q}}\left[b\left(\frac{q - q_j}{q^2}\right)\right]. \tag{7.18}$$

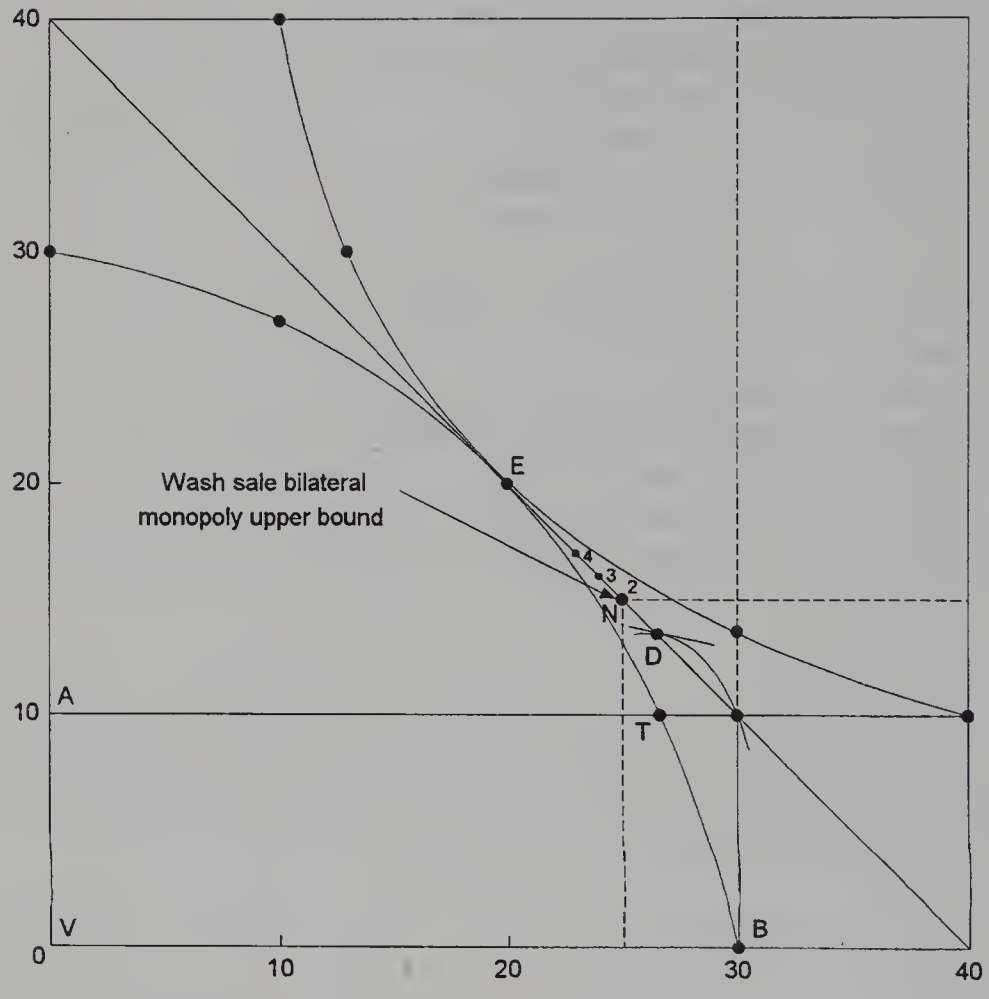


Figure 7.3
Equilibrium and thin markets.

From (7.17) and (7.18), it can be seen that

$$q_i = b_j = \frac{10(2n-3)}{(2n-1)}. \tag{7.19}$$

Figure 7.3 shows the Edgeworth box and the noncooperative equilibria on the main diagonal. The limiting NE is the competitive equilibrium shown in the center of the box at point E, where the price line is tangent to the indifference contours of both traders, and where their final allocation sets, curved as shown in figure 7.1, both flatten out and approach the main diagonal as $n \to \infty$.

Because the final allocation sets for traders of each type pass through the initial allocation point T and the equilibrium D at the NE for $n = 2$, the allocation at D is attainable by all simultaneously and the final allocation sets of traders i and j are tangent to their respective indifference contours. This Edgeworth box is drawn to illustrate only symmetric strategies by traders of all types with the possible deviation by one trader. The diagram should be more properly viewed as being in $2n$ dimensions.

Suppose that instead of restricting themselves to either bids or offers (this is equivalent to staying on the L-shaped boundary ATB of the rectangle $VATB$), traders make both bids and offers, thereby creating "wash sales" (i.e., selling and buying back the same commodity) and creating gross trade that is more voluminous than net trade.

We can write the payoffs to the traders as follows:

$$u^1 = \log\left(10 + b_i\frac{q}{b} - q_i\right) + \log\left(30 - b_i + q_i\frac{b}{q}\right) \tag{7.20}$$

and

$$u^2 = \log\left(30 + b_j\frac{q}{b} - q_j\right) + \log\left(10 - b_j + q_j\frac{b}{q}\right). \tag{7.21}$$

Hence from (7.21), we obtain

$$\frac{1}{10 + \dfrac{b_i q}{b} - q_i}\left\{\frac{q(b - b_i)}{b^2}\right\} = \frac{1 - \dfrac{q_i}{q}}{30 - b_i + \dfrac{q_i b}{q}} \tag{7.22}$$

and

$$\frac{1 - \dfrac{b_i}{b}}{10 + \dfrac{b_i q}{b} - q_i}\left\{\frac{b(q - q_i)}{q^2}\right\} = \frac{1 - \dfrac{q_i}{q}}{30 - b_i + \dfrac{q_i b}{q}}, \tag{7.23}$$

but multiplying (7.22) by b/q, we see that it gives (7.23).

Similarly, from (7.21), we obtain

$$\frac{1 - \dfrac{b_j}{b}}{30 + \dfrac{b_j q}{b} - q_j} = \frac{b\left(\dfrac{q - q_j}{q^2}\right)}{10 - b_j + \dfrac{q_j b}{q}}. \tag{7.24}$$

If (7.23) and (7.24) had a solution other than the one previously obtained with $b_j = q_i = 0$, wash sales would play a role in our equilibrium. The equations have many solutions, and an example is calculated in section 7.1.5.

Wash sales can create "thicker markets" and hence lower the impact of change by a single individual. Paradoxically here, the market with heavy wash sales is best for all traders. For the model with a continuum of traders, this trade-off between creating the effect of more traders by wash sales attenuates. The market is always thick.

When there are many traders (more formally a continuum), if we impose a "cash-flow minimization"condition, the indeterminacy is removed. For a market with atomic traders, however, there is no indeterminacy. There may be continuum of equilibrium points each with a different level of wash sales and a different value. Because it is possible that an equilibrium point with wash sales is better for all than one without, even with positive transaction costs there may be noncooperative equilibria with wash sales that are better for all than an equilibrium with gross trade equal to net trade. (For the general proof of the existence and behavior of the noncooperative equilibria, see Dubey and Shubik, 1978.)

7.1.5 Bilateral monopoly with wash sales

One can readily infer that the possibility of wash sales permits the existence of active noncooperative equilibria for bilateral monopoly, provided both sides want both goods and both are endowed with both goods. In table 7.1, where it was assumed that there are no wash sales (i.e., indi-

Table 7.1
Minimal bid equilibria

				Trader of type 1	
n	$n-1/n$	b	q	x	m
1	0	—	—	10	30
2	1/2	10/3	10/3	13/33	26/67
3	2/3	6	6	16	24
4	3/4	50/7	50/7	17/14	22.86
5	4.5	70.9	70.9	17.78	22.22
⋮	⋮	⋮	⋮	⋮	⋮
∞	1	10	10	20	30

viduals either buy or sell a given commodity), the NE for $n = 1$ has no active trade. The two traders merely maintain their initial endowments. Here suppose that both traders offer to sell up to their capacity the stock of the good they wish to buy. Computation based on equations (7.22) and (7.23) gives the existence of an NE with the most highly "inflated" strategies (or the thickest trade), where the strategy for trader i is (10, 15.6155) and for trader j is (15.6155, 10). The final allocation for trader i is (15.62, 24.38) and for trader j is (24.38, 15.62), noted on figure 7.3 by the point N. It can be seen that the outcome from the use of fully inflated strategies in bilateral monopoly noncooperative trade is better than the outcome from uninflated (i.e., no wash sales) trade with two traders on each side of the market. All traders obtain higher payoffs in this instance, an outcome denoted by the point D.

Trade in bilateral monopoly has a continuum of NEs illustrated in this example by all points in the interval NT. The point N, with the largest amount of wash sales, shows the highest outcome that can be achieved by the traders; the point T indicates the outcome with no wash sales. There will be a different range for each value of n; as n increases, both the upper and the lower bound will approach the point E.[6]

7.1.6 Multiple markets

In the offer for sale model studied in this section, given $(m + 1)$ commodities with the $(m + 1)$st serving as money, it has been assumed that there are m markets and that a strategy by an individual is a vector (s^i, b^i), of dimension $2m$. We could consider a model in which there are k trading posts for each good,[7] thus a strategy would involve not merely allocating money in m markets for different goods, but also allocating money in each of k markets for the same good. A strategy for each individual will be of dimension $2mk$.

Although the introduction of the extra markets enlarges the strategic freedom of the traders, for a fixed number of traders, the extra markets imply thinner trade on the whole. The existence of equilibria for this model follows from Dubey and Shubik, 1978, by reinterpreting the extra markets as markets for new goods which happen to be perfect substitutes.

In the actual economy, where transportation costs are usually important, multiple markets for real goods (as opposed to markets for financial instruments, where communication rather than transportation dominates) arise naturally from the trade-off between market thinness (caused by

adding markets) and transportation costs (lowered by a minimization of transportation distance).

7.1.7 The price level

Although, with very small traders and enough money, the NEs of the strategic market game coincide with the CEs of the related exchange economy both in the final distribution of resources and in relative prices (Dubey and Shapley, 1994), there is a natural normalization in prices in the game with a commodity money. Prices are determined relative to the worth of a unit of commodity money and are not homogeneous of order zero with respect to doubling the physical amount of money in the economy. There is a clear difference between actually having twice as much gold available and merely redefining the size of the unit of account.

In the general equilibrium formulation, money is only a unit of account. Here money is a physical entity. To be nonzero, trade requires the physical existence of money, which provides an upper bound to prices for a fixed velocity.

We begin our investigation of money by considering a commodity money because it helps to make precise the physical meaning of how much money there is in the system as well as clarify the important distinctions between money and credit.

7.2 The Double-Auction Market

When Mr. Jones sells 100 shares of General Motors at 43 and then goes short at 10 A.M. and the stock goes to 47 by 2 P.M. the friendly, kindly auctioneer does not step in and run a Walrasian tâtonnement, telling Mr. Jones that his sale at 10 A.M. was only "for fun." A week or so later, when Mr. Jones gets his check for somewhere around $4,100 and finds that General Motors has gone to 62, he has plenty of time to reflect about when to cover his short position.

When constructing a static theory of distribution and production—with no interest in process or control—one can abstract out of the actual process of price formation. When writing rules for the operation of a stock market, auctioning off cattle or used cars, or pricing groceries or dresses, this is not the case.

There are a host of different institutions and market mechanisms for moving goods at a price in a modern economy. Some involve the trans-

portation of the goods to a single point of sale, where buyers and sellers actually meet. Others involve goods stored in warehouses, where buyers and sellers anonymously, exchanging ownership claims through brokers.

To the outsider, the ongoing market appears to be a parallel process, with a price given mysteriously by the instantaneous matching of supply and demand. Individual traders see a sequential process where they make or fail to make advantageous trades. Both views may be consistent if individual behavior is independent and aggregated appropriately.

7.2.1 A price-quantity, buy-sell mechanism

In the sell-all and bid-offer market mechanisms, a trader's strategy for trading in m commodities using another commodity as a money consisted, respectively, of m numbers $(b_1^i, \ldots, b_m^i)$ and $2m$ numbers $(b_1^i, q_1^i, b_2^i, q_2^i, \ldots, b_m^i, q_m^i)$, where the numbers were quantities of goods or money. Here, where quantities of goods are bid for or offered together with prices bid or asked, our trader's strategy involves $4m$ numbers, and both prices and quantities appear as strategic variables.

We first describe a mechanism for a single market and then generalize for m markets. Suppose that n traders have endowments of two commodities, one of which serves as a means of payment and a numéraire. We fix the price of a unit of numéraire at 1.

All traders are required to move simultaneously without knowledge of each other's actions by bidding and offering in a market. The endowment of trader i is given by (a_1^i, a_2^i); a move by trader i (which is also i's strategy) is described by four numbers $(p^i, q^i, \tilde{p}^i, \tilde{q}^i)$. The number p^i is the maximum price (in terms of the numéraire) that i will pay to buy q^i or fewer units of the first good. We require that $p^i q^i \leq a_2^i$ (i.e., trader i cannot bid more money than i has on hand).[8] We assume $q^i \geq 0$. The number $\tilde{p}^i$ is the minimum price that i will accept to sell $\tilde{q}^i$ or fewer units of the first good. We require that $0 \leq \tilde{q}^i \leq a_2^i$ (i.e., i cannot offer for sale what i does not possess).

Although it may be unlikely that a trader will wish both to sell and buy the same item at the same time, there is no a priori reason to rule this behavior out. Hence the strategies employed here permit a trader to be on both sides of the market if the trader so chooses.

Figures 7.4a, 7.4b, and 7.4c show possible configurations of supply and demand for the market in aggregate. The supply schedules are obtained

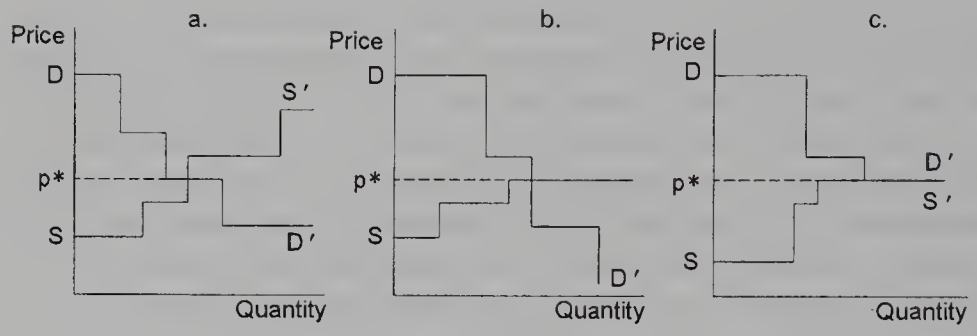


Figure 7.4
Bids and offers.

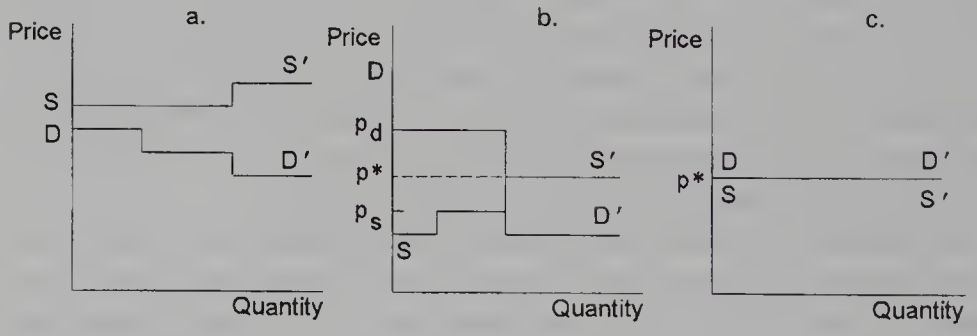


Figure 7.5
Indeterminate bids and offers.

by ranking the offer prices in ascending order and cumulating supply; similarly demand prices are ranked, but in descending order.

The market mechanism works as follows. The aggregate supply and demand schedules are calculated, and the market price is fixed at the price given by their intersection. All suppliers who require a higher price sell nothing; all buyers who require a lower price buy nothing. At the margin, as shown in figures 7.4a and 7.4c, there may be some buyers willing to pay the market price of p^* for whom there is not enough supply. Similarly there may be sellers willing to supply more at p^* than the buyers wish to buy. This is shown in figure 7.4b.

Figure 7.5 illustrates three further possibilities. In figure 7.5a, there will be no trade as there is no p^* for which both sides of the market will be active. Figure 7.5b illustrates an instance where there is a closed interval $[p_s, p_d]$ of prices at which effective demand will equal supply. This range is given by $p_s \leq p^* \leq p_d$. Figure 7.5c shows an instance where all buyers and sellers have their demands met.

When, there is excess demand at p^* (figures 7.4a and 7.4c) or there is excess supply (figure 7.4b), we adopt the convention that the marginal buyers or sellers are rationed in proportion to their demands or offers. In a market such as that shown in figure 7.5a, no trade will take place. In a market such as that shown in figure 7.5b, we assume that a price interior to the range $p_s p_d$ is selected randomly. The bid offer queue in a continuous double auction market will look like figure 7.5a.

7.2.2 The noncooperative equilibria

Let there be n individuals trading in m commodities using an $(m+1)$st commodity as a means of payment. Each trader i has an endowment of $(a_1^i, a_2^i, \ldots, a_{m+1}^i)$, where each is assumed to have enough money[9] and $a_j^i \geq 0$ for all $j = 1, \ldots, m+1$ and $i = 1, \ldots, n$; and $a_j (= \sum_i a_j^i) > 0$.

Each trader i has a preference ordering, which can be represented by a utility function $\phi_i(x_1^i, \ldots, x_{m+1}^i)$, assumed to be concave increasing and differentiable. It is further assumed that for any distribution of resources there are always at least two traders who would desire more than they have of some particular resource.[10] It is also assumed that in the initial distribution of resources at least two traders hold supplies of any particular resource.

A strategy by a trader i consists of a vector of $4m$ numbers $(p_1^i, q_1^i, \tilde{p}_1^i, \tilde{q}_1^i, \ldots, p_m^i, q_m^i, \tilde{p}_m^i, \tilde{q}_m^i)$ where

$$\sum_{j=1}^{m} p_j^i q_j^i \leq a_{m+1}^i \quad \text{and} \quad 0 \leq \tilde{q}_j^i \leq a_j^i \quad \text{for } j = 1, \ldots, m.$$

The market for each commodity $j, j = 1, \ldots, m$ operates as described in section 7.2.1.

THEOREM 7.1 The CEs of the trading market have associated[11] NEs in the price-quantity bid-offer game.

Proof The CEs are obtained by solving the trading economy consisting of n traders trading in $(m+1)$ goods. At a CE, aggregate supply will equal aggregate demand in each market. Consider all traders at a CE and allow some trader i to deviate. As can be seen from figure 7.4c, i can only influence price in a way that leaves i's payoff reduced or as before. (For example, i could indulge in a "wash sale" by offering k units of a

commodity at a price lower than p^* and simultaneously bidding for k units at that price. ■[12]

The size and smoothness of the gap will depend on the thickness of the market and the preferences of the traders. Competition among the intermediaries cannot eliminate the influence of a grid. If the market has a high money volume, the spread could easily compensate the intermediaries. Thus the selection of the fineness of the unit of price to be utilized is not an inconsequential assumption to those running the trading mechanism.

There remains the case shown in figure 7.5b. Consider the inactive trades first. Either they are not in equilibrium (hence this cannot be an equilibrium point) or they are. If they are, then the existence of an equilibrium depends on the active traders. Although the moves of the active traders are such that over the range $p_s p_d$ supply equals demand, excess supply or demand can be zero only at competitive equilibrium prices. Thus a case such as figure 7.5b can be in equilibrium only if there is a continuous range of CEs from p_s to p_d.

We must now show that if there is only one trader with excess demand or supply, the system may be in equilibrium. Suppose, for example, there were a single trader with excess demand Δq_j at p_j^* and suppose further that the trader would have to raise his price (and hence p_j^*) by Δp_j to have it satisfied. If the payoff

$$\phi_i(x_1^i, \ldots, x_j^i, \ldots, x_{m+1}^i) > \phi(x_1^i, \ldots, x_j^i + \Delta x_j^i, \ldots; x_{m+1}^i - p_j^* \Delta x_j^i),$$

then the marginal trade with excess demand is not motivated and the system is in equilibrium. Similar statements cover excess supply and demand and supply in more than one market.

7.2.3 On "enough money"

If we impose the condition that $\sum_{j=1}^{m} p_j^i q_j^i \leq a_{m+1}^i$ on the traders' bids, we are calling for a limitation that is highly restrictive if multiple bids are made in several markets. After all, whether you are bidding for oil leases or art or new faculty members simultaneously, the odds are usually high that you will not buy all that you bid for. To require 100 percent backing for every bid is thus ultraconservative; in some markets involving bids only 10 percent is required as earnest money (i.e., enough money to show that the bid is serious).

If we relax the 100 percent cash requirement, although it may well be rare to have a trader committed to buy more than the trader has money to finance, this event is nevertheless possible. Accordingly, to keep the process well defined, we must specify settlement rules for when a trader buys more than he can pay for. (This is closely related to insolvency and bankruptcy conditions and is treated in chapter 11.)

A fundamental distinction between the bid-offer (Cournot) market and the double-auction (Edgeworth-Bertrand) market is that in the former the actions are noncontingent,[13] whereas in the latter they are highly contingent. Personal prices and quantities are stated, the market then determines how much it will take and at what price. Individuals have the opportunity to lie. They may not have the stock they have offered. In the Cournot mechanisms, the move is a physical act: goods and money are shipped and quantities, not statements, about quantities form price. We stress the physical distinction between an offer to sell q units and the physical delivery of q units for sale.

7.2.4 Contingent bids

In our general discussion on bidding, we have frequently suggested that individuals could and should use contingent bids. For example, they might submit a complete demand schedule dependent on all prices in all markets. A trader's strategy might therefore be a general mapping that indicates how much the trader will buy or sell, given any m market prices.

The device of having contingent bids does not enable us to avoid specifying a market mechanism. Price must still be formed and rules provided for tie breaking and for resolving intederminate situations. One simple mechanism might be to aggregate all contingent bids and offers at the same price and to calculate and excess supply or demand function. If points of zero excess were discovered, the market manager would announce the price associated with one of them as the market price. Alternatively, we might select whatever price minimizes excess supply or demand as a market price and ration all individuals in proportion to their supply or demand at that price, should rationing be necessary.[14]

Unlike the simpler bids suggested in section 7.2, complex contingent bids require a more elaborate or complicated market mechanism to determine price; moreover, they may permit the signaling of threats and cause a proliferation of equilibria. In actual economic behavior, when buying a

pound of bacon, even if the purchasers were to base their act on the closing price of platinum, they would not in general convey this information to the butcher.

On retail markets Trade in retail markets is strategically and informationally different from that described above. Because this type of trade requires multistage consideration, more relevant to marketing than finance, we do not consider it further.

The CE solution for many multistage models is unsatisfactory because it implicitly ignores all information and credit conditions and the possibility that there may be large economic units. The NE solutions are also not totally satisfactory because, as information is taken into account, the strategy spaces may become enormous and the noncooperative equilibria may proliferate and include many alternative outcomes that do not appear to be reasonable.

The key element, central to resolving this unsatisfactory state, is the correct portrayal of the low information and communication conditions that characterize much of mass-market activity. Many of the ad hoc models of macroeconomics at least appear to offer a dynamics of behavior based on more realistic information conditions than are implicit in the Walrasian system; but these conditions are described less precisely than is required by a game-theoretic analysis.

7.2.5 Competitive and noncooperative equilibria without trust

We have been concerned with showing the conditions under which the NEs of a strategic market game coincide with the CEs of the related exchange economy. Although it is always desirable to see if the results of one approach match the accepted results of another, if that were all, there would be no "value added" in the adoption of the new approach.

A basic reason for using the strategic market game approach, in contrast with the competitive equilibrium, is that in the former, the need for trust is replaced by cash payment, whereas in the latter the need for trust is implicit.

THEOREM 7.2 Consider the class of all exchange economies defined by a set N of trader types, each composed of a continuum of small individuals, a utility function for each, and a set M of goods with a given supply of each distributed in any manner whatsoever. In association with each member of this class of exchange economies, we may construct m related

strategic market games, (where m is the number of elements of M) distinguishing each market game by using a different single commodity as money. If there is at least one commodity satisfying the double coincidence of wants with all other commodities and it is in sufficient supply, there will always exist a robust set of strategic market games whose NEs will coincide with the CEs of the related exchange economy.

The proof is simply that the strategic market games must satisfy not merely the budget constraint but also a cash payment constraint, which in essence is a no-trust or sufficient double-coincidence-of-wants constraint. In the class of all exchange economies, there will always be a set whose initial endowment points are on or near the Pareto optimal surface in the sense that they require little or no trade. If the net trade (denominated in terms of the money) in all commodities other than the commodity money is less than or equal to the desired change in money holdings for all types, then there will be enough money. In other words, it will be possible to satisfy the cash payment constraint.

When the distribution of the money is such that efficient trade cannot be attained, the society may wish to substitute trust, credit, and law to provide a workable substitute to behave as though the double coincidence of wants was being met at each trade.

In summary, we may consider the set of all CEs of the class of exchange economies obtained by considering all distributions of the total endowments. We may then examine the set of all NEs that are solutions to the class of all associated strategic market games. A subset of these NEs will coincide with the CEs of the related exchange economies—the class of efficient trades that can be reached with no trust, credit, or exogenous laws required beyond the rules of market exchange or the commercial codes governing direct payments and delivery.

7.3 Trade with Two or More Monies and Multiple Markets

7.3.1 Bimetallism

On occasion, various countries have used two metals for their monies and have attempted to fix the exchange rate between the two. There is no difficulty in using more than one good as a medium of exchange. Interference with the optimal functioning of a price system occurs only when

an attempt is made to fix the exchange rates between the monies. The reductio ad absurdum observation is that if all goods could serve as a money (as is the case with complete markets) and the exchange rates were fixed between all monies, then there would be no free price system because all prices would be frozen. When only two goods are used as money, the effect of pegging their exchange ratio is to take away a free market and cut down the strategic freedom of the individuals. If we were to consider the government mint as an active special player in the game, it could import or export gold and silver in such a way that the price ratio between the two could be stabilized. The usual pair of metals for a bimetal standard are gold and silver, but prior to 1772 Sweden used copper and silver (Buist, 1974). The history of bimetallism (see Giffen, 1898; Schumpeter, 1955, pp. 288–299; Kindleberger, 1984, pp. 55–70) is filled with examples of the international flows of gold or silver as different countries by law, custom or bureaucratic inadvertence tried to maintain a fixed ratio between the two.

The reason for utilizing two metals such as gold and silver is that one needs coins of reasonable size for both large and small payments.[15] Thus a ratio of around 13 to 1 was handy. When large price differentials developed, as happened between Europe and China, the arbitrage possibilities resulted in a flow of silver to the East. (Figure 7.6a shows an economy with two monies, and figure 7.6b shows one with complete markets or all commodities serving as a direct means of payment.)

7.3.2 Complete markets with trading posts

It is possible to build and analyze strategic market games with any number of markets. But the two most interesting and extreme or bounding

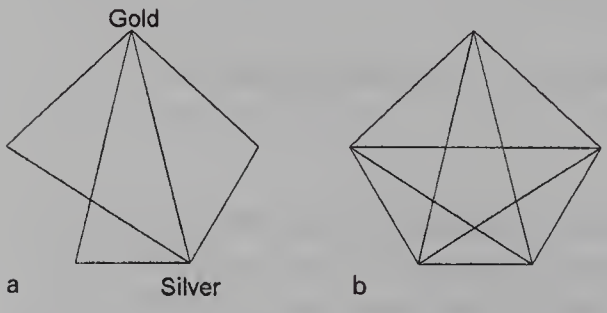


Figure 7.6
Two moneys or complete markets.

structures are when there is exactly one money and when there are complete markets. Several models with only one money have already been constructed and analyzed. Here the other extreme where all markets exist is considered.

The strategy set can be formally described as follows. Let $I_n = \{1, 2, \ldots, n\}$ and $I_m = \{1, 2, \ldots, m\}$ be the sets of traders and commodities, respectively; where both m and n are at least 2. Each trader α has a positive initial endowment $a_i^\alpha > 0$ of each commodity i. The traders' utility functions u^α are assumed to be strictly concave, increasing functions from the nonnegative orthant $\mathbb{R}_+^m$ to $\mathbb{R}_+$; which are continuously differentiable in the interior of the nonnegative orthant.

A bid by trader α is an $m \times m$ matrix B^α such that

$$b_{ij}^\alpha \geq 0 \quad i, j \in I_m \tag{7.25}$$

and

$$\sum_i b_{ij}^\alpha \leq a_j^\alpha \quad j \in I_m,\ a \in I_n. \tag{7.26}$$

The ith column of the matrix B^α is considered the vector of commodities that trader α offers in exchange for commodity i. The *strategy set* of α is the set of all matrices B^α satisfying (7.25) and (7.26), and is denoted by S^α.

It has been proved that for a strategic market game with complete markets and bid-offer strategies, a pure strategy noncooperative equilibrium will exist (see Amir et al., 1990). For a finite number of individuals, it is possible (but not necessary) that all could be made better off by restricting the number of markets they use. Given a fixed number of individuals with fixed resources, the more markets they use, the thinner must be the level of trade in the markets.

The proof of existence of a pure strategy NE when there are complete markets poses difficulties not encountered when there is only one money. In particular, when the strategies of all other traders are taken as fixed and there is only one market for each commodity, the optimization problem faced by the remaining trader is of one dimension in each good. When all markets exist, then in the purchase of any good the trader has m degrees of freedom.

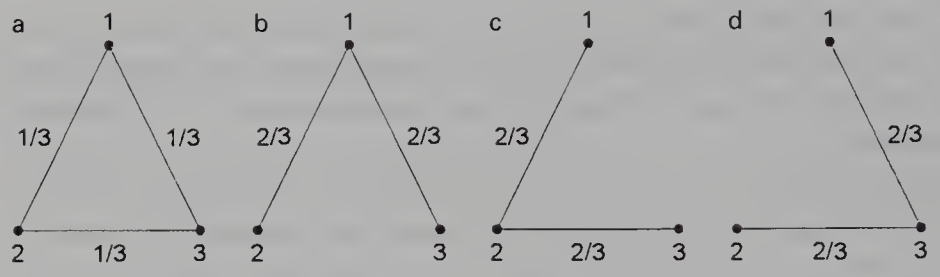


Figure 7.7
Equilibria using different markets.

Even with a continuum of traders and complete markets, there may be a set of NEs with different volumes of trade associated with each CE. This can be illustrated in a simple example. Suppose all traders have utility functions of the form $(x_1 x_2 x_3)^{1/3}$. Let the endowments of traders of type 1, 2, and 3 be $(2,1,1)$, $(1,2,1)$, and $(1,1,2)$, respectively. The (unique from the viewpoint of final holdings) CE is $(4/3,4/3,4/3)$ for each with prices of $(1,1,1)$. This can be achieved by four different NEs, each of which gives the same final distribution but has different markets active and different volumes of trade. In figure 7.7a, all markets are active with offers of all traders that can be described as follows (for types 1, 2, and 3): $(1/3,0,1/3,0,0,0)$, $(0,1/3,0,0,1/3,0)$, and $(0,0,0,1/3,0,1/3)$. The first two numbers represent trade in the 12 market, the next two in the 13, and the last in the 23 market. Thus the strategy for a trader of type 1 indicates an offer of 1/3 of good 1 in the 12 market, 1/3 of good 1 in the 13 market, and nothing in the 23 market.

In figure 7.7b, although final allocations are symmetric, the strategies of the traders are no longer symmetric. They are $(1/3,0,1/3,0)$, $(0,2/3,1/3,0)$, and $(1/3,0,0,2/3)$, where the first two numbers are for the market 12 and the next two for 13. There is no 23 market.

In figures 7.7a–7.7d, the numbers next to each arc represent the volume of trade using commodity 1 as the numéraire. (In this instance, because prices are $(1,1,1)$, all measures give the same numbers). The volume of trade is least with all markets active; it is 2, as compared to 8/3 when only two markets exist.

If the initial endowments were $(3,0,0)$, $(0,3,0)$, and $(0,0,3)$, the final distributions would be $(1,1,1)$ and prices would be $(1,1,1)$, but only the NE with all markets active could achieve this. If trade were restricted to

fewer markets, at least one type of trader could not purchase optimally without credit.

7.3.3 Enough money in the trading post model

When everything exchanges for everything, there is no cash-flow constraint to trade. All goods serve as a means of payment. Although it would appear that a trader's net worth and liquidity coincide, we have not yet formally defined liquidity, and as indicated by the second example below, some difficulties appear. When only one good serves as a means of exchange then a trader's liquidity may differ considerably from the trader's wealth. If there is to be enough of a commodity money available so that efficient trade can be achieved, we require not merely the usual budget constraint that

$$\sum_{j=1}^{m} p_j(a_j^i - x_j^i) = 0 \quad \text{for all } i \tag{7.27}$$

but also the cash flow constraint that

$$\sum_{j=1}^{m-1} p_j \max[(x_j^i - a_j^i), 0] \leq a_m^i, \tag{7.28}$$

where $p_1, \ldots, p_{m-1}$ are prices with $p_m = 1$, the a_j^i and x_j^i are initial and final endowments, and a_m^i can be interpreted as the holding of the monetary commodity.

It is possible that if an inappropriate commodity were chosen as a money, there would never be enough, even if more were added. The price of money relative to other commodities might fall sufficiently fast that the extra money would not increase its total value in trade. An example is as follows. All traders have utility functions of the form

$$u^i = x_1^{i^{2/5}} x_2^{i^{2/5}} x_3^{i^{1/5}}.$$

Traders of type 1 have endowments of $(2, 0, a)$, and those of type 2 have $(0, 2, a)$. The third commodity is used as the money. At the CE, prices are $(2a, 2a, 1)$ and the final distribution will be $(1, 1, a)$. Each trader needs a purchasing power of $2a$ but never has more than a.

Generally, a sufficient condition to ensure that by adding money to the economy, enough money can be reached for any endowment of the mth good is

$$\frac{\partial u^i/\partial x^i_m}{\partial u^i/\partial x^i_j} > \varepsilon,$$

where $j = 1, \ldots, m-1$, and ε is some small positive number.

In the example given,

$$\frac{\partial u^i/\partial x^i_m}{\partial u^i/\partial x^i_j} = \frac{1}{2}\left\{\frac{x^i_j}{x^i_3}\right\},$$

which is arbitrarily small for x^i_3 sufficiently large.

Even when all pairwise markets exist, it is possible that efficient trade cannot be achieved in one trade without credit. Consider three types of traders characterized by utility functions and endowments of the following form:

$U_1(x^1_1, x^1_2, x^1_3) = 4x^1_1 + 2x^1_2 + x^1_3$ and $(0, 3, 0)$,

$U_2(x^2_1, x^2_2, x^2_3) = x^2_1 + 4x^2_2 + 2x^2_3$ and $(0, 0, 3)$,

$U_3(x^3_1, x^3_2, x^3_3) = 2x^3_1 + x^3_2 + 4x^3_3$ and $(3, 0, 0)$.

The CE has prices $p_1 = p_2 = p = 1$ and final allocations of (3, 0, 0), $(0, 3, 0)$, and (0, 0, 3). In this game, the pairwise prices are $p_{12} = p_{23} = p_{31} = 1$, but at those prices, trader 1 offers to trade good 2 for 1, but no supply of good 1 is offered for good 2 (and similarly for the others).

The market cannot achieve the CE. Thus we observe that trade with complete markets is not the same as total liquidity. Even with complete markets, the distribution of holdings may limit trading possibilities. As is shown in section 9.3, when there is only one money both its quantity and distribution influence trade. When all goods serve as a money, their distribution may limit trade. Thus, in particular, (some or all of) the CEs of an exchange economy may not be attainable by trade in the associated strategic market game. If there are only two types of traders, m goods, and all pairwise markets, or alternatively, if there are n types of traders but only two goods, there will always be enough money that is well distributed.

7.3.4 Nonequivalence of competitive and noncooperative equilibria

As already noted in section 3.2, there can be many NEs with different volumes of trade that give rise to the same CE. A natural question to ask

is, are there NEs in the game with complete markets that are not CEs? The answer is yes and is established by the example below, where there are four types of traders, a continuum of each type, and three commodities.[16]

Let the utility functions and endowments be

$u_1(x_1, x_2, x_3) = (x_2 x_3^2)^{1/3}$ and $(0, 3, 0)$,

$u_2(y_1, y_2, y_3) = (y_1^2 y_3)^{1/3}$ and $(0, 0, 3)$,

$u_3(z_1, z_2, z_3) = (z_2 z_2^2)^{1/3}$ and $(3, 0, 0)$,

$u_4(w_1, w_2, w_3) = (w_1 w_2 w_3)^{1/3}$ and $(1, 1, 1)$,

where x_j, y_j, z_j, and w_j are the holdings of good j by traders 1, 2, 3, and 4 respectively.

Traders 1, 2, and 3 are willing to exchange 2 units with trader 4 for their more desired good. Trader 4 trades to the full capacity in each market trading completely around the arbitrage circuit. The final NE allocations are

$(x_1, x_2, x_3) = (0, 1, 1)$,

$(y_1, y_2, y_3) = (1, 0, 1)$,

$(z_1, z_2, z_3) = (1, 1, 0)$,

$(w_1, w_2, w_3) = (2, 2, 2)$.

However, the unique CE is easily seen to consist of the price vector $(1, 1, 1)$ and gives rise to final allocations of $(0, 1, 2)$, $(2, 0, 1)$, $(1, 2, 0)$ and $(1, 1, 1)$. It can be checked that the CE is not feasible in this trading mechanism with the given endowments.

The NE violates the no-arbitrage condition[17] that $p_{ij} p_{jk} p_{ki} = 1$. It would be destroyed as an NE if credit were available. It depends on there being only one round of trade so that an individual trader can take only limited advantage of the full arbitrage circuit for $p_{ij} p_{jk} p_{ki} > 1$.

It might be thought that with a continuum of traders and complete markets there would be an equivalence between CEs and NEs. Surprisingly, this is not the case. There may be CEs that are not NEs, and there may be NEs that are not CEs.

A key point to observe in this trading post model is that trade is completely decentralized. There is no clearinghouse or central agency of any

variety, each market clears independently, thus a double coincidence of wants is a necessity if efficiency is to be achieved in one round of trade. The model suggested here is a strategic extension of the Cournot mechanism. We could equally well have set up a double-auction mechanism in each market and obtained similar results.

7.3.5 Complete markets: The window model

Shapley has suggested a somewhat different model than the one discussed in section 7.3.2. In this variation, the bids and offers of the traders are the same as previously, but rather than consider clearance in all pairs of markets, there are m "windows," one for each good. All offers to sell and bids to buy are aggregated at each window, and a central agency calculates prices to clear all markets. The traders' demand and supply in each market will be equated in terms of the overall economy clearing prices.

This is a highly centralized and sophisticated mechanism with a central clearinghouse carrying out a massive coordinated computation. Because consistent prices are imposed by assumption, the nature of the mechanism rules out all arbitrage.[18] Sahi and Yao (1989) have established existence of pure strategy NEs for this model. For a continuum of traders, the NEs and CEs coincide.

An important distinguishing feature to note between the trading post model and the window model is that the latter *requires* the existence of a clearinghouse and the concomitant bookkeeping, trust, and accounting that goes with the netting out of trade, while the former does not. In essence, the mechanism grants all the short-term credit of being able to use the proceeds of sales to cancel out purchases even if made simultaneously or slightly before or after each other.

7.3.6 Market structures and prices

Markets and trading methods are peculiarly institutional. The detailed dynamics of price formation may depend on many special variables. The approach adopted here is to concentrate on the logically simplest complete mechanisms and to add complications one at a time. Thus we must stress that the realism or relevance of any specific price formation mechanism is not asserted. We first concentrate on a simple, fully defined mechanism and then consider general properties of markets, attempting axiomatic descriptions for general classes of mechanisms (see section 5.4).

But, for the purposes of understanding the roles of money and credit, this is more of a luxury than a necessity.

Notes

1. With a continuum of traders, a coincidence between the noncooperative equilibria and the competitive equilibria can be established. See Dubey and Shapley, 1994.

2. For simplicity in describing the strategies of a single trader, the superscript i has been dropped.

3. By $S_j^i(w_j^i, Q_j, B_j)$ we denote the set of all strategies that give w_j^i of commodity j to trader i, assuming that the total supply (bid) of the others in the jth market is $Q_j(B_j)$. This is a convex set and is clearly continuous in its variables.

4. Our discussion assumes either that money is sufficiently plentiful not to require credit to finance the bids or that credit is costlessly granted and repaid at the end of the period.

5. This is true only if the indifference sets are not "thick."

6. Both Yao (1987) and Peck and Shell (1986) have utilized the concept of an inflated bidding strategy in establishing the existence of equilibria.

7. This model was suggested by Shapley.

8. This condition rules out submitting false bids, which may not be reasonable. After all, it is administratively easy to punish those caught with bids they cannot honor, but it is unlikely that one can punish those who submitted false bids and had them rejected. The information that they were infeasible may never be known.

9. "Enough money" has been defined in chapter 6, but for this market, some additional observations are called for See section 7.2.3.

10. Precisely, a trader i desires good j if $\phi^i(x^i)$ is an increasing function of x_j^i for any fixed choice of the other variables.

11. An NE is associated with a CE if it has associated with it the same relative prices, distribution of goods, and payoffs. For "CE" read "competitive equilibrium" and for "NE," "noncooperative equilibrium."

12. Note that if the bidding grid is coarse, for example, 1/8 or 1/16 of a point, as in many stock exchanges, there will exist a continuum of NEs that differ from the CEs and hence manifest an excess supply or demand. This excess will call for rationing and possibly a buy-sell spread in prices with revenues that can be absorbed by the intermediaries making the market.

13. Except when a market is completely inactive on one side and the goods on the other side must be returned or otherwise accounted for.

14. The tâtonnement process can be regarded as one type of contingent demand mechanism, possibly practical for stockmarket trading, but little else.

15. Marshall (1923, p. 64) has suggested the use of both gold- and silver-backed paper, where the government fixes the ratio of gold to silver and offers redemption of notes in fixed proportions of gold to silver, as though the backing were a single commodity.

16. Similar examples hold true for a game with a finite number of traders. The needed computations are naturally much more tedious. Yao, 1987, provides a twelve-person example.

17. The no-arbitrage condition is based on the implicit assumption that there will always be credit available to close any arbitrage gap.

18. There does not appear to be any simple natural extension of the Edgeworth-Bertrand or double-auction market model when all prices are to be determined centrally.

8 Auctions, Markets, and Trade Fairs

8.1 Auctions and Tâtonnement

8.1.1 Tâtonnement

In a variant of the double-auction market, tâtonnement, an auctioneer calls out a price and lists all bids and offers made at that price. If these fail to match, the auctioneer raises or lowers the price until by bidding and offering again a sufficient number of times supply and demand are equated. At that point, the price is fixed and trades take place. The Walrasian auctioneer appears to have been based on this method of trading, which was once used at the Pairs Bourse, where a list of stocks is processed one at a time by an auctioneer.

We may apply this procedure to the economic structure described in chapter 7.1. Each trader i has an endowment of $(a_1^i, \ldots, a_m^i, a_{m+1}^i)$ and preferences represented by a utility function $\varnothing_i(x_1^i, x_2^i, \ldots, x_{m+1}^i)$. To fully define the procedure, we must specify the details of the bidding mechanism. One possible way is as follows. Goods are traded in order $1, 2, \ldots, m$. The auctioneer lowers or raises the price in a fixed finite increment Δp_j and begins by randomly selecting an initial price from a range 0 to p_j. If supply equals demand, trade in that commodity takes place. If excess supply or demand is not zero, the price is moved until they match. The procedure stops after all m commodities have been offered for trade.

If we limit bids and offers to discrete units, a strategy by a trader will be complicated but finite. It is easy to see that in general the outcome will not be Pareto optimal, although in special cases, such as when the utility for all items is separable and additive $\varnothing_i(x_1^i, \ldots, x_{m+1}^i) = \sum_{j=1}^{m} a_j^i x_j^i + x_{m+1}^i$, it will be optimal. This method of auctioning is fast and easy to administer.

There is a growing literature on the theory of auctions. The stress is on uncertainty, nonsymmetric information, and the fact that auctions often involve indivisible and unique items for sale (see Engelbrecht-Wiggans, 1980, and Milgrom and Weber, 1982, for surveys). Because, however, much of this work is only of secondary importance to understanding money and financial institutions, it is not pursued further here.

8.1.2 A history of auctions

As a method of selling, auctions have only appeared in comparatively civilized societies meeting two necessary conditions: an adequate concentration of the population to provide sufficient numbers of buyers and

sellers, on the one hand, and a coinage so that the values of bids are immediately recognizable on the other.[1] Thus, before the seventeenth century, there were few auction sales anywhere that were not sporadic, and many were on an annual basis. Auctions found little favor in the Orient, where bargaining has been traditional, little importance being attached to the time taken to make a sale. Consequently, the rise of the auction appears to have followed economic development in the West.

The earliest reference to a regular organized auction market is probably Herodotus's description of the annual Babylonian marriage market (Rawlinson, 1935, p. 128), unique in that bidding often started at a negative price:

> Once a year in each village the maidens of age to marry were collected all together into one place, while the men stood round them in a circle. Then a herald called up the damsels one by one, and offered them for sale. He began with the most beautiful. When she was sold for no small sum of money, he offered for sale the one who came next to her in beauty. All of them were sold to be wives. The richest of the Babylonians who wished to wed bid against each other for the loveliest maidens, while the humbler wife-seekers who were indifferent about beauty, took the more homely damsels with marriage-portions. For the custom was that when the herald had gone through the whole number of the beautiful damsels, he should then call up the ugliest—a cripple if there chanced to be one—and offer her to the men, asking who would agree to take her with the smallest marriage-portion. And the man who offered to take the smallest sum had her assigned to him. The marriage-portions were furnished by the money paid for the beautiful damsels, and thus the fairer maidens portioned out the uglier. No one was allowed to give his daughter in marriage to the man of his choice, nor might any one carry away the damsel whom he had purchased without finding bail really and truly to make her his wife; if, however, it turned out that they did not agree, the money might be paid back. All who liked might come even from distant villages and bid for the women. This was the best of their customs, but it has now fallen into disuse.

Auctions were used to dispose of mine concessions in ancient Greece and became a regular event in Roman times, when they were used mostly for selling shared and confiscated property. It is from Latin that the trading words "auction," "subasta," and "encan" are derived. "Auction" comes from *auctio*, (increase); "encan" comes from *incanto*, (I cry); "subasta" is a shortened form of *sub hasta*, (under spear). Roman soldiers would auction off surplus spoils of war on the battlefield by driving a spear into the ground and selling off the booty gathered round it to the highest bidders. This led to slave auctions being held under the sign of a

spear in the forum, thus perpetuating the word. Often the auctioned slaves were captives from a Roman military victory, and the money obtained was used to help pay for the campaign.[2]

One of Cicero's court cases circa 80 B.C. concerned the hasty auction of property left as a surety by his client, who had not expected it to be sold. It was customary for debtors' property to be confiscated and auctioned, and Sulla extended it to enemies of his regime, eliminating all opposition and gaining enormous amounts of money for his treasury at the same time. Ovid, in A.D. 10, mentions the practice of selling the rights to the state revenues to the highest bidder, relieving the government of collecting them itself.

The status of auctioneers is indicated by Juvenal who, in his third satire in A.D. 117, classes them with various unsavory types. He is more charitable in his seventh satire:

> In these sorrowful days, when poets of high reputation in order to make a few cents, think of a bakeshop here in Rome, or perhaps try crying the sales at an auction. None too great a disgrace, after all; any one of the Muses Starving, could hardly be blamed for leaving Helicon's fountains in hot haste for a job in the auction rooms of the city, more or less content with Machaera's trade and his profits. Bawling "Going! Gone!" over wine casks, bookcases, tripods, copies of plays, for example the "Thebes" of "Tereus" of Faustus, Paccius' masterpiece, the one about Minyas' daughter changed to a bat. A better career, to be sure, than appearing in some police-court case, claiming you saw what you did not. Leave all that to our new-made knights, the ex-slaves from Asia, Cappadocia, Gaul, with the chalk marks still on their ankles!

After the murder of the Emperor Pertinax in A.D. 193, the Praetorians auctioned off the empire between two contenders and elected emperor Didius Juliannus, who offered the larger bribe and who stayed in power for two months until overthrown and executed by Septimus Severus.

Between the end of the Roman Empire and the eighteenth century, there were few auctions. In the Middle Ages, self-sufficiency was the rule and the great mass of the population had little money to spend. The feudal peasants owed part of their time and crops to their lords and otherwise grew and made only what they needed. Any necessities they could not fabricate for themselves they were able to buy at the annual fairs or from others by barter. The low density of the population and the small quantity of circulating coinage made it virtually impossible to hold auctions. In France, the office of Huissier-Priseur was created in 1556 to appraise and

sell property left by death or taken in execution. In the Charter of London granted by Charles I in there was only one auctioneer for the whole of London. In 1649, the household belongings of Cardinal Mazarin were sold "au plus offrant" (to the highest bidder). Matthias Nicoll was appointed "vender master" in New York in 1676, and given £2,000 security, which indicates the importance of the post at the time.

Auctions became commonplace in the seventeenth century. There were four distinct types—auctions using the hammer, the hourglass, or the candle, and Dutch auctions, which originated in Holland and were not much used. The hammer auctions were of the type normally used today. The hourglass and candle auctions allowed the bidders only a limited time for bidding and were very similar, except that the candle auctions afforded greater opportunity to use skill. An account of one is given by Pepys in his diary for November 6, 1660:

> To our office, where we met all, for the sale of two ships by an inch of candle (the first time that ever I saw any of this kind), where I observed how they do invite one another, and at last how they all do cry, and we have much to do to tell who did cry last. The ships were the Indian, sold for £1300, and the Half-moon, sold for £830.
>
> After dinner we met and sold the Weymouth, Success and Fellowship hulks; where pleasant to see how backward men are at first to bid, and yet, when the candle is going out, how they bawl and dispute afterwards who bid the most first. And here I observed one man cunninger than the rest, that was sure to bid the last man, and to carry it; and enquiring the reason, he told me that, just as the flame goes out, the smoke descends, which is a thing I never observed before, and by that he do know the instant when to bid last.

The earliest catalog extant is that for a sale of books belonging to Ebenezer Pemberton at Boston in 1717. Auctioneers soon obtained a bad reputation for dishonest practices, and in 1799, the Select Society of Auctioneers was formed in England for the purpose of training them and making the occupation respectable. This body still exists today, but only in token form.

In 1844, Garraways in London was operating in a specially designed building, where auctions for various kinds of goods could be held by appointment. Tattersall's, also in London, auctioned horses of all kinds at that time. Nowadays, it holds its auctions in Newmarket and specializes in racehorses. Founded in 1886, the Auctioneers' and Estate Agents' Institute of the United Kingdom organized British auctioneers and did much to improve their reputation.

In 1852, the notorious Hôtel Drouot was built in Paris amid general acclaim, replacing the Hôtel Bullion as the auction center in Paris; called the "Temple du Bric-à-Brac," it still retains its position and deals in paintings, furniture, collections, and large quantities of junk. The prices obtained are influenced by the time of day, and considerable profits have been made by buying at 2 P.M., when the auctions start, and selling at 4 P.M., when more people attend. Although auctions in France follow the same pattern as elsewhere, the post of auctioneer must be bought or inherited, their being a fixed number of them. The auctioneer is known as the "commisseur-priseur". The *Revue de Paris* of 1934 notes that 'On nait commisseur-priseur prestigieux comme on nait rotisseur' and that a sale by a master-auctioneer may realize up to 30 percent more than one held by "un collègue apathique, distrait, à la digestion laborieuse" (an indifferent, absentminded, dyspeptic colleague).

French law in the 1950s still demanded the use of the inch of candle method for auctioning real estate. The object is to allow the bidding to be unhurried. The auction is not closed until three candles lit successively after the last bid have gone out. If in that time more bids are made, the auction is not closed until two candles have gone out after the new "final" bid. An alternative timing mechanism has been sought, but none more satisfactory has been found. Anyone may bid during the eight days following this initial auction, but such a bid must be at least one-sixth higher than the one reached then.

Although auctions may predate the reference given by Herodotus, it appears unlikely they predate coinage. While auctions may have started in locales other than Babylon, Greece, and Rome, it is surprising that the earliest known reference in the Far East is to the sale of the belongings of deceased monks in the seventh century A.D. (Yang, 1950). Yet we must place in a separate class, auctions where goods have to be sold, either by order of law or through necessity, to the highest bidder at no matter what price. In such auctions, time or, in the case of government auctions, justice is the primary factor, and profit is of secondary importance. Examples include auctions of the belongings of deceased or bankrupt persons and the sale of surplus government equipment, which, by law, have to take place by auction to discourage collusion and to ensure an equal chance for all to buy.

Recent auctioning and bidding Cassady (1967) offers a useful description of the conditions for efficient auction markets together with observations

Table 8.1
Use of various contracting methods, fiscal year 1980

Contract type	Amount (thousands)	Sealed bid
Two-step formal advertising	346,128	yes
Other formal advertising	9,440,400	yes
Negotiated competitive	25,699,369	[a]
Negotiated noncompetitive	53,678,668	
Acquired from foreign government	8,098,376	
Tariff or regulated	2,398,471	
Total	99,661,412	9,786,528

[a] "Contracting is generally viewed as being accomplished by one of two processes: (1) the formally advertised process—'sealed bid' and (2) the negotiations process." Office of Federal Procurement Policy, *A Guide for the Acquisition and Distribution of Commercial Products*, April 1980, p. 27.

on the types of commodities sold and the structure of the trade in the United States. He indicates that in the 1950s in the United States there were between 20,000–35,000 auctioneers and about 1,900 wholesale auction houses selling some $3.4 billion in goods, and that 1,600 retail firms earned over $22 million in commissions.[3] The types of items heavily sold at auction are tobacco, livestock, fruits and vegetables, used cars, furs, fish, used goods, antiques, and real estate.

Possibly of more importance to the U.S. economy than auctions are sealed bids. These are used in the purchase of large items such as transformers and steam turbines and many military contracts for specific weapons such as planes or tanks or for whole weapons systems. The sealed bid appears to be called for where the item is large or valuable and there is one seller or buyer or at most only a few. In such an instance, the realities of oligopolistic competition must be taken into account. Some indication of the role of bidding in economic life is given in table 8.1.

While sealed bidding for large contracts in general does not involve middlemen, open auctions and double-auction markets such as the New York Stock Exchange do. Some indication of the economic importance or power of the brokers is given by their fees, which in London, as Cassady (1967, p. 109) notes, were 10 percent for pictures, rugs, jewels, and the like in lots more than £100; 12.5 percent for smaller lots; 12.5 percent on old coins and medals; and 15 percent for books and magazines. He suggests that rates are still higher in New York and Los Angeles. Purebred cattle at auction may yield 5 percent to the auctioneer, comparable figures exist for realtors; in the stock market, commissions may range from a fraction

of a percent to three or four percent depending on the size and type of transaction. (For a valuable bibliography of modern references on bidding see Cassady, 1967; for more historical information, see Talamanca, 1954; for references on mathematic bidding models, see Stark and Rothkopf, 1979, Engelbrecht-Wiggans, 1980, and Milgrom and Weber, 1982.)

8.1.3 Variants of auctions

One of the better-known variants of auctions is the one-sided open, or English, auction, where open bids among a group of buyers increase the price until bidding ceases voluntarily or time runs out. The sealed bid is clearly related to the one-sided auction, but all bidders move simultaneously. There are variants in payoff such as awarding the object to the highest bidder, but at the price selected by the second highest bidder.

In the double auction, as described in section 7.2, there are many bidders and offerers on both sides of the market. Bidding or offering is done simultaneously. This contrasts with the previously noted tâtonnement process of the French Bourse.

The Dutch auction is a single auction in which price decreases until someone bids. A clock starts with a high price and moves down until it is stopped by a bidder pressing a button.

There are considerable variations in the details of bidding rules, such as conditions on terminating an auction and limits on the smallest incremental bid allowable. In most auctions or bidding situations, the bid is required to be in money and a minimum unit is specified. In general, when we construct a mathematical model of an economic phenomenon, we would like to show that the stability of the solution we use does not depend in any important way on what appear to be minor details of the actual situation. This is not quite so, however, in a simple model of bidding to which we apply the noncooperative equilibrium solution concept.

Suppose two bidders may each bid in units of \$1 for an item worth \$10 to the first and \$8 to the second. If they tie, the auctioneer breaks the tie by randomizing to see who obtains the object. The payoffs to the two bidders are given by

$$p_1 = \begin{Bmatrix} 0, & \text{if } b^1 < b^2 \\ 5 - (b^1)/2, & \text{if } b^1 = b^2 \\ 10 - b^1, & \text{if } b^1 > b^2 \end{Bmatrix}, \qquad p_2 = \begin{Bmatrix} 0, & \text{if } b^2 < b^1 \\ 4 - (b^2)/2, & \text{if } b^2 = b^1 \\ 8 - b^2, & \text{if } b^2 > b^1 \end{Bmatrix}.$$

The pair of bids $(9, 8)$ form an equilibrium point with payoffs of $P_1 = 1$ and $P_2 = 0$. If we cut the size of the minimum unit of bid to $\frac{1}{2}$, then the equilibrium points become $(10, 9.5)$, $(9.5, 9)$, $(9, 8)$, and $(8.5, 8)$, with payoffs of $(0, 0)$, $(.5, 0)$, $(1, 0)$, and $(1.5, 0)$, respectively. If we make the minimal bidding increment some arbitrarily small number ε, then there is a set of equilibrium points ranging from $(10, 10 - \varepsilon)$ to $(8 + \varepsilon, 8)$. But if we allow the two bidders to select any number on the closed interval $0 \leq b^i \leq 10$, then there is no pure strategy equilibrium whatsoever. Even with the finite grid there are "spoiler" equilibria at which the second player bids above the worth of the item in the belief that he will not get it because the other will bid higher.

8.1.4 Axioms for Cournot, Edgeworth, and auction mechanisms

Assuming a continuum of small economic agents, Dubey, Mas-Colell, and Shubik (1980) were able to show that the five simple assumptions are all that are required to describe the properties of Cournot markets having an efficient price system as a noncooperative equilibrium. Their omission of the Bertrand-Edgeworth mechanisms is important. Fortunately, the mathematical investigations of Dubey (1982) and Benassy (1986) and the new axioms of Dubey, Sahi, and Shubik (1993) cover the Bertrand-Edgeworth mechanism.

Giving up the assumption of continuity of the payoffs as strategies are varied paradoxically lets one obtain NEs that are efficient and that coincide with CEs when there are as few as two traders on each side of any market. This result should not be interpreted too generally. Although it indicates that it is logically possible for efficient competition to begin with only two agents on each side of the market, it does not guarantee it. When a multiperiod dynamics is considered with very few agents in the market, threat strategies and quasi-collusive arrangements are to be expected.

Auctions are often used to sell unique and indivisible items. Thus in any investigation concerned with the whys and wherefores of market mechanisms, it is reasonable to inquire about the virtues of the auctions as an efficient trading mechanism where the commodities on one side (goods being sold) are unique and indivisible and the good on the other side is divisible, as is money. The general assignment game (see Shapley and Shubik, 1971; Shubik, 1984, chapters 1, 9), of which the horse market studied by Böhm-Bawerk (English trans., 1923) is a special case, provides examples of prices existing with indivisibilities. Mas-Colell (1977) has

also considered the existence of prices with indivisible goods in large economies.

It appears that transaction considerations such as speed of sale play a central role in the use of the auction for differentiated indivisible goods. As yet, no axiom system including transaction considerations has been offered to justify the use of the auction mechanism. If there is a lack of credit available and an item of high value is to be sold, a lottery might be considered as a better alternative to an auction (Shubik, 1985b).

8.2 Markets, Trade Fairs, and Middlemen

8.2.1 A historical sketch

So far as we have any knowledge, they (the Lydians) were the first nation to introduce the use of gold and silver coin, and the first who sold goods by retail.
—Herodotus (Rawlinson, 1935)

From a brief sketch of the institutions surrounding barter and the development of trade via markets, we may obtain the first glimmerings of the emergence of exchange rates that eventually led to the formation of competitive prices. In contrast to the noninstitutional description of the open competitive market, the origins of price appear to have come about in trade heavily controlled by government agencies. Polanyi, Arensberg, and Pearson (1957, p. 242) note: "We must rid ourselves of the ingrained notion that the economy is a field of experience of which human beings have necessarily always been conscious." Market exchange in which transactions are not an expression of social, religious, or kinship obligations but are purely economic acts among essentially anonymous individuals may well be the exception rather than the rule of economic history.

Whether in the primitive economies of the South Seas or in the ancient economies of Babylon (Polanyi, 1944), Egypt (Janssen, 1975), Persia (Olmstead, 1948), China (Fitzgerald, 1954), or Rome (Jones, 1974), the mechanisms for internal trade and those for external trade appear to have developed in different ways. The great empires of the ancients appeared to have their major economic bases more in administration, taxation, and law than in open markets. The New World empires of the Aztecs, Mayas, and Incas appear to have followed a similar pattern.

Societies lie within the broader context of a civilization. Polities are contained within societies and economies require as their setting, polities. The human race appears to have emerged from a nomadic hunting existence some 10,000–15,000 years ago, with the first villages appearing possibly 7,000–8,000 years ago, the first writing some 5,000 years ago, and the use of metal 5,000–6,000 years ago. Although coinage may be traced back to ancient Greece and Asia Minor, and formal marketplaces may have existed in ancient Babylonia, this does not mean that price formation was primarily economic.

Polanyi (1944) suggests that the Babylonian economy may best be characterized by governmental taxation, storage of vital commodities such as grain, fixed prices, and direct redistribution. The merchant class may have been hereditary. The economy of Egypt in the Twentieth Dynasty appears to have been essentially the same. The "deben" of copper emerges as a unit of account (i.e., a numéraire) when prices of copper or bronze objects are recorded (Janssen, 1975), but there is no indication of money for exchange. Wages appear to have been paid in kind and there are few price lists of foodstuffs. The overall economic picture suggested by Janssen is one of an administration involving both the monarch and the priesthood which fixes and pays much of the wages in a society where many basic prices were controlled and supplies obtained for redistribution by direct taxation.

While Jones (1974, pp. 138, 139) suggests a somewhat more advanced economy for Rome, with at least minted coinage playing a role as a means of exchange, he is quick to point out:

> I have deliberately ignored industry and trade because I think that their importance was marginal. For the later Empire, at any rate, there are figures which suggest that trade and industry combined contributed something like 5 per cent of the national income, and agriculture the remaining 95 per cent ... and that the capital of a great Alexandrian merchant was about a twentieth of that of a typical senatorial landowner.
>
> I have said nothing of inflation ... partly because I doubt if the effects of inflation were very important. Most wealth was invested in land, which maintained its real value. Since coins however inferior in quality, had to be minted by hand, the rate of the depreciation of the currency was slow and its extent limited until the device of retariffing coins was invented by Aurelian. Private transactions in the third century as the papyri show, always continued to be carried on in money, and prices, wages and rents were adjusted. The main losers by the inflation were charitable trust funds,... and many civic endowments invested in mortgages or fixed rent charges.

The taxation system was critical. Many taxes were originally farmed out by auction. "The ancient world," Jones (1974, p. 175) suggests, "never achieved the notion of an income tax. Even if they had thought of it, their accounting methods were too primitive to distinguish income from capital."

Banking was rudimentary, involving mainly money changing with some acceptance of deposits and short-term loans. There was bottomry, lending on ship voyages that yielded high rates of return if the voyage were successful but little or nothing if the voyage failed. The Roman state did not normally borrow from its citizens, operating essentially on a cash basis. The empire formed a closed economy except for an external trade in luxury goods, which caused a somewhat unfavorable balance of trade (and an outflow of coinage, especially to India).

As opposed to modern economies with more or less free markets for land, labor, and capital, in the ancient empires effective supply and demand were heavily under the control of the state, and the mechanisms for adjustment were essentially administrative.

Trade and industry in classical Greece (Hopper, 1979, pp. 61, 64) appear to have had an active small retail component, where moderately free markets were to be found:

> A contrast to our limited knowledge of foreign trading on a large scale (the sphere of the empires) is provided by retail trade at Athens. Here "trade" is involved, not "trades." By this is meant the function of distributing in small amounts to individual purchasers the food and other natural products and manufactured goods which they require for their daily life. The retail trader was either the actual producer, such as the country people who came to market in the town, often travelling considerable distances, to sell their own fruit and vegetables and the like, or the true middleman, whose function appears to have been the handling not only of natural commodities which he purchased from the primary producer (as, for instance, Pericles found it convenient to sell the produce of his estate, which must imply the existence of traders) but also of what might be called prepared commodities, such as bread, and, we might add, the wares of Aristophanes' Sausage-seller, for to this category of retail trade belong not only the cook-shop but also the itinerant vender.

Hopper goes on to describe the attitude toward trade:

> In an Ideal State such activities were better left to foreigners! The word *kapeleuein*, "to engage in retail trade," was not one Plato or Aristotle used with approval. Plato clearly defines in the Laws his attitude to retail trade. In this Ideal State he seeks to confine this activity to metics and foreigners and to control trade and

promote standards of integrity, though this is not a matter on which he believed success to be attainable, as he makes clear from his discussion of the faults of retail traders and their origins. After this discussion, however, he returns again to questions of the restriction, regulation and "illiberality" of retail trade. Incidentally Plato would also exclude from the market the principle of credit. Exchange shall be of money for goods and goods for money, and there shall be no redress for the trader foolish enough to give credit. One of the corrupting factors, in his opinion, was the profit motive. Hence the proposal to fix profit margins in retail trade. There is an air of impracticability and unreality about these ideas, but it would be a mistake to suppose that they did not, though perhaps in an exaggerated manner, reflect common notions and prejudices relating to small-scale trading.

Hopper (1979, pp. 102, 104) notes the evidence on the size of Athenian industry. A couch factory employing 20 slaves and a sword factory with 32–33 slaves were regarded as large. A shoe factory with 9 or 10 workers might be of more normal size, although another estimate puts the average factory size at 20–30 workers.

Rudimentary banking existed in Greece by 400 B.C. for money changing, acceptance of deposits, verification of quality of coinage, and some loans in the style of pawnbroking. There was also a primitive form of giro banking, where a sum was deposited by one individual with payments made by the banker to another individual on instructions from the first. Beyond the retail markets and rudimentary banking there is little evidence of more than the minor private capital formation required to service this level of trade. The import of grain, control of taxes, and dealings with slaves and foreigners were concerns of the city-state itself and far removed from the modern market.

The splitting of the Roman Empire and the invasion of the Germanic tribes in the west did not destroy the trade and markets that existed (Pirenne, English trans., 1956, p. 31). It was the Moslem sweep across North Africa and into Spain and the Mediterranean islands that effectively destroyed the maritime trade of centuries.

By the ninth century, according to Pirenne, the West was a closed domestic economy more or less without markets. The empire of Charlemagne was essentially agricultural, and the merchant classes had disappeared from it. The power of the bishops increased; in their towns of residence, they began to control taxation, general administration, and through their granaries and storehouses, even the food supply. There were annual fairs and almost always weekly markets for produce. Taxes were levied in kind and in labor. Along with the towns of the bishops, fortified

burgs of the counts and princes were built. It was around these towns and burgs that the cities of the tenth to eleventh century grew.

In the south, Venice was still part of the Byzantine Empire. Although somewhat isolated, it still traded with Constantinople, which in the eleventh century was a city with around a million inhabitants and a center of trade and manufacturing as well as political activity. Pirenne suggests that it was largely through Venice in the south and Flanders in the northwest that commerce revived in Europe.

In the twelfth century, a trade and a merchant class developed around the burgs in societies where the attitude of the church could be summarized (Hotchkiss, 1938) as follows:

1. Direct marketing from producer to consumer is the ideal.
2. Trade must be restricted to appointed markets.
3. There is a "just price" for everything.
4. No rate of interest on money should be charged.

Whether a peddler with a pack or a long-distance trader traveling in an armed convoy of ships or a protected caravan, the merchant was in most contexts regarded as a foreigner answerable to public jurisdiction.

Even before the commercial revival there were markets, often around a market cross or at a market post signifying the protection and monopoly of the local aristocracy or clergy. These markets met weekly for the most part, and in the highly agricultural economies of the feudal period served as relatively rudimentary exchanges for agricultural and craft products. They also provided the aristocracy and clergy with important opportunities for taxation, be it stallage (for setting up one's stall), tronage (for weighing), pontage (for crossing a bridge), or any of a myriad of other taxes.

There were important annual fairs often tied in with religious festivities such as Stourbridge Fair, dating from 1211, and St. Audrey's Fair[4] in England, Messines in Flanders, and Bar-sur-Aube in France. At these fairs, luxury goods from afar were exchanged by middlemen. Nevertheless, Pirenne explains, neither the local markets nor major fairs appear to have served as the basis for the growth of towns where the middle class and economic activity forged ahead.

The "portus" of the Roman Empire, places dedicated to the protection and transfer of merchandise, had a permanent commercial population

from the tenth century on. Pirenne suggests that some of the old "portus" revived and that new ones were created at places such as Bruges and Ghent.

The growth of the basis for modern trade and the growth of free towns appear to have been closely related. The experience in England is illustrative. Originally, there were penalties for "regrating" (i.e., performing an arbitrage function of buying to sell again in the same market) and for "engrossing" (i.e., buying in quantities larger than retail). The first middlemen, to maintain the appearance of being producers, may have added finishing operations to their trading, as the pepperers did by grinding spices.

By the end of the thirteenth century in England there were over 100 free towns, where the townspeople enjoyed the right to trade freely and to govern themselves. Originally composed of different craftsmen and tradesmen who produced and sold directly to consumers, the merchant guild of the town controlled the market. At the start of the fourteenth century, there were over 100 guilds in London and around 200 in other English towns (Hotchkiss, 1938, p. 37). These included grocers, mercers, drapers, goldsmiths, saddlers, butchers, and ironmongers. The craft guilds encouraged the growth of shops or stores opened by their members. They also tried to police prices.

In 1313, the Staple market for wool was created by royal statute. This insured that by centralizing this important export, market taxes would not escape the king. With Bruges and Calais as locations for the market, the merchants of the Staple, enjoying the king's monopoly, could charge a "just price" for wool, one that covered taxes, costs, and a comfortable profit.

The pressures of foreign trade and nationalism gave rise to companies such as the Merchant Adventurers and the Muscovy Company. Other chartered companies followed, such as the Eastland Company, chartered in 1579 for trade in the Baltic. By the start of the seventeenth century, monopolies had been granted to guilds, to chartered companies, and to court favorites on items such as salt or wine.

The history of the market economy appears to have been one of slow emergence against the power of the state and of changes in socio-political organization, technology and population vastly accelerated the predominance of the market economy in the western world.

The stage had been set for the industrial revolution in England by an earlier, financial revolution (Dickson, 1967). The Bank of England was founded in 1694. Although ship voyages had been organized on a joint stock basis where the organization lasted one voyage, the East India Company, chartered in 1600, became a continuous joint stock corporation. Nicholas Barbon is attributed with the founding of modern fire insurance in 1641; fire insurance facilities increased rapidly in England after the Sun Fire Office was established in 1710. The financing of the English public debt in the early eighteenth century helped to develop financial markets and provide securities for the merchant class.

In the United Provinces tontine loans were used as a form of life insurance.[5] England used lottery loans in public finance in the late seventeenth and early eighteenth century. By 1749, the number of public creditors was put at 30,000–40,000 (Dickson, 1967, p. 251).

By 1690, the major features for stock transfer had been settled (Dickson, 1967, p. 460). Turnover of stock as a percentage of nominal capital has been estimated for the Bank of England at 44 percent for 1718 and at 13 percent for 1755. For the East India Company, stock turnover in 1704 was estimated at 102 percent and at 26.8 percent in 1755 (Dickson, 1967, p. 466).

In 1692, John Houghton started a specialized list of prices for shares; by 1694, he quoted shares of nearly sixty companies (Dickson, 1967, p. 487). John Castaign published *Course of the Exchange* in the 1690s. The brokers concentrated on Exchange Alley at several coffeehouses. The earliest "put" noted is dated 29 July 1687 by Sir Bazill Firebrass to deliver £1,000 of East India stock on or before 1 March 1688 at 200 for a premium of 150 guineas (Dickson, 1967, p. 441). In 1697, Parliament limited the number of brokers to 100; sworn brokers were forbidden to deal in stock themselves; of the 100 brokers, twelve could be aliens and twelve Jews.

In 1752, stockholders of the Bank of England were estimated at 4,750, those of the East India Company at 2,140, and those of the South Sea Company at 4,000 (Dickson, 1947, p. 285). Dutch, Swiss, and German holdings of total stock were 14.2, 15.9, and 7.6 percent, respectively, in 1723–1724 (Dickson, 1967, p. 312).

The growth of a press and other means for the quick communication of information to large and essentially anonymous groups is an important factor in the development of efficient mass markets. In England, the first examples of periodical advertising appeared in 1647–1648; these were

announcements of the publication of books or pamphlets (usually religious). The first paid advertisement, for a lost horse, appeared in the *Moderate* for March 20–27, 1649 (Hotchkiss, 1938, p. 107). *The Publick Adviser*, which contained only advertisements, was published in 1657, but was not successful. Started in 1692, Houghton's publication *Collections* built up a substantial advertising section (Hotchkiss, 1938, p. 124).

By the early eighteenth century in England, with the breakdown of the guilds, trade was carried on by wholesale merchants, tradesmen, shopkeepers, and handicraftsmen. Some fairs had become wholesaling markets, setting the price for commodities. An example is the Stourbridge Fair (Hotchkiss, 1938, p. 123), as a quotation by Defoe in 1724 illustrates: "There is scarce any price fix'd for hops in England till they know how they sell at Sturbridge."

The industrial revolution and its aftermath may thus be considered in terms of three technological revolutions—in production, in distribution, and in finance. The growth of the sources of nonhuman power revolutionized the nature and size of manufacturing. The growth of population, means of communication, and the power and legal protection afforded by the nation-states provided the basis for the revolutions in both distribution and finance that had to accompany the successful use of large manufacturing.

It has been suggested that much of the Western world in the late twentieth century is entering a new, "postindustrial" phase (Bell, 1976). The stress in postindustrial society may be more on communication, information, distribution, and finance than on power sources and the more physical aspects of manufacturing.

Technological, political, social, and economic changes are all delicately interwoven. Causes and effects are seldom simple. Thus in this brief sketch no attempt has been made to determine whether the growth of mass manufacturing caused the development of mass markets and population or vice versa. It is in my estimation a bad question because it poses a probably false dichotomy. Indeed, it appears that all three conditions are necessary for the twentieth-century economies of Europe, United States, and Japan.

Given Japan's isolation from the West in earlier times, one might expect to find that its markets and economy developed in radically different ways. The difference appears to have been in the conditions under which money, mass markets, and price mechanisms were imported from the West and modified by Japanese social and political institutions.

The early history of Japan is marked by the borrowing and modification of administrative methods from China. The economy per se appears to have been of minor importance. This is evinced in the role of money. Sansom (1952) notes:

"There had been silver and copper money as far back as the seventh century, but its circulation was very limited and it was not until the fifteenth and sixteenth centuries that is obtained even a moderately wide currency."

The growth of city life (by 1800, Yedo had a population of probably over a million and Japan as a whole, 28–30 million), size of population, better internal long-distance transportation (see, for example, Hiroshige's fifty-three stages of the Tokaido) and pressures from foreign trade meant that rice as a medium of exchange could not last long into the Tokugawa regime even though the regime tried to maintain a feudal Japan.

With the breakup of feudalism during the Tokugawa regime and the reopening of foreign trade, Japan absorbed European technology, crafts, and science, setting the stage for its modern economic development.

An epilogue on markets Even given the necessary conditions, free markets have not had much success in supplying care for the young and old or in supplying some labor, legal governmental, and other public needs, and it is unlikely that price mechanisms alone will ever be able to do so. The pharaohs and the bishop-princes have gone, but many of the services they provided have not been replaced by the market. Mass-market technology, so highly efficient in guiding the production and distribution of many economic goods, cannot readily be extended to the political economy. There is no simple price system for votes.[6]

8.2.2 The language of trade

In chapter 5, it was suggested that one might construct a highly restricted language for trade, where the nouns are names of goods and services, adjectives are qualifications on amounts or quantities, verbs comprise a narrow list of actions such as to bid, offer, buy, and sell.

When we consider trade with credit possibilities and with complex ownership conditions and the sociopolitical as well as economic context of bargaining, the extension of the vocabulary signals nuances to the student that may be hard to portray in a mathematical model. Fine distinctions in the concept of property or contracts, which are accepted as a

matter of course in the law, begin to appear. A few key nouns and verbs associated with trade are noted below, together with allied nouns and verbs, which illustrate differences in nuance:

Offer (n.): proffer, tender, bid, proposal, overture, proposition

Request (n.): requisition, claim, petition, suit, prayer, plea, motion, application, appeal, proposal, rogation, imprecation

Request (v.): ask, beg, crave, sue, petition, entreat, press, dun, tax, cadge, panhandle, canvass, tout

Promise (n.): undertaking, word, pledge, oath, assurance, warranty, guarantee, insurance, obligation, contract

Compact: contract, agreement, bargain, deal, transaction, bond, covenant, indenture, stipulation, convention, treaty

Security: guarantee, gage, warranty, bond, pledge, mortgage, debenture, hypothecation, lien, pegnoration, bottomry, vadium, stake, covenant

Acquisition: gaining, procuration, purchase, gift, retrieval, profit, earnings, winnings, income, proceeds

Possession: holding, ownership, occupancy, tenure, villenage, socage, monopoly, corner, retention

Property: possession, proprietorship, seignority, empire, interest, stake, estate, right, title, claim, holding, trust, benefit, assets, belongings, means, resources, wealth

Transfer (n.): coveyance, assignment, alienation, transmission, enfeoffment, lease, sale

Giver: grantor, donor, feoffor, settlor, testator, investor, subscriber, contributor

Receiver: accipient, assignee, devisee, legatee, feoffee, donee, lessee

Lend: advance, loan, accommodate, let, put out, place

Borrow: hire, rent, form demise, lease

Barter (v.): exchange, truck, scorse, swap, trade traffic, negotiate, haggle, higgle, dicker, bargain

Mart: marketplace, forum, fair, bazaar, staple, stock exchange, bourse, rialto, shop, stall, booth, depot, store

Credit (n.): trust, tick, tally, account, score

Debt: obligation, liability, indebtment, debit, score

Payment: discharge, defrayment, quittance, settlement, clearance, liquidation, satisfaction, remittance

And at the basis of markets and trade:

Money: currency, circulating medium, specie, coin, cash, lucre, shekels, roll, jack, rhino, blunt, gelt, dust, brass, dibs, dough, mopus, tin, salt, oofish, spondulicks, pile.

Wealth: riches, fortune, opulence, capital, means, resources, property

The study of markets and financial institutions calls for considerable abstraction to provide the simplification needed to analyze in depth. Simplification results in the loss of delicate distinctions (which may or may not be important to the question at hand). This foray into verbal comparison is offered as a simple way of seeing whether a mathematical or diagrammatic description of an economic phenomenon matches the verbal description.

Notes

1. Part of the material in section 8.1.2 is based on translation by Hasan Ozbekhan of a work in Italian on the history of auctions by Talamanca (1954); and also upon the research assistance of Christopher Dougherty.

2. Antonius Matthaeus, in his work *De Auctionibus* quotes Damhonderos, who believed that *subhastatio* was a corruption of *ab hastario*, Hastarium being the name of a place used as a market. Matthaeus, however, does not agree.

3. Quoted from *Newsweek*, February 3, 1958, pp. 78–80. Numbers have been rounded here.

4. From which we have the word "tawdry" (Hotchkiss, 1938, p. 17).

5. Lorenzo Tonti (1630–1695) is attributed with devising this scheme. A tontine is formed by a group of subscribers forming a joint investment pool used to buy, say, shares. All income is paid pro rata to the survivors until the last survivor obtains all.

6. See Shubik and van der Heyden, 1978, for an attempt to recast voting in terms of market prices, together with an "impossibility theorem." There is a large literature on this subject, where the argument is still on models and context. But an elementary consideration of public goods and political processes should be sufficient to indicate that the analogy between competitive markets and the price of votes is more a matter of political rhetoric than scholarship.

9 Trade with Gold, Credit, and Inside Money

My credit is so bad,
They won't even accept my cash.
—New York folk saying

9.1 The Transaction Demand for Money and the Float

Chapters 5, 6, and 7 proposed a series of different one-move markets and market structures as mechanisms for trade, each requiring different amounts of money for transactions. (In chapter 13, we estimate how much money is required given different mechanisms.) Chapters 6 and 7 noted that when a commodity money is used, there may be an insufficient supply or inappropriate distribution of the commodity in the hands of at least some of the traders, which prevents optimal trade from taking place without credit, a complex clearing mechanism, or more than one round of trade.

9.1.1 The float

The idea of a single round of trade with velocity constrained to one or less is unrealistic, but in keeping with the approach of dividing difficulties, it enables us to understand the role of currency as a *strategic decoupling device*. In all of the previous models there has been no time discount. The only role of commodity money is in transactions where value is given for value. Trade is trustless and immediate. Money is *currency*.

Because only one period is considered, no distinction is made between consumable goods and assets. The final holder of a good derives the ultimate utility from the good. For example, if the commodity money were a brick of Mongolian tea, individual i could pay individual j, who could then consume it. If the commodity money were gold, which as a consumer durable can be used either as jewelry or as money, it would seem unreasonable to credit the final owner with its use for the whole period.[1]

Transactions take time. Even in a single-period model (as can be seen clearly when trying to play a game based on the model), at any point in time the money must be located in one of four places: with individual i (the payor), with individual j (the ultimate recipient), with a third party (the bank, clearinghouse, or other intermediary), or in float (in transit, in the post, or being otherwise delivered).

Let the unit of time be 1 and let the fraction of time the money is with i, j, and the intermediary be a, b, and c, respectively. Then the length of

time in float is $1 - a - b - c$. An important but difficult distinction to make is between a strategic and a truly nonstrategic float. The float at the post office is essentially nonstrategic. The postal service does not take possession of a check sent through the mails. A stockbroker, on the other hand, does take possession of a check sent in payment. In those magic few days between when i has paid and j receives payment, the broker had the strategic power to live off the customers' money (Baruch, 1971). This opportunity is not available in the one-period models we have presented, whose nonstrategic float time is thus $1 - a - b$.

In this chapter, where we are concerned with trade during a single period, money has only a transaction use. It features as part of the technology of trade wherever one is required to pay in money. Because payments are often not carried through instantly, there will always be a pool of payments in transit or in limbo between buyers and sellers. The *float* is the amount of money or credit instruments in transit in the system (i.e., in clearance but not yet cleared).[2] The size of the float, which varies with the volume of transactions, and the transaction costs of handling the paper are ever present features of trade.

Table 9.1 indicates the size of the float over a considerable period of time. The change between 1975 and 1985 in spite of considerable inflation can be accounted for by a revolution in computer and communications technology. In the models presented in chapters 6 and 7, a float is created in the time lost by all parties between the delivery of both goods and money and their receipt. Because the models are extremely simplified, there are no float losses indicated. In an economy without credit instruments, the float would consist of currency in transit.

9.1.2 How to finance transactions

If there is enough commodity money in the system, transactions are financed implicitly in the process of exchange, using, say, salt bars or tea bricks as the commodity money. If one is going to study exchange models with a commodity money, historical precedent makes gold the natural commodity. But one of the key attractive features of gold is its durability, hence it is more suited to the study of multistage models. For this reason, in considering only the one-stage models, we might wish to select a consumable money. Historically, bricks of tea and bars of salt have been used. In an experimental game, one could use chocolates.

Table 9.1
Monthly average of daily float figure (millions of $)

Month	1985	1975	1965	1955
1	1,203	2,456	2,126	805
2	1,248	2,079	1,650	710
3	540	1,994	1,659	804
4	503	2,061	1,658	838
5	587	1,877	1,633	798
6	600	2,046	1,714	878
7	703	1,911	1,831	940
8	488	1,691	1,581	746
9	779	1,823	1,922	924
10	669	1,945	1,869	926
11	1,203	2,480	1,965	1,055
12	1,176	3,029	2,349	1,389
Average	808	2,116	1,830	901
Standard deviation	293	354	222	173

Source: *Federal Reserve Bulletin*.

If we use gold as the commodity money, then in the ideal, but counter factual model, trade would take place instantly at the start of the period, and the ultimate recipient of the gold would have consumption use of the gold as jewelry for the whole period.[3] In this instance, where $a = 0$, $b = 1$, and $c = 0$, the length of time in float is of measure zero.

If there is not enough commodity money to finance transaction needs, there are essentially four ways this shortage can be overcome. Optimality can be restored (1) by the issue of inside credit; (2) by the issue of outside credit; (3) by increasing the supplies of the commodity money; or (4) by changing the transaction technology and velocity.

Because in this chapter we stay with only one round of trade and no further production of the commodity money is considered, only two institutional alternatives are considered: (1) the issue of "paper gold" and (2) the direct use of personal credit. Both of these can be considered as involving the issue of *inside money*, namely, debt against which the claim is held by some other private economic agent (see Gurley and Shaw, 1960); the government issue of fiat or *outside money* is considered in chapter 11 and in Volume 2, chapters 3 and 4).

9.1.3 Cash and anonymity

Anonymity in one context is not necessarily anonymity in another. As we have repeatedly stressed, anonymity in economic exchange differs from anonymity in government surveillance for taxes or criminal or political investigation. Small coins and paper currency cost little, relative to their expected economic life. And cash payment leaves no paper trail. It is anonymous both in economic trade and in government or other investigation. It is also reasonably efficient, cheap, and fast.

In the context of trade alone, small change is here to stay. The trade-off between privacy rights of the individual and criminal behavior must be considered, (like the blackboard and cockroach) the odds are that currency may survive for a long time as a means of payment for small, secret, or illegal items (see Walter, 1985, on secret money and Denning, 1987, on coding to avoid Big Brother).

9.2 From Commodity Money to Paper

When a shortage of a commodity money exists, optimality can be restored by the issue of credit or paper money. To separate out difficulties, we keep details about the nature and functioning of the agents who supply the various forms of credit to a bare minimum. These agents may be merchant bankers, moneylenders, commercial banks, or other. Their roles as active players in the game can be seen more clearly when multistage economies are considered. Part of their role as nonsymmetric players begins to emerge even in one period, when a playable game is designed that includes borrowing.

9.2.1 Personal credit monetized

A credit mechanism can be grafted onto any of the market structures described in chapters 6 and 7. Although for expository purposes the easiest might be the sell-all model described in section 6.4 (see especially figure 6.13), a more general model is the bid-offer model.

As soon as any form of credit is introduced into a strategic market game, two factors must be determined: (1) the worth to an individual at the end of trade of having credit left over; and (2) the cost or penalty for ending up in debt. These factors require a formal discussion of the treat-

ment of creditors and debtors in bankruptcy or default (delayed until chapter 11).

If, for the sake of simplicity in the design of a playable game, the game "ends" after one period of trade, then positive amounts of paper and leftover durables such as gold can be declared worthless. In actuality, all economies are open in time, and hence the rules governing the use of the financial instruments in a future game may serve to maintain an ending value. This can be modeled rigorously by assigning a salvage value to leftover money and credit and by studying how solutions are influenced by the salvage value.

It is straightforward to construct a one-period experimental trading game where the players are instructed that no extra payoff will be given to those who end as net creditors, but that a penalty will be levied against those who end as net debtors. Trade may be modeled as follows. The initial endowment for each trader i is still given by m quantities of goods, but a new utility function related to the old one is defined, which has one more dimension.[4] Thus

$$U_i(x_1^i, \ldots, x_{m+1}^i) = u^i(x_1^i, \ldots, x_m^i) \qquad \text{if } x_{m+1}^i \geq 0$$

$$= u^i(x_1^i, \ldots, x_m^i) + \lambda^i x_{m+1}^i \quad \text{if } x_{m+1}^i = \sum_{j=1}^{m} p_j(a_j^i - x_j^i) < 0.$$

The meaning of this function is illustrated in figure 9.1, where it can be observed that credit is defined in both the positive and negative orthants,[5] whereas ordinary goods are only defined in the positive orthant. If holding credit at the end of the game is of no positive worth, this can be reflected in the vertical indifference curves portrayed in figure 9.1 by the line AE and others. The severity of the penalty for net debtors at game's end can be illustrated by continuing the definition of the indifference curves into the negative orthant, as shown by AEC. The level of punishment is reflected in the curvature of the segment EC. The slope of the line ED provides a simple description of the marginal disutility of the punishment at the point E.

A strategy in this game is to bid a set of numbers $b_j^i, \ldots, b_m^i$, where $b_j^i \geq 0$ for $j = 1, \ldots, m$, but where there is no constraint on the sum of these numbers. This differs from the model in figure 6.10, where the bids were required to sum to no more than a_{m+1}^i, the amount of commodity money in possession of each. The interpretation of this game is that

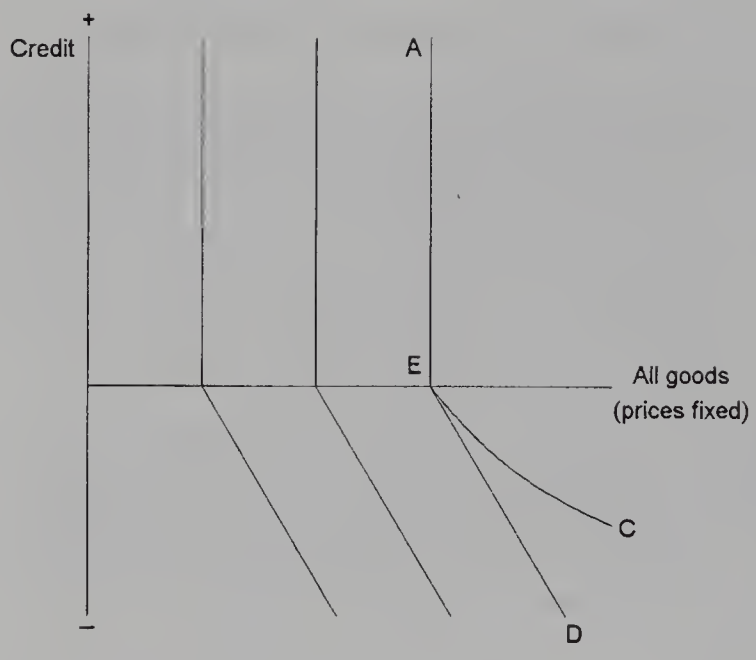


Figure 9.1
Bankruptcy penalty.

traders can monetize their personal credit without restriction. Of the several institutionally different mechanisms consistent with this level of specification, the following can be considered:

1. A clearinghouse for personal IOU notes.
2. A central computerized facility where the bookkeeping is supplied for a clearing system without the actual issue of financial instruments.
3. A mutual bank that provides all traders with banknotes or credit line at a price, so that each trader issues bank credits, not personal IOU notes (i.e., indebtedness is centralized; neither markets nor individuals accept personal IOUs, but all accept bank checks or bank notes).
4. One or more for-profit banks that lend their money to the traders.

At the level of abstraction considered here, these systems differ only in what may appear to be fine points. When we consider multiperiod trade, however, even without going into detail concerning the credit granting agencies, several more differences appear.

In particular, if physical entities such as specific IOU notes are actually issued, then the rules and forms of contract must be specified, and the

body of law and custom governing how they are issued and cleared becomes part of the rules of the game. When transaction costs and exogenous uncertainty are considered further distinctions emerge.

Jaynes, Okuno and Schmeidler (1978) and Dubey and Shapley (1994) have investigated exchange where all traders are insignificantly small with respect to the market and where credit is unlimited. Postlewaite and Schmeidler (1978) have treated a finite version of a similar noncooperative trading game studying asymptotic behavior. Although the Dubey and Shapley (1994) results are for the bid-offer market structure, Dubey, Mas-Colell, and Shubik (1980) have shown that a broad class of mechanisms will yield equivalent results for nonatomic markets.

Let E stand for a trading economy with the bid-offer market structure described in section 7.1. Let there be a default penalty parameter $\lambda^t > 0$ associated with a trader t where, in this continuum version, $U^t(x^t) = u^t(x_1^t, \ldots, x_m^t) + \lambda^t \min(0, x_{m+1}^t)$.

The game with market structure E and penalties λ^t is denoted by Γ_λ. An active noncooperative equilibrium (NE) is one that produces positive total bids and offers at each post. An *active NE allocation* of Γ_λ is an allocation produced by an active NE of Γ_λ.

THEOREM 9.1 (Dubey and Shapley) Every active noncooperative equilibrium allocation of Γ_λ that leaves almost all traders unsatiated is competitive for E, provided that the penalties for λ^t are positive.

(For complete technical details and their interpretation, see Dubey and Shapley, 1993.)

Assume there are many traders trading in an economy where none has any market power and where each can obtain unlimited credit. Then, given certain reasonable conditions on the structure of preferences and trade and given that there is some positive level of penalty for ending in debt, there will be noncooperative equilibria that yield the same allocations of physical goods in the game as the competitive equilibria. The milder the penalty is made, the more credit will be brought in, so that prices will rise until the marginal disutility of the penalty for ending in debt is equal to or greater than the marginal utility of the extra purchasing power obtained from borrowing.

Where Hicks (1946) suggested that the Lagrangian multiplier in a consumer's constrained optimization problem could serve as the marginal utility of income, here it is suggested that the penalty for ending in debt

must be at least as large as the marginal utility of income to have optimal trade. When this is the case in the one-period game, the rate of interest paid on credit is zero. In other words, if running the credit facility to finance transaction needs is essentially costless, then credit can be supplied as a free good by organizing an inside bank or money market.

Although specifying default penalties immediately imposes a lower bound on the equilibrium price level, it does not imposes an upper bound; thus equilibrium prices can be so high that the marginal utility of the purchasing power of an extra dollar is far lower than the penalty for defaulting on a dollar of debt. But because at the given prices no one is going to default, all borrow enormous amounts and the system is in equilibrium.

9.2.2 A detour on clearing mechanisms and markets

A simple way we can appreciate the importance and the nature of clearing mechanisms in mass trade is to try to design a playable game using credit. In the previous models of trade presented in chapters 6 and 7, bids and offers were made to markets, which then shipped goods and money to the traders. At each post a broker dealer, or mechanism was needed to list the bids and offers. Even at this level of simplicity, we can make several distinctions:

1. Does the operator of the market act merely as an agent, noting and matching bids and offers, or does the operator actively buy to sell?

2. Are the goods sold from individual i to individual j delivered directly or do they actually go through the market, sold to a dealer or sold on consignment to a broker or other agent?

3. In the course of trade, are ownership papers, warehouse receipts, or both required.[2] If they are, are any of these items negotiable paper, that is, can the holder assign them to a third parties?

With credit, we can make other distinctions:

4. Are payments made with personal IOU notes, with bank checks, credit cards, or banknotes?

5. Does a trader buying from a market pay the market (or the agent running the market) or does the trader pay the seller directly?

Trade with IOU notes Suppose that each trader bids using personal IOU notes addressed to the market the trader is bidding in. Thus messages sent

by bidders may have the name of n senders and m addresses. Each market must be permitted to issue IOU notes to the sellers. If goods are cleared through all markets, each market actually receives all the goods; it must record ownership, possibly unpacking, breaking bulk, repacking, and shipping.

Although each market clears trade, we have not yet described the final settlement mechanism for the settlement of credit claims. After prices have been established, the markets can issue their IOUs to the sellers. There is twice as much paper as there has been trade. All buyers have issued IOUs to the markets, which in turn have issued IOU notes to all sellers. At this point, we may introduce a clearinghouse. All markets and individuals holding IOU notes send them to the clearinghouse for matching. In equilibrium, the trade of each trader should net to zero; otherwise, each trader and market is informed of its net (positive or negative) balances.

The rules of the game are not complete without specifying the institution or device that enforces the penalties on those who end in debt. In this formulation, the rules call for the agents or dealers in the markets to accept personal notes and to issue their own notes to virtually all traders; the agents or dealers act as rudimentary merchant bankers.

Warehouses and record keeping Because we concentrate on the borrower aspects of the credit system in this chapter, the modeling of central banks, commercial banks, brokers, and dealers is kept to a minimum, although we cannot totally avoid it.

We also cannot avoid describing some of the details of trading. If, for example, we tried to build a playable game for the model noted above, we would need to issue warehouse receipts in order to keep track of goods delivered to a central distribution point (unless a clearinghouse arranged for direct shipping) and we would need a clearinghouse to balance out all of the checks written. As soon as we have warehouses and clearinghouses rather than direct one-on-one trade, the conditions are ripe for accounting and auditing.

Warehousing was already important in Sumeria, Babylon, and Egypt, as can be seen from the nature of the tax records. Hence, it is likely that warehouses, treasuries and countinghouses came in with government taxation and booty collection. Record keeping and auditing were likely to have been among the first financial activities.

Double-entry bookkeeping comes in far later, with the growth of private trade in Genoa, Tuscany, Lombardy, and Venice in the fourteenth and fifteenth centuries. The classical reference to bookkeeping is the *Summa de Arithmetica* of Luca Pacioli[6] published in 1494, although Izutani (1980) dates the ledger of the Genoa government at 1340 and the ledgers of two Tuscan merchants, at the end of the thirteenth century.

Trade with banknotes, credit cards, or checks If trade is in banknotes, then the markets may be relieved of all banking tasks. We may consider a banking system, which consists of a single bank. It provides banknotes, which can be used as a direct and anonymous means of payment; it takes in return IOU notes written on the bank.

Individuals bid for goods using banknotes. The markets flow through the banknotes (with the minor problem of making small change) and pay the suppliers directly. After all traders have been paid, they go to the bank to settle up all outstanding credit.

If checks rather than banknotes are used, then the markets do not have the direct means to pay suppliers and there must be $n + m$, rather than n, bank accounts. A new set of checks must be written from the markets to suppliers. We must impose rules on brokers to make sure either that they are forbidden to incur net debt or that they are penalized; otherwise, they may have an opportunity to borrow without constraint.

After trade has taken place, goods shipped, and payment made by check, then we may consider that the bank acts as a clearinghouse for all the checks. We note whether the market acts as broker or dealer; if a two-name monetary instrument such as a check or IOU note is used, then the market must issue paper to the same amount as it takes in. Where there is more than one bank, the banks will require a central clearinghouse to balance accounts after trade.

In a world with time lags, random failures in data processing, and errors in paperwork, the costs of running the clearing systems, whether direct expenses such as data processing or indirect expenses such as auditing and carrying the float, may be nontrivial and will vary considerably with different organizations. Thus in the United States it was estimated that in 1972 the cost for the clearance of a check was around 14 to 16 cents (Mayer, 1974, p. 161). In parts of Europe Gyro banking utilizes a different clearing system (Mayer, 1974, pp. 171–172). Credit card transactions clearing is different from both cash and checks (see Hester, 1972).

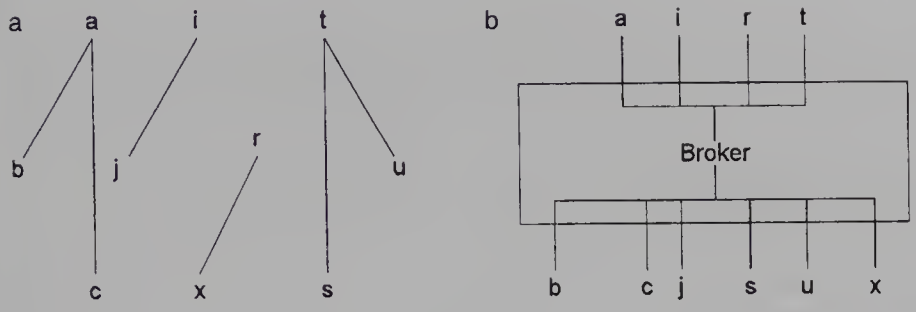


Figure 9.2
Borrowing with and without an intermediary.

In 1998 the clearinghouse for interbank payments charged its members 17 cents for a transfer using its computer system.

9.2.3 Elementary credit mechanisms and institutions

To completely specify how borrowing and lending takes place, we must describe a mechanism. Four elementary mechanisms are suggested, each of which has an institutional counterpart. The first is pairwise borrowing and lending. A pair of individuals make a private agreement, one gives the other some amount of the monetary commodity in return for a non-negotiable IOU note, which is held by the lender i until the borrower j pays back the loan. Any default arrangement is between i and j, although society as a whole through the courts may be required for enforcement.

The second is matched pairwise borrowing and lending via a broker. The function of the broker as is indicated in figure 9.2b is to serve as an information focus and clearinghouse so that otherwise anonymous agents can be matched. The broker is not a principal and bears no role after i and j have been matched. Figure 9.2a shows straight matching of lenders and borrowers without any intermediary.

The role of the broker is in pooling information and matching principals, not pooling principals. We might expect brokers to fix or to compete for fees. (See Nti and Shubik, 1984). In this regard, we note but do not discuss clubs that call for formal membership and dues, but provide insurance, rule making, clearing, enforcement of contracts holding of earnest money, and other activities that turn a loosely associated group of individuals into a formal financial institution.

The third mechanism (which splits into two submechanisms) can be described as a rudimentary form of money market or as a commercial

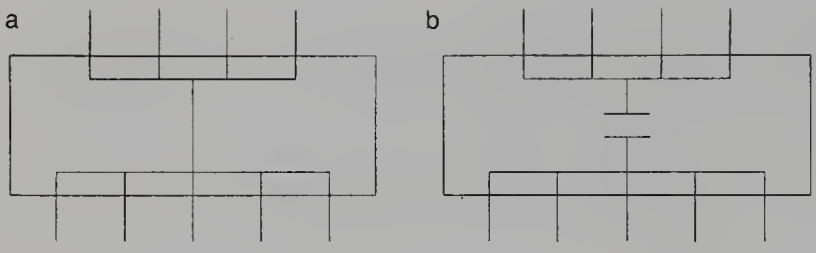


Figure 9.3
Mutual fund or principal agent.

bank. Both involve aggregation but there is a difference in which principals bear the responsibility for absorbing losses in the event of a failure to pay back a loan. Figures 9.3a and 9.3b help illustrate the distinction. In the rudimentary pooled money market fund, the fund serves merely as an aggregating device for both sides of the market. All loanable funds are pooled and then matched against the demand for funds to determine a rate of interest. There are two financial instruments involved. Money or "gold" from the lenders and IOU notes from the borrowers. But these are no longer of the form "*i* owes *j*" but are "*i* owes 'the fund.'" If *i* fails to pay back the fund, then the rules of the game require that some form of prorating of the loss be assigned to all creditors. This could involve the specification of seniority, or as in the model here the losses could be required to be in proportion to the amounts involved.

If, instead of modeling a pooled money market where the market is merely an aggregating device that determines the price of money, we wished to model an inside or commercial bank, we would need three, not two, financial instruments. The first is the commodity money or gold deposited by the agents with surplus gold (they can be called "lenders", "savers," or "depositors"). The second is the IOU note given by the bank to these individuals. "The bank owes individual *i*." The third instrument is the IOU note between borrowers and the bank "Individual *j* owes the bank." The problem of 100 percent reserve banking appears at this point in the rules. Does the bank issue paper or gold to the borrowers? If it issues gold or is required to issue 100 percent gold-backed paper, then (leaving aside any capital of the bankers) it can only issue as loans an amount equal to its deposits. If this is the case, then the difference between this model and the money market model is that one more financial instrument is needed and that liability has been changed. Because a borrower's failure to pay the bank may cause the bank to fail against the

depositors, we need not only bankruptcy rules but bank failure rules. In a world without exogenous uncertainty, this may appear to be much ado about nothing (in a well-designed system, there is no reason for anyone to default), but the complete definition is required for a playable game.

A fourth mechanism can be constructed by imagining a central or government bank that issues "paper gold" in return for IOU notes from individuals. Unless the government itself has gold, this cannot be a 100 percent reserve system but involves the creation of government money (see chapter 12). We begin with the third mechanism, that of a money market.

9.3 Enough Gold in a Society with and without Moneylenders

9.3.1 Quantity and distribution of money

An exchange economy modeled as a strategic market game with one commodity serving as a money, with no credit and all traders insignificant in size will have its noncooperative equilibria (NEs) coincide with the competitive equilibria (CEs) if there is enough money, well distributed, to facilitate trade. The meaning of "enough money" is that the NEs are interior. In other words, the constraint that an individual cannot spend more of the means of payment than the individual holds is not binding.

The condition of enough money is characterized both by the total amount of money in the system and by its distribution (taking the transaction technology as given). As already noted in section 7.3.3, an economy may not have enough money, no matter how it is distributed. It is also possible that a redistribution will give rise to interior solutions. These statements can be made precise and the general theorems illustrated by means of specific examples. If there is enough money but it is maldistributed, it can be shown that a loan market for gold will bring efficiency.

9.3.2 Enough money in an economy with one money

Let $E(n, \phi^i, a^i)$ be an exchange economy with n types of trader where each trader of type i has an endowment a^i of $m+1$ goods and a utility function ϕ^i. Associated with this economy is a strategic market game where the $(m+1)$st commodity serves as a money. The game can be generally denoted by $\Gamma(n, \phi^i, a^i, s^i)$, where s^i denotes the strategy set for a player of type i. In any specific kind of trade, the need for money in trade will

depend on the nature of the strategies employed. For example, if we use the game suggested by Dubey and Shubik (1978), a strategy s^i by a trader of type i is of the form (b^i, q^i), where $\sum b^i_j \leq a^i_{m+1}$, all $b^i_j \geq 0$ and $0 \leq q^i_j \leq a^i_j$. A strategy consists of a vector of money bids, where the sum of the bids cannot exceed cash on hand, and an amount offered of every other commodity, where the amount offered cannot exceed one's initial endowment of the commodity.

We can now specify the three conditions concerning the sufficiency and distribution of money. Associated with the exchange economy E will be a set of competitive equilibria (CEs).

Enough money, well distributed The game Γ will have enough money that is well distributed with respect to a CE k if

$$\sum_{j=1}^{m} p_{jk} \max[0, (x^i_{jk} - a^i_j)] \leq a^i_{m+1} \tag{9.1}$$

for all i, where p_{jk} are the prices at the kth CE and x^i_{jk} is the final holding of good j by individual i at CE k.[7]

Enough money, badly distributed The game Γ will have enough money that is badly distributed with respect to a CE k if

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p_{jk} \max[0, (x^i_{jk} - a^i_j)] \leq \sum_{i=1}^{n} a^i_{m+1} \tag{9.2}$$

and for some i

$$\sum_{j=1}^{m} p_{jk} \max[0, (x^i_{jk} - a^i_j)] > a^i_{m+1}. \tag{9.3}$$

Not enough money The game Γ will not have enough money with respect to a CE k if

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p_{jk} \max[0, (x^i_{jk} - a^i_j)] > \sum_{i=1}^{n} a^i_{m+1}. \tag{9.4}$$

9.3.3 Exchange without a loan market

A simple example illustrates the three cases noted above. Let there be m types of trader trading in $(m + 1)$ commodities. All traders have the same

utility function of the form

$$\phi = \prod_{j=1}^{m} x_j^{1/m} + x_{m+1}. \tag{9.5}$$

The separability of the $(m+1)$st commodity is for ease of calculation in the example and is not critical to the general argument.

Let each trader of type i have an endowment density of $(0, 0, \ldots, mk, 0, 0, \ldots, 1)$, where the mk is for the ith good. By inspection, the unique CE of this economy is at the symmetric distribution $(k, k, \ldots, k, 1)$ for all. If we set $p_{m+1} = 1$, then from

$$\frac{\partial \phi}{\partial x_j} \Big/ p_j = 1,$$

we obtain

$$\frac{1}{m} \frac{k^{m-1/m}}{k^{m-1/m}} = p_j \quad \text{or} \quad p_j = \frac{1}{m}. \tag{9.6}$$

The value of the purchases of each individual equals the value of sales and is $\frac{1}{m}(m-1)k$.

If the exchange economy is modeled as a bid-offer game (see Dubey and Shubik, 1978) using the $(m+1)$st commodity as a means of payment, then a strategy by a player of type i is of the form $(q_1^i, b_1^i, q_2^i, b_2^i, \ldots, q_m^i, b_m^i)$, where $q_j^i \leq a_j^i$ for $j = 1, \ldots, m$ and $b_j^i \geq 0$ for $j = 1, \ldots, m$, and $\sum_{j=1}^{m} b_j^i \leq a_{m+1}^i$. (We omit a subscript or superscript to identify a specific trader; the symbol a_j^i indicates the amount of good j held by an individual of type i). At equilibrium in this game, if

$$\frac{1}{m}(m-1)k \leq 1, \tag{9.7}$$

then a strategy of a trader of type i will be of the form $(0, k/m, 0, k/m, 0, k/m, \ldots, 0, k/m, (m-1)k, 0, k/m, \ldots)$, where $(m-1)k$, 0 is the ith pair.

If $(m-1)k > m$, then no one has enough money for efficient trade. There will be an NE with constraints that will not coincide with the CE.

We are now in a position to describe a class of different but related games that differ only in the distribution of the means of payment. Instead of giving all m types of trader 1 unit of money each, we give type i, x^i,

where $\sum_{j=1}^{m} x_m^j = m$. If $(m-1)k > m$, then no matter how the money is distributed there is not enough. For every game in this class, the NEs will be on the boundary for some traders. This is not true for *some* of the games if $(m-1)k \leq m$.

Whenever there is enough money so that for some distribution there is an interior NE, it is easy to construct a distribution with the same amount of money where some individuals will be at a boundary. For example, give all traders of one type all of the money. However special our example above, the comments thus far are completely general, except that there might be a commodity money that was never in sufficient supply because its marginal worth could fall fast enough to more than offset any increase in supply.[8]

9.3.4 Exchange with a loan market

In this section, a money market is introduced and an endogenous rate of interest is obtained and interpreted. The introduction of loans requires at least one new financial instrument, the IOU note. A trader of type i who wishes to borrow, bids an amount b_{m+2}^i, which is interpreted as follows. The specific trader promises to pay the money market at the end of all trade an amount b_{m+2}^i in gold. The trader of type j who wishes to lend offers the market q_{m+1}^j in gold. The price of a loan is fixed by

$$\frac{\sum_{i=1}^{m} b_{m+2}^i}{\sum_{j=1}^{m} q_{m+1}^i} = 1 + \rho,$$

which states that one plus the rate of interest is determined by dividing the sum of all promises to pay by the sum of all offers of loanable funds.

In the game that has a pooled money market and a continuum of traders, a strategy becomes

$$(b_1^i, q_1^i, \ldots, b_m^i, q_m^i, q_{m+1}^i, b_{m+2}^i), \tag{9.8}$$

where the first m pairs represent a bid of an amount of money and an offer of an amount of good j in the market $(j, m+1)$, where j is exchanged for money. The final market $(m+1, m+2)$ is where gold is exchanged for IOU notes.

Because IOU notes can be created at will by those who wish to borrow money, we face a game design problem concerning the unbounded creation of debt. In actuality, if a private firm or individual goes to a bank or any other source of lending, the borrower's IOU note becomes more and more suspect as its size grows in relation to the real resources owned by the borrower. A more complicated game could distinguish between "prime names" and "lesser names" in their ability to create debt, but this is not a logical necessity.

To keep the game well defined and playable, we need to introduce rules that have the effect of bounding rational strategic behavior. This means that if an individual acting in a rational optimizing manner were informed of the action of all others, no matter how irrational or bizarre, that individual's best reply would always be bounded in the amount of IOU notes issued.

A natural way to keep the amount of debt issued bounded is to limit the issue by an individual of IOU notes. Clearly, if an individual promises to pay in gold more gold than there is in the whole economy, that individual will be unable to his full obligations. Thus an upper bound exists.

When a default occurs, society must make adjustments concerning two sets of individuals, the debtors and the creditors. Discouraging default or punishing defaulters is therefore only part of the problem. The rules must be such that lenders are not discouraged from lending; furthermore, they must specify what is to be done in case of default. This could be payment of ten cents in the dollar," the right to confiscate the debtor's remaining property, or any other noneconomic but societally sanctioned arrangement.

There appear to be two fundamentally different situations involving default. The first is where there is no exogenous uncertainty present in the economy, as is the situation here. The second is where there is exogenous uncertainty present. In the first instance, society need only guard against purposeful strategic default. It is a problem of moral hazard. The individual goes into default because it pays to do so.

When there is exogenous uncertainty present and the money market treats all individuals in aggregate regardless of the possibility that they may have different risk aversion, the possibility of default involves luck as well as intent. A Draconian default penalty could penalize borrowers so badly that no one would dare to borrow. Too light a penalty could cause the lenders to refuse to lend. If penalties are not individual but are the

same for all then the selection of a single anonymous penalty is the equivalent to the creation of a public good that sets the societal acceptable level of default. Individual risk attitudes are only implicitly reflected in the overall rules.

When no exogenous uncertainty is present, if a single anonymous penalty is suficiently harsh to discourage strategic default for all, then creditors do not need to be concerned with seniority of claims and settlement arrangements. There will be no default in equilibrium.

Even though a rational solution will not involve default when there is no exogenous uncertainty, a playable game (as opposed to its solution) requires settlement rules be specified in case of default. Shubik and Wilson (1977) and Dubey and Shubik (1979) have discussed the optimal default punishment for any economy without exogenous uncertainty where individuals can borrow fiat money from an outside or government bank. In essence, the problem is the same here as far as a borrower is concerned.

An easy way to construct an optimal default penalty for the economy with a money market is to imagine that at the end of trade, when the IOU notes are due, all individuals pay up whatever they can and the debt still outstanding is transmuted into holdings of negative amounts of gold. Thus we adjust the definition of the utility function so that it is defined not merely in $\mathbb{R}_+^{m+1}$ but in $\mathbb{R}_+^m \times \mathbb{R}$. We consider the utility of negative final holdings of money directly. Figure 9.4 shows the indifference curves for gold drawn against an aggregate bundle of all other goods. Negative gold is IOU notes in default.

The continuation of the convex contours of the indifference curves into the negative orthant indicates in a general way how badly going into default hurts. The line $M_1 I_1 E_1$, is to be interpreted as follows. Suppose that the price system at an equilibrium is known. The value of an individual's initial resources is indicated by I_1; in terms of gold, they are worth M_1. An individual facing this price system is willing to choose a strategic default at point E_1. In a general equilibrium system with initial resources of I_2 and prices indicated by $M_2 I_2 E_2$, the individual would not default but would trade to E_2.

If computation were costless, the referee (government) in this game could pick precisely the least harsh default penalty to guarantee that any CE in the exchange economy could be attained as an NE of the strategic market game. This can be done by solving the exchange economy for its

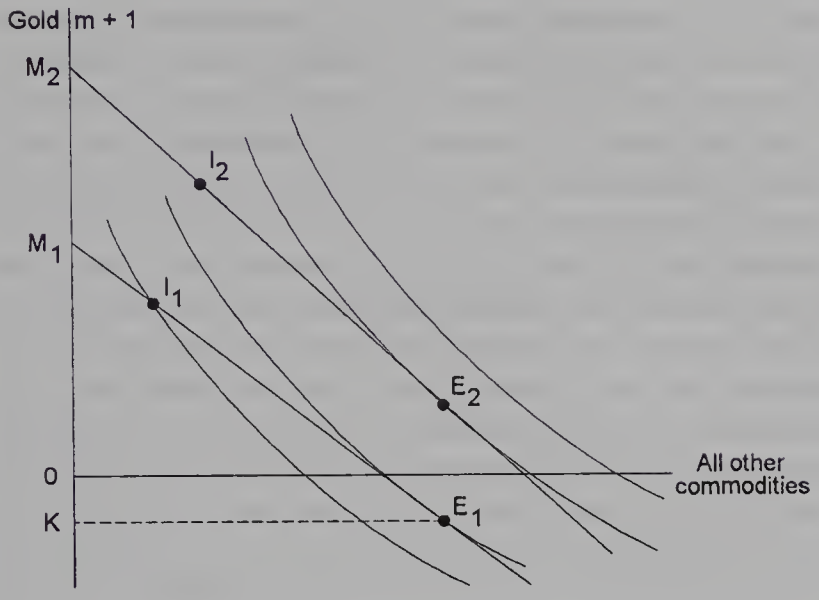


Figure 9.4
Default penalty.

set of CEs and selecting any CE you wish to have as an NE of the game. There are m types of traders; hence at the CE, there will be m different Lagrangian multipliers $(\lambda_1, \lambda_2, \ldots, \lambda_m)$, where each can be interpreted as the marginal utility of an extra unit of gold at equilibrium. Select the maximum of these:

$$\lambda^* = \max(\lambda_1, \lambda_2, \ldots, \lambda_m).$$

We now extend the definition of the utility functions of all individuals as follows:

$$\phi_i = \phi_i(x_1^i, \ldots, x_{m+1}^i) \quad \text{for all } x_j^i \geq 0,\ j = 1, \ldots, m+1$$
$$= \phi_i(x_1^i, \ldots, x_m^i, 0) + \lambda^* x_{m+1}^i \quad \text{where } x_{m+1}^i \leq 0,$$

which merely states that the marginal disutility of default is always greater than or equal to the marginal utility of income at equilibrium.

Because neither computation nor knowledge is free, the referee may not be able to carry out this calculation. Fortunately, when no exogenous uncertainty is involved, the referee does not need to calculate, only to make the penalty harsh enough. This is not true when exogenous uncertainty is present (see Dubey, Geanakopols, and Shubik, 1988).

When the game has no exogenous uncertainty and lasts for only one period, a harsh enough penalty should stop strategic bankruptcy and obviate the need to worry about the rules for treating the claims of creditors who are not being fully repaid. Unfortunately, a well-defined game must specify the payoffs for all feasible states, even though it might appear to be foolish ex post to have attained them.

When the loan market involves the pooling of lenders' funds, the rules concerning the treatment of default must specify how the lenders share in losses and in dividing any assets that are recovered. The simplest rule is that they share in a pro rata manner in the losses, recovering nothing from those in default, and that the penalties on defaulters are exogenous to the economic activity (such as debtors' prisons). A more complicated rule might involve garnishing the possessions of the debtors, selling them off, and paying the proceeds to the creditors (see section 11.3.5 and Dubey and Shubik, 1988). In fact, most societies have both garnishing rules and seniority rules concerning debt settlements. These depend on both the physical properties of the goods (such as durability) and social customs (such as leaving a bankrupt craftsman enough basic tools to continue his craft). Because our main purpose is to consider the money rate of interest that emerges from the loan markets, we do not further discuss justifying the specific structure of the loan default penalty, seniority of creditors, and garnishing rules. The key point is that they are *logical necessities* for a well-defined game and are not merely institutional ephemera.

9.3.5 The extensive and strategic forms

The extensive form of the game with a money market is shown in figure 9.5. The label $P_{1,\ldots,m}$ indicates that all players of all types move simulta-

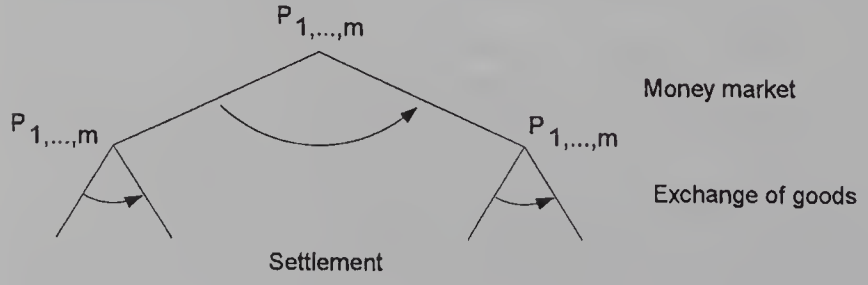


Figure 9.5
Exchange with a money market.

neously in the money market. The arrows indicate that the strategies are continuous and cannot be fully described by a finite tree. The information conditions indicate one-element sets, or simultaneous moves with otherwise full information. In a game without exogenous uncertainty and nonatomic players, the distinction between perfect equilibria and nonperfect equilibria disappears (see Dubey and Shubik, 1981a). In essence, if all know something but all others choose to ignore the knowledge, then being too small to influence the market does not do the individual any good. The important technical observation is that equilibria exist for strategies in this two-stage game, even where it is known that the strategies do not become complex functions of the previous moves of everyone but depend only on the state the player is in. The strategies that are simply vectors, as shown in (9.8), yield perfect NEs.

9.3.6 The money rate of interest in one-period exchange

In section 1.2, we considered $E(n, \phi^i, a^i)$ to be an exchange economy, with which we associated a strategic market game, denoted by $\Gamma(n, \phi^i, a^i, s^i)$, where the $(m+1)$st commodity serves as a money. We now consider a somewhat larger game with a money market, denoted by $\Gamma(n, \phi^i, a^i, s^i, \mu)$, where the μ symbolizes the default penalty. A simple example of a strategic market game with a commodity money and a loan market is sufficient to explore the emergence of the endogenous money rate of interest.

We consider an example with two types of traders trading with two goods and a commodity money. The utility functions of all traders are the same and are of the form $\phi = \sqrt{x} + \sqrt{y} + z^+ + \mu z^-$, where the z^+ and z^- stand for the positive and negative amounts of the final holding of gold, and μ is the default or bankruptcy penalty parameter.

The initial endowment of traders of type 1 is assumed to be $(2, 0, 0)$ and that of traders of type 2 is $(0, 2, A)$, where the third commodity is used as money. A strategy by a trader of type i is of the form (u^i, v^i, b^i_j, q^i_j), where $j = 1, 2$, u^i is the amount of IOU notes bid in the money market, and v^i is the amount of gold offered.

Because of the special structure and the assumption that all traders are of insignificant size we can simplify the notation even further. We assume that the price of money at the end of trade is set at $p_3 = 1$. The other prices are denoted by p_1 and p_2. Traders of the first type only will borrow and those of the second type only will lend. Similarly, because of the

initial endowments, only those of the first type can sell the first good, and only those of the second type can sell the second good. The writing of superscripts and subscripts can be cut down in this example by setting $q_1^1 = x$ and $q_2^2 = y$.

We assume that there is a type-symmetric noncooperative equilibrium[9] without strategic bankruptcy in the range of $A \geq .5$. This calls for selecting a bankruptcy penalty of $\mu \geq 1.179419$, which provides a sufficiently harsh penalty to avoid strategic bankruptcy. We furthermore assume that traders of type 1 borrow only to spend on buying the second good (i.e., they do not borrow to buy the first good or to hoard). Given these simplification, we can write the payoff functions as follows:

$$\Pi_1 = \sqrt{(2-x)} + \sqrt{(u/(1+\rho)p_2)} + (p_1 x - u)^+ + \lambda(p_1 x - u)^- \tag{9.9}$$

$$\Pi_2 = \sqrt{(A-v)/p_1)} + \sqrt{(2-y) + v(1+\rho)} + p_2 y, \tag{9.10}$$

where $p_1 = (Av)/x$ and $p_2 = u/(1+\rho)y$. Also, $1 + \rho = u/v$; and λ in (9.9) is a Lagrangian multiplier that reflects the shadow price of the capacity constraint on money needed for transactions. If bankruptcy is to remain strategically unattractive, $\lambda \leq \mu$.

In essence, the simple structure reduces the strategy of each trader to two variables: for type 1, the amount of money to borrow (and bid) and the amount of good 1 to sell; and for type 2, the amount of money to lend and the amount of good 2 to sell—(u, x) and (v, y).

Even an example as simple as this, where the boundary inequalities are to be treated in full generality, involves considerable computation. If treated game-theoretically in the fullest sense, with a finite number of traders each influencing price, the calculation is hardly tractable. By treating of p_1 and p_2 as parameters, however, we can consider a modified general-equilibrium approach.

Taking first-order conditions from (9.9) and (9.10), we obtain the following relations:

$$\sqrt{x} = 2u^2/v, \tag{9.11}$$

$$v = y/2\sqrt{(2-y)}, \tag{9.12}$$

$$y = x^2/(1x). \tag{9.13}$$

Table 9.2
Trade, money supply, and interest rate

A	ρ	x	y	u	v	p_1	p_2
≥ 1	0	1	1	.5	.5	.5	.5
.9	.0619	.9691	.9109	.4635	.4365	.4783	.4791
.8	.1356	.9329	.8155	.4254	.3746	.4560	.4594
.5	.4789	.7748	.4950	.2983	.2017	.3850	.4075

We also have

$$u + v = A, \tag{9.14}$$

and obtain

$$1 + \lambda = 1/2\sqrt{(2 - \rho)}p_2 u. \tag{9.15}$$

Fortunately, because this example is simple enough, we can solve these equations for various values of A with a simple program. Four values are shown in table 9.2.[10] The first entry indicates a solution region not covered by the equations, namely, when there is more than enough money. The next three give values for A ranging from just enough ($A = 1$) to half that amount. Table 9.2 displays some values of all relevant variables as the amount of money is varied from more than enough to less than that needed for efficient trade.

We observe that when there is enough money, however badly it is distributed (without a money market, trade would be zero), the rate of interest is zero. There is no time discount in the model and because the money figures in the utility function at full worth at the end, this implies that no utility is lost in lending one's gold and using it for consumption, provided it is repaid in the same period. In reality, if gold were truly a consumer durable, then there would be a utility loss attached to the amount of time the gold was in transit or float.

When there is not enough money, a positive interest rate appears and this has a clear interpretation. It is the shadow price of the capacity constraint on the optimization with the cash-flow constraint. In essence, the ρ represents a price wedge between the consumption worth of the gold and the extra value it has as a medium of exchange in short supply.

If we assume that there will be strategic bankruptcy, trade will still proceed, and the payoff to each type will be as follows:

$$\Pi_1 = \sqrt{2 - x} + \sqrt{(u/(1+\rho)p_2)} + \mu(px - u) \tag{9.16}$$

$$\Pi_2 = \sqrt{(A - v)/p_1)} + \sqrt{(2 - y)} + (1 + \rho^*)v + p_2 y, \tag{9.17}$$

where the ex ante rate of interest is

$$1 + \rho = u/v, \tag{9.18}$$

the ex post rate is

$$1 + \rho^* = (A - v)/v, \tag{9.19}$$

and $p_1 = (A - v)/v$ and $p_2 = u/(1 + \rho)y = v/y$.

From the first-order conditions, we obtain

$$v = A - \frac{1}{\mu}\left(\frac{x}{2\sqrt{2 - x}}\right) \tag{9.20}$$

$$\mu = \frac{1}{2\sqrt{u(1+\rho)p_2}} \tag{9.21}$$

$$(1 + \rho^*) = \frac{1}{2\sqrt{(A - v)p_1}} \tag{9.22}$$

$$p_2 = \frac{1}{2\sqrt{2 - y}}. \tag{9.23}$$

Table 9.3 shows the solutions for values of A from $A = .5$ to $A = .1$, where the bankruptcy penalty is set at $\mu = 1.179419$.

The value of $A = .5$ is on the borderline of strategic bankruptcy for $\mu = 1.179419$. For A smaller than .5, bankruptcy is active. There is a

Table 9.3
Trade with bankruptcy

A	$1+\rho$	$1+\rho^*$	x	y	u	v	p_1	p_2	Π_1	Π_2
.5	1.478	1.479	.778	.495	.298	.202	.384	.408	1.810	2.608
.4	1.740	1.651	.676	.384	.263	.151	.369	.394	1.750	2.494
.3	2.137	1.899	.557	.272	.221	.104	.353	.381	1.980	2.361
.2	2.837	2.306	.414	.164	.172	.061	.337	.369	1.632	2.200
.1	4.580	3.201	.239	.066	.110	.024	.319	.360	1.552	1.938

spread between ex ante and ex post interest rates and the bankruptcy penalty is a deadweight loss to welfare.

The example selected above was simple enough that it could be calculated simply, albeit tediously. Any more complicated example calls for a general computation procedure. Fortunately, the Scarf algorithm can be modified to include monetary control variables. Hence using a version of this algorithm constrained by a commodity money, Rutherford and Shubik were also able to compute the solution to this simple example (see also Shubik, 1990).

Notes

1. A distinction must be made between a *consumable*, which yields its value at a point in time, and a *durable*, which provides its value as a stream of services over the time interval. (This is considered in detail in chapter 13.)

2. *The McGraw-Hill Dictionary of Modern Economics* (1965) defines *Float* as follows:

The amount of money outstanding at any specific time in checks which have been issued but not collected. The float arises from the time differential between the crediting and collection of payments. If, for example, the Federal Reserve Bank of New York credits a New York City member bank on its reserve account for a check drawn on a California bank in advance of the actual collection of the check, total reserve balances temporarily increase, since the New York bank has been given a credit while the California bank has not yet been charged. This type of credit extension is necessary because of the huge task of check collection and the large volume of checks that are handled each year. Instead of keeping track of each check and crediting the proper account only at the time of collection, the reserve banks grant credit accounting to a specified time schedule, based primarily on the location of the drawee bank in relation to its Federal Reserve bank office. Under present time schedules, credit is granted in two business days for checks drawn on banks in areas where no Federal Reserve banks or branches are located, even if it takes three days to make the collection. The inability of the system to make the collection within a specific time schedule is due to the impossibility of meeting time requirements even though no unusual delays have occurred, to bad weather, and to strikes or similar factors which delay the delivery or payment of checks. Figures for the float appear in the Federal Reserve report and are published weekly in most national newspapers.

See "Forecasting Float," *Monthly Review*, Federal Reserve Bank of New York, February, 1963, pp. 30–35. In 1974, Federal Reserve float provided about 6 to 7 percent of number bank reserves (Hoel, 1975).

3. Or, more precisely, if the period stretches from [0, 1], all trade takes place at time $t = 0$ and the final user has the gold for (0, 1].

4. We adjust the definition of the utility function to change the domain from $\mathbb{R}^m_+$ to $\mathbb{R}^m_+ \times \mathbb{R}$.

5. Note we are discussing *credit*, not fiat money. This credit could be granted by an inside or outside bank.

6. There is some debate on his precise name, Pacioli or Paciolo.

7. Price is given by $\sum b^i_j / \sum q^i_j$. It is also important to note that the inequalities are specified for the instance where the individuals are assumed either to buy or to sell; buying *and* selling in the same market would call for more money.

8. Note if we had chosen a utility function of the form $\Phi = \prod_{j=1}^{m+1} x^{1/m+1}$, then given endowments $(0, \ldots, 0, mk, 0, \ldots, 0, x)$ for traders of type i, no matter what size x is selected, there never would be enough money. The price elasticity for extra money is such that an increase in supply lowers price in a way that purchasing power is never sufficient for the efficient exchange of the other resources.

9. A *type-symmetric equilibrium* is an equilibrium where all agents of the same type take the same action.

10. These calculations were also checked independently by Jingang Zhao.

10 Trade with Gold and Fractional Reserves

10.1 One-Period Rates for Interest

10.1.1 The three sources of the money rate of interest

In the conventional literature on the interest rate, there really is no interest rate to be expected in the one-period exchange model. One might argue that the interest rate appears as an artifact of imposing a cash-in-advance constraint, but this would miss the more important point, namely, that we have a *money rate of interest* and have scarcely begun to explore how it relates to what we might call "a real rate of interest."

Leaving out physical transaction costs, there are two inside money rates of interest possible in a one-period money exchange economy without exogenous uncertainty: (1) the "trust constraint" premium in financing the float; and (2) the default premium in recouping loan losses. (A third rate, related to a "societal risk premium" when there is socially desirable active bankruptcy and incomplete markets, requires the introduction of exogenous uncertainty.)

In essence, *the money rate of interest is the price of trust*. In an economy where all trades are for gold, no trust is needed in the goods exchange. The money market, however, requires that a set of individuals (lenders) temporarily exchange their gold for individual IOU notes or promises to pay. Trust may be ameliorated by laws imposing unpleasant consequences on those who default, but even so, the act of lending requires faith by the lenders in the laws, the borrowers, or both.

When the level of honesty is high or the laws sufficiently harsh[1] that individuals will pay back if they can, then the money rate of interest is only the shadow price of the transaction constraint on trade. If there is "enough gold," then the shadow price is zero, even if there is lending.

If the formal penalties are too low, then it will pay some individuals to go bankrupt strategically. But with symmetric information this can be foreseen[2] and translated into a loan interest rate that reflects a loss reserve premium for strategic bankruptcy. The lower the penalty, the higher the likelihood of bankruptcy and the higher the loss premium. The ex ante money rate of interest goes up as the severity of the default penalty goes down.

10.1.2 Interest with exogenous uncertainty

In an economy with exogenous uncertainty, the presence of incomplete markets introduces another factor into the one-period money interest rates. Dubey, Geanakoplos, and Shubik (1988) have shown that, with

incomplete markets, a default penalty conducive to default in some states may be better for all than a penalty so harsh that no one is willing to risk default, and little or no credit is granted. But a default penalty is not uniquely selected on the basis of efficiency (see section 12.4.1). Some may favor one class of traders over others. For any given default penalty and strategy in an economy with exogenous uncertainty, an individual may default if unlucky. The point here is that without exogenous uncertainty, and given enough money, one can obtain optimal exchange without bankruptcy and a zero rate of interest, whereas in a society with incomplete markets, efficiency considerations call for active bankruptcy and a positive rate of interest to compensate for default. Furthermore the interest charged must rise as the variance in uncertainty increases.

10.1.3 The level of commercial trust

We can model the intrinsic level of trust for short-term transactions by introducing one extra parameter into the system. The overall level of trust in any society allows some individuals to grant short-term credit to each other without the need to use a bank. Thus four or five days are granted for settlement, or thirty days on some bills. This can be modeled as follows. Instead of requiring the condition

$$\sum_{j=1}^{m} p_{jk} \max[0, (x_{jk}^i - a_j^i)] \leq a_{m+1}^i,$$

we add one term with the parameter w, where $0 \leq w \leq 1$ to obtain the condition

$$\sum_{j=1}^{m} p_{jk} \max[0, (x_{jk}^i - a_j^i)] \leq w \sum_{j=1}^{m} p_{jk} \max[0, (a_j^i - x_{jk}^i)] + a_{m+1}^i,$$

which has the property that if $w = 1$, full credit is given to everyone for all sales. If $w = 0$, the cash constraint is as has been analyzed. The effect of a $w \geq 0$ is to lower the rate of interest. The w may be interpreted as the index of commercial confidence.

10.2 Alternative Loan Models

There are several alternatives to a money market, including banks that can expand the money supply. This section outlines the two most impor-

tant alternative institutions: the mutual not-for-profit bank and the for-profit bank. In a one-period model without exogenous uncertainty, a clearinghouse would serve adequately as a credit device, but, as indicated below, there are difficulties in the analysis of default in a clearinghouse not dealt with in this volume.

10.2.1 The mutual clearinghouse

In a clearinghouse arrangement with gross settlement the debt is specifically between two individuals. Trader i buys from trader j at the market price; j may deliver the good only to find out after the delivery that the clearinghouse cannot honor the paper issued by i. When this happens, the default is specific, j suffers the loss rather than sharing it with others. In a netting system an individual default may set off a chain of defaults. A mutual bank provides a means for pooling default. We turn to the analysis of such a bank, although the clearinghouse model merits future consideration.[3]

10.2.2 The mutual bank with fractional reserves

Experience ... shows that neither a state nor a bank ever have had the unrestricted power of issuing paper money without abusing that power; in all states therefore, the issue of paper money ought to be under some check and control; and none seems so proper for that purpose as that of subjecting the issuers of paper money to the obligation of paying their notes either in gold coin or bullion.
—Ricardo, [1821] 1965, p. 241

The mutual bank avoids some of the difficulties of designing either a clearinghouse system or individual for-profit commercial banking, in particular, it nicely centralizes the granting of credit and the accumulation of debt. Instead of having to be concerned with individual indebtedness between traders i and j, the bank is a full intermediary. If i has a line of credit from the bank and pays j, i's default is against the bank, not against j. The financial damage to j by i's default may come in an aggregated form through j's part ownership of the bank.

Because a mutual bank is designed to facilitate trade, it is not necessarily a profit maximizer. Furthermore, because it is a mutual bank, its losses may be prorated among the depositors.

The extensive form of the game is as follows. At the first move, all individuals who wish to do so deposit gold in the mutual bank. The bank

ownership is in proportion to the amount deposited. The amount of gold deposited is multiplied by a gearing ratio, which determines the total amount of credit offered to all borrowers. At the second move, all borrowers bid for bank money. At the third move, they exchange goods for bank money.

As before, there are n trader types, each with an endowment of $(m+1)$ commodities. Representative individual i has resources $(a_1^i, a_2^i, \ldots, a_{m+1}^i)$ and preferences represented by a concave utility function $\phi^i(x_1^i, x_2^i, \ldots, x_{m+1}^i)$. Our variables are as follows:

u^i = the amount of gold deposited in the bank;

k = the gearing ratio for banknote issue;

μ = the bankruptcy penalty;

M = the total note issue;

v^i = the amount of IOU notes bid by i for banknotes.

The ownership of the mutual bank is determined by the deposits of gold:

$$0 \leq u^i \leq a_{m+1}^i.$$

An individual i's share of ownership is specified by

$$u^i \Big/ \sum_{j=1}^{n} u^j.$$

The second move of the game has the mutual bank (a strategic dummy) offer to issue banknotes up to the level of $M = ku$, where k is the gearing ratio and u is the total of all gold deposits. All individuals bid for their share of the banknotes (which along with gold constitute legal tender). Thus an endogenous rate of interest is formed and is given by

$$1 + \rho = \sum_{i=1}^{n} v^i \Big/ M,$$

where v^j is the amount (in terms of banknotes or gold promised at the end of trade) offered by individual i for banknotes to be obtained before trade. As i promises more and more, it is less likely that i will be able to pay back. A natural extreme upper bound to impose is that i can never offer more than the largest amount of money that can be created, thus

$$0 \leq v^i \leq k \sum_{j=1}^{n} a_{m+1}^j.$$

The presence of a default or bankruptcy penalty will also supply a motivation against unreasonably high bidding, but the physical bound is sufficient.

The third move by the agents is to trade. To study the simplest set of strategies, we may imagine that all agents are completely uninformed at each stage; thus strategies remain a set of numbers rather than complex functions of information. This can be done by having the players name ratios for the amount they intend to spend in different markets. Thus a strategy is of the form

$$(u^i, v^i, b_1^i, q_1^i, \ldots, b_m^i, q_m^i, b_{m+1}^i, q_{m+1}^i),$$

where $\sum b_1^i \leq 1$ (less than one if there is hoarding).

After trade, there is a settlement of all accounts, which involves repayment of all loans to the bank. The bank accepts either its own notes or gold at par in repayment. After all loans have been repaid (including adjustment for default), the bank is liquidated, paying out its gold to the depositors pro rata.

The actual amount of banknotes in possession of i after the second move is given by

$$w^i = Mv^i \Big/ \sum_{j=1}^{n} v^j.$$

The final holdings of all of the first m goods are given by

$$x_j^i = a_j^i - q_j^i + b_j^i w_i / p_j.$$

The final amount of gold if no one went bankrupt is given by

$$x_{m+1}^i = a_{m+1}^i - u^i + u^i + (b_{m+1}^i w^i / p_{m+1} - q_{m+1}^i),$$

where the first two terms are the amount of gold retained. The next term is the return of the original deposit to a depositor in the mutual bank. The final two terms are the change in holdings of gold through purchase and sale.

Several items must be noted. If we restrict the use of gold so that it is not a direct means of payment but can be used only as bank reserves or for consumption purposes, then we can have a market for gold, where it is

bought for banknotes. If gold can be used for payments, then the money supply in the hands of the public includes both the banknotes issued against the gold deposited and the gold held by the public.

The interest rate formed in the second stage and denoted by ρ is the contractual (or ex ante) rate used to determine how much any borrower has agreed to pay. It may be that some borrowers are unable to meet their obligations and default. In this case, the depositors of the mutual will receive less than promised from the mutual. There will be an effective rate ρ^* where $\rho^* < \rho$, which is the ex post rate of interest taking into account the default of various traders.

The calculation of ρ^* is complicated by a possible "domino effect" in default among traders who are both depositors and debtors. If individual i goes into default, i lowers the dividend to be received by individual j, but this could cause j to default. Figure 10.1 shows a simple flow diagram for an iterative process to calculate the terminal interest rate.

On first pass through, we assume that all depositors will be credited with full payments. We note who will default on this assumption and calculate the total of the default. We then use this to calculate a new interest rate. The computation is repeated with the new interest rate and this is repeated until there are no additional defaults. This determines the final effective interest rate. Implicit in this computation are considerations of accrual accounting. For individuals who are both stockholders and debtors to the same bank and who owe a payment and are owed a dividend at approximately the same time, whether they will go bankrupt may depend on the sequencing of the payments.

THEOREM 10.1 The strategic market game with a continuum of traders of n types and a mutual bank for any (nonzero) amount of gold and for any gearing ratio k and default penalty μ has an active noncooperative equilibrium.

THEOREM 10.2 For a sufficiently high gearing ratio k and a sufficiently harsh default penalty μ the strategic market game with a mutual bank will have noncooperative equilibria that do not involve bankruptcy, that produce an endogenous rate of interest $\rho = 0$, and that give the same relative prices and distribution as the CEs of the related exchange economy.

(The proofs of these theorems are given in Shubik and Tsomocos, 1992; and proofs for a related model with a finite number of agents are given in Shubik and Zhao, 1991.)

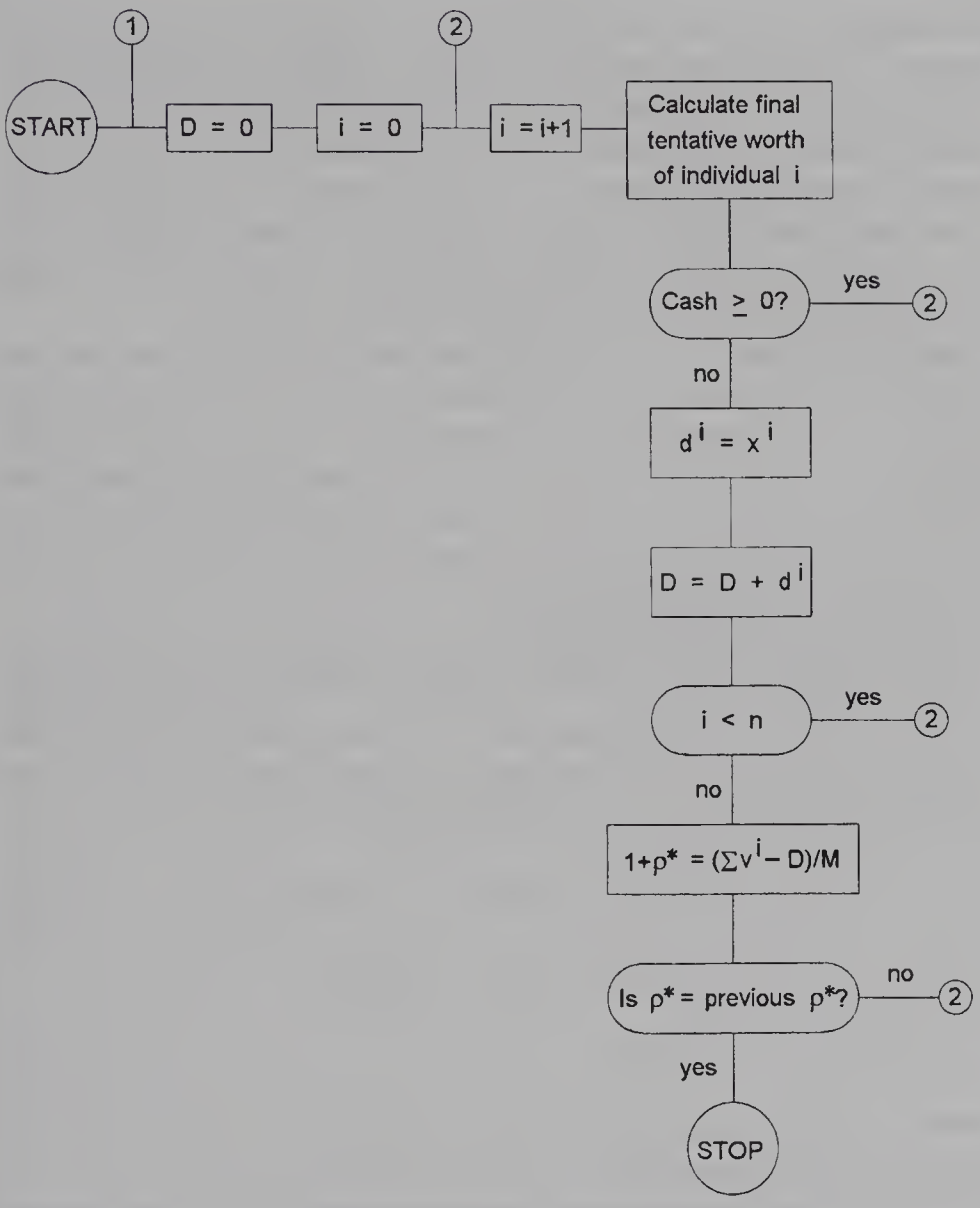


Figure 10.1
Calculating the ex post interest rate.

An example We extend our example here to a bank with fractional gold reserves. The initial endowment of traders of type 1 is assumed to be $(2,0,0)$ and that of traders of type 2 is $(0,2,A)$, where the third commodity is gold and can be used as money. A strategy by a trader of type 1 is of the form (v, b_j^1, q_j^1), where $j = 1,2$ and v is the amount of IOU notes bid to obtain banknotes. A strategy by a trader of type 2 is of the form (u, w, b_1^2, q_2^2), where u is the amount of gold deposited in the mutual bank and w is the amount of IOU notes bid to obtain banknotes.

Because all traders are of insignificant size in this example, we can simplify the notation even further. We assume that the price of money is set at $p_3 = 1$. The other prices are denoted by p_1 and p_2. Traders of the first type will only borrow and those of the second type will form the mutual bank and then borrow from it. Because of the initial endowments, only those of the first type can sell the first good and only those of the second type can sell the second good.

We assume that there is a type-symmetric noncooperative equilibrium and that setting $\mu = 10$ provides a sufficiently harsh penalty to avoid strategic bankruptcy. We furthermore assume that traders borrow only to spend on buying goods (i.e., they do not borrow to hoard). Given these simplifications we can write the payoff functions as follows:[4]

$$\Pi_1 = \sqrt{(2 - q_1^1)(v/(1+\rho)p_2)} + (p_1 q_1^1 - v)^+ + \mu(p_1 q_1^1 - v)^- \tag{10.1}$$

$$\Pi_2 = \sqrt{(2 - q_1^1)(v/(1+\rho)p_1)} + (p_2 q_2^2 - w + k\rho A)^+ + \mu(p_2 q_2^2 - w + k\rho A)^-, \tag{10.2}$$

where $p_1 = w/(1+\rho)q_1^1$ and $p_2 = u/(1+\rho)q_2^2$. Also, $1 + \rho = (v + w)/ku$ and $v = p_1 q_1^1$ and $w = p_2 q_2^2$.

Treating p_1 and p_2 as parameters and noting that if $k > 1$ and the rate of interest is greater than zero all gold will go into the bank, we find that the first-order optimization conditions give

$$\frac{1}{2}\sqrt{v/(1+\rho)p_2(2 - q_1^1)} = p_1 \quad \text{and} \tag{10.3}$$

$$\frac{1}{2}\sqrt{v/(2 - q_1^1)/(1+\rho)p_2 v} = 1 \tag{10.4}$$

Table 10.1
Different gearing ratios for $A = .5$

k	ρ	q_1^1	q_2^2	w	v	p_1	p_2
4	0	1	1	1	1	1	1
2	0	1	1	.5	.5	.5	.5
1.8	.0619	.9691	.9109	.4734	.4266	.4783	.4791
1	.4789	.7748	.4950	.2983	.2017	.3850	.4075

$$\frac{1}{2}\sqrt{w/(1+\rho)p_1(2-q_2^2)} = p_2 \quad \text{and} \tag{10.5}$$

$$\frac{1}{2}\sqrt{(2-q_2^2)/(1+\rho)p_1 w} = 1. \tag{10.6}$$

We can check that for $A = .5$ the values shown in table 10.1 satisfy. In the first instance, there will be a continuum of efficient NEs with a price range from .5 to 1. There is no a priori reason to select a specific price level.

10.2.3 The for-profit bank with fractional reserves

Let us subject the model given in section 10.2.2 to a few modifications. We assume the bank, instead of being a strategic dummy, is a profit maximizer. We distinguish between deposits in a mutual and ownership of shares in a for-profit organization. The bidding of gold in the first move is now interpreted as buying shares in the bank. This difference is not without consequences. In modeling the mutual, the losses due to bad loans were personal; even though aggregated and prorated, they were flowed through to the depositors. In a for-profit bank, one must distinguish between an institution where stockholders have limited liability and one where their liability is unlimited. If there is unlimited liability, there is no need for bank failure rules because losses are flowed through. If there is limited liability, however, the bank as legal person, can fail, and bank failure laws become a logical necessity.

The question of competitive banking is an important side issue to understanding the money supply and rate of interest; but an adequate discussion requires at least a three-period model. (We return to this in Volume 2, chapters 3 and 4.) For the one-period model, the relatively straightforward results are that with a Cournot (or quantity-of-money-control) model of bank competition for profit, efficiency cannot be achieved without a continuum of private banks because the oligopoly effect

attenuates slowly (see Shubik, 1976). If the oligopoly price drops as $1/n$, then having more than twenty banks might be regarded as "good enough." If banks are modeled as competing through price (i. e., naming a rate of interest), and given enough money, the classical price duopoly result obtains: competition begins at two. But this result is to some extent an artifact of the model. With only two banks competing, one may suspect that over time they might collude.

10.3 Comments and Interpretation

10.3.1 Ricardo and trust

Section 10.2.2 begins with a quotation from Ricardo, who argues that an attractive reason for having redemption in gold is that otherwise governments or private banks may cheat. The key element is the perception of trust, or the lack of trust. The threat of redemption discourages cheating.

Ricardo's remarks can be given operational and mathematical meaning in an attempt to construct a playable game with a mutual bank. In particular, by having redemption, the banknotes are tied into gold and the bank becomes a strategic dummy, thereby obviating the need to trust in its behavior.

10.3.2 Credit control as a public good

When there is not enough gold to use it directly as money, the members of the economy as a whole might wish to find an efficient, more or less anonymous way to extend the domain of the essentially trustless value-for-value trading with gold. This can be done with a mutual bank whose role is not to maximize profit, but to facilitate trade. By constructing the bank as a strategic dummy, we make the question of its goals irrelevant. In reality, every bank must judge credit risk, and hence does have an active strategy. At this level of simplicity, however, the bank is strictly a dummy. The government or the referee outside of the game can be regarded as setting the two key parameters, the default penalty and the gearing ratio.

10.3.3 Gold and the numéraire

By tying the issue of banknotes to gold and by relating the default penalty to the value of gold, both upper and lower bounds to the price system are

provided.[5] The quantity of gold provides the upper bound and the penalty linked to gold provides the lower. When there are enough banknotes and the default penalty is correctly set, the price of gold in terms of banknotes if k, and hence the price level, for a fixed k, is anchored to a commodity. We can use either gold or banknotes as the numéraire.

10.3.4 On uniqueness

In section 10.2.2, we observed that there exists a pair of values μ^* and k^* which define a strategic market game which has a single CE as an attainable NE (but see chapter 11, note 17). In essence, if we use gold as the numéraire, then in any exchange economy we can solve for all CEs and order them in terms of the amount of trade required to attain them (each measured in terms of $p_{m+1} = 1$). If the amount of gold in existence is less than the amount required to fiance transaction at the least trade CE, then to attain the CE as an NE we will require a $k > 1$. If we select $k = k^*$ such that

$$\sum_{i=1}^{n}\sum_{j=1}^{m} p_{jk} \max[0, (x_{jk}^i - a_j^i)] = k \sum_{i=1}^{n} a_{m+1}^i,$$

then only that CE is feasible as an NE in the strategic market game.

10.3.5 Hundred percent reserve banking will not work

In the current model, one could have $k = 1$, which would imply 100 percent reserve banking. But if gold were in short supply, the chief difference between using banknotes backed 100 percent by gold and using gold itself as currency is that the gold would remain in the bank instead of being in circulation. In terms of wear and tear and avoiding the dangers of theft, there may be a small societal gain. But unless there is enough gold, there is no expansion of the money supply and optimality cannot be achieved (see section 9.3.2).

10.3.6 Financing transaction demand

The whole structure of trade using banknotes from a mutual bank allows us (as a reasonable first-order approximation) to decouple trade in a mass-market mechanism—by substituting a system of technology and law for individual trust. The time needed for this substitute is the time required for transactions to clear. The demand is the *transaction demand*

for cash. If we extend the model to several time periods, the corresponding need to extend credit calls for a different and more complex mechanism. As the payment technology improves, the need for transaction loans diminishes, but the need for longer term production and consumption loans is scarcely influenced.

10.3.7 Banking and the money rate of interest

In the one-period model, there is no reason why the charge for providing banknotes should exceed their marginal cost. If, as has been done here, we accept the fiction that as a first-order approximation, we can regard the cost of supplying the notes to be negligible, then the cost of borrowing is zero if there is a large enough supply of money. Otherwise, there will be an interest rate even for the short-term transaction loan, reflecting the shadow price of its capacity constraint on trade.

10.3.8 Multistage interest rates and trust

In God we trust, all others pay cash.

The existence of a float has always posed problems to the economic theorist. In the one-period model without exogenous uncertainty, if simultaneous trades can be netted out, the need for cash in advance disappears with the construction of a central clearinghouse to keep track of all trades and call for the settlement of net differences. The amount of trust one needs to have in the honesty, speed, and efficiency of a clearinghouse is relatively small, but it is still more than gold coin on the barrelhead. Trade for a commodity money still involves some trust in the rule of law as it applies to commerce, but it is nevertheless less than the trust in a clearing system.

The existence of the "constraint" rate of interest depends on a lack of trust in the clearing system. Thus we have a cash-in-advance requirement. But as soon as we consider exchange involving two or more periods, the float or cash-in-advance constraint is of less importance. It is one matter to trust a central clearing mechanism that may take a few minutes or days, and quite another to extend credit for a period of several months or years. If the level of short-term trust in a society is high enough, we may dispense with cash in advance for intratemporal trading, but require cash settlement for any intertemporal balances.

When many periods are considered, then we need to consider (1) the presence of time preferences; (2) production and growth; and (3) the nature of the durability and depreciation of assets including the money (see Dubey, Geanakoplos, and Shubik, 1992). Furthermore, there are opportunities to vary the velocity of circulation. These influence the size of the money supply required for a given level of trust, although the variability of velocity can be treated as a logically separate problem (see Shubik, 1987).

In one-period exchange economies without exogenous uncertainty, the money rate of interest is the shadow price for the marginal increment of the trust substitute provided by commodity money in a world with little trust. As individual trust or commercial confidence increases, the need for a commodity money or for trust in a fiat money decreases, until no explicit store of value is needed in exchange and a numéraire or pure accounting money will serve to cover the credit exposure in the central clearing system.

In multiperiod economies, transaction financing appears to be less important and to pose different trust and evaluation problems than production or consumption financing.

10.3.9 Time, trust, money, and general equilibrium

In a timeless trustful world, there is no need for money. In a world where consumption desires and income streams do not match, where production and trade is anonymous, transactions take time, and trust a rare commodity, a surrogate for trust is needed. A commodity used as a money can serve. But for it to permit efficient trade, the commodity must satisfy the double coincidence of wants for all and also be in sufficient supply. Not even gold appears to have both of these properties, although for many centuries it has served as an approximation to the ideal.

The ideal is a societally created financial instrument that is accepted by all citizens as a surrogate for individual trust. Pure fiat is discussed in chapter 11, but before considering a government money, where the cost of a substitute for individual trust is trust in government, we note the properties of the "halfway house," a paper currency with fractional gold reserves. We compare this with the general equilibrium system for the same economy.

In the general-equilibrium system, prices are homogeneous of order zero. They are defined on the open interval $0 < p < \infty$, with a singularity

at zero. The strategic market game with fractional reserves, ties down the price system. It indicates how harsh the default penalty must be to avoid strategic default. The reserve ratio determines the amount of banknotes, hence both a lower and upper bound are placed on price. The bankruptcy penalty and the reserve ratio, both tied in with the gold, fully determine a closed range of prices. Thus the control mechanism is provided to bound prices.

If regarded as truly timeless, the one-period general-equilibrium exchange economy needs no monetary coverage of transactions. If regarded as a strategic market game, with markets taking some small finite amount of time to clear, it does require coverage of transactions, which can be provided by introducing commodity money, a fractional reserve paper money, or (as considered in chapter 11) a pure fiat money.

10.4 Historical Notes

10.4.1 A historical sketch of gold

Gold is both highly malleable and highly ornamental, inert, and easily alloyed to vary color and durability. It is heavier than lead (atomic weight 197), with a cubic foot weighing around half a ton. The total world mined supply is estimated at around 90,000–100,000 tons, although the estimated amount of gold on the planet (much of it dissolved in the sea) is 10,000,000,000 tons (Collins, 1975). Alluvial or placer gold appears to have been the first source of the metal. Currently, South Africa and Russia are the two main sources of gold. Pure gold is 24 carat.[6]

Gold bow tips from southern Nubia have been dated as predynastic, over 5,000 years old (Marx, 1978, p. 39). Gold as jewelry was present in dynastic Egypt and Sumeria. A mural from the tomb of Mereruka at Saqqara dating from 2300 B.C. shows the activities of goldsmithing (Vicker, 1975, p. 23).

Gold is used principally in jewelry, industry, and money. Even before coinage, gold in ingot form used in trade is attributed to the 14 gram bars of Pharoah Narmer or Menes (with the gold-silver ratio fixed at 1:2.5, Marx, 1978, p. 44).[7] The Babylonian shekel weighed 8.3 grams (Vicker, 1975, p. 144). The invention of coinage, attributed to the Lydians, provided a guarantee of quality and standardization of size useful to trade.[8] Croesus (ruled ca. 560–546 B.C.) established a bimetallic system, using

gold and silver coins with a ratio of 13:3 for silver to gold (Vicker, 1975, p. 144). Darius (ruled 522–486 B.C.) appears to have been the first to have his portrait stamped on his coins, and the Persians, to have been the first to use coinage for tax payments and as part of governmental control on the economy (Vicker, 1975, p. 46; for a fine popular history of gold and its uses through the ages, see Jennifer Marx, 1978). Without digressing into details on art, display, ostentation, avarice, lust, greed, and fear, the overall historical message is clear. Aesthetics, industrial and trade uses, and the relatively high cost of acquiring new supplies from Nature have enabled gold to play a significant economic role for around five thousand years.

Much of modern microeconomics and finance is based on static models and considerations of equilibrium in the context of a given set of laws. This utterly misses the significance of gold as a basic political social, and economic device where trust is vested, not in the kings or dictators, but in the substance itself. In a world where national law is far from perfect and international law is even less perfect, gold is an exemplar of the saying that "possession is nine-tenths of the law."

The history of Asia Minor and Persia from 500 B.C. onward appears to have gold being made into jewelry, jewelry into bullion, and bullion into coin in every permutation. The importance of gold as a store of wealth and means of liquidity in times of crisis even today is indicated by the use of jewelry in India, where the difference between the value of the jewelry and its gold content may be less than 10 percent, and where the customs and technology of the society are such that the family jewels can be turned into cash with reasonable ease. In the United States, by contrast, the gold value may be a relatively small fraction of the jewelry's price and the transaction costs for an individual not in the trade to buy and sell are large.

The breakdown of the forms and uses of the world's gold in the late 1970s is given in table 10.2.

10.4.2 Mints, treasuries, coining, and seigniorage

The needs of trade call for cognizability, standardization, minimization of individual trust, divisibility, durability, and portability—all of which can be satisfied reasonably well by gold coins (although for minor trades coins of less worth are called for). The minting of coins calls for an available supply of metal, the technical capacity to produce standard coins of a

Table 10.2
Breakdown of gold stock (millions of troy ounces)

Use	Amount	% of total
Lost or undetermined	311	10.94
Jewelry, decorative and religious	650	22.86
Private hoarding and investment	677	23.81
China and former USSR	72	2.53
Central banks	937	32.96
Emcf, imf	196	6.89
Total	2,843	100.00

Source: Coyne (1982).

guaranteed quality, the political power to maintain standards against counterfeiting—and the political will to abstain from debasing the coinage. These features are primarily properties of a public good and the authority in control of an acceptable coinage can profit by its manufacture.

Custom, trading habits, and law intermix. Since the start of coinage, some coins have been more popular than others. The trust appears to have been based on the behavior of the monarchs. The Roman Empire, from Nero onward tended to debase its coinage (Marx, 1978, chapter 9; Sutherland, 1974, chapter 7). By contrast, the bezant, nomisma, or solidus, first coined by Constantine, maintained its quality until the reign of Michael VII in 1071 (Carson, 1970, p. 195). The Venetian ducat, the Florentine florin, the Marie Theresa taler, the Spanish piece of eight and the British sovereign, among others, have had their popularity and wide circulation. Vicker (1975, p. 56) notes that various dies were still recently available in Beirut to strike fascimilies of the more popular coins.

The control of coinage to some extent depends on the technology. Until the sixteenth century coins were "hammer-struck"—a metal blank was placed between two dies and then hammered. Both speed and accuracy were relatively poor. Leonardo da Vinci had suggested means for mechanizing coin production and in 1580, a French moneyer Eloye Mestrelle[9] produced coins using a screw press in England for Elizabeth, (Carson, 1970, p. 241). He was later hanged as a forger, and the Royal Mint did not adopt machinery until 1662. In 1790, Matthew Boulton and James Watt built a self-powered coining machine. In 1839, a reducing machine that could copy a design in miniature was used in die making. In 1800, 3 million coins were struck in the United States; in 1850, 25 million were

Table 10.3
Cost of coins (in cents)

Penny	0.61
Nickel	2.32
Dime	1.22
Quarter	2.64
Half-dollar	5.45
Eisenhower dollar	8.32
S. B. Anthony dollar	2.97

Source: A Cosgarea, Associate Director for Operations, U.S. Mint, personal correspondence, November 18, 1986.

struck at the Royal Mint. In 1966, the Royal Mint produced 1.4 billion coins; in 1967, U.S. mints produced 25 billion (Becker, 1969, chapter 8).[10]

The power to mint was in general vested with the monarch. In republican Rome, the power was vested with the Roman Senate,[11] although Roman generals could obtain the right to mint to pay their troops. Kindleberger (1984) notes that the sovereign either owned the mint directly or farmed out the production of coins. While a profit was available in seigniorage charges, if the issuer priced the metal content too low in international trade, the coins might be melted down and exported as bullion.

The United States Mint was established by an act of law of April 2, 1792. Free coinage was provided. Individuals could bring metals to the mint and have them returned as coins with the same bullion value, with the mint bearing the coinage expense (Hobson, 1965, p. 73). The free minting of silver was omitted from the Coinage Act of 1873 (Wemple in Butts and Coxe, 1967, p. 36). The Bland Allison Act of 1878 required the Treasury to buy $24 million of silver, which produced 28 million silver dollars, leaving a sizable seigniorage spread (Hobson, 1965, p. 105). In March 1964, the U.S. Treasury discontinued the issue of silver coinage; between 1961 and 1964, its stock of silver dollars dwindled from 130,000,000 to 3,000,000 (Wemple in Butts and Coxe, 1967, p. 48).

Table 10.3 shows the cost per coin for United States coinage including manufacturing and metal costs was.

The estimated attrition rates for cents are 6.0–7.5 percent per annum, for nickels 5.2 percent; for dimes 4.6 percent and for quarters 2.5 percent.[12]

10.4.3 Mines and the cost of gold

Historically gold and silver mines were the property of the crown (see Wilcox, 1967, on silver production). The mines of the pharaohs, in Nubia and on the Red Sea coast (Marx, 1978, p. 43), were worked by slaves.[12] In Athens, mines were leased out and, except for a few one-man operations, generally worked by slaves (Finley, 1981, p. 101). The Lydians mined placer gold from the local rivers. A prime source of Roman gold was Spain, where the introduction of hydraulic mining considerable improved productivity. The Roman mines were worked almost entirely by slaves (Marx, 1978, p. 194).

In the Middle Ages, the supply of gold in Europe was relatively small. The total annual production in the thirteenth and fourteenth centuries has been estimated at between one and four metric tons (Day, 1978). This contrasts with world production of around 1,000 metric tons today.

The exploitation of the gold and silver of the Americas by the Spaniards is a monument to plunder, bloodshed, and slavery. The crown owned the mines and the local populace was enslaved to work them. In 1744, gold was discovered on the eastern slopes of the Urals and later in Siberia, mined first by serfs and later by political prisoners.

Free enterprise in gold and silver mining waited for the nineteenth century, when the world of gold was both "democratized" and expanded. Major find followed major find in the United States, Canada, South Africa, Australia, and elsewhere. In 1848, gold was discovered at Sutter's Mill in California by James Marshall; in 1850, near Bathurst in New South Wales, Australia, by Edward Hargraves. In 1859, gold and silver were discovered at the Comstock lode in Nevada. The Main Reef in the Witwatersrand basin, South Africa, was discovered by George Harrison in 1886. In 1896, gold was discovered in the Klondike in the Yukon; in 1892, Baily and Ford struck gold at what is now Coolgardie in Western Australia (Holmes, 1947; Curle, 1905; Green, 1968).

In the United States, with government permission, a grub stake, and rudimentary equipment, a "49er" or other placer miner could stake out and work a claim and have the San Francisco or Carson City mint coin money for him at no extra cost. By contrast, in the great underground mines of South Africa, working the reefs requires high capital investment, technology and plentiful labor.[13] It has been suggested that possibly the first three metals used by men were gold, copper, and silver in that order

Table 10.4
Major producers of silver and copper, 1985

Silver	Troy ounces (×1,000)	% of total
Mexico	69,000	16.74
Peru	60,395	14.65
USSR	47,200	11.45
U.S.	39,257	9.55
Canada	38,887	9.43
Other	157,434	38.19
Total	412,273	100.00
Copper	**Metric tons (×1,000)**	**% of total**
U.S.	1,007	12.41
USSR	600	7.39
Zaire	560	6.90
Poland	431	5.32
Peru	397	4.90
Other	5,118	63.08
Total	8,114	100.00

Source: Miksell (1970, p. 344).

(McDonald in Butts and Coxe, 1967). All three have been monetary metals. The world supply of silver has been estimated as around 1,000,000 metric tons and of copper as 90,000,000 metric tons (Mikesell, 1970, p. 344).[14] Recent production figures are given in table 10.4 (see also Wicksell, 1962).

Due to the relation between weight and value, copper never attained international monetary importance and is now essentially a nonmonetary metal. The monetary use of silver is much diminished, although it has vied with gold. One of the earliest references to the payments role of silver is given in Genesis 23:16, when Abraham buys land for Sarah's grave: "Abraham weighed to Ephrom the silver ... four hundred shekels of silver, current money with the merchant" (King James Version).

The gold-to-silver exchange ratios are a function of standard commodity supply costs and a demand with both economic and political components as well as considerable transaction costs. In Egypt, as has been noted, the ratio of gold to silver at one time was 1:2.5; at some points in early African trade gold to salt has been reported at around 1:1 (Marx, 1978, p. 252). McDonald (in Butts and Coxe, 1967, p. 12) gives

the ratios of gold to silver as Rome 1:10; then 1492, 1:11; 1760, 1:14; 1860, 1:15. In 1954 it was 1:39 and in 1984 it was 1:44 (see also Del Mar, 1968, on the ratios of gold to silver).

The supply of gold (and other metals) is highly dependent on the technology that controls cost. It is suggested that beyond a given level short-term costs are relatively inelastic. Edie (1983, chapter 4) provides a study of costs from the late nineteenth century to the nineteen twenties. The richness of the mines, geology, technology, and cost of labor result in great differentiations among mines. Edie notes (1983, p. 44) that between 1915 and 1925 costs per fine ounce at Old Rand rose from \$14.88 to \$17.16 while they fell in the New Rand from \$11.57 to \$10.75. A commercial estimate in 1983 of supply costs for gold (Gold 1983, Consolidated Gold Fields Plc., p. 19) suggest that the marginal ounce was at around 350 U.S. dollars, with steeply rising costs for any expansion of production unless it comes about by the discovery of new low-cost mines.

Notes

1. We do not formally model social custom, moral codes, or the intrinsic honesty of individuals, which may be regarded as implicitly included in the formal deadweight loss of the penalty.

2. In actuality, many of the problems in assessing credit risk involve nonsymmetric information, another extremely important topic supplementary to the observations here.

3. It is straightforward but unrealistic to modify the "window model" of Shapley, which was solved for its NEs by Sahi and Yao (1989), to allow for loan markets in all pairs of commodities by introducing $m(m-1)/2$ markets. For every pair, there are two markets: a loan of apples against oranges and a loan of oranges against apples. In essence this amounts to introducing two rounds of trade, which can overcome the failure of the double coincidence of wants.

4. The notation $(\ )^+$ or $(\ )^-$ stands for the positive or negative part of the item in the brackets, or zero.

5. Abstracting from the variable velocity of transactions.

6. The word "carat" is derived from the name of the seed of the carob tree (Greek *keration*; Arabic *qirat*).

7. Keynes (1930a, pp. 11–12) attaches more significance to the state designation of "the objective standard which shall correspond to the money-of-account" than actual coinage, which he suggests might be a mere piece of local vanity.

8. Whether coinage was independently invented by the Chinese, possibly earlier than in the West, is still an open question (see Carson 1970, p. ix).

9. Becker (1969, p. 121) gives this date as 1561 and the name as Mestrel.

10. The stamping machine at the Philadelphia mint can produce 10,000 coins per minute (Becker, 1969, p. 157).

11. The Roman mint was in the Temple of Juno Moneta (Marx, 1978, p. 204).

12. Other ancient gold sources are Ophir and Punt, although their precise location still appears open to question.

13. In the 1960s, the South African mines employed around 400,000; all but around 40,000 were black (Green, 1968, p. 48).

14. Of which the United States accounted for 38 million, Western Europe 41 million, and other countries 11 million metric tons.

11 One-Period Trade with Fiat Money

11.1 Inside and Outside Money

Gurley and Shaw (1960, pp. 72, 73) have helped highlight the distinction between inside and outside money. The distinction stressed here, however, differs somewhat from theirs. Inside money is a credit instrument issued by nongovernment agent used as a means of payment for which there is an offsetting financial instrument held by a nongovernment or government agent. Outside money is commodity money, coin, or fiat used as a means of payment for which there is not necessarily an offsetting financial instrument.

Although Wicksell ([1935] 1962, p. 8) noted that in the modern world a commodity money is not regarded as a true store of value, I suggest that the original use of gold, first uncoined and later as coins, was based on its properties as a transportable, durable store of value. The dynamics of trade has laid stress on the use of gold as a means of exchange. The buildup of governmental power and the development of formal rules of trade over the centuries have enabled governments to substitute law, custom, force, and expectations as a surrogate for the store-of-value aspects of money.

In this book, for all one-period models, commodity money is considered and studied as a means of exchange that is a full store of value: all exchanges between commodity money and any other goods are value for value, not value for expected value.[1] Although commodity money has not served as a full-bodied store of value for most times and most economies, it can be well defined and studied as an important extreme case in a playable game. The other extreme is a fiat money, which can be regarded as a commodity money of no worth as a commodity, but of worth as a means of exchange. The use of a commodity money as a national money has attenuated over the centuries. With the building up of the state and the law, the means of exchange function has fully dominated the store of value function. But the store of value function has been transmuted into the acceptance of expectations that the money will have purchasing power tomorrow. But this implicitly gives it the property of being a surrogate for a direct store of value.

A clean distinction between fiat money and commodity money as a store of value blurs as soon as we consider more than one period of exchange. Expectations enter not only in considering the support for paper money as a store of value, but also in deciding whether gold will hold up as a store of value, given new sources or changes in taste.

Table 11.1
Inside and outside money

	Means of payment?	Offsetting instrument?	Anonymity?	Store of value?
Outside money:				
Commodity money	Yes	No	Yes	Yes
Coin	Yes	No	Yes	Part
Fiat	Yes	No	Yes	Surrogate
Book entry	Yes	No	No	Surrogate
Inside money:				
Bank notes	Yes	Yes	Yes	Yes if 100% backed; usually surrogate
Checks	Yes	Yes	No	Surrogate
Credit cards	Yes	Yes	No	Surrogate
Book entry	Yes	Yes	No	Surrogate

A money, whether commodity or fiat, is the only financial instrument that exists in positive supply. All other financial instruments are always created in pairs that net to zero. Bank debt, which in Gurley and Shaw (1960) terminology is inside money, comes close to being fully acceptable as a means of payment, but it must be considered in terms of itself and the offsetting instruments in its creation. They net to zero.

In terms of the analysis here, a necessary condition for an instrument to be a money is its use as a means of payment, but there are at least three other properties an ideal money might have: that there is no offsetting financial instrument against it; that its use can be anonymous; and that it is a store of value. (Ideally, the most perfect money would have constant marginal utility under all circumstances. It would be the equivalent of transferable "utiles".)

Table 11.1 shows eight instruments considered to be types of money at various levels in a modern economy. The second column makes the point that the inside-outside distinction depends on the existence of an offsetting instrument. The cryptic "no/yes" concerning the existence of an offsetting instrument is due to the fiction that personal IOUs are issued as an offset to fiat or book entry money in a finite game. But as we go to infinite limit, the IOUs are never redeemed. The third column indicates the anonymity property. Paradoxically, when money does not have a physical existence, it loses anonymity because its ownership must be recorded. The fourth column notes the shadings in the store-of-value property.

The evolution of money in any society presupposes the existence of organization and is intimately related to the history and institutions of the society. Thus any attempt to model the introduction of a money into an economic system existing in vacuo must appear unrealistic and contrived. With this caveat in mind, we shall try to construct playable games that illustrate the bare-bones structure of the system.

Although it is logically possible, no major society appears to have used fiat or paper money without having previously used coins. The progression appears to have been from barter or noneconomic means of the distribution of resources (see chapter 5) to the use of precious metals in trade, to the coinage of metals, and finally to the use of various forms of paper money.

It is tempting to indulge in a philosophical debate as to "what money really is"[2] (see Tobin, 1980, for a broad survey of the properties of money). This approach is resolutely avoided. Instead rules of the game are given and the behavior of the financial instruments are studied. For example, one can debate whether gold as nuggets or bullion can be counted as a money, even though either might be used in trade. Does "moneyness" begin with coinage and the full power of the state? If so, then gold coin is money and gold bullion is not. But if a property of a money is that it can be exchanged for all economic[3] goods and services, if the rules of the game are such (be they determined by custom, law or whim) that gold bullion is as acceptable as coin,[4] then both have that property of a money. Be that as it may, our chief concern is with the rules of the game and the behavior of financial instruments.

Although coin and credit are called for by the necessities of trade, policing the rules that govern them calls for a rule of law and the power of government. But the power of the government over the financial system offers the means for control of the economy.[5] Seigniorage and other charges, taxes, money, credit and debt management become tools of the nation-state. If our concern were limited only to the theory of a finite period, purely private exchange, or an exchange-and-production economy without public goods, an internal mutual banking system could provide the credit structure (see Volume 2, chapters 3 and 4). But even, when modeled as a game, exchange requires a rudimentary governmental structure that must be powerful enough to enforce default and bankruptcy laws.

Without the need to finance public expenditures or to provide insurance and stability in times of crises or extreme disequilibrium, there is little

need for the government to exercise an elaborate financial control over the economy. But (even ignoring consideration of overlapping generations as indicated in chapter 10) economic efficiency may call for borrowing and lending, and loan markets require rules of default and their enforcement. A government is the natural referee or enforcer of the rules.

In chapters 9 and 10, the key observation was that even in one-period exchange with a price system, if the mechanics of trade involve time, either both sides must put up value for value or they must trust each other by granting short-term credits until all markets have cleared and settled. A way of dispensing with trust is to have all traders always pay immediately in a commodity money.

The device of a mutual bank with fractional reserves provides a way for economizing on gold. Although there may be no need for the government to supply financial services, at this level of abstraction, there is no economic reason to rule out government banking as an alternative to mutual banking. In this chapter, we are concerned with overcoming the inefficiencies of a low supply of commodity money by having government introduce the appropriate financial mechanism, fiat money.

11.1.1 A note on financial instruments

In general, many financial instruments and institutions appear to be redundant at a high level of abstraction. The differences among them appear when more relevant detail is taken into account (such as data processing costs, the expense of extra markets, the need for public revenues, the costs of collection, and so forth). Because our concern is here confined to one-period exchange, there is no logical necessity for the use of fiat money, but it can serve as an alternative to having every seller grant inside credit to every buyer until the clearing mechanism has settled all accounts. (In Volume 2, chapter 9, it is argued that, with overlapping generations, the existence of fiat money becomes a necessity for some economies to achieve efficiency using a competitive price system.)

11.1.2 On fiat money and the government's choices

"The enemy was repelled. But victory was not won. The war dragged on for a year and there was no decision. Gold grew scarce, and again the Government was in despair.

I easily relieved them. 'Write,' I said, 'promises on paper to be repaid in gold.' They did as I advised—paying me (at my request) a trifle of half a million for the

advice. I handled the affair—on a merely nominal profit. I punctually met for another year every note that was paid in. But too many were presented, for the war seemed unending and entered a third year.

Then did I conceive yet another stupendous thing. 'Bid them,' said I to the Sultan, 'take the notes as money.' Cease to repay. Write not 'I will on delivery of this paper pay a piece of gold,' but 'this is a piece of gold.'

He did as I told him. The next day the vizier come to me with the story of an insolent fellow to whom fifty such notes had been offered in payment for a camel for the war and who had sent back, not a camel, but another piece of paper on which was written 'This is a camel.'

'Cut off his head!' said I.

It was done and the warning sufficed. The paper was taken and the war proceeded."

—Hilaire Belloc, *The Mercy of Allah*[6] (1922)

In chapters 9 and 10 the creation of personal credit was considered. Some discussion of commercial banks or the institutions internal to the private sector of the economy was given, the modeling was of private individuals lending to private individuals in a mass money market or via a mutual bank where all "names" were treated equally. In this chapter, attention is turned to the possibility of a government issuing the fiat money supply of a society. From one point of view (as already been noted in chapter 2), the creation of a governmental money is a debt contract, the same as the creation of a commercial bank loan or a private loan. Two parties, two instruments, and two time periods are involved. But in actuality, the asymmetry between the government and a private individual is such that a quantitative difference in size and power leads to a qualitative difference in the relationship. Thus when a government retires bonds, two financial instruments are involved; when it prints money to buy real assets, no other financial instrument is involved.

For virtually all operational purposes, the individual holder of government coin or paper money need not regard it as a normal debt contract. One might be able to take a dollar bill into the local federal reserve bank and ask for payment. In which case, a new dollar bill might be offered in exchange. But the currency does not even have any promise to pay. It merely says "In God We Trust" and "This note is legal tender for all debts, public and private." Currently, the largest U.S. note is $100. But the largest denomination for public circulation was a $10,000 note, first authorized on March 3, 1863, although it bore the legend "This note is legal tender for all debts public and private, except duties on imports and

interest on the public debt."[7] The largest notes issued were for $100,000, but these were not for public circulation and required registration. In general, however, paper currency is anonymous. It is a bearer instrument. It is not redeemable.

We may wish to indulge in an accounting fiction "in order to make the books balance" that has fiat money regarded as a debt instrument, where the government owes the bearer the sum in question. This can be carried further by arguing that the debt on the government's balance sheet is offset by the assets of the government, such as public lands. In strategic fact, however, fiat money appears on only one side of the balance sheets. It appears as an asset of individuals, but not as a debt of government. It is the only financial asset for which there is no strategically relevant offsetting debt. All other financial instruments are created as contracts involving the pairing of a debt or equity position with assets.

The main model in this chapter is not a realistic representation of an economy because it is not generally possible to use an intrinsically worthless paper as a surrogate for trust in one-period trade without having some means for enforcing its acceptability. To support the existence of a money with no offsetting financial instrument, we must consider exchange with overlapping generations, an infinite horizon, and the appropriate conditions on expectations. Or otherwise we must introduce a positive interest rate or tax to be paid to the government (this relatively cryptic statement will be made clear below). At this stage, the investigation is limited to one period.

How can a fiat money come into circulation? It is a product of government, and a government can use many different rules of the game to place it in circulation. For example, during a war a government might run the printing presses and pay the troops with crisp new bills. The accounting convention would have money in the hands of the public and the counterbalancing asset on the government's books would be the services of the troops. Thus the tangible costs of wars could be accrued as an intangible asset of the "value of freedom" in order to make the books balance. Or a government could issue paper money to buy gold or other goods from the citizens. Thus there would be real assets offsetting the issue of the paper. In some instances, as with early U.S. paper money, some of the paper might be silver certificates or gold certificates that actually promise to pay the bearer an amount of precious metal.

Table 11.2
Government fiat

Before exchange			
Government assets	Debt	Private assets	Debt
		Goods	
			Ownership

After exchange			
Government assets	Debt	Private assets	Debt
Private IOUs	Government fiat	Government fiat, goods	Private IOUs
			Ownership

Another way in which fiat money could enter the economy is by an exchange of debt between private individuals and the government. This amounts to a governmental monetization of private credit. The accounts for this are shown in table 11.2. All citizens get to turn their IOU notes into circulating government money. As is the case with seigniorage for coins, the government could exact a fee, or it might charge a fixed rate of interest, or it might auction off a quantity of its currency for individual IOU notes in such a manner that an interest rate might be created by the bidding. If, in the initial conditions, there exists a government debt owned by the private agents, the government then has a mechanism to vary its supply of money by selling or buying debt, paying or receiving payment in fiat.[8]

The simplest model where individuals obtain and use government money has only one time period. Rather than pretend that it is a close approximation of any particular economy, the stress is on looking at it as a playable game and making use of a completely defined structure to study how the money gets into and out of this game. When multiperiod models are considered, new problems appear, and it may become administratively and technologically desirable (though not logically necessary) to introduce a commercial or inside banking system to help vary the credit and money supply.

Given a one-period model and hence no long-term goods or financial instruments, there are seven simple ways fiat or outside money can be introduced into a strategic market game.[9]

1. An amount of fiat M is specified and auctioned off in return for personal IOU notes, to be redeemed in fiat at the end of the game; the money rate of interest is formed endogenously.

2. The government (referee) specifies a money rate of interest ρ, and individuals can obtain any amount of fiat in exchange for their IOU notes promising to return the amount borrowed with interest.

3. The government arbitrarily distributes an amount M of fiat without requiring that it be returned.

4. The government arbitrarily distributes an amount M of fiat requiring that it be returned at the end of the game, or a penalty is inflicted.

5. The government issues M units of fiat by buying services with them (such as paying the army).

6. The government issues M units of fiat by buying goods.

7. The government arbitrarily distributes an amount M of fiat without requiring that it be returned, but then offers to lend all players more money in return for a specified rate of interest.

There are several variations on the nature of the fiat and the penalties for failure to repay. While the money issued could take many different physical forms, for the sake of specificity, we assume that it looks like a current paper fiat note such as a dollar bill or pound note, or that it is a bearer certificate with a fixed positive or negative rate of money interest. If the interest were positive, it might have coupons attached, or if negative it might require the affixing of a stamp (as proposed and illustrated by Gessel, English trans., 1929).

There is no logical a priori reason why fiat money should not be interest bearing (where interest could be positive or negative). But if the purpose of fiat money is to provide an efficient substitute for trust in trade (i.e., a costless way of providing circulating capital), then we may establish whether it should be interest bearing or not. (This question is examined in chapter 13.) The history of paper money has included interest-bearing circulating notes (Bereziner and Narbeth, 1973): various monetary reform schemes, pejoratively called "funny money" proposals, have suggested this possibility, (see especially Gessel, English trans., 1929).[10] Although treasury notes and bills are government issues and can be classified as interest-bearing near-monies, they cannot generally be used for the direct purchase of most economic goods and services.

The penalties for failure to pay one's debt at the end of trade could be a blend of the economic and extraeconomic, such as the possibility of confiscating assets after trade but prior to consumption. There is a consider-

able body of law on claims to the assets of a debtor. At this level, of abstraction we do not specify the details of recapture or punishment.

11.1.3 A simple mechanism

In chapter 9, a model with unrestricted credit was noted and could be mathematized by the simple expedient of throwing away the cash holdings constraint.[11] The model was kept well defined by introducing a default penalty. Here a somewhat different model is introduced. A constraint is placed on the price system by pegging in advance the total amount of the money supply. This is method 1 of the different ways to introduce fiat noted in section 11.1.2.

An outside bank or government agency (or the referee of the game) fixes the numéraire and auctions off M units of money denominated in the numéraire. It accepts bids in personal IOU notes. A two-stage game is considered. First, all individuals bid to obtain a share of the fixed money supply. Thus if i bids u^i, i obtains

$$a^i_{m+1} = \frac{u^i}{\sum_{j=1}^{n} u^j} M.$$

In the second stage, i then bids $(b^i_1, \ldots, b^i_m)$, where $\sum_{j=1}^{m} b^i_j \leq a^i_{m+1}$ in order to buy quantities of the m commodities for sale.

After trade has taken place, i is called upon to settle up i's debt of u^i with the outside bank. This can be done if

$$u^i \leq a^i_{m+1} - \sum_{j=1}^{m} b^i_j + \sum_{j=1}^{m} p_j a^i_j.$$

If i is unable to pay back i's debt, i will end with $x^i_{m+1} < 0$, having to pay a default penalty whose unpleasantness is directly related to the size of that debt.

A key difference between this model and the one involving direct granting of credit is that the quantity of money is specified in advance, which means that the bankruptcy penalties are no longer more or less arbitrary because leniency or harshness cannot be compensated for by changes up or down in the money supply.

Even though the model suggested here covers only one period of time, it becomes meaningful to consider an endogenous rate of interest as

defined by

$$1 + \rho = \frac{u}{M}, \quad \text{where} \quad u = \sum_{j=1}^{n} u^i.$$

If $\rho > 0$, this implies that someone will go bankrupt. The variable ρ is understood to represent a *loss reserve* charged by the bank such that even if various traders go bankrupt the bank nevertheless obtains a return of capital.

In the models presented in chapters 9 and 10, the money is gold or gold-backed banknotes. Here, by contrast, the money is totally separated from a commodity of intrinsic value. Like the money of Knapp (see Schumpeter, 1954, p. 1090), it is totally a creation of the state.

As before, however, the bankruptcy penalty places a lower bound on the potential price level, and the amount of fiat places an upper bound. With a gold-backed currency, the default penalty and price system are all linked in terms of gold. With fiat money, the key element linking the money to value is the nature of the utilitarian worth of default.

With the lending of gold, it was possible to have a positive rate of interest without bankruptcy. This was because if the borrowers promised to pay back more gold than they had borrowed this was feasible by the transfer of real resources (i.e., they sold more than they consumed). Here, as money comes from the outside, by conservation if there are M units, no more than M can be returned. A more realistic model must consider payment in real resources to the outside to compensate for any shortfall if there is a positive rate of interest without default.

11.2 Fiat Money and Bankruptcy Conditions

11.2.1 A trading economy without money

Dubey and Shubik (1979) provide a detailed general mathematical analysis of one-period trade with fiat money and a default penalty.[12] The procedure adopted here, following Shubik and Wilson (1977), is to begin by taking a simple model of trade, formulated and solved for the standard competitive equilibrium solution. The same economic model is then formulated as a game in strategic form, with the referee passively auctioning fiat money to be used as the means of payment to finance trade.

The game is solved for a type-symmetric noncooperative equilibrium (TSNE), the point at which traders of the same type obtain equal treatment. Equal treatment is a property of the competitive equilibrium, but not necessarily of the noncooperative equilibrium. We then study the conditions under which the TSNE coincides or fails to coincide with the CE in terms of market prices and distribution of resources.

Consider n traders of type 1 with endowments of $(A, 0)$ and n traders of type 2 with endowments of $(0, B)$. Traders of the first type have utility functions of the form

$$u^1 = \log(x^\alpha y^{1-\alpha}), \tag{11.1}$$

and those of the second type,

$$u^2 = \log(x^\beta y^{1-\beta}). \tag{11.2}$$

11.2.2 The competitive equilibrium and Pareto optimal surface

To solve for the competitive equilibrium, we may consider two functions of the following form:

$$G_1 = \log[(A - x)^\alpha y^{1-\alpha}] + \lambda_1[p_1 x - p_2 y] \tag{11.3}$$

$$G_2 = \log[x^\beta (B - y)^{1-\beta}] + \lambda_2[p_2 y - p_1 x], \tag{11.4}$$

where λ_1 and λ_2 are Lagrangian multipliers. Taking derivatives with respect to x, y, λ_1 and λ_2, we obtain

$$\frac{\alpha}{A - x} = \lambda_1 p_1 \Leftrightarrow \frac{\partial G_1}{\partial x} = 0, \tag{11.5}$$

$$\frac{1 - \alpha}{y} = \lambda_1 p_2 \Leftrightarrow \frac{\partial G_1}{\partial y} = 0, \tag{11.6}$$

$$p_1 x = p_2 y \Leftrightarrow \frac{\partial G_1}{\partial \lambda_1} = 0, \tag{11.7}$$

$$\frac{\beta}{x} = \lambda_2 p_1 \Leftrightarrow \frac{\partial G_2}{\partial x} = 0, \tag{11.8}$$

and

$$\frac{1 - \beta}{B - y} = \lambda_2 p_2 \Leftrightarrow \frac{\partial G_2}{\partial y} = 0. \tag{11.9}$$

We may set

$$p_1 = 1.$$

From (11.5) and (11.6), we obtain

$$p_2 = \left(\frac{1-\alpha}{\alpha}\right)\left[\frac{A-x}{y}\right]. \tag{11.10}$$

or

$$x = (1-\alpha)(A-x). \tag{11.11}$$

From (11.8) and (11.1),

$$\lambda_2 = \frac{\beta}{A(1-\alpha)}. \tag{11.12}$$

From (11.5) and (11.1),

$$\lambda_1 = 1/A. \tag{11.13}$$

From (11.9), (11.10), and (11.12),

$$y = B\beta, \tag{11.14}$$

and from (11.10), (11.11), and (11.14),

$$p_2 = \left(\frac{1-\alpha)}{\beta}\right)\left(\frac{A}{B}\right). \tag{11.15}$$

In this extremely simple model, it is easy to calculate the Pareto optimal surface. It is given by

$$\frac{\alpha(1-\beta)}{(A-x)(B-y)} = \frac{\beta(1-\alpha)}{xy}$$

or

$$\begin{vmatrix} \dfrac{\partial u^1}{\partial x} & \dfrac{\partial u^1}{\partial y} \\ \dfrac{\partial u^2}{\partial x} & \dfrac{\partial u^2}{\partial y} \end{vmatrix} = 0, \tag{11.16}$$

which gives

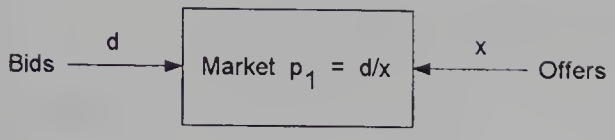


Figure 11.1
Bids, offers, and price.

$$\left\{\frac{\alpha-\beta}{\beta(1-\alpha)}\right\}xy = AB - Bx - Ay. \tag{11.17}$$

A check shows that (11.11) and (11.14) satisfy (11.17).

11.2.3 The money game

The trading mechanism Let the amount bid by a trader i of type 1 by b^i and by a trader j of type 2 be d^j. As traders of type 1 only have the first commodity, we may assume that they can offer only it for sale. Let the amount of the first commodity offered for sale by a trader i of the first type be x^i. A trader j of the second type offers an amount y^j of the second commodity.

Let the symbol $b = \Sigma_{j=1}^{n} b^i$, and similarly, for the others.

We may assume in this simple market that traders of type 1 bid only for the second good and traders of type 2 bid only for the first good. It has been shown elsewhere that there is no loss of generality in making this assumption in large markets.[13]

The markets for the goods are extremely simple. Figure 11.1 illustrates the market for the first good. All bids are aggregated (d), all offers are aggregated (x), and the price is fixed by dividing the amount of money bid by the amount of good offered. (The several other price formation mechanisms that could have been selected have been discussed in chapters 6 and 7.)

When the price has been formed, individual j, who has bid d^j for the first good, obtains

$$z^j = \frac{d^j}{p_1} = \frac{d^j}{d}x, \tag{11.18}$$

where

$$p_1 = \frac{d}{x}. \tag{11.19}$$

Similarly, trader i of the first type obtains

$$w^i = \frac{b^i}{p_2} = \frac{b^i}{b} y, \tag{11.20}$$

where

$$p_2 = \frac{b}{y}. \tag{11.21}$$

Financing exchange The currency or means of payment for goods must be specified. We assume that all bids are made using a fiat or central bank money that must be obtained from a government or referee "outside bank," modeled as a mechanism or "strategic dummy" in the sense that it is given a fixed strategy. In particular, the referee announces in advance a fixed amount of money to be offered at auction. In an economy with $2n$ traders, let this amount be nK.

The loan mechanism to lend out the nK units will be a simple Cournot mechanism. Each trader i of type 1 is permitted to create a financial instrument of i's own, to wit, a nonnegotiable promissory note of size or denomination u^i. Similarly, a trader j of type 2 bids v^j.

The traders use the promissory notes to bid for the money supply. A promissory note is understood to be a contract between an individual i and the referee that promises to return to the referee an amount of fiat (central bank, government or referee money) u^i at the end of the period in return for an amount obtained at the start of the period. The amount to be returned is determined by the supply of loanable funds nK and the offers of promissory notes, that is, if s^i and t^j are the loans obtained by i and j, they are respectively

$$s^i = \frac{u^i}{u+v}(nK) \tag{11.22}$$

and

$$t^j = \frac{v^j}{u+v}(nK). \tag{11.23}$$

The model is naively simple. Banks would generally be suspicious of borrowers wishing to promise to pay back enormous sums for loans whose size they do not know in advance. Nevertheless, the mechanism is

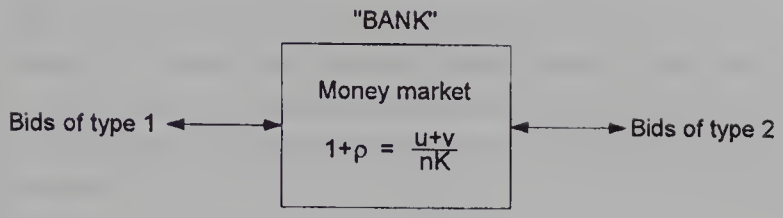


Figure 11.2
Formation of an interest rate.

well defined and in a mass market where individuals have estimates of the aggregate money supply and demand, such a promise can be interpreted more reasonably; an individual i might, for example, have an estimate of equilibrium price and not expect the price of money to move very much in response to i's actions.

As can be seen from figure 11.2, the price of money is given by

$$1 + \rho = \frac{u + v}{nK}, \tag{11.24}$$

where ρ can be interpreted as a money rate of interest.[14]

Settlement and bankruptcy After trading is over, individuals obtain their final allocations of goods, obtain their money incomes, and must make their final settlements at the bank. If traders end up with a positive amount of money after having paid the bank, this has no positive value to them. If, on the other hand, they are unable to honor their debts in full, a penalty is leveled against them. This penalty does not necessarily have to be the same for all individuals. This model introduces two parameters, μ_1 and μ_2, associated with a linear term that measures the "punishment" leveled against all traders unable to repay their debts. The specific structure of the payoff functions is shown below.

We may consider the penalty for bankruptcy as purely economic or not directly so. An example of the former might involve sale of the debtor's assets; an example of the latter would be social disgrace, a prison sentence, or death. In a one-period model, we may imagine confiscated assets are sold off in a subsequent period.

A bankruptcy law should adequately discourage willful strategic bankruptcy. This can be done in a world without exogenous uncertainty and incompetence by making the penalty sufficiently harsh. But such a law

does not take care of exogenous uncertainty with incomplete markets, nor does it cover incompetence. When these features are present, the bankruptcy law must try to serve three somewhat conflicting goals. It must balance the rehabilitation of the debtors with restitution to creditors and the overall well-being of society. In keeping with the policy of dividing difficulties, this section shows that without exogenous uncertainty and incompetence, given a strict enough penalty, the equilibrium outcomes will not involve anyone going bankrupt. Because the interests of the creditors and the public are automatically satisfied, they need no special consideration.

Regardless of how we regard the punishment, its presence is needed if we wish to influence individuals against making exorbitant repayment commitments to increase the current loans, knowing that they will be unable to meet their obligations.

The payoff functions We have seen that the utility functions of the traders, shown in (11.1) and (11.2), involve only the two consumer commodities. We define the payoff functions in terms of strategies and outcomes including the final disposition of money. We may write the payoff to a trader i of type 1 as:

$$\Pi_1^i = \log\left[(A - x^i)^\alpha \left(\frac{b^i y}{b}\right)^{1-\alpha}\right] + \mu_1 \min\left[0, \left(\frac{d}{x}x^i - b^i + \frac{u^i nK}{u+v} - u^i\right)\right] \tag{11.25}$$

and the payoff to a trader j of type 2 is:

$$\Pi_1^j = \log\left[\left(\frac{d^j x}{d}\right)^{1-\beta}(B - y^j)^{1-\beta}\right] + \mu_2 \min\left[0, \left(\frac{b}{y}y^j - d^j + \frac{v^j nK}{u+v} - v^j\right)\right], \tag{11.26}$$

where the x^i and y^j are considered as functions of the u^i and v^j, and where $0 \leq b^i \leq u^i nK/(u+v)$, and $0 \leq d^j \leq v^j nK/u + v$.

Prices are given by

$$p_1 = d/x \quad \text{and} \quad p_2 = b/y. \tag{11.27}$$

We note that the bankruptcy conditions enter as linear terms, or as zero. We assume $0 < \alpha, \beta < 1$ to ensure a solution with some trade. The preferences of the individuals are represented by concave utility functions.

The introduction of the bankruptcy condition can be considered as though the utility function for an individual had been defined to include money holdings where nothing is added for positive holdings, but where debt (which may be interpreted as negative holdings) has a negative worth.[15]

11.2.4 The noncooperative equilibrium with no information

Equilibria may be extremely sensitive to information conditions. In multistage games such as this, the less information the traders have, the easier it may be to examine the equilibrium points of the game.[16] The simplest situation is where all traders are required to announce simultaneously not only u^i and v^j but[17] also b^i, d^j and x^i, y^j.[18]

We may consider that a trader i of the first type attempts to maximize

$$G_1^i(u^i, b^i, x^i) = \alpha \log(A - x^i) + (1 - \alpha)\log(yb^i/b) + \lambda_1^i\{dx^i/x - b^i + (M-1)u^i\} + \gamma_1^i(Mu^i - b^i). \quad (11.28)$$

There will be a similar expression for a trader j of the second type. We define

$$M = \frac{nK}{u+v} \geq 0, \quad u = \sum_{i=1}^{n} u^i \quad \text{(and similarly for the others)}, \quad (11.29)$$

where λ_1^i and γ_1^i can be regarded as Lagrangian multipliers with the following conditions:

$$\text{If } (dx^i/x) - b^i + (M-1)u^i \begin{cases} > 0, & \text{then } \lambda_1^i = 0 \\ = 0, & \text{then } 0 \leq \lambda_1^i \leq \mu_1 \\ < 0, & \text{then } \lambda_1^i = \mu_1 \end{cases} \quad (11.30)$$

$$\text{If } (Mu^i - b^i) \begin{cases} > 0, & \text{then } \gamma_1^i = 0 \\ = 0, & \text{then } \gamma_1^i \geq 0. \end{cases} \quad (11.31)$$

First-order maximization conditions give

$$\frac{\partial G_1^i}{\partial x^i} = \frac{-\alpha}{A - x^i} + \lambda_1^i d\left(\frac{1}{x} - \frac{x^i}{x^2}\right) = 0 \quad (11.32)$$

and from symmetry:

$$\frac{\alpha}{A - x^i} = \lambda_1^i \left(\frac{n-1}{n}\right) \frac{d}{x} \Leftrightarrow \frac{\partial G_1^i}{\partial x^i} = 0. \tag{11.33}$$

Similarly, from $\partial G_1^i / \partial b^i = 0$ and $\partial G_1^i / \partial u^i = 0$, we obtain

$$(1 - \alpha)\left(\frac{n-1}{n}\right) = b^i(\lambda_1^i + \gamma_1^i) \Leftrightarrow \frac{\partial G_1^i}{\partial b^i} = 0 \tag{11.34}$$

and

$$\lambda_1^i = M(\lambda_1^i + \gamma_1^i)\left\{\frac{v + \left(\frac{n-1}{n}\right)u}{u + v}\right\}. \tag{11.35}$$

From (11.33) and $x = nx^i$,

$$x\alpha = \lambda_1^i \left(\frac{n-1}{n}\right) d(A - x^i)$$

gives

$$x^i = \frac{\lambda_1^i dA\left(\frac{n-1}{n}\right)}{n\alpha + \lambda_1^i d\left(\frac{n-1}{n}\right)}, \quad \lambda_1^i > 0. \tag{11.36}$$

If $\lambda_1^i = 0$, the only solution consistent with equation (11.36) is $x^i = 0$. But this means that each type i trader receives 0, rather than d^j units of money from type j traders. As a consequence, it is easy to verify that no solution exists for $\lambda_1^i = 0$. Therefore, (11.30) implies

$$d - b + (M-1)u \leq 0 \quad \text{and} \quad b - d + (M-1)v \leq 0 \tag{11.37}$$

or

$$(M-1)(u+v) \leq 0.$$

Hence if $u + v > 0$, then

$$M \leq 1. \tag{11.38}$$

From (11.35) and (11.38) and symmetry, $\gamma_1 > 0$ and, similarly, $\gamma_2 > 0$; hence

$$Mu = b \quad \text{and} \quad Mv = d. \tag{11.39}$$

From (11.34), (11.35), and (11.39) and symmetry:

$$u^i = \frac{1-\alpha}{\lambda_1}\left(\frac{n-1}{n}\right)\left(\frac{v+\left(\frac{n-1}{n}\right)u}{u+v}\right) \tag{11.40}$$

$$v^i = \frac{\beta}{\lambda_2}\left(\frac{n-1}{n}\right)\left(\frac{u+\left(\frac{n-1}{n}\right)v}{u+v}\right).$$

From (11.30) and (11.39):

$$\lambda_1 < \mu_1 \Rightarrow Mv = u \quad \text{and} \quad \lambda_2 < \mu_2 \Rightarrow Mu = v \tag{11.41}$$

and

$$Mv < u \Rightarrow \lambda_1 = \mu_1, \quad Mu < v \Rightarrow \lambda_2 = \mu_2. \tag{11.42}$$

We now confine our attention to the limiting behavior in markets as $n \to \infty$. From (11.40), we obtain

$$u^i \to \frac{1-\alpha}{\lambda_1} \quad \text{and} \quad v^j \to \frac{\beta}{\lambda_2}. \tag{11.43}$$

We wish to consider all λ_1 and λ_2 consistent with (11.41) or (11.42).

From (11.29) and (11.43),

$$M = \frac{\lambda_1\lambda_2 K}{\lambda_2(1-\alpha)+\lambda_1\beta},$$

hence

$$Mu^i = \frac{K\frac{1-\alpha}{\lambda_1}}{\frac{1-\alpha}{\lambda_1}+\frac{\beta}{\lambda_2}} \quad \text{and} \quad Mv^j = \frac{K\frac{\beta}{\lambda_2}}{\frac{1-\alpha}{\lambda_1}+\frac{\beta}{\lambda_2}}. \tag{11.44}$$

There are four cases arising from (11.41) and (11.42):

A. $u^i = Mv^j$ and $v^j = Mu^i$

B. $u^i = Mv^j$ and $v^j < Mu^i$

C. $u^i > Mv^j$ and $v^j = Mu^i$ and

D. $u^i > Mv^j$ and $v^j > Mu^i$.

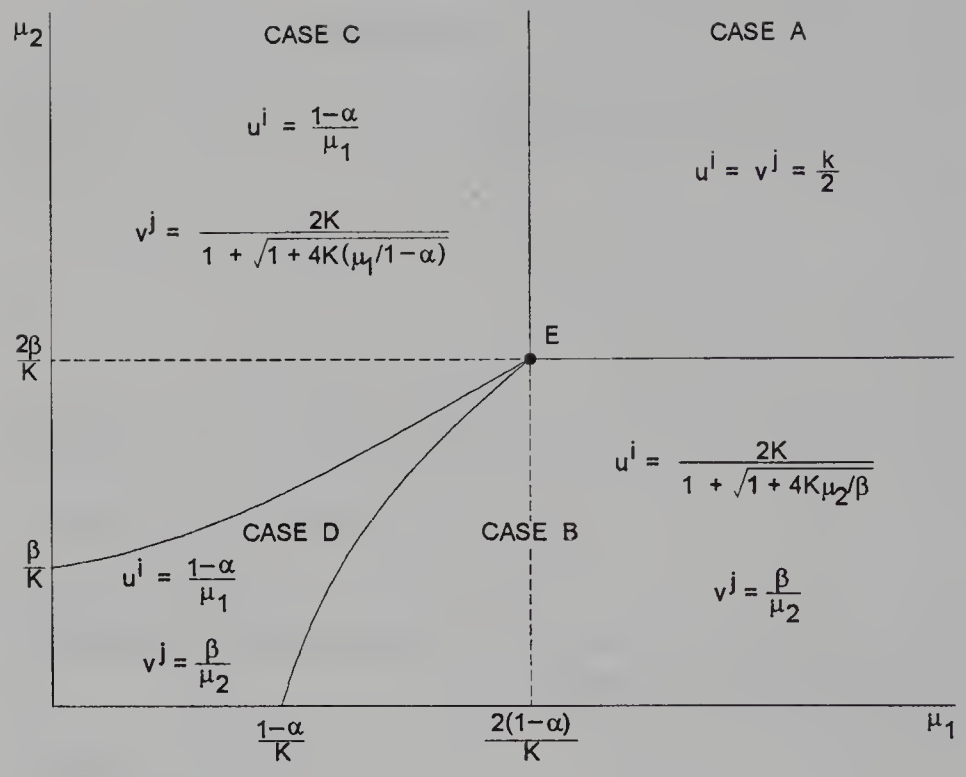


Figure 11.3
Default zones.

Figure 11.3 shows the four cases as μ_1 and μ_2 are varied. The calculations of the values are given in appendix A. In cases A and B traders of type 1 stay solvent; in cases A and C traders of type 2 stay solvent.

We observe that at the point E, $\mu_1 = 2(1-\alpha)/K$ and $\mu_2 = 2\beta/K$, where $K = 2A(1-\alpha)$. Hence $\mu_1 = 1/A = \lambda_1$ and $\mu_2 = \beta/A(1-\alpha) = \lambda_2$, which are the same as the Lagrangian multipliers obtained from solving the competitive equilibrium. Thus, for these bankruptcy penalties, the limit noncooperative equilibrium coincides with the competitive equilibrium. For other bankruptcy penalties, we see that if the μ_1, the μ_2, or both are set to be less than λ_1 or λ_2, one or both types will elect bankruptcy and trade will not be efficient.

A finite-infinite game solution distinction We can see from (11.35) and (11.38) that, for a finite n, hoarding cannot take place, that is, $\gamma_1, \gamma_2 > 0$. Since $\mu \leq 1$, if $u + v > 0$ and the Kuhn-Tucker complementary conditions are satisfied. This would not be so if there were a continuum of nonatomic

traders. If all traders felt that their influence on the aggregates were negligible, a new solution would appear.

Suppose each trader i takes x, y, b, d, and M as given, the chooses i's optimal x^i, b^i, and u^i accordingly. Then the first-order conditions for a maximum, given by equations (11.32), (11.34), and (11.35), simplify to

$$x^i = \frac{A\lambda_1^i d - \alpha x}{\lambda_1^i d} \tag{11.45}$$

$$b^i = \frac{1-\alpha}{\lambda_1^i + \gamma_1^i} \tag{11.46}$$

$$\gamma_1^i = \frac{1-M}{M}\lambda_1^i. \tag{11.47}$$

Using (11.47), (11.46) becomes

$$b^i = M\frac{1-\alpha}{\lambda_1^i}. \tag{11.48}$$

Adding (11.45) n times and simplifying yields

$$x^i = \frac{A\lambda_1^i d}{n\alpha + \lambda_1^i d}.$$

From equation (11.28), $b^i > 0$ implies $\lambda_1^i > 0$, and therefore (11.47) requires $M \leq 1$ to keep $\lambda_1^i \geq 0$. Note that $\gamma_1^i = 0$ if and only if $M = 1$.

Suppose $M = 1$, then $u^i + v^j = K$. From (11.48) and an equivalent expression for type 2 traders, $b^i \leq Mu^i$ and $\lambda_1^i \leq \mu_1$, which imply

$$\frac{1-\alpha}{\mu_1} + \frac{\beta}{\mu_2} \leq K. \tag{11.49}$$

Furthermore, (11.37) implies that $b^i = d^j$. Therefore,

$$\frac{1-\alpha}{\lambda_1} = \frac{\beta}{\lambda_2}, \tag{11.50}$$

which does not violate (11.49) only if

$$\frac{1-\alpha}{\mu_1}, \frac{\beta}{\mu_2} \leq \frac{K}{2}. \tag{11.51}$$

Following the argument in appendix A, one can verify that $M = 1$ if and only if (11.51) holds. If (11.51) does not hold, then the limiting results of the previous section hold. However if (11.51) is satisfied, not only do both types stay solvent, but the possibility of hoarding occurs. In this case the following values of x^i, y^j, b^i, d^j, u^i, v^j satisfy the conditions for an equilibrium

$$x^i = A(1-\alpha); \quad y^i = \beta B \tag{11.52}$$

$$b^i = d^j \geq \max\left(\frac{1-\alpha}{\mu_1}, \frac{\beta}{\mu_2}\right) \tag{11.53}$$

$$u^i + v^j = K; \quad u^i, v^j \geq b^i = d^j. \tag{11.54}$$

Not only may there be an indeterminate solution for u and v, but even when u and v are specified, any values of b and d satisfying (11.53) and (11.54) will generate an equilibrium.

The indeterminacy is to some extent an artifact of modeling the game as a one-period model. Matters may change when several periods and the possibility of a positive rate of interest are considered. But it is important to note that a zero rate of interest wipes out the distinction between hoarding and saving.

11.3 Two-Stage Noncooperative Equilibria

11.3.1 The extensive form, information, and strategies

Suppose that all individuals simultaneously bid for loans. They obtain no information beyond the size of the loan they have secured. After they have obtained their loans, they then all bid simultaneously in the markets for goods.

If the individuals know the size of their own loans, as they know their own bids, they can calculate the price of money. And if they all know the size of the total money supply, they can calculate the aggregate amount of loans obtained by their competitors.

Let $\bar{u}^i$ signify $u + v - u^i$ and $\bar{v}^j$ signify $u + v - v^j$. Then a strategy for a trader i of type 1 is a number u^i and two functions $x^i = \varnothing_1^i(u^i, \bar{u}^i)$, $b^i = \varnothing_2^i(u^i, \bar{u}^i)$ and, similarly, for a trader j of type 2.

If we wished, we might assume that trader i is given complete information as to who has borrowed how much. In this case, a strategy would

be a number u^i and two functions $x^i = \varnothing_1^i(u^1, u^2, \ldots, u^n; v^1, v^2, \ldots, v^n)$ and $b^i = \varnothing_2^i(u^1, u^2, \ldots, u^n; v^1, v^2, \ldots, v^n)$.

In the first instance, the trader bases his plan on *macroeconomic statistics*; in the second, on *microeconomic detail*. The refinement of information in a general noncooperative game could easily create many new equilibrium points associated with a noncooperative solution, although we will not consider this possibility here. In general, the competitive equilibrium points of the trading economy associated with the current noncooperative game can be attained as limit noncooperative equilibrium points making either the macroeconomic or microeconomic assumption about information.

Fortunately, there is a simple way to pick the functions. Assume that u^i and v^j, $i = 1, \ldots, n, j = 1, \ldots, n$ are given. Any individual i will select the pair (x^i, b^i) by maximizing i's payoffs in the one-stage game. Any individual j will select the pair (y^j, d^j) in the same manner. We may solve for x^i, y^j, b^i, d^j as functions of the u^i and v^j, then solve the bidding-for-loans stage of the market by maximizing with respect to u^i and v^j.

11.3.2 Threats and two-stage equilibria

In section 3.1, we noted that a strategy in a two-stage game can consist of a number in the first period and a function depending on the information about the moves of all others in the first period. The extreme generality of this function enables traders to convey highly implausible threats, which may nevertheless give rise to new (and improbable) equilibrium points. Thus in a two-stage game, the problem is generally not with existence of noncooperative equilibria[19] but with a surfeit of them.

Is it possible to distinguish "plausible" or nice equilibria from the others? As yet, there does not seem to be a completely satisfactory general way to do so. We might use the type of backward solution used in dynamic programming, but there are two basic difficulties in doing so. One concerns the information conditions. We need perfect information between the stages so that subgames can be well defined. It is likely that by making use of the special structure of these economic games where it is possible to aggregate moves, we could weaken the information requirements; however, this is not explored further here. The second difficulty concerns uniqueness. If we wish to replace the second-stage games by the values of their noncooperative equilibria, either we need them either to be unique or, at the least, we need some sort of indexing scheme to associate

the equilibrium points we select by some natural property such as continuity. By appealing to continuity in the simple example considered here, we can choose a unique value of the payoff function so that a backward solution is well defined. The calculations have been carried out elsewhere (Shubik and Wilson, 1977) and serve primarily to warn us as to how complicated matters can become in a multistage game with information available where the players influence the market.

When all players are nonatomic and there is no exogenous uncertainty, all noncooperative equilibrium points are perfect and the strategies for all players are greatly simplified. But a central bank is not small with respect to the economy. Fortunately, a game-theoretic distinction can be made that overcomes this difficulty. We can take away the monopoly power of the bank as a player by fixing its strategy in advance and announcing it, much in the spirit of the Simons (1948) suggestions for an automatic mechanism for the money supply role of the central bank. To examine this possibility fully, however, a multiperiod model is required, which is not considered in this chapter.

11.3.3 So what?

We have labored through much tedious calculation on a special example to at least begin to disclose the fine structure of a general class of economic problems we believe are of importance. In particular, a modern economy using a paper money or some other symbolic credit instrument for payments must have rules to specify what happens when debts cannot be repaid. Any positive penalty provides some support for the value of the original paper. As we have shown here, and as is generally true, however, penalties equal to the Lagrangian multipliers in the nonmonetary Walrasian model lead to limit noncooperative equilibria in the monetary model that converge to the same distribution and prices as the Walrasian model. Any other bankruptcy penalties may give different results—particularly if set too low. These different results are illustrated in the welter of cases worked out in this example (and in the two-stage calculations in Shubik and Wilson, 1977).

There are two items of note concering this type of trade model. The first is that the price level is not quite homogeneous of order zero with respect to the money supply. While fixing the amount of money together with the bankruptcy penalty sets both upper and lower bounds on price for a fixed velocity, this still leaves a range. The whole system cannot use more

money than there is; hence for any finite volume of trade, the price level is bounded above, however it may be specified. If the price level drops too much, then for any fixed bankruptcy penalty, eventually it would pay some individual to elect bankruptcy; hence for a solution without bankruptcy, there is a lower bounding price.

If the bankruptcy penalties are picked so that they coincide with the Lagrangian multipliers associated with some particular equilibrium in the Walrasian model, the limit equilibria of the noncooperative game will include that particular equilibrium point. Thus it appears that the money supply and penalty settings provide a way to guide a market economy to a subset of competitive equilibria, and possibly (with some more conditions) a specific equilibrium.[20] However when this is considered from the viewpoint of computation and information content, this appears to indicate that a market economy with an outside bank requires as much calculation to guide it as does a centrally planned economy using computational methods to arrive at an efficient price system (Scarf, 1973). The calculation of the appropriate bankruptcy penalties appears as the dual of the calculation of the prices.

The current model appears strange to those accustomed to the one-period general-equilibrium model, where a money or means of payment and a cost of money or interest rate do not seem to have any role. Here because of decentralized or simultaneous bidding, a float equal in size to the transaction needs must be financed. We in fact obtain a "zero rate of interest" when the penalties are set appropriately. Thus we arrive at the observation that the government can finance the float at no cost when the money rate of interest is zero.

Only the transaction need for financing has appeared. If we wish to capture intertemporal trade and the role of money or credit in financing this trade, we need a model that involves at least two periods, with both issue and deposit banking.

11.3.4 A comment on bankruptcy and insolvency

In chapter 7, it was shown that even with complete and anonymous mass markets, efficient trade cannot necessarily be achieved in a single round unless some form of credit is available. As soon as credit is introduced, the definition of a mechanism that must be described for positions of disequilibrium requires that we specify what happens if some individual cannot pay back what is owed.

There is one further distinction concerning bankruptcy that cannot be made in a one-period model, namely, the distinction between bankruptcy and insolvency. Individuals or firms are bankrupt when their net worth is negative, and insolvent when they cannot meet current obligations even though they may have enough assets to cover the obligations valued at some fraction of current market price.

There are many institutional and legal fine points in bankruptcy and insolvency laws that vary from country to country. As our discussion abstracts from these features, perhaps a more neutral word such as "default" should be considered in preference to "bankruptcy," which may imply greater institutional detail than is needed here.

11.3.5 An endogenous bankruptcy penalty

Rather than imposing an exogenous penalty to discourage bankruptcy, we could consider an endogenous penalty inflicted by garnishing the goods of the bankrupt. There are two simple rules that might be applied: the forgiving rule and the no-information rule.

Suppose an individual i goes bankrupt for an amount A and obtains a final endowment of $\bar{x}$. The referee removes a vector of resources b such that

$$\max \varphi_i(\bar{x}_1^i - b_1^i, \bar{x}_2^i - b_2^i, \ldots, \bar{x}_m^i - b_m^i)$$

subject to

$$\bar{p} \cdot b = A,$$

where $\bar{p}$ are the equilibrium prices. This is the forgiving rule and is exactly the same as though i were required to adjust to a loss of income of A. Thus i is indifferent between not defaulting and defaulting subject to this penalty.[21] With such penalties, the Nash Equilibria of strategic market games with a continuum of traders will be Walrasian.

The no-information rule has the important advantage that it requires no knowledge of the individual preferences by the referee. A number k is selected where $0 \leq k \leq 1$ such that

$$k\bar{p} \cdot \bar{x} = A \quad \text{for } k < 1; \text{ otherwise, } k = 1.$$

If the referee is able to garnish assets before they have been dissipated, this is at least as harsh as the forgiving rule and is an easy economic rule that will discourage strategic bankruptcy. The resources obtained from a

bankrupt go to the referee and could be used for redistribution (see Dubey and Shubik, 1988). The no-information rule, which works perfectly in a world without exogenous uncertainty, many small traders, and the ability to garnish without expense, has to cope with the design practicalities of the real world, with all its "imperfections."[22]

11.4 Outside Money and Currency

11.4.1 Free currency

In a one-period exchange model, introducing money or exchange seems to be much ado about nothing until one tries to play an exchange game. At that point, one begins to appreciate the role of currency, money, or credit as a means of executing transactions.

This can be easily illustrated by constructing a simple game using method 4 for the government's issue of currency (see section 11.1.2). The referee or government supplies a fixed amount of currency to all players at no charge. The only condition required is that after the game all players return the amount of currency they borrowed. If anyone fails to do so, a sufficiently high penalty exists to discourage strategic default. The individuals may lend each other the outside money before trading. The rate of interest in this model is endogenously determined to be zero.

11.4.2 Paying the troops

The fifth way the government can introduce currency is by printing it and buying goods or services. Paying the troops is a classical way government money comes into circulation in return for services purchased. The currency in this instance has no offsetting debt balance because the recipients are not exchanging a debt instrument to obtain it. This holds true for methods 3, 5, and 6. In method 3, the money is just given away; in methods 5 and 6, it is exchanged for goods or services. In all of these instances, any attempt to build a playable game with active trade where the players have the option of not selling their initial resources fails. Because there is no need to pay back the money at the end and it is worthless after trade, at equilibrium, no one will accept it in trade even though the convention requires that trade is for money. This is not necessarily true for games with two or more periods (see, for example, Shubik and Whitt, 1973).[23]

11.4.3 The rate of interest and the money supply

The common phrase "the rate of interest" is a catchall aggregate for a complex phenomenon involving many rates that reflect differentials in time and the quality and quantity of risk as well as transaction costs. In the simple, stripped-down, one-period models considered here, where there is no time discount, the only purpose of money or credit is to finance transactions. There is no need to carry capital stock. There are no distinctions between the long and the short term. In the model presented in section 11.2, consideration was given to the quantity of money to be issued, which was given exogenously. We proved that, with the appropriate selection of bankruptcy penalties, an endogenous interest rate of zero emerges. In section 11.4.2, a model was sketched where the exogenous interest rate was fixed at zero, the money supply was endogenous, and the government would meet any demand. We could equally well consider a game where the government fixes a positive interest rate exogenously. If it merely fixed a positive rate and a bankruptcy penalty, either no borrowing and no trade would take place or by simple laws of conservation, because more money would be owed than issued, some traders would go bankrupt unless the government was willing to take real resources in the settlement of money debt. We may construct a game in which the government fixes a rate of interest and a bankruptcy penalty and also announces a vector of buying prices $(p_1, p_2, \ldots, p_m)$ at which it will accept goods in lieu of debt owed.

In this formulation, the rate of interest is a control variable and is essentially a tax by the government. To define the model well, it is necessary to specify what the government does with the interest it earns, which is achieved by having it obtain real resources. But at this level of simplicity, the government is truly outside the game, and no motivation is imputed to its selection of parameters. If we were to build a model with inside money and oligopolistic bankers, we could obtain an endogenous positive rate of interest caused by oligopolistic bankers appropriating real resources. This point can be dealt with formally in the investigation of banking (Shubik, 1976).

The rate of interest in a one-period model is the shadow price of the rental of money for transaction purposes. The amount of money or the rate of interest may be specified exogenously, and the other is determined endogenously. But one can view both of them strategically as control

Table 11.3
Seigniorage accounts

Before minting		After minting	
Govern-ment assets Debt	Private assets Debt	Govern-ment assets Debt	Private assets Debt
	Goods Gold Ownership	Gold coin Government ownership	Gold Goods Gold coin Ownership

Note: The cost of producing and replacing worn-out money is nontrivial. In a separate study, Shubik and Tsomocos (1993) consider production costs explicitly and note that a direct lump sum tax on the society to pay for production costs permits for efficient trade. But a seigniorage charge or "percentage markup" does not.

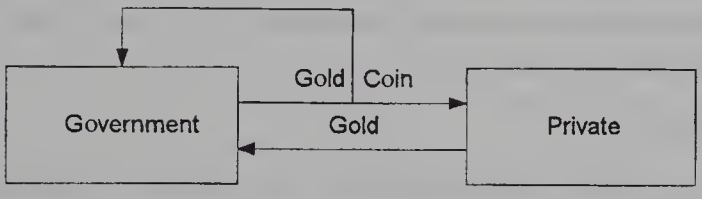


Figure 11.4
Government-controlled coinage.

variables of an outside agency, to be used simultaneously as control variables in much the same way that both price and quantity can be considered part of a strategy in a Bertrand-Edgeworth oligopoly model.[24]

11.4.4 Seigniorage on gold and other coinage

The management of money renders a service, requires governmental force, and provides many opportunities to raise revenues. One of the oldest sources of revenue is seigniorage.

A simple model where the government raises its revenue by profits from seigniorage or coining money is as follows. Initially, the private sector owns everything and brings gold to the government, which coins the gold at a profit. The T accounts are shown in table 11.3. The gold is produced through private discovery, industry, and trade. (See figure 11.4 for the flows of gold and coin). If we viewed this game as reality, the government has some control on the economy by spending its seigniorage profits.[25] Even if the public has plenty of gold, the government monopoly controls the amount that is turned into money. In reality, if the government

overreached, the public would soon resort to trade in unminted gold, barter, or credit.

If the government (or the king) owns all the gold mines (as in medieval Spain), it profits from a double monopoly. But unless it borrows, the monetary system remains "outside" because if the government pays in coin, no claims on or by the government are created. Furthermore, in any attempt to define Pareto optimality, it is worth noting a difference between the treatment of a government that consumes resources and is owned by a single active player (the king) and that of a government acting as a strategic dummy or mechanism owned by all of the agents.

In fact, the cost of producing and replacing worn-out money is nontrivial. In a separate study, Shubik and Tsomocos (1993) consider production costs explicitly and note that a direct lump sum tax on the society to pay for production costs permits for efficient trade. But a seignorage charge or "percentage mark up" does not.

11.4.5 A sell-all model

In chapter 6, the sell-all model was considered to be the tax collector's dream because all goods are monetized each period. Although it is not institutionally true, we could conceive of a society where individuals are required by law to sell at least some of their assets each period. In such a society, even though a paper money may be ultimately worthless, because traders do not have the strategic choice to refuse to offer for sale, all traders will spend all their money, and the rules of the game will transform the worthless paper into an economic asset. Furthermore, the CEs associated with any distribution of real resources can be obtained as NEs by handing out the appropriate amount of fiat.

This can be stated more formally, modifying chapter 7. Let there be n types of individuals, each with an endowment of m commodities and an amount of fiat money given by the referee. Individual i has resources $(a_1^i, a_2^i, \ldots, a_{m+1}^i)$ and preferences represented by a concave utility function $u^i(x_1^i, x_2^i, \ldots, x_m^i)$.

Each individual i is required by the rules of the game to offer all of i's endowments of the m commodities for sale at the m markets. The fiat money supplied by the referee is used to bid on the m real goods. Because the total supplies of all goods except money must pass through the markets, the meaning of ownership is modified to imply that ownership of a

commodity is title to the money receipt from the sale of the commodity rather than the commodity itself.

Because there is only one period of trade, the money received by each trader for the sale of the trader's ownership claims is of no further value, although this would not be the case if there were more periods of trade. Furthermore, with an infinite horizon, the paper money would maintain its worth even if we permitted individuals to hold back assets (see Shubik and Whitt, 1973). Because there are no loans between the referee and the players, there is no need for default or bankruptcy laws, which will be called for when inside loans or borrowing between the government and traders is permitted.

11.4.6 Interest as a tax

There are two variations of method 7 suggested in 11.1.2. In the first, each individual i starts with an initial endowment of money m_i, where $\Sigma m^i = M$, which need not be returned. The government then auctions off an extra amount of money B, which must be returned, possibly with interest. Failure to return debt owed to the government results in a penalty. For the appropriately harsh penalties, this game has two types of NEs. No trade is an NE, where all individuals refuse to take paper money for goods of value. But there exist other equilibria with active trade and a positive rate of interest given by $1 + \rho = (M + B)/M$, where all individuals bid promissory notes to obtain a share of the extra B units of money and bid the rate of interest up to the point where all of the M is paid back in interest.

A second variation has the government name the rate of interest, but not the amount available for borrowing. The basic effect is the same. All of the M is used up as interest on the amount borrowed. Here the rate of interest is purely an artifact of government policy in controlling the money supply and how it is to be accounted for. In essence all of the "free money" is taxed away by the payments entailed in borrowing from the government.

11.5 Historical Notes

11.5.1 A historical sketch of paper money

The Chinese appear to have first simultaneously supplied the three magic ingredients for the successful use of paper money: paper, printing

or engraving, and governmental force. Paper was invented in China around 200 B.C. (Tsien, 1986);[24] printing from wooden blocks by the eighth century A.D., and the use of governmental force considerably earlier than either.

A reference is made to a paper issue during the Tang dynasty around A.D. 650, and an illustration of a 10-kwan note exists. The Mongols under Kublai Khan were apparently the first to issue notes in substantial number, possibly for the purpose of paying their soldiers, around 1270. The Ming dynasty issued notes in 1368, and paper money was abolished in China around 1450. Marco Polo (Vissering, 1968) writes glowingly on "how the great khan causeth the bark of trees made into something like paper, to pass for money over all his county." He also notes, "When any of those pieces of paper are spoilt—not that they are so very flimsy neither—the owner carries them to the Mint, and by paying 3 per cent on the value he gets new pieces in exchange."

Prior to the issue of paper money, there is a record of the issue of "white hart currency" around 120 B.C. Fitzgerald (1954, p. 166) notes:

> The Emperor, always ready to try a new experiment, now attempted to grapple with this currency crisis, his first efforts were far from successful. In the imperial park at Ch'ang An the Emperor had a white stag, a very rare beast, which had no fellow in the empire. On the advice of a minister, the Emperor had this animal killed, and made a kind of treasury note out of its skin, which he believed could not be copied. These pieces of skin were a foot square, and were made with a fringed border and decorated with a pattern. Each piece was assigned the arbitrary value of 400,000 copper coins. The princes, when they came to pay their respects to the throne, were compelled to buy one of these pieces of skin for cash, and present their gifts to the Emperor upon it. This precaution ensured the circulation of the "White Stag Notes." The skin of the white stag was, however, a limited quantity and the time soon came when this device ceased to supply the treasury with much needed money.

Whether the Chinese notes were issued by the central government or by the emperor himself, little is known of the details of their issue.

The first banknotes were those of the Stockholms Banco issued in July of 1661. Goldsmith notes were endorsed and circulated in mid-seventeenth-century England, but the exchequer order was the first note recognized as legal tender. The Bank of England issued notes from the time it was established in 1694, but its notes did not achieve a limited legal tender status until 1833. The Bank of Scotland, established in 1695, issued one-pound notes in 1699. Under the guidance of John Law, the Banque

Royale of France issued notes in 1719, which were suspended in October 1720.

In the United States, the Continental Congress authorized the issue of paper money in 1775, suspending the notes in 1781. The Bank of North America, in Philadelphia, was the first of a series of state-chartered banks to issue notes redeemable in specie. The first Bank of the United States, chartered for twenty years in 1791, played the role of a central bank, but owing to political opposition, it was unable to renew its charter. The second Bank of the United States had a similar fate. During its one and only term (1816–1836), it issued notes, which circulated at more or less on par with specie. Until the National Bank Act of 1863, a welter of state-chartered and private banks issued banknotes, which were soon sold at a discount at any distance from the bank of issue. The Bureau of Engraving the Printing of the United States, established in 1863, became an entity within the Treasury in 1869. From August 1866, a 10 percent tax on notes issued by banks heralded the era of the issue of a uniform U.S. currency and the growth of checks as the main form of bank money.

We note that the first paper money was government money. Banknotes (and for that matter private banks) are a relatively recent invention. Resembling, but not quite the same as, paper money, are three types of paper instruments, each with a history of its own: the letter of credit, the bill of exchange, and the warehouse receipt.

Fiat money, whether paper or coinage of less intrinsic worth than its exchange value, has been subject to forgery since the time of its invention. Fitzgerald (1954, p. 167) reports the execution of forgers in China for turning out counterfeit coins in the second century B.C., Bereziner and Narbeth (1973, p. 23) refer to an execution as late as 1821 in Glasgow of a woman, Ann Wilson, for passing a forged note.

According to Griffiths (1884, p. 166):

> The heaviest penalties did not check this crime. The offence was high treason; men sentenced for it were hanged, drawn, and quartered, and women were burnt. In 1683 Elizabeth Hare was burnt alive for coining in Bunhill Fields. Special legislation could not cope with this crime, and to hinder it the Lords of the Treasury petitioned the Queen (Mary in the absence of Willian III) to grant no pardon to any sentenced for clipping unless before their conviction they discovered their accomplices.

During the Second World War, the Germans used inmates of the Sachenhausen concentration camp to forge British currency with a face

value of £140 million, to be used to pay spies and to disrupt the British economy, although economic warfare via massive forgery was never put into full force.

One considerable advantage of paper money, from the viewpoint of a government is that it is far cheaper than coinage to produce. In the United States current costs of production in the 1980s were estimated at \$26 per thousand bills of any denomination.[31] The average life of a \$1 bill is estimated at eighteen months, a \$5 bill at two years, a \$10 bill at three years, a \$20 bill at five years, a \$50 bill at nine years, and a \$100 bill at twenty-three years.[32] In 1986 the Federal Reserve System had to scrap around \$20 billion in worn notes. Most are withdrawn by automatic sorters at the Federal Reserve and shredded.[33] If U.S. currency is partially eaten by termites, soaked, shredded, partially burned, or otherwise mutilated, the Examining and Redemption Division of the Treasury Department will try to reconstruct and redeem the currency (see *Insight*, November 9, 1987, p. 44).

It must be stressed that when trying to comprehend the properties of a money, custom, law, technology, and economics are closely intertwined. Thus, for example, the advent of the computer and the impetus of giro banking (see Thomson, 1968) influence velocity. In a lighter vein, appendix B provides some observations on the paradoxes of the dismal science in reference to the value of worthless paper money.

Appendix A: Case Calculations

To compute u^i and v^j, there are four cases to consider

A. $u^i = Mv^j; \quad v^j = Mu^i,$

B. $u^i = Mv^j; \quad v^j > Mu^i,$

C. $u^i > Mv^j; \quad v^j = Mu^i,$

D. $u^i > Mv^j; \quad v^j > Mu^i.$

CASE A: $u = Mv = Mu \Rightarrow M = 1 \Rightarrow u^i + v^j = K$. Therefore

$$u^i = v^j = K/2. \tag{11.A1}$$

Therefore

$$\frac{1-\alpha}{\mu_1} = \frac{\beta}{\lambda_2} = \frac{K}{2}.$$

But $\lambda_1 \leq \mu_1$ and $\lambda_2 \leq \mu_2$ require

$$\frac{1-\alpha}{\mu_1} \leq \frac{K}{2}; \quad \frac{\beta}{\lambda_2} \leq \frac{K}{2}. \tag{11.A2}$$

Thus (11.A1) holds only if (11.A2) holds.

CASE B: $Mu^i < v^j \Rightarrow \lambda_2 = \mu_2$ and from (11.41) in 2.4 $u^i = Mv^j$ implies

$$K\left(\frac{\beta}{\mu_2}\right) = \left(\frac{1-\alpha}{\lambda_1}\right)\left(\frac{1-\alpha}{\lambda_1} + \frac{\beta}{\mu_2}\right)$$

or

$$\lambda_1 = \frac{1-\alpha}{2K}(1 + \sqrt{1 + 4K\mu_2/\beta}).$$

Therefore

$$u^i = \frac{2K}{1 + \sqrt{1 + 4K\mu_2/\beta}} \tag{11.A3}$$

$$v^i = \beta/\mu_2$$

But $\lambda_1 \leqq \mu_1$. Therefore, if (11.A3) holds,

$$\mu_1 \geqq \frac{1-\alpha}{2K}(1 + \sqrt{1 + 4K\mu_2/\beta})$$

and

$$M(u^i + v^j) < (u^i + v^j) \Rightarrow K < \frac{1-\alpha}{\lambda_1} + \frac{\beta}{\mu_2} \tag{11.A4}$$

Thus

$$\mu_2 < \frac{2\beta}{K}$$

CASE C: By symmetry, $\lambda_1 = \mu_1$; $v^j = Mu^i$;

$$v^i = \frac{2K}{1 + \sqrt{1 + 4K(\mu_1/1-\alpha)}}; \quad u^i = \frac{1-\alpha}{\mu_1} \tag{11.A5}$$

only if

$$\mu_2 \geq \frac{\beta}{2K}(1+\sqrt{1+4K(\mu_1/(1-\alpha))}); \quad \mu < \frac{2(1-\alpha)}{K}. \tag{11.A6}$$

CASE D:

$$\mu^i = \frac{1-\alpha}{\mu_1}; \quad v^j = \frac{\beta}{\mu_2}, \tag{11.A7}$$

where $\mu_1 = \lambda_1$; $\mu_2 = \lambda_2$.

But for (11.A7) to hold, (11.41) and (11.A7) imply

$$\frac{\mu_1 \beta K}{\mu_2(1-\alpha)+\mu_1\beta} < \frac{1-\alpha}{\mu_1}$$

$$\frac{\mu_2(1-\alpha)K}{\mu_2(1-\alpha)+\mu_1\beta} < \frac{\beta}{\mu_2}$$

or

$$\begin{aligned} \mu_1 &< \frac{1-\alpha}{2K}(1+\sqrt{1+4(\mu_2/\beta)K)}) \\ \mu_2 &< \frac{\beta}{2K}(1+\sqrt{1+4(\mu_1/1-\alpha)K}) \end{aligned} \tag{11.A8}$$

It is easy to check that (11.A8) implies

$$\begin{cases} \mu_1 < \dfrac{1-\alpha}{2K} \Rightarrow u^i > \dfrac{K}{2} \\ \mu_2 < \dfrac{\beta}{2K} \Rightarrow v^j > \dfrac{K}{2} \end{cases} \tag{11.A9}$$

Using the expression for x^i in (11.36) of 4.2 and letting $n \to \infty$, we can show

$x^i(\mu_1, \mu_2) = A(1-\alpha)$ if $(\mu_1, \mu_2) \in$ area A or B figure 11.3

$$\frac{\partial x^i}{\partial \mu_2} < 0; \quad \frac{\partial x^i}{\partial \mu_1} > 0 \quad \text{if } (\mu_1, \mu_2) \in \text{area } D$$

$$x^i(\mu_1, \mu_2) = A \frac{\mu_1^2 K(1 + \sqrt{1 + 4K(\mu_1/1 - \alpha)})}{(\alpha(1 - \alpha) + \mu_1^2 K)(1 + \sqrt{1 + 4K(\mu_1/1 - \alpha)} + \alpha\mu_1)}$$

$$< A(1 - \alpha) \quad \text{if } (\mu_1, \mu_2) \in \text{area } C.$$

Similar results hold for y^j.

Appendix B: The Value of Worthless Money

A quote from Bresciani-Turroni (1937) appears apposite:

In the past money which has suffered the greatest depreciation was probably the so-called "continental money," issued by the American Colonies during the War of Independence. In 1781 that money was worth a thousandth part of its original value. The depreciation of the famous assignats [of the French Revolution] was also less than the mark. On June 1, 1796, a metal franc was worth 533 assignat francs. Among currencies, other than the German mark, which depreciated enormously after the [First World] war were the Austrian crown, the Hungarian crown, the Polish mark, and the Russian rouble. In January 1923 the depreciation of the German mark had already exceeded that of the Austrian and Hungarian crowns; towards the end of the first half of that year it overtook and passed that of the Polish mark; and at the beginning of October the German mark was also more depreciated than the Russian rouble, a gold mark being worth 709 million paper marks, while a gold rouble was equivalent to 505 million old paper roubles.... In the day preceding the monetary reform the quotations of the dollar at Berlin were as follows:

November 13, 1923	840.	milliard marks
November 14, 1923	1260.	" "
November 15, 1923	2520.	" "
November 20, 1923	4200.	" "

... Actually in the open foreign market the dollar reached much higher rates ... [reaching] 11,200 milliards on November 20 ... on November 11, 1923, notes for 1,000 milliards were issued. There followed notes for 2,000, 5,000, 10,000 and 100,000 milliards denominations. These last were the highest issued. They were stamped on one side only.

Although "inflation money" is still basically worthless (one dealer quoted twenty-five cents per bill as a typical price),[28] "Continental" money from the American Colonies is more valuable, running in the $15 to $50 range for genuine bills of good quality. Money issued by the

Confederate States can be quite valuable, commanding as much as twenty times its face value.[29]

Notes

1. This is true for one-period exchange, but as soon as multistage exchange and production are considered, even with a commodity money the exchange may involve expectations concerning both the future worth of the money and the goods being bought, especially if the goods are intermediate goods in a production process with no intrinsic value as a final product.

2. The adjective "economic" is added because even in the most individualistic utilitarian society there are many items that money cannot buy in the direct sense of the market.

3. There may be differences in costs of production, assaying, weighing, and transporting that somewhat differentiate coin and bullion.

4. It is not merely the money supply that a government could use for financial leverage. For example, as in India, it could own a controlling block of shares in major corporations and thus influence the economy with the issue or purchase of corporate shares.

5. I am indebted to Ross Starr, who called my attention to this passage from Hilaire Belloc.

6. See the article by J. Stevenson in the Coins section of the *New York Times*, Sunday, June 17, 1990, for a description of the $10,000 bills.

7. Once an economy is in existence, there is both inside and outside money and, in fact, the books do not balance. There is a basic nonsymmetry caused by the presence of a positive amount of outside money for which there is no offset. Although as a rule I am highly skeptical of analogies between the social and physical sciences, the lack of balance between matter and antimatter in modern physics (for an layman's account, see Adair, 1988) suggests the possibility that the basic nonsymmetry is a manifestation of the existence of time.

8. Left out is the entry of money through international trade. In the early history of the United States, foreign currency played an important role. An adequate modeling of this phenomenon would require an model including international trade.

9. Gessel's "free money" proposal was for a negative-interest-bearing currency that would quicken its use as a means of payment and discourage hoarding.

10. This can be interpreted as method 2 noted for the introduction of fiat (where $\rho = 0$). We can imagine all individuals with checkbooks able to draw any amount on a central bank, which will clear all trades.

11. Sections 11.2.1 and 11.2.2 are based on Shubik and Wilson, 1977.

12. When numbers are few, "wash sales" may be of importance, as we have seen in chapter 6.

13. In a one-period model, the concept of an interest rate seems somewhat strange without time preferences. However, the price of money must cover not only time preference but also loss reserves, exogenous uncertainty, monopolistic competition, and inflation. Here, as already observed, only loss reserves are covered.

14. A natural question for the economist to ask is why we have not specified the bankruptcy penalty in "real" terms, that is, in terms corrected for the price level rather than just in terms of the paper units. There are two reasons for our choice, the first concerns the logic and the second concerns the law. There is a need for a bankruptcy law to be given to define the game well. As prices are formed within the game, it is not possible to write a noncontingent law. Hence if we wished to consider a penalty in "real terms," the law would have to be contin-

gent and would need to contain a clause for the treatment of unbounded bidding. If the penalty were homogeneous of order 1 in the price system, the strategy sets would need bounding. In fact, the bankruptcy laws are *not* immediately contingent. They change at a pace far slower than the markets and after years of inflation will be adjusted, although at any point in time they are in essence noncontingent on the price level.

15. In section 11.2.4's analysis, only the first-order conditions for utility maximization are used to describe the equilibrium. In particular, we have not provided a complete proof that any of our solutions are in fact noncooperative equilibria. Conscientious readers are encouraged to provide their own proofs where necessary.

16. As an individual i may not be sure of i's money supply before i's bid $b^i(d^j)$ is fixed, we require a feasibility or acceptability condition in the game. For example, a nonfeasible bid is replaced by a zero bid or some convention is given that transforms the bid into a feasible act. It is straightforward to check that degenerate equilibria always exist—that is, equilibria with x^i, $y^j = 0$, and no bankruptcy. Here we will be restricting our attention to nondegenerate equilibria.

17. Although it is well known that perfect equilibria may not exist (see Roberts and Sonneschein, 1977).

18. In particular, the condition required appears to be as follows. Suppose that an exchange economy has K competitive equilibria. Given a normalization obtained by fixing the amount of money at M, we can then calculate the Lagrangian multipliers at every CE. Consider any two CEs denoted by j and k. If we can show that λ_j^i cannot be greater than or equal to λ_k^i for all i, then it appears possible to construct a strategic market game with a unique CE as an NE for each CE. Shapley (personal correspondence) and Chipman and Moore (1976) have examples where a strict domination of one set of Lagrangian multipliers by another is feasible.

19. This rule has, however, the considerable inconvenience of requiring knowledge of agents' utility functions for its implementation.

20. The term *imperfection* appears to have been used in economics to stand for a nasty fact that ruins a simplistic theory. An understanding of money and financial institutions requires that one recognize that process and the design of institutions that carry process are central and not mere "imperfections."

21. The backward induction does not wipe out trade in a sell-all model of two or more periods.

22. If both the quantity of money and the rate of interest are treated as control variables, the need for accuracy and fine-tuning becomes extremely high as they interact.

23. A feasible seignorage charge is around 3 percent, as noted by Marco Polo in his observations on the charges for the replacement of paper money in China.

24. The earliest Chinese notes still in existence appear to be an issue from A.D. 960, according to Bereziner and Narbeth (1973).

25. Estimate provided by personal correspondence from the U.S. Treasury, 1988.

26. Private communication from Paul R. Frey, Department of Treasury, Bureau of Engraving and Printing, September 19, 1986.

27. According to the *Wall Street Journal*, August 20, 1986, p. 2, Money Promotions Inc. paid $10 a ton for shredded money. Indeed, I possess a transparent ballpoint pen filled with shredded money bearing the legend "Go for the green stuff."

28. I am grateful for the research efforts of D. Stoddard in obtaining this information.

29. A perusal of the *Official 1987 Blackbook Price Guide of United States Paper Money* shows that for a $100 Confederate note in mint condition the price may be as high as $2,000.

12 Money, Credit, and Bankruptcy

12.1 The Policing of Credit

In one-period games without exogenous uncertainty, the use of commodity or fiat money greatly facilitates trade. Currency may be preferred to credit because it is more anonymous and involves less bookkeeping and clearing. There is no need for the extension of credit either for intertemporal consumption loans or speculation. In the one-period model without uncertainty, financing is needed only for transactions. When anonymous trade is contemplated, a commodity money or fiat appears to offer a somewhat better means of exchange than granting individual credit, which may be more convenient for dealing with known customers. Thus we have accounts receivable and thirty-day or other payment periods offered by many firms.

Payments in money or credit instruments denominated in money (such as bank checks) provide for a great simplification to trade. A commodity money such as gold provides the extra benefit of self-insurance. As value is given for value, the noncooperative equilibrium is self-enforcing. One trusts the gold, not the government or the not directly economic rules of the game.

In most societies, however, the honoring of credit relationships is not enforced by transfers of gold but by the laws of the land combined with custom and social pressure. When the money in use is fiat and the credit arrangements are manifested in paper promises of money, the enforcement of contracts comes primarily through the laws and customs. We could argue economic determinism for each of the laws, but here they are assumed to be given as part of the rules of the game. We attempt no political, economic, and social explanation of the foundations of the law, observing instead that rules and penalties are required if the system is to achieve certain economic goals deemed desirable.

12.1.1 Statistical justice

The definition of what is meant by an optimal bankruptcy law and its design are both fraught with difficulties. How do we distinguish the unfortunates from the fools, and the fools from the scoundrels? Many of the most attractive results in economic theory depend on the presence of large-scale anonymous competition. Yet many of the more important features of credit relationships appear to depend on trust, the ability to evaluate, and the understanding of special situations. As the technology

of trade and communications becomes faster and more sophisticated along with the growth of nationwide or worldwide financial institutions, it may be that the "boutiques" that specialize in knowing their business and their customers' business may also flourish.

Does "economic efficiency" weigh more than honesty? when laws are to be administered to a vast anonymous mass of traders, such a question poses deep moral problems for the society. A simple example is provided by the behavior in the United States with respect to parking tickets. To many individuals, parking tickets are a cost paid on a statistical basis, with little concern for social purpose. Going to jail for a white-collar crime appears to have a completely different gestalt from paying a fine. Indeed, there appears to be little justification for a smooth utilitarian scale between going to jail and being fined.

In the modeling of default and bankruptcy discussed below, there is an open question as to what social and political price a society is willing to pay in its design of a mass (and in general expensive) enforcement system. This cannot be answered merely by arguing an economic justification for every aspect of a legal system. To do so would imply that the economic values are either independent from or more important than the other values of the society within which the economy is embedded. (We return to this point in Volume 2, chapter 7.)

12.2 Bankruptcy with Incomplete Markets

Many of the more interesting aspects of default and bankruptcy occur in multiperiod economies with production exchange and uncertainty. In keeping with our policy of separating out difficulties, we present the first set of problems in a one-period model with incomplete markets.

When markets are incomplete and there are securities with unlimited short sales permitted, it is well known since Arrow's paper (1964) that, with many traders, an efficient competitive equilibrium outcome can be attained as though there were complete markets. Thus the condition of spanning means that by using a weighted combination of the available securities, all bets can be attained.

Arrow (1964) observed that for large numbers of goods and states ms could be large compared to $m + s$. But by appropriately constructing s lottery tickets, bets, financial instruments, or stocks, the same outcome can

be achieved with $m + s$, instead of ms, markets. In essence, each security enables the individual to transfer income across states. Implicit in Arrow's treatment is the proposition that individuals own shares or stock paying out in income or money, although there is no explicit consideration of any monetary mechanism.

A complete set of security markets will exist if there are enough securities so that all individuals can achieve any transfer of income across states they wish, consistent with their endowments. Suppose that there are s securities, one for each state. Security j pays one unit of income in state j, and zero otherwise. A combination of purchase and sale of these securities can move an individual's income into any state.[1] A set of securities with this property is said to span the uncertainty.[2]

When spanning fails, the complete market's competitive equilibrium cannot be attained. Furthermore, if we require that traders balance their budgets in every state, this highly constraining requirement may result in state-dependent equilibria that are highly inefficient.

The requirement that an individual must balance his budget in every state can be modeled as though there is a bankruptcy penalty of unbounded disutility imposed on each trader. It is a natural question to ask if such a draconian penalty is in the interests of society as a whole? Before considering specific examples, let us turn to the general purposes of laws on default, insolvency, bankruptcy, and reorganization.

12.2.1 Default, insolvency, bankruptcy, and reorganization

In a one-period exchange economy without exogenous uncertainty, with the granting of credit, insolvency and bankruptcy need not be distinguished from each other, and reorganization is irrelevant. Rules of the game are needed to prevent individuals from electing strategic bankruptcy. The rules will be harsh enough if there are no circumstances under which a Pareto improvement can be achieved by any individual electing for active bankruptcy. Matters become considerably more complicated when uncertainty and more than one period are considered.

There are at least five competing (and often conflicting) factors that go into the design of failure laws in a society: (1) overall societal efficiency; (2) protection of creditors; (3) deterrence of debtors, especially if the failure involves elements of purposeful strategic choice; (4) rehabilitation of debtors, especially if their state appears to have been reached more through bad luck than bad judgment; and (5) "fairness" and equity for all

parties involved, including the public. Two other desiderata may be revelation of truth and administrative efficiency.

In a complex society, the failure of corporations may have a broad impact on many constituencies. There may be a social stigma attached to bankruptcy that damages the bankrupt's future rehabilitation. Not only must creditors and the public be protected, but the interests of employees, whether owed back wages or not, must also be considered. The seniority of claims poses legal and societal problems. Liquidation or reorganization can be lengthy and time-consuming. The possibilities for concealment of a weakened financial condition are many, as are the opportunities for an incompetent or dishonest management to delay the day of reckoning through the wasting of assets. Or there may be criminal looting of assets, abetted by special inside knowledge. And the fees of accountants, trustees, courts, and litigators can swallow whatever assets that remain.

Because we are limited to one period in this chapter, many of the more complex phenomena do not appear. But even at this level of simplicity with incomplete markets, the first four of the five factors listed above can be considered. In particular, it is argued that there will exist equilibria in which all members of society will be better off with the correctly chosen failure penalty rather than one that is either too harsh or too lenient. This may mean that an economy with incomplete markets and laws such that there are failures, even in equilibrium, may be more efficient for all, than one in which budget constraints are always met.

When individuals have to make a choice under exogenous, as well as strategic, uncertainty, it may be in the interest of society as a whole that they select a strategy whose outcome depends on chance. If penalties for bad luck are high, an intelligent borrower may choose always to play it safe; if they are too low, an intelligent lender will not be willing to lend. But either way, the strategies of all players involve the strategic selection of a level of risk, which will be influenced by the nature of the default laws. These laws essentially reflect a societal acceptance of risk. Furthermore, where investigation and data gathering are expensive from the point of view of economic efficiency, the settlement rules may not necessarily involve the revelation of truth.

The bankruptcy and reorganization laws of a society control the level of mutation and innovation in the economy. They are critical to the control of the birth, mutation and death processes of a dynamic economy.

12.2.2 A digression on one-period multistage models

All but the simplest of strategic market game models involve more than one move because, unlike general-equilibrium models with complete markets, the sequencing of moves and the availability of information are critical. The decision to distinguish between models with many moves but only one period and those with many periods is to a great extent arbitrary. Yet it is suggested here that in keeping with the separating out of difficulties and new phenomena, it may be worth doing, at least for some purposes. The particular distinction we may wish to make is at the level of the utility function. A one-period multimove model may be distinguished from a multiperiod model by the arguments in the utility function. In the one-period model, the evaluation of the consumption of any good is given as the sum of its consumptions at various moves. If we regard each move as a new period, we might wish to indicate a difference in the evaluation of consumption by introducing a new argument each period for each good. This might entail a discount factor to model impatience (Koopmans, 1960), although in spite of elegant axioms, the empirical evidence for time discount is still not settled.

When exogenous uncertainty is introduced, it is reasonable but not logically necessary to model trade as involving at least two periods. A full recognition of more than one period requires careful distinctions among perishables, storable consumables, and durables. We defer this discussion to subsequent chapters.

12.2.3 Modeling incomplete markets with money or credit

When there are incomplete markets, we may require that budgets be met in every state. An equilibrium that satisfies this condition is called a "general equilibrium with incomplete markets," or GEI.

An even more important reason than those given above for dividing a single period into more than one stage is to be able to differentiate the act of trading from that of consumption. Implicit in one-period models of exchange is the idea that exchange and consumption are temporally linked together.

A simple example serves to illustrate GEI. Consider two types of traders trading in two commodities in two states. Each trader has a utility function of the form

$$u_i = \frac{1}{2}\sqrt{x_1^i y_1^i} + \frac{1}{2}\sqrt{x_1^i y_1^i}, \quad i = 1, 2. \tag{12.1}$$

There are four contingent commodities, good 1 if state 1 or 2 and good 2 if state 1 or 2. The initial endowments of the first type of trader are denoted by $(6, 2; 0, 4)$ and of the second type by $(2, 6; 4, 0)$. For incomplete markets (i.e., no trades in future markets or securities), if the economy goes to state 1, endowments will be $(6, 2)$ and $(2, 6)$; if state 2, $(0, 4)$ and $(4, 0)$; in both instances, spot trade may take place subject to the budget constraint of each trader in that state. They both achieve $1/2\sqrt{4 \cdot 4} + 1/2\sqrt{2 \cdot 2}$. If futures trading or securities were introduced, they could obtain $1/2\sqrt{3 \cdot 3} + 1/2\sqrt{3 \cdot 3}$ each.

A strategic market game with incomplete markets will have strategies similar to those described in chapters 6 and 7, but extended to account for contingencies and individual information about them. When there is a continuum of traders, it may be mathematically easier to consider GEI than the related game. In many instances, but not all, the two approaches appear to give the same results.

Results concerning bankruptcy with incomplete markets are in general beyond the mathematical level of this book. (These results are presented in detail by Dubey, Geanakoplos, and Shubik, 1988; Karatzas, Shubik, and Sudderth, 1994; and Dubey and Geanakoplos, 1989.)[3] Here we advance only a qualitative heuristic argument, together with several examples.

The formulation of a GEI model or a strategic market game in which agents default is not completely straightforward. In the first place, we must consider what lenders know about the particular borrowers with whom they are dealing. Here we preserve the anonymity of markets, and also perfect equilibrium, by supposing that when a borrower defaults on some security the loss is spread out proportionally to all creditors of that class. This sharing of losses will tend to be true in economies with a high level of financial intermediation.

In equilibrium, all lenders will rationally anticipate what fraction of each kind of loan will be repaid as a function of the state of nature. If we assume that there is a continuum of agents, all agents will rationally anticipate that these functions are independent of their own actions.

One might suppose that it is a simple matter to describe a GEI with bankruptcy. One need only specify the securities according to what is actually delivered, as opposed to what is promised. But what is delivered

is determined endogenously and cannot be predicted without solving for the equilibrium. Moreover, different agents may make different deliveries on the same asset even though the lenders receive the same aggregated payoffs.

When an asset market is active, the informational requirements for the equilibrium state in our model are roughly the same as in competitive equilibrium. Traders need not bother about the identities of those with whom they are trading. When the market is not active, however, then potential lenders cannot be guided by the macro variables in anticipating what fraction of their loans would be repaid were they to offer them. In this case, we suppose that the lenders anticipate in a sense we make precise.

As the endogenous prices change, borrowers will change what they decide to deliver. Hence the actual span of the assets can change. This change can be discontinuous and even involve a robust change in dimension. As pointed out in the works referred to above, this change in dimension is one of the key reasons for the desirability of active bankruptcy.[4]

12.3 Incomplete Markets and Commodity Money

In chapter 10, we explored an economy using commodity money with a loan market that, given enough money, could bring about efficient trade. Here we wish to examine the meaning and implications of enough money when there are incomplete markets.

12.3.1 Enough commodity money and efficiency

A simple example serves to describe the fundamental aspects of considering the nature of trade with a commodity money when there are incomplete markets. Suppose that there are two types of trader, each with a utility function of the form

$$u(x, y, z) = .5\sqrt{(x)} + .5\sqrt{(y)} + z, \tag{12.2}$$

where the endowments are $(2, 0, A)$ and $(0, 2, A)$. These signify that in the first state, traders of type 1 have an endowment of 2 and traders of type 2 have 0, and in the second state it is vice versa. In both states, each player has an amount A of commodity money. With little calculation, we note that for any value of A, the complete market competitive equilibrium

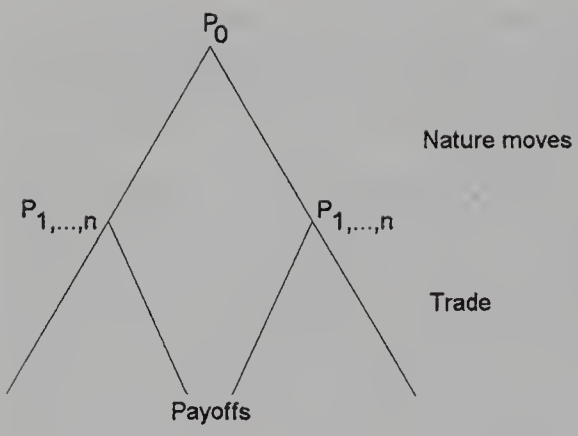


Figure 12.1
Incomplete market with commodity money.

solution to this exchange introduces futures contracts to eliminate the uncertainty, and each individual obtains $(1, 1, A)$, which yields $u = 1 + A$ to each.

The strategic market game without futures markets or borrowing has an extensive form, as indicated in figure 12.1. There are many variations in information conditions and in the type of randomness present. For instance, is randomness correlated for members of the same type? Do individuals find out the outcome of their random variable or randomness over the whole economy before they move? Each leads to a different game. But as our prime purpose is to illustrate the features of the role of money, not the delicacy of information, we select the simplest game.

As can be seen from the game tree in figure 12.1, we assume that Nature moves and all individuals are then informed of the subgame they are in. The meaning of enough money now becomes enough money to cover desired optimal trade in any state (or subgame). In the example, there is enough money in all states if $A \geq .5$ for traders of each type. The solution to the strategic market game is Pareto optimal, but *not the same* as the competitive equilibrium. In the complete market competitive equilibrium, the solution gives each trader $(1, 1, A)$ in state 1 and $(1, 1, A)$ in state 2. The final endowments in the game are $(1, 1, A + .5)$ and $(1, 1, A - .5)$ in state 1 and $(1, 1, A - .5)$ and $(1, 1, A + .5)$ in state 2. Because we selected individuals who are risk neutral to money, the ex ante or expected value of trade is the same as the CE, but the ex post or realized trade is not.

The possibility for efficient self-insurance exists only if individuals are risk neutral in the monetary good and it is in sufficient supply. The simple change in the utility function shown in equation (12.2) illustrates that as soon as individuals are risk averse in the monetary good, it is no longer possible to self-insure efficiently. We change the utility functions to

$$u(x, y, z) = .5\sqrt{(x)} + .5\sqrt{(y)} + \sqrt{(z)}. \tag{12.3}$$

With a little manipulation, we obtain the following solution:

$$x = (A + 2\sqrt{(A)})/(A + \sqrt{(A)})$$

and

$$p = \sqrt{(A)}, z = p^2 x.$$

We observe, for example, that for $A = 1$, the CE expected payoff for both trader types is 2, but the game gives

$$u = .5(\sqrt{1.5} + \sqrt{1.5}) + .5(\sqrt{.5} = \sqrt{.5}) = 1.93185.$$

12.3.2 Not enough commodity money and no loans

In the example above, if A is less than .5, there is not enough money to finance efficient transactions in each state. In a manner similar to that in chapter 9, we must consider in each state whether there is enough money, and if so, whether it is well or badly distributed.

The idea of enough money, well distributed appears to generalize easily from complete to incomplete markets only for a linear separable utility for money, even then the amount of money required by each trader must be enough to cover the trader's worst state, and this may become unreasonably large.

With complete markets, the use of money to cover transactions provides an efficient substitute for trust and some record keeping. If there is not enough money or trust, trade is constrained and hence inefficient. When there is exogenous uncertainty present, enough money would call for enough in the worst state for each individual. A less cumbersome and efficient alternative is to introduce futures contracts and insurance. But both futures and insurance contracts in essence involve two points of time, the point of contract before the drawing of a random variable and

the point of delivery after the drawing of the random variable. (We return to this in section 12.4.2.)

12.3.3 Commodity money and loans

A simple extension of the examples illustrated in chapter 10 and in section 12.3.1 indicates the problems of money supply and incomplete markets. In particular, with a linear separable money, the results of chapter 9 hold. That is, if there is enough money in each state but it is badly distributed, and if the loan market meets after the random variables have been selected, then money will be lent at an interest rate of zero and the final outcome will be Pareto optimal.

What happens when loans are feasible, markets are incomplete, and money has no linear utility? Figures 12.2a and 12.2b show the two simplest extensive forms. In figure 12.2a, traders borrow and lend, the random variable is selected, and then they trade in spot markets. Qualitatively, we note that traders may run a surplus of money in some states and have a shortage in others. Because traders do not know the state before the loan market is active, their lack of knowledge prevents them from correcting what would be a maldistribution under complete information.

In figure 12.2b, the loan markets take place after the state has been selected. Qualitatively, the results in each state are the same as in chapter 9. When there is not enough money, a positive rate of interest emerges. The presence of loans calls for a default penalty to be specified. At this level of abstraction, it should be noted that any security put up for a secured loan is a form of precautionary garnishing, or what the creditors can claim back with certainty (see also section 11.3.5).

12.4 Incomplete Markets and Other Structures

12.4.1 A basic example with short sales

In this section, our concern is with active bankruptcy and the size of the penalty, not with the amount of commodity money serving as insurance. Here we consider a two-period model with fiat money and with one state in period 0 and two states in period 1. This is solved in three ways, first with complete markets for its competitive equilibrium, then with incom-

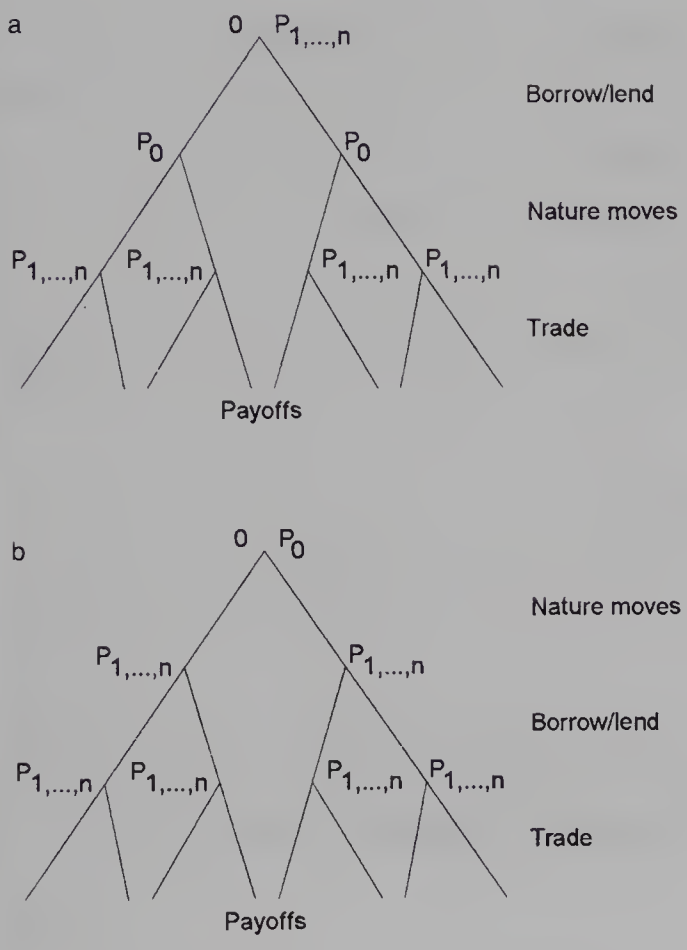


Figure 12.2
Incomplete markets with commodity money and loans.

plete markets for its incomplete market competitive equilibrium (IMCE) with all budgets balanced, and finally with a finite failure rule.

Solution 1: Complete markets We consider two types of traders, each with the same utility function. Individual i has a utility function of the form

$$\varnothing(x_0^i, x_1^i, x_2^i) = \sqrt{x_0^i} + .9\sqrt{x_1^i} = .01\sqrt{x_2^i}, \tag{12.4}$$

where x_0^i = consumption by i of initial holdings prior to chance's move, x_1^i = consumption by i in state 1, and x_2^i = consumption by i in state 2. Traders of type 1 have as initial endowments $(100, 1, 1)$, and those of type 2 have $(0, 100, 0)$ (i.e., traders of type 1 have 100 units of period 0 good, 1 in each of the two states of period 1. Traders of type 2 have 100 units in state 1; otherwise, nothing).

Traders of type 1 maximize

$$\sqrt{x_0^1} + .99\sqrt{x_1^1} + .01\sqrt{x_2^1}, \tag{12.5}$$

subject to

$$100 + p_1 + p_2 = x_0^1 + x_1^1 + x_2^1.$$

Traders of type 2 maximize

$$\sqrt{x_0^2} + .99\sqrt{x_1^2} + .01\sqrt{x_2^2}, \tag{12.6}$$

subject to

$$100p_2 = x_0^1 + p_1 x_1^1 + p_2 x_2^1,$$

where

$$x_0^1 + x_0^2 = 100, \tag{12.7}$$

$$x_1^1 + x_1^2 = 101, \tag{12.8}$$

$$x_2^1 + x_2^2 = 1. \tag{12.9}$$

If we let μ_1, μ_2 be the Lagrangian multipliers, we obtain

$$\frac{1}{2\sqrt{x_0^1}} = \mu_1, \tag{12.10}$$

$$\frac{.99}{2\sqrt{x_1^1}} = \mu_1 p_1, \tag{12.11}$$

$$\frac{.01}{2\sqrt{x_2^1}} = \mu_1 p_2, \tag{12.12}$$

$$\frac{1}{2\sqrt{x_0^2}} = \mu_2, \tag{12.13}$$

$$\frac{.99}{2\sqrt{x_1^2}} = \mu_2 p_2, \tag{12.14}$$

$$\frac{.01}{2\sqrt{x_2^2}} = \mu_1 p_2. \tag{12.15}$$

Solving (12.4)–(12.12), we obtain

$p_0 = 1, \quad p_1 = .985086, \quad p_2 = 0.1,$

$x_0^1 = 50.64541, \quad x_1^1 = 51.15186, \quad x_2^1 = .506454,$

$x_0^2 = 49.35459, \quad x_1^2 = 49.84814, \quad x_2^2 = .493546,$

$\mu_1 = .070258, \quad \mu_2 = .071171.$

Total utility to traders of type 1 $\prod_1 = 14.20420,$

Total utility to traders of type 2 $\prod_2 = 14.02202.$

Solution 2: Incomplete markets We reconsider the model in example 1 without futures markets, but with a single lottery ticket, with pays 1 unit in state 1 or 2 in exchange for some of the good in state 0. Trade in a single lottery offers a way to carry out a limited exchange between goods in state 0 and in states 1 and 2 linked together.

Solution 2a: No short sales Selling a lottery ticket where there is a chance of failure to deliver is equivalent to allowing a limited short sale. If short sales are not permitted, there will be no trade in this model—all traders will consume their own endowments:

$\prod_1 = 11,$

$\prod_2 = .99.$

Before we discuss failure, we can dispense with two extreme cases: when the default penalty is zero and when it is prohibitively high.

Solution 2b: Short sales (0 or ∞ penalties) If the penalty for failure to deliver is zero, a single debtor i in a mass anonymous market may not have an economic incentive to pay. There are two cases: (1) i can choose whether to in either state; and (2) i is forced to pay if able to do so, but is not penalized if unable to pay.

To make these cases unambiguous, we assume for case 2 that creditors know the state and can claim payment prior to the debtor consuming. If this is so, then it may still pay creditors to lend even though they know they will not be paid back in some state and the debtor will not pay any penalty.

If, as in case 1, payment is at the choice of debtor i, no buying of short sales will take place. If, in contrast, the penalty is extremely harsh and i has utility that is bounded, then there will be a penalty (such as exile, execution, prison, loss of citizenship, slavery) sufficiently high that if i faces a finite probability of ever being unable to repay, i will not risk going short.

In summary, a zero penalty lowers or kills the demand in the market for lottery tickets, and a large enough penalty kills the supply. In the second instance, there are no short sales; thus in the example

$$\prod_1 = 11,$$

$$\prod_2 = .99.$$

Solution 2c: Short sales, intermediate penalties In specifying the default penalty, we must answer the following questions:

1. Should the penalty be individual?
2. Should the penalty be state-dependent?
3. Should or can the penalty be strictly economic, such as confiscation of other assets, garnishing salary, and so forth?
4. Should the penalty be designed to punish strategic default or to improve economic efficiency?

We can answer the first two questions directly and simply. As a reasonable first approximation to economic and administrative reality, the answer is no. In practice, a distinction may be made in some societies between personal and corporate default. Furthermore, there may be an "escape clause" for an act of God, war, or other occurrence that might be construed as "force majeure." At most, a society will have two or three essentially anonymous, more or less state-independent default conditions, modified by a disaster clause and qualified by a host of special considerations decided ad hoc in the courts of law. We suggest that the most reasonable economic model calls for a single individual and a single corporate rule that is both anonymous and state independent.[5]

The third question concerns the nature of the penalty. At this level of simplicity and abstraction, we avoid several details of importance to the functioning of a modern enterprise economy, such as whether the default

is denominated in stock, money, goods, or fixed alternatives between them. For example, a failure to deliver 100 shares might be covered by the payment of three times the average price per share during the last five days of trading. Here, for simplicity, we consider the presence of a separable term in the utility function that can be interpreted as an economic or noneconomic loss (pay up enough or go to jail).

The economist and moralist must ask what the penalty is designed to accomplish. There may be a difference between promoting honesty (at least in the short run) and improving economic efficiency.

Returning to the original example, we reformulate it as a game with a default penalty. A strategy by an individual i of type 1 is to offer an amount of the good in state 0 in exchange for a promise to pay in states 1 and 2. Let b^i be the amount of good 0 offered and z^j be the amount of stock or lottery ticket offered by type 2. The trading price of lottery tickets will be

$$v = \int b^i \Big/ \int z^j, \tag{12.16}$$

while the actual price is

$$v^* = \int b^i \Big/ \int w^j, \tag{12.17}$$

where w^j are the payments made.[6]

In contrast with (12.5), we may write the payoff to a trader of type 1 as

$$\Pi_1^i = \sqrt{100 - b^i} + .99\sqrt{b^i/v^* + 1} + .01\sqrt{1}. \tag{12.18}$$

A strategy for a trader j of type 2 is a pair of numbers (z^j, w^j), where w^j is the amount that a short seller delivers in state 1. This may be less than j has promised:

$$0 \leq w^j \leq z^j.$$

In contrast with (12.6), the payoff to a trader of type 2 is

$$\Pi_2^j = \sqrt{vz^j} + .99\{\sqrt{100 - w^j} + \mu \min[(w^j - z^j), 0]\} - .01\mu z^j. \tag{12.19}$$

The last term is the penalty per unit for failure to deliver. If (in a type-symmetric equilibrium) an individual j has sold short z^j, because j cannot

cover in state 2, j's penalty will be μz^j. In state 1, j has 100 units, which j could use to cover the z^j owed, or j might choose to pay only $w^j < z^j$.

If there is an interior solution, where $w^j < z^j$, then

$$\frac{\partial \prod_1}{\partial b^i} = 0 \quad \text{or} \quad \frac{.99(1/v^*)}{2\sqrt{b/v^* + 1}} = \frac{1}{2\sqrt{100 - b^i}}, \tag{12.20}$$

$$\frac{\partial \prod_2}{\partial z^j} = 0 \quad \text{or} \quad \frac{v}{2\sqrt{vz^j}} = \mu, \tag{12.21}$$

$$\frac{\partial \prod_2}{\partial w^j} = 0 \quad \text{or} \quad \frac{1}{2\sqrt{100 - w^j}} = \mu, \tag{12.22}$$

and (12.16). At a type-symmetric NE, all traders of the same type may select the same strategy; thus we can consider the five equations to be

$$v = b/z, \tag{12.23}$$

$$v^* = b/w, \tag{12.24}$$

$$\frac{b}{\sqrt{100 - b}} = \frac{.99w}{\sqrt{w + 1}}, \tag{12.25}$$

$$z = \sqrt{b}/2\mu, \tag{12.26}$$

and

$$w = 100 - 1/4\mu^2, \tag{12.27}$$

where all $b^i, z^j, w^j \geq 0$, $0 \leq \mu \leq \infty$.

For the utility functions selected, the range of this solution is as shown in table 12.1.

Table 12.1
Default penalty ranges

μ	w	b	z	Payoff 1	Payoff 2
0–.05	0	0	—	11	9.9
.0505	1.970	10.696	32.381	11.166	11.532
.0705	49.700	49.235	49.764	14.184	13.999

Above $\mu = .0705337$, z exceeds w,[7] and we must consider the equations

$$\frac{b}{\sqrt{100-b}} = \frac{.99z}{\sqrt{z+1}} \tag{12.28}$$

and

$$\sqrt{b} = z\left\{\frac{.99}{\sqrt{100-z}} = .02\mu\right\}, \tag{12.29}$$

which can be solved numerically for a range of the parameter μ (see table 12.1).

Figure 12.3 shows the Pareto surface for the traders in the exchange economy with complete markets. The point CE is the competitive equilibrium point. The point NE is the best incomplete market equilibrium, in the sense that the game with penalty $\mu = .0705337$ gives an NE that has higher payoffs for all traders than any game with a different penalty. The point NT is the no-trade point, where $\mu = 0$ or ∞.

Figure 12.3 also shows the failure payoffs as a loop in the payoff space. Penalties that are too lenient or too harsh are not optimal in the sense that for a different penalty there is an NE with all traders better off. The harsh

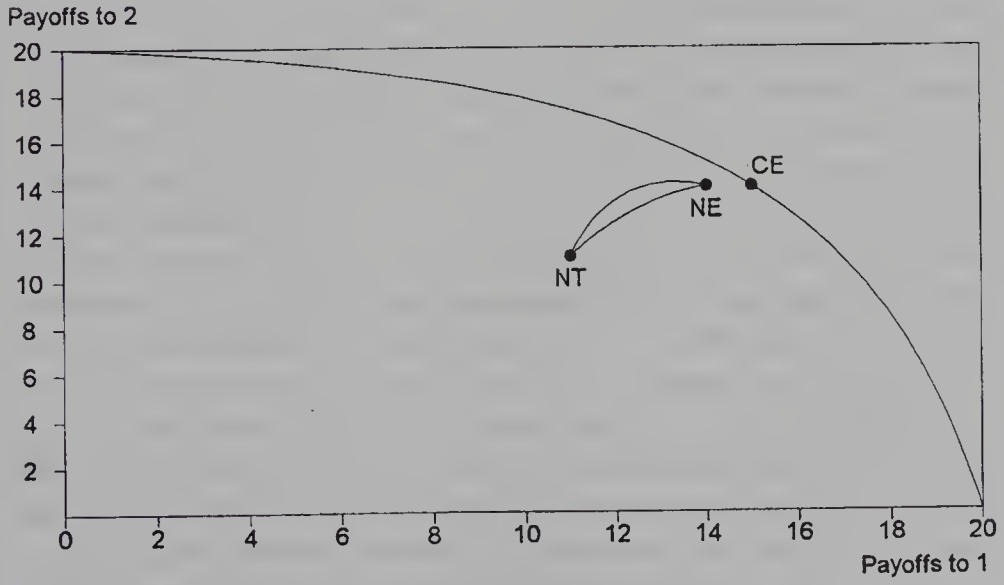


Figure 12.3
Pareto surface and penalty curve.

nonoptimal range is $.0705337 < \mu \leq \infty$, and the lenient range is $0 \leq \mu \leq .0705337$. In this example, the range is bounded by the point at which $w^j = z^j$.

In this example, there is a unique optimal penalty at $\mu = .0705337$. In general, this might be a range and the selection of the penalty, a matter of public choice. In an example with only two types and a clear distinction between borrowers and lenders, a voting mechanism where, for example, a simple majority chooses μ would lead to a debtor-dominated economy selecting one end of the range and a creditor-dominated economy selecting the other. Unfortunately, this simplicity appears to be an artifact of an example with only two types, although the broad characterization into debtors and creditors might yield an interesting class of economies with single-peaked preferences for the size of μ.

12.4.2 Enough money with goods and financial instruments

If we extend our concept of trustless transactions to trade with futures contracts or insurance, we must decide who pays whom and when. The meaning of trustless trade in one-period exchange is clear—all traders pay cash. But suppose we extend the analysis of trustless trade to exchange with random endowments, but with futures markets available. What do we mean by trustless trade in futures contracts?

In the simple example below, we may imagine that each of the two types of traders whose endowments are determined randomly is in a position to sell a state-contingent futures contract. But with such a contract, if delivery is to depend on neither law nor trust, who pays whom and when? If the seller of the contract does not trust the buyer, the seller should require full payment in advance. But if the buyer does not trust the seller, the buyer also should require that the seller deliver some form of hostage in advance. At the minimum, we need to invent some form of escrow arrangement for the enforcement of contracts. Thus at the minimum, the enforceability of an escrow arrangement is called for.

Historically, at the Chicago Board of Trade, earnest money was required from both parties to discourage backing out of the contract. But as the timing could involve a considerable fraction of a year, the question arises as to who collects the interest on the earnest money in escrow.

The two polar extremes for futures trading are (1) no money is put up by either side until the contract is exercised, at which point it is, as before,

cash for goods; (2) each side must deposit cash in an escrow account, to the full amount of the contract, with any interest paid to the owners. The second arrangement requires twice as much cash as the first.

A quick perusal of the *Wall Street Journal* shows that futures markets scarcely exist for more than two or three years into the future. Furthermore, even with the great growth of commodity futures in the past twenty years, anonymous mass trading is still limited to relatively few goods.

12.4.3 Futures markets and transactions needs

A reasonable question in practical economics as well as in economic theory is, why are there incomplete markets? A plausible answer is that the probability of the occurrence of many future events is difficult to estimate and hence to insure. Furthermore, markets and contracts are expensive, often involving high setup costs. Chance events such as the imposition of special taxes or the creation of temporary needs provide the opportunity to create a new market, which, once in existence, may continue to be profitable even if the original stimulus is no longer present.

At the level of abstraction considered here, both an insurance policy and a futures contract are lottery tickets. The basic difference between them is that an insurance policy is a lottery ticket that is bought for money and pays out in money, where the size of the payoff may vary with the state. A futures contract is paid for in money and pays out in a commodity at a later time. It is apparently state independent. But this is false on two counts. In particular, the money value of the commodity paid out will vary with the state. Furthermore, even though the odds are small, it is possible that the futures seller may not be able to deliver; thus a failure rule must be specified.

We can complete markets in different ways, by adding futures contracts, securities, or insurance opportunities. In this section, futures contracts are considered. The simple example with utility functions given in section 12.3.1 can be extended to include two futures contracts that complete the spanning of markets. The extensive form is shown in figure 12.4, where futures are exchanged before the random variable is selected, and then spot trade takes place. If we follow an extreme no-trust convention, then both buyer and seller should put up in escrow the full value in commodity money of the futures contract until delivery, at which point the seller gets his own "trust units back" together with the buyer's payment.

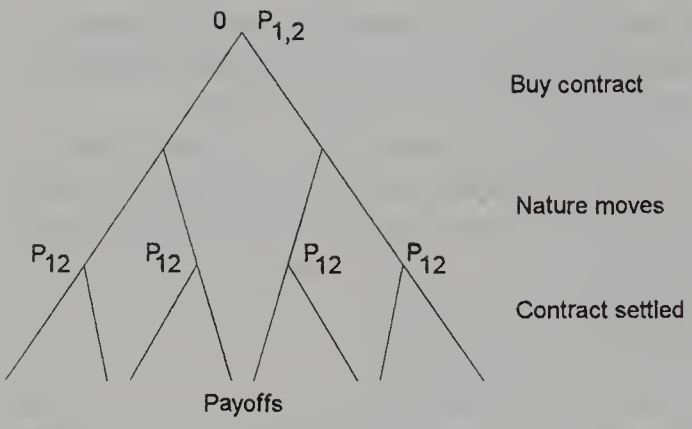


Figure 12.4
Trade with futures markets.

The two additional markets are for contracts of the variety

"Pay 1 unit of good 1 if state 1; 0 if state 2."

"Pay 0 if state 1; 1 unit of good 2 if state 2."

Suppose that the original endowments are $(2, 0, A)$ and $(0, 2, A)$. It is straightforward to check that the solution is the same as the CE with final endowments being $(1, 1, A)$ and $(1, 1, A)$ and each individual buying one unit of one of the futures contracts. If there is enough money, prices are $(1/4\sqrt{A}, 1/4\sqrt{A}, 1/2\sqrt{A}, 1/2\sqrt{A}, 1)$, where the first two prices are the prices of futures, the second two the spot prices of the goods, and the last the price of money. In trading futures, all buyers must pay cash in escrow for the full sale price and all sellers must cover their maximum exposure if there is no trust at all; thus each requires $3/4\sqrt{A}$ and this will be enough if it is smaller than A. In this simple example there will be no trade in the spot markets because each will just take delivery of the future.

12.4.4 The minimum cash needed to complete markets

Suppose there is an economy with incomplete markets. There will be many ways in which markets can be completed. Each way of completing the markets will generally, require a different level of transactions. There will be one or more ways of completing the markets in such a manner that the amount of financing to achieve efficient trade is kept to a minimum. A simple example illustrates this.

Consider the example in section 12.3.1. There is a commodity money whose supply is state independent. which each individual holds, and another commodity whose distribution depends on two states. The distribution is described by $(2, 0, A)$ for traders of the first type and $(0, 2, A)$ for traders of the second type, where with probability .5 type 1 traders have 2 units of the good and with probability .5 they have 0. Regardless of state, each trader type has A units of commodity money. The utility function of each is

$$u(x, y, z) = .5\sqrt{x}\ .5\sqrt{y} + \sqrt{z}.$$

Two ways of completing markets are considered. The first market is completed by adding two contingent financial instruments or contingent futures: the first pays 1 unit of the uncertain good if state 1 occurs, and pays 0 units otherwise; the second pays 1 unit in state 2 and 0 units otherwise. A trader of the first type sells a unit of the first bet and buys a unit of the second (the second types does vice versa).

Instead of introducing these two instruments, the markets can be completed with two contracts, the first of which pays 1 unit in state 1 and .9 of a unit in state 2; the second, vice versa. Traders of type 1 sell 10 units of the first contract and buy 10 units of the second. Traders of type 2 do vice versa. In both instances, with A large enough, the outcome for both types is $(1, 1, A)$, but the volume of trade and escrow requirements are far larger in the second than in the first case.

A small amount of algebra shows that the requirement to minimize the volume of trade in financial instruments is fulfilled by selecting one instrument to pay out in one state and pay zero otherwise. If there are s states, s such instruments are needed.[8] Basically, if no short sales are called for trade will be minimized. Short sales are never needed if there are s instruments of the variety noted above.

12.4.5 Insurance and transaction needs

If, instead of introducing futures, we were to introduce insurance or money-for-money lottery tickets, the money quantity conditions amount to the same as previously, but the interpretation of price prediction changes. We consider two lottery tickets of the variety:

"Pay 1 unit of money if state 1; 0 if state 2."

"Pay 0 if state 1; 1 unit of money if state 2."

The price of each lottery is .5 There will be enough money if $A \geq 1$.

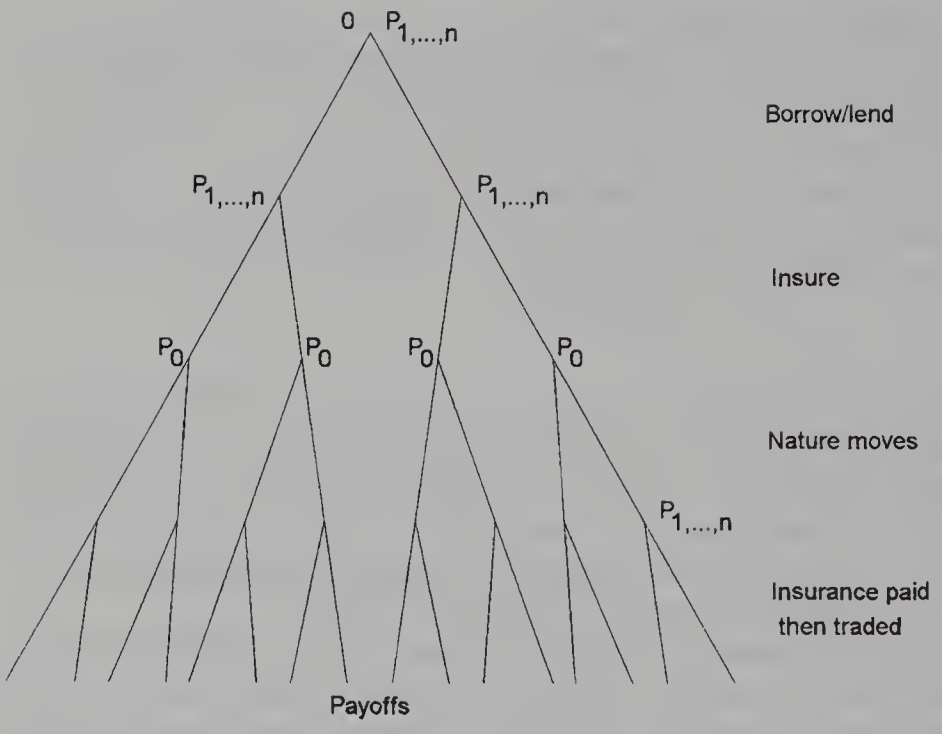


Figure 12.5
Insurance with fiat money.

12.4.6 Fiat instead of commodity money

In the one-period model, we may replace commodity money with fiat money it the following manner. Select an amount of fiat M and a single bankruptcy penalty μ such that it is sufficiently high that there is no state in which an individual will strategically go bankrupt. The extensive form is as shown in figure 12.5. Individuals first bid for fiat, then buy and sell money bets. After the random variable is determined, they are paid their insurance and trade in spot markets. After trade, they are required to redeem their IOUs from the referee. This game will produce optimal exchange without bankruptcy, with a zero outside rate of interest Modifying a previous simple example, where the utility functions are

$$u_i = \frac{1}{2}\sqrt{x^i} + \frac{1}{2}\sqrt{y^i} + \mu \min[0, \text{final debt}]$$

and endowments are $(2,0)$ and $(0,2)$. The following notation is introduced:

M = the amount of outside money available;

v^i = bid by i for outside money;

$u_1^1(u_2^2)$ = the offer of 1 (2) to sell lottery tickets which pay $(1,0)((0,1))$ (i.e., 1 unit in state 1 and 0 units in state 2, and vice versa);

$v_2^1(v_1^2)$ = the bid by 1 (2) for the second lottery ticket;

s_j = the price of the jth lottery ticket;

$q_1^1(q_2^2)$ = the offer for sale by 1 (2) of the state 1 (2) good;

$b_2^1(b_1^2)$ = the bid to buy by 1 (2) of the state 2 (1) good;

p_j = the spot price of the good in state j;

μ = the default penalty;

ρ = the money rate of interest.

A strategy by 1, (2) is described by $(v_1, u_1^1, v_2^1, q_1^1, b_2^1), ((v_1^2, u_2^2, v_1^2, q_2^2, b_1^2))$.

The payoff to a trader of the first type (and similarly for the second type) in terms of the strategic variables is

$$\prod_1 = \frac{1}{2}\left\{\sqrt{2-q_1^1} + \mu \min\left[0, \left(\frac{v^1}{1+\rho} - v^1 + u_1^1 s_1 - u_1^1 - v_2^1 + q_1^1 p_1\right)\right]\right\}$$

$$+ \frac{1}{2}\left\{\sqrt{b_2^1/p_2} + \mu \min\left[0, \left(\frac{v^1}{1+\rho} - v^1 + u_1^1 s_1 - v_2^1 + v_2^1/s_2 - b_2^1\right)\right]\right\},$$

where

$$0 \le q_1^1 \le 2,$$

$$1 + \rho = \int (v^1 + v^2)/M,$$

and where

$$\int v^i = \text{the sum of all bids of traders of type } i,$$

$$s_1 = \int v_1^2 \Big/ \int v_1^1,$$

$$s_2 = \int v_2^1 \Big/ \int v_2^2,$$

$$p_1 = \int b_1^2 \Big/ \int q_1^2,$$

$$p_2 = \int b_2^1 \Big/ \int q_2^2,$$

and $0 \leq b_2^1 \leq v^1/1 + \rho - v_2^1 + v_2^1/s_2$.

If escrow money is required, the seller of the bet needs to place the sale price of the bet plus reserves to cover the bet fully, thus

$$\frac{v^1}{1+\rho} - u_1^1 + \frac{u_1^1}{s_1} \geq 0,$$

that is, the cash requirements of escrow limit the sale of the bets.

If sale and purchase of all bets are simultaneous, then the cash requirements are

$$\frac{v^1}{1+\rho} - u_1^1 + \frac{u_1^1}{s_1} - v_2^1 \geq 0.$$

A buyer of a bet does not require more escrow than its price because the buyer has no opportunity to renege for more.

For $M = 2$ and $\mu \geq 1/2$, a solution is provided by $v^i = 1$; $\rho = 0$; $u_1^1 = u_2^2 = 1$; $v_2^1 = v_1^2 = 1/2$; $s_j = 1/2$; $b_2^1 = b_1^2 = 1$; $q_1^1 = q_2^2 = 1$; $p_j = 1$.

This simple but somewhat tedious example is provided to help us observe that the velocity of money here, used once to cover the exchange of paper and once for goods, is 2. The escrow conditions are needed to stop the seller from taking strategic advantage of the buyer by failing to deliver. The bankruptcy or default penalty is required (as in chapter 11) to have the outside money returned.

12.5 Bankruptcy: Firms and Individuals

12.5.1 Bankruptcy and failure

Table 12.2 presents a few statistics on bankruptcy filings in the United States (see Jevons 1909, pp. 9–10, on bankruptcies in England from 1806 to 1860). Although the business filings are only a fraction of personal filings, they account for a preponderant amount of the indebtedness. Many of the nonbusiness bankruptcies are for minor sums. Table 12.3 indicates

Table 12.2
Bankruptcy filings, 1960–1985

Year	Business	Other	Total
1960	12,284	97,750	110,034
1965	16,618	163,405	180,323
1970	16,362	178,037	194,399
1975	30,130	224,354	254,484
1980	36,499	241,431	277,880
1985	666,651	297,885	364,536

Source: *Statistical Abstract of the United States* (1977, p. 571, 1987, p. 510).

Table 12.3
Business population and failures, 1960–1985

Year	Total concerns in business (thousands)	Net worth (billions of $)	Failures	Current liabilities (millions of $)
1960	2,708	409	15,445	939
1965	2,527	536	13,514	1,322
1970	2,442	763	10,748	1,888
1975	2,679	1,438	11,432	4,380
1980	2,780	2,602	11,742	4,635
1985	4,990	3,216	57,067	333,376

Source: *Statistical Abstract of the United States* (1987, table 863).

the total industrial and commercial population and the failures, giving failure liabilities and the net worth of nonfinancial corporate business.[9] Failure rates and a crude index of their financial importance are indicated in table 12.4. Table 12.5 compares the total population with new incorporations and all exits as well as failures, which enables us to compare exit by failure with all forms of retirement.[10]

The last four tables suggest that with the bankruptcy laws the actual economic system implicitly recognizes that incomplete markets and errors in economic behavior abound. The bankruptcy laws are not completely state dependent and relatively few contingent markets exist. Instead, there is a positive level of actual bankruptcy that enables the system to run with highly aggregated rules and that takes care of ill fortune, incompetence, and bad forecasting at the moral cost of being unable on occasion

Table 12.4
Failure rates, 1960–1985

Year	Failure rate (×10,000)	Liabilities (% of net worth)[a]
1960	57	22.7
1965	57	24.6
1970	44	24.6
1975	43	30.5
1980	42	17.8
1985	114	104.0

a. The calculation of percentage of liability is based on *Statistical Abstract of the United States* (1997, p. 57, 1987, tables 863, 871). Data for 1980 and 1985 appear to involve a database change and should be regarded with caution.

Table 12.5
Entries exits, and failures, 1965–1983

Year	Total firms in business	New incorporations	Exits	Failures	Failures/ exits (%)
1965	2,527,000	204,000	N/A	13,500	NA
1966	2,520,000	200,000	211,000	13,000	6.16
1967	2,519,000	207,000	208,000	12,400	5.96
1968	2,481,000	234,000	245,000	9,600	3.92
1969	2,444,000	274,000	271,000	9,200	3.39
1970	2,442,000	264,000	276,000	10,700	3.88
1971	2,466,000	288,000	240,000	10,300	4.29
1972	2,490,000	317,000	264,000	9,600	3.64
1973	2,567,000	329,000	240,000	9,300	3.88
1974	2,591,000	319,000	305,000	9,900	3.25
1975	2,679,000	326,000	231,000	11,400	4.94
1976	2,782,000	376,000	2,323,000	9,600	4.30
1977	2,793,000	436,000	365,000	7,900	2.16
1978	2,786,000	478,000	443,000	6,600	1.49
1979	2,708,000	525,000	556,000	7,600	1.37
1980	2,780,000	534,000	453,000	11,700	2.58
1981	2,745,000	581,000	569,000	16,800	2.95
1982	2,806,000	567,000	520,000	24,900	4.79
1983	2,851,000	600,000	522,000	31,300	6.00

Source: *Statistical Abstract of the United States* (1987, table 863).

to differentiate among the scoundrels, the fools, and the misfortunates (see Weiss, 1986, on bankruptcy in the Victorian novel).

With the formal introduction of exogenous uncertainty into a model of exchange without complete markets, it can be shown that a bankruptcy law that invites some number of bankrupts may dominate a draconian law that discourages ever risking bankruptcy (Dubey, Geanakoplos, and Shubik, 1988). When administrative costs are considered, it is fairly clear that economic efficiency in general will call for a trade-off between failures and complexity. The ultimate in complexity, from the perspective of strategic market games, would have the penalty depend both on the individual and the state.

12.5.2 The Modigliani-Miller result

Modigliani and Miller (1958) observed that if no firm went bankrupt then the mix between the issue of debt or equity of corporate financing would not matter. Intuitively, whatever mix of bond and lottery ticket (stock) a firm offers, if there are no constraints on borrowing, an individual can undo the leveraging offered by the firm by borrowing and selling short. When firms can and do go bankrupt, this result is in general no longer true. But in section 12.4.1, it has been shown that, in an economy with incomplete markets, active bankruptcy may be desirable. Thus the correct question to ask can be specified as follows. Given k firms limited to the issue of bonds and stock that do not fully span all markets, and given a default penalty, is there an optimal debt equity mix for the firm? The answer appears to be yes, although it may not be unique. To consider this in detail, we need to describe an economy with money, firms, stocks, and bonds. A detailed general study of this must be delayed until firms, production and a stockmarket are formally introduced in a subsequent volume.[11] However, a simple example shows some of the difficulties with the Modigliani-Miller proposition even without bankruptcy.

Consider an economy with two types of agents. Type 1 own the input good in period 0. Type 2 also have the input commodity at period 0 and own a stochastic production process whose output depends on chance and the amount of input. In particular, the utility function for traders of type 1 is

$$U_1 = .5 \min[x_0, x_1] + .5 \min[x_0, x_2],$$

with initial endowments of $(2, 0, 0)$. The utility function for traders of type 2 is

$$U_2 = .5\sqrt{x_1} + .5\sqrt{x_2},$$

and their initial endowments are $(3, 0, 0)$.

The lottery ticket (or the stochastic production process owned by traders of type 2 is as follows:

$z_1 = .3125w$ with probability .5

$z_2 = 100\sqrt{w}$ with probability .5,

where w is the input and the z_1 and z_2 are the outputs in state 1 and 2.

If traders of type 2 can issue a commodity-denominated bond (i.e., one that pays a unit of commodity in both states), it can be shown that there is a noncooperative solution where all traders of type 1 sell one unit of input in return for 1 unit of the bond. Traders of type 2 maintain 100 percent ownership of the "firm" and go into production with 4 units of input. The final consumptions are (1, 1, 1) and (0, .25, 199), with utility of 1 for type 1 and 7.4242 for type 2, as contrasted with no trade utility of 0 for type 1 and 6.341 for type 2.

In this example, type 1 traders are completely risk averse, while type 2 traders can use their resources plus the purchases from type 1 to produce enough under all circumstances, given the stochastic outcomes, that they can put together a riskless bond to finance their purchases from type 1.

The Modigliani-Miller result appears to depend on unspoken assumptions concerning an open model. In this example, which is completely closed, there is only one optimal financial structure for the bonds-stock ratio. There is now a considerable literature on this topic, which, being tangential to the main theme of this book, is not considered here.

12.6 Historical Notes

12.6.1 A historical sketch of bankruptcy

It is probable that the word "bank" is derived from *banca*, the Italian word for the bench set up in the marketplace to aid a moneychanger in transacting business.[12] A moneychanger whose business failed, it is sug-

gested, had his bench broken, hence the word "bankruptcy" may have emerged from *banca rotta.*[13]

The principal purpose of bankruptcy legislation has been to arrange for an equitable and orderly liquidation of assets of an insolvent debtor. In early times, bankrupts were subject to penalties and possible loss of civil rights. Under ancient Roman law, an unpaid creditor with a favorable judgment could force the sale of the debtor's estate and the debtor might lose his civil rights. The Italian medieval cities had harsh statutes calling for seizure of the assets of debtors, especially those of merchants absconding or fraudulently causing insolvency. Thus at an early date, the two strands of strategic and involuntary bankruptcy were recognized.

In the earlier laws, there was a tendency to treat bankrupts as criminals, absconders, and defrauders. There were debtors' prisons, criminal penalties, loss of civil rights, and liquidation of assets. The trend in the twentieth century has been toward voluntary proceedings and reorganizations rather than criminal actions and liquidations, unless fraud is proved. There are considerable differences from country to country in the treatment of individuals and corporations, manifested in which assets are exempt from and which subject to seizure for liquidation or distribution to creditors.

Bankruptcy law in the United States attempts to serve the ends of economic efficiency. Early law was otherwise, as the chronicles of Newgate prison indicate (Griffiths, 1884, pp. 2, 9).

> For centuries a large section of the permanent population of Newgate, as of all gaol, consisted of offenders against commercial laws. While fraudulent bankrupts were hanged, others more unfortunate than criminal were clapped into gaol to linger out their lives without a chance of earning the funds by which alone freedom could be recovered. Debtors of all degrees were condemned to languish for years in prison, often for the most paltry sums. The perfectly innocent were also detained. Gaol deliveries were rare, and the boon of arraignment and fair trial was strangely and unjustly withheld, while even those acquitted in open court were often hauled back to prison because they were unable to discharge the gaoler's illegal fees....
>
> As a general rule the movement of offenders through Newgate was pretty rapid. The period of imprisonment for debtors might be often indefinitely prolonged, and there was the well known case of Major Bernardi and his companions who were detained for forty years in Newgate without trial or the chance of it.

"As early as 1700 a pamphlet preserved in the 'Harleian Miscellany' and entitled 'Labour in Vain' anticipates modern feeling and modern

legislation." Griffiths protests against the imprisonment of debtors, which he compares to shutting up a cow from herbage when she gives no milk. "In England we confine people to starve, contrary to humanity, mercy, or policy. One may reasonably expect his dog," he says, "when chained to a post should catch a hare, as that poor debtors when in gaol should get wherewithal to pay their debts" (Griffiths, 1884, p. 70).

The earliest known recorded bankruptcy legislation appears to be in the Code of Hammurabi of Babylon in 1750 B.C.:

> Let the farmer who owes the corn, but cannot pay, produce all his goods before witnesses and give them to the merchant. The merchant shall take them as full payment of the debt. But if the witnesses value the debtor's goods to be more than what is owed, then the merchant shall take only enough goods to satisfy the debt. If the merchant takes more than is due him or illegally seizes the property of the farmer, then the merchant shall be punished according to law. If the debtor conceals his property, or fails to produce all of it, then he shall be punished.

In Deuteronomy 15:1, 2, the release of debts is noted: At the end of every seven years thou shalt make a release. And this is the manner of the release: Every creditor that lendeth ought unto his neighbor shall release it; he shall not exact it of his neighbour, or of his brother; because it is called the Lord's release.

Virtually all societies of the West and Middle East had laws regarding commercial contracts and insolvency, illustrating that it was recognized that even early trade could not function solely on trust. An apparent exception is India, where community pressure against default appeared to be sufficient to ensure that the creditor would be paid. There was the custom of "sitting *d'harna*" or fasting on the debtor's doorstep until paid (Manu, VIII, 49). The law of Emperor Justinian in Rome in A.D. 533 contained a *cessio bonorum* clause where debtors, by voluntarily relinquishing their assets, could circumvent the creditors' right to put them to death or subject them to torture. But no indication was given of valuation or exemption of any of the debtors' property.

In 1215, King John of England included valuation and exemption provisions in England's first recorded law of insolvency. The Siete Partidas of Don Alfonso X, the Wise, of Castille and Leon in 1250 distinguished between the insolvency of merchants and that of nonmerchants and included the voluntary provision whereby debtors could declare themselves unable to pay their creditors while avoiding prosecution through the surrender of all personal assets. In 1542, under Henry VIII, Statute 34

was an "acte againste suche persones as doo make Bankrupte" applied to absconding and concealed debtors.

Appendices A and B to this chapter contain a brief chronology of the development of international and United States bankruptcy laws. The basic theme indicated by this brief survey is that the bankruptcy laws and customs are a public good required by the needs of trade in a world with both strategic and exogenous uncertainty. Furthermore, as the legal persons of the individual and the corporation have evolved, so has the need to distinguish them in bankruptcy. Over the ages, bankruptcy laws have come to distinguish between fraud and ill fortune, and the attitude behind the laws, to change from punishment to economic settlement.

What an optimal bankruptcy law is in a world with uncertainty and incomplete markets depends to some extent on the overall level of risk a society wishes to adopt. In general, it is likely (as a virtual corollary of inventory theory) that an optimal bankruptcy law will always have a certain number of individuals going bankrupt (see Dubey, Geanakoplos, and Shubik, 1988).

12.7 Concluding Remarks: An Optimal Bankruptcy Rule?

When there are complete markets (or their equivalent) in a strategic market game, the concept of an optimal bankruptcy rule is easily defined in a strategic market game, although a default penalty may be required if borrowing takes place. It is the rule that is harsh enough to always discourage strategic bankruptcy.

When there are incomplete markets, this definition is no longer sufficient because there is a trade-off between strategic behavior and ill fortune. In essence, the default laws become a public good that indicates the public's aggregate preference for risk taking in that society. The trade-off between strategic motivations for default and ill fortune is determined by the strength of the bankruptcy laws.

Law is on a longer time horizon than everyday economic life. It does or should reflect the evolution of consensus in a society. As noted in section 12.4, one way we can reflect society's overall preference for risk via the default penalty is to have the default penalty set by political process.

At an even more fundamental level, in a world with uncertainty, the bankruptcy laws are the key factor in the dynamics or the bioeconomic

evolution of the firms and new products of society. Bankruptcy is a controlling factor on risk taking. Paradoxically, the stress in microeconomic general equilibrium theorizing ignores risk taking, which is central to any viable theory of growth.

(We return to the question of an optimal bankruptcy rule in the presence of exogenous uncertainty in Volume 2, chapter 8, where we investigate the possibility that a bankruptcy rule may serve to complete markets and that, at least theoretically, there may be an optimal bankruptcy rule under uncertainty, but where we find that considerations of information processing and disclosure make it impractical.)

Appendix A: Chronology of International Bankruptcy Law

1750 B.C. Code of Hammurabi (Babylonia)

Let the farmer who owes the corn, but cannot pay, produce all his goods before witnesses and give them to the merchant. The merchant shall take them as full payment of the debt. But if the witnesses value the debtor's goods to be more than what is owed, then the merchant shall take only enough goods to satisfy the debt. If the merchant takes more than is due him or illegally seizes the property of the farmer, then the merchant shall be punished according to law. If the debtor conceals his property, or fails to produce all of it, then he shall be punished.[14]

623 B.C. Criminal Code of Draco (Greece)

Under this code, indebtedness was classified with murder, sacrilege, and other capital crimes. During Solon's reign (late seventh and early sixth centuries B.C.) the code was revised to allow that, after restitution, debts would be forgiven, and debtors and their heirs suffer only loss of citizenship.

450 B.C. The Laws of the Twelve Tables (Rome)

Under these laws, debtors pledged their own persons for repayment of loans. Hence the unpaid creditors could take debtors as their slaves, sell them, or punish them through torture or death.

326 B.C. Lex Poetalia (Rome)

Under Lex Poetalia, the debtors could, through swearing that their assets equalled their total liabilities and immediately surrendering all assets to their creditors, escape the horrors of the Laws of the Twelve Tables.

100 B.C. Missio in bona; cessio bonorum (Rome)

Missio in bona provided that creditors could take possession of debtors' property as security against nonpayment. *Venditio bonorum* (provided that the debtors') estates could then be sold for the benefit of all creditors, resulting in a loss of civil rights for the debtors. *Cessio bonorum* allowed debtors to voluntarily relinquish their assets to their creditors by petitioning a magistrate; such action circumvented the creditors' right to put the debtors to death or subject them to bodily torture.

A.D. 476 Rumpentes et falliti (Italy)

Italian medieval cities subjected bankrupts to severe penalties and liquidated their estates; such bankrupts were called "*rumpentes et falliti*."

A.D. 533 Emperor Justinian's Code (Rome)

Earliest known law providing charitable treatment of debtors.

A.D. 1215 King John (England)

Under King John, England's first insolvency law was put into effect. The law provided that it was illicit to seize debtors' land and property when the debtors' personal assets were sufficient to pay creditors and when the debtors offered them in payment. The law further stipulated that debtors could not be imprisoned; only their property could be taken to satisfy a claim.

A.D. 1250 Siete Partidas (Spain)

These laws, *Siete Partidas*, are attributed to the reign of Don Alfonso X, the Wise, King of Castile and Leon. It included detailed provisions relating to debtors and creditors. Notable features of the laws were that they applied to both merchants and nonmerchants and included a voluntary provision whereby debtors could declare themselves unable to pay their creditors while avoiding prosecution through the surrender of all personal assets.

A.D. 1400–1500 Western Europe

Bankruptcy laws relating to fraudulent debtors and the property of absconders appeared in Western Europe largely during the fifteenth century. Such laws were modeled after the statutes of medieval Italian cities. Best documented are the early laws of France, Brabant, and Flanders.

A.D. 1531 Decree of Administration of Justice and Good Order (Flanders)

This decree, under the Emperor Charles V (Count of Flanders), set forth harsh provisions for the curtailment of bankruptcies.

A.D. 1542 Statute 34, Henry VIII (England)

This was the first well-known English "acte againste suche persones as doo make Bankrupte" and applied to absconding and concealed debtors.

A.D. 1571 Extension of Statute 34 (England)

This more detailed law provided rules governing merchants and other traders. No voluntary proceedings were included in the law.

A.D. 1582 Customs of Antwerp

These customs included comprehensive provisions for the treatment of bankrupts and the distribution of their estates.

A.D. 1600 Privy Council (England)

During the period that the Privy Council functioned as the supreme British legislative body, laws were enacted allowing deserving debtors to restructure their liabilities with the agreement of their creditors.

A.D. 1673 Ordonnance du Commerce (France)

Under this ordinance, national rules for insolvency and bankruptcy relating to merchants were set forth. A voluntary provision was included and limited restructuring of debts permitted.

A.D. 1705 Chapter 17, Queen Anne (England)

Under Queen Anne, this statute provided for the release of all debts owed by bankrupts as well as freedom from future attachment as long as the bankrupts complied with their statutory duties.

A.D. 1706 Revision of Chapter 17 (England)

Under George II, Queen Anne's statute was liberalized to allow bankrupts to retain certain possessions. This is thought to be the beginning of the principle of exemption.

Early Labyrinthus Creditorum (Spain)

Based on Siete Partidas, Salgado de Somoza, a Spanish jurist, formulated in the 1700s explicit rules for the voluntary liquidation proceedings applicable to both merchants and nonmerchants. The breadth of *Labyrinthus creditorum* influenced the bankruptcy laws enacted by Germany, Portugal, Argentina and Brazil.

A.D. 1737 Ordinances of Bilbao (Spain)

Modeled after the Ordonnance du Commerce of France, these bankruptcy ordinances included clauses relating to both voluntary and involuntary liquidations. The Ordinances of Bilbao served as the basis for much of the bankruptcy law enacted in Latin America.

A.D. 1787 Article I, U.S. Constitution

A bankruptcy clause making national bankruptcy law legal in the United States was passed and included in Article I, Section 8, of the Constitution.

A.D. 1807 The Commercial Code (France)

This law was enacted to change what was considered to be an overly lenient clause in the Ordonnance du Commerce relating to preventive sanctions for bankruptcy. Rather than preventing bankruptcy proceedings, it allowed for a method of termination, thus providing a release from the dire consequences historically faced by debtors.

A.D. 1825 Statute 6, George IV (England)

This was the first British legislation to recognize the doctrine of voluntary bankruptcy. It provided that debtors could solicit and procure liquidation of their assets of could seek the adjustment of their obligations through private settlements.

A.D. 1850–1980

During the 1850s and the 1860s, the bankruptcy laws of most countries were restructured to include voluntary provisions and reorganization clauses. The dates of the operating bankruptcy laws for several countries up to the 1970s are set forth below (chosen date is that of the most recent *major* legislation or amendment).

1877 Germany
1881 Spain
1889 Switzerland
1914 Austria
1914 England
1920 India
1921 Sweden
1922 Japan
1931 Chile
1932 Peru
1933 Argentina
1942 Italy
1942 Mexico
1945 Brazil
1949 Canada
1961 Portugal
1966 Australia
1967 France
1967 New Zealand
1969 Colombia
1978 United States

Appendix B: Chronology of United States Bankruptcy Law

1607 Under the "Laws of England as known and practiced in Virginia," satisfaction of creditors was enforced through deprivation and torture of debtors. Imprisonment, the confiscation of all assets, and the attachment of all future property were common.

1638 The first formulated bankruptcy laws in America was enacted in Maryland. It provided rules for the recovery of debts and corresponded to the English law of the period.

1647 Revisions of the Maryland law were included in the new "Act Touching Payment of Debts."

1667 Law (Maryland) compelling merchants to accept tobacco in lieu of money as payment for debts was enacted in order to relieve the great number of unpaid creditors.

1682 "Frame of Government of the Province of Pennsylvania in America: Together with certain laws agreed upon in England by the Governor and divers Freemen of the aforesaid Province" was the first specific legislation to afford a remedy for debt.

1684 Legislation passed in Pennsylvania stipulating the manner and form of paying debts.

1692 First legislation mentioning the valuation of the debtors' estates: "The creditor shall take goods at appraised value, if at public sale they will not advance to more."

1701 The Charter of Philadelphia dictated that imprisonment would no longer be the form of punishment for bankruptcy.

1712 First legislation (Philadelphia) on bankruptcy to include a doctrine of limitations. Under the law, creditors could only seek recourse within six years of a debtor's default.

1729 Pennsylvania adopted the first code for relief of insolvent debtors. Amended in 1735 to make recovery of small debts easier and given a three-year term. Made a standing law in 1748 after two renewals.

1776 "An ordinance passed for the Relief of Prisoners of the Several Gaols in the State of Pennsylvania." (dealing in part with debtors)

1787 A bankruptcy clause making national bankruptcy law legal in the United States was passed and included in Article I, Section 8, Clause 4, of the Constitution. In *The Federalist* (no. 42), James Madison stated," The power of establishing uniform laws of bankruptcy is so intimately connected with the regulation of commerce and will prevent so many frauds where the parties or their property may be or be removed into different States that the expediency of it seems not likely to be drawn into question."

1789 A committee established in the House of Representatives to draft a bankruptcy bill was disbanded "until the public debt [could be] funded and banks established, without which it was difficult to conceive how arrangements could be made to facilitate the payment of debts or the operation of such law."

1790 Speculation in government scrips and bank stocks led the Second Congress to introduce a bankruptcy bill, although it was not enacted.

1797 Widespread land speculation led to increased financial chaos and the Fifth Congress again tried to introduce a national bankruptcy law. At this point, most of the states did not have bankruptcy laws. The bill was voted down due to a split between the industrial North (in favor) and the agricultural South (opposed).

1800 First national bankruptcy law was introduced in 1799; it passed by one vote in the House (49 to 48) and by four votes in the Senate (16 to 12). The law was patterned after English law, was limited to traders, and provided only for involuntary bankruptcy proceedings.

1803 After being applied to fewer than 500 bankruptcies, the first national bankruptcy law was repealed by strong majority votes in both the House and Senate.

1806 The French Decrees and English Orders of Council allowing confiscation of American shipping caused widespread insolvency. States with bankruptcy laws found their jails overflowing with debtors, and again legislators turned toward drafting a national bankruptcy law.

1814 Pennsylvania's insolvency law ruled unconstitutional.

1819 Now York's insolvency law ruled unconstitutional.

1820 Legislation for a national bankruptcy law, including provisions for voluntary bankruptcy, was introduced in the Senate but voted down by the South.

1821 Kentucky repealed legislation making prison mandatory for debtors.

1822 National bankruptcy legislation voted down in the Senate.

1826 Modified version of the 1822 bill voted down in the Senate.

1827 Rulings on the insolvency laws of Pennsylvania and New York reversed.

1837 President Jackson's Specie Circular, requiring payment in gold or silver for all public lands; the Deposit Act, providing distribution of Treasury surplus funds to depository banks (causing contracting

of credit); severe American crop failures; and the "Great Panic" led Congress to again consider national bankruptcy legislation. President Van Buren urged passage of a national bankruptcy bill in a special message to Congress, but the bill was defeated.

1840 Daniel Webster introduced a bankruptcy bill including both voluntary and involuntary provisions. The bill passed the Senate, but the House adjourned before considering the bill. Passage of Webster's bill became a major issue in the platforms for the election of 1840, with the Whigs in support of the bill.

1841 Webster's bankruptcy bill passed in the House and Senate. The law was limited to traders, bankers, factors, brokers, underwriters, and marine insurers; provisions for both voluntary and involuntary proceedings were included. This is the first U.S. bankruptcy law to include voluntary proceedings for debtors.

1848 After being applied to 33,379 cases ($43 million in surrendered property was given to creditors in settlement for $441 million in outstanding debts), the second national bankruptcy law was repealed in favor of the states' insolvency laws.

1857 The Panic of 1857 again encouraged legislators to consider national bankruptcy legislation, only to be pushed aside by the Civil War.

1864 National bankruptcy legislation was introduced in the House; although it passed the House, the Senate did not act.

1866 Legislation of 1864 was reintroduced in 1865 and passed both the House and the Senate.

1867 The third national bankruptcy law was broader in scope than the second national law and again included both voluntary and involuntary provisions. It was largely enacted to deal with Southern debtors as a result of the Civil War, but was frequently abused.

1873 The Law of 1867 was amended to include rehabilitation clauses.

1878 Senator J. R. McCreary (Kentucky) introduced legislation to repeal the third national bankruptcy law. Repeal was Senate approved by the after only two days of debate, and by the House (205 to 40). In McCreary's words, the act was "an assault upon public morals in its violation of good faith in its craft, its falsehoods and frauds."

1889 The Conventions of National Commercial Organizations began a movement encouraging the introduction of a new national bankruptcy law, drafted by Jay L. Torrey, a St. Louis lawyer, in 1890. But as late as 1893, the *Congressional Record* stated "the prejudices created by the abuses under the Act of 1867 make a fair discussion of any bankrupt act difficult."

1893 The Panic of 1893 caused hundreds of banks to close, and business failures numbered in the thousands. Although this encouraged rumblings with regard to a national bankruptcy law, senators were reluctant to pass legislation which de facto affected such large numbers of their constituents.

1898 In the form essentially proposed by Jay L. Torrey, the fourth national bankruptcy law was enacted. With amendments, it governed bankruptcy in the United States until 1978.

1978 The first major revision of bankruptcy law in around forty years, the Bankruptcy Reform Act, among other things, made it easier for a trustee to recover property transferred to a third party.

Appendix C: The Word "Bankrupt"

Different forms of the word "bankrupt" found in the English language, arranged chronologically:

1533 "bankcke rouptes"

1548 "banqueroutes"

1552 "bankerowten or make banckerowte"

1562 "bankeroute"

1563 "bankrouts prevented"

1577 "It is a doore and windowe into pourtie and bankrupting"

1577 "bankerupt"

1580 "banqueroute"

1580 "bankeroute"

1593 "bankrupts"

1600 "bankerout"

1601 "Banquerupt"
1613 "bankrout"
1613 "bankrouts"
1617 "bankruptures of religion"
1620 "banker-up"
1630 "He is in danger of breaking, or bankruptisme"
1656 "The most important Bankruptship . . ."
1668 "The Sieur Tiller . . . being lately bankrupted and fled"
1698 "Can the bankrumpting of one of the obligees quit"
1698 "With danger to make banke rota"
1700 "bankruptsie"
1709 "bankrupt"

Other forms used are "banke rota," "banckroupt," "banquerowpte," "banqweroote," "banquerupt," "bankrup," "bankrupture," and "bankruptship."
Source: F. Regis Noel (1921).

Appendix D: Biblical References to Debtors

2 Kings 4:1

A certain woman of the wives of the prophets came to Elisha saying: "Thy servant, my husband is dead and thou knowest that thy servant was only that feared God, and behold the creditor has come to take away my two sons to serve him." Elisha told the woman, "Go outside, borrow vessels of all your neighbors. . . ." Then from a single vessel of oil she filled all the vessels that had been gathered. Then Elisha said to her, "Go sell the oil and pay thy debt. . . ."

Matthew 18:24–25

Heaven is compared to a master settling accounts with his servants: "When he began the reckoning, one was brought to him who owed him ten thousand talents; and as he could not pay, his lord ordered him to be sold, with his wife and children and all that he had, and payment to be made."

Deuteronomy 15:1–2

At the end of every seven years you shall grant a release. And this is the manner of release: every creditor shall release what he has lent his neighbor; he shall not exact it of his neighbor, his brother, because the Lord's release has been proclaimed.

Appendix E: Select Bibliography on Bankruptcy

Books

Bays, Alfred W. *The Law of Bankruptcy and Debtors and Creditor*. Chicago: Callaghan, 1917.

Bellerby, J. R. *Control of Credit as a Remedy for Unemployment*. London: P. S. King and Son, 1923.

Fraser, Sir Drummond, *International Credits: The "Ter Meulen" Scheme.* London: Harrison and Sons, 1921.

MacGuigan, March R., *Cases and Materials on Creditors' Rights*. Toronto: University of Toronto Press, 1967.

Noel, F. Regis. *A History of the Bankruptcy Clause of the Constitution of the United States of America*. Gettysburg, PA: Complier Print, 1921.

Sullivan, George, *The Boom in Going Bust*. New York: Macmillan, 1968.

Tilden, Freeman. *A World in Debt*. New York: Funk and Wagnalls, 1936.

Pamphlets

Seasonal Considerations Relating to Insolvent Debtors, 1729.

A Proposal for Rendering Bankruptcies Less Frequent in and about London, 1732.

Debtor and Creditor, 1810.

Speech of Mr. Hayne, of South Carolina, against the Bankrupt Bill Extending a System of Voluntary Bankruptcy to Persons Other than Traders, 1827.

Speech of Mr. Tallmadge, of New York, on the Bankrupt Bill, 1840.

Opening the Debate on the Bankrupt Bill, 1841.

Statement on the Present Condition of Bankruptcy Legislation in Great Britain, France and the United States, 1882.

A Review of the Present Position of Bankruptcy Legislation in England and the United States, 1883.

Do the United States Require a Bankrupt Law? 1884.

Proceedings of the National Bankruptcy Convention, 1884.

The New Bankruptcy Law: Its Inadequacies and Shortcomings, 1899.

The Bankruptcy Law, 1900.

The Credit System and its Dependence upon Wise Bankruptcy Legislation and Administration, 1901.

Bankruptcy Legislation, 1904.

Amendment of the National Banking Laws, 1908.

Is There a Receivership Racket? 1942.

Notes

1. This may involve short sales.

2. If there are m securities ($m \geq s$) and s states and if security i pays $d_{i1}, d_{i2}, \ldots, d_{is}$, we require that the matrix $[d_{ij}]$ have rank s—that there be s linearly independent securities.

3. Much of section 12.2.3 is based on works referred to here.

4. In GEI models the possibility of a dimension change in the asset span creates the possibility of nonexistence of equilibrium. Yet it can be shown that no matter what the bankruptcy penalty, as long as it is strictly positive, if all assets promise delivery in a single commodity, then equilibrium always exists, the discontinuity in asset span notwithstanding.

5. Allowing for force majeure or "acts of God" and usually having some condition to increase penalties for fraud or other clearly illegal actions.

6. Pro forma, a lottery ticket pays $(1, 1)$; in actuality, it pays $(w, 0)$.

7. The solution where a short seller actually pays w involves a modeling detail concerning sequence of payment. The game with simultaneous moves is well defined, but if traders of type 1 does not pay until delivery offering b for z, they may not pay b for w; hence there would be no pure strategy NE of this type.

8. The instruments form a basis spanning the s dimensions.

9. Failures are a larger category than filed bankrupts, including voluntary withdrawals with known losses to creditors. A full definition is given in *Statistical Abstract of the United States* (1987, table 863, note 3).

10. Exits were calculated from information on total numbers and new incorporations in *Statistical Abstract of the United States* (1987, table 863).

11. There are further problems with the effect of mergers on spanning and with incomplete markets, where the goals of the firm are not well defined.

12. Much of this sketch was prepared with the able assistance of Tanya Styblo.

13. Indeed, one business bankruptcy newsletter published in the United States in 1982 was called the *Broken Bench Review*.

14. Dates of main entries in appendices A–C are approximate and have been verified only by single sources. These appendices (except for minor updating) were prepared by Tanya Styblo.

13 Money as a Numéraire, Means of Payment, Store of Value, and Substitute for Individual Trust

13.1 Major Properties of a Money

Lovers of epigram will perhaps find pleasure in the following: Money is the football of economic life.
—Silvio Gesell (English trans. 1929)

Money is an efficient coding device, a language, a societal substitute for individual trust, and a means for running a loosely coupled system. But before we can fully comprehend its more complex features, there is still much we need to understand about its more mundane features.

Among the major properties of a money already noted are that it can serve as (1) a numéraire, (2) a means of exchange, (3) a store of value, and (4) a source of liquidity. Desirable lesser properties are that it be easy to transport and identify, durable, easily divisible, hard to counterfeit, and easy to store. A possibly desirable property is that it be an anonymous "bearer instrument," but the price of anonymity is that such an instrument is hard to recover if stolen. On a personal check, which is bank money, payment can be stopped; on a $100 bill, it cannot.

Chapters 6–10 have analyzed money as a store of value and a means of exchange. Here its other properties and its central role of a means of exchange are considered.

13.1.1 Money as a numéraire

As a rule, where a single general means of exchange is used, the unit of the means of exchange also serves as the numéraire. Thus both wealth and prices in the United States are measured in dollars. The basic unit of the currency, whether gold or paper, is arbitrary—a troy ounce of a gram could serve equally well to denote price in terms of gold. A pound or guinea is equally adequate to denominate the value of transactions (although a guinea use to be somewhat more honorific).

When one solves a general equilibrium system for its equilibria, all variables are treated as symmetric a priori. Once the price system has been calculated, because the prices $p_1, p_2, \ldots, p_m$ are homogeneous of order zero, one can replace them with new prices by dividing through by any (nonzero) price p_i.[1] In the strategic market game models, any attempt to instruct the players how to play a simple experimental game raises problems concerning the ex ante attempt to select a numéraire. This can be illustrated for the simple model where there are three commodities in trade.

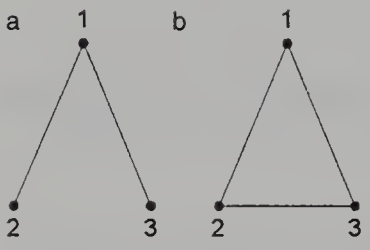


Figure 13.1
One or three numéraires?

When there are only two markets, as in figure 13.1a, where good 1 serves as a money, only two prices are formed by whatever mechanism the market uses to form price. These prices are exchange ratios between 1 and 2 and 1 and 3, respectively. It is natural to quote all prices in terms of good 1, the money that enters into all transactions.

In figure 13.1b, where markets are complete and all goods serve as money, there are three markets, and three prices are formed. In particular, the third market involves only goods 2 and 3. But, ex ante, there is no guarantee that in a playable game a no-arbitrage condition will hold. No arbitrage is a possible equilibrium condition,[2] not an overall condition on the set of feasible strategic outcomes. With complete markets, the ex ante selection of any commodity as numéraire has no meaning because there is always at least one market where the commodity chosen as numéraire is not involved in the exchange.

In controlled or premonetary nonmarket economies, because price is not being formally formed by any strategic market mechanism the selection of a *unit of account* presents no problem. Thus well before the appearance of a competitive monetary economy, the Egyptian and Babylonian civilizations appear to have used a unit of account for their warehouses and other economic activity.

We may summarize as follows: With m goods, if a money is selected in advance and there are only $(m-1)$ markets, then only $(m-1)$ prices are to be determined, hence ex ante normalization is natural. When all markets are available without ex ante imposing a no-arbitrage condition on prices, no natural normalization exists.

Another important feature of the selection of the numéraire is that in virtually all societies the bankruptcy penalty is related to the size of default measured in terms of the numéraire. Furthermore, at least for

most societies, the penalties do not change continuously with inflation; thus the selection of a numéraire that bears a fixed relationship to the means of payment ties the money to the individuals' preferences for default.

13.1.2 Money as a means of exchange

In the previous chapters, the models have concentrated on the highly important function of money as a means of exchange, in other words, its use in transactions. This is not its only function, but in keeping with the approach suggested here, if it is possible to isolate a property and study it separately, it is probably worth doing so.

The models required to illustrate the means of payment property were studied primarily for only one time period. The investigation of commodity money gave some insight into its store of value aspects, but an attempt to investigate fully the role of money as a store of value requires the examination of a multistage process. This is done from chapter 15 onward.

Can we, as Gesell seems to suggest, design an efficient monetary system where fiat and consumer deposit money serve primarily as numéraire and means of exchange and are unattractive as an intrinsic store of value? The answer depends on whether we wish to consider dynamics or statics. The function of means of exchange can be performed by a money without an intrinsic store of value property, if individuals *expect* that it will buy more or less the same in the future. A money with a high intrinsic value offers a guarantee of worth even if the system breaks down. But in a modern economy the trust in the future worth of money depends heavily on governmental control of its supply and the continuity of law and order. The need for a means of exchange to be a direct store of value varies inversely as the trust of the individuals in the stability and continuity of the economy.

13.2 Money as a Store of Value and as a Source of Liquidity

Money is by no means the only store of value, or necessarily even the prime store of value. To examine its relative role as a store of value, we need to study systems with more than one durable. This is deferred until Volume 2, chapter 2. Here we limit the investigation to a system where

the money has one of two forms. Either it is the only durable in the system, say gold, and can be used as a means of exchange or for direct utilitarian purposes, such as jewelry. (In each instance, it is assumed as a first-order approximation that is does not wear out,[3] always has some use, and is a store of value.) Or it is a consumer storable, such as a brick of tea, a package of cigarettes, or a can of sardines. In actuality, there does not appear to exist any perfect consumer storable. Tea decays, cigarettes dry out, canned goods have a limited shelf life. But the ideal type can be defined clearly and strategically. It is an item that can be stored indefinitely at no cost and then instantly consumed at a point of time chosen by its owner.

13.2.1 Gold and liquidity

Chen shih won neng-te
With gold anything is possible.
—Chinese proverb

Although a full treatment of multiperiod models follows in later chapters, here in order to illustrate the liquidity and store of value aspects of gold, an infinite-horizon model is considered. In this section, money is modeled as an ideal gold, which, if used solely as jewelry, yields an infinite stream of services. The model is extremely simple. It can be viewed as a game where gold is used as a means of payment and where, at the start of every period, all individuals can divide their holdings into gold used for money and jewelry. Jewelry can be converted to coin and vice versa, costlessly and timelessly, at the start of each period. In this formulation, all transactions take place at time $t = 0$, and gold used for transactions is in float for the full period. Thus, in terms of the parameters suggested in section 9.1.1, $a = b = c = 0$, and hence the time in float is 1 because neither the traders nor any middleman is in a position to use the gold for purposes other than transactions for the whole period.

The full formulation of a general game requires considerable detail and (together with formal proofs) is presented elsewhere (Shubik and Yao, 1989). To illustrate the properties of the supply of money, liquidity, and price levels, our comments are confined to simple example.

Gold is the means of exchange and is durable store of value, but here its use as a means of exchange involves a loss in its consumption services. Thus, is this game, staying liquid entails keeping gold in monetary form to use for imminent transactions.

We maintain the fiction that traders live as long as the economy. This is patently false, although a "dynasty" interpretation can be used for a justification. The reason suggested here is methodological. The infinitely lived trader with a discount is the easiest extension beyond the finite model. before we address overlapping generations and the new phenomena they bring.

There are two types of traders and three goods. All traders have a utility function of the form

$$\phi_i = \sum_{t=1}^{T} \beta^{t-1} \left\{ \sqrt{x_t^i y_t^i} + k \log z_t^i \right\}, \quad i = 1,2; k > 0, 0 < p < 1, 0 < \beta < 1. \tag{13.1}$$

The first two commodities are perishable, but are replenished each period. The third is infinitely durable, and each trader is given an initial quantity denoted by A.

Traders of types 1 and 2 obtain at the start of each period amounts $(2,0)$ and $(0,2)$, respectively, of the first two goods. By inspection, the general-equilibrium solution is that each period all traders consume $(1,1,A)$. Thus the payoff to each is

$$\phi_i = \frac{1-\beta^T}{1-\beta}\{1 + k \log A\}. \tag{13.2}$$

For $T \to \infty$, ϕ_i approaches

$$\frac{1}{1-\beta}\{1 + k \log A\}. \tag{13.3}$$

We now compare the solution for the infinite-horizon general-equilibrium model with the type-symmetric noncooperative equilibrium[4] solution to a strategic market game with two types of traders, all of whom are small. A strategy by a trader i is a vector $(\tilde{q}_i^i, \tilde{q}_2^i, \tilde{b}^i)$, where $\tilde{q}_j^i = (q_{j1}^i, q_{j2}^i, q_{j3}^i, \ldots,)$, $\tilde{b}_i = (b_1^i, b_2^i, \ldots,)$; for a trader of type 1, $0 \le q_{1t}^1 \le 2$, $q_{2t}^1 = 0$, $0 \le b_t^1 \le M_t^1$, where M_t^1 is the amount of gold held by a trader of type 1 at the start of t. Similarly, for a trader of type 2.

The payoff to a trader of type 1 is given by

$$\pi(\tilde{q}_1^1, \tilde{q}_2^1, \tilde{b}_1) = \sum_{t=1}^{\infty} \beta^{t-1} \left\{ \sqrt{(2 - q_{1t}^1) b_t^1 / p_{2t}} + k \log(M_t^1 - b_t^1) \right\}, \tag{13.4}$$

where

$$p_{2t} = \int b_t^1 \Big/ \int q_{2t}^2 \tag{13.5}$$

$$p_{1t} = \int b_t^2 \Big/ \int q_{1t}^1 \tag{13.6}$$

and $M_1^1 = A$ and

$$M_{t+1} = M_t^1 - b_t^1 + p_{1t} q_{1t}^1. \tag{13.7}$$

There are similar expressions for traders of type 2.

Let $V_i(A)$ be the value of the infinite-horizon game to a trader of type i. If there is a solution, it should satisfy the two functional equations:

$$V_1(A) = \max_{\tilde{q}_{1t}^1, \tilde{b}_t^1} \left[\sqrt{(2 - q_{1t}^1)(b_t^1 / p_{2t})} + k \log(A - b_t^1) + \beta V(A - b_t^1 + p_{1t} q_{1t}^1) \right] \tag{13.8}$$

and

$$V_2(A) = \max_{\tilde{q}_{2t}^2, \tilde{b}_2^2} \left[\sqrt{(b_t^2 / p_{1t})(2 - q_{2t}^2)} + k \log(A - b_t^2) + \beta V(A - b_t^2 + p_{2t} q_{2t}^2) \right] \tag{13.9}$$

If we assume that (1) there is a stationary-state solution and (2) there is enough money to finance trade (a point we shall consider later), then we can guess that $q_{jt}^i = 1$ and, by symmetry, that traders of each type will have the same solution except for an interchange of the roles of goods 1 and 2.

Thus the solution would be of the form

$$V(A) = \frac{1}{1 - \beta} [\sqrt{b/p} + k \log(A - b)], \tag{13.10}$$

where b and p remain to be determined.

At equilibrium, we have $q = 1$, and hence $b/q = b/1 = p$; and at equilibrium

$$\frac{1}{2\sqrt{bp}} = \frac{k}{A - b} \quad \text{or} \quad A - b = 2kb, \tag{13.11}$$

hence

$$p = A/(1 + 2k). \tag{13.12}$$

Thus the value is

$$V = \frac{1}{1-\beta}\left[1 + k \log \frac{2kA}{1+2k}\right], \tag{13.13}$$

in contrast with the payoff at the CE, which is

$$\frac{1}{1-\beta}[1 + k \log A]. \tag{13.14}$$

It is important to note, however, that if there were no lag or float loss so that all recipients could immediately receive the full consumption use of income, then the CE and NE would coincide. We can call the commodity money equilibrium a "gold-backed CE."

Liquidity and inefficiency The total utility loss from using a consumer durable commodity money to cover the full value of trade in an infinite-horizon model is the difference between (13.14) and (13.13) or

$$\frac{k}{1-\beta} \log \frac{1+2k}{2k}. \tag{13.15}$$

This assumes a full float loss of one period.

The finite-horizon model approaches the infinite-horizon model smoothly. In the finite-horizon model, because gold left over at the end has no value, there is no stationary state: there cannot be any trade in the last period because no traders will accept gold in payment if they can derive no utility from it during that period. In the finite-horizon model, prices inflate because the value of gold as an asset decreases over time even though there is no physical depreciation.

Enough money The total value of money in trade in any period at a stationary equilibrium is given by

$$\frac{d\phi_i}{db} b \text{ or, in this example, } \frac{k}{A-b} b. \tag{13.16}$$

There will always be enough money to support a gold-backed CE if b is feasible (i.e., is in the range $0 \leq b \leq A$) and satisfies

$$\text{Total purchases} \equiv \frac{A}{1+2k} \equiv \frac{bk}{A-b} \equiv \text{the value of money spent,} \tag{13.17}$$

or

$$b = A^2/(A + k(1+2k)) \leq A. \tag{13.18}$$

Hence, in this instance, there is always enough money.

This is not generally true, as can be seen from the example with a linear separable money of the form kz_t^i rather than $k \log z_t^i$. Equation (13.17) becomes

$$\frac{A}{1+2k} = bk \quad \text{or} \quad b = \frac{A}{k(1+2k)}, \tag{13.19}$$

but for $k < 1/2$, $b > A$. Hence the gold-backed CE is not feasible.

The price level In our example, the stationary state includes the price level. Real and nominal cash balances are the same, and because the economy repeats itself every period, enough money to finance one period is enough to finance the economy forever.[5] The key point to observe here is that the utilization of a finite amount of commodity money at finite velocity (not necessarily one) bounds prices. Unfortunately, as we shall see in chapter 16, even with zero float loss the efficient employment of a commodity money in a multistage economy is highly unlikely without strong and unreasonable conditions or with the addition of rental and loan markets.

The rate of interest Our model does not include money market, banking, or loan facilities. There is one intertemporal link, the relationship between the current and subsequent period's spot price of gold. In this specia example with a fixed (and sufficient) quantity of gold, a shadow or implicit rate of interest of

$$1 + \rho = 1/\beta \tag{13.20}$$

would emerge at the stationary state if we were to set up a money market so that individuals could lend each other gold, even though no borrowing or lending takes place.[6] In this extremely special stationary state, the need to vary the money supply does not arise. We note that even though gold is

a durable of infinite length, in a finite-horizon model (and to a reasonably good approximation in reality), a consumer cannot speed the consumption value derived from gold as the end approaches. When the game is over, the infinite-horizon durable left over is of no utility.[7]

The rate of interest depends on the time discount and the length of the horizon and is different from that in chapter 9, where the rate of interest was the price of a capacity constraint.

13.2.2 A perfect store of value

In the best of all possible imaginary worlds of the economist, it is at least logically possible that there might exist some magic commodity, which we could call manna or "utility pills" or trading sardines, such that it enters the utility functions of all as a linear and separable term, does not deteriorate, has no time discount, and can be stored costlessly. Consider an economy with n types of traders, trading for T periods in m perishable commodities. Suppose that utility function of a representative trader of type i is given by

$$\phi_i(x^i_{1,1}, x^i_{2,1}, \ldots, x^i_{m,T}) + \mu_i \sum_{t=1}^{T} x^i_{m+1,t}. \tag{13.21}$$

At every period t, a trader of type i obtains new endowments of $(a^i_{1,t}, \ldots, a^i_{m,t})$, but at the first period, i is also given an amount M^i of the storable consumable which is also used as the money in an exchange economy with only spot markets.

Individual i can consume i's money anytime because it yields i the same utility at any point. In particular, because it is a perfect store of value, i might as well consume it at the end. But this means that if i has enough money, i is financing trade at no loss because no services are forgone in executing transactions. All that individuals are doing is transferring utility among themselves at fixed rates.

The solution will provide an endogenous comparison of utility. If there is enough money and it is appropriately distributed, then there will be a unique CE, which is an NE of the game, and one could solve for this CE by maximizing the sum[8]

$$\sum_{t=1}^{T} \sum_{i=1}^{n} \{(1/\mu_i)\phi_i(x^i_{i1,1}, \ldots, x^i_{im,T}) + M^i\}. \tag{13.22}$$

Suppose that final endowments (and consumptions) are denoted by $(z^i_{1,t}, \ldots, z^i_{m,t})$ in period t; in period T, there will also be the consumption of $z^i_{m+1,T}$. Let the equilibrium prices be $(p_{11}, \ldots, p_{mT})$. Because the marginal utility of the money does not change from period to period, we can set $p_{m+1,t} = 1$ for all t.

At any period t, the amount of all purchases amounts to

$$\sum_{j=1}^{m} \max[p_{ij}(z^i_{jt} - a^i_{jt}), 0] \tag{13.23}$$

for an individual i. If we assume that the amount of money remains constant until the end of the game, then the amount of money held by i at the start of time $t+1$ will be

$$x^i_{m+1,t+1} = x^i_{m+1,t} + \sum_{j=1}^{m} \max[p_{jt}(a^i_{jt} - z^t_{jt}), 0] - \sum_{j=1}^{m} \max[p_{jt}(z^i_{jt} - a^i_{jt}), 0] \tag{13.24}$$

for $t = 1, \ldots, T$ and $x^i_{m+1,1} = M^i$.

The money needed during period t will be

$$\sum_{j=1}^{m} \max[p_{jt}(z^i_{jt} - a^i_{jt}), 0].$$

An individual i will have enough at period t if $x^i_{m+1,t}$ is greater than or equal to this amount.

We may ask, what are the lower and upper bounds on the amount of money needed for trade? As before, the amount needed falls between zero, if the initial endowments are Pareto optimal, and all nonmonetary wealth, if everyone possesses only items desired by others. In a modern economy, with income for many being composed primarily of wages, monetary needs are closer to the upper than lower bound.

In the instance of "utility pills," no one has to trust anyone. Every trade is of value for value. The ideal money provides total insurance at every stage. Unlike fiat or gold, neither law nor custom, convention nor fear, supports the worth of the perfect money.

The intuitive desire for a money of constant purchasing power can be made precise. It enters as a linear separable term in consumption. But if we distinguish, first, between a durable good providing a constant stream

of services and a storable good providing its value at an instant in time, and second, between a good whose consumption worth is independent of time and one whose consumption value is subject to a time preference discount, we arrive at four kinds of commodity money.

1. *A durable without discount.* The utility structure is of the form

$$\phi_i(x_{1,1}, x_{2,1}, \ldots, x_{m,T}) + \mu_i \sum_{t=1}^{T} x_{m+1,t}, \tag{13.25}$$

where the $x^i_{m+1,t}$ are the levels of consumption of the services of the commodity money, which does not physically depreciate. With a sufficient fixed amount of money, if the utility functions and other endowments are such that they permit a stationary state, then futures and spot prices will inflate at the rate of $T/(T-t)$ in terms of the durable.[10] Futures and spot prices will be constant in terms of a one-period use of the money, but because the asset (not its one-period service) is the money, in a finite-length economy, it is a wasting asset whose value is approaching zero, which is its disposal value at the end of time. In an infinite-horizon economy, the inflation would cease because the money would not be a wasting asset.

If there were a natural discount on all other goods but not the commodity money, then the change in spot prices would be modified by the rate of the natural discount. Thus the change in spot prices per period in a T-period economy with a β natural discount would be $\beta^{(t-1)}T/(T-t)$.

2. *A durable with discount.* The utility function is of the form

$$\phi_i(x_{1,1}, x_{2,1}, \ldots, x_{m,T}) + \mu_i \sum_{t=1}^{T} \beta^{t-1} x^{t-1}_{m+1,t}. \tag{13.26}$$

In a stationary state, if the natural discount on the other goods is also β, then spot prices will be inflating at $T/(T-t)$, but futures prices will be changing by $\beta^{(t-1)}T/(T-t)$.

3. *A storable consumable without discount.* The utility structure is as in (13.25), but no value is yielded unless the money is consumed.[11] Without discount, no value is gained by early consumption, hence it can be stored and consumed at the end without loss of efficiency, thus providing a store of value until the end of trade.

4. *A storable consumable with discount.* The utility structure is as in (13.26). With discount, value is lost by delaying consumption. Thus, if

the money appeared as a linear separable term, optimality would call for immediate consumption of all of the storable consumable, hence there would be no store of value for future transactions. This shows that a storable consumable that enters the utility function as in (13.26) can never serve as an efficient money if there if there is a natural discount.

I suggest that the appropriate interpretation of the casual verbal call for a money that is a "perfect store of value" is that its value is covariant with any natural discount.

13.2.3 Gold, tea, or fiat?

In a society with a time discount and trust, if (as is usually true) the resource cost of a fiat money is far less than a consumer durable or storable, it should dominate both. If we had a magic linear separable "utility radiating" consumer durable money (or, alternatively, a storable without discount) in adequate supply, all spot exchanges would be of value for value. There would be a unique CE of the exchange economy, which could be attained as an NE of a strategic market game with a continuum of traders using a commodity money. Because the exchange is always value for value, and thus fully self-policing, there is no need for bankruptcy laws.

In a multiperiod economy modeled as a strategic market game, the full meaning of "enough money" is that the economy using only spot markets has no need for rules of the game that involve futures contracts, bankruptcy laws or any trading to goods for paper or of paper for paper. Trade involves only goods for a commodity money,[12] exchanging value for value.

Economic activity is to its political environment as a game within a game. The outside game sets the rules for the inside game. (There may be feedback but it is relatively slow.) If the agents engaged in economic activity do not trust the rules, then an imperfect gold even in short supply may be more attractive than paper with pictures of Big Brother.

Viewing the world more optimistically, I suggest that, because the ideal commodity money does not exist, the evolution of paper money represents an attempt to devise a minimal set of rules for constructing a paper substitute that approximates the mathematical properties of a perfect store of value. (Volume 2 is devoted to developing some of these properties and sketching others that can only be fully specified by considering production and overlapping generations.)

13.3 The Ratio of Trade to Capital

How much money is required for trade? What is the ratio of trade to total capital? In the one-shot trading of the one-period strategic market game model of exchange, these questions can be made precise. In reality, there are many money substitutes and both explicit and implicit forms of credit that cut down on the need for money. Furthermore, there can be hoarding and the velocity of money influences trade. All of these factors require separate treatment. Here we concentrate on exchange games with a single money and one round of simultaneous exchange.

13.3.1 The trade to capital index

In a exchange economy with n traders and m goods, where $a^i = (a_1^i, a_2^i, \ldots, a_m^i)$ is the initial endowment of trader i and $x^i = (x_1^i, x_2^i, \ldots, x_m^i)$ is the final endowment, given the ruling prices $p = (p_1, p_2, \ldots, p_m)$ and assuming a velocity of one, the amount of money required by i to finance trade is

$$\sum_{j=1}^{m} p_j \max[(x_j^i - a_j^i), 0].$$

Thus the ratio of the value of trade to all endowments is

$$\frac{\sum_{j=1}^{m} p_j \max[(x_j^i - a_j^i), 0]}{\sum_{j=1}^{m} p_i a_j^i}.$$

In an actual economy, the means of production and other capital stock account for much of the assets. Thus (as noted in section 13.3.2) we would expect that this ratio is smaller than in the hypothetical economy involving only exchange. A factor in a modern economy tending to raise the size of this ratio is that most individuals are not traders. They primarily sell their services and buy everything else except the inputs they obtain from their own assets such as houses and automobiles.

The familiar Edgeworth box has been used to illustrate the elementary aspects of exchange. In particular, it is easy to display the basic features of the contract curve of an exchange economy or the core of the associated

market game as well as the competitive equilibrium or efficient price system. We can utilize this diagram to answer some questions concerning an economy which uses money to finance trade.

Although the general mathematical problem that must be posed appears to be of considerable complexity for complete solution, a simple example illustrates the nature of the question and its relevance to economies using money or credit. For ease of exposition, we consider an exchange economy with two types of traders trading in two goods. The utility function of a trader of type i is denoted by a $f_i(x_i, y_i) = \sqrt{x_i y_i}$ and i's endowments by (a_i, b_i), where $a_i + a_j = A$ and $b_i + b_j = A$. The Edgeworth box is a square of size A by A. The Pareto optimal surface is the diagonal O_1O_2, where O_i is the point of endowment (O, O) for a trader of type i.

We know that for any distribution of resources the exchange economy will have a unique competitive equilibrium point, which can be illustrated by drawing a line from the initial distribution point to the Pareto optimal surface. In figure 13.2, two examples are illustrated. If the initial resource distribution were at T, where a trader of type 1 has (A, O) and of type 2 has (O, A), then the competitive equilibrium is at the point E, with a distribution of $(A/2, A/2)$. If the initial resource distribution were given by S, then the equilibrium distribution is at F. Using the price system given by the slope of TE (SF), we can construct an index to measure the amount of trade relative to total wealth in trading form T to E (S to F), It is given in the Edgeworth box by

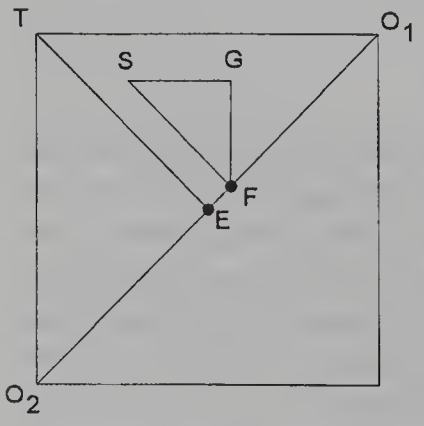


Figure 13.2
Amount of trade needed.

$$q = \sum_{i=1}^{n} \max[p(a^i - x^i), 0]/pa,$$

which is the value of all trade divided by the value of all initial resources at the price system at equilibrium.

The largest that this index could be is 1 in the extremely rare instance that all individuals wish to trade all of the resources they possess. The minimum amount is 0, which could occur if the initial point were to lie on the Pareto optimal surface, in which case there would be no motive for trade, although if the surface were differentiable unique shadow prices could still be defined.

Given the utility functions and total endowments as illustrated in figure 13.2, suppose that the initial endowments of the traders were selected randomly. Could we then calculate the average amount of money or credit required to facilitate trade in the economy. This relatively sloppy concept can be made precise as follows. Suppose that we were to actually set up trade as a playable game and to require that all actual trade that takes place is paid for in a money, "blue chips" or "coin of the realm." We can imagine a game where the resources are selected and announced, and where all individuals first go to the bank, the government, or the referee to obtain whatever the unit of payment is meant to be, and then bid. We can also make the unit of payment the unit of account; thus we may denominate its price as 1. After trade has taken place and all account have been settled, all individuals are required to return the amount of chips they borrowed to the referee. Furthermore, in order to make the game even more specific, we assume that all decisions to buy or sell are processed simultaneously; thus the velocity of the credit unit in the game is limited to be at most 1.

We may imagine that the initial distribution of resources can fall anywhere in the square with equal probabilities for any point. In the example given we can now calculate with relative ease that the relationship between the value of total resources of the economy and the expected amount of trade is $\sqrt{2}/6$.

The Pareto surface divides the box in half hence the endowments can fall with equal probability on either side. Thus we need only consider the average distance of a point in the triangle TO_1O_2 from its base. This is given by

$$2\int_0^{1/\sqrt{2}}\left(\frac{1}{\sqrt{2}}-y\right)ydy\Big/\int_0^{1/\sqrt{2}}dy=1/6.$$

Note that the measure is not SF, which indicates the change in resources, but $SG+GF$, which gives the actual trades. Hence the money required is $\sqrt{2}/6$.

13.3.2 The minimal trust equilibrium

In general, as proved by Debreu (1970), the CEs of an exchange economy are not unique, but are generically finite in number. The question arises, is there any way to select among them?

It is suggested here that there is a natural extra condition we can impose on a multiperiod exchange economy to select among the equilibria. This condition is to minimize the need to extend credit, while maintaining efficiency.

In a K-period exchange economy with m goods and n trader types and with only spot markets, there are two reasons for the use of money and credit. Neither appears explicitly in the general-equilibrium model. The first is a transaction need, if spot transactions are to be paid for in money. The second is a need to finance intertemporal trade when the period-by-period budget constraints do not match.

Qualitatively, these two needs involve considerably different levels of trust. The first concerns extremely short-term financing to take care of the mismatch that occurs until a clearinghouse has cleared all transaction and balanced its books. The second involves the extension of trust between individuals for considerable lengths of time.

In general equilibrium, the transaction need for financing is not present. Here, if we assume that all individuals are willing to credit spot purchases with the income derived from spot sales (i.e., they trust the immediate clearing mechanism), we may ignore the transaction need. This leaves the need to finance intertemporal trade.

Suppose that we have solved the general equilibrium for its set of CEs, and there are s CEs. We now consider the selection of the CE that minimizes credit exposure. This criterion is independent of the choice of the normalization for the price system. It is a pure number, as can be seen below.

Let w_{jt} be the new endowment of good j of individual i at time t, and let x_{jgt} be the final endowment after trade,[13] where the second subscript, g, is

an index indicating a specific equilibrium point $g = 1, 2, \ldots, s$. The value of all goods (or total wealth) if given by

$$W_g = \sum_{i=1}^{n} \sum_{j=1}^{m} \sum_{t=1}^{K} p_{jgt} w_{jt}^{i},$$

where p_{jgt} are the prices at the gth equilibrium.

Let D_{gt} be the debt outstanding required to achieve the equilibrium resource distribution at equilibrium g.

$$D_{gt}^{*} = \max_{t}[D_{g1}, D_{g2}, \ldots]$$

This is the largest credit exposure of the system.

The ratio $R_g = D_{gt}^{*}/W_g$ is pure number, indicating maximum exposure as a percentage of total wealth.

We select $R^{*} = \min_{g}[R_1, R_2, \ldots, R_s]$. This identifies the least credit exposure equilibrium. Because the CEs are generically a finite point set, they will generically have a unique minimum R^{*}.[14]

The mathematical problem of uniqueness conditions for a CE is different from the economic problem, namely, are there additional reasonable economic constraints that can be added to the original problem to yield uniqueness?

An immediate question to consider about the selection proposed here is, can this equilibrium be calculated? The various algorithms for the calculation of CEs are not exhaustive; thus at first glance there does not appear to be any immediate gain in computational possibilities. The extra conditions on optimization should, we can hope, provide more structure that might be used.[15]

Notes

1. More precisely, provided there is at least one money, that is, for m goods, a tree with $(m-1)$ branches or a graph with up to $m(m-1)/2$ edges.

2. But not a necessary condition, as is shown by example in chapter 9.

3. It is likely that loss through wear, war, and accident could be anywhere from around 5 to 15 percent. An interesting discussion in this regard is given in Blake 1869, chapter 8.

4. The assumption of a continuum of traders implies that a single individual cannot influence price. Type symmetry is the assumption that all players of the same type choose the same strategy.

5. It is conjectured that the active type-symmetric noncooperative equilibrium (TSNE) in this example is unique. If there were more gold than just "enough," some of it would be used for jewelry and the price level would rise.

6. If we changed endowments so that, for example, traders of type 1 had only (2, 0) and no gold and traders of type 2 had (0, 2) and all the gold, say, 2 units, a little calculation shows an active loan market with the rate of interest as in (13.20) and consumption in favor of those with gold. Traders of type 1 obtain a utility of $(1/(2+\rho))$ per period, while those of type 2 obtain $((1+\rho)/(2+\rho))$. See Housman, 1983.

7. Or, as the saying on Wall Street goes, "He who dies with the most chips wins."

8. Without loss of generality, we divide the utility function of individual i by μ_i, thus arranging for the $(m+1)$st good to be valued at 1:1 by all.

9. As Keynes (1930b, vol. 2, p. 27) guessed for England.

10. The value of the service is constant every period; hence in a finite-horizon model, the value of the asset is depreciating in a linear manner with time.

11. With a consumable, however, we must distinguish consumption from the stock, which would require a new symbol in the updating equations.

12. This is not quite accurate because mass trade even at one trading post may require records to clear trade that is not directly face to face. Warehouse receipts may be issued and money paid after sale.

13. To save notation, we consider only perishables, but the observation is the same with durables.

14. Alternatively, we might select $R_g = M_g/W_g$ where $M_g = \max D_{gt}$, which amounts to minimizing the maximum amount of credit needed in any period.

15. We might enlarge the problem by introducing an extra agent whose preferences are defined on the size of total loans. This new problem would have a unique Pareto optimal point.

14 Trade and Paper

14.1 The Historical Relationship of Trade to Money and Capital

There is no definitive history of traders, middlemen, and the abandonment of barter. But from the records of ancient Egypt and Babylon, well-defined international trade has existed for over four thousand years, and bars of precious metal were used for payment long before the invention of coinage. We might well ask, What, were the ratios of values of market barter, market exchange for a money or a means of exchange, and administrative or bureaucratic distribution of goods and services at different times and in different empires or countries? Indeed, what are the ratios of exchange to gross national product and to overall national wealth?

Estimates of these items for dates before A.D. 1700 would be more or less for historical curiosity. But since 1700, the modern financial and corporate world has been developing with breakneck speed. In this world, we must have some feeling for the way payments are made, the quantity of means of exchange required, and the bounds on velocity if we are to understand the role of money. In his *Treatise on Money*, Keynes (1930a, 1930b) already stressed the need to distinguish between the velocity of income deposits and business deposits. He also raised basic questions concerning the significance of the relationship of different velocities to price levels. Friedman and Schwartz (1970) discuss the many problems faced in deriving an adequate measure of the money supply. Through the use of simple games, we can bring in financial instruments in an isolated manner, thereby avoiding the many pitfalls in the practical definition and measurement of the quantity of money and its velocity. The approach here attempts to skirt the morass of causal interpretation of the Cambridge equation in the $PQ = MV$ or any other form, except as a by-product of a well-defined formal game.

Tables 14.1–14.9 provide a crude intuitive feeling for the nature of the orders of magnitude involved in transactions, while keeping in mind the many institutional differences in means of payment such as coin, fiat, checks, banknotes and credit cards. (In Volume 2, where stocks and bonds and other financial instruments are considered explicitly, our concerns will turn more to contrasting the totality of all paper claims against real or physical goods and services, rather than merely money of institutionally defined near-monies.)

Table 14.1 indicates the ratio between GNP and internal wholesale and retail trades and manufacturing in the United States over approximately

Table 14.1
GNP and internal trade (billions of $)

Year	GNP (a)	Trade[a] (b)	Ratio (b/a)
1985	3,988.5	5,104.2	1.3
1980	2,631.7	3,927.0	1.5
1975	1,549.2	2,186.8	1.4
1970	992.7	1,289.4	1.3
1965	684.9	963.4	1.4
1960	503.7	729.0	1.4
1955	398.0	620.3	1.6
1950	284.8	463.2	1.6
1945	211.9	286.2	1.4
1940	99.7	145.6	1.5

a. The volume of internal trade was approximated as the sum of manufacturing, wholesale and retail sales.
Source: *Business Statistics.*

fifty years. This gives a crude low estimate of the volume of production and exchange. Including transportation, mining, construction, and agriculture would add around 25 percent. Financial transactions and second-hand exchanges of ownership are not included. We note a ratio of around 1.3 to 1.6. It might be as high as between 2 to 3. Table 14.2 gives the three major measures of money supply, while tables 14.3 and 14.4 indicate the ratio of GNP and trade to the different money supplies. Tables 14.5 and 14.6 provide some older estimates of velocity for the United States and the United Kingdom. Table 14.7 indicates the results of a recent survey of the nature of the means of payment utilized by U.S. families. Table 14.8 provides some historical data on means of payment in England for both firms and individuals. Table 14.9 provides an international comparison of the ratio of currency to GNP for eleven developed countries for the year 1983. It also shows the per capita amount of currency (in U.S. dollars) held in the various countries.

The question of how much money a nation needs has been considered by economists for well over 200 years. Marshall (1923, p. 47), in a footnote observes:

> Petty thought that the money "sufficient for" the nation is "so much as will pay half a year's rent for all the lands of England and a quarter's rent of the Houseing, for a week's expense of all the people, and about a quarter of the value of all

Table 14.2
Various money stocks (billions of $)

Year	M1	M2	M3
1985	582.0	2,445.0	3,076.0
1980	415.0	1,631.0	1,989.0
1975	291.0	1,023.0	1,172.0
1970	220.0	424.0	656.0
1965	163.8	285.9	442.1
1960	141.2	210.7	304.0
1955	134.4	183.7	240.5
1950	114.1	150.8	183.8
1945	99.2	126.6	147.8
1940	39.7	55.2	70.0

Sources: *Statistical Abstract of the United States* (1987) and *Historical Statistics of the United States, Colonial Times to 1970.*

Table 14.3
Velocity, GNP/M

Year	GNP/M1	GNP/M2	GNP/M3
1985	6.9	1.6	1.3
1980	6.3	1.6	1.3
1975	5.3	1.5	1.3
1970	4.5	2.3	1.5
1965	4.2	2.4	1.5
1960	3.6	2.4	1.7
1955	3.0	2.2	1.7
1950	2.5	1.9	1.5
1945	2.1	1.7	1.4
1940	2.5	1.8	1.4

Sources: *Statistical Abstract of the United States* (1987) and *Historical Statistics of the United States, Colonial Times to 1970.*

Table 14.4
Velocity, trade/M

Year	Trade/M1	Trade/M2	Trade/M3
1985	8.8	2.1	1.7
1980	9.5	2.4	2.0
1975	7.5	2.1	1.9
1970	5.9	3.0	2.0
1965	5.9	3.4	2.2
1960	5.2	3.5	2.4
1955	4.6	3.4	2.6
1950	4.1	3.1	2.5
1945	2.9	2.3	1.9
1940	3.7	2.6	2.1

Sources: *Statistical Abstract of the United States* (1987) and *Historical Statistics of the United States, Colonial Times to 1970.*

Table 14.5
Money stock (millions of $) in the United States, 1870–1930 and velocity

Year	M2	Velocity
1870	1,390	4.12
1880	2,042	4.97
1890	3,908	2.93
1900	6,378	2.53
1910	13,120	2.20
1920	34,708	2.20
1930	45,303	1.70

Source: Friedman and Schwartz (1970).

exported commodities" (Quantulumcunque, Queries 23 and 25: see also his *Political Arithmetic*, ch. IX and *Verbum Sapienti*, ch. VI). Locke estimated that "one-fiftieth of wages and one-fourth of the landowner's income and one-twentieth part of the broker's yearly returns in ready money will be enough to drive the trade of any country". Cantillon (A.D. 1755) after a long and subtle study, concludes that the value needed is a ninth of the total produce of the country, or, what he takes to be the same thing, a third of the rent of the land. Adam Smith has more of the skepticism of the modern age and says: "it is impossible to determine the proportion," though "is has been computed by different authors at a fifth, at a tenth, at a twentieth and at a thirtieth part of the whole value of the annual produce."

Table 14.6
Money stock (millions of £) in Great Britain, 1750–1875 and velocity

Year	M2	Velocity
1750	40.0	2.5
1775	63.0	2.1
1800	165.0	1.2
1811	215.0	1.2
1821	151.0	1.6
1831	166.0	1.8
1844	220.0	1.8
1855	374.7	1.3
1875	578.6	1.5

Table 14.7
Use of cash and transaction accounts by U.S. families, 1985

Means	% of families using this method	% of total expenditures
Cash	100	37
Check[a]	83	57
Credit card	52	6
Total		100

a. Checks include personal checks, bank checks, automatic payments, and electronic payments. *U.S.A.—1984 Survey* data.
Source: *Federal Reserve Bulletin* (February 1986, vol. 72, pp. 87–108).

Table 14.8
Money stock and means of payment (% of total)

Period	Specie	Bank notes	Deposits	Bills of exchange	Total
1688–1689	50.8	10.0		40.0	100.0
1750	37.5	12.5	12.5	37.5	100.0
1775	25.4	15.9	11.1	47.6	100.0
1800–1801	12.1	15.2	3.05	69.7	100.0
1811	7.0	20.9	7.0	65.1	100.0
1821	11.9	21.2	16.6	50.3	100.0
1831	18.1	17.4	24.1	40.4	100.0
1844	16.4	12.9	36.6	34.1	100.0
1855	13.3	7.1	38.7	40.9	100.0
1875	18.1	5.6	70.7	5.6	100.0

Source: Cameron et al. (1967, p. 42).

Table 14.9
International comparison of currency/GNP ratios, 1984

	% of GNP in cash or currency	Money holdings per capita ($ U.S.)	% of GNP covered by money supply (M1)
Belgium	9.9	750.0	180.7
Canada	3.2	399.0	46.8
France	5.1	416.8	75.1
Germany	6.0	577.0	53.0
Italy	6.4	392.0	95.9
Japan	7.6	740.9	150.0
Netherlands	6.5	602.0	68.4
Sweden	6.0	677.0	60.0
Switzerland	10.0	1,545.0	117.0
U.K.	4	308	33
U.S.	4.2	595.5	75.4
Average	6.6		

Source: Survey of payment systems prepared by the Bank for International Settlements, 1985.

14.2 A Digression on the Gold Standard

Although this volume does not deal with international trade, the history of money from its inception appears to be intimately related to international trade. In particular, the role and concept of the gold standard as an automatic control on international trade is about as pure an example of formal game design in theory as one can find. The reasons why political reality and practice differ from economic theory merit explanation. Eichengreen (1985, pp. 3–4) specifies the three central features of a gold standard regime: (1) interconvertibility between every domestic money and gold at an official price; (2) freedom for private citizens to import and export gold; and (3) rules linking the quantity of a country's money to its gold stock. But although England had more or less adopted a gold standard in the eighteenth century, the era of the gold standard in Europe was from around 1880 to 1914, with the United States restoring gold convertibility in 1879.

The attraction of the gold standard was that it could provide automatic discipline and smoothness in price changes and exchange rates. In essence, if applied strictly, the gold standard is a device for limiting the strategic

control of the nation-state over its monetary policy and substituting instead the competitive (or oligopolistic) forces of gold mining for international law. In his *Essays in Persuasion*, Keynes (1925) used the phrase "the rules of the game," and this is a formally apt description.

The debate on both the theory and interpretation of the history of the gold standard has been excellently covered in the edited volume of Eichengreen (1985). The Austrian perspective indicating that proponents for a gold standard still exist has been given in an edited volume by Rockwell (1985). McClosky and Zecker (1976) offer an interpretation of the workings of the standard. In particular, they note the integration of the commodity markets of the British and American economies. But underlying all of the economics—and the key point of the "Austrian" position—is the limitation of political control over the monetary system (see, for example, the concluding remarks of Barro, 1979).

Because this volume deals only with the monetary and financial system of a single closed economy, we cannot construct a model of the gold standard without exceeding its scope. Nevertheless a few observations are in order. The fundamental theme developed here is that while money and financial instruments arise naturally to satisfy the needs of trade and production, they offer a natural control system for the state. Setting aside questions of private oligopolistic power (such as the actual gold-mining industry) a reasonable model of the single economy is as a game with a single atomic or large artificial player, "the government," and with a continuum or mass of small players. This formulation makes it imperative to specify the goals of the government and its agents prior to commencing the analysis (for example, what is it that a central bank tries to maximize, a subject considered in Volume 2).

When n countries and currencies are to be considered then at least the model must be a game of an n atomic players, with n types of nonatomic players representing the mass of economic agents in each country. But this game in its very essence raises questions about the euphemisms used in the discussions on economic efficiency and free trade. The political economy of the nation-state, especially in international affairs is dominated by political considerations. What rules of this game are all countries willing to accept in international trade? How much sovereignty over their control of their monetary and financial instruments are nation-states willing to surrender?

Table 14.10
Guesstimate on world product

Period	World GDP		World population		World per capita GDP ($)[b]
	Amount (billions of $)[a]	Growth rate (%)[b]	Number (millions)	Growth rate (%)[b]	
1980	7,184.27	3.462	4,432	1.833	1,621
1970	5,111.57	5.074	3,696	1.983	1,383
1960	1,747.50	5.954	3,037	1.965	699
1900	681.64	2.099	1,625	0.865	381
1850	229.68	2.001	1,200	0.608	191
1800	155.29	1.388	900	0.577	128
1700	60.94	0.640	610	0.390	100
1600	47.80	0.243	545	0.113	88
1500	32.72	0.380	425	0.249	77
1000	10.64?	0.225	265	0.095	77?
0	3.09?	0.124	170	0.044	18?
1000 B.C.	0.82?	0.132	50	0.122	16?
2000 B.C.	0.40?	0.072	27	0.062	15?
3000 B.C.	0.19?	0.076	14	0.066	13?

a. In 1970 real U.S. dollars.
b. Average percentage growth rate between periods.
Sources: For population data, McEvedy and Jones, (pp. 343–351); for GDP, see Appendix.
? = wild extrapolation; really hard to compose before 1600 because of basic differences in social organization, products, and monetization.

The appeal of the gold standard, to libertarian and peasant alike, is that faith was to be placed in gold not politicians or the state. As primitive a store of value as gold may be, in around five thousand years of history the evidence appears to favor trust in gold over politicians, kings, or international law.

14.2.1 The growth of paper

Tables 14.10 and 14.11 are more or less conjectural and laced with conceptual difficulties.[1] Nevertheless, they are presented, if only to stress the questions concerning fundamental changes in the role and importance of trade and the nature of the means of exchange. Table 14.10 offers estimates of the world population since 3000 B.C., together with some "guesstimates" and conjectures concerning average per capita income. Table 14.11 presents guesstimates on trade.[2]

Table 14.11
Guesstimate on world trade

Period	World trade		World GDP (billions of $)[a]	Trade/GDP (ratio)
	Amount (billions of $)[a]	Growth rate (%)[b]		
1980	10,417.200		7,184.27	1.45
1970	7,411.800	3.462	5,111.57	1.45
1960	4,518.100	5.074	3,115.96	1.45
1950	2,533.900	5.954	1,747.50	1.45
1900	556.800	3.077	618.64	0.90
1850	183.740	2.242	229.68	0.80
1800	69.170	1.973	115.29	0.60
1700	30.470	0.823	60.94	0.50
1600	19.120	0.467	47.80	0.40
1500	13.090	0.380	32.72	0.40
1000	4.260?	0.225	10.64	0.40
0	0.928?	0.152	3.09?	0.30?
1000 B.C.	0.165?	0.173	0.82?	0.20?
2000 B.C.	0.040?	0.141	0.40?	0.10?
3000 B.C.	0.019?	0.076	0.19?	0.10?

a. In 1970 real U.S. dollars
b. Average percentage growth rate between periods.
Sources: See table 14.12.

It appears that at around the start of the Egyptian dynasties the prime means of distribution of goods were administrative or by fiat and by barter. Around 2000 B.C., there is some evidence that gold by weight began to appear as a means of exchange; around 650 B.C., coinage appears to supplement and possibly supplant bullion in international trade. The bill of exchange appears for certain by the twelfth century (Dean, 1884, p. 6; Braudel, 1967, pp. 357–372; Marks, 1924, pp. 1–21), but there is an open question as to whether an instrument similar to the bill of exchange might have existed far earlier. "Flying money" of the Chinese administration in the Tang dynasty played a role similar to the bill of exchange. A form of fiat money appears briefly in China around A.D. 100 (Fitzgerald, 1954), then from the ninth century on (Braudel, 1967) until A.D. 1100, after which it is banned for many years. The banknote appears in Sweden in 1661 (Kindleberger, 1984, p. 50), then in England with the issue of the Bank of England and in France in 1788 with the issue of assignats

approved by Louis XVI and backed by land based on John Law's earlier precepts. The concept of the use of the check appears to have been established by the late eighteenth century as is evinced by a quotation of Petiah Webster in 1786: "The advantage would still be greater if instead of bank bills the owner would take a bank credit and draw checks on the bank whenever he needed his money; this would enable him to pay any sum exactly without the trouble of making change..." (quoted by Hammond, 1957,).[3] Hendrickson (1972) notes that the charge plate system was used by Filene's in 1927. Credit cards make their appearance in the United States at the beginning of the twentieth century. "By 1914 large department stores and gas chains were issuing cards. It was not until the 1950's that third party cards began first as travel and entertainment cards and then as bank cards" (Garcia, 1980, p. 327). The pioneer in the universal credit card was Diners Club in 1950.

A full appreciation of the growth and change in the nature of the means of payment requires also an investigation of the growth of near-monies such as government bills and notes. The last twenty years has seen an explosion in the production of electronic near monies.

Appendix

The method and justification for these estimates given in tables 14.12 and 14.13 are sketched below.

I. World per capita GDP—literature survey

A. World per capita GDP, 1950–1980

Year	Index number	$ 1970 real
1980	109	1,621
1970	93	1,383
1960	69	1,026
1950	47	699

Data source:

a. 1970 real per capita GDP: Kravis, Heston, and Summers (1978, p. 240).

b. Index number: United Nations, *Statistical Yearbook* 1982, 1966.

B. England per capita GDP, 1900–1980

Year	GDP (1980 billion pounds)	Population (million)	Per capita GDP	
			1980 pound	1970 U.S.$
1980	199.3	56.30	3,540	3,551
1970	166.4	55.42	3,003	3,012
1960	124.2	52.37	2,372	2,379
1950	95.5	50.57	1,888	1,894
1930	64.6	45.87	1,408	1,412
1910	58.3	44.92	1,298	1,302
1900	51.5	41.16	1,251	1,269

Data source: Economic statistics for 1900–1983 in the United Kingdom, United States, France, Germany, Italy, and Japan were compiled by Thelma Liesner. Conversion rate into 1970 U.S.$ is based on the adjustment factor given by Kravis, Heston, and Summers (1978) and used by Crafts (1983).

C. Europe per capita GDP, 1700–1910 (in 1970 U.S. $)

	1700	1760	1800	1820	1830	1840	1850	1860	1870	1880	1890	1900	1910
Great Britain	333	399	427		478	567	660	804	904	979	1,130	1,269	1,302
Belgium				358	383	402	534	637	738	832	932	1,013	1,110
Denmark					343	392	489	497	563	617	708	850	1,050
Germany							432	481	579	602	729	868	958
France								474	567	602	668	784	883
Sweden								292	351	419	469	597	763
Norway								420	441	486	548	605	706
Finland								300	390	407	458	529	561
Italy								451	467	466	466	502	548
Russia								236	252	253	279	342	398

Data source: Crafts (1983), gross national product in Europe, 1870–1910.

D. Roman Empire per capita GDP

Data source: Goldsmith (1984).

Note: Per capita GDP = 31 grams or 1 troy once of gold (conversion to 1970 real $: $36.4 at 1970 U.S. price of gold per troy ounce).

E. Ancient periods (B.C.) per capita GDP

Data source: Goldsmith (1984, p. 274) implied that the growth rate of per capita GDP before A.D. 14 was below 0.01 percent.

II. Estimation of per capita GDP of the world

Period	Leading country per capita GDP (1970 $ real)		Conversion factor	World per capita GDP (1970 $ real)	
1980	3,551	1.660%	0.456	1,621	1.601%
1970	3,012	2.387	0.459	1,383	3.031
1960	2,379	2.306	0.431	1,026	3.912
1950	1,894	0.804	0.369	699	1.223
1900	1,269	1.316	0.300	381	1.385
1850	660	0.875	0.290	191	0.806
1800	427	0.249	0.300	128	0.249
1700	333	0.130	0.300	100	0.130
1600	292	0.130	0.300	88	0.130
1500	257	0.130	0.300	77	0.130
1000	134	0.130	0.300	40	0.079
0	36	0.130	0.500	18	0.010
1000 B.C.	33	0.010	0.500	16	0.010
2000 B.C.	30	0.010	0.500	15	0.010
3000 B.C.	27	0.010	0.500	13	0.01

Notes: % is average growth rate between periods.

Method:

a.

Period	Leading country
1700–1980	England
0	Rome

b. Period 0–1500: interpolation.

c. 1000–3000 B.C.: I guess 0.01 percent growth rate of per capita GDP.

d. Conversion factor

1950–1980: actual data (refer to the literature survey A. and B.).

1900: ratio between England and Russia (refer to the literature survey C).

Before 1900 is a guesstimate.

III. Trade estimation

1. Reference data sources

A. Recent periods

Period	Country	Trade/GNP
1980	U.S.	1.5
1970	U.S.	1.3
1960	U.S.	1.4
1950	U.S.	1.6
Average		1.45

B. India in 1950s

Per capita GDP: $152–223

Monetization ratio: proportion of cash farm expenditure close to 60 percent of total expenditure.

Source: Ghush (1964, pp. 29–30).

C. Naples in 1750

Monetary stocks in motion settled half the transactions of the Kingdom of Naples in 1751.

Source: Braudel (1967, p. 356).

D. Roman Empire

Goldsmith implied that monetization of the Roman economy amounted to 50 percent, but, he mentioned that the ratio fell to 30 percent if he followed Hopkins's estimate in his footnote (Goldsmith, 1984, p. 275). Thus we may regard the trade/GNP ratio as 30 percent.

E. Greece

Early in the sixth century B.C. dependent farmers in Attica had to yield one-sixth of their crops to their creditors. This ratio was the maximum available from an Attica farm provided that the surplus were totally subject to mobilization.

Source: Starr (1977, p. 105).

Before 1000 B.C.

The Near Eastern economies were dominated by large palace or temple complexes, which owned the greater part of the arable land and virtually monopolized whatever could be called industrial production as well as foreign trade, and which organized the economic military and political life of the society through a single complicated, bureaucratic, record-keeping operation.

Source: Finley (1985, p. 28).

Notes

1. Tables 14.12 and 14.13 were prepared with the able research assistance of Guytag Oh.
2. Not represented in the table 14.11 but needed to account for the upward shift in the volume of trade relative to income and wealth is the growth of the corporation and the vast change in production and distribution since 1700.
3. I have been unable to determine with sufficient accuracy when savings accounts were first differentiated from deposit accounts. It appears to be in the nineteenth century.

15 Liquidity and the Velocity of Money

15.1 Liquidity: Bridging Finance or Speculation?

Before leaving one-period models of exchange, we need to clarify the important role of the velocity of money and how it relates to liquidity and the quantity of money. Liquidity is a concept basically associated with incomplete markets and transaction costs. But even in a world with complete markets, commodity money and spot trade, it can be usefully defined. When payment in cash is required, liquidity conditions can be defined. In such a setting, enough money and sufficient liquidity are the same.

In a world without uncertainty, but one that uses a means of payment, individuals can calculate their respective cash needs at each point in time. They will be completely liquid if they always have enough cash on hand to meet their cash-flow constraints. Without uncertainty, there is no need to hold either speculative or precautionary sources of cash as there is nothing unforseen.

Let us turn now to markets with uncertainty.

15.1.1 The speculative motive for liquidity

Wicksell (1935, p. 71) noted two basic reasons for holding cash: for transactions and for unforeseen needs. Keynes (1936, chapter 13) distinguished transactions as well as precautionary and speculative motives for keeping liquid. He also noted the importance of the difference between the cash needs of consumers and those of firms. Three categories of agents may be called for: consumers, traders, and producers. In modern economies, especially with the growth of financial intermediaries and markets for firms, the liquidity needs of the financial sector are different from those of manufacturing.

The speculative need for money is an extension of the transaction need. This can be seen easily by modifying the model presented in section 13.2.1 above, where instead of the traders each knowing the resources available for trade prior to dividing their gold into currency and jewelry, each only is informed of a probability of the different amount of goods that might be available. One can interpret the decision to split gold into jewelry and currency as equivalent to splitting the use of fiat money into currency and savings, where currency earns no interest but is available as a *masse de manoeuvre* to take immediate advantage of any buying opportunity, while savings earn interest but are not available immediately for purchasing.

Figure 15.1 shows the extensive form of a one-period game, where the individuals must select their levels of liquidity prior to knowing what opportunities will be available to them.

Although, given a more or less constant velocity for currency, it is reasonable to suspect that the transaction need for currency varies in a linear manner with the volume of trade, the shape of the speculative demand for money is not so evident. One can produce plausible examples where the speculative demand for liquidity increases as the square root of the expected volume of opportunity, or other functional forms depending specifically on the interest rates, nature of uncertainty, other financing opportunities, and ad hoc details of the transaction technology of a specific society.

15.1.2 The precautionary motive for liquidity

In a world without bankruptcy or insolvency, it is difficult to know how to operationally describe the precautionary motive. Precautionary against what? At this level of abstraction without bankruptcy, precautionary and speculative motives must be the same. They are manifested in the need for liquidity to maximize in situations involving incomplete markets.

Keynes's concern for precautionary motives for the holding of money and near-liquid assets is reflected in even casual reading of business literature such as the *Wall Street Journal*. "Downside analysis" and survival play an important role in business decision making.

Can or should failure, insolvency, and bankruptcy be considered as representing a continuous change in the valuation of possible states or do they represent a fundamental discontinuity? If there is a discontinuity in valuation on the downside, but none on the upside, then there is an operational distinction between the speculative and precautionary motive even in a highly stripped down or abstracted representation of reality. In accounting analysis, there is a distinction made between the accounting for an ongoing business and the accounting for liquidation. This would seem to imply a discontinuity caused possibly by setup and transaction costs. Another possible distinction involves the nature of "upside" and "downside" subjective probabilities in the assessment of risk and uncertainty. Psychologically, winning and losing may not be treated symmetrically.

Keynes (1936, p. 171) suggests that the precautionary motive (and the transaction motive) are not particularly sensitive to the interest rate. This

may have been empirically true of England in the 1930s, but an attempt to produce a formal strategic game in which a precautionary motive can be identified requires incomplete markets, randomness, the presence of outstanding commitments, and the selection of the level of liquidity before the disclosure of the random outcome. If in the multistage game there is an interest rate present, the selection of the level of liquidity called for by the precautionary and speculative motives will depend parametrically on the interest rate and the nature of the randomness. Furthermore, the precautionary and speculative motives will in general be confounded. Elementary consideration of portfolio analysis calls for the treatment of these two motives together (Markowitz, 1959; Tobin, 1956).

15.1.3 Bridging or long-term finance

One view of commercial banking is that its prime function is to provide the financing of riskless production and trade. The time and coordination features of exchange involve the hiatus between when A writes a check on A's account and B finally gets it credited to B's. This may take anywhere from a few seconds to several weeks. An importer in New York buys shoes from Italy. The time between the various rounds of paper transactions and the delivery of the shoes may be out of phase by several weeks.[1]

With production, new timing problems appear. The gap between outlays and income may be considerable (the central theme of Böhm-Bawerk 1923). A productive process may take several months or years before the cash flow turns positive. All of these activities require the appropriate financing to bridge the time between receipts and payments. But in a world of uncertainty there is a considerable qualitative difference between short-term transactions financing and long-term production or capital investment financing (arguably, not a function of commercial banking).

Section 13.2.1 defined a game in which there was a physical loss of the use of gold. By specifying an exogenously given interest rate of $1 + \rho = 1/\beta$, we can reinterpret this game in terms of the use of fiat money kept in circulation to pay for transactions, rather than invested at a positive rate of interest. With a positive rate of interest and a known need for transactions, no traders will keep extra money in univested cash because they can always improve by investing. When there are incomplete markets, matters change, as in shown in the example below.

15.1.4 The *masse de manoeuvre*

In contrast with these relatively prosaic considerations, if uncertainty is present, an individual may hold cash balances to take advantage of random mispricing, the spotting of bargains, the opportunities provided by technological or political change, the opportunities created by the insolvency, lack of credit, or even bankruptcy of others. Furthermore, reserves of relatively high liquidity may be held for precautionary reasons, to meet unforeseen payments.

Any general theory of money must involve uncertainty. The presence of exogenous uncertainty implies incomplete markets. Common sense, an elementary knowledge of business, and Keynes's description (1936) of the speculative and precautionary reasons to hold cash all point to the critical role of uncertainty. Yet to a great extent the roles of transaction cost and uncertainty are worth separating in a first analysis, in order to be able to clearly distinguish the various properties of money.

A single-period partial equilibrium game with two types of trader and an exogenously given rate of interest and valuation of the future worth of fiat money serves to illustrate the meaning of and need for liquidity. The extensive form with disclosure after the random variable is shown in figure 15.1. We assume that with a probability of .5 the endowments of

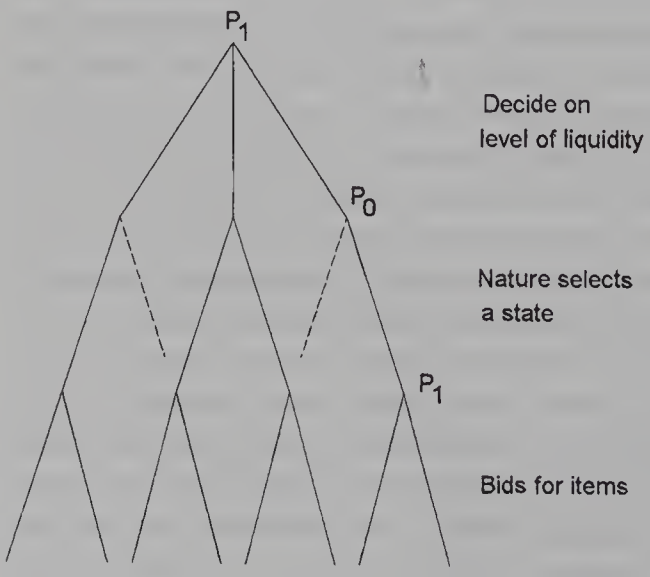


Figure 15.1
Choice of liquidity.

traders of types 1 and 2 are, respectively, $(200, 0, A)$ and $(0, 200, A)$ in state 1, and $(20, 0, A)$ and $(0, 20, A)$ in state 2.

The utility functions are of the form

$$u_i = .5\sqrt{x_1^i y_1^i} + .5\sqrt{x_2^i y_2^i} + .5\sum_{s=1}^{2}[(z_s^i - b_{s1}^i + p_{s1}q_{s1}^i) + (1+\rho)(A - z_s^i)],$$

$$i = 1, 2 \tag{15.1}$$

for complete information, where the last terms of the utility function imply that net revenues are evaluated linearly. This can be interpreted as assigning a worth to fiat money in a next period. The penultimate term indicates that, rather than being used for transactions, the money could earn an interest rate during the period.

This simple model can be easily solved in three ways: (1) the competitive equilibrium;[2] (2) maximization where an individual i names a percentage of the good i will sell and the amount of money i is willing to bid before knowledge of the random variable; this is the "no-information" scenario; and (3) maximization where the amount of gold for money is selected after knowledge of the outcome of the random variable (we call this "full information"). Suppose $\rho = 10$ percent and $A = 100$, with some calculation we obtain

1. The CE: payoff = 165.
2. No-information NE: $b = 23.256$ and $q = .4762$; and payoff = 158.837.
3. Full-information NE: $b_1 = 45.403$, $b_2 = 4.5403$, and $q = .4767$; and payoff = 162.44.

The intuitively satisfactory results are obtained. The CE has the highest payoff obtained by assuming no interest is forgone in trading. The full-information NE obtains less than the CE where the financing requires that money be locked up, but the amount can be precisely tailored to the state the individual is in. In the low-information trade, the appropriate amount of money needed for trade in all states cannot be selected in advance. The no-information model is well defined, but less realistic than the low-information model. Traders usually (though not always) know what they have to sell before deciding how much to offer. The basic reason for holding cash turns out to be equivalent to that of *masse de*

manoeuvre at this level of abstraction. The marginal value of staying liquid must equal the opportunity cost of using the fiat to make purchases.

15.1.5 The rate of interest

In chapter 9, the rate of interest appears as a capacity constraint. In chapter 11, a rate of interest emerges in a one-period model as a banking loss reserve charge, that is, a way in which the bank recoups its bankruptcy losses. In section 11.4.6, a rate of interest on outside money emerges as a tax that balances the books. In section 13.2, the rate of interest appears as a shadow price of a unit of the durable gold at time t in terms of units of the durable at $t+1$; the rate of interest is linked to a "natural" time discount in the utility functions of the individuals. But the money rate of interest as a control variable has not yet appeared because the models discussed so far have not been sufficiently complex.

A more realistic model of the economy is open-ended in time and reflects both the asset structure of economic goods and the presence of overlapping generations. In such a model, the economic system is underdetermined. Either the quantity of money or a money rate of interest must be supplied exogenously. In a modern state these are governmental control variables. Separating out problems as we have before, we proceed from the single-period models of the economy with commodity and fiat money, to multiperiod finite economies, and then to economies with infinite horizons and overlapping generations.

In various growth models or models with time discount, the rate of interest assumes "natural economic" meanings. But a government, by setting an "unnatural money rate of interest," may be able to change the direction of growth and hence the real rate of interest.

Money, politics, and war are far more intimately related than many treatises on microeconomics indicate. Causality is intertwined. The magic real or natural rate of interest is a figment of the imaginations of those who confuse the preinstitutional structure of a von Neumann or other linear growth model with a society having variable population growth, a diverse preference structure and a complex production technology. On top of these are politics and public goods; even with the slightest governmental structure, these do not fit comfortably into the host of counterfactual assumptions that enable the economic theorist to find a one-dimensional measure for the dynamics of a multivariate process.

The various rates of interest in a modern society are the outcome of a complex interaction among government manipulation of control variables, the expectations of businessmen and bureaucrats on noneconomic factors such as elections, wars, and social upheaval, and the underlying technology of production. Before the appropriate questions can even be formulated about the various rates of interest and their relationship to each other, we must proceed from the one-period models of exchange to multiperiod models of exchange and production, with both the role of government and a concrete representation of expectations built in. The approach here is to try to identify the minimum level of complexity at which various aspects of the rates of interest appear. Until government is introduced explicitly, the control feature of interest rates cannot be fully modeled, and until uncertainty and an age structure of assets are considered, the term rates cannot be explained.

15.2 Strategic Control of Velocity and Hoarding

A criticism that can be leveled against models of exchange with or without production, where the assumptions made require all actions to take place simultaneously, is that the velocity of money is restricted to be one (or less) each period. Up to a point all economic agents have some control over the speed of payment. This varies from next to no control of the worker living in a nineteenth-century company-owned town, to the control of the king short of cash, executing his bankers and confiscating their estates.

The classical discussion of velocity is given by Fisher (1931, p. 79), who lists the following factors:

1. Habits of the individuals
a. As to thrift and hoarding,
b. As to book credit,
c. As to the use of checks.
2. Systems of payments in the community
a. As to frequency of receipts and disbursements,
b. As to regularity of receipts and disbursements,
c. As to correspondence between times and amounts of receipts and disbursements.

3. General causes

a. Density of population,

b. Rapidity of transportation.

These conditions are an intermix of the state of society, technology, custom, habit, and conscious optimization. The task of formal game models is to attempt to isolate what can be considered as susceptible to individual or governmental control.

15.2.1 Variations on the theme of liquidity

An asset is completely liquid if in a market economy with markets that are thick enough a single small individual i does not influence price; i can trade i's asset directly for all goods and instruments offered for sale with essentially no loss in time or resources. While even this definition is laden with various imprecision, for most everyday purposes these do not matter. Having discussed both using money as an instant means of payment and keeping it on hand as a *masse de manoeuvre*, we now consider the influence of velocity on liquidity.

Some simple models of exchange are presented to illustrate various conceptual problems in the microeconomic modeling of the velocity of money and the cost of liquidity. The celebrated equation linking the amount of money, M, its velocity, V, the price level, P, and quantity of goods traded, Q, has the reassuring specificity of an equation in an elementary physics textbook:

$$MV = PQ.$$

The terms can be rearranged and modified, but a microeconomic rather than macroeconomic approach is adopted here. No attempt is made at spurious realism. Well-defined, but clearly overly simple models are considered in order to try to isolate and examine the complex of factors that make up the various aspects of velocity and liquidity.

There is a basic distinction between two types of commodity money, which can even be made for the one-period model, namely, between money as a consumer durable and money as a storable consumable. In the one-period model, the distinction is between an item that yields value at a single consumption point and one that yields value continuously over an interval.

If there is no time discount in the single-period game, no value is lost in using a storable consumable money in exchange for almost all of the period and then consuming it at the end of the interval. In contrast in the same economy a consumer durable money should yield its consumption value continuously hence the time it is used in trade represents a consumption loss.

We consider and contrast two economies, one with a consumer durable money (say gold) and one with a storable consumable (say cans of sardines or bars of salt).

MODEL 1: *A k-trading session exchange with a storable consumable money* Suppose each individual i has initial endowments of $m+1$ goods of $a_0^i, a_1^i, \ldots, a_m^i$, where a_0^i is the initial amount of money. Each individual can trade k times in the single period. Let $\phi(x_0^i, x_1^i, x_2^i, \ldots, x_m^i) = \phi(\sum_{t=1}^k x_{0t}^i, \sum_{t=1}^k x_{1t}^i, \ldots, \sum_{t=1}^k x_{mt}^i)$ be the value of final consumption to individual i at the end of the period. Without loss of generality because all goods are storable consumables, we may let the consumption take place at the end of the period.

Suppose that the markets meet for k trading sessions in a single period. Then a low-information[3] strategy of an individual i is

$$(b_{11}^i, q_{11}^i, \ldots, b_{mk}^i, q_{mk}^i), \quad \text{where } b_{jt}^i \geq 0,$$

$$\sum_{j=1}^m b_{jt}^i = 1, \quad q_{jt}^i \geq 0, \quad \sum_{j=1}^m q_{jt}^i = 1.$$

Let the amount of money held by i at the end of a trading session be

$$a_{01}^i = a_0^i \quad \text{and} \quad a_{0t}^i = a_{0t-1}^i - \sum_{j=1}^m b_{jt}^i a_{0t-1}^i + \sum_{j=1}^m p_{jt} q_{jt}^i a_{jt-1}^i.$$

The amount of good j is

$$a_{jt}^i = a_{jt-1}^i - q_{jt}^i a_{jt-1}^i + b_{jt}^i a_{0t-1}^i / p_{jt}^i,$$

where $a_{jt}^i =$ the amount of j held by i at the start of t, $j = 0, \ldots, m$, $t = 1, \ldots, k$.

Each individual i bids a percentage of i's amount of money and offers a percentage of i's holdings of each good each session. At the end of trading session k, all consume all goods, hence $x_{jk}^i = a_{jk+1}^i$.

Suppose that all the money is bid each time. If the number of trading sessions k is sufficiently large, then any amount of the commodity money will be sufficient for trade. Consider any CE, suppose that the CE prices are p_j, $j = 1, \ldots, m$ (where $p_0 = 1$) with one round of trade the transaction needs are $\sum_{i=1}^{n} \sum_{j=1}^{m} p_j \max[(x_j^i - a_{j1}^i), 0] = T$. If we wish to use an amount of money M, where $M < T$, there will be two integers $k - 1$ and k such that $k - 1 \leq T/M \leq k$. If we select the number of trading sessions k to be large enough, any amount M of money will be enough.

The distinction between the game with many sessions of trade and the games with credit described in chapters 9, 10, and 11 is that, with credit, rules or laws are required to police the credit conditions, whereas here all transactions are *value for value*. The many trading sessions (although in reality a nuisance and a cost) enable efficient trade to take place *without trust*. The trade is efficient because there is no consumption loss in delaying all consumption until after the end of trade.

MODEL 2: *A k-session trading game with a consumer durable money*
The distinction that we now have to make is fundamentally empirical. How do we distinguish storable consumables from durables? Fairly obviously there is a continuum of distinction, but at the extremes, a perfect storable consumable involves no maintenance and has point consumption. A perfect consumer durable yields value or utility to its owner continuously and requires no maintenance. Model 1 had everything as a storable consumable. As a good approximation, the money in an experimental game would be chocolate bars. In this model everything is a consumer durable. Thus all trade (if it involves a finite amount of time) must involve a utility loss.

In this formulation, trade involves a utility loss to all parties. The new radio wrapped in its box is not available to its seller and is not available to its buyer until bought and set up for use. We could call the time value loss of the alternative use of money tied up in trade a "float loss." But that expression is more specifically used to refer to money in the bank clearing system. The loss here (which includes the float) is the overall transaction loss of any alternative use of money while it is tied up in trade, although if we are accounting for loss of alternative use for the durable money we should also account for the loss of use of every other durable that yields continuous services.

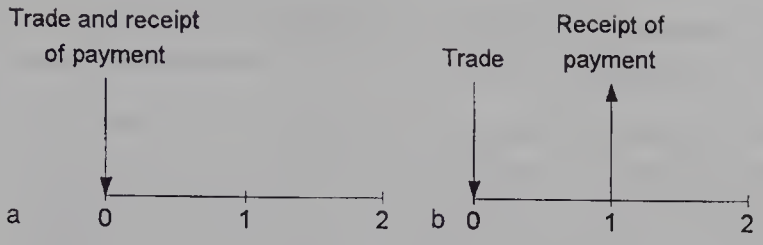


Figure 15.2
Point trade with and without lags.

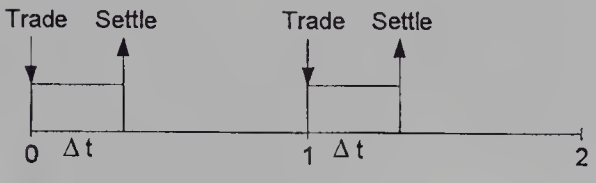


Figure 15.3
Trade with a finite clearing time.

Figure 15.2a shows the magic timeless world of perfect exchange. All exchanges, including payments and receipts, take place at $t = 0$. There is no loss of the use of any durable. Figure 15.2b shows a nonsymmetric world where all buyers get immediate delivery of their purchases, but everyone manages to lag payments by one period.

A more realistic model for many trades does not have point trade, but recognizes that there may be a finite time loss of resource use in actual trade. In figure 15.3, it is assumed that trading takes a finite amount of time, which we denote by Δt. The lags in delivery of both goods and money can go in any direction. The goods could be delivered first and the money later or vice versa. An approximation not too far from the truth in many transactions is that both are delivered essentially simultaneously in a small, but finite time. Thus there will be a finite upper bound to the number of trading sessions that can be held in any period.

Dubey, Sahi, and Shubik (1993) have shown generally that a fast enough velocity makes any amount of money adequate, but this was without transaction losses from forgone uses. It must be stressed that a faster velocity cuts down on the need to borrow, and hence may remove the need for trust.

MODEL 3: *A strategic market game with a finite number of traders* The device of considering a continuum of traders simplifies the mathematics

but obliterates the opportunity to study the possibility that there could be a trade-off between the frequency in which markets meet and their thickness.

Implicit in model 1 is that price may be regarded as a parameter. This can be justified when there is a continuum of traders, but with a finite number n, this is not so. Suppose that there are k trading sessions and n traders available. With a convergence to market efficiency at the speed of $1/n$, then at a 5 percent (1 percent) level, 20 (100) individuals are required on each side of the market in order to approximate competition.

From the first-order conditions for an optimum, if we solve and treat the number of trading sessions as a strategic choice, we can find a finite optimal number of trading sessions where the oligopoly effect is being traded off against the cost of liquidity. The basic point can be made without the calculation. The liquidity cost may decrease with more trading sessions, but with a finite number of traders, the oligopolistic solution may call for less trade than with more traders, but this may decrease velocity.

MODEL 4: *How to have your gold and eat it* A simple variant of model 3 is to imagine that, instead of trading gold, all individuals were to trade paper claims to the gold they own. In this manner, the gold is used for the utilitarian purpose, and the paper is the means of exchange. This type of distinction occurs with the use of one's house, although there is a mortgage outstanding on it, in contrast with depositing one's jewels at the pawnbroker, where they remain out of use until the debt is paid. Or in contrast with keeping gold in Fort Knox.

This device can increase value *if* the cost of the administrative arrangements validating and policing the paper claims is less to all than the cost of the direct use of gold. But once more, the problem of the cost of trust or law arises.

MODEL 5: *Trade using fiat with and without a rate of interest* We can consider the issue of fiat money by the "referee" or government. But in order to well define such a model, so that it is actually playable, we must specify how the fiat enters and leaves. If it is as a debt that must be repaid to the government[4] then logic demands a bankruptcy penalty. Without going into detail, it is easy to see that if fiat money bears a positive rate of interest, then its use in exchange generates forgone losses of interest in the

same way that the use of gold as a money generates losses of forgone nonmonetary use.

15.2.2 On time lags, details, technological facts, and velocity

A general model of gold production can be developed in a fairly straightforward manner. Such a model is very much in the tradition of basic operations research, where the equilibria depend heavily on the details of the production technology. Is the production time period a day, a week, a year? At what point does the consumption of a consumable make a difference to its yield? If a durable yields a stream of services, how do we describe this stream and the possibility of sharing it? What are the average and minimum levels of time required for various transactions? How much strategic leeway do individuals actually have in changing the time length of transactions?

A key element in integrating velocity into a strategic game is the recognition that exchange is a form of the productive process and involves both the use of time and other resources. Time has an opportunity cost in terms of the rate of interest on fiat money, on the one hand, and both interest and alternative production or consumption uses on commodity money, on the other.

The partial equilibrium writings of Allais (1947), Tobin (1956), Baumol (1952), Miller and Orr (1966), and others deal in a matter-of-fact way with money as a factor of production. Shubik and Yao (1990b) have extended the models above with production technology for the use of money incorporated into a closed model of the economy.

15.2.3 On search and randomness

Lippman and McCall (1986) consider the process of selling in the context of search, defining "an asset's liquidity as the expected time until it is sold while pursuing an optimal policy." The introduction of search gives recognition to the imperfection of markets. If a market is thick and atomless, and if all know where it is, then ideally all can use it costlessly. In reality, many markets require search, transportation, and other inputs; furthermore, the clearance process may have randomness in it. The mails are late, the check is lost, clearance procedures are delayed, the store is closed, and so forth. Although search is undoubtedly important, as a practical consideration, it may not be logically necessary for a basic understanding of

liquidity, and hence should be treated separately as an extra factor in determining the optimal size of liquidity.

15.2.4 An aside on hoarding

Hoarding is a relatively ill-defined concept. The word is usually used with pejorative overtones. The phenomenon of hoarding does not exist in a general-equilibrium model. But with a positive rate of interest and incomplete markets, the concept that an individual i will keep i's gold as cash, rather than earn interest at a bank or a government, can be made operational. But even here, it is may be difficult to sort our precautionary motives from speculative motives or from miserly distrust.

15.2.5 Velocity and liquidity in strategic market games

The general-equilibrium mathematical structure, because it is essentially nonprocess, is not suited to considering velocity or liquidity. Strategic market games provide a more general structure, which is consistent with the general equilibrium models, but which encompasses process. Thus items such as velocity can be accommodated. Liquidity of assets and the strategic selection of the velocity of a money depend on the structure of markets, and depend heavily on the technology of trade, durability, and other properties of goods.

It is empirically reasonable to believe that there is a finite upper bound to the speed at which both goods and money can be moved in most transactions. Buying shares of IBM, an ice cream cone, or an elephant require different times. If there is a minimum transaction time, there is a maximum velocity to money and hence a finite minimum amount required. In the course of trade, leaving out bankruptcy, individuals are paid sooner or later within a finite period; thus a strategic model will have some lower bound to velocity that is greater than zero.

As a matter of practical policy and control, variation in velocity may well be of considerable importance. The strategic ability of individuals to influence velocity may have great influence in modifying or negating government policy.

In an ideal imaginary world, with all markets thick and with velocity approaching infinity, so that every trade can be regarded as virtually instantaneous, all goods are totally liquid. Every good can be changed into money instantaneously at the market price quoted, and vice versa.

In the real world, velocity is not infinite,[5] and liquidity is highly variable. While the idealizations of perfect liquidity and instantaneous trade are adequate for the development of general-equilibrium theory, a theory of money and financial institutions is concerned with process. Time enters explicitly, and both limited velocity and liquidity may be important features (see Wicksell 1935 for a perceptive discussion of currency and velocity).

Another aspect of the rate of interest—and of the loose coupling of monetary control—is that a change in the rate influences the cost of trust. Thus when "money is tight," we may expect firms to extend more credit to each other than before.

15.2.6 On efficiency in exchange

When transactions take time and there is an opportunity cost to the use of money, we must redefine the concept of efficiency or Pareto optimality to take it into account. Efficiency becomes dependent on the initial conditions as the cost of trade appears. (See Volume 2, chapters 10 and 11, for an explicit discussion of transactions costs.)

15.2.7 Liquidity, marketability, and their costs

Liquidity and marketability are in essence concepts to help us understand a highly imperfect world. They are elusive and difficult to define precisely. In most of the models we have constructed so far, perfect marketability has been implicitly assumed for all goods. Given a continuum of traders, markets that clear in a single session exist for all goods, anyone of which can be shipped as quickly and costlessly as any other to every market. In order to establish operational differences in marketability among goods, we need to know time and transaction costs and to appreciate that some markets are thin.

Perfect marketability A good or financial instrument has *perfect marketability* if there exists a mass anonymous market with a continuum of agents that can be used costlessly to convert the good into money at the current market price by any set of agents of measure zero, given that the moves of all others are fixed. We might wish to add the condition that the time for completion of the transaction is of measure zero and all agents know that the market exists.

It is reasonable to assume that there is a minimum transaction time, which varies from market to market. Thus a market that requires two days or weeks to clear may be regarded as less "perfect" than one that requires two minutes, ceteris paribus.

In the one-period strategic market games, liquidity can be operationally defined in terms of the relative completeness of markets. Liquidity requires marketability, but marketability does not imply liquidity.

Perfect liquidity A good or financial instrument has *perfect liquidity* if it there exist perfect mass anonymous markets between it and every other good or instrument, that is, it is a full means of exchange. An ideal money has perfect liquidity. In the exchange economy with complete markets analyzed in chapter 7, all goods are ideal money. Implicit in the way strategies were defined and markets clear in the one-period strategic market games with a continuum of traders, all markets clear costlessly and take the same time. In actuality, even if an individual has money that can be used to buy any economic good, the lack of perfect marketability between it and some goods means that, even with cash in hand, buying an estate with doubtful title may be considerably slower than buying a can of beans.

Leaving aside differences in marketability, there are only two levels of liquidity in an economy with a single money. This is illustrated in figures 15.4a and 15.4b. In both figures, good 5 is the money. But figure 15.4b is drawn to call attention to the possibility that if there were two trading sessions, a trader i could liquidate all of i's goods and then buy what i wants. Thus money is one trade from every good and all other goods are two trades from every other good.

The expression "near-money," which one may use to describe a treasury bill or other highly liquid financial instrument, implies that it is easy, fast, and cheap to convert it into money and then use the money. The expression "I am totally illiquid" can be interpreted as meaning that an individual has no money and does not wish to liquidate assets at their expected next price.

Table 15.1 presents a summary of some of the features of the major sources of liquidity and goods and services. They are classified with respect to liquidity, market efficiency, and other features. We assume that there are m goods and services and the four financial instruments noted. The first column of table 15.1 indicates that the first three instruments are all direct means of exchange, and thus each has $(m + 3)$ markets because it

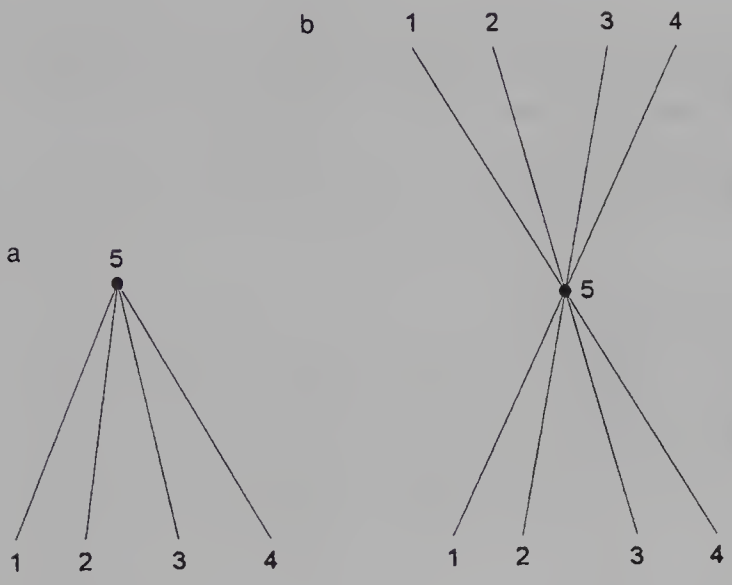


Figure 15.4
Liquidity and trade.

can be exchanged directly for all other items. There may be some markets where a check may not be accepted, thus bank money (in both check form or private banknotes) may not have all $(m + 3)$ markets. Government notes and bills, although near-money may only have three markets, that is, three markets that convert them into money. Goods and services, ruling out barter, also have only three markets, each for the various forms of direct means of payment present.

The column entitled "Market Efficiency" provides a crude summary of information on thickness and transaction technology. Money markets may be close to the dream of an efficient competitive market. The market for goods such as paintings or antiques or rare birds is far from the ideal.

Anonymity is an important property of an efficient market and a useful feature in capital flight. Cash, gold bars, dollar bills, and bearer bonds are more anonymous than checks or owner-registered bonds.

Liquidity is complete when an instrument can be exchanged for all other items in the economy and the transaction cost generated on the payments side of the market are low. Thus money buying art is liquid, but not vice versa.

The cost of liquidity is measured in terms of alternative values forgone. If an interest rate ρ exists and Δt units of time that could earn interest must be given up, then the forgone cost is $\rho\Delta t$. If monetary gold can be

Table 15.1
Liquidity and other properties

	Number or markets	Market efficiency	Anonymous?	Liquidity	Cost of liquidity	Sources of store of value
Commodity money	$m+3$	High among all of these financial instruments	Yes	Complete	Forgone utility or interest	Commodity use, custom, and law
Fiat money	$m+3$		Yes	Complete	Forgone interest	Custom and law
Bank money	$m+3 \leq$		Not quite	Almost complete	Forgone interest	Custom and law
Government notes and bills	$\simeq 3$		Not quite	Near-moneys	Forgone interest	Custom and law
Commodities	$\simeq 3$	Variable, from high to low	Variable, from not quite to individual	Some are near-moneys; other have high liquidity	Forgone use while being marketed and cost of exchange	Commodity use if durable
Services	$\simeq 3$					

alternately used for jewelry, and if the rate of interest is of less worth, then consumer use forgone is a measure of the cost of liquidity.

The store of value aspect of a commodity money is broken down into its commodity worth and an extra value due to its use as a money. Thus fiat money has no commodity value, although as long as the rules enable it to be of use in further exchange, it does have worth induced by a game solution.

Life is not pure economics, and although some purists may want a strictly short-term economic explanation of what sustains the value of a fiat money, the answer lies elsewhere. Institutions, law, and custom involving taxes, prison, social pressure, torture, death, and other unpleasantnesses provide enough incentives for most of us to refrain from counterfeiting, on the one hand, and from refusing to accept legal tender, on the other.

15.3 Enough Money, Uncertainty, and Financial Instruments

When markets are complete, few financial instruments are called for. As uncertainty is eliminated by contracts that pay on contingencies, we need only loans and futures contracts. These instruments involve, in general, two periods, two parties, and an exchange of money against a good or paper of some variety. The presence of financial intermediaries and the extra layers of paper they create increases the number of exchanges and the velocity of or need for money.

The details of the functioning of the many financial instruments and the specific reasons for their existence are not dealt with in this volume. The points to be made here are that when markets are incomplete, a host of instruments involving time and uncertainty are introduced to improve trade. But the introduction of these new instruments brings with it the need for (1) generating more trade, (2) resolving problems of information and verification, and (3) creating mechanisms for the fulfillment of contracts.

A brief list of some of the instruments usually found in an economy with exogenous uncertainty includes loans, bonds and debentures, bills of exchange, common stock, short sales, options, warrants, sale buybacks, insurance policies of various sorts, futures contracts and lottery tickets, bets, and wagers. For virtually all of these, we must consider the nature of

the payment and the payout, the roles of time, randomness, information, and the need for enforcement mechanisms supplied by the law or other aspects of the rules of the game.

At a high level of abstraction in an uncertain world, all financial instruments[6] may be regarded as some form of lottery ticket. But in the development of a process model of multiperiod trade, many differentiating features must be noted.

The instrument we may most readily compare to a lottery ticket is common stock. Unlike the others listed, it is purely an ownership instrument. In a world without exogenous uncertainty, as an ownership claim for a part of a real asset, it may still provide a means of dividing the ownership of firms. From one point of view, as when we develop Arrow's model (1953) for completing markets by introducing securities, in an economy with uncertainty, a share of common stock appears to be a lottery ticket. But strategically, when trying to incorporate common stock into a playable game, one cannot avoid the need to make minimal assumptions concerning the nature of the firm, limited liability, and how the economic strategy set is changed by the actions of management and stockholders.

A lottery ticket, by contrast, is a wager sold for money by A to B at time t in return for a promise to pay p_j per unit if state j occurs. In general, the enforcement mechanism guaranteeing the honoring of contract for a lottery ticket is the law. Escrow arrangements are rarely called for on an individual basis, although the issuer of a lottery may be required to keep reserves. Most lottery tickets pay out once in a few days and are rarely resold. This contrasts with common stock, which may pay out every few months and may be often resold.

When a lottery ticket is bought, the transaction is almost always between the lottery agency and a buyer, whereas when a stock is bought, buyer and seller are usually third parties. The firm with common stock is run by agents who are fiduciaries for the stockholders, who are meant to represent the stock holders' best interests even if the composition of the stockholders changes. The lottery agency, although not owned by the ticket buyers, has the duty to live up to contract, and this is often enforced by law.

In both instances, there is an exchange of money for an ownership right involving a single payment or a stream of contingent future payments.

In general, the payments will be in money at a future time and state, but payments can also be in goods, such as a champagne company paying out a case of champagne as a dividend, or a lottery agency awarding an automobile as a prize.

Other instruments cannot as readily be compared to a lottery ticket. A futures contract involves the sale of a right to make a purchase of a specific commodity at a specified price and time. The actual purchase will take place at some point in the future. A short sale involves the current sale of stock at a specified price, together with a promise to deliver the stock within a specified time after being required to do so. If a seller does not own the stock (i.e., has made a "naked sale"), the seller will have to make a purchase when called upon to deliver. Options and warrants have the extra feature that the choice to exercise lies with the owner of the instrument. While insurance policies vary considerably in term, contingencies covered, and patterns of payment, both in purchase and payout, virtually all involve the exchange of current money for a contingent payout of future money. As the policy terms are often long, the details concerning who holds the payments and how interest earnings are accounted for become of paramount concern in evaluating the profitability of an insurance enterprise.

This foray into brief (and somewhat incomplete) institutional descriptions of various instruments may, at first glance, appear to be a digression; but it is done to call attention to two fundamental conceptual questions in strategic analysis and the mathematical modeling of finance. What rules and enforcement mechanisms are we willing to accept as arising from the rules of the game? What forces for the enforcement of contracts arise from the way the game is actually played?

This distinction has already been made in the instance of bankruptcy and default. One can impose rules that are realistic, but more or less economic in their structure. A rule for garnishing assets is more in tune with economic behavior than rules involving prison, exile, or death for default. Although, as will be discussed further in Volume 2, chapter 2, if the individual has no assets, garnishing them offers no deterrence. The devising of rules for honoring contracts calls for taking into account information, administrative costs, the ability of individuals to lie or otherwise break their contractual promises, and the sociopolitical context of the law (and its enforcement) concerning escrow arrangements and liability resulting from broken contracts.

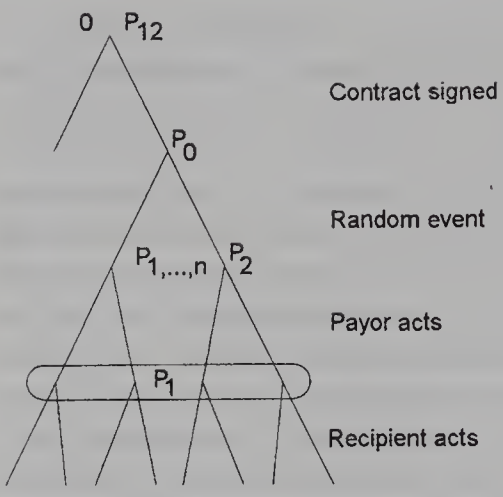


Figure 15.5
Enforcement of contracts.

At a more abstract level, many of the financial instruments noted above arise in situations that require modeling by at least four levels of a game tree. This is shown in figure 15.5. The first level is where the traders simultaneously agree to a contract; the second level is where a random event occurs. The third and fourth levels involve contract settlement. In general, it is safe to assume that the individual called on to honor a contract is informed of the outcome of the random event; the other party will often not be as well informed. The farmers who have sold wheat futures may know better than the buyers the state of their stocks after harvest.

Although the payment of a dividend on a common stock fits the above game description as well as the sale of a futures contract, there is a basic difference between the two. The corporate management and directors who determine the dividends are meant to be acting as fiduciaries for the stockholders, whereas the seller of a futures contract is not an agent for the buyer.

In section 12.4.3, a simple example for the pricing of a futures contract was given. From a strategic point of view, the principle is simple. A payment by both parties, to be held in escrow, can be designed so as to deter both sides from breaking the rules. The pledging of specific assets or revenues against specific bonds is much in the same spirit.

At a high and simple level of abstraction where all fiduciaries are perfect, there is no bankruptcy, and all contracts are honored, most of the

instruments noted can be regarded as some form of lottery ticket. In appropriate combination, they are used to try to complete markets in an uncertain world. But human assessment of risk is imperfect and transaction costs are high. At a more contextual and process-oriented level of abstraction, all of these instruments differ. They represent the products of individuals with limited capacity and limited perceptions trying to cope with subjective uncertainty, as well as reasonably measured risk.

The level of abstraction in any theory should be considered in relationship to the questions being asked. For general-equilibrium theory and some parts of finance, it can be argued that regarding all the investments noted as some form of lottery ticket is sufficient. This is not true when strategic behavior is considered concerning concealment of information, default, bankruptcy, and less than ideal fiduciary behavior. For the immediate purposes of this volume, there is no need to deal with the many distinctions between the instruments. The features most relevant to constructing a game-theoretic theory of money are the sequencing of payments, escrow conditions, secured loans, and default rules.

Table 15.2 presents a sketch of some of the main distinguishing features of the instruments noted. The first column indicates whether the party on the other side of the instrument is a fiduciary or a separate party. The asset holdings represented by a stock are in the hands of fiduciaries, the directors and management of the firm. Likewise, a mutual insurance company is a fiduciary to its policy holders. The second column covers the timing of payouts. Dividends can be varied both in size and timing, although many firms try to pay them regularly over the year. Lottery tickets are frequently of only a few days' duration. Bonds pay regular amounts at specific instances. The third column notes exchange conditions. Most instruments can be sold, but on occasion restrictions are placed on various instruments, such as warrants. Most insurance policies cannot be transferred by individuals, and some loans cannot be sold. The fourth column covers the choice to exercise. This amounts to a specification on the strategy sets of the individuals. Some instruments may allow either side to initiate settlement, others lock in both sides to specific dates. The fifth column indicates escrow or security conditions. These have direct bearing on the possibility for creating games with equilibria that are self-enforcing at each stage (perfect equilibria). They also are of importance in determining the need for money in the system. The sixth and seventh columns deal with the type of payment to be made by the purchaser of an

Table 15.2
Financial instruments

Instrument	Nature	Choice in timing of payouts	Choice to sell or exercise instrument	Initiator	Escrow or security	Type of payment	Type of payout
Stock	F	Irregular stream	Y	—	—	$	$, $, ...
Short sale	A	Irregular stream	—	Various	$	$	$
Warrant	F	One zone of settlement	Y/N	Holder's option	—	$	Stock
Lottery ticket	A	Short-term payout	Y/N	—	Legal reserves	$	$
Bet	A	Short-term payout	Y/N	—	—	$	$
Insurance policy	A/F	Indefinite sequence of contingent payments	N	Holder initiates	Legal reserves	$, $, ...	$, $, ...
Future	A	Specific settlement	Y	N	$, $	$, $	Goods
Bond	A	Regular sequence of payments with final repayment	Y	Y/N; some may permit the issuer to call	Specific goods or expected income	$	$, $, ...
Debenture	A	Regular sequence of payments with final repayment	Y	Y/N; some may permit the issuer to call	$	$, $, ...	
Loan	A	One or more specified payments	Y/N	Y/N; buyer, sell, or both may have early settlement option	Y/N, secured or unsecured	$	$, $, ...

Key:
A = arm's = length
F = fiduciary
N = no
Y = yes
$ = one money payment
$, $ = two money payments
$, $, ... = series of money payments.

instrument and the payouts made to the owner of the instrument. Most of the payments and payouts involve money or streams of money, although warrants pay out stock and futures pay out goods.

In an economy with uncertainty and the need for loans, unless the legal system by itself is assumed to be sufficient to enforce contractual obligations, either money or goods must be put up to serve as "hostages" to guarantee compliance with contract.

15.4 Summary of One-Period Models of Exchange

These first fifteen chapters have been devoted to setting up the context and then studying many variations of one-period exchange. The approach adopted has been to use *strategic market games*, which provide simple playable and mathematically tractable models of exchange with price formation well defined. Our solution concept has been the *noncooperative equilibrium*, although it must be stressed that there are many difficulties with this solution concept, especially when applied to multistage processes. Nevertheless, in the context of exchange with many small traders, the noncooperative equilibrium appears to be a natural generalization of the competitive equilibrium: given the appropriate conditions on strategic game models, any of the results of general-equilibrium theory, can be obtained as special instances of noncooperative equilibria.

General-equilibrium theory, as noted in chapter 4, is essentially nonprocess and preinstitutional in structure. Macroeconomics and the role of money and financial institutions call for a study of process. But the dangers of a process and institutional approach are that one becomes prematurely bogged down in institutional ephemera and irrelevant detail. To avoid this trap, the approach adopted here is to explicitly stress playable games of minimal complexity sufficient to illustrate the need for an institution or financial instrument. Historical and statistical detail is supplied to sweeten the intuition concerning economic reality, but the models themselves are minimal, and the test they must meet is one of logical completeness and consistency, and of playability as experimental games, not one of realism.

Our argument thus far can be summarized as follows. The strategic market games provide an easy way to model the details of process in mass exchange. If we restrict ourselves to economically plausible restrictions

concerning the properties of market mechanisms (see chapter 7), there are only a few basic mechanisms and among the simplest are those essentially implied by Cournot and Bertrand-Edgeworth. The way to parsimoniously embed the Cournot model into a closed system of exchange is to strategically distinguish one commodity as a money (see chapter 6). But doing this introduces problems concerning cash flow and the meaning of "enough money."

The concept of enough money has three different components when the money is a commodity of intrinsic value: (1) the overall supply, (2) the distribution of the supply, and (3) what happens when the supply is increased (all other things being held constant). In essence, for a commodity money to be in sufficient supply to permit optimal trade, everyone must hold enough of it so that at equilibrium its market value is at least equal to the market value of all purchases (a cash-flow condition). If this condition is not satisfied, then whether this failing can be cured by introducing more commodity money depends upon the change in the marginal utility of the money as more is introduced.

All of the above observations are made on the assumption that the velocity of money is one or less. The constraint on velocity can be removed by considering a trading model with several trading opportunities. This has been illustrated above in section 15.5. In essence a transaction cost must be introduced when considering trade. This can be done quite generally without delving into the details of setup costs and other practically important institutional features such as indivisibilities. This work shares with Niehans (1978) the view that, logically, transaction costs rather than exogenous uncertainty provide the necessary and sufficient conditions for money and financial institutions.[7]

When there is not enough commodity money, the efficiency of trade may be improved by the introduction of more markets. It has been shown in chapter 7, however, that even with complete markets, where any commodity can be exchanged directly for any other commodity, the distribution of goods may be such that it is not possible to achieve the trade called for by competitive equilibrium in a single round of trade.

When there is not enough commodity money, or its distribution is skewed, the rules of the game can be changed to overcome the shortage. In particular there are an array of fixes, which include (1) mining more gold; (2) increasing the velocity of circulation; (3) introducing loans; (4) introducing banking and fractional reserves; (5) introducing fiat money;

(6) using other credit instruments such as bills of exchange; and (7) increasing the number of markets, or any combination of the above.

The introduction of fiat money or credit requires the specification of enforcing conventions, laws, or rules of the game. Chapters 11 and 12 define a game in which the fiat of the government is introduced in exchange for private debt. This is clearly not realistic, although it is perfectly feasible. Fiat, or outside money, like commodity money is not debt, although a way to increase or decrease its supply is to have it exchanged for debt. This model serves to stress the logical necessity of a bankruptcy law when there is the possibility of default. But the strength of the default laws will be intimately related to the quantity of money issued, thus the usual comments on the homogeneity of order one of a price system with fiat money will not hold when the default law is specified.

When a money market is set up, even in a one-period model, and endogenous interest rate appears as the ratio of money offered for loan to the quantity of IOU notes offered in return. Without exogenous uncertainty, one can define an optimal bankruptcy rule under which strategic bankruptcy does not pay, hence no one goes bankrupt and the rate of interest in a one-period game with enough money and no residual values after trade has taken place turns out to be zero. The rate is positive if there is a money shortage. Once there is enough money, further increases produce only a "liquidity trap," where the rate of interest stays at zero and either prices rise or individuals hoard or hold back money.

As soon as an economic value to the introduction of credit appears, not only must default conditions be introduced but the economic purpose and distinctions between different types of borrowers and lenders must be specified. Only in chapter 9 has some distinction been made among a money market, various types of bank, and other lenders.

Because the strategic market games are well defined for a continuum of traders or for finite numbers, they are suitable for the study of oligopolistic phenomena (chapter 7 shows the effect of thin markets, together with the possibility that there can be equilibria involving wash sales).

The strategic market game formulation of economic exchange has for the most part been achieved with the use of games having single simultaneous moves. As trade takes place using a money, playing the game may create a float or a sequencing in the income and expenditure patterns of all individuals. If trade could take place with unlimited speed, velocity could be arbitrarily high and the float could be made arbitrarily close to zero.

A key distinction between the general-equilibrium and strategic market game approaches to exchange is that the latter lays stress on process, transactions, and trust. Although under the appropriate circumstances it is possible to prove that the CEs of an exchange economy coincide with the NEs of a strategic market game having enough commodity or fiat money, it is precisely the presence of enough commodity or fiat money that enables an efficient NE to be supported without trust in more than rules of trade. If credit is introduced, the economy *logically requires* rules to enforce contracts or substitutes for trust.

Although, for the economic analysis of one-period exchange, the gap between expenditures and receipts may be regarded as minor, this is certainly not the case for multiperiod economies and production processes involving considerable lengths of time. Money and financial instruments are needed not only to finance the float, but also to finance intertemporal trade and capital stock.

The economic requirements of a transaction technology call for the invention of markets, money, and credit. These factors are so central and valuable to trade in virtually all societies that they provide the instruments for control and for revenue for the state. An appreciation of the role of both private and state financial institutions requires more elaborate models than even those of multiperiod exchange. Not only do production, durable and storable assets, and the indefinite horizon have to be introduced, but also the explicit role of overlapping generations and government must be considered. In the following volume, we turn to the study of multistage exchange with durables and storables, briefly commenting on production and overlapping generations, but leaving detailed consideration of production, financial institutions, and government for Volume 3.

Notes

1. See Kindleberger (1984, p. 39), for example, on grace periods in clearing international trade trasactions for trade between different locations in Europe in the sixteenth century.

2. We could also consider maximization where the amount of gold for money must be selected before knowledge of the random variable, then after the random variable is disclosed individuals decide what to bid and offer; we call this "low information."

3. "Low-information" in the sense that all decisions for each trading session must be made without knowledge of the outcome from a previous session. Individuals can bid a percentage of their (unknown) wealth.

4. Pesek (1976) would argue against this. It is clearly dependent on how government introduces money into the economy.

5. Even in the German hyperinflation there was a bound as to how fast one could run with billions of marks to the local store.

6. Even fiat money's purchasing power may change as the state of the system changes and bonds can become worthless.

7. Niehans (1978) not only stresses the importance of transaction costs, but he develops several general open transaction models.

References

Adair, R. K., 1988. A flaw in a universal mirror. *Scientific American* 258(2): 50–56.

Allais, M., 1947. *Économie et interêt*. Paris: Imprimerie Nationale.

Amir, R., S. Sahi, M. Shubik, and Yao, 1990. A strategic market game with complete markets. *Journal of Economic Theory* 32: 124–143.

Arrow, K. J., [1951] 1963. *Social choice and individual values*, 2d ed. New York: Wiley.

Arrow, K. J., 1953. Le rôle de valeurs boursières pour la répartition de la meilleure des risques. In *Économétrie*, pp. 41–48. Paris: CNRS.

Arrow, K. J., 1964. The role of securities in the optimal allocation of risk bearing. *Review of Economic Studies* 31: 91–96.

Arrow, K. J., and G. Debreu, 1954. Existence of an equilibrium for a competitive economy. *Econometrica* 22: 265–290.

Arrow, K. J., and F. H. Hahn, 1971. *General competitive analysis*. San Francisco: Holden Day.

Aumann, R. J., 1987. Correlated equilibrium as an expression of Bayesian rationality. *Econometrica* 55: 1–18.

Bagehot, W., [1873] 1962. *Lombard street*. Reprint, with a new introduction by Frank C. Genovese. Homewood, IL: Irwin.

Barro, R. J., 1979. Money and the price level under the gold standard. *Economic Journal* 89: 13–31.

Baruch, H., 1971. *Wall street: Security risk*. Washington, DC: Acropolis.

Baumol, W. J., 1952. The transactions demand for cash: An inventory theoretic approach. *Quarterly Journal of Economics* 56: 545–556.

Baumol, W. J., 1959. *Economic dynamics*. New York: Macmillan.

Becker, T. W., 1969. *The coin makers*. Garden City, NY: Doubleday.

Becker, G. S., 1975. *Human capital*, 2d ed. New York: National Bureau of Economic Research and Columbia University Press.

Bell, D., 1976, *Coming of postindustrial society: A venture in social forecasting*. New York: Basic Books.

Belloc, H., 1922. *The Mercy of Allah*, pp. 293–294. London: Chatto and Windus.

Benassay, J. P., 1986. On competitive market mechanisms. *Econometrica* 541: 95–108.

Bereziner, Y., and C. Narbeth, 1973. *The story of paper money*. Newton Abbot, England: David and Charles.

Bertrand, J., 1883. Théorie mathematique de la richesse sociale. *Journal des Savants* (Paris) 68: 499–508. (Review.)

Bewley, T., 1986b. Knightian uncertainty, pt. 1. Cowles Foundation Discussion Paper 807, Yale University.

Bewley, T., 1987. Knightian uncertainty, pt. 2. Cowles Foundation Discussion Paper 835, Yale University.

Blake, W. P., 1869. *The production of the precious metals*. New York: Putnam.

Böhm-Bawerk, E. von, [1891] 1923. *Positive theory of capital*. Translated and with a preface and analysis by William Smart. New York: Stechert-Hafner and Co.

Bohlen, F. H., 1941. *Cases on the law of torts*. 4th ed. Indianapolis: Bobbs-Merrill.

Boulding, K. E., 1950. *Reconstruction of economics*. New York: John Wiley.

Bradley, G. H., 1973. Trading rules for a decentralized exchange economy. In S. E. Elmaghraby, ed., *Symposium on the theory of scheduling and its applications*, pp. 224–241. New York: Springer.

Brainard, W. C., and J. Tobin, 1968. Pitfalls in financial model building. *American Economic Review Proceedings* 58: 303–324.

Branson, W. H., 1989. *Macroeconomic theory and policy*. 3d ed. New York: Harper and Row.

Braudel, F., 1967. *Capitalism and material life, 1400–1800*. New York: Harper and Row.

Bresciani-Turroni, C., 1937. *The economics of inflation*. Translated by M. E. Sayers. London: Allen and Unwin.

Brooks, J., 1975. *The telephone*. New York: Harper and Row.

Brown, R. A., 1955. *The law of personal property*. Chicago: Callaghan.

Buist, M. B., 1974. *At spes non fracta: Hope & Co., 1700–1815, Merchant bankers and diplomats at work*. The Hague: Nijhoff.

Butler, S., 1878. *Life and habit*. London: Trubner.

Butts, A., and C. D. Coxe, eds., 1967. *Silver, economics, metallurgy, and use*. Princeton, NJ: Van Nostrand.

Cameron, R. O. Crisp, H. T. Patrick, and R. Tilly, 1967. *Banking in the early stages of industrialization*. Oxford: Oxford University Press.

Carson, R. A. G., 1970. *Coins*. 2d ed. London: Hutchinson.

Cassady, R., 1967. *Auctions and auctioneering*. Berkeley and Los Angeles: University of California Press.

Cassel, G., 1923. *The theory of social economy*. Translated by Joseph McCabe. London: Fisher Unwin.

Chamberlin, E. H., 1933. *Theory of monopolistic competition*. Cambridge, MA: Harvard University Press.

Chipman, J., and J. C. Moore, 1976. Why an increase in GNP need not imply an improvement in potential welfare. *Kyklos* 29: 391–418.

Chow, G., 1986. Development of a more market-oriented economy in China. *Science* 235: 295–299.

Clower, R., 1967. A reconsideration of the microfoundations of monetary theory. *Western Economic Journal* 6: 1–8.

Collins, R. S., 1975. *Gold, mineral resources consultative committee*. London: Her Majesty's Stationery Office.

Cournot, A. A., [1838] 1897. *Researches into the mathematical principles of the theory of wealth*. Translated by Bacon. New York: Macmillan.

Crafts, N. F. R., 1983. Gross national product in Europe, 1870–1910: Some new estimates. *Exploration and Economic History* 2, 4: 387–401.

Cross, J., 1969. *The economics of bargaining*. New York: Basic Books.

Curle, J. H., 1905. *The gold mines of the world*. 3d ed. London: Routledge.

Davidson, P., 1980. Post-Keynesian economics. *Public Interest* 61: 151–173. Special issue.

Day, J., 1978. The great bullion famine of the fifteenth century. *Past and Present* 79: 3–54.

Dean, S., ed., 1884. *History of banking and banks from the Bank of Venice to the Year 1883*, pp. 6–8. Boston: Phelan Studios.

Debreu, G., 1959. *Theory of value*. New York: Wiley.

Debreu, G., 1970. Economies with a finite set of equilibria. *Econometrica* 38: 389–392.

Debreu, G., 1975. The rate of convergence of an economy. *Journal of Mathematical Economics* 2: 1–7.

Debreu, G., 1976. Regular differentiable economies. *American Economic Review* 66: 280–287.

Debreu, G., and H. E. Scarf, 1963. A limit theorem on the core of an economy. *International Economic Review* 4: 235–246.

Deleeuw, F., and E. M. Gramlich, 1969. The Federal Reserve-MIT model. *Federal Reserve Bulletin* 55: 11–40.

Del Mar, A., [1880] 1986. *A History of Precious Metals*. London: G. Bell & Sons.

Denardo, E., 1981. *Dynamic programming and allied topics: Definite planning horizons*. Englewood Cliffs, NJ: Prentice-Hall.

Denning, P. J., 1987. The science of computing: Baffling Big Brother. *American Scientist* 75: 464–466.

Dickson, P. G. M., 1967. *The financial revolution in England: A study in the development of public credit*, 1688–1756. New York: St. Martin's Press.

Dornbush, R., and S. Fisher, 1981. *Macroeconomics*. 2d ed. New York: McGraw-Hill.

Downs, A., 1957. *An economic theory of democracy*. New York: Harper and Row.

Dubey, P., 1982. Price quality strategic market games. *Econometrica* 50: 111–126.

Dubey, P., A. Mas-Colell, and M. Shubik, 1980. Efficiency properties of strategic market games: An axiomatic approach. *Journal of Economic Theory* 22: 339–362.

Dubey, P., J. Geanakoplos, and M. Shubik, 1987. Revelation of information in strategic market games: A critique of rational expectations. *Journal of Mathematical Economics* 16: 105–137.

Dubey, P., J. Geanakoplos, and M. Shubik, 1988. Bankruptcy and efficiency in a general equilibrium model with incomplete markets. Cowles Foundation Discussion Paper 879, Yale University.

Dubey, P., and J. Geanakoplos, 1989. *Liquidity and bankruptcy with incomplete markets*. Cowles Foundation Discussion Paper 900, Yale University.

Dubey, P., J. Geanakoplos, and M. Shubik, 1992. Is gold an efficient store of value? Cowles Foundation Discussion Paper 1031, Yale University.

Dubey, P., and S. Sahi, 1992. Price mediated trade with quantity signals: An axiomatic approach. Institute for Decision Sciences, State University of New York, Stony Brook.

Dubey, P., S. Sahi, and M. Shubik, 1993. Repeated trade and the velocity of money, *Journal of Mathematical Economics* 22, 2: 125–137.

Dubey, P., and L. S. Shapley, 1994. Noncooperative exchange with a continuum of traders: Two models. *Journal of Mathematical Economics* 23: 253–293.

Dubey, P., and M. Shubik, 1978. The noncooperative equilibria of a closed trading economy with market supply and bidding strategies. *Journal of Economic Theory* 17: 1–20.

Dubey, P., and M. Shubik, 1979. Bankruptcy and optimality in a closed trading mass economy modelled as a noncooperative game. *Journal of Mathematical Economics* 6: 115–134.

Dubey, P., and M. Shubik, 1980. A strategic market game with price and quantity strategies. *Zeitschrift für Nationalöconomie* 40: 25–34.

Dubey, P., and M. Shubik, 1981a. Information conditions communication and general equilibrium. *Mathematics of Operations Research* 6: 186–189.

Dubey, P., and M. Shubik, 1981b. The profit maximizing firm: Managers and stockholders. *Économies et Sociétés* 14: 1369–1388.

Dubey, P., and M. Shubik, 1986. "On monopolistic competition and perfect equilibria." In D. Cantarelli, A. Agnati, and A. Montesano, eds. *Essays in honor of Tullio Bagiotti*, pp. 413–424 Padua: Casa Editrice Dott. Antonio Milani (CEDAM).

Dubey, P., and M. Shubik, 1988. A note on an optimal garnishing rule. *Economic Letters* 27: 5–6.

Duesenberry, J. S., G. Fromm, L. R. Klein, and E. Kuh, eds., 1965. *The Brookings quarterly econometric model of the United States*. Chicago: Rand McNally.

Eckstein, O., 1983. *The DRI model of the U.S. economy*. New York: McGraw-Hill.

Eckstein, O., E. W. Green, and A. Sinai, 1974. The data resources model: uses, structure, and analysis of the U.S. economy. *International Economic Review* 15, 3: 595–615.

Edgeworth, F. Y., 1925. *Papers relating to political economy*, vol. 1. London: Macmillan.

Edgeworth, F. Y., [1881] 1932. *Mathematical psychics: An essay on the application of mathematics to the moral sciences*. Reprint, London: London School of Economics.

Edie, L. D., 1983. *Gold production and prices before and after the World War*. New York: Garland. (Reprint of Ph.D. diss., Indiana University, 1928.)

Eichengreen, B., 1985. *The gold standard in theory and history*. New York: Methuen.

Einzig, P., 1948. *Primitive money*. London: Spottiswoode.

Engelbrecht-Wiggans, R., 1980. Auctions and bidding models: A survey. *Management Science* 26: 119–142.

Engelbrecht-Wiggans, R., M. Shubik, and R. Stark, 1983. *Auctions, bidding, and contracting: Uses and theory*. New York: New York University Press.

Evans, H. K., and L. R. Klein, 1967. *The Wharton econometric forecasting model*. Philadelphia: Economics Research Unit, University of Pennsylvania.

Fabozzi, F. J., 1985. *Handbook of mortgage-backed securities*. Chicago: Probus.

Fair, R. C., 1974. *A model of macroeconomic activity*. Vol. 1, *The theoretical model*. Cambridge, MA: Ballinger.

Fair, R. C., 1976. *A model of macroeconomic activity*. Vol. 2, *The theoretical model*. Cambridge, MA: Ballinger.

Fair, R. C., 1984. *Specification, estimation, and analysis of macroeconometric models*. Cambridge, MA: Harvard University Press.

Farquharson, R., 1969. *Theory of voting*. New Haven and London: Yale University Press.

Finley, M. I., 1981. *Economy and society in ancient Greece*. New York: Viking.

Finley, M. I., 1985. *The ancient economy*. London: Hogarth Press.

Fisher, I., 1931. The purchasing power of money. 2d ed. New York: Macmillan.

Fitzgerald, C. P., [1935] 1954. *China: A short cultural history*. Rev. ed. London: Cresset Press.

Friedman, M., 1969. *The optimum quantity of money and other essays*. Chicago: Aldine.

Friedman, M., and A. J. Schwartz, 1970. *Monetary statistics of the United Statcs*. New York: National Bureau of Economic Research and Columbia University Press.

Garcia, G., 1980. Credit cards: An interdisciplinary survey. *Journal of Consumer Research* 6: 327–331.

Garvey, G., 1977. *Money, financial flows and credit in the Soviet Union*. Cambridge, MA: National Bureau of Economic Research and Ballinger.

Geanakoplos, J., and H. Polemarchakis, 1986. Walrasian indeterminacy and Keynesian Macroeconomics. *Review of Economic Studies* 53: 755–779.

Gesell, S., [1916] 1929. *The natural economic order*. Translated by Philip Pye from *Die Naturliche Wirtschaftsordnung durch Freiland und Freigeld*. Berlin-Frohnau: Neo-Verlag.

Ghush, S., 1964. *Monetization of an economy*. Calcutta: World Press Private.

Giffen, R. G., 1898. *The case against bimetallism.* 5th ed. London: George Bell.

Goldsmith, R. W., 1984. An estimate of the size and structure of the national product of the early Roman Empire. *Review of Income and Wealth* 30: 263–288.

Goldsmith, R. W., 1987. *Premodern financial systems: A historical comparative study.* Cambridge: Cambridge University Press.

Grandmont, J. M., 1982. Temporary general equilibrium. In K. J. Arrow and M. Intriligator, eds., *Handbook of Mathematical Economics*, vol. 2, pp. 879–922. Amsterdam: Elsevier.

Grandmont, J.-M., and Y. Younes, 1972. On the role of money and the existence of a monetary equilibrium. *Review of Economic Studies* 39: 355–372.

Green, B., 1971. *Horse tradin'.* New York: Knopf.

Green, J. R., 1973. Temporary general equilibrium in a sequential trading model with spot and futures transactions. *Econometrica* 41: 1103–1123.

Green, T., 1968. *The world of gold.* New York: Walker.

Greismer, L. H., R. Levitan, and M. Shubik, 1967. Towards a study of bidding processes, pt. 4. *Naval Research Logistics Quarterly*, 10: 11–21, 151–173, 193–247.

Griffiths, A., 1884. *The chronicles of Newgate*, vol. 1. London: Chapman and Hall.

Gurley, J. G., and E. S. Shaw, 1960. *Money in a theory of finance.* Washington, DC: Brookings.

Hahn, F. H., 1965. On some problems of proving the existence of an equilibrium in a monetary economy. In Hahn and F. Brechling, eds., *The theory of interest rates.* New York: Macmillan.

Hammond, B., 1957. *Banks and politics in America from the Revolution to the Civil War.* Princeton: Princeton University Press.

Hansen, L. P., and T. Sargent, 1980. Formulating and estimating dynamic linear rational expectations models. *Journal of Economic Dynamics and Control* 2: 7–46.

Hendrickson, R. A., 1972. *The cashless society.* New York: Dodd and Mead.

Herskovits, M. J., 1940. *Economic anthropology.* New York: Knopf.

Hester, D. D., 1972. Monetary policy in the 'checkless" economy. *Journal of finance* 27: 279–293.

Heyman, D. P., and M. Sobel, 1982. *Stochastic models in operations Research*, vol 1. New York: Macmillan.

Hicks, J. R., 1937. Mr. Keynes and the "classics": A suggested interpretation. *Econometrica* 5: 147–159.

Hicks, J. R., 1946. *Value and capital.* 2d ed. Oxford: Oxford University Press.

Hobson, B., 1965. *Coins and coin collecting.* New York: Dover.

Hoel, A., 1975. A primer on Federal Reserve float. *The Monthly Review of the Federal Reserve Bank of N. Y* 57, 10: 245–253.

Holmes, I. M., 1947. *Great gold rushes.* London: Peter Lunn.

Hopper, R. S., 1979. *Trade and industry in classical Greece.* London: Thames and Hudson.

Horn, A. E., 1896. A history of banking in the Russian Empire. In *A history of banking in all nations*, pp. 342–435. New York: Journal of Commerce and Commercial Bulletin.

Hotchkiss, G. B., 1938. *Milestones in marketing.* New York: Macmillan.

Hotelling, H., 1929. Stability in competition. *Economic Journal* 39: 41–57.

Housman, D., 1983. Some noncooperative game models of exchange. Ph.D. diss., Cornell University.

Iklé, F. C., 1964. *How nations negotiate.* New York: Harper and Row.

Izutani, K., 1980. The origin of double-entry bookkeeping. *Osaka Keidai Ronshu* 136: 1–34.

Janssen, J. J., 1975. *Commodity prices from the Ramasid period.* London: Brill.

Jaynes, G., M. Okuno, and D. Schmeidler, 1978. Efficiency in atomless economy with fiat money. *International Economic Review* 19: 149–157.

Jevons, W. S., [1871] 1970. *The theory of political economy.* Reprint, with introduction by R. D. C. Black. London: Penguin.

Jevons, W. S., 1875. *Money and the mechanism of exchange.* London: Macmillan.

Jevons, W. S., 1909. *Investigations in currency and finance.* 2d ed. London: Macmillan.

Johnson. H. G., 1967. *Essays in monetary economics.* Cambridge, MA: Harvard University Press.

Jones, A. H. M., 1974. *The Roman economy.* Oxford: Basil Blackwell.

Karatzas, Y., M. Shubik, and W. Sudderth. 1994. Construction of stationary Markov equilibria in a strategic market game. *Mathematics of Operations Research* 19, 4: 975–1006.

Keynes, J. M., 1925. *Essays in persuasion.* London: Macmillan.

Keynes, J. M., [1930a] 1958. *A treatise on money.* Vol. 1. Reprint, London: Macmillan.

Keynes, J. M., [1930b] 1960. *A treatise on money.* Vol 2. Reprint, London: Macmillan.

Keynes, J. M., [1936] 1957. *The general theory of employment, interest and money.* Reprint, London: Macmillan.

Kindleberger, C. P., 1984. *A financial history of Western Europe.* London: Allen and Unwin.

Kisch, B., 1965. *Scales and weights.* New Haven: Yale University Press.

Kiyotaki, N., and R. Wright, 1989. On money as a medium of exchange. *Journal of Political Economy* 97: 927–934.

Klein, L. R., 1950. *Economic fluctuation in the United States, 1929–1944.* New York: Wiley.

Klein, L. R., and A. S. Goldberger, 1955. *An econometric model of the United States, 1929–1952.* Amsterdam: Elsevier.

Knapp, G. F., 1905. *Staatliche Theorie des Geldes.* Leipzig: Duncker and Humblot.

Koopmans, T. C., 1960. Stationary ordinal utility and impatience. *Econometrica* 28: 287–309.

Koopmans, T. C., 1977a. Concepts of optimality and their uses. *American Economic Review* 67: 261–274.

Kravis, I. B., A. Heston, and R. Summers, 1978. Real GDP per capita for more than one hundred countries. *Economic Journal* 88: 215–242.

Kroos, H. E., and M. R. Blyn, 1971. *A history of financial intermediaries.* New York: Random House.

Lange, O., 1936. On the economic theory of socialism. *Review of Economic Studies* 4: 53–71.

Leijonhufvud, A., 1968. *On Keynes in economics and the economics of Keynes.* New York: Oxford University Press.

Leijonhufvud, A., 1983. Keynesianism, monetarism, and rational expectations: Some reflections and conjectures. In R. Frydman, and E. S. Phelps, eds., *Individual forecasting and aggregate outcomes,* chapter 10. Cambridge: Cambridge University Press.

Lenin, V. I., 1960. *Collected works,* vol. 26. London: Lawrence and Wishart.

Lerner, A. P., 1944. *The economics of control: Principles of welfare economics.* London: Macmillan.

Lerner, A. P., 1947. Money as a creature of the state. *American Economic Review* 37: 312–317.

Levitan, R. E., and M. Shubik, 1971. Noncooperative equilibria and strategy spaces in an oligopolistic market. In H. Kuhn and G. Szego, eds., *Differential games and related topics*, pp. 429–447. Amsterdam: Elsevier.

Levitan, R. E., and M. Shubik, 1972. Price duopoly and capacity constraints. *International Economic Review* 13: 111–122.

Lippman, S. A., and J. J. McCall, 1986. An operational measure of liquidity *American Economic Review* 76, 1: 43–55.

Little, I. M. D., 1950. *A critique of welfare economics.* Oxford: Oxford University Press.

Lucas, R. E., Jr., 1976. Econometric policy evolution: A critique. In K. Brunner and A. H. Meltzer, eds., *The Phillips curve and labor markets*, pp. 19–46. Amsterdam: Elsevier.

Lucas, W., 1963. On solutions to *n*-person games in partition function form. Ph.D. diss., University of Michigan.

MacKay, C., [1841] 1980. *Extraordinary delusions and the madness of crowds.* Reprint, New York: Crown, (Original title *Memoirs of extraordinary popular delusions.*)

Manas, M., 1972. A linear oligopoly game. *Econometrica* 40: 917–922.

Markowitz, H. M., 1959. *Portfolio selection.* New York: Wiley.

Marks, M., 1924. *Credit and commercial paper.* New York: J. W. Block.

Marshall, A., 1920. *Principles of economics.* London: Macmillan.

Marshall, A., 1923. *Money, credit and commerce.* London: Macmillan.

Marx, J., 1978. *The magic of gold.* Garden City, NY: Doubleday.

Mas-Colell, A., 1977. Indivisible commodities and general equilibrium theory. *Journal of Economic Theory* 16: 443–445.

Mauss, M., 1954. *Le don.* Glencoe, IL: Free Press.

Mayer, M., 1974. *The bankers.* New York: Ballantine.

McClosky, D. N., and J. R. Zecker, 1976. How the gold standard worked, 1880–1913. In J. A. Frenkel and H. G. Johnson, eds., *The Monetary approach to the balance of payments*, pp. 184–208. London: Allen and Unwin.

McDonald, D., 1967. The history of silver. In A. Butts and C. D. Coxe, eds., *Silver, Economics, Metallurgy, and Use*, pp. 1–15. Princeton, NJ: Van Nostrand.

McEvedy, C., and R. Jones, 1978. *Atlas of world population history*, Penguin Books.

Mikesell, R. F., 1970. *The world copper Industry.* Baltimore: Johns Hopkins University Press.

Milgrom, P. R., and R. J. Weber, 1982. A theory of auctions and competitive bidding. *Econometrica* 50: 1089–1122.

Miller, M., and D. Orr, 1966. A model of demand for money by firms, *Quarterly Journal of Economics* 80: 413–435.

Miller, M., 1986. Financial innovation: The last twenty years and the next. *Journal of Financial and Quantitative Analysis* 21: 459–471.

Modigliani, F., and M. H. Miller, 1958. The cost of capital, corporation finance and the theory of investment, *American Economic Review* 48: 261–297.

Muth, J. F., 1961. Rational expectations and the theory of price movements. *Econometrica* 29: 315–335.

Nash, J. F., Jr., 1951. Noncooperative games. *Annals of Mathematics* 54: 286–295.

Nelson, R. R., and S. G. Winter, 1982. *An evolutionary theory of economic change.* Cambridge, MA: Harvard University Press.

Niehans, J., 1978. *The theory of money*. Baltimore: Johns Hopkins University Press.

Nti, K., and M. Shubik, 1984. Noncooperative exchange using money and broker-dealers. *International Journal of Mathematical Social Sciences* 7: 59–82.

New York Stock Exchange (NYSE) Fact Book. 1977, 1987. New York: NYSE.

Olmstead, A. T., 1948. *History of the Persian Empire*. Chicago: University of Chicago Press.

Ostroy, J. M., and R. M. Starr, 1974. Money and the decentralization of exchange. *Econometrica* 42: 1093–1090.

Owen, G., 1975. On the core of linear production games. *Mathematical Programming* 9: 358–370.

Pacioli, L., [1494] *Summa de Arithmetica, Geometria, Proportioni, et Proportionalita*. Originally printed in Venice by Paganino de Paganini. Reprinted by Industrial Grafica Pistolesi Snc., Siena. Preliminary offprint of *Economic Notes*, special issue for the conference on "Accounting and Economics," held in Siena, November 18–19, 1992.

Pesek, B. P., 1976. Monetary theory in the post-Robertson "Alice in Wonderland" era. *Journal of Economic Literature* 14: 856–884.

Pirenne, H., [1925] 1956. *Medieval cities*. Translated by Frank D. Halsey. Garden City, NJ: Doubleday.

Plott, C. R., and G. George, 1988. Marshallian vs. Walrasian stability in an experimental market. Social Science Working Paper 683, California Institute of Technology.

Polanyi, K., C. M. Arensberg, and H. W. Pearson, eds., 1957. *Trade and market in the early empires*. Glencoe, IL: Free Press.

Polanyi, M., 1944. *The great transformation*. New York: Rinehart.

Postlewaite, A., and D. Schmeidler, 1978. Approximate efficiency of non-Walrasian Nash equilibria. *Econometrica* 46: 127–136.

Radner, R., 1970. Problems in the theory of markets under uncertainty. *American Economic Review* 60: 454–460.

Radner, R., 1968. Competitive equilibrium under uncertainty. *Econometrica* 36: 31–58.

Radner, R. 1979, Rational expectations equilibrium: Generic existence and the information revealed by prices. *Econometrica* 47: 655–678.

Raw, C., B. Page, and G. Hodgson, 1971. *Do you sincerely want to be rich?* New York: Viking.

Rawlinson, G., trans., 1935. *The history of Herodotus*. London: Nonesuch Press.

Ricardo, D., [1821] 1965. The principles of political economy and taxation. 3d ed. Reprint, London: J. M. Dent.

Roberts, J., and H. Sonneschein, 1977. On the foundations of monopolistic competition. *Econometrica* 45: 101–113.

Rockwell, L. H., 1985. *The gold standard: An Austrian standard*. Lexington, MA: Lexington Books.

Rosefielde, S., 1986. Regulated market socialism: The semi-competitive Soviet solution. *Soviet Union/Union Sovietique* 13(1): 1–20.

Rubinstein, A., 1982. Perfect equilibrium in a bargaining model. *Econometrica* 50: 97–109.

Ruffin, M. D., and K. J. Tippett, 1975. Service life expectancy of household appliances: New estimates from the USDA. *Home Economics Research Journal* 3: 159–170.

Sahi, S., and S. Yao, 1989. The noncooperative equilibria of a closed trading economy with market supply and bidding strategies. *Journal of Mathematical Economics* 28: 325–346.

Samuelson, P. A., 1947. *Foundations of economic analysis.* Cambridge, MA: Harvard University Press.

Sanghvi, A., and M. Sobel, 1976. Bayesian games as stochastic processes. *International Journal of Game Theory* 5: 1–22.

Sansom, G. B., 1952. *Japan: A short cultural history.* 2d ed. London: Cresset Press.

Sargent, T. J., 1979. *Macroeconomics theory.* New York: Academic Press.

SAUS. See United States Department of Commence, *Statistical Abstract of the United States.*

Scarf, H. S., 1960. Some examples of global instability of competitive equilibria. *International Economic Review* 1, 3: 157–172.

Scarf, H. S., with T. Hansen, 1973. *The Computation of Economic Equilibria.* New Haven, CT: Yale University Press.

Schmidt, M., 1984. *Albert Speer: The end of a myth.* London: St. Martin's Press.

Schumpeter, J. A., 1954, *History of economic analysis.* Edited from manuscript by Elizabeth Boody Schumpeter. London: Allen and Unwin.

Schumpeter, J. A., 1955. *The theory of economic development.* Cambridge, MA: Harvard University Press.

Sen, A. 1986. Social choice theory. In K. J. Arrow and M. D. Intrilligator, eds. *Handbook of Mathematical Economics III.* Amsterdam: North Holland, pp. 1073–1181.

Shapley, L. S., 1975b. An example of a slow converging core. *International Economic Review* 16: 345–351.

Shapley, L. S., 1976. Noncooperative general exchange. In S. A. Y. Lin, ed., *Theory and measurement of economic externalities*, pp. 155–175. New York: Academic Press.

Shapley, L. S., and Shubik, M., 1969a. On the core of an economic system with externalities. *American Economic Review* 59: 678–684.

Shapley, L. S., and M. Shubik, 1969b. Pure competition, coalitional power, and fair division. *International Economic Review* 10: 337–362.

Shapley, L. S., and M. Shubik, 1971. The assignment game: 1. The core. *International Journal of Game Theory* 1: 111–130.

Shapley, L. S., and M. Shubik, 1977. Trade using one commodity as a means of payment. *Journal of Political Economy* 85: 937–968.

Shapley, L. S., and M. Shubik, 1966. Quasi-cores in a monetary economy with nonconvex preferences. *Econometrica* 34: 805–827.

Shubik, M., 1959a. Edgeworth market games. In R. D. Luce and A. W. Tucker, eds., *Annals of mathematical studies*, Vol. 40, pp. 267–278. Princeton: Princeton University Press.

Shubik, M., 1959b. *Strategy and market structure.* New York: Wiley.

Shubik, M., 1968a. Extended Edgeworth bargaining games and competitive equilibrium. *Metroeconomica* 20: 299–312.

Shubik, M., 1968b. A two-party system, general equilibrium, and the voter's paradox. *Zeitschrift für Nationalökonomie* 28: 341–354.

Shubik, M., 1970. Voting or a price system in a competitive market structure. *American Political Science Review* 64: 179–181.

Shubik, M., 1971. Pecuniary externalities: A game-theoretic analysis. *American Economic Review* 61: 713–718.

Shubik, M., 1973. Commodity money, oligopoly, credit and bankruptcy in a general equilibrium model. *Western Economic Journal* 11: 24–38.

Shubik, M. 1975. On the eight basic units of a dynamic economy controlled by financial institutions. *Review of Income and Wealth* 21: 183–201.

Shubik, M., 1976. A noncooperative model of a closed trading economy with many traders and two bankers. *Zeitschrift für Nationalökonomie* 36: 49–60.

Shubik, M., 1984. *Game theory in the social sciences.* Vol 2, *A game-theoretic approach to political economy.* Cambridge, MA: MIT Press.

Shubik, M., 1985b. A note on auctions and lotteries. Cowles Foundation Preliminary Paper 850730, Yale University.

Shubik, M., 1987. Silver, gold and liquidity. Cowles Foundation Discussion Paper 841, Yale University.

Shubik, M., 1990. The transactions trust demand for money. *Zeitschrift für Nationalökonomie* 52: 211–232.

Shubik, M., and D. Tsomocos, 1992. A strategic market game with a mutual bank with fractional reserves and redemption in gold. *Zeitschrift für Nationalökonomie* 55: 123–150.

Shubik, M., and D. Tsomocos 1993. A strategic market game with seignorage costs of fiat money. Cowles Foundation Discussion Paper 1043, Yale University.

Shubik, M., and L. van der Heyden, 1978. Logrolling and budget allocation games. *International Journal of Game Theory* 7: 151–162.

Shubik, M., and S. Yao, 1989, Gold, liquidity and secured loans in a multistage economy: 1. Gold as money. *Zeitschrift für Nationalökonomie* 49: 245–277.

Shubik, M., and S. Yao, 1990a. The transactions cost of money: A strategic market game analysis. *International Journal of Mathematical Social Sciences* 22: 99–114.

Shubik, M., and W. Whitt, 1973. Fiat Money in an economy with one nondurable good and no credit: A noncooperative sequential game. In A. Blaquière, ed., *Topics in differential games*, pp. 401–449. Amsterdam: Elsevier.

Shubik, M., and C. Wilson, 1977. The optimal bankruptcy rule in a trading economy using fiat money. *Zeitschrift für Nationalökonomie* 37: 3–4.

Shubik, M., and J. Zhao, 1991. A strategic market game of a finite exchange economy with a mutual bank. *Mathematical Social Sciences* 22: 257–274.

Simon, H. A., 1984. On the behavioral and rational foundations of economic dynamics, *Journal of Economic Behavior and Organization* 5: 35–55.

Simons, H. C., 1936. Rule versus authorities in monetary policy. *Journal of Political Economy* 44: 1–30.

Simons, H. C., 1948. *Economic policy for a free society.* Chicago: University of Chicago Press.

Skinner, F. G., 1976. *Weights and measures.* London: Her Majesty's Stationery Office.

Smith, V. L., 1965. Experimental auction markets and the Walrasian hypothesis. *Journal of Political Economy* 73: 387–393.

Spence, M., 1974. Competitive and optimal responses to signals. *Journal of Economic Theory* 7: 246–332.

Stark, R. M., and M. H. Rothkopf, 1979. Competitive bidding: A comprehensive bibliography. *Journal of the Operations Research Society of America* 27: 364–390.

Starr, C. G., 1977. *The economic and social growth of early Greece, 800–500 B.C.* New York: Oxford University Press.

Starr, R. M., 1974. The price of money in a pure exchange monetary economy with taxation. *Econometrica* 42: 45–54.

Starr, R. M., 1976. Decentralized nonmonetary trade. *Econometrica* 44, 5: 1087–1090.

Stigum, M., 1978. *The money market: myth, reality and practice*. Homewood, IL: Dow Jones–Irwin.

Suits, D. B., 1962. Forecasting and analysis with an econometric model. *American Economic Review* 52, 1: 104–132.

Sutherland, C. V. H., 1974. *Roman coins*. New York: Putnam.

Talamanca, M., 1954. *Contributi allo studio delle vendite all'asta nel mondo classico*. series 8, vol. 6, fascicolo 2. Rome: Academia Nasionale dei Lincei.

Thomson, F. P., 1968. *Money in the computer age*. London: Pergamon Press.

Tindbergen, J., 1959. *Selected papers*. Amsterdam: Elsevier.

Tobin, J., 1956. The interest elasticity of the transactions demand for cash. *Review of Economics and Statistics* 38: 241–247.

Tobin, J., 1982. Money and finance in the macroeconomic process. *Journal of Money, Credit, and Banking* 14, 2: 171–204.

Tobin, J., 1980. Explaining the existence and value of money: Comment. In *Models of monetary economies*, pp. 83–90. J. H. Karaken and N. Wallace, eds., Minneapolis: Federal Reserve Bank of Minneapolis.

Tsien, T.-H., 1986. *Science and civilization in China*. Vol. 5 of *Chemistry and chemistry technology*, pt. 1: *Paper and printing*, ed. J. Needham. Cambridge: Cambridge University Press.

United States Department of Commerce. *Statistical Abstract of the United States*, 1984, 1994.

Vaillant, G. C., [1944] 1950. *The Aztecs of Mexico*. Reprint, New York: Penguin.

Vicker, R., 1975. *The realms of gold*. New York: Scribner.

Vickrey, W. 1961. Counter speculation, auctions, and competitive sealed bids. *Journal of Finance* 19: 8–37.

Viner, J., 1931. Cost curves and supply curves. *Zeitschrift für Nationalökonomie* 3: 23–46.

Vissering, W., 1968. *On Chinese currency, coin, and paper money*. Taipei: Ch'Eng Wen.

von Neumann, J., [1937] 1945. A model of general economic equilibrium. *Review of Economic Studies* 13: 1–9.

Wald, A., 1951. On some systems of equations of mathematical economics (translated from German; original 1936). *Econometrica* 19: 368–403.

Walras, L., [1874] 1954. *Elements of pure economics*. Translated by W. Jaffe. Reprint, Homewood, IL: Irwin.

Walter, I., 1985. *Secret money*. Lexington, MA: Lexington Books.

Weintraub, E. R., 1978. *The microfoundations of macroeconomics*. Cambridge: Cambridge University Press.

Wemple, F. H., 1967. The silver market. In A. Butts and C. D. Coxe, eds., *Silver, Economics, Metallurgy, and Use*, pp. 36–56. Princeton, NJ: Van Nostrand.

Weiss, B., 1986. *The Hell of the English*. Lewisburg, Bucknell University Press.

Whitin, T. M., 1953. *The theory of inventory management*. Princeton: Princeton University Press.

Wicksell, K., [1935] 1962. *Lectures on political Economy*. Vol. 2, *Money* Reprint, London: Routledge and Kegan Paul.

Wilcox, R. L., 1967. Sources of silver. In A. Butts and C. D. Coxe, eds., *Silver, Economics, Metallurgy, and Use*, pp. 16–35. Princeton, NJ: D. Van Nostrand.

Williamson, O. E., 1975. *Markets and hierarchies: Analysis and antitrust implications.* New York: Free Press.

Wold, H. O., 1969. Mergers of economics and philosophy of science. *Synthèse* 2: 427–482.

Yang, L.-S., 1950. Buddhist monasteries and four money-raising institutions in Chinese history. *Harvard Journal of Asiatic Studies* 13: 174–191.

Yao, S., 1987. Strategic market games with complete markets. Ph.D. diss., University of California at Los Angeles.

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